

# Finite Excess in Transfinite Nim Multiplication

## Exact reductions, order bounds, and the 0/1/4 frontier

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### Abstract

Conway’s field of ordinal nimbers below  $\omega^{\omega^{\omega}}$  is organized by finite Kummer carries  $\alpha_p = \kappa_{f(p)} + m_p$ , where  $f(p) = \text{ord}_p(2)$  and  $m_p$  is Lenstra’s finite excess. The observed rule is  $m_p \in \{0, 1, 4\}$ . We prove that this rule is equivalent to four distinguished finite-field order statements: a primitive-support norm condition, maximal projective order on the ordinary odd Kummer spines, primitivity of a Gaussian period on the cubic spine, and a current-factor condition on the exceptional  $2 \cdot 3^k$  spine.

For each arm we derive exact norm, resultant, character, and polynomial criteria, together with unconditional order lower bounds that settle all sufficiently large primary factors relative to explicit regulators. The zero arm is proved at the first two power-of-two two-spine levels. The ordinary arm is reduced to selected trace and mixed-Jacobi coordinates; the cubic arm to a selected reciprocal Wendt factor and prescribed-trace character value; and the exceptional arm to nonvanishing of a selected cyclotomic antiunit index, equivalently exclusion of a named reciprocal cubic factor. We also prove that the principal generic alternatives— degree, norm, reciprocity, unweighted character products, bounded-support arguments, and several natural descent schemes—do not determine these selected values.

Thus the universal 0/1/4 rule remains conjectural, but its proof frontier is reduced to four explicit nonvanishing problems. A separate affine-line criterion characterizes Lenstra’s weaker absolute-boundedness conjecture, and the exceptional column admits the effective bound  $m_p < 16(3^k)^4$ .

**Scope of the results.** Statements presented as theorems, propositions, lemmas, and corollaries are proved here or deduced from cited results. Finite computations are identified as such and are used for certification and falsification only; they are not promoted to uniform theorems. Values imported from the published A380496 table are source-pinned inputs. The four universal order assertions are conjectures throughout.

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# 1 Introduction and main results

Conway's nim-multiplication turns the ordinals into a field of characteristic two. Below  $\omega^{\omega^\omega}$ , the multiplication law is organized by a Kummer tower. At an odd prime  $p$ , its new carry has the form

$$\alpha_p = \kappa_{f(p)} + m_p, \quad f(p) = \text{ord}_p(2),$$

where the transfinite term  $\kappa_{f(p)}$  is structural and the finite integer  $m_p$  is Lenstra's *excess*. Lenstra proved general lower bounds but left absolute boundedness open. All presently tabulated values suggest the following much sharper rule.

**Conjecture 1.1** (the 0/1/4 rule). *For every odd prime  $p$ ,*

$$m_p = \begin{cases} 0 & \text{if } \mathcal{Q}(f(p)) \text{ is not a singleton odd prime-power,} \\ 4 & \text{if } f(p) = 2 \cdot 3^k \text{ for some } k \geq 1, \\ 1 & \text{otherwise.} \end{cases}$$

The principal result of this note is a case-exhaustive reduction of the conjecture to four explicit order problems. The notation  $\Pi_h, T_h, \mathcal{L}_{r,a}, \gamma_k, \Psi_k, M_k$  is developed in the corresponding arm sections below.

**Theorem 1.2** (four-arm reduction). *Conjecture 1.1 is equivalent to the conjunction of the following four universal statements.*

(Z) *For every  $h > 1$  for which  $\mathcal{Q}(h)$  is not a singleton odd prime-power,*

$$\Pi_h \mid \text{ord}\left(N_{\mathbb{F}_{2^{d(\kappa_h)}/\mathbb{F}_{2^h}}(\kappa_h)}\right).$$

(O) *For every odd prime  $r \neq 3$  and every  $a \geq 1$ , the class of  $y_a = \kappa_{r^a} + 1$  in the projective quotient  $G_{r,a} = F_a^\times / F_{a-1}^\times$  has maximal prime-to- $r$  order: for every prime  $p \neq r$  dividing  $\mathcal{L}_{r,a}$ ,*

$$v_p(\text{ord}(y_a F_{a-1}^\times)) = v_p(\mathcal{L}_{r,a}).$$

(C) *The Gaussian period  $\gamma_k \in \mathbb{F}_{2^{3^k}}^\times$  is primitive for every  $k \geq 1$ .*

(D) *The current exceptional factor satisfies  $\Psi_k \mid \text{ord}(M_k)$  for every  $k \geq 1$ .*

*Proof.* If  $\mathcal{Q}(h)$  is not a singleton odd prime-power, the conjecture asks for  $m_p = 0$  at every prime with  $f(p) = h$ . Proposition 4.1 identifies this exactly with (Z).

If  $\mathcal{Q}(h) = \{r^a\}$  with  $r \neq 3$ , then Proposition 5.3 partitions the eligible row primes among the primary divisors of  $\mathcal{L}_{r,a}$  and identifies  $m_p = 1$  with the equality in (O).

For  $r = 3$ , one has  $d(\alpha_3) = 2$ , so the possible residue degrees are  $3^a$  and  $2 \cdot 3^a$ . Proposition 6.74 identifies the first family with (C). The unconditional lower bound  $m_p \geq 4$ , together with Proposition 8.3, identifies the second family with (D). These cases are disjoint and exhaust the definition of Conjecture 1.1.  $\square$

## **1.1 Main results and conjectural boundary**

The theorem is a reduction, not a proof of the four universal arms. The resulting boundary is summarized below; finite certifications are listed separately in Appendix A.

| Arm | Exact target                     | Proved here                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Remaining universal assertion                                                                                               |
|-----|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| (Z) | full primitive-support order     | synchronized-phase reduction; exact two-component resultant; nontrivial support in the power-of-two two-spine family; complete $h = 12, 24$ ; exact root-ancestry, Dickson dichotomy, and reverse-Dickson trace-divisor and Kummer-coset criteria; two-step selected quartic and all-or- $\ell$ -quartic Kummer dichotomy; an exact one-step countermodel whose predecessor and successor retain every earlier Fermat order factor; exact $3/2$ Kummer-tail and parallel-ray transport; canonical global lift, weighted reciprocity, selected first-Witt polynomial, named Artin-class and dihedral reflection-norm reductions, the complementary factor/cofactor resultant duality and its canonical CRT-projector/trace-annihilation boundary, the hypothetical-failure shifted-Fibonacci block collapse, and the selected Euler-factor/root-number boundary; a nonlinear lower-semiconjugacy resultant flag for the surviving odd Hasse collision, with a degree-range exclusion | generation for composite two-spine quotients, the general synchronized phase, and the singleton-even Conway–Fermat quotient |
| (O) | maximal primary projective order | transverse norm and relative-unit formulas; nontrivial order; absolute-Frobenius signed-ball factor range; weighted-selector and mixed-Jacobi formulas; projective minimal polynomial and binary-section ancestry; tower-faithful common-section no-go; uniform safe-prime trace detector; full-conductor truncated-log regulator and exact conductor- $p^2$ ray/Jacobi reciprocity; invariant realification and selected local reflection norm; the exact sparse marked full-conductor polynomial at $p = 359$ , its checked Hilbert-root certificate proving $m_{359} = 1$ , the exact crossed-tower computation proving $m_{719} = 1$ , the direct sparse certificate proving $m_{727} = 1$ , and a fixed-history audit of two corroborating source-pinned executions                                                                                                                                                                                                            | uniform selected-Conway nondivisibility across the ordinary spine                                                           |
| (C) | $\gamma_k$ primitive             | proof for $k \leq 3$ ; partition order bounds; exact derivative regulator, exceptional ray-character quotient, real-square resultant, reciprocity boundary, cyclic-parity obstruction, and augmentation-ideal and block normalization; Singer–Wendt factorization and selected $S_3$ -trace quotient; exact Frobenius orbit ranks and the complete normality                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | exclusion of the selected Wendt/trace fibre at the remaining current factors                                                |

The extensive non-determination results below serve one purpose: they separate these four selected-value problems from statements that follow only from degree, trace, norm, reciprocity, or generic character-sum bounds. They are obstructions to standard proof strategies, not additional conjectural arms.

## 1.2 Organization

Section 2 fixes the tower notation, and Section 3 gives the exact power and norm criteria. Sections 4, 5, 6, 7, and 8 treat the zero, ordinary, cubic, singleton-even, and exceptional structures. The singleton-even section is part of arm (Z), but is separated because its quadratic tower and generalized-Jacobian reformulations are substantial. Section 9 records the boundedness consequence and the four exact remaining theorem targets. Computational and implementation evidence is isolated in Appendix A.

## 2 Mathematical setup

### 2.1 Tower notation

For an odd prime  $p$ :

- $f(p) = \text{ord}_p(2)$ , the multiplicative order of 2 modulo  $p$ ;
- $d(x) = [\mathbb{F}_2(x) : \mathbb{F}_2]$ , the least finite-field degree containing the algebraic number  $x$ ;
- $\kappa_h$  is the Lenstra/DiMuro tower element indexed by  $h$ ;
- $\mathcal{Q}(h)$  is Lenstra's set of prime-power components appearing in  $\kappa_h$ ;
- the *excess*  $m_p$  is the least finite  $m$  such that  $\kappa_{f(p)} + m$  has no  $p$ -th root in  $\mathbb{F}_{2^{d(\kappa_{f(p)} + m)}}$ ;
- the Kummer carry is  $\alpha_p = \kappa_{f(p)} + m_p$ .

For the ordinal tower, a row with  $\mathcal{Q}(f(p)) = \{q\}$  and finite excess  $m$  gives the ordinal sum corresponding to  $\kappa_q + m$ . Addition of numbers is XOR;  $\otimes$  is the nim product.

The implementation boundary, external table provenance, and finite certifications are collected in Appendix A; none is used as evidence for an unbounded theorem.

## 3 The order criterion and structural norm reduction

### 3.1 The exact criterion

Throughout,  $E$  denotes the degree of the component field containing  $\beta = \kappa_{f(p)} + m$ , so  $f = f(p) \mid E$  and hence  $p \mid 2^E - 1$ .

**Proposition 3.1** (power criterion).  $\beta \in \mathbb{F}_{2^E}^\times$  has no  $p$ -th root in  $\mathbb{F}_{2^E}$  iff  $\beta^{(2^E-1)/p} \neq 1$ . Hence

$$m_p = \min\{m \geq 0 : (\kappa_{f(p)} + m)^{(2^E-1)/p} \neq 1\}.$$

*Proof.*  $\mathbb{F}_{2^E}^\times$  is cyclic of order  $2^E - 1$  and  $p \mid 2^E - 1$ ; in a cyclic group of order  $N$ ,  $\beta$  is a  $p$ -th power iff  $\beta^{N/p} = 1$ .  $\square$

**Corollary 3.2** (order form). If  $v_p(2^E - 1) = 1$ , then  $\beta$  has no  $p$ -th root in  $\mathbb{F}_{2^E}$  iff  $p \mid \text{ord}(\beta)$ , so  $m_p$  is the least  $m$  with  $p \mid \text{ord}(\kappa_{f(p)} + m)$ .

**Remark 3.3** (Wieferich caveat). In general “no  $p$ -th root” is equivalent to  $v_p(\text{ord } \beta) = v_p(2^E - 1)$ , which collapses to  $p \mid \text{ord}(\beta)$  only when  $v_p(2^E - 1) = 1$ . By lifting the exponent,

$$v_p(2^E - 1) = v_p(2^{f(p)} - 1) + v_p(E/f(p)).$$

Thus the hypothesis can fail either at base-2 Wieferich-type primes with  $p^2 \mid 2^{f(p)} - 1$  or when  $p \mid E/f(p)$ . The only known base-2 Wieferich primes are  $p = 1093$  and  $p = 3511$ , both of which lie inside the contributor’s extended A380496 auxiliary range. Its calculator uses the exact test of Proposition 3.1, so the reported zeroes assert full  $p^2$ -primary order; Appendix A records their thinner certificate boundary. For those rows, and for any row with  $p \mid E/f(p)$ , the exact power test (or the full- $p$ -part order condition) must be used. The local certifications of Section 6 compute orders exactly and include squarefreeness checks. The research-note predecessor of this document stated only the order form; the power criterion is the correct general statement.

### 3.2 Norm reduction

**Proposition 3.4** (norm reduction). *For  $\beta \in \mathbb{F}_{2^E}^\times$  with  $f = f(p) \mid E$ ,*

$$\beta^{(2^E-1)/p} = N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^f}}(\beta)^{(2^f-1)/p}.$$

*Proof.*  $N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^f}}(\beta) = \beta^{(2^E-1)/(2^f-1)}$ , and  $((2^E - 1)/(2^f - 1)) \cdot ((2^f - 1)/p) = (2^E - 1)/p$ .  $\square$

Frobenius conjugacy preserves multiplicative order, and relative norms are Frobenius invariant. Consequently each later selected-value test depends only on the Conway conjugacy class fixed by the irreducible Kummer steps, not on a choice of root or field presentation.

## 4 The zero arm

### 4.1 Primitive-support norm criterion

For  $h > 1$ , write

$$\Pi_h := \prod_{\substack{\ell \text{ odd prime} \\ \text{ord}_\ell(2)=h}} \ell^{v_\ell(2^h-1)}.$$

This is the full prime-power part of  $2^h - 1$  supported at primes whose order of 2 is exactly  $h$ ; using the full valuations, rather than only the radical, incorporates the Wieferich cases.

**Proposition 4.1** (exact-order support). *Let  $x = \kappa_h$ , let  $E = d(x)$ , and put*

$$T_h = N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^h}}(x).$$

*Then  $m_p = 0$  for every odd prime  $p$  with  $f(p) = h$  if and only if*

$$\Pi_h \mid \text{ord}(T_h).$$

*For an individual such  $p$ ,  $m_p = 0$  iff  $v_p(\text{ord } T_h) = v_p(2^h - 1)$ .*

*Proof.* The definition of  $\kappa_h$  gives  $h \mid E$ . Proposition 3.4 with  $\beta = x$  identifies the  $p$ -power test in  $\mathbb{F}_{2^E}$  with the same test on  $T_h$  in  $\mathbb{F}_{2^h}$ . In the cyclic group  $\mathbb{F}_{2^h}^\times$ , failure to be a  $p$ -th power is exactly the presence of the full  $p$ -primary part of the group order in  $\text{ord}(T_h)$ . Taking the product over the distinct eligible primes proves the claim. No hypothesis on  $p \nmid E/h$  is hidden here: if  $c = v_p((2^E - 1)/(2^h - 1))$ , the norm exponent gives  $v_p(\text{ord } T_h) = \max\{0, v_p(\text{ord } x) - c\}$ , so equality with  $v_p(2^h - 1)$  is equivalent to  $v_p(\text{ord } x) = v_p(2^E - 1)$ .  $\square$

**Proposition 4.2** (multicomponent Kummer no-collapse). *Suppose  $\mathcal{Q}(h) = \{q_1, \dots, q_t\}$  with  $t \geq 2$ . Put  $x_i = \kappa_{q_i}$ ,  $x = \kappa_h = \sum_i x_i$ ,  $E_i = d(x_i)$ ,  $E = d(x)$ , and  $D = \text{lcm}(E_1, \dots, E_t)$ . For a prime  $p$  with  $\text{ord}_p(2) = h$ , let*

$$T_h = N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^h}}(x).$$

*Then  $p > h$ ,  $p \nmid D$ , and, in additive notation for  $\mathbb{F}_{2^E}^\times/(\mathbb{F}_{2^E}^\times)^p$ ,*

$$[T_h] = \frac{E}{h}[x]. \quad (\text{MC}_1)$$

*The scalar  $E/h$  is nonzero modulo  $p$ , so norm reduction is an isomorphism on the relevant Kummer line. On the other hand every  $x_i$  is a  $p$ -th power in the common compositum  $\mathbb{F}_{2^D}$ . Thus no product of the individual component Kummer classes, or of their one-component norms, can produce the obstruction carried by the additive sum  $x$ .*

*If  $P_h(Y) \in \mathbb{F}_2[Y]$  is the minimal polynomial for  $\kappa_h$ , then*

$$m_p = 0 \iff P_h(Y^p) \text{ is irreducible over } \mathbb{F}_2. \quad (\text{MC}_2)$$

*In contrast, every individual component has trivial  $p$ -Kummer class. Hence the multicomponent zero arm is a genuinely new family of additive-resultant/Capelli irreducibility assertions, not a multiplicative consequence of the singleton components.*

*Proof.* Since  $h = \text{ord}_p(2)$  divides  $p - 1$ , one has  $p > h$ ; this is conditional on such a  $p$  and uses no Zsigmondy assertion. Write  $q_i = r_i^{a_i}$ . Lenstra gives  $E_i = q_i b_i$  with every prime divisor of  $b_i$  smaller than  $r_i$ . Since  $r_i \mid h$ , every prime divisor of  $D$  is at most  $h$ , so  $p \nmid D$ . Also  $x \in \mathbb{F}_{2^D}$ , hence  $E \mid D$  and  $p \nmid E/h$ .

Writing  $n = E/h$ , the norm exponent satisfies

$$\frac{2^E - 1}{2^h - 1} = 1 + 2^h + \dots + 2^{(n-1)h} \equiv n = \frac{E}{h} \pmod{p},$$

because  $2^h \equiv 1 \pmod{p}$ . This proves  $(\text{MC}_1)$ ; the natural map from the  $p$ -Kummer quotient of  $\mathbb{F}_{2^h}$  to that of  $\mathbb{F}_{2^E}$  is an isomorphism because the same scalar  $n$  is nonzero modulo  $p$ .

For each  $i$ ,  $q_i \mid h$  gives  $\kappa_{q_i} \leq \kappa_h$ . If  $h \mid E_i$ , the defining minimality of  $\kappa_h$  gives the reverse inequality, hence  $\kappa_h = \kappa_{q_i}$ , contradicting uniqueness of the multicomponent expansion. Thus  $h \nmid E_i$ , so  $p \nmid 2^{E_i} - 1$ . The  $p$ -power map is consequently an automorphism of  $\mathbb{F}_{2^{E_i}}^\times$ , and  $x_i$  is a  $p$ -th power there and hence in  $\mathbb{F}_{2^D}$ . Because  $p \nmid D/E$ , extension from  $\mathbb{F}_{2^E}$  to  $\mathbb{F}_{2^D}$  is injective on  $p$ -Kummer quotients, so this comparison loses no class of  $x$ .

Finally Capelli's lemma says  $P_h(Y^p)$  is irreducible iff  $Y^p - x$  is irreducible over  $\mathbb{F}_{2^E}$ . Since  $\mu_p \subset \mathbb{F}_{2^E}$ , the latter holds iff  $x \notin (\mathbb{F}_{2^E}^\times)^p$ , which is exactly  $m_p = 0$ . This proves  $(\text{MC}_2)$ .  $\square$

**Proposition 4.3** (synchronized additive resultants and the missing phase). *Put  $Q = 2^h$ ,  $K = \mathbb{F}_Q$ , and suppose  $p \mid Q - 1$ . For  $1 \leq i \leq t$ , let  $L_i = \mathbb{F}_{Q^{r_i}}$  and choose  $u_i \in L_i$  of full degree  $r_i$  over  $K$ . Set*

$$R = \text{lcm}(r_1, \dots, r_t), \quad M = \mathbb{F}_{Q^R}, \quad u = \sum_i u_i \neq 0,$$

*and assume  $p \nmid R$ . Exponents in the following formula are read modulo the corresponding  $r_i$ . The diagonal translation action on*

$$\prod_i \mathbb{Z}/r_i \mathbb{Z}$$



has orbit set

$$\Omega = \left( \prod_i \mathbb{Z}/r_i\mathbb{Z} \right) / \langle (1, \dots, 1) \rangle.$$

For a representative  $\mathbf{d} = (d_i)$  of an orbit, define

$$F_{\mathbf{d}}(Z) = \prod_{j=0}^{R-1} \left( Z + \sum_i u_i^{Q^{j+d_i}} \right). \quad (\text{MC}_3)$$

Then  $F_{\mathbf{d}} \in K[Z]$ , it depends only on the class of  $\mathbf{d}$ , and the ordinary componentwise additive composed product factors as

$$\prod_{0 \leq j_i < r_i} \left( Z + \sum_i u_i^{Q^{j_i}} \right) = \prod_{\mathbf{d} \in \Omega} F_{\mathbf{d}}(Z). \quad (\text{MC}_4)$$

The distinguished common-embedding sum is the zero phase, and

$$F_{\mathbf{0}}(0) = N_{M/K}(u), \quad [F_{\mathbf{0}}(0)] = R[u] \quad \text{in } M^\times/M^{\times p}. \quad (\text{MC}_5)$$

Consequently

$$u \notin M^{\times p} \iff F_{\mathbf{0}}(0) \notin K^{\times p}. \quad (\text{MC}_6)$$

If the  $r_i$  are pairwise coprime,  $\Omega$  is a singleton and the ordinary additive resultant is the exact obstruction. Otherwise it gives only the product of several synchronized Frobenius phases. Independently replacing  $u_i$  by  $u_i^{Q^{e_i}}$  preserves every separate minimal polynomial and merely permutes these phases. Thus separate component polynomials do not identify the distinguished factor  $F_{\mathbf{0}}$ .

For the Conway components of Proposition 4.2, take

$$r_i = \frac{E_i}{\gcd(E_i, h)}, \quad R = \frac{D}{h}, \quad L_i = K(x_i) = \mathbb{F}_{2^{\text{lcm}(h, E_i)}}, \quad M = \mathbb{F}_{2^D}.$$

Then  $p \nmid R$ , and

$$F_{\mathbf{0}}(0) = N_{\mathbb{F}_{2^D}/\mathbb{F}_{2^h}}(x) = T_h^{D/E}.$$

Since  $p \nmid D/E$ , this is exactly the same Kummer obstruction as (MC<sub>1</sub>), now expressed as one synchronized additive-resultant factor rather than the product over all relative embeddings.

*Proof.* The diagonal element  $(1, \dots, 1)$  has order  $R$ , so its action is free and its orbits partition the tuples  $(j_i)$  into sets of size  $R$ . The  $Q$ -Frobenius cycles the factors in (MC<sub>3</sub>), proving that  $F_{\mathbf{d}}$  has coefficients in  $K$ ; changing the representative translates the index  $j$ . Partitioning the factors on the left of (MC<sub>4</sub>) by these diagonal orbits proves the factorization.

For the zero phase, the factors are precisely the  $M/K$ -conjugates of  $u$ , so the constant term is its norm (signs disappear in characteristic two). Under the natural inclusion of  $K^\times/K^{\times p}$  into  $M^\times/M^{\times p}$ , the norm exponent is

$$1 + Q + \dots + Q^{R-1} \equiv R \pmod{p}.$$

The assumption  $p \nmid R$  proves (MC<sub>5</sub>)–(MC<sub>6</sub>). The description of  $\Omega$  proves the conditional pairwise-coprime case, and independent Frobenius shifts translate the corresponding phase class, proving the information-loss statement.

In the Conway application,  $K(x_i) = \mathbb{F}_{2^{\text{lcm}(h, E_i)}}$ , whose degree over  $K = \mathbb{F}_{2^h}$  is the displayed  $r_i$ . Taking least common multiples gives  $R = D/h$ . Proposition 4.2 proves  $p \nmid D$ , hence  $p \nmid R$ , and norm transitivity gives the last displayed equality. It also gives  $p \nmid D/E$ , completing the comparison. No pairwise-coprimality assertion for the actual  $r_i$  is used.  $\square$

**Proposition 4.4** (strict phase no-go). *Separate component degrees, minimal polynomials, and trivial  $p$ -Kummer classes do not determine the Kummer class of an additively composed sum, even when both possible sums have full compositum degree.*

*Proof.* Here is an exact example. In this proof, hexadecimal bit  $j$  denotes the coefficient of  $T^j$ . Put

$$k = \mathbb{F}_2[T]/(m), \quad m = 11\mathbf{b} = T^8 + T^4 + T^3 + T + 1.$$

Rabin's criterion is certified by

$$T^{256} = T \pmod{m}, \quad T^{16} + T = 5\mathbf{c} \pmod{m},$$

and the exact Bézout identity

$$13m + 515\mathbf{c} = 1,$$

so  $k = \mathbb{F}_{256}$ . Let  $K = \mathbb{F}_4 \subset k$ , put  $\gamma = \mathbf{bc}$ , so  $\gamma^2 + \gamma + 1 = 0$ , and take

$$a = 0\mathbf{c} = T^3 + T^2, \quad b = 2\mathbf{e} = T^5 + T^3 + T^2 + T.$$

Their 4-Frobenius data are

$$\begin{aligned} a^4 &= \mathbf{b}0, & a^{16} &= a, \\ b^4 &= 37, & b^{16} &= \mathbf{c}3, & b^{64} &= 67, \end{aligned}$$

so  $a$  and  $b$  have degrees two and four over  $K$ , respectively. Moreover

$$a^{85} = b^{85} = 1,$$

and hence both are cubes in  $k^\times$ . The elements  $b$  and  $b^4$  are  $K$ -conjugate and have the same minimal polynomial; explicitly

$$\begin{aligned} \min_K(a) &= U^2 + \gamma U + 1, \\ \min_K(b) &= \min_K(b^4) = U^4 + (\gamma + 1)U^3 + \gamma U^2 + 1. \end{aligned}$$

Nevertheless

$$a + b = 22, \quad (a + b)^{85} = \gamma \neq 1,$$

whereas

$$a + b^4 = 3\mathbf{b}, \quad (a + b^4)^{85} = 1. \tag{MC_7}$$

Both sums have degree four over  $K$ : their 4-Frobenius orbits are

$$(22, 87, \mathbf{cf}, \mathbf{d7}), \quad (3\mathbf{b}, 73, 6\mathbf{b}, 9\mathbf{e}).$$

Since  $k^\times$  has order 255, a nonzero element is a cube exactly when its 85-th power is one. Thus the first sum is a noncube and the second is a cube, although the separate component data are identical. As  $\mu_3 \subset K$ , Capelli's lemma says that composing their irreducible degree-four minimal polynomials with  $X^3$  gives, respectively, an irreducible and a reducible polynomial. This proves the claimed strict no-go. It is a counterexample to a componentwise inference, not to the distinguished Conway statement.  $\square$

**Corollary 4.5** (two-component intersection resultant). *In Proposition 4.3, suppose  $t = 2$ . Put*

$$A = K(u_1) = \mathbb{F}_{Q^{r_1}}, \quad B = K(u_2) = \mathbb{F}_{Q^{r_2}}, \quad g = \gcd(r_1, r_2), \quad J = A \cap B = \mathbb{F}_{Q^g},$$

*and let  $A_i(U)$  be the minimal polynomial of  $u_i$  over  $J$ . Then*

$$\text{Res}_U(A_1(U), A_2(U)) = N_{AB/J}(u_1 + u_2), \quad (\text{MC}_8)$$

*and*

$$u_1 + u_2 \notin (AB)^{\times p} \iff \text{Res}_U(A_1, A_2) \notin J^{\times p}. \quad (\text{MC}_9)$$

*Thus the exact positive additive invariant is a single resultant over the intersection field, whose coefficients retain the relative Frobenius alignment omitted by the separate  $K$ -polynomials.*

*Suppose in addition that  $p \nmid \text{ord}(u_1)\text{ord}(u_2)$ , as for the Conway components  $u_i = x_i$ , which lie in the original fields  $\mathbb{F}_{2^{E_i}}$  with  $p \nmid 2^{E_i} - 1$ . Then each  $u_i$  is a  $p$ -th power and*

$$[u_1 + u_2] = \left[ 1 + \frac{u_1}{u_2} \right]. \quad (\text{MC}_{10})$$

*Writing  $N = \text{ord}(u_1/u_2)$ , one has  $p \nmid N$ . After taking the Teichmüller lift  $u_1/u_2 \leftarrow \zeta_N$ , the remaining assertion is exactly nontriviality of the fixed power-residue symbol*

$$\left( \frac{1 + \zeta_N}{\mathfrak{P}} \right)_p$$

*at the prime above two selected by the common Conway embedding. Hence the two-component zero arm is a cyclotomic-unit Artin-symbol problem after the intersection resultant has removed all generic additive data.*

*Proof.* The extensions  $A/J$  and  $B/J$  have coprime degrees  $r_1/g$  and  $r_2/g$ , so they are linearly disjoint and their pairs of relative conjugates are exactly the  $AB/J$ -conjugates of  $u_1 + u_2$ . Taking their product gives (MC<sub>8</sub>). Here  $\mu_p \subset K \subset J$ , and

$$[AB : J] = \frac{R}{g}$$

is prime to  $p$ , because  $p \nmid R$ . After inclusion in  $(AB)^{\times}/(AB)^{\times p}$ , the resultant class is therefore  $[AB : J][u_1 + u_2]$ , proving (MC<sub>9</sub>).

Under the additional order hypothesis, division by the  $p$ -th power  $u_2$  gives (MC<sub>10</sub>), and the ratio has order prime to  $p$ . The standard definition of the power-residue symbol identifies nontriviality of the residue class  $1 + u_1/u_2$  with the displayed symbol of its Teichmüller lift. This proves the final formulation.  $\square$

**Proposition 4.6** (a power-of-two two-spine family has primitive support). *Let  $r$  be an odd prime for which*

$$b_r := d(\alpha_r) = 2^t.$$

*Take  $a \geq t + 1$ ,  $b \geq 1$ , and  $h = 2^a r^b$ . Then*

$$\mathcal{Q}(h) = \{2^a, r^b\}, \quad d(\kappa_h) = h, \quad \Pi_h = \Phi_h(2). \quad (\text{MC}_{11})$$

*Put  $x = \kappa_{r^b}$ ,  $c = \kappa_{2^a}$ , and  $z = \kappa_h = x + c$ , and define the two maximal proper subfields*

$$A = \mathbb{F}_{2^{h/2}}, \quad B = \mathbb{F}_{2^{h/r}}, \quad J = A \cap B = \mathbb{F}_{2^{h/(2r)}}.$$

The primitive-support quotient is cyclic of order  $\Pi_h$ :

$$\mathcal{T}_h := \mathbb{F}_{2^h}^\times / (A^\times B^\times), \quad |\mathcal{T}_h| = \Phi_h(2) = \Pi_h. \quad (\text{MC}_{12})$$

The class of  $z$  in  $\mathcal{T}_h$  is nontrivial. Consequently  $\text{ord}(z)$  is divisible by at least one prime  $p$  with  $\text{ord}_p(2) = h$ . More precisely, the full zero-support assertion at this  $h$  is equivalent to

$$\langle z(A^\times B^\times) \rangle = \mathcal{T}_h. \quad (\text{MC}_{13})$$

The full zero-support assertion in particular holds whenever  $\Phi_h(2)$  is prime. This gives, for example, a proof without an order computation of the complete multicomponent zero arm at  $h = 12$  and  $h = 24$ , whose quotients have orders 13 and 241, respectively.

*Proof.* By the odd Kummer-spine degree formula,

$$d(\kappa_{r^b}) = b_r r^b = 2^t r^b.$$

Lenstra's component recursion first constructs  $\mathcal{Q}(r^b) = \{r^b\}$ , and then adjoins the  $2^a$ -component exactly when  $2^a \nmid d(\kappa_{r^b})$ . The hypothesis  $a \geq t + 1$  therefore gives the first assertion in  $(\text{MC}_{11})$ . Moreover

$$d(\kappa_{2^a}) = 2^a, \quad \text{lcm}(2^a, 2^t r^b) = h.$$

Thus  $\kappa_h$  lies in  $\mathbb{F}_{2^h}$ ; its defining degree is divisible by  $h$ , so  $d(\kappa_h) = h$ .

Write  $q = 2^{h/(2^r)}$ . The standard prime-power substitution identity for cyclotomic polynomials gives

$$\Phi_h(2) = \Phi_{2^r}(q) = \frac{q^r + 1}{q + 1}.$$

If a prime  $\ell$  divides this value but  $\text{ord}_\ell(2) < h$ , the cyclotomic-divisor criterion forces  $\ell \mid h$ . The only possible odd exception is therefore  $r$ . The Kummer construction contains  $\mu_r$  in  $\mathbb{F}_{2^{b_r}}$ , so  $f(r) \mid b_r = 2^t$ ; write  $f(r) = 2^u$  with  $u \leq t$ . The exceptional case would require

$$\frac{h}{\text{ord}_r(2)} = 2^{a-u} r^b$$

to be a power of  $r$ , which it is not because  $a \geq t + 1 > u$ . Hence every prime divisor of  $\Phi_h(2)$  has exact order  $h$ , and all of its valuation in  $2^h - 1$  occurs for the first time in  $\Phi_h(2)$ . This proves  $\Pi_h = \Phi_h(2)$ .

The intersection and compositum rules for finite fields give  $A \cap B = J$  and  $AB = \mathbb{F}_{2^h}$ . Since the multiplicative group of a finite field is cyclic,

$$|A^\times B^\times| = \text{lcm}(q^r - 1, q^2 - 1) = \frac{(q^r - 1)(q^2 - 1)}{q - 1}.$$

Its index in  $q^{2^r} - 1$  is

$$\frac{(q^{2^r} - 1)(q - 1)}{(q^r - 1)(q^2 - 1)} = \frac{q^r + 1}{q + 1} = \Phi_h(2),$$

which proves  $(\text{MC}_{12})$ .

It remains to show that  $z \notin A^\times B^\times$ . The degree formulas and  $a \geq t + 1$  give

$$x \in A \setminus J, \quad c \in B \setminus J.$$

Because  $[A : J] = r$  and  $[B : J] = 2$  are coprime, multiplication induces an isomorphism

$$A \otimes_J B \xrightarrow{\sim} \mathbb{F}_{2^h}.$$

Under it,  $z$  is the tensor

$$x \otimes 1 + 1 \otimes c.$$

The pairs  $\{x, 1\} \subset A$  and  $\{1, c\} \subset B$  are both  $J$ -linearly independent, so this tensor has rank two. Every element of  $A^\times B^\times$ , on the other hand, is the image of a nonzero rank-one tensor  $u \otimes v$ . Thus  $z \notin A^\times B^\times$ , and its quotient class is nontrivial.

Every prime dividing the order of the quotient class divides  $\Pi_h$ , hence has exact order  $h$ . Conversely no such prime divides  $|A^\times B^\times|$ , so the full  $\Pi_h$ -part of  $\text{ord}(z)$  is exactly the order of its class in  $\mathcal{T}_h$ . This proves (MC<sub>13</sub>). If  $\Phi_h(2)$  is prime, nontriviality of the class already means generation. Finally  $\Phi_{12}(2) = 13$  and  $\Phi_{24}(2) = 241$ , so the complete zero arm holds at those two levels.  $\square$

**Proposition 4.7** (cyclic-variation sieve for the two-spine cross ratio). *Retain the hypotheses and notation of Proposition 4.6. Put  $q = 2^{h/(2r)}$ ,  $L_0 = (q^r - 1)(q + 1)$ ,  $P = z^{L_0}$ , and  $c' = c^q$ . The same Hilbert–90 calculation gives the exact representative*

$$P = \left( \frac{x + c'}{x + c} \right)^{q+1} = \frac{(x + c')(\zeta x + c)}{(x + c)(\zeta x + c')}, \quad (\text{MC}_{14})$$

where  $x^q = \zeta x$  and  $\zeta$  is a primitive  $r$ -th root in  $J$ . Set

$$R = \frac{r-1}{2}, \quad \mathcal{D}_R = \left\{ \mathbf{d} \in \mathbb{Z}^r : \sum_{j=0}^{r-1} d_j = 0, \quad \sum_{j=0}^{r-1} |d_j| \leq R \right\}, \quad B_r^{\text{cyc}} = |\mathcal{D}_R|. \quad (\text{MC}_{15})$$

Then

$$\text{ord}(P) = \text{ord}(z(A^\times B^\times)) \geq B_r^{\text{cyc}}. \quad (\text{MC}_{16})$$

In particular,

$$B_r^{\text{cyc}} \geq \sum_{\substack{k \geq 0 \\ 2k \leq R}} \binom{r}{2k} \binom{2k}{k}, \quad (\text{MC}_{17})$$

so  $B_r^{\text{cyc}} \geq 1 + r(r-1)$  for  $r \geq 5$ . Consequently, if a prime  $p \mid \Phi_h(2)$  satisfies

$$p > \frac{\Phi_h(2)}{B_r^{\text{cyc}}}, \quad (\text{MC}_{18})$$

then the class of  $z$  contains the full  $p$ -primary part of the primitive-support quotient. This is an unconditional lower bound and large-prime sieve; it does not prove generation when the quotient is composite.

*Proof.* The order equality in (MC<sub>16</sub>) follows first from the preceding proposition. Indeed,  $|A^\times B^\times| = L_0$ , and every prime divisor of  $\Phi_h(2)$  has exact order  $h$  for the base 2, so none divides  $L_0$ . Thus exponentiation by  $L_0$  kills the subgroup  $A^\times B^\times$  and is an automorphism on the quotient of order  $\Phi_h(2)$ . Also  $P^{\Phi_h(2)} = z^{|\mathbb{F}_{2^h}^\times|} = 1$ , so  $\langle P \rangle$  intersects  $A^\times B^\times$  trivially; projection to the quotient is injective on  $\langle P \rangle$ .

Index all subscripts modulo  $r$ , and put

$$f_j = \frac{\zeta^j x + c'}{\zeta^j x + c}, \quad P_j = \frac{f_j}{f_{j+1}}.$$

Since  $c' = c^q$ ,  $c'^q = c$ , and  $\zeta \in J = \mathbb{F}_q$ , one has

$$f_j^q = f_{j+1}^{-1}, \quad P_0 = P, \quad P_j = P^{(-q)^j}. \quad (\text{MC}_{19})$$

The last identity follows inductively from  $P_j^q = P_{j+1}^{-1}$ .

For  $\mathbf{d} \in \mathcal{D}_R$ , there is a unique cyclic integer vector  $\mathbf{e} = (e_j)$  with  $e_0 = 0$  and

$$d_j = e_j - e_{j-1}.$$

Map  $\mathbf{d}$  to  $W_{\mathbf{d}} = \prod_j P_j^{e_j}$ . By (MC<sub>19</sub>), every  $W_{\mathbf{d}}$  is a power of  $P$ . We show that these powers are distinct.

Suppose  $W_{\mathbf{d}} = W_{\mathbf{d}'}$ , put  $g_j = e_j - e'_j$ , and let  $\delta_j = g_j - g_{j-1} = d_j - d'_j$ . Telescoping the  $f_j$ -exponents gives

$$1 = \prod_j P_j^{g_j} = \prod_j f_j^{\delta_j}. \quad (\text{MC}_{20})$$

After clearing denominators, the two sides of (MC<sub>20</sub>) are products of the linear polynomials

$$\zeta^j X + c, \quad \zeta^j X + c' \quad (0 \leq j < r)$$

over  $B$ , each of degree

$$\sum_j |\delta_j| \leq \sum_j |d_j| + \sum_j |d'_j| \leq 2R = r - 1.$$

Their values agree at  $x$ . Since  $[B(x) : B] = r$ , the two polynomial products are therefore identical.

The displayed  $2r$  linear factors are pairwise nonassociate. This is clear within either family. A cross-family association would give  $c'/c \in \mu_r$ . But  $(c'/c)^{q+1} = 1$ , while  $r \mid q - 1$  and  $r$  is odd, so  $\gcd(r, q + 1) = 1$ ; hence it would force  $c' = c$ , contrary to  $c \in B \setminus J$ . Unique factorization now forces  $\delta = 0$ . Thus  $\mathbf{g}$  is constant, and  $g_0 = 0$  forces  $\mathbf{g} = 0$ , proving  $\mathbf{d} = \mathbf{d}'$ . This proves the injection  $\mathcal{D}_R \hookrightarrow \langle P \rangle$  and hence (MC<sub>16</sub>).

For (MC<sub>17</sub>), take the vectors with  $k$  entries equal to 1,  $k$  entries equal to  $-1$ , and all other entries zero. There are  $\binom{r}{2k} \binom{2k}{k}$  of them and their  $\ell^1$ -norm is  $2k$ . Finally, write  $D = \text{ord}(P) \mid \Phi_h(2)$ . If  $D$  lacked the full  $p$ -primary part, then  $D \mid \Phi_h(2)/p$ , contradicting (MC<sub>16</sub>) under (MC<sub>18</sub>). This proves the sieve.  $\square$

**Remark 4.8** (sharp boundary of the tensor argument). Write more generally

$$d(\alpha_r) = 2^t c, \quad c > 1 \text{ odd},$$

and take  $a > t$ ,  $b \geq 1$ . Over  $J = \mathbb{F}_{2^{2t}}$ , the fields generated by  $\kappa_{r^b}$  and  $\kappa_{2^a}$  have coprime relative degrees

$$m = cr^b, \quad n = 2^{a-t}.$$

Tensor rank therefore still proves that their sum is nontrivial in the product quotient of their compositum. If  $q = |J|$ , however, that quotient has order

$$\frac{(q^{mn} - 1)(q - 1)}{(q^m - 1)(q^n - 1)} = \prod_{\substack{d_1 \mid m, d_1 > 1 \\ d_2 \mid n, d_2 > 1}} \Phi_{d_1 d_2}(q). \quad (\text{MC}_{21})$$

The desired  $h = 2^a r^b$  support occupies only the factor  $\Phi_{nr^b}(q)$ . Tensor nontriviality may instead be supported on a factor whose  $d_1$  uses a nontrivial divisor of  $c$ ; norming the degree- $hc$  compositum down to  $\mathbb{F}_{2^h}$  kills every such prime-order component that does not divide  $2^h - 1$ . Thus the rank-two proof does not project to the required zero phase when  $c > 1$ . The hypothesis  $d(\alpha_r) = 2^t$  is exactly where the present tensor/divisor argument isolates a single primitive-support factor; the general case still needs a theorem controlling the synchronized norm.

## 5 Ordinary odd Kummer spines

### 5.1 Transverse and projective reductions

The singleton-odd case admits a sharper field square. Let  $r$  be an odd prime,  $q = r^a$ , and set  $b_r = d(\alpha_r)$ , where  $\kappa_q^q = \alpha_r$ . Since  $\alpha_r < \kappa_r$ , Lenstra's field description gives  $r \nmid b_r$ , while the successive Kummer equations have degree  $r$  [22, Proposition 1.8]; hence

$$d(\kappa_q) = qb_r, \quad \gcd(q, b_r) = 1.$$

**Proposition 5.1** (singleton transverse-norm criterion). *Suppose  $\mathcal{Q}(h) = \{q\}$  with  $q = r^a$  odd. Then  $h = qs$  for a divisor  $s \mid b_r$ . In the field square*

$$F = \mathbb{F}_{2^{qb_r}}, \quad B = \mathbb{F}_{2^{b_r}}, \quad K = \mathbb{F}_{2^{qs}}, \quad C = \mathbb{F}_{2^s},$$

one has  $B \cap K = C$  and  $F = BK$ . Put  $x = \kappa_q$  and

$$\Theta_{q,s} := N_{F/K}(x+1) \in K^\times.$$

For every prime  $p$  with  $f(p) = h$ :

(i)  $x$  is a  $p$ -th power in  $F$ , so  $m_p \geq 1$ ;

(ii)  $m_p = 1$  iff

$$v_p(\text{ord } \Theta_{q,s}) = v_p(2^{qs} - 1);$$

(iii) the easy Kummer norm is

$$N_{F/B}(x+1) = 1 + \alpha_r,$$

but  $p \nmid |B^\times|$ , so this norm cannot see the required  $p$ -primary obstruction. The norm to  $K$  is genuinely transverse.

*Proof.* Lenstra's component theorem gives  $\kappa_h = \sum_{t \in \mathcal{Q}(h)} \kappa_t$  and, for each component  $t$ ,  $t \mid h$  and  $\gcd(t, h/t) = 1$  [22, Corollary 2.2]. Thus the singleton assumption identifies  $\kappa_h = \kappa_q$ . Since  $h \mid d(\kappa_h) = d(\kappa_q) = qb_r$ , one has  $h = qs$  with  $s \mid b_r$ . The intersection and compositum statements follow from the gcd/lcm rules for finite subfields and  $\gcd(q, b_r) = 1$ .

Since  $p > h \geq r$ , one has  $p \nmid q$ . Also  $f(p) = qs \nmid b_r$ , so  $p \nmid 2^{b_r} - 1$ . The relation  $x^q = \alpha_r \in B$  therefore gives  $p \nmid \text{ord}(x)$ , proving (i), including the full  $p$ -primary statement. Part (ii) is Proposition 3.4 applied to  $x+1$ . Finally,  $X^q + \alpha_r$  is the minimal polynomial of  $x$  over  $B$ , and evaluating it at 1 gives (iii); the invisibility follows again from  $p \nmid 2^{b_r} - 1$ .  $\square$

**Corollary 5.2** (Hilbert–90 cross-ratio for the transverse class). *Retain Proposition 5.1, put  $y = x+1$ , and define*

$$z = x^{2^{b_r}-1}, \quad U = y^{2^{b_r}-1} = \frac{zx+1}{x+1}.$$

Then  $z$  has exact order  $q$ , and

$$U \in \ker(N_{F/B} : F^\times \longrightarrow B^\times).$$

For every prime  $p$  with  $f(p) = h = qs$ ,

$$v_p(\text{ord } U) = v_p(\text{ord}(x+1)) = v_p(\text{ord } \Theta_{q,s}). \quad (\text{T})$$

Thus the singleton-one arm is equivalently the assertion that the explicit Hilbert–90 cross-ratio  $(1 + zx)/(1 + x)$  contains the full  $p$ -primary part. In particular, the large transverse norm may be replaced, for the unresolved primary component, by one norm-one cyclotomic cross-ratio. On the one-dimensional quotient

$$\mathcal{V} = F^\times / (F^\times)^p$$

the automorphism  $\sigma = \text{Frob}^{b_r}$  acts by  $\lambda = 2^{b_r} \pmod{p}$ , an element of exact order  $q$ , and

$$[U] = (\lambda - 1)[y], \quad [\Theta_{q,s}] = (b_r/s)[y]. \quad (\text{T}_1)$$

Both displayed scalars are nonzero, whereas

$$(1 + \sigma + \cdots + \sigma^{q-1})[y] = 0. \quad (\text{T}_2)$$

Finally the same target has the exact translated-composition form

$$m_p = 1 \iff ((T^p + 1)^q + \alpha_r) \text{ is irreducible over } B. \quad (\text{T}_3)$$

*Proof.* The  $2^{b_r}$ -Frobenius generates  $\text{Gal}(F/B)$  and sends  $x$  to  $zx$ . Since  $x^q = \alpha_r \in B$ , the class  $xB^\times$  has exact order  $q$ : if  $x^d \in B$  for  $0 < d < q$ , then  $x$  would satisfy a polynomial of degree  $d$  over  $B$ , contradicting  $[B(x) : B] = q$ . In the cyclic group  $F^\times$ , the map

$$F^\times / B^\times \longrightarrow F^\times, \quad gB^\times \longmapsto g^{|B^\times|}$$

is injective, because  $B^\times$  is the unique subgroup of order  $|B^\times| = 2^{b_r} - 1$ . Its value at  $xB^\times$  is  $z$ , so  $z$  has exact order  $q$ . The displayed formula for  $U$  is immediate, and its form  $y^{\text{Frob}^{b_r} - 1}$  gives norm one by Hilbert 90.

Now  $f(p) = qs \nmid b_r$ , so  $p \nmid 2^{b_r} - 1$ . Exponentiation by  $2^{b_r} - 1$  therefore preserves the  $p$ -primary part of the order of  $y$ , proving the first equality in (T). On the other hand

$$\Theta_{q,s} = y^A, \quad A = \frac{2^{qb_r} - 1}{2^{qs} - 1} \equiv \frac{b_r}{s} \pmod{p}.$$

Every prime divisor of  $b_r = d(\alpha_r)$  is smaller than  $r$ , by Lenstra’s field description and  $\alpha_r < \kappa_r$  [22, Proposition 1.8]; since  $p > h \geq r$ , one has  $p \nmid b_r/s$ . Thus  $p \nmid A$ , so this exponentiation also preserves the  $p$ -primary part, proving the second equality.

The quotient  $\mathcal{V}$  has order  $p$ . Since  $\text{ord}_p(2) = qs$ ,  $b_r = s(b_r/s)$ , and  $\gcd(q, b_r) = 1$ , the order of  $2^{b_r}$  modulo  $p$  is exactly  $q$ . The definitions give  $[U] = (\sigma - 1)[y] = (\lambda - 1)[y]$ , while the preceding congruence for  $A$  gives the second equality in (T<sub>1</sub>). The geometric sum in (T<sub>2</sub>) vanishes because  $\lambda$  has exact order  $q$ . This is the representation-theoretic reason that the easy  $B$ -valued norm sees none of the target class.

The minimal polynomial of  $y = x+1$  over  $B$  is  $(Y+1)^q + \alpha_r$ . Since  $p$  is odd, Capelli’s lemma says that its composition with  $T^p$  is irreducible exactly when  $y$  is not a  $p$ -th power in  $F$ . Proposition 5.1 identifies this with  $m_p = 1$ , proving (T<sub>3</sub>).  $\square$

**Proposition 5.3** (one projective-order target for each odd Kummer spine). *Fix an odd prime  $r$ , put  $b = d(\alpha_r)$ , and define*

$$x_0 = \alpha_r, \quad x_a = \kappa_{r^a} \ (a \geq 1), \quad y_a = x_a + 1, \quad F_a = \mathbb{F}_{2^{r^a b}}.$$

Then, for every  $a \geq 1$ ,

$$N_{F_a/F_{a-1}}(y_a) = y_{a-1}. \quad (\text{P}_1)$$



The projective quotient

$$G_{r,a} = F_a^\times / F_{a-1}^\times$$

is cyclic of order

$$\mathcal{L}_{r,a} = \frac{2^{r^a b} - 1}{2^{r^{a-1}b} - 1} = \Phi_{r^a}(2^b). \quad (\text{P}_2)$$

If  $p \neq r$  is a prime divisor of  $\mathcal{L}_{r,a}$ , then

$$\text{ord}_p(2^b) = r^a, \quad f(p) = r^a s \text{ for some } s \mid b, \quad \mathcal{Q}(f(p)) = \{r^a\}. \quad (\text{P}_3)$$

Conversely every prime  $p$  in a singleton row  $\mathcal{Q}(f(p)) = \{r^a\}$  occurs in this way. For every such  $p$ ,

$$m_p = 1 \iff v_p(\text{ord}(y_a F_{a-1}^\times)) = v_p(\mathcal{L}_{r,a}). \quad (\text{P}_4)$$

Consequently every singleton-one test for a fixed  $r$  is a primary-order test on this one projective class, simultaneously for all residue-degree divisors  $s \mid b$ . For  $r \neq 3$ , the 0/1/4 rule asks that its prime-to- $r$  part be maximal. For  $r = 3$ , the two cyclotomic halves instead split as in the following remark.

*Proof.* The Kummer tower equations are  $x_1^r = \alpha_r$  and  $x_a^r = x_{a-1}$  for  $a > 1$ . The conjugates over  $F_{a-1}$  are  $\xi x_a$ ,  $\xi \in \mu_r$ ; evaluating the corresponding binomial at one gives

$$N_{F_a/F_{a-1}}(x_a + 1) = x_{a-1} + 1,$$

which proves (P<sub>1</sub>). The field degrees give (P<sub>2</sub>).

For  $p \neq r$ , the standard cyclotomic-divisor criterion applied to  $p \mid \Phi_{r^a}(2^b)$  gives  $\text{ord}_p(2^b) = r^a$ . If  $h = f(p)$ , then

$$\frac{h}{\gcd(h, b)} = r^a.$$

Since  $h \mid r^a b$ , this writes  $h = r^a s$  with  $s \mid b$ . Now  $h \mid d(\kappa_{r^a}) = r^a b$ , so the defining minimality gives  $\kappa_h \leq \kappa_{r^a}$ . But  $r^a \mid h$ , and  $\kappa_{r^a}$  is the least element whose degree is divisible by  $r^a$ , so the reverse inequality holds. Hence  $\kappa_h = \kappa_{r^a}$ , and uniqueness in Lenstra's component theorem gives  $\mathcal{Q}(h) = \{r^a\}$ . Conversely Proposition 5.1 writes every singleton order as  $h = r^a s$ ,  $s \mid b$ , and then  $\text{ord}_p(2^b) = r^a$ , so  $p \mid \mathcal{L}_{r,a}$ .

Such a  $p$  does not divide  $|F_{a-1}^\times|$ , because the order of  $2^b$  modulo  $p$  is  $r^a$ . Its entire primary contribution to  $|F_a^\times|$  therefore lies in the projective quotient. The exact power criterion now identifies non- $p$ -power of  $y_a$ , equivalently  $m_p = 1$ , with the full  $p$ -primary order of its projective class. This proves (P<sub>4</sub>).  $\square$

**Proposition 5.4** (exact two-half decomposition of the cubic projective quotient). *Let  $a \geq 1$ , put  $h = 3^a$ ,  $\zeta = \kappa_h$ ,  $y = \zeta + 1$ , and  $\gamma_a = \zeta + \zeta^{-1} \in \mathbb{F}_{2h}^\times$ . Then*

$$\Phi_h(4) = A_a B_a, \quad A_a = \Phi_h(2), \quad B_a = \Phi_{2h}(2),$$

with  $\gcd(A_a, B_a) = 1$  and  $v_3(B_a) = 1$ . The projective order of  $y$  in  $\mathbb{F}_{2h}^\times / \mathbb{F}_{2^{2h/3}}^\times$  is

$$3 \frac{\text{ord}(\gamma_a)}{\gcd(\text{ord}(\gamma_a), 2^{h/3} - 1)}. \quad (\text{CPS})$$

Consequently every prime  $\ell \neq 3$  dividing  $B_a$  is absent from the translate-one projective class, unconditionally; it satisfies  $\text{ord}_\ell(2) = 2h$ . The  $A_a$ -part is full exactly when the new projective part of  $\gamma_a$  is full, and the all-level assertion is equivalent to Gaussian-period primitivity  $C_a$ . There is no overlap between the ordinary and exceptional halves. The missing  $B_a/3 = \Psi_a$  part is precisely the translate-four residue-symbol problem  $D'_a$ .

*Proof.* The identity  $\Phi_{3^a}(X^2) = \Phi_{3^a}(X)\Phi_{2 \cdot 3^a}(X)$  gives the factorization. With  $X = 2^{3^{a-1}}$ , the two factors are  $X^2 + X + 1$  and  $X^2 - X + 1$ ; a common odd divisor divides their difference  $2X$ , so the gcd is one. Since the exponent in  $X$  is odd, LTE applied to  $(X^3 + 1)/(X + 1)$  gives  $v_3(B_a) = 1$ .

The half-angle identity gives

$$\text{ord}(y) = 3^{a+1} \text{ord}(\gamma_a), \quad \text{ord}(\gamma_a) \mid 2^h - 1.$$

The base group has 3-valuation  $a$ , and

$$\gcd(2^h - 1, 2^{2h/3} - 1) = 2^{h/3} - 1.$$

In a cyclic group, the order modulo its unique base subgroup is the element order divided by its gcd with the base-group order. This proves (CPS). If  $\ell \neq 3$  divides  $B_a$ , the cyclotomic-divisor criterion gives  $\text{ord}_\ell(2) = 2h$ , so  $\ell$  divides neither  $3^{a+1}$  nor  $2^h - 1$ . The remaining assertions follow from the two displayed factors and the definitions of  $C_a$  and  $D'_a$ .  $\square$

**Remark 5.5** (the first nontrivial split). At  $a = 2$ , one has  $\text{ord}(\gamma_2) = 2^9 - 1 = 7 \cdot 73$ , and therefore

$$\text{ord}((\kappa_9 + 1)\mathbb{F}_{26}^\times) = 3 \cdot 73, \quad \Phi_9(4) = 3 \cdot 19 \cdot 73.$$

Thus the missing prime 19 is exactly the exceptional  $\Phi_{18}(2)/3$  half, not a failure of the projective reduction.

## 5.2 The selected relative cyclotomic unit

**Proposition 5.6** (selected-prime cyclotomic Kummer form). *Fix an odd prime  $r$ , put  $b = d(\alpha_r)$  and  $M_r = \text{ord}(\alpha_r)$ , and retain  $x_a = \kappa_{r^a}$ ,  $y_a = x_a + 1$ ,  $F_a = \mathbb{F}_{2^{r^a b}}$ . Set  $N_a = r^a M_r$ . Then*

$$\text{ord}(x_a) = N_a, \quad \text{ord}_{N_a}(2) = r^a b. \quad (\text{CK}_1)$$

Let  $p \neq r$  divide  $\mathcal{L}_{r,a} = \Phi_{r^a}(2^b)$ , choose a primitive complex  $N_a$ -th root  $\zeta_{N_a}$ , and choose a prime  $\mathfrak{P} \mid 2$  of

$$K_{a,p} = \mathbb{Q}(\mu_{N_a}, \mu_p)$$

whose residue map sends  $\zeta_{N_a}$  to  $x_a$ . Its residue field is  $F_a$ . For  $\varepsilon_a = 1 - \zeta_{N_a}$ , the following are equivalent:

- (i) the class  $y_a F_{a-1}^\times$  has full  $p$ -primary order  $p^{v_p(\mathcal{L}_{r,a})}$ ;
- (ii)  $y_a \notin (F_a^\times)^p$ ;
- (iii)  $\left(\frac{\varepsilon_a}{\mathfrak{P}}\right)_p \neq 1$ ;
- (iv)  $\text{Frob}_{\mathfrak{P}}$  is nontrivial in  $K_{a,p}(\sqrt[p]{\varepsilon_a})/K_{a,p}$ .

Thus, for  $r \neq 3$ , the predicted maximal prime-to- $r$  projective order is exactly a simultaneous non-splitting assertion at the Conway-selected primes over two for these cyclotomic  $S$ -unit Kummer extensions. If  $N_a$  is not a prime power,  $\varepsilon_a$  is a global cyclotomic unit; in every case it is a unit at  $\mathfrak{P}$ .

*Proof.* The binomial  $X^r - \alpha_r$  is irreducible over  $B = \mathbb{F}_{2^b}$ . Hence  $\alpha_r \notin (B^\times)^r$ . In particular  $r \mid 2^b - 1$ , and cyclicity of  $B^\times$  gives

$$v_r(M_r) = v_r(2^b - 1).$$

If an element  $u$  has order  $r^c m$ , with  $c \geq 1$  and  $r \nmid m$ , and  $v^r = u$ , then  $\text{ord}(v) = r^{c+1}m$ . Applying this first to  $x_1^r = \alpha_r$  and then to  $x_a^r = x_{a-1}$  proves  $\text{ord}(x_a) = r^a M_r$ . Since  $\mathbb{F}_2(x_a) = F_a$ , the degree of a root of unity of order  $N_a$  gives  $\text{ord}_{N_a}(2) = r^a b$ .

For  $p \neq r$  dividing  $\Phi_{r^a}(2^b)$ , the cyclotomic-divisor criterion gives  $\text{ord}_p(2^b) = r^a$ . Hence  $p$  divides  $|F_a^\times|$  but not  $|F_{a-1}^\times|$ . Also  $p \nmid N_a$ : if  $p \mid M_r$ , then  $\text{ord}_p(2) \mid b$ , contradicting  $\text{ord}_p(2^b) = r^a > 1$ . Therefore the entire  $p$ -Sylow subgroup of the projective quotient comes from  $F_a^\times$ , and in a cyclic group an element has full  $p$ -primary order iff it is not a  $p$ -th power. This proves (i)  $\Leftrightarrow$  (ii).

The prime above two exists because  $x_a$  is a primitive  $N_a$ -th root with degree  $r^a b$ . The completion of  $\mathbb{Q}(\mu_{N_a})$  at this prime is the unramified extension of  $\mathbb{Q}_2$  of degree  $\text{ord}_{N_a}(2) = r^a b$ . Since  $\text{ord}_p(2) \mid r^a b$ , its residue field already contains  $\mu_p$ , so adjoining  $\mu_p$  does not enlarge that residue field. Its reduction sends  $\varepsilon_a$  to  $1 + x_a = y_a$ . The usual power-residue formula gives

$$\left( \frac{\varepsilon_a}{\mathfrak{P}} \right)_p = \overline{\varepsilon_a}^{(|F_a| - 1)/p},$$

which is nontrivial exactly when the residue is not a  $p$ -th power, proving (ii)  $\Leftrightarrow$  (iii). Finally,  $p$  is odd and the residue characteristic is two, so the  $p$ -power map is an automorphism on the principal-unit group at  $\mathfrak{P}$ . The local Kummer extension is therefore unramified of degree  $p$  when the residue is not a  $p$ -th power, and split when it is; its Frobenius acts on  $\sqrt[p]{\overline{\varepsilon_a}}$  by the displayed residue symbol. This proves (iii)  $\Leftrightarrow$  (iv).  $\square$

**Proposition 5.7** (relative cyclotomic unit at the selected prime). *Retain the notation of Proposition 5.6, and put*

$$q_0 = 2^{r^{a-1}b}, \quad Q = q_0^r = 2^{r^a b}, \quad N = N_a.$$

*The automorphism*

$$\sigma(\zeta_N) = \zeta_N^{q_0}$$

*generates  $\text{Gal}(\mathbb{Q}(\mu_N)/\mathbb{Q}(\mu_{N/r}))$ . For  $\varepsilon = 1 - \zeta_N$ , the relative cyclotomic unit*

$$\rho = \frac{\sigma(\varepsilon)}{\varepsilon} = \frac{1 - \zeta_N^{q_0}}{1 - \zeta_N} \tag{RU_1}$$

*is a global unit and has relative norm one. If  $p \neq r$  divides  $\mathcal{L}_{r,a}$ , extend  $\sigma$  to  $K_{a,p} = \mathbb{Q}(\mu_N, \mu_p)$  by fixing  $\mu_p$ , and let  $\mathfrak{P}$  be the Conway-selected prime of the preceding proposition. Then*

$$\bar{\rho} = y_a^{q_0-1}, \quad \left( \frac{\rho}{\mathfrak{P}} \right)_p = \left( \frac{\varepsilon}{\mathfrak{P}} \right)_p^{q_0-1}. \tag{RU_2}$$

*The residue class  $\lambda = q_0 \pmod{p}$  has exact order  $r$ . Consequently  $\lambda - 1 \neq 0$ , and the four equivalent conditions of Proposition 5.6 are also equivalent to*

$$\left( \frac{\rho}{\mathfrak{P}} \right)_p \neq 1. \tag{RU_3}$$

*Thus the ordinary target is already a residue-symbol problem for an explicit norm-one global unit; the inherited lower-layer factor has been removed.*

*Proof.* Since  $x_a^r = x_{a-1} \in F_{a-1}$ , one has  $q_0 \equiv 1 \pmod{N/r}$ . The  $q_0$ -Frobenius has exact order  $r$  on  $F_a/F_{a-1}$ , so  $q_0$  has exact order  $r$  modulo  $N$ . Moreover  $r \mid M_r$  by the preceding proof, so  $[\mathbb{Q}(\mu_N) : \mathbb{Q}(\mu_{N/r})] = r$ . This proves the assertion about  $\sigma$ . The quotient in (RU<sub>1</sub>) is a global unit: if  $N$  is not a prime power, both numerator and denominator are cyclotomic units, while if  $N$  is a prime power their principal ideals are the same unique prime above the conductor prime. Telescoping gives

$$N_{\mathbb{Q}(\mu_N)/\mathbb{Q}(\mu_{N/r})}(\rho) = \prod_{j=0}^{r-1} \frac{\sigma^{j+1}(\varepsilon)}{\sigma^j(\varepsilon)} = 1.$$

Reduction at  $\mathfrak{P}$ , in characteristic two, gives

$$\bar{\rho} = \frac{1 + x_a^{q_0}}{1 + x_a} = \frac{(1 + x_a)^{q_0}}{1 + x_a} = y_a^{q_0-1},$$

and the power-residue formula gives (RU<sub>2</sub>). Write  $h = \text{ord}_p(2) = r^a s$ , where  $s \mid b$ , as in (P<sub>3</sub>). Every prime divisor of  $b = d(\alpha_r)$  is smaller than  $r$ , so  $r \nmid b$ . Therefore

$$\text{ord}_p(q_0) = \frac{r^a s}{\gcd(r^a s, r^{a-1}b)} = r.$$

Raising an element of  $\mu_p$  to  $q_0 - 1$ , equivalently to  $\lambda - 1$ , is therefore an automorphism. This proves (RU<sub>3</sub>).  $\square$

**Proposition 5.8** (ordinary realification and the selected local reflection). *Retain Proposition 5.7, put*

$$E = \mathbb{Q}(\mu_N, \mu_p),$$

*and let  $j$  be simultaneous complex conjugation on  $E$ . If*

$$\rho = \frac{1 - \zeta_N^{q_0}}{1 - \zeta_N}, \quad u = \rho j(\rho),$$

*then*

$$j(\rho) = \zeta_N^{1-q_0} \rho, \quad \boxed{u = \zeta_N^{1-q_0} \rho^2}. \quad (\text{OR}_1)$$

*Since  $p \nmid N$ , the root-of-unity factor in (OR<sub>1</sub>) is a global  $p$ -th power. Hence  $u$  and  $\rho$  have the same  $p$ -Kummer class, globally and after every completion.*

*Put  $B = E^{\langle j \rangle}$ . In the nondegenerate case  $u \notin (E^\times)^p$ , choose  $\theta^p = u$ , extend  $j$  by  $j(\theta) = \theta$ , and put*

$$L_O = B(\theta).$$

*Then  $E(\theta)/B$  is dihedral of order  $2p$ , and at the contraction  $\mathfrak{r}$  of the Conway-selected prime  $\mathfrak{P} \mid 2$ ,*

$$\boxed{\text{ordinary failure at } \mathfrak{P} \iff 2 \in N_{(L_O \otimes_B B_{\mathfrak{r}})/B_{\mathfrak{r}}}((L_O \otimes_B B_{\mathfrak{r}})^\times)}. \quad (\text{OR}_2)$$

*The global norm equation is strictly less selective:*

$$2 \in N_{L_O/B}(L_O^\times) \quad (\text{OR}_3)$$

*holds exactly when the corresponding  $p$ -power condition holds at every prime of  $\mathbb{Q}(\mu_N)$  above two. Thus it detects simultaneous failure on every irreducible conductor- $N$  cyclotomic factor, not the one Conway-selected factor. Passing to the decomposition field does not repair this loss: its decomposition group contains  $\langle q_0 \rangle$ , while  $N_{\langle q_0 \rangle}(\rho) = 1$ . Isolating one two-adic prime would require precisely the mixed principal/ray selector already left open by the ordinary arm.*

*Proof.* Direct simplification gives

$$\frac{1 - \zeta_N^{-q_0}}{1 - \zeta_N^{-1}} = \zeta_N^{1-q_0} \frac{1 - \zeta_N^{q_0}}{1 - \zeta_N},$$

and hence (OR<sub>1</sub>). Because  $\gcd(p, N) = 1$ , exponentiation by  $p$  is an automorphism of  $\mu_N$ ; because  $p$  is odd, squaring is an automorphism of every exponent- $p$  Kummer quotient. This proves the Kummer-class assertion. Both algebraic steps are kernel-checked in `0gdoad.Excess`.

If  $\sigma(\theta) = \zeta_p \theta$ , then  $j\sigma j^{-1} = \sigma^{-1}$ , so the stated fixed field is the degree- $p$  reflection field. The completionwise norm equivalence (HD<sub>5</sub>) below applies verbatim. At  $\mathfrak{P}$ , local Kummer theory says that the cyclic algebra is split exactly when the selected residue of  $\rho$ , equivalently of  $u$ , is a  $p$ -th power. In the other case it is the unramified degree- $p$  field, whose norm valuations are divisible by  $p$ ; since  $v_{\mathfrak{r}}(2) = 1$ , this proves (OR<sub>2</sub>).

For the global assertion, every place away from  $2p$  is tame-unit trivial. At a place over  $p$ , the totally ramified cyclotomic group fixes  $u$ ,  $2$ , and the place while acting faithfully on  $\mu_p$ , so covariance forces the Hilbert symbol to one. Cyclic Hasse and the reflection symmetrization therefore turn (OR<sub>3</sub>) into the conjunction of the local conditions over two. Generally  $\varphi(N) > \text{ord}_N(2) = r^a b$ , so these include several distinct cyclotomic factors. Finally the telescoping relative norm of  $\rho$  is one by Proposition 5.7; hence descent to the two-decomposition field erases rather than isolates its class.  $\square$

**Proposition 5.9** (ordinary projective polynomial and binary sections). *Retain Proposition 5.7, and write*

$$F_0 = F_{a-1} = \mathbb{F}_{q_0}, \quad F = F_a, \quad x = x_a, \quad A = x_{a-1} = x^r, \quad y = 1 + x.$$

Put

$$\zeta = x^{q_0-1} \in F_0, \quad \rho = y^{q_0-1} = \frac{1 + \zeta x}{1 + x}, \quad L = \frac{q_0^r - 1}{q_0 - 1}.$$

Then  $\zeta$  has exact order  $r$ , and the monic irreducible polynomial of  $\rho$  over  $F_0$  is

$$H_{r,a}(T) = \frac{(T+1)^r + A(T+\zeta)^r}{1+A}. \quad (\text{OB}_1)$$

In particular,

$$\begin{aligned} H_{r,a}(0) &= 1, & \text{Tr}_{F/F_0}(\rho) &= \frac{1 + A\zeta}{1 + A}, \\ H_{r,a}(1) &= \frac{A(1 + \zeta)^r}{1 + A}, & H_{r,a}(\zeta) &= \frac{(1 + \zeta)^r}{1 + A}. \end{aligned} \quad (\text{OB}_2)$$

Also  $N_{F/F_0}(\rho) = 1$ , so  $H_{r,a} \mid T^L - 1$ .

Let  $p \neq r$  be a prime divisor of  $L$ , and put  $e = L/p$ . The ordinary arm succeeds at  $p$ , with its full  $p$ -primary valuation, exactly when

$$H_{r,a}(T) \nmid T^e - 1. \quad (\text{OB}_3)$$

For  $0 \leq k < r$ , define the binary section polynomial

$$S_{e,k}(Z) = \sum_{\substack{t \geq 0 \\ k+tr \leq e}} \left( \binom{e}{k+tr} \bmod 2 \right) Z^t.$$

Then failure at  $p$  is exactly the simultaneous selected-value system

$$S_{e,k}(A) = 0 \quad (1 \leq k < r). \quad (\text{OB}_4)$$

This system descends all the way to the original Conway carry. Put  $n = r^{a-1}$ , let  $P_{\alpha_r}$  be the minimal polynomial of  $\alpha_r$  over  $\mathbb{F}_2$ , and, for  $1 \leq c < r^a$  with  $r \nmid c$ , define

$$R_{e,c}(Z) = \sum_{\substack{v \geq 0 \\ c+r^a v \leq e}} \left( \binom{e}{c+r^a v} \bmod 2 \right) Z^v.$$

Then

$$\text{failure at } p \iff P_{\alpha_r}(Z) \mid R_{e,c}(Z) \quad \text{for every } c \pmod{r^a} \text{ with } r \nmid c. \quad (\text{OB}_5)$$

Thus the unresolved inverse-Fourier phase has an equivalent purely binary form, but it is still a divisibility assertion at the selected minimal polynomial  $P_{\alpha_r}$ , not a generic Kummer-step identity.

*Proof.* The relative Frobenius sends  $x$  to  $\zeta x$ , and its order is  $r$ ; hence  $\zeta \in F_0$  has exact order  $r$ . Solving

$$T = \frac{1 + \zeta x}{1 + x}$$

for  $x$  gives  $x = (T + 1)/(T + \zeta)$ . Substitution in  $x^r = A$  gives  $(\text{OB}_1)$ . The inverse Möbius formula shows that  $F_0(\rho) = F_0(x) = F$ , so this degree- $r$  polynomial is irreducible. Its constant term, next-to-leading coefficient, and the two displayed values give  $(\text{OB}_2)$ . Telescoping the Frobenius conjugates of  $y^{q_0-1}$  proves its norm is one.

The prime  $p$  does not divide  $q_0 - 1$ , because  $\text{ord}_p(q_0) = r$ . Therefore the full  $p$ -primary order of  $\rho$  equals that of the projective class of  $y$ . In a cyclic group of order  $L$ , an element contains the full  $p$ -primary part exactly when its  $L/p$ -th power is nonidentity, with no squarefreeness hypothesis. This proves  $(\text{OB}_3)$ .

In the  $F_0$ -basis  $1, x, \dots, x^{r-1}$ , reduction by  $x^r = A$  gives

$$(1 + x)^e = \sum_{k=0}^{r-1} x^k S_{e,k}(A). \quad (\text{OB}_6)$$

Now  $\rho^e = 1$  exactly when  $y^e \in F_0^\times$ . The nonconstant coordinates in  $(\text{OB}_6)$  therefore prove both directions of  $(\text{OB}_4)$ .

Finally  $A^n = \alpha_r$ , and  $1, A, \dots, A^{n-1}$  is a basis over  $\mathbb{F}_{2^b}$ . Writing  $t = u + nv$  in each section gives

$$S_{e,k}(A) = \sum_{u=0}^{n-1} A^u R_{e,k+ru}(\alpha_r).$$

As  $k$  and  $u$  vary, the residues  $k+ru$  are exactly the nonmultiples of  $r$  modulo  $r^a$ . Basis independence, followed by the minimal-polynomial criterion over  $\mathbb{F}_2$ , proves  $(\text{OB}_5)$ .  $\square$

**Proposition 5.10** (full cyclotomic ancestry of an ordinary carry). *Let  $\beta$  be an element of exact odd order  $M$  over  $\mathbb{F}_2$ , put  $b = \text{ord}_M(2)$ , and let  $r$  be an odd prime such that*

$$v_r(M) = v_r(2^b - 1) \geq 1. \quad (\text{FC}_1)$$

*For  $R = r^s$ , every root  $\theta$  of  $\theta^R = \beta$  has exact order  $RM$ , and*

$$\text{ord}_{RM}(2) = bR. \quad (\text{FC}_2)$$

Consequently, if  $P_\beta$  is the minimal polynomial of  $\beta$ , then

$$P_\beta(X^R) \text{ is irreducible of degree } bR, \quad \Phi_{RM}(X) = \Phi_M(X^R). \quad (\text{FC}_3)$$

In the ordinary Conway arm, put

$$b = d(\alpha_r), \quad M = \text{ord}(\alpha_r), \quad R = r^{a-1}, \quad C = RM, \quad A = x_{a-1}.$$

The Kummer carry supplies  $(\text{FC}_1)$ , and hence

$$P_A(X) = P_{\alpha_r}(X^R). \quad (\text{FC}_4)$$

Thus the failure system in Proposition 5.9 is located at the actual full-conductor factor  $P_{\alpha_r}(X^R) \mid \bar{\Phi}_C(X)$ . At the top conductor  $N = r^a M$ , failure at a current prime  $p$  is equivalently

$$P_{\alpha_r}(X^{r^a}) \mid \gcd(\bar{\Phi}_N(X), (1+X)^{(2^{r^a b}-1)/p} - 1). \quad (\text{FC}_5)$$

In particular a model with the wrong lower conductor cannot represent the selected full-conductor factor.

*Proof.* Write  $\text{ord}(\theta) = r^u D$ , with  $r \nmid D$ . Since  $\theta^R = \beta$ , comparison of the prime-to- $r$  parts gives  $D = M/r^{v_r(M)}$ , while the  $r$ -part gives  $u = s + v_r(M)$ . Hence  $\text{ord}(\theta) = RM$ . LTE and  $(\text{FC}_1)$  show that adjoining one further factor of  $r$  to the modulus multiplies the order of 2 by  $r$ ; iteration gives  $(\text{FC}_2)$ . Therefore every root of  $P_\beta(X^R)$  has an absolute Frobenius orbit of its full degree  $bR$ , proving irreducibility. The cyclotomic identity in  $(\text{FC}_3)$  is the standard prime-divisor identity for  $r \mid M$ .

For the Conway carry, the exact  $r$ -valuation is part of the Kummer construction of  $M = \text{ord}(\alpha_r)$ , and  $A^R = \alpha_r$ . The degree just proved identifies its minimal polynomial, giving  $(\text{FC}_4)$ . Finally  $x_a$  has exact order  $N$  and minimal polynomial  $P_{\alpha_r}(X^{r^a})$ . Failure says precisely  $(1+x_a)^{(2^{r^a b}-1)/p} = 1$ ; the minimal-polynomial criterion and the order- $N$  cyclotomic factor give  $(\text{FC}_5)$ .  $\square$

**Proposition 5.11** (singleton dependency-section descent). *Let  $Q = s^v$  be an odd prime power, with  $s$  an odd prime, and suppose a singleton Conway dependency has*

$$\theta^Q = \alpha_s, \quad \alpha_r = \theta + \varepsilon, \quad \varepsilon \in \mathbb{F}_2, \quad [\mathbb{F}_2(\theta) : \mathbb{F}_2(\alpha_s)] = Q. \quad (\text{DD}_1)$$

Then

$$P_{\alpha_r}(Z) = P_{\alpha_s}((Z + \varepsilon)^Q) = \text{Res}_Y(P_{\alpha_s}(Y), (Z + \varepsilon)^Q + Y). \quad (\text{DD}_2)$$

Every  $R(Z) \in \mathbb{F}_2[Z]$  has a unique section decomposition

$$R(Z) = \sum_{j=0}^{Q-1} (Z + \varepsilon)^j R_j((Z + \varepsilon)^Q), \quad (\text{DD}_3)$$

and the selected divisibility descends exactly:

$$P_{\alpha_r} \mid R \iff P_{\alpha_s} \mid R_j \quad (0 \leq j < Q). \quad (\text{DD}_4)$$

Thus every odd-Kummer singleton 0/1-excess edge satisfying  $(\text{DD}_1)$  can be pushed to its unique predecessor. At a two-primary Artin-Schreier node one must instead use its quadratic ancestry relation. At a multicomponent or excess-4 node, the synchronized orbit resultant is essential: separate univariate compositions discard the phase. Even along an all-singleton chain,  $(\text{DD}_4)$  is an exact ancestral reformulation, not a degree-forced nondivisibility theorem.

*Proof.* The first two relations in (DD<sub>1</sub>) make the right side of (DD<sub>2</sub>) vanish at  $\alpha_r$ ; the degree hypothesis shows that both monic polynomials have the same degree and proves the identity. Grouping the powers of  $Z + \varepsilon$  by their residue modulo  $Q$  gives the unique decomposition (DD<sub>3</sub>). After evaluation at  $Z = \alpha_r$ , it becomes

$$R(\alpha_r) = \sum_{j=0}^{Q-1} \theta^j R_j(\alpha_s).$$

The elements  $1, \theta, \dots, \theta^{Q-1}$  are a basis over  $\mathbb{F}_2(\alpha_s)$ , so this value is zero exactly when all the coefficients vanish. Applying the two minimal-polynomial criteria proves (DD<sub>4</sub>).  $\square$

**Proposition 5.12** (norm compression of the dependency sections). *Retain Proposition 5.11. For  $R(Z) \in \mathbb{F}_2[Z]$ , define*

$$\mathcal{K}_{Q,\varepsilon}(R)(Y) = \text{Res}_T(T^Q + Y, R(T + \varepsilon)) \in \mathbb{F}_2[Y]. \quad (\text{DN}_1)$$

*Then*

$$\mathcal{K}_{Q,\varepsilon}(R)(\alpha_s) = N_{\mathbb{F}_2(\theta)/\mathbb{F}_2(\alpha_s)} R(\alpha_r), \quad (\text{DN}_2)$$

*and hence*

$$\boxed{P_{\alpha_r} \mid R \iff P_{\alpha_s} \mid \mathcal{K}_{Q,\varepsilon}(R).} \quad (\text{DN}_3)$$

*Thus the  $Q$  simultaneous conditions in (DD<sub>4</sub>) may equivalently be compressed to one lower resultant. Applied to every actual  $R_{e,c}$  in (OB<sub>5</sub>), iteration through an all-odd-Kummer singleton dependency chain ending at  $\mathbb{F}_2$  produces one terminal scalar for each  $c$ . The compression stops at the same declared boundaries as the section descent: a two-primary node uses its quadratic ancestry, while a multicomponent or excess-4 node retains a synchronized phase.*

*This norm compression is lossless, not a nonvanishing theorem. Its terminal scalar is zero exactly when the original selected binary section is zero; the norm identity alone forces no leading, constant, or reciprocal coefficient condition that excludes that event.*

*Proof.* After specializing  $Y = \alpha_s$ , the first polynomial in (DN<sub>1</sub>) is the monic minimal polynomial  $T^Q + \alpha_s$  of  $\theta$  over  $\mathbb{F}_2(\alpha_s)$ . Its roots are the relative conjugates of  $\theta$ , and translation by  $\varepsilon \in \mathbb{F}_2$  sends them to the conjugates of  $\alpha_r$ . The product formula for the resultant therefore gives (DN<sub>2</sub>). A field norm vanishes exactly when its argument vanishes. Applying the two minimal-polynomial criteria proves (DN<sub>3</sub>), and iteration proves the final assertions.  $\square$

**Proposition 5.13** (the universal order-indicator word). *Let  $D > 1$  be odd, and put*

$$\epsilon_D(e) = \begin{cases} 0, & D \mid e, \\ 1, & D \nmid e. \end{cases}$$

*Then  $\epsilon_D$  has least period  $D$ , and its generating function over  $\mathbb{F}_2$  is*

$$\sum_{e \geq 0} \epsilon_D(e) T^e = \frac{T + T^2 + \dots + T^{D-1}}{1 + T^D}. \quad (\text{OIW}_1)$$

*After cancellation this fraction has denominator*

$$H_D(T) = 1 + T + \dots + T^{D-1}. \quad (\text{OIW}_2)$$

*Consequently the word has exact linear complexity  $D - 1$ . Its 2-kernel, and equivalently its minimal binary DFAO, has exactly  $D$  states.*



*Proof.* The zero set is exactly  $D\mathbb{Z}_{\geq 0}$ , which proves the least-period claim and the geometric-series formula. A nontrivial  $D$ -th root of unity is not a root of the numerator: the complete geometric sum vanishes there, so the sum with the constant term removed is one. At  $T = 1$ , the denominator has a simple root and the numerator vanishes. Thus their greatest common divisor is exactly  $T + 1$ , and the reduced denominator is  $H_D$ , proving the linear-complexity assertion.

Every sequence in the 2-kernel tests one residue class modulo  $D$ , because multiplication by 2 is invertible modulo the odd integer  $D$ . All  $D$  residue classes occur once the section depth satisfies  $2^j \geq D$ , and they are distinguished by the location of their next zero. The same residue transition  $r \mapsto 2r + b \pmod{D}$  is the minimal DFAO.  $\square$

**Proposition 5.14** (closed ordinary section words and their aggregate). *Retain Proposition 5.9, and put*

$$R_0 = r^a, \quad K = \mathbb{F}_2(\alpha_r), \quad u_c(e) = R_{e,c}(\alpha_r) \quad (0 < c < R_0, r \nmid c).$$

*Then, in  $K[[T]]$ , every individual selected section has the reduced rational generating function*

$$\boxed{\sum_{e \geq 0} u_c(e) T^e = \frac{T^c(1+T)^{R_0-c-1}}{(1+T)^{R_0} + \alpha_r T^{R_0}}.} \quad (\text{OSW}_1)$$

*Its least period is  $\text{ord}(1+x_a)$ .*

*Let*

$$t_c(e) = N_{K/\mathbb{F}_2} u_c(e), \quad \mathcal{T}_{r,a}(e) = 1 + \prod_{\substack{0 < c < R_0 \\ r \nmid c}} (1 + t_c(e)),$$

*and let*

$$d_{r,a} = \text{ord}((1+x_a)F_{a-1}^\times)$$

*be the actual projective order. Then the aggregate word is exactly*

$$\boxed{\mathcal{T}_{r,a}(e) = \epsilon_{d_{r,a}}(e).} \quad (\text{OSW}_2)$$

*It therefore has least period  $d_{r,a}$ , linear complexity  $d_{r,a} - 1$ , and  $d_{r,a}$  states in both its 2-kernel and minimal binary DFAO. In particular, for a current prime  $p \mid \mathcal{L}_{r,a}$ ,*

$$\boxed{\mathcal{T}_{r,a}(\mathcal{L}_{r,a}/p) = 1 \iff v_p(d_{r,a}) = v_p(\mathcal{L}_{r,a}).} \quad (\text{OSW}_3)$$

*Thus the entire family of dependency-section terminal scalars has a closed automatic description, but its minimal state size is the unknown selected projective order itself. The compression is an exact order oracle, not a smaller universal nonvanishing problem.*

*Proof.* The standard binomial generating function gives

$$\begin{aligned} \sum_{e \geq 0} u_c(e) T^e &= \sum_{v \geq 0} \alpha_r^v \frac{T^{c+R_0v}}{(1+T)^{c+R_0v+1}} \\ &= \frac{T^c(1+T)^{R_0-c-1}}{(1+T)^{R_0} + \alpha_r T^{R_0}}. \end{aligned}$$

The numerator can vanish only at zero or one, while the denominator is nonzero at both points. Hence the fraction is reduced. Its denominator is the reciprocal of the translated irreducible polynomial

$$(X+1)^{R_0} + \alpha_r$$

up to a nonzero scalar; its roots are the inverses of the conjugates of  $1 + x_a$ . It is separable because  $R_0$  is odd. Therefore the least period of the reduced recurrence is the common multiplicative order  $\text{ord}(1 + x_a)$ .

The norm  $t_c(e)$  is zero precisely when the selected section vanishes. Hence  $\mathcal{T}_{r,a}(e) = 0$  precisely when every nonconstant section vanishes. Regrouping the binary expansion of  $(1 + x_a)^e$  modulo  $R_0$  shows that this happens exactly when  $(1 + x_a)^e \in F_{a-1}^\times$ , equivalently  $d_{r,a} \mid e$ . This proves (OSW<sub>2</sub>), and Proposition 5.13 supplies the period, complexity, and automaton assertions. Since  $d_{r,a} \mid \mathcal{L}_{r,a}$ , evaluation at  $\mathcal{L}_{r,a}/p$  proves (OSW<sub>3</sub>), including the full  $p$ -primary valuation.  $\square$

**Corollary 5.15** (norm and complement are automatic in the ordinary sections). *Retain Proposition 5.9. Write  $e = rE$ . Then*

$$\frac{L}{r} = pE \equiv 1 \pmod{q_0 - 1}. \quad (\text{ON}_1)$$

*Under failure, the sole scalar coordinate in  $(\text{OB}_6)$  is forced to be*

$$S_{e,0}(A) = (1 + A)^E, \quad (\text{ON}_2)$$

*and the sections satisfy the reciprocal symmetry*

$$S_{e,r-k}(Z) = Z^{E-1} S_{e,k}(Z^{-1}) \quad (1 \leq k < r). \quad (\text{ON}_3)$$

*Thus relative norm and binary complementation are exactly compatible with the simultaneous vanishings in  $(\text{OB}_4)$ .*

*Proof.* LTE gives  $v_r(L) = 1$ , since  $r \mid q_0 - 1$ , while  $p \neq r$ ; hence  $e = rE$ . Write

$$L - r = (q_0 - 1)D.$$

Modulo  $r$ , the usual geometric-sum quotient gives

$$D \equiv \sum_{j=1}^{r-1} j = 0,$$

so division by  $r$  proves (ON<sub>1</sub>). If failure holds,  $(1 + x)^e = S_{e,0}(A) \in F_0^\times$ . Raising to  $p$  and using

$$(1 + x)^L = N_{F/F_0}(1 + x) = 1 + A$$

together with (ON<sub>1</sub>) proves (ON<sub>2</sub>). Finally  $e$  is odd, and complementing a binary submask  $j = k + tr$  sends it to

$$e - j = (r - k) + r(E - t - 1).$$

Pairing the coefficients gives (ON<sub>3</sub>).  $\square$

**Corollary 5.16** (the ordinary lower norm is current-Kummer blind). *Retain the ordinary current-prime notation, so  $x^r = A$ ,  $y = 1 + x$ , and  $\text{ord}_p(q_0) = r > 1$ . Then*

$$N_{F/F_0}(y) = 1 + A \in (F_0^\times)^p. \quad (\text{ON}_4)$$

*Thus the only additive lower norm available from the selected binomial step is trivial in the current  $p$ -Kummer quotient, independently of whether  $y$  is a  $p$ -th power upstairs. In particular no induction through this norm can transmit either success or failure to a lower ordinary coordinate.*

*Proof.* The binomial norm is  $N(1+x) = 1+x^r = 1+A$ . Since  $\text{ord}_p(q_0) = r > 1$ , one has  $p \nmid q_0 - 1 = |F_0^\times|$ . Exponentiation by  $p$  is therefore an automorphism of the lower cyclic group. This is the generic coprime-order power lemma kernel-checked in `Ogdoad.Excess`.  $\square$

**Proposition 5.17** (reduced reciprocal divisor of ordinary failure). *Retain the ordinary current-prime notation, and write*

$$x = x_a, \quad A = x_{a-1} = x^r, \quad L = \Phi_{r^a}(2^b), \quad e = L/p = rE.$$

Then  $E$  is odd. Put

$$\begin{aligned} F_{r,E}(X) &= (1+X)^{rE} + (1+X^r)^E, \\ U(X) &= (1+X)^r, \quad V(X) = 1+X^r, \\ W_{r,E}(X) &= \sum_{j=0}^{E-1} U(X)^{E-1-j} V(X)^j. \end{aligned} \tag{RD_1}$$

Let

$$P_x(X) = P_{\alpha_r}(X^{r^a})$$

be the actual full-conductor minimal polynomial of  $x$ , of degree  $D = r^a b$ . Then

$$\boxed{\text{failure at } p \iff P_x \mid F_{r,E} \iff P_x \mid W_{r,E}.} \tag{RD_2}$$

The second polynomial is monic and self-reciprocal of degree  $e - r$ . Consequently its reciprocal selected factor  $P_x^*$  also divides it, and

$$P_x \neq P_x^* \implies 2D \leq e - r. \tag{RD_3}$$

Assume failure for the remaining assertions. Then  $E > 1$ , since  $E = 1$  would make  $W_{r,E} = 1$ . The hypothetical intersection is nevertheless transverse. With  $\theta = (1+A)^E$ ,

$$\boxed{F'_{r,E}(x) = \theta \frac{1+x^{r-1}}{(1+x)(1+A)} \neq 0,} \tag{RD_4}$$

so

$$P_x^2 \nmid F_{r,E}, \quad P_x^2 \nmid W_{r,E}. \tag{RD_5}$$

Its exact transverse norm is

$$\boxed{N_{F_a/F_{a-1}} F'_{r,E}(x) = (1+A)^{e-r-1} (1+A^{r-1}).} \tag{RD_6}$$

Iteration through the actual  $r$ -tower gives

$$\boxed{N_{F_a/\mathbb{F}_2(\alpha_r)} F'_{r,E}(x) = (1+\alpha_r)^{e-r-1} (1+\alpha_r^{r-1}) \neq 0.} \tag{RD_7}$$

Its further absolute norm to  $\mathbb{F}_2$  is therefore one. Wherever the odd-Kummer dependency norm of Proposition 5.12 continues to the bottom, it computes this same terminal bit.

Thus a hypothetical ordinary failure puts every original nonconstant binary section on its zero terminal branch while the canonical transverse derivative lies on a one branch. This proves that the selected full-conductor intersection is reduced. The displayed derivative is a transversality certificate, not an additional condition excluding that intersection.

*Proof.* Corollary 5.15 gives  $e = rE$ , and all the integers involved are odd. Failure is equivalent to

$$(1+x)^e = (1+A)^E = (1+x^r)^E,$$

which is exactly  $F_{r,E}(x) = 0$ . The full-conductor minimal-polynomial criterion proves the first equivalence in (RD<sub>2</sub>).

Since  $E$  is odd, the geometric-sum identity in characteristic two gives

$$F_{r,E} = U^E + V^E = (U+V)W_{r,E}. \quad (\text{RD}_8)$$

Both  $U$  and  $V$  are monic reciprocal polynomials of degree  $r$ . Hence  $W_{r,E}$  is monic and reciprocal of degree  $r(E-1) = e-r$ . The leading terms in  $U+V$  cancel, while the next coefficient is  $r=1$  in  $\mathbb{F}_2$ , so  $\deg(U+V) = r-1 < D$ . Thus the irreducible polynomial  $P_x$  cannot divide  $U+V$ , and (RD<sub>8</sub>) proves the second equivalence. Reciprocity proves (RD<sub>3</sub>).

Differentiate (RD<sub>1</sub>). Since  $r, E, e$  are odd,

$$F'_{r,E}(X) = (1+X)^{e-1} + X^{r-1}(1+X^r)^{E-1}.$$

Under failure, substitute  $\theta = (1+A)^E = (1+x)^e$  and  $A = x^r$ ; bringing the two terms to a common denominator gives (RD<sub>4</sub>). Its numerator is nonzero because the exact order of  $x$  contains  $r$ , so  $x^{r-1} \neq 1$ . Also  $(U+V)(x) \neq 0$  by the degree argument above. Differentiating (RD<sub>8</sub>) at  $x$  now proves both assertions in (RD<sub>5</sub>).

Taking the relative norm of (RD<sub>4</sub>) uses

$$N(1+x) = 1+A, \quad N(1+x^{r-1}) = 1+A^{r-1}.$$

The base scalar  $\theta$  has norm  $\theta^r$ ; collecting the powers of  $1+A$  gives (RD<sub>6</sub>). The two displayed norm identities retain the same form at every preceding Kummer step, so iteration gives (RD<sub>7</sub>). Its factors are nonzero because  $\alpha_r \neq 1$  and the exact order of  $\alpha_r$  contains  $r$ .  $\square$

**Proposition 5.18** (four exact blind spots for the relative unit). *With the preceding notation, let*

$$\nu(T) = 1 + T + \cdots + T^{r-1}, \quad \mathcal{L} = \nu(q_0) = \Phi_{r^a}(2^b), \quad \lambda = q_0 \pmod{p}.$$

*The following standard structural inputs do not determine  $\left(\frac{\rho}{\mathfrak{P}}\right)_p$ .*

(i) *Relative norm and distribution give only*

$$\prod_{j=0}^{r-1} \left( \frac{\sigma^j(\varepsilon)}{\mathfrak{P}} \right)_p = A^{\nu(\lambda)} = 1, \quad A = \left( \frac{\varepsilon}{\mathfrak{P}} \right)_p, \quad (\text{RUB}_1)$$

*and  $\nu(\lambda) = 0$  has a simple zero.*

(ii) *For the finite-field norm  $N : F_a^\times \rightarrow F_{a-1}^\times$ , every nonempty norm fibre maps surjectively to*

$$F_a^\times / (F_a^\times)^p. \quad (\text{RUB}_2)$$

*In particular, fixing the exact lower norm—even norm one—permits both the trivial and every nontrivial  $p$ -Kummer class.*

(iii) Put  $H = \langle \sigma \rangle$ ,  $R = \mathbb{Z}_p[H]$ , and  $\mathfrak{m} = (p, \sigma - q_0)$ . In  $R_{\mathfrak{m}}$ , if  $u = \mathcal{L}/p$ , then the complete first-order group-ring relation is

$$0 = pu + \nu'(q_0)(\sigma - q_0) \pmod{\mathfrak{m}^2}, \quad (\text{RUB}_3)$$

where  $\nu'(q_0)$  is a unit modulo  $\mathfrak{m}$ . This Bockstein/cotangent relation depends only on the cyclotomic layer and contains no coordinate of  $\rho$ .

(iv) Global  $p$ -Hilbert reciprocity for the pair  $(\rho, 2)$  is an identity independently of the selected local symbol. More precisely, if  $\mathfrak{q} \mid 2$  is the selected prime of  $\mathbb{Q}(\mu_N)$ , its  $p-1$  lifts  $\mathfrak{P}_c$  to  $K_{a,p}$ , indexed by  $c \in \mathbb{F}_p^\times$ , satisfy

$$(\rho, 2)_{\mathfrak{P}_{c,p}} = R_2^c, \quad \prod_{c \in \mathbb{F}_p^\times} (\rho, 2)_{\mathfrak{P}_{c,p}} = R_2^{p(p-1)/2} = 1 \quad (\text{RUB}_4)$$

for an arbitrary  $R_2 \in \mu_p$ .

Hence all four displayed consequences are compatible with every value of the selected symbol. Without an additional identity peculiar to  $\rho$ , they do not determine  $(\text{RU}_3)$ . An argument must evaluate the distinguished additive unit  $1 - \zeta_N$ , or an equivalent explicit  $p$ -adic or Jacobi-sum congruence, beyond ideal factorization.

*Proof.* The conjugates of  $\varepsilon$  in the current degree- $r$  layer have reductions

$$1 + x_a^{q_0^j} = (1 + x_a)^{q_0^j} = y_a^{q_0^j}.$$

Their residue symbols are therefore  $A^{\lambda^j}$ , proving  $(\text{RUB}_1)$ . Equivalently, their product is the reduction of  $1 - \zeta_N^r$ , which lies in  $F_{a-1}^\times$ ; it is automatically a  $p$ -th power because  $p \nmid |F_{a-1}^\times|$ . Differentiating  $(T-1)\nu(T) = T^r - 1$  at  $T = \lambda$  gives

$$(\lambda - 1)\nu'(\lambda) = r\lambda^{r-1} \neq 0,$$

so the vanishing is simple but supplies no value of  $A$ .

The kernel of the finite-field norm has order  $\mathcal{L} = (Q-1)/(q_0-1)$ . Since  $p \mid \mathcal{L}$  but  $p \nmid q_0-1$ , the full  $p$ -Sylow subgroup of  $F_a^\times$  lies in this kernel. Its image in the cyclic order- $p$  quotient  $F_a^\times / (F_a^\times)^p$  is nontrivial, hence surjective. Every norm fibre is a coset of the kernel, which proves  $(\text{RUB}_2)$ .

In the group ring,  $(\sigma - 1)\nu(\sigma) = 0$ . The element  $\sigma - 1$  is a unit after localization at  $\mathfrak{m}$ , because its residue is  $\lambda - 1 \neq 0$ ; hence  $\nu(\sigma) = 0$  there. Taylor expansion about the integer  $q_0$ , together with  $\nu(q_0) = \mathcal{L} = pu$ , gives  $(\text{RUB}_3)$ . The derivative is a unit by the preceding simple-root calculation. In particular this is a universal relation in  $\mathbb{Z}_p[H]$ , not an evaluation of a Kummer class.

Finally set  $K = \mathbb{Q}(\mu_N)$  and  $E = K(\mu_p)$ . Since  $p \nmid N$ , one has  $K \cap \mathbb{Q}(\mu_p) = \mathbb{Q}$ . The local field  $K_{\mathfrak{q}}$  already contains  $\mu_p$ , because its residue degree is  $r^a b$  and  $\text{ord}_p(2) \mid r^a b$ . Thus  $\mathfrak{q}$  splits completely in  $E/K$ . Galois equivariance of the Hilbert symbol under the automorphisms  $\zeta_p \mapsto \zeta_p^c$  proves  $(\text{RUB}_4)$ ; at a prime above two,  $R_2$  is the residue symbol of  $\rho$ , up to the harmless choice of inversion convention.

For completeness, all remaining local factors are one. Away from  $2p$ , both entries are units and the tame Hilbert symbol is trivial. At a prime above  $p$ ,  $K/\mathbb{Q}$  is unramified locally and both entries come from that unramified subfield. Their Teichmüller parts are  $p$ -th powers; their principal parts in  $E$  lie in  $U_{p-1} = 1 + p\mathcal{O}_E$ , and the standard unit-filtration orthogonality gives  $(U_{p-1}, U_{p-1})_p = 1$ , since  $2p-2 > p$ . The archimedean places are complex. Global reciprocity therefore reduces to the already trivial products  $(\text{RUB}_4)$ , one for each prime of  $K$  above two. Moreover  $(\rho) = 1$ , so ideal-theoretic Stickelberger or Jacobi-sum factorizations alone retain no datum that can distinguish the selected value.  $\square$

**Remark 5.19** (every unweighted base-field selector cancels). The cancellation in (RUB<sub>4</sub>) is not peculiar to the second argument 2. Retain  $K = \mathbb{Q}(\mu_N)$ ,  $E = K(\mu_p)$ , and the selected prime  $\mathfrak{q} \mid 2$ . Label its  $p-1$  lifts  $\mathfrak{P}_c$ ,  $c \in \mathbb{F}_p^\times$ , so that the automorphism carrying  $\mathfrak{P}_1$  to  $\mathfrak{P}_c$  sends  $\zeta_p$  to  $\zeta_p^c$ . For every  $w \in K^\times$ , if

$$R_w = (\rho, w)_{\mathfrak{P}_1, p},$$

then Galois equivariance gives

$$(\rho, w)_{\mathfrak{P}_c, p} = R_w^c, \quad \prod_{c \in \mathbb{F}_p^\times} (\rho, w)_{\mathfrak{P}_c, p} = R_w^{p(p-1)/2} = 1. \quad (\text{RUB}_5)$$

Thus no unweighted selector from the base cyclotomic field can make global reciprocity isolate one chosen lift above two. A weighted eigenspace identity or a Jacobi-sum selector outside  $K$  may evade (RUB<sub>5</sub>), but ideal factorization alone then leaves local unit symbols to evaluate. The same cancellation applies to the exceptional arm with the current prime  $\ell$  in place of  $p$ : since  $\ell \nmid 15h$  and  $\text{ord}_\ell(2) = 2h \mid 4h$ , each of the two base primes above two splits into  $\ell-1$  lifts and its unweighted orbit product is one. Thus merely distinguishing the two base primes does not distinguish their individual lifts.

**Proposition 5.20** (Teichmüller-weighted selector reciprocity). *Let  $M$  be odd, let  $p \nmid M$  be an odd prime, and put*

$$K = \mathbb{Q}(\mu_M), \quad E = K(\mu_p), \quad f = \text{ord}_p(2), \quad d = \text{ord}_M(2).$$

*Assume  $f \mid d$ . For each prime  $\mathfrak{q} \mid 2$  of  $K$ , choose  $\mathfrak{P}_{\mathfrak{q},1} \mid \mathfrak{q}$  in  $E$ , and write*

$$\mathfrak{P}_{\mathfrak{q},c} = \tau_c \mathfrak{P}_{\mathfrak{q},1}, \quad \tau_c(\zeta_p) = \zeta_p^c \quad (c \in \mathbb{F}_p^\times).$$

*Normalize every valuation above two by  $v_{\mathfrak{P}}(2) = 1$ . For a global unit  $u \in \mathcal{O}_K^\times$ , set*

$$A_{\mathfrak{q}} = (u, 2)_{\mathfrak{P}_{\mathfrak{q},1}, p}, \quad W_{\mathfrak{q}}(w) = \sum_{c \in \mathbb{F}_p^\times} c v_{\mathfrak{P}_{\mathfrak{q},c}}(w) \in \mathbb{F}_p.$$

*Then, for every  $w \in E^\times$ ,*

$$\prod_{c \in \mathbb{F}_p^\times} (u, w)_{\mathfrak{P}_{\mathfrak{q},c}, p} = A_{\mathfrak{q}}^{W_{\mathfrak{q}}(w)}. \quad (\text{WS}_1)$$

*If the divisor of  $w$  is supported above two, global reciprocity becomes the exact weighted identity*

$$\prod_{\mathfrak{q} \mid 2} A_{\mathfrak{q}}^{W_{\mathfrak{q}}(w)} = \prod_{\lambda \mid p} (u, w)_{\lambda, p}^{-1}. \quad (\text{WS}_2)$$

*Moreover*

$$W_{\mathfrak{q}}(\tau_e w) = e W_{\mathfrak{q}}(w), \quad (\text{WS}_3)$$

*and*

$$W_{\mathfrak{q}}(w) = 0 \quad \text{if } w \in K^\times \quad \text{or } w \in \mathbb{Q}(\mu_p)^\times. \quad (\text{WS}_4)$$

*Thus a selector which isolates a chosen lift above two must be genuinely mixed between the  $M$ - and  $p$ -cyclotomic directions. Even then, (WS<sub>2</sub>) retains the local factors above  $p$ : they are not determined by the ideal-theoretic weights.*

*Proof.* The completion  $K_{\mathfrak{q}}$  has residue degree  $d$ , so  $f \mid d$  implies  $\mu_p \subset K_{\mathfrak{q}}$ . Hence  $\mathfrak{q}$  splits completely in  $E/K$ . At a place above two, the tame  $p$ -Hilbert symbol of two local units is one. Since  $u$  is a unit, bilinearity therefore gives

$$(u, w)_{\mathfrak{P}, p} = (u, 2)_{\mathfrak{P}, p}^{v_{\mathfrak{P}}(w)}.$$

Galois equivariance gives

$$(u, 2)_{\mathfrak{P}_{\mathfrak{q}, c}, p} = \tau_c(A_{\mathfrak{q}}) = A_{\mathfrak{q}}^c,$$

which proves (WS<sub>1</sub>). If  $(w)$  is supported above two, then away from  $2p$  both entries are units and their tame symbol is one; the archimedean factors are also one. The global product formula therefore gives (WS<sub>2</sub>). Notice that no triviality is asserted at the wild places above  $p$ .

Reindexing the valuations proves (WS<sub>3</sub>). If  $w \in K^{\times}$ , its valuations are constant in  $c$ , and  $\sum_{c \in \mathbb{F}_p^{\times}} c = 0$ . If  $w \in \mathbb{Q}(\mu_p)^{\times}$ , the restrictions of  $\mathfrak{P}_{\mathfrak{q}, c}$  and  $\mathfrak{P}_{\mathfrak{q}, c2^j}$  to  $\mathbb{Q}(\mu_p)$  agree for  $0 \leq j < f$ . The valuations are therefore constant on cosets of  $\langle 2 \rangle \subset \mathbb{F}_p^{\times}$ , while each weighted coset sum is

$$c \sum_{j=0}^{f-1} 2^j = c(2^f - 1) = 0 \pmod{p}.$$

This proves (WS<sub>4</sub>). □

**Proposition 5.21** (the first mixed Jacobi phase). *Let  $Q = 2^D$ , let  $M$  be odd, and let  $p \nmid M$  be an odd prime such that  $Mp \mid Q - 1$ . Choose a generator  $g \in \mathbb{F}_Q^{\times}$ , put*

$$z = g^{(Q-1)/M}, \quad \text{ind}_p(g^v) = v \pmod{p},$$

*and choose characters, extended by zero at zero, by*

$$\chi_M(g) = \zeta_M, \quad \chi_p(g) = \zeta_p.$$

*For  $1 \leq j < M$ , define*

$$J_j = \sum_{x \in \mathbb{F}_Q} \chi_M(x)^j \chi_p(1 - x).$$

*Fix a place above  $p$  of  $\mathbb{Q}(\mu_M, \mu_p)$ , put  $\pi = \zeta_p - 1$ , and let an overline denote reduction at that place—not complex conjugation. Then  $J_j \equiv -1 \pmod{\pi}$ , and*

$$C_j := \frac{\overline{J_j + 1}}{\pi} = \sum_{a=1}^{M-1} \overline{\zeta_M}^{ja} \text{ind}_p(1 - z^a). \tag{MJ_1}$$

*With  $J_0 = -1$  and  $C_0 = 0$ , Fourier inversion gives*

$$\text{ind}_p(1 - z^a) = M^{-1} \sum_{j=0}^{M-1} \overline{\zeta_M}^{-ja} C_j \quad (1 \leq a < M). \tag{MJ_2}$$

*Consequently the following are equivalent:*

- (i) *every  $1 - z^a$ ,  $1 \leq a < M$ , is a  $p$ -th power in  $\mathbb{F}_Q^{\times}$ ;*
- (ii)  *$C_j = 0$  for every  $j$ ;*
- (iii)  *$J_j \equiv -1 \pmod{\pi^2}$  for every  $1 \leq j < M$ .*

For prime  $M$ , the first-order equivalence is the  $k = 2$  layer of Arul's Jacobi congruence [1, Theorem 1.5]; the elementary argument here also permits composite  $M$ .

*Proof.* Modulo  $\pi^2$ , one has  $\zeta_p^v = (1 + \pi)^v \equiv 1 + v\pi$ . The constant term of  $J_j$  for  $j \neq 0$  is

$$\sum_{x \in \mathbb{F}_Q^\times \setminus \{1\}} \chi_M(x)^j = -1,$$

while the trivial-character identity gives  $J_0 = -1$  exactly. The coefficient of  $\pi$  is

$$\sum_{v=1}^{Q-2} \zeta_M^{jv} \text{ind}_p(1 - g^v).$$

Partition  $v = a + tM$ , and put  $N = (Q - 1)/M$ . For  $a \neq 0$ ,

$$\prod_{t=0}^{N-1} (1 - g^{a+tM}) = 1 - g^{aN} = 1 - z^a,$$

so the sum of the indices on that block is  $\text{ind}_p(1 - z^a)$ . In the  $a = 0$  block the omitted zero factor leaves

$$\prod_{t=1}^{N-1} (1 - g^{tM}) = N = 1 \quad \text{in characteristic two,}$$

because  $N$  is odd; its index is zero. This proves (MJ<sub>1</sub>). The Fourier matrix is invertible because  $p \nmid M$ , giving (MJ<sub>2</sub>). Finally,  $\text{ind}_p(y) = 0$  exactly when  $y$  is a  $p$ -th power in the cyclic group  $\mathbb{F}_Q^\times$ , which proves the three equivalences.  $\square$

**Corollary 5.22** (ordinary target as one inverse-Fourier coordinate). *Retain Proposition 5.6. Put*

$$M = N_a, \quad Q = 2^{r^a b}, \quad z = x_a,$$

*and let  $p \neq r$  divide  $\Phi_{r^a}(2^b)$ . Choose  $g$  in the preceding proposition so that  $g^{(Q-1)/M} = z$ , and form its mixed Jacobi sums  $J_j$  and coefficients  $C_j$ . Then the ordinary condition at  $p$  is exactly*

$$\text{ind}_p(y_a) = M^{-1} \sum_{j=0}^{M-1} \overline{\zeta_M}^{-j} C_j \neq 0. \quad (\text{OMJ})$$

*Indeed  $y_a = 1 - z$  in characteristic two, and*

$$\text{ind}_p(\bar{\rho}) = (q_0 - 1) \text{ind}_p(y_a),$$

*where  $q_0 - 1 \not\equiv 0 \pmod{p}$ . Thus mixed Jacobi sums evade the zero weight in (WS<sub>4</sub>), but their required first local phase is exactly the original selected Kummer coordinate. Formula (WS<sub>2</sub>) relates its ideal weights to the still-essential local symbols above  $p$ ; it does not itself prove nonvanishing.*

**Proposition 5.23** (full-conductor Mirimanoff-ray reciprocity). *Retain Propositions 5.7 and 5.22. Thus*

$$N = r^a \text{ord}(\alpha_r), \quad q_0 = 2^{r^{a-1}b}, \quad \rho = \frac{1 - \zeta_N^{q_0}}{1 - \zeta_N}, \quad E = \mathbb{Q}(\mu_N, \mu_p),$$



where  $p \neq r$  is a current prime, and the Jacobi specialization is

$$M = N, \quad Q = 2^{r^a b}, \quad z = x_a.$$

Let  $\mathfrak{q}_C \mid 2$  be the Conway-selected prime of  $\mathbb{Q}(\mu_N)$ , characterized by  $\zeta_N \mapsto x_a = \kappa_{r^a}$ . Choose a generator  $g \in \mathbb{F}_Q^\times$  with  $g^{(Q-1)/N} = x_a$ , and choose the lift  $\mathfrak{P}_0 \mid \mathfrak{q}_C$  to  $E$  for which  $\bar{\zeta}_p = g^{(Q-1)/p}$ . Then  $\mathfrak{q}_C$  restricts to the actual lower ancestor:

$$\zeta_N^{r^a} \mapsto x_a^{r^a} = \alpha_r. \quad (\text{MR}_0)$$

In particular this construction uses the full conductor  $N$ , not only the degree- $r$  Kummer step. With this compatible normalization of the  $p$ -power residue symbol,

$$\left( \frac{\rho}{\mathfrak{P}_0} \right)_p = \zeta_p^{(q_0-1) \text{ind}_p(1+x_a)}, \quad (\text{MR}'_0)$$

and  $q_0 - 1 \not\equiv 0 \pmod{p}$ . Thus this symbol retains the full  $p$ -primary ordinary target.

Let  $\lambda \mid p$ , put  $f = \text{ord}_N(p)$ , and let  $E_\lambda^0$  be the maximal unramified subfield of  $E_\lambda$ , with residue field  $k_\lambda$ . The current-prime condition gives

$$p \nmid N, \quad |k_\lambda| = p^f, \quad E_\lambda = E_\lambda^0(\mu_p), \quad \rho \in (E_\lambda^0)^\times.$$

Write  $\tau$  for the Teichmüller image of  $\zeta_N$ , and put

$$u = \tau^{p^{f-1}}, \quad u^p = \tau, \quad \mathcal{L}_p(T) = \sum_{i=1}^{p-1} \frac{T^i}{i} \in \mathbb{F}_p[T].$$

For  $v \in \mathcal{O}_{E_\lambda^0}^\times$ , define the Fermat quotient

$$q_\lambda(v) = \frac{\overline{v^{p^f-1} - 1}}{p} \in k_\lambda.$$

Then the full-conductor local regulator is

$$A_\lambda := q_\lambda(\rho) = \frac{\mathcal{L}_p(u)}{(1-u)^p} - \frac{\mathcal{L}_p(u^{q_0})}{(1-u^{q_0})^p}. \quad (\text{MR}_1)$$

It detects the complete local Kummer class:

$$\rho \in (E_\lambda^\times)^p \iff A_\lambda = 0. \quad (\text{MR}_2)$$

If  $A_\lambda \neq 0$ , the character  $w \mapsto (\rho, w)_{\lambda, p}$  has conductor exactly  $\lambda^2$ ; if  $A_\lambda = 0$ , it is locally trivial. Hence  $E(\sqrt[p]{\rho})/E$  is unramified away from  $p$ , with exact finite conductor

$$\mathfrak{m}_\rho = \prod_{\substack{\lambda \mid p \\ A_\lambda \neq 0}} \lambda^2. \quad (\text{MR}_3)$$

Fix  $\pi = \zeta_p - 1$ . For each  $\lambda \mid p$  there is a nonzero normalization scalar  $c_\lambda \in k_\lambda^\times$ , fixed by the Hilbert-symbol convention, such that a local unit

$$w = [\bar{w}](1 + \pi B_\lambda(w) + O(\pi^2))$$

satisfies Iwasawa's critical-pairing formula

$$(\rho, w)_{\lambda, p} = \zeta_p^{\text{Tr}_{k_\lambda/\mathbb{F}_p}(c_\lambda A_\lambda B_\lambda(w))}. \quad (\text{MR}_4)$$

Normalize local reciprocity so a uniformizer maps to arithmetic Frobenius. If  $w \in E^\times$  has divisor

$$(w) = \prod_{\mathfrak{p}|2} \mathfrak{p}^{n_{\mathfrak{p}}}, \quad \left(\frac{\rho}{\mathfrak{p}}\right)_p = \zeta_p^{\ell_{\mathfrak{p}}},$$

then global reciprocity is the exact bilinear identity

$$\boxed{\sum_{\mathfrak{p}|2} n_{\mathfrak{p}} \ell_{\mathfrak{p}} = - \sum_{\lambda|p} \text{Tr}_{k_\lambda/\mathbb{F}_p}(c_\lambda A_\lambda B_\lambda(w)) \pmod{p}.} \quad (\text{MR}_5)$$

For the specialized mixed Jacobi sums of Proposition 5.21, write

$$C_{j,\lambda} = \frac{\overline{J_j + 1}}{\pi}$$

at  $\lambda$ , where  $1 \leq j < N$ . The choice  $w = -J_j$  has  $B_\lambda(w) = -C_{j,\lambda}$ , so

$$\boxed{\sum_{\mathfrak{p}|2} v_{\mathfrak{p}}(J_j) \ell_{\mathfrak{p}} = \sum_{\lambda|p} \text{Tr}_{k_\lambda/\mathbb{F}_p}(c_\lambda A_\lambda C_{j,\lambda}) \pmod{p}.} \quad (\text{MR}_6)$$

Thus Stickelberger's explicit valuation vector and the first Jacobi coefficient meet in one exact ray-class formula, but neither side has disappeared.

*Proof.* The restriction statement (MR<sub>0</sub>) follows from the exact order  $N$  of  $x_a$  and the defining Kummer ancestry. It also shows that replacing  $N$  by a generic conductor changes the residue of  $\tau$ , and therefore can change the regulator in (MR<sub>1</sub>). Formula (MR'<sub>0</sub>) is (RU<sub>2</sub>), with the chosen generator of  $\mu_p$ ;  $q_0 - 1 \neq 0$  follows because  $q_0$  has exact order  $r > 1$  modulo  $p$ .

Put  $m = p^f - 1$ . Expanding  $(1 - \tau)^{p^f} - (1 - \tau)$  modulo  $p^2$ , the only interior binomial coefficients of  $p$ -valuation one occur at exponents  $ip^{f-1}$ ,  $1 \leq i < p$ , and

$$\frac{1}{p} \binom{p^f}{ip^{f-1}} \equiv \frac{(-1)^{i-1}}{i} \pmod{p}.$$

After including the sign from  $(-\tau)^{ip^{f-1}}$  and dividing by  $1 - \tau$ , this gives

$$q_\lambda(1 - \tau) = -\frac{\mathcal{L}_p(u)}{(1 - u)^p}.$$

The congruence  $(vw)^m - 1 \equiv (v^m - 1) + (w^m - 1) \pmod{p^2}$  makes  $q_\lambda$  additive on products. Applying it to the quotient defining  $\rho$  proves (MR<sub>1</sub>).

Let  $\tilde{A}_\lambda \in \mathcal{O}_{E_\lambda^0}$  be the Teichmuller lift of  $A_\lambda$ . The Teichmuller decomposition has the sharper form

$$\rho = [\bar{\rho}](1 - p\tilde{A}_\lambda + O(p^2)). \quad (\text{MR}_7)$$

Its Teichmuller factor is a  $p$ -th power. In  $E_\lambda = E_\lambda^0(\mu_p)$ , the absolute ramification index is  $p - 1$ . If  $A_\lambda \neq 0$ , the principal factor in (MR<sub>7</sub>) has nonzero class in the critical graded piece  $U_{p-1}/U_p$ , so it is not a  $p$ -th power. If  $A_\lambda = 0$ , it lies in  $U_{2p-2}$ , which is the  $p$ -th power of  $U_{p-1}$  by the  $p$ -adic logarithm.

This proves (MR<sub>2</sub>). The critical Kummer pairing is perfect between the displayed graded piece and  $U_1/U_2$ , and is trivial on  $U_2$ . This gives the exact conductor dichotomy and the local formula (MR<sub>4</sub>); it is the standard explicit norm-residue formula of Iwasawa [19]. Since  $\rho$  is a global unit and  $p$  is invertible away from the primes above  $p$ , the Kummer extension is unramified at every other finite place, proving (MR<sub>3</sub>).

At a place  $\mathfrak{P} \mid 2$ , bilinearity and the tame unit formula give

$$(\rho, w)_{\mathfrak{P}, p} = \left( \frac{\rho}{\mathfrak{P}} \right)_p^{n_{\mathfrak{P}}}.$$

At every place away from  $2p$ , both entries are units and the tame symbol is one. The global product formula, followed by (MR<sub>4</sub>), proves (MR<sub>5</sub>).

Finally  $\chi_N^j$ ,  $\chi_p$ , and their product are all nontrivial for  $1 \leq j < N$ , because  $(N, p) = 1$ . Thus  $J_j$  is integral and the standard identity

$$J_j \overline{J_j} = Q = 2^{r^{ab}}$$

(with complex conjugation in this display) shows that its divisor is supported only above two. Also  $J_j = -1 + \pi C_{j, \lambda} + O(\pi^2)$ . Hence  $-J_j = 1 - \pi C_{j, \lambda} + O(\pi^2)$ , so  $B_{\lambda}(-J_j) = -C_{j, \lambda}$ . The two minus signs in (MR<sub>5</sub>) cancel and give (MR<sub>6</sub>).  $\square$

**Remark 5.24** (the remaining ordinary ray coordinate). Proposition 5.23 makes the obstruction conductor-sensitive and explicit, but does not prove its nonvanishing. Stickelberger determines the valuations on the left of (MR<sub>6</sub>), not the  $p$ -adic unit coefficients on the right. Mixed Jacobi sums supply those coefficients, but their inverse Fourier transform, at the compatible  $p$ -adic place and generator used in (OMJ), is exactly  $\text{ind}_p(1 + x_a)$ , the original selected coordinate. If every  $A_{\lambda}$  vanishes, the Kummer extension is everywhere unramified; it is then either trivial or, if nontrivial, defines a  $p$ -class-group character. If some  $A_{\lambda}$  is nonzero, reciprocity gives a conductor- $p^2$  ray equation but still requires the selected Jacobi phase. Rohrlich reduced the ramified Jacobi local component to an explicit Hilbert-symbol coefficient, and Coleman–McCallum subsequently computed the odd-prime local components and conductors [35, 7]. Those formulas solve Rohrlich’s stated local-component problem, but do not evaluate the separate Conway-selected regulator (MR<sub>1</sub>): applying them here still requires isolating the compatible prime above two, equivalently the missing mixed ray-class coordinate.

### 5.3 Unconditional projective order bounds

**Proposition 5.25** (unconditional nontrivial projective order). *Let*

$$d_{r,a} = \text{ord}(y_a F_{a-1}^{\times}) \mid \mathcal{L}_{r,a}.$$

*Then*

$$d_{r,a} \notin \{1, r\}, \quad v_r(\mathcal{L}_{r,a}) = 1. \quad (\text{PO}_1)$$

*Consequently  $d_{r,a}$  has a prime divisor  $p \neq r$  at every level. This asserts only  $p \mid d_{r,a}$ : when  $v_p(\mathcal{L}_{r,a}) > 1$ , it does not by itself give the full  $p$ -primary equality in (O). If the chosen divisor satisfies  $v_p(\mathcal{L}_{r,a}) = 1$ , however, its instance of (O) follows. In particular, if  $r \neq 3$  and  $\mathcal{L}_{r,a}/r$  is prime, then the entire ordinary target (O) at this layer holds.*

*Proof.* If  $d_{r,a} = 1$ , then  $y_a$ , and hence  $x_a = y_a + 1$ , lies in  $F_{a-1}$ , contrary to  $[F_a : F_{a-1}] = r$ . If  $d_{r,a} = r$ , then  $y_a^r = c \in F_{a-1}^\times$ . Since  $y_a$  has degree  $r$  over  $F_{a-1}$ , its minimal polynomial would be  $Y^r - c$ . On the other hand  $x_a^r = x_{a-1}$  shows that its monic degree- $r$  minimal polynomial is

$$(Y + 1)^r + x_{a-1}.$$

The coefficient of  $Y^{r-1}$  in the latter is  $r \equiv 1 \pmod{2}$ , whereas it is zero in  $Y^r - c$ , a contradiction.

Because  $r \mid 2^b - 1$ , also  $q_0 \equiv 1 \pmod{r}$ . LTE gives

$$v_r(\mathcal{L}_{r,a}) = v_r(q_0^r - 1) - v_r(q_0 - 1) = v_r(r) = 1.$$

Thus a divisor of  $\mathcal{L}_{r,a}$  other than 1 and  $r$  has a prime divisor different from  $r$ . The final assertions follow from  $(P_4)$ , with the stated full-valuation caveat.  $\square$

**Proposition 5.26** (absolute-Frobenius signed-ball order sieve). *Retain the notation of Proposition 5.3. Put*

$$t_r = \text{ord}_r(2), \quad \nu_{r,a} = \frac{r^a b}{t_r}, \quad R = \frac{r-1}{2},$$

and define

$$B_{r,a}^{\text{abs},\pm} = \sum_{u=0}^R \sum_{v=0}^R [X^u Y^v] \left( 1 + \frac{X}{1-X} + \frac{Y}{1-Y} \right)^{\nu_{r,a}}. \quad (\text{CS}_0)$$

Then

$$d_{r,a} = \text{ord}(y_a F_{a-1}^\times) \geq B_{r,a}^{\text{abs},\pm}. \quad (\text{CS}_1)$$

Moreover  $\nu_{r,a} \geq r^a$ . Thus this contains the bound obtained from the  $r$  relative conjugates, is strictly stronger at every  $a > 1$ , and is also strictly stronger at  $a = 1$  whenever  $b > t_r$ . More explicitly, if  $B_r^\pm$  denotes the same coefficient sum as  $(\text{CS}_0)$  with exponent  $r$ , then

$$B_{r,a}^{\text{abs},\pm} \geq B_r^\pm + 2(\nu_{r,a} - r), \quad B_{r,a}^{\text{abs},\pm} \geq \binom{\nu_{r,a}}{R} \binom{\nu_{r,a} - R}{R}. \quad (\text{CS}'_1)$$

The same construction gives a selected line-saturation statement. Define

$$\mathcal{C}_{r,a} = \left\{ x_{a-1}^{(2^{t_r j} - 1)/r} : 0 \leq j < \nu_{r,a} \right\} \subset F_{a-1}^\times.$$

Then  $|\mathcal{C}_{r,a}| = \nu_{r,a}$ , and failure at a primary factor  $p$  forces

$$1 + cx_a \in (F_a^\times)^p \quad \text{for every } c \in \mathcal{C}_{r,a}. \quad (\text{CS}_{\text{sat}})$$

Consequently, if  $p \neq r$  divides  $\mathcal{L}_{r,a}$  and

$$p > \frac{\mathcal{L}_{r,a}}{B_{r,a}^{\text{abs},\pm}}, \quad (\text{CS}_2)$$

then the Conway-selected  $p$ -th power-residue symbol is nontrivial and

$$v_p(d_{r,a}) = v_p(\mathcal{L}_{r,a}). \quad (\text{CS}_3)$$

Thus every prime factor above the sharper explicit complementary threshold is proved in the ordinary arm, including its full primary valuation.

*Proof.* Write  $D = r^a b = d(x_a)$ ,  $t = t_r$ , and  $\nu = \nu_{r,a} = D/t$ . Since  $r^a \mid \text{ord}(x_a)$ , one has  $t \mid D$ . For  $0 \leq j < \nu$ , put

$$n_j = \frac{2^{tj} - 1}{r}, \quad c_j = x_{a-1}^{n_j} \in F_{a-1}^\times.$$

The equation  $x_a^r = x_{a-1}$  gives the absolute-Frobenius linearization

$$y_a^{2^{tj}} = 1 + x_a^{2^{tj}} = 1 + c_j x_a. \quad (\text{CS}_4)$$

These  $\nu$  coefficients are pairwise distinct. Indeed, if  $c_i = c_j$ , then  $x_a^{2^{ti}} = x_a^{2^{tj}}$ . The absolute Frobenius orbit of  $x_a$  has exact length  $D$ , so  $D \mid t(i - j)$ . Since  $0 \leq i, j < D/t$ , this forces  $i = j$ . This proves the cardinality assertion for  $\mathcal{C}_{r,a}$ . If  $y_a = z^p$ , then (CS<sub>4</sub>) gives  $1 + c_j x_a = (z^{2^{tj}})^p$  for every  $j$ , proving (CS<sub>sat</sub>).

The relation  $r \mid 2^b - 1$  gives  $t_r \mid b$ . Hence

$$\nu_{r,a} = \frac{r^a b}{t_r} \geq r^a. \quad (\text{CS}_5)$$

For  $\mathbf{e} = (e_0, \dots, e_{\nu-1}) \in \mathbb{Z}^\nu$ , write  $e_j^+ = \max(e_j, 0)$  and  $e_j^- = \max(-e_j, 0)$ , and consider

$$\mathcal{S}_R = \{\mathbf{e} \in \mathbb{Z}^\nu : |\mathbf{e}^+| \leq R, |\mathbf{e}^-| \leq R\}.$$

The single-coordinate generating series

$$1 + X + X^2 + \dots + Y + Y^2 + \dots = 1 + \frac{X}{1 - X} + \frac{Y}{1 - Y}$$

shows that  $|\mathcal{S}_R| = B_{r,a}^{\text{abs}, \pm}$ .

For  $\mathbf{e} \in \mathcal{S}_R$ , form the nonzero signed product

$$Z_{\mathbf{e}} = \prod_{j=0}^{\nu-1} (1 + c_j x_a)^{e_j} \in F_a^\times.$$

Suppose  $Z_{\mathbf{e}}$  and  $Z_{\mathbf{f}}$  represent the same class modulo  $F_{a-1}^\times$ , and put  $\mathbf{g} = \mathbf{e} - \mathbf{f}$ . Then

$$\prod_j (1 + c_j x_a)^{g_j^+} = c \prod_j (1 + c_j x_a)^{g_j^-} \quad \text{for some } c \in F_{a-1}^\times. \quad (\text{CS}_6)$$

The two polynomial products in (CS<sub>6</sub>) have degrees at most

$$|\mathbf{g}^+| \leq |\mathbf{e}^+| + |\mathbf{f}^-| \leq 2R = r - 1,$$

and the same bound holds for  $|\mathbf{g}^-|$ . Their difference therefore has degree less than  $r$ , whereas  $x_a$  has degree  $r$  over  $F_{a-1}$ ; it must be the zero polynomial. Both constant terms are one, so  $c = 1$ . The normalized linear factors  $1 + c_j X$  are pairwise distinct. Unique factorization gives  $\mathbf{g}^+ = \mathbf{g}^-$ , hence  $\mathbf{e} = \mathbf{f}$ .

Finally, (CS<sub>4</sub>) gives

$$[Z_{\mathbf{e}}] = \prod_{j=0}^{\nu-1} [y_a]^{e_j 2^{tj}} = [y_a]^{\sum_j e_j 2^{tj}}.$$

Thus the  $B_{r,a}^{\text{abs}, \pm}$  elements of  $\mathcal{S}_R$  give distinct powers of the single class  $[y_a]$ , proving (CS<sub>1</sub>). Embedding the old  $r$ -coordinate ball and then adding the two signed unit vectors on each new coordinate

proves the first inequality in  $(\text{CS}'_1)$ . Taking vectors with exactly  $R$  entries  $+1$ , exactly  $R$  disjoint entries  $-1$ , and all other entries zero proves the second. In particular, strictness when  $\nu_{r,a} > r$  follows at once, and the preceding inequality for  $\nu_{r,a}$  gives the two stated sufficient conditions.

If the selected  $p$ -th residue symbol were trivial, then  $y_a$  would be a  $p$ -th power in  $F_a$ . Since the full  $p$ -part of  $|F_a^\times|$  lies in the projective quotient, this would give  $d_{r,a} \mid \mathcal{L}_{r,a}/p$ . Under  $(\text{CS}_2)$  that contradicts  $(\text{CS}_1)$ . Nontriviality is equivalent to the full primary equality by Proposition 5.6, which proves  $(\text{CS}_3)$ .  $\square$

## 5.4 Limits of generic criteria

Thus the singleton-one arm is not an unspecified root search. It is the uniform assertion that the explicitly defined transverse norm  $\Theta_{q,s}$  contains the full  $p$ -primary part for every prime  $p$  of exact order  $qs$ . The  $C_k$  norms below are the special case  $q = 3^k$ ,  $b_r = 2$ ,  $s = 1$ ; the  $m_{359}$  obstruction is the same square at  $q = 179$ ,  $b_r = 19580$ ,  $s = 1$ . Lenstra's use of Carlitz's primitive-root distribution theorem gives an effective bound at which *some* finite translate works [5, 22]; it does not evaluate this fixed first transverse norm.

There is a sharp reason that this effective argument cannot be replaced, for arbitrary finite-field elements, by any fixed initial translate window.

**Proposition 5.27** (finite-window Kummer no-go). *Let  $p$  be an odd prime and let*

$$A = \{b_1, \dots, b_r\} \subset \overline{\mathbb{F}_2}$$

*be a finite set of distinct elements. Choose  $k_0 = \mathbb{F}_{2^d}$  containing both  $A$  and  $\mu_p$ . There are arbitrarily large integers  $n$  and elements  $x_n \in \mathbb{F}_{2^{dn}}$  of exact degree  $dn$  over  $\mathbb{F}_2$  such that*

$$x_n + b_i \in (\mathbb{F}_{2^{dn}}^\times)^p \quad (1 \leq i \leq r).$$

*Consequently no absolute truncation of Lenstra's translate-generation lemma, nor any argument using only full degree, an order- $p$  character, and a fixed affine translate set, can prove an absolute bound for arbitrary elements. This does not decide the distinguished elements  $\kappa_{f(p)}$ : any uniform excess theorem must use their special Conway-tower identities.*

*Proof.* Over  $k_0(X)$  form the Kummer extension

$$\mathcal{K} = k_0(X)(Y_1, \dots, Y_r), \quad Y_i^p = X + b_i.$$

The classes of the  $X + b_i$  are independent in  $\overline{k_0}(X)^\times / \overline{k_0}(X)^{\times p}$ : in a relation  $\prod_i (X + b_i)^{e_i} = G^p$ ,  $0 \leq e_i < p$ , the valuation at  $X = -b_i$  forces  $e_i = 0$ . Thus the smooth projective curve  $C$  with function field  $\mathcal{K}$  is geometrically integral and

$$[\mathcal{K} : k_0(X)] = p^r, \quad \text{Gal}(\mathcal{K}/k_0(X)) \cong (\mathbb{Z}/p\mathbb{Z})^r.$$

The cover  $C \rightarrow \mathbb{P}_X^1$  is ramified at the  $r$  points  $-b_i$  and at infinity, each with inertia order  $p$ . Hence Riemann–Hurwitz gives

$$2g(C) - 2 = -2p^r + (r+1)p^r \left(1 - \frac{1}{p}\right).$$

Put  $Q = 2^d$ . Hasse–Weil, followed by deletion of the finitely many branch points, supplies

$$Q^n + O_{p,r}(Q^{n/2})$$

unramified  $\mathbb{F}_{Q^n}$ -points. Since the map to the  $X$ -line has degree  $p^r$ , the number of resulting  $X$ -coordinates is at least

$$\frac{2^{dn} - O_{p,r}(2^{dn/2})}{p^r}.$$

The union of all proper subfields of  $\mathbb{F}_{2^{dn}}$  has at most

$$\sum_{\substack{e|dn \\ e < dn}} 2^e \leq dn 2^{dn/2}$$

elements. For all sufficiently large  $n$  the former lower bound is larger. Choose an unramified  $X$ -coordinate  $x_n$  outside that union. It has exact degree  $dn$ , and its lifted coordinates satisfy  $x_n + b_i = Y_i^p \neq 0$  for every  $i$ , as required.  $\square$

**Proposition 5.28** (Kummer-eigenline translate no-go). *Let  $r$  and  $p$  be distinct odd primes, let  $q$  be a prime power with*

$$r \mid q - 1, \quad \text{ord}_p(q) = r,$$

*and put  $K = \mathbb{F}_{q^r}$ . Fix  $\zeta \in \mathbb{F}_q^\times$  of order  $r$ , and let*

$$L_\zeta = \{x \in K : x^q = \zeta x\}.$$

*Every nonzero  $x \in L_\zeta$  has degree  $r$  over  $\mathbb{F}_q$ , the element  $x^r$  lies in  $\mathbb{F}_q^\times$ , and*

$$X^r - x^r$$

*is irreducible over  $\mathbb{F}_q$ . Nevertheless, if*

$$q - (p - 1)(r - 1)\sqrt{q} > p, \tag{KE}$$

*there is a nonzero  $x \in L_\zeta$  such that*

$$x + 1 \in (K^\times)^p.$$

*Thus irreducibility of the Kummer binomial, exact degree, and the Frobenius eigenrelation still do not force the ordinary singleton translate to have a nonzero  $p$ -Kummer class.*

*For every odd prime  $r$ , such characteristic-two countermodels occur over arbitrarily large base fields. Indeed, choose a prime  $p$  with  $\text{ord}_p(2) = r$ , put  $t = \text{ord}_r(2)$ , and take*

$$q = 2^{t(1+kr)} \quad (k \geq 0).$$

*Then the two displayed congruence conditions hold, and (KE) holds for all sufficiently large  $k$ .*

*Proof.* By the normal-basis theorem, the  $q$ -Frobenius on  $K$ , viewed as an  $\mathbb{F}_q$ -linear operator, is the regular representation of the cyclic group of order  $r$ . Since  $\mu_r \subset \mathbb{F}_q$  and the characteristic does not divide  $r$ , its  $r$  eigenspaces are one-dimensional. Hence  $L_\zeta$  is a line. A nonzero  $v \in L_\zeta$  does not lie in  $\mathbb{F}_q$ , and, as  $r$  is prime, it has degree  $r$ . Moreover

$$(v^r)^q = v^r.$$

If  $v^r = a^r$  for some  $a \in \mathbb{F}_q^\times$ , then  $v/a \in \mu_r \subset \mathbb{F}_q$ , a contradiction. Kummer's prime-degree criterion therefore makes  $X^r - v^r$  irreducible. The same argument applies to  $x = cv$  for every  $c \in \mathbb{F}_q^\times$ .

Let  $\chi : K^\times \rightarrow \mu_p$  be a character of exact order  $p$ ; its kernel is  $(K^\times)^p$ . No element  $1 + cv$ , with  $c \in \mathbb{F}_q$ , vanishes. Character orthogonality gives

$$\begin{aligned} N &:= \#\{c \in \mathbb{F}_q : 1 + cv \in (K^\times)^p\} \\ &= \frac{1}{p} \left( q + \sum_{j=1}^{p-1} \sum_{c \in \mathbb{F}_q} \chi^j(1 + cv) \right). \end{aligned}$$

The element  $v^{-1}$  has degree  $r$ , so Katz's affine-line estimate [20] gives, for  $1 \leq j < p$ ,

$$\left| \sum_{c \in \mathbb{F}_q} \chi^j(1 + cv) \right| = \left| \chi^j(v) \sum_{c \in \mathbb{F}_q} \chi^j(c + v^{-1}) \right| \leq (r-1)\sqrt{q}.$$

Consequently

$$N \geq \frac{q - (p-1)(r-1)\sqrt{q}}{p} > 1.$$

The value  $c = 0$  accounts for only one solution, so another has  $c \neq 0$ ; the corresponding  $x = cv$  proves the first assertion.

For the final assertion, Zsigmondy's theorem (whose exponent-six exception is irrelevant for the odd prime  $r$ ) supplies a prime divisor  $p$  of  $2^r - 1$  with  $\text{ord}_p(2) = r$ . Since  $t = \text{ord}_r(2) \mid r-1$ , the exponent  $t(1 + kr)$  is divisible by  $t$  and coprime to  $r$ . Hence  $r \mid q-1$  and  $\text{ord}_p(q) = r$ . Letting  $k$  grow proves (KE).  $\square$

**Proposition 5.29** (tower-faithful common-section countermodels). *Let  $r$  and  $p$  be distinct odd primes, let  $\nu \geq 1$ , put  $R = r^\nu$ , and let  $q$  be even with*

$$r \mid q-1, \quad \text{ord}_p(q) = R.$$

*Choose any*

$$A_* \in \mathbb{F}_q^\times \setminus (\mathbb{F}_q^\times)^r, \quad \theta^R = A_*,$$

*in  $K = \mathbb{F}_{q^R}$ . Then  $X^R - A_*$  is irreducible over  $\mathbb{F}_q$ . If*

$$\mathcal{N} = \#\{c \in \mathbb{F}_q : 1 + c\theta \in (K^\times)^p\},$$

*then*

$$\mathcal{N} \geq \frac{q - (p-1)(R-1)\sqrt{q}}{p}. \tag{TN}_1$$

*For every nonzero scalar counted by  $\mathcal{N}$ , put*

$$x_\nu = c\theta, \quad x_i = x_\nu^{r^{\nu-i}} \quad (0 \leq i < \nu), \quad \alpha_c = x_0 = c^R A_*.$$

*Then*

$$x_i^r = x_{i-1}, \quad [\mathbb{F}_q(x_i) : \mathbb{F}_q] = r^i, \tag{TN}_2$$

*and every binomial  $X^{r^i} - \alpha_c$  is irreducible over  $\mathbb{F}_q$ . If  $M = \text{ord}(\alpha_c)$ , then*

$$v_r(M) = v_r(q-1), \quad \text{ord}(x_i) = r^i M. \tag{TN}_3$$

*Put*

$$q_0 = q^{R/r}, \quad L = \frac{q^R - 1}{q_0 - 1}, \quad e = \frac{L}{p}.$$



The projective class of  $1 + x_\nu$  in  $K^\times/\mathbb{F}_{q_0}^\times$  misses the full  $p$ -primary part of its order, with no assumption on  $v_p(L)$ . More precisely, for the section polynomials of Proposition 5.9,

$$P_{\alpha_c}(Z) \mid R_{e,d}(Z) \quad (1 \leq d < R, \ r \nmid d), \quad (\text{TN}_4)$$

where  $P_{\alpha_c}$  is the minimal polynomial over  $\mathbb{F}_2$ .

These countermodels occur with every Conway-like degree constraint. Fix  $r, \nu$ , let  $p$  be a primitive prime divisor of  $2^R - 1$ , put  $\tau = \text{ord}_r(2)$ , and set

$$b = \tau 2^j, \quad q = 2^b.$$

For all sufficiently large  $j$ , there is a choice above for which  $\alpha_c$  has exact absolute degree  $b$ . Then

$$[\mathbb{F}_2(x_i) : \mathbb{F}_2] = br^i, \quad \text{ord}_{r^i M}(2) = br^i, \quad (\text{TN}_5)$$

$r \nmid b$ , and every prime divisor of  $b$  is smaller than  $r$ . Thus the full Kummer ancestry, smooth degree, binomial irreducibility, primary order, Lucas sections, and exact  $p$ -primary bookkeeping all admit failure. The remaining distinction is the unscaled Conway selector  $c = 1$ .

*Proof.* Because  $A_*$  is an  $r$ -nonresidue, its order contains the full  $r$ -primary part of  $q - 1$ . The finite-field binomial criterion therefore makes  $X^R - A_*$  irreducible. Let  $\chi : K^\times \rightarrow \mu_p$  have exact order  $p$ . Character orthogonality and Katz's affine-line estimate, applied to the degree- $R$  element  $\theta^{-1}$ , give

$$\mathcal{N} = \frac{1}{p} \left( q + \sum_{h=1}^{p-1} \sum_{c \in \mathbb{F}_q} \chi^h(1 + c\theta) \right),$$

$$\left| \sum_{c \in \mathbb{F}_q} \chi^h(1 + c\theta) \right| \leq (R-1)\sqrt{q}.$$

This proves (TN<sub>1</sub>).

Multiplication by  $c^R$  does not change the  $r$ -power coset of  $A_*$ , so every  $\alpha_c$  is again an  $r$ -nonresidue with the full  $r$ -part in its order. Applying the same binomial criterion to  $X^{r^i} - \alpha_c$  proves (TN<sub>2</sub>); the usual order induction proves (TN<sub>3</sub>).

Since  $\text{ord}_p(q_0) = r$ , one has  $p \mid L$  and  $p \nmid q_0 - 1$ . If  $1 + x_\nu = z^p$ , then in the cyclic group  $K^\times/\mathbb{F}_{q_0}^\times$ , of order  $L$ ,

$$[1 + x_\nu]^e = [z]^L = 1.$$

Thus the full  $p$ -primary valuation of the projective order is strictly deficient even when  $p^2 \mid L$ .

Let  $B = x_{\nu-1}$  and  $n = R/r$ . Expanding in the  $\mathbb{F}_{q_0}$ -basis  $1, x_\nu, \dots, x_\nu^{r-1}$  gives

$$S_{e,k}(B) = 0 \quad (1 \leq k < r).$$

Now  $B^n = \alpha_c$ , and  $1, B, \dots, B^{n-1}$  is an  $\mathbb{F}_q$ -basis. The ancestry expansion

$$S_{e,k}(B) = \sum_{u=0}^{n-1} B^u R_{e,k+ru}(\alpha_c)$$

therefore proves every divisibility in (TN<sub>4</sub>).

For the final family, Zsigmondy's theorem supplies  $p$  because  $R$  is odd. The choice  $b = \tau 2^j$  gives  $\gcd(b, R) = 1$ ,  $r \mid q - 1$ , and  $\text{ord}_p(q) = R$ . Take  $A_*$  primitive. The union of the proper subfields

of  $\mathbb{F}_q$  has at most  $b\sqrt{q}$  elements, while the map  $c \mapsto c^R A_*$  has fibres of size at most  $R$ . Hence a counted nonzero  $c$  with  $\alpha_c$  of exact degree  $b$  exists whenever

$$\frac{q - (p-1)(R-1)\sqrt{q}}{p} > 1 + Rb\sqrt{q}, \quad (\text{TN}_6)$$

which holds for all sufficiently large  $j$ . Since  $\mathbb{F}_2(x_i)$  contains  $\mathbb{F}_2(\alpha_c) = \mathbb{F}_q$ , (TN<sub>2</sub>) gives the degree in (TN<sub>5</sub>); (TN<sub>3</sub>) gives its order statement. Finally  $\tau \mid r-1$ , so  $r \nmid b$  and every prime divisor of  $b$  is smaller than  $r$ .  $\square$

**Remark 5.30** (the common zero locus is large). In Proposition 5.29, define

$$D_e(Z) = \gcd_{\substack{1 \leq d < R \\ r \nmid d}} R_{e,d}(Z), \quad g_R = \gcd(R, q-1).$$

Every nonzero scalar counted by  $\mathcal{N}$  gives the common root  $\alpha_c = c^R A_*$ , and each such value has at most  $g_R$  preimages. Thus

$$\deg D_e \geq \left\lceil \frac{q - (p-1)(R-1)\sqrt{q} - p}{pR} \right\rceil. \quad (\text{TZ}_1)$$

In the binary family of the proposition, let  $M_b(D_e)$  denote the number of distinct degree- $b$  irreducible factors. Excluding the proper-subfield values in the same count gives

$$M_b(D_e) \geq \left\lceil \max \left\{ 0, \frac{q - ((p-1)(R-1) + pRb)\sqrt{q} - p}{pRb} \right\} \right\rceil. \quad (\text{TZ}_2)$$

This tends to infinity with  $j$ . Even the exact simultaneous section pattern is therefore abundant among tower-faithful selectors; only membership of the actual Conway value in this common zero locus remains unresolved.

**Proposition 5.31** (a fixed-translate minimal-selector countermodel). *Let*

$$f(T) = T^{20} + T^3 + 1, \quad F = \mathbb{F}_2[T]/(f), \quad t = T \bmod f, \quad M = 2^{20} - 1.$$

*The polynomial  $f$  is irreducible and  $t$  is primitive in  $F^\times$ . Let  $K = \mathbb{F}_{16} \subset F$ , and put*

$$A = t^{M/5}, \quad x = t^{M/25}, \quad y = 1 + x. \quad (\text{FT}_1)$$

*Then*

$$\text{ord}(A) = 5, \quad \text{ord}(x) = 25, \quad x^5 = A, \quad F = K(x), \quad (\text{FT}_2)$$

*so  $X^5 - A$  is irreducible over  $K$ . Nevertheless the literal fixed translate fails at  $p = 41$ :*

$$y^{M/41} = y^{25575} = 1. \quad (\text{FT}_3)$$

*Indeed*

$$\Phi_5(16) = 69905 = 5 \cdot 11 \cdot 31 \cdot 41, \quad e = \frac{\Phi_5(16)}{41} = 1705,$$

*and  $y^e \in K$ . Therefore the exact binary-section criterion gives*

$$S_{1705,k}(A) = 0 \quad (1 \leq k < 5). \quad (\text{FT}_4)$$

The relative-unit and reciprocity data fail simultaneously. The element

$$\rho = y^{15}$$

has norm one from  $F$  to  $K$  and is a 41-st power. In the cyclotomic lift of conductor 25, at the prime  $\mathfrak{P}$  above two which sends  $\zeta_{25}$  to  $x$ , one has

$$\left(\frac{1 - \zeta_{25}}{\mathfrak{P}}\right)_{41} = 1, \quad \left(\frac{(1 - \zeta_{25}^{16})/(1 - \zeta_{25})}{\mathfrak{P}}\right)_{41} = 1. \quad (\text{FT}_5)$$

The model also has the ordinal-minimal selector property. Let  $s = t^{M/3}$ , of order three. The fifth powers in  $K$  are

$$\{0, 1, s, s^2\}, \quad s^2 = 1 + s.$$

Extend  $(1, s, A)$  to an ordered  $\mathbb{F}_2$ -basis of  $K$ . In the induced binary labeling, the labels 0, 1, 2, 3 are precisely the fifth powers and  $A$  has label 4. Thus  $A$  is literally the least fifth-power nonresidue even though the fixed translate  $1 + x$  fails.

This is not a counterexample to the Conway arm. Its exact missing datum is multiplicative ancestry:

$$\text{ord}(A) = 5, \quad \text{ord}(\alpha_5) = 15.$$

Accordingly the model has conductor 25, whereas the Conway-selected Kummer step has conductor 75. Hence fixed translate, exact Kummer degree, simultaneous binary sections, norm-one Hilbert-90, reciprocity, and ordinal-minimal nonresidue status do not by themselves prove the ordinary arm; the proof must use the full multiplicative order of the lower Conway ancestor and its selected prime above two.

*Proof.* All finite assertions have short polynomial certificates. In the bit encoding of  $\mathbb{F}_2[T]$ ,  $f = 0x100009$ . Rabin's criterion gives

$$\begin{aligned} t^{2^{20}} &= t, \\ R_{10} &:= t^{2^{10}} + t = 0x46101, \quad 0x237d4 f + 0x81c75 R_{10} = 1, \\ R_4 &:= t^{16} + t = 0x10002, \quad 0xec7d f + 0xec7ca R_4 = 1. \end{aligned} \quad (\text{FT}_6)$$

Since the prime divisors of 20 are two and five, these two gcd certificates prove irreducibility. Moreover

$$M = 3 \cdot 5^2 \cdot 11 \cdot 31 \cdot 41,$$

and the residues  $t^{M/p}$ , for  $p = 3, 5, 11, 31, 41$ , are respectively

$$0xc5deb, \quad 0xff200, \quad 0x4a5df, \quad 0x1e639, \quad 0xa00ea, \quad (\text{FT}_7)$$

none equal to one. Thus  $t$  is primitive.

The orders in  $(\text{FT}_2)$  follow immediately. The element  $A$  has exact degree  $\text{ord}_5(2) = 4$ , hence generates the unique  $K = \mathbb{F}_{16}$ . The element  $x$  has degree  $\text{ord}_{25}(2) = 20$ , so  $F = K(x)$ ; its degree-five minimal polynomial is  $X^5 - A$ . Direct reduction gives

$$x = 0xc2073, \quad y = 0xc2072, \quad y^{25575} = 1, \quad y^{1705} = 0x85029, \quad (\text{FT}_8)$$

and the last element is fixed by the sixteenth-power Frobenius. This proves  $(\text{FT}_3)$ , membership  $y^e \in K$ , and hence  $(\text{FT}_4)$  by  $(\text{OB}_6)$ .

Since  $y$  is a 41-st power, so is  $y^{15}$ ; telescoping its five relative conjugates gives norm one. Reduction of the two cyclotomic units in  $(\text{FT}_5)$  gives  $y$  and  $y^{15}$ , respectively, so both residue symbols are trivial.

Finally  $K^\times$  has order fifteen, and its fifth-power image is the order-three subgroup generated by  $s$ . The relation  $s^2 + s + 1 = 0$  identifies its four affine elements with the first four binary labels. Since  $A$  has order five, it does not lie in  $\mathbb{F}_4 = \langle 1, s \rangle$ ; thus  $(1, s, A)$  is independent and extends to the asserted basis.  $\square$

## 5.5 The selected phase at $p = 359$

Set

$$d = 19,580, \quad E = 179d = 3,504,820, \quad B = \mathbb{F}_{2^d}, \quad F = \mathbb{F}_{2^E}, \quad L = \mathbb{F}_{2^{179}}.$$

For an independent derivation of  $m_{359} = 1$ ,  $f(359) = 179$ ,  $\mathcal{Q}(f(359)) = \mathcal{Q}(179) = \{179\}$ , and the candidate element is  $\beta = \kappa_{179} + 1$ . To represent  $\kappa_{179}$ , the tower uses the lower prime row  $f(179) = 178$ ,  $\mathcal{Q}(178) = \{89\}$ ,  $m_{179} = 1$ . Thus in the tower model  $F = B(x)$  with  $x = \kappa_{179}$  and

$$x^{179} = \kappa_{89} + 1 \in B,$$

because the lower row  $m_{179} = 1$  has already been certified.

The cheap Kummer norm down the top tower step is not the 359-test norm. Indeed,

$$N_{F/B}(x + 1) = 1 + (\kappa_{89} + 1) = \kappa_{89}$$

(evaluate the monic polynomial  $T^{179} + (\kappa_{89} + 1)$  at  $T = 1$ , in characteristic 2). But  $359 \nmid |B^\times| = 2^{19580} - 1$ , since  $\text{ord}_{359}(2) = 179$  and  $179 \nmid 19580$ . Thus the structurally cheap norm lands in exactly the field where the 359-primary obstruction is invisible.

The norm forced by Proposition 3.4 is the transverse one

$$N_{F/L}(\beta) = \prod_{i=0}^{19579} \text{Frob}^{179i}(\beta) \in L.$$

Here  $\gcd(179, 19580) = 1$ , so  $B \cap L = \mathbb{F}_2$  and  $F = BL$ . The Frobenius  $\text{Frob}^{179}$  preserves  $B$  as a set but acts nontrivially on it; it is not the relative Kummer action  $x \mapsto \zeta x$  over  $B$ . Consequently the known lower certificates for  $m_{89}$  and  $m_{179}$  do not propagate to  $m_{359}$  by taking the easy top-step norm.

The Wieferich caveat is absent at the next two pressure points: direct modular checks give  $2^{179} \not\equiv 1 \pmod{359^2}$  and  $2^{359} \not\equiv 1 \pmod{719^2}$ . The order form of Corollary 3.2 is therefore equivalent to the full power criterion at  $p = 359$  and at the first row beyond the table,  $p = 719$ , which is evaluated below.

**Proposition 5.32** (safe-prime trace detector). *Let  $p = 2r + 1$  be prime, where  $r$  is an odd prime and  $\text{ord}_p(2) = r$ . For every  $A \in \mu_p(\mathbb{F}_{2^r})$ ,*

$$A \neq 1 \iff \text{Tr}_{\mathbb{F}_{2^r}/\mathbb{F}_2}(A + A^{-1}) = 1. \quad (\text{SPH}_0)$$

*Thus every safe-prime instance of the ordinary arm reduces its selected power-residue obstruction to one absolute trace bit; the proposition detects that bit but does not evaluate it.*

*Proof.* The subgroup  $H = \langle 2 \rangle \subset \mathbb{F}_p^\times$  has order  $r$ , so it is the quadratic-residue subgroup. Since  $p = 2r + 1 \equiv 3 \pmod{4}$ , the cosets  $H$  and  $-H$  partition  $\mathbb{F}_p^\times$ . If  $A = \zeta^u \neq 1$ , its Frobenius exponents and those of  $A^{-1}$  are respectively  $uH$  and  $-uH$ . Hence, in characteristic two,

$$\mathrm{Tr}_{\mathbb{F}_{2^r}/\mathbb{F}_2}(A + A^{-1}) = \sum_{j=1}^{p-1} \zeta^{uj} = 1.$$

For  $A = 1$ , the traced element is zero. □

**Proposition 5.33** (the selected phase at  $p = 359$ ). *Retain the notation above and put*

$$\Theta = N_{F/L}(x + 1), \quad c = \frac{2^{179} - 1}{359}, \quad A = \Theta^c \in \mu_{359}(L).$$

*Then*

$$2^{19580} \equiv 3 \pmod{359}, \quad v_{359}(2^{179} - 1) = 1,$$

*and the independent order condition has the exact binary form*

$$m_{359} = 1 \iff A \neq 1 \iff \mathrm{Tr}_{L/\mathbb{F}_2}(A + A^{-1}) = 1. \quad (\mathrm{SPH}_1)$$

*There is also an intrinsic selected-phase resultant description. Let  $a = x^{179} = \kappa_{89} + 1$ , and let  $P_a(T) \in \mathbb{F}_2[T]$  be its degree-19580 minimal polynomial. Over  $L$  one has*

$$P_a(X^{179}) = \prod_{j=0}^{178} G_j(X), \quad G_j(X) = \prod_{i=0}^{19579} (X - x^{2^{j+179i}}), \quad (\mathrm{SPH}_2)$$

*where the  $G_j$  are the 179 Frobenius-conjugate irreducible factors of degree 19580. The common-embedding factor is  $G_0$ , and*

$$G_0(1) = \Theta, \quad \prod_{j=0}^{178} G_j(1) = P_a(1) = 1. \quad (\mathrm{SPH}_3)$$

*Thus the full resultant symmetrizes the required class away; the only remaining datum is the selected value in  $(\mathrm{SPH}_1)$ .*

*Proof.* Since  $19580 \equiv 69 \pmod{179}$ , repeated squaring gives  $2^{69} \equiv 3 \pmod{359}$ , proving the first congruence. The same calculation one power higher gives

$$2^{179} \equiv 10412 = 1 + 29 \cdot 359 \pmod{359^2},$$

so the displayed valuation is one. Proposition 3.4 now identifies the 359-power test on  $x + 1$  with the corresponding test on  $\Theta \in L^\times$ . Since the 359-part of  $|L^\times|$  has order exactly 359, that test is nontrivial exactly when  $A \neq 1$ .

Proposition 5.32, with  $(p, r) = (359, 179)$ , proves the final equivalence in  $(\mathrm{SPH}_1)$ .

The lower-row condition says that  $a = \kappa_{89} + 1$  is not a 179-th power in  $B$ , while  $\mu_{179} \subset B$ . The standard prime Kummer criterion therefore makes  $X^{179} - a$  irreducible over  $B$ ; equivalently  $P_a(X^{179})$  is irreducible of degree  $179 \cdot 19580$  over  $\mathbb{F}_2$ . Over  $L = \mathbb{F}_{2^{179}}$ , its roots split into the 179 orbits in  $(\mathrm{SPH}_2)$ , each of length 19580; this proves the asserted factorization and irreducibility. Evaluating the common-embedding orbit at one gives

$$G_0(1) = \prod_{i=0}^{19579} (1 + x^{2^{179i}}) = N_{F/L}(1 + x) = \Theta.$$

Finally, evaluation of the full factorization at one gives  $P_a(1)$ . This is  $N_{B/\mathbb{F}_2}(1+a)$ , up to the irrelevant characteristic-two sign, and  $1+a = \kappa_{89} \neq 0$ ; its norm lies in  $\mathbb{F}_2^\times = \{1\}$ . Hence (SPH<sub>3</sub>) follows.  $\square$

**Proposition 5.34** (the fixed full-conductor polynomial at  $p = 359$ ). *Define polynomials over  $\mathbb{F}_2$  by*

$$\begin{aligned} F_5(Z) &= Z^4 + Z + 1, \\ F_{11}(Z) &= F_5((Z+1)^5), \\ F_{89}(Z) &= F_{11}((Z+1)^{11}), \\ F_{179}(Z) &= F_{89}((Z+1)^{89}), \\ \mathcal{P}_{359}(X) &= F_{179}(X^{179}). \end{aligned} \tag{OFC_1}$$

*The actual Conway dependencies*

$$\alpha_{179} = \kappa_{89} + 1, \quad \alpha_{89} = \kappa_{11} + 1, \quad \alpha_{11} = \kappa_5 + 1, \quad \alpha_5 = \kappa_4$$

*and their degree data identify these as the minimal polynomials of  $\alpha_5, \alpha_{11}, \alpha_{89}, \alpha_{179}, \kappa_{179}$ , respectively. Their exact degree/term-count pairs are*

$$(4, 3), \quad (20, 5), \quad (220, 55), \quad (19580, 3447), \quad (3504820, 3447). \tag{OFC_2}$$

*Thus the actual full-conductor polynomial  $\mathcal{P}_{359}$  is sparse, but it retains both the complete ancestry and the mark  $X \mapsto \kappa_{179}$ .*

*Put  $G = 2^{3504820} - 1$ . The unresolved selected assertion has the equivalent exact forms*

$$\begin{aligned} m_{359} = 1 &\iff \gcd(\mathcal{P}_{359}(X), (1+X)^{G/359} + 1) = 1 \\ &\iff \mathcal{P}_{359}(T^{359} + 1) \text{ is irreducible over } \mathbb{F}_2 \\ &\iff \text{Tr}_{\mathbb{F}_{2^{179}}/\mathbb{F}_2}(A + A^{-1}) = 1, \end{aligned} \tag{OFC_3}$$

*where  $A$  is the selected phase of Proposition 5.33. Under failure the gcd in (OFC<sub>3</sub>) is  $\mathcal{P}_{359}$ , the composition is reducible, and the trace is zero.*

*There is consequently no full-ancestry fixed-mark countermodel to this test. Any two roots of the irreducible polynomial  $\mathcal{P}_{359}$  give isomorphic marked finite-field realizations, and a field isomorphism preserves whether  $1 + \kappa_{179}$  has a 359-th root. Once (OFC<sub>1</sub>) and the mark are retained, the displayed gcd or trace must actually be evaluated; changing models cannot alter the answer.*

*Proof.* Proposition 5.11, applied successively with  $Q = 5, 11, 89$ , gives the first four minimal polynomials in (OFC<sub>1</sub>). Proposition 5.10 at the final 179-Kummer step gives  $\mathcal{P}_{359}(X) = F_{179}(X^{179})$  and its irreducibility. The degrees follow multiplicatively. The term counts are exact finite polynomial arithmetic; the maintained certificate computes the canonical little-endian coefficient-bit encoding of  $\mathcal{P}_{359}$ , which has 438103 bytes and SHA-256

7f1d6d5cb36c2c233b5b67282540ef1a238f0b2374b81171e9b726cc2c559978.

The first equivalence in (OFC<sub>3</sub>) is the irreducible-factor criterion applied to the selected root  $\kappa_{179}$ . Since  $\mu_{359} \subset \mathbb{F}_{2^{3504820}}$ , the prime Kummer–Capelli criterion identifies nonvanishing of that Euler test with irreducibility of  $T^{359} - (1 + \kappa_{179})$ ; taking the norm of its coefficients to  $\mathbb{F}_2$  gives the displayed composition. The last equivalence is Proposition 5.33. Finally a field isomorphism carries

$y^{359} = 1 + \kappa_{179}$  to the same equation for the image of the marked root, proving the no-countermodel assertion.

The exact sparse compositions and fingerprints are reproduced by `experiments/ordinary_359_full_conductor.py`. Lean checks the first dependency composition and the invariance of marked power status under a ring equivalence. The full finite-field dependency chain, degree identifications, and gcd/trace evaluation remain paper-level; in particular this proposition packages but does not prove  $m_{359} = 1$ .  $\square$

**Proposition 5.35** (a compact factor certificate at the fixed  $p = 359$  mark). *Let  $h(Z) \in \mathbb{F}_2[Z]$  be irreducible of degree 179, put  $k = \mathbb{F}_2[Z]/(h)$ , and regard  $\mathcal{P}_{359}$  as a polynomial in  $k[X]$ . Suppose that a monic polynomial  $g(X) \in k[X]$  satisfies*

$$\deg g = 19580, \quad g \mid \mathcal{P}_{359}. \quad (\text{OFCert}_1)$$

Put

$$\theta = g(1), \quad c = \frac{2^{179} - 1}{359}.$$

Then  $g$  is one of the irreducible degree-19580 factors of  $\mathcal{P}_{359}$  over  $k$ , and

$$m_{359} = 1 \iff \theta^c \neq 1. \quad (\text{OFCert}_2)$$

Consequently, an explicit pair  $(h, g)$  satisfying  $(\text{OFCert}_1)$  and  $g(1)^c \neq 1$  is a compact exact certificate for the fixed-mark assertion. Packed coefficient bits for  $g$  require only  $19581 \cdot 179 = 3504999$  bits, or 438125 bytes, before framing. One deterministic coefficient-field choice is

$$h(Z) = Z^{179} + Z^4 + Z^2 + Z + 1;$$

the Rabin equalities  $Z^{2^{179}} \equiv Z \pmod{h}$  and  $\gcd(h, Z^2 + Z) = 1$  certify its irreducibility.

*Proof.* Write  $r = 179$ ,  $d = 19580$ , and  $F = \mathbb{F}_{2^{rd}}$ . Since  $\mathcal{P}_{359}$  is irreducible of degree  $rd$  over  $\mathbb{F}_2$ , the standard finite-field base-change theorem factors it over  $k = \mathbb{F}_{2^r}$  into exactly  $r$  irreducible factors, each of degree  $d$ . Thus  $(\text{OFCert}_1)$  makes  $g$  one such factor. If  $y$  is a root of  $g$ , then

$$g(1) = N_{F/k}(1 + y) = (1 + y)^{(2^{rd}-1)/(2^r-1)}.$$

Therefore

$$g(1)^c = (1 + y)^{(2^{rd}-1)/359}. \quad (\text{OFCert}_3)$$

Every root of  $\mathcal{P}_{359}$  is an absolute-Frobenius conjugate of the Conway-marked root  $\kappa_{179}$ . Frobenius preserves the identity or nonidentity of the right side of  $(\text{OFCert}_3)$ . Proposition 5.34 now identifies its nonidentity with  $m_{359} = 1$ .  $\square$

**Proposition 5.36** (a Hilbert-root certificate at the fixed  $p = 359$  mark). *Put*

$$h(Z) = Z^{179} + Z^4 + Z^2 + Z + 1, \quad k = \mathbb{F}_2[Z]/(h),$$

let  $f(A) = F_{179}(A)$  be the degree-19580 polynomial in  $(\text{OFC}_1)$ , and set  $E = k[A]/(f)$ . Suppose  $y \in E$  satisfies

$$y^{179} = A,$$

let  $\tilde{y}(A) \in k[A]$  be its unique representative of degree less than 19580, and put

$$\theta = \text{Res}_A(f(A), 1 + \tilde{y}(A)) = N_{E/k}(1 + y). \quad (\text{OHC}_1)$$

Then the minimal polynomial  $g$  of  $y$  over  $k$  is an irreducible degree-19580 factor of

$$\mathcal{P}_{359}(X) = f(X^{179}),$$

$\theta = g(1)$ , and

$$\boxed{m_{359} = 1 \iff \theta^{(2^{179}-1)/359} \neq 1.} \quad (\text{OHC}_2)$$

The checked-in v1 certificate encodes  $y$  as a 438103-byte little-endian bitstream. Bit  $179i + j$  is the coefficient of  $Z^j$  in the  $A^i$ -coefficient of  $\tilde{y}$ ; its SHA-256 digest is

$$\text{c62433e428e6b0942c210b2df2543fcff6a9e444b835ceb5390db1c9e433bd9e.}$$

In the polynomial basis  $1, Z, \dots, Z^{178}$  of  $k$ , exact computation gives

$$\begin{aligned} \theta &= 0x28494a54795638cf203b2404fe5133e2ae0e13c2ca4e6, \\ \theta^{(2^{179}-1)/359} &= 0x7f81a92c0c3b04e42a5d3aa423e1bc4d0082a3522e87e \neq 1. \end{aligned} \quad (\text{OHC}_3)$$

The latter element has order 359, and the safe-prime trace of it plus its inverse is one. Therefore the selected value is

$$\boxed{m_{359} = 1.}$$

*Proof.* The polynomial  $h$  is irreducible by its two Rabin residues, which the maintained verifier checks. Since  $\gcd(19580, 179) = 1$ , the standard finite-field base-change theorem shows that  $f$  remains irreducible over  $k$ . Hence  $[E : k] = 19580$ . The identity  $A = y^{179}$  gives

$$k(A) \subseteq k(y) \subseteq E = k(A),$$

so equality holds and  $y$  also has degree 19580 over  $k$ . Moreover  $f(y^{179}) = f(A) = 0$ , so its minimal polynomial  $g$  is one of the degree-19580 factors of  $\mathcal{P}_{359}$  over  $k$ . The norm identity in (OHC<sub>1</sub>) gives  $\theta = g(1)$ , and Proposition 5.35 proves (OHC<sub>2</sub>).

The maintained verifier `experiments/ordinary_359_hilbert_certificate.py` reconstructs  $f$  from the full selected dependency chain, checks the certificate hash and  $y^{179} = A$ , and in authoritative `--full` mode recomputes the 19579-step Euclidean resultant in (OHC<sub>1</sub>). It then checks both masks in (OHC<sub>3</sub>), nonidentity, order 359, and the safe-prime trace. Its default quick mode checks the root and stored phase arithmetic but explicitly does not recompute the norm, so only full mode verifies the complete certificate.  $\square$

*Source-pinned fixed-mark computation audit.*

The independently maintained calculator of Peeters [32], at commit `427d0db3d40fdfaf4345deb14b160a00cf5250a1`, performs this exact fixed-mark test in the mixed-radix algebra with components

$$\{2, 4, 5, 11, 89, 179\}, \quad (1, 2, 4, 20, 220, 19580, 3504820)$$

as basis strides. Its defining reductions are

$$u_2^2 = u_2 + 1, \quad u_4^2 = u_4 + u_2, \quad u_5^5 = u_4, \quad u_{11}^{11} = u_5 + 1, \quad u_{89}^{89} = u_{11} + 1, \quad u_{179}^{179} = u_{89} + 1.$$

Thus its candidate indices `[0, 19580]` are literally  $1 + \kappa_{179}$ , with the complete ancestry of (OFC<sub>1</sub>). The code computes the exact exponent  $(2^{3504820} - 1)/359$ , records `{359, 1}`, and reports 177379 seconds for that row. Its timing control flow records this nonzero duration only for an uncached calculation. This pins an executed exact computation, not merely the copied OEIS integer.



The audited current upstream commit has an empty calculation log, but the repository history retains substantially stronger evidence. At historical commit

7a26543f11f5319d04da5402840beecc9e5b35fe

two separate complete runs start from one result term and finish all 3504812 exponent bits with the same final support of 1743227 terms. Their stored elapsed times are 177379 and 177717 seconds. The maintained provenance checker `experiments/ordinary_359_source_audit.py` reads those historical git blobs, verifies their complete-file SHA-256 digests, and checks both endpoint pairs and both result rows. Since basis one has support size one, the logged support size proves nonidentity inside the pinned exact sparse implementation; the repeat run is a reproducibility check of that implementation, not an independent backend.

The upstream history itself retains neither the final coefficient vector nor a factor  $g$  as in Proposition 5.35. Its component comparator also ties the (2)- and (4)-components and therefore relies on the published input order being preserved; the recorded GNU/Linux run has the stated order, but arbitrary source replay is not formally portable without pinning it. This caveat does not affect the checked-in Hilbert-root certificate of Proposition 5.36; the historical runs now provide independent source-pinned corroboration of that local exact computation.

**Proposition 5.37** (nested compression of the next carry  $p = 719$ ). *Put*

$$d = 19580, \quad r = 179, \quad s = 359, \quad p = 719, \quad b = rd,$$

*and retain the selected fields*

$$a_0 = \kappa_{89} + 1, \quad B_0 = \mathbb{F}_{2^d}, \quad B = B_0(x) = \mathbb{F}_{2^b}, \quad x^r = a_0.$$

*Let*

$$K = \mathbb{F}_{2^s}, \quad C = B_0K = \mathbb{F}_{2^{ds}}, \quad E = BK = C(x) = \mathbb{F}_{2^{bs}}.$$

*The relevant degrees are*

$$[B : B_0] = r, \quad [C : B_0] = s, \quad [E : C] = r, \quad [E : B] = s, \quad [C : K] = d. \quad (\text{O719}_1)$$

*Write  $\beta = 1 + x = \alpha_{359}$ . The certified  $p = 359$  phase is*

$$\eta = \beta^{(2^b-1)/s} \neq 1.$$

*Consequently  $T^s + \beta$  is irreducible over  $B$ . If  $y^s = \beta$ , then  $B(y) = E$ , and every such  $y$  is a relative Frobenius conjugate of the Conway-marked  $\kappa_{359}$ . Put*

$$z = N_{E/C}(1 + y), \quad \theta = N_{C/K}(z) = N_{E/K}(1 + y).$$

*Then the next excess row has the exact tests*

$$\boxed{m_{719} = 1 \iff \theta^{(2^s-1)/p} \neq 1 \iff z^{(2^{ds}-1)/p} \neq 1.} \quad (\text{O719}_2)$$

*The construction has two exact lower norm shadows. If  $w = N_{E/C}(y)$ , then*

$$\boxed{w^s = \kappa_{89}, \quad w = \kappa_{89}^{e_0} \in B_0, \quad e_0 = \frac{r(2^d - 1) + 1}{s}.} \quad (\text{O719}_3)$$

Moreover

$$\boxed{N_{C/B_0}(z) = a_0 = \kappa_{89} + 1.} \quad (\text{O719}_4)$$

Neither shadow determines the target. If  $G = 2^{ds} - 1$ , then  $\mu_p \subset K \subset C$ , and for every  $\zeta \in \mu_p$ ,

$$N_{C/B_0}(\zeta) = 1, \quad (z\zeta)^{G/p} = z^{G/p}\zeta^{G/p}. \quad (\text{O719}_5)$$

Here  $v_p(G) = 1$ , so  $\zeta \mapsto \zeta^{G/p}$  permutes  $\mu_p$ . Thus, for every  $z$  in the fixed fibre  $N_{C/B_0}^{-1}(a_0)$ , its  $\mu_p$ -twist orbit contains exactly one element with trivial  $p$ -phase and  $p - 1$  with nontrivial phase. This is a no-go for the scalar norm shadows only; it does not vary the selected Conway factor.

There is nevertheless an exact finite certificate route. The polynomial

$$\mathcal{P}_{719}^{\text{nest}}(T) = (T^s + 1)^r + a_0 \in B_0[T] \quad (\text{O719}_6)$$

is irreducible of degree  $rs = 64261$  over  $B_0$ . Over  $C$  it is the product of  $s$  irreducible factors of degree  $r$ . For any one monic factor  $g$ ,

$$z_g = g(1) = N_{C(y_g)/C}(1 + y_g)$$

and

$$\boxed{m_{719} = 1 \iff g(1)^{(2^{7029220}-1)/719} \neq 1.} \quad (\text{O719}_7)$$

The factors are  $B_0$ -Galois conjugate, so identity or nonidentity is independent of the chosen factor. A naive packed table of all 180 coefficients of  $g$  would occupy 158157450 bytes before framing (the known monic coefficient may of course be omitted).

Thus, before the crossed certificate below is evaluated, the next unsupported carry is reduced from a direct degree-1258230380 Euler test to one degree-179 factor and one Euler test in  $C = \mathbb{F}_{2^{7029220}}$ . The proposition supplies the equivalence and a concrete exact algorithm, but this proposition alone does not evaluate  $g(1)$ . Proposition 5.40 performs the equivalent smaller crossed-tower test exactly.

*Proof.* The gcds

$$\gcd(d, r) = \gcd(d, s) = \gcd(b, s) = 1$$

give (O719<sub>1</sub>). Since  $\text{ord}_s(2) = r$ , one has  $\mu_s \subset B$ . Proposition 5.36 identifies the displayed  $\eta$  with the certified nontrivial  $s$ -residue of  $\beta$ . As  $s$  is prime, the Kummer criterion makes  $T^s + \beta$  irreducible over  $B$ , and gives all assertions about  $y$ .

For completeness,  $y$  also has a Hilbert–90 construction from the certified phase. Choose a normal element  $t$  of  $K/\mathbb{F}_2$ , let  $\sigma = \text{Frob}^b$ , and set

$$v = \sum_{i=0}^{s-1} \eta^{-i} \sigma^i(t).$$

Linear disjointness makes the conjugates of  $t$   $B$ -linearly independent, so  $v \neq 0$ , while  $\sigma(v) = \eta v$ . Hence  $c = v^s \in B$ . With

$$M = \frac{2^b - 1}{s},$$

one has  $c^M = v^{2^b-1} = \eta = \beta^M$ . The direct check  $2^{179} \not\equiv 1 \pmod{359^2}$ , together with  $s \nmid d$ , gives  $\gcd(s, M) = 1$ . If  $se \equiv 1 \pmod{M}$ , then

$$y = (\beta/c)^e v$$

satisfies  $y^s = \beta$ .

The singleton transverse criterion at  $p = 719$  gives the first equivalence in (O719<sub>2</sub>). Norm transitivity gives

$$\theta^{(2^s-1)/p} = z^{(2^{ds}-1)/(2^s-1) \cdot (2^s-1)/p} = z^{(2^{ds}-1)/p},$$

which gives the second. The direct congruence

$$2^{359} \equiv 1 + 696 \cdot 719 \pmod{719^2} \quad (\text{O719}_8)$$

shows that this Euler test also has the full 719-primary strength.

For the first shadow,

$$w^s = N_{E/C}(y^s) = N_{E/C}(1+x) = 1 + a_0 = \kappa_{89}.$$

Because  $r \nmid ds$ ,  $s$ -powering is bijective on  $C^\times$ ; it is also bijective on  $B_0^\times$ . The root is therefore unique and belongs to  $B_0$ . The congruence  $2^d \equiv 3 \pmod{s}$  gives

$$se_0 = r(2^d - 1) + 1,$$

which proves (O719<sub>3</sub>). In the other tower direction,

$$N_{E/B}(1+y) = 1 + y^s = x,$$

and therefore

$$N_{C/B_0}(z) = N_{E/B_0}(1+y) = N_{B/B_0}(x) = a_0,$$

proving (O719<sub>4</sub>).

Since  $p \nmid 2^d - 1$  but  $p \mid 2^{ds} - 1$ , the relative norm exponent  $(2^{ds} - 1)/(2^d - 1)$  is divisible by  $p$ , proving the first identity in (O719<sub>5</sub>). Applying the lifting-the-exponent formula to (O719<sub>8</sub>) gives

$$v_p(G) = v_p(2^s - 1) + v_p(d) = 1.$$

Thus  $p \nmid G/p$ , which proves the permutation and the exact phase-blindness assertion.

Finally  $x = y^s + 1$ , so  $y$  is a root of  $\mathcal{P}_{719}^{\text{nest}}$ . Since  $[B_0(y) : B_0] = rs$ , that polynomial is its minimal polynomial. The standard finite-field base-change theorem, with extension degree  $s$ , splits it over  $C$  into  $\gcd(rs, s) = s$  irreducible factors of degree  $r$ . For a root  $y_g$  of any factor  $g$ , the monic evaluation formula gives

$$g(1) = N_{C(y_g)/C}(1 + y_g).$$

All factors are Frobenius conjugate over  $B_0$ , so Frobenius preserves identity or nonidentity of the Euler phase. This proves (O719<sub>7</sub>).

The finite-field factorization, norm shadows, and phase-blindness argument remain paper-level. The compact crossed-tower construction below supplies a smaller exact certificate route, and Proposition 5.40 instantiates its concrete selected evaluation.  $\square$

**Proposition 5.38** (a compact crossed-tower certificate for the fixed  $p = 719$  mark). *Retain Proposition 5.37, and put*

$$r = 179, \quad d = 19580, \quad s = 359, \quad p = 719, \quad D = \mathbb{F}_{2^r}, \quad K = \mathbb{F}_{2^s}.$$

Let  $B = D[A]/(f(A))$ , where  $f = F_{179}$  has degree  $d$ , and retain the checked  $p = 359$  root  $x \in B$  with  $x^r = A$ . Put  $\beta = 1 + x$ ,

$$M = \frac{2^{rd} - 1}{s}, \quad \eta = \beta^M \in \mu_s(D),$$

where the certificate proves  $\eta \neq 1$ . The exact congruences

$$M \equiv 241 \pmod{s}, \quad 241^{-1} \equiv 216 \pmod{s}$$

allow us to set  $c = \eta^{216}$ . Then  $c^M = \eta$ ,  $c \notin (D^\times)^s$ , and

$$H := D[V]/(V^s + c) \cong \mathbb{F}_{2^{rs}}$$

is a field; write  $v$  for the class of  $V$ , and identify the unique subfield of  $H$  of order  $2^s$  with  $K$ . With compatible embeddings in an algebraic closure, this is the compositum  $H = DK$ .

Suppose that  $a \in B$  and a monic  $f_a(T) \in D[T]$  satisfy

$$a^s = \beta/c, \quad \deg f_a = d, \quad f_a(a) = 0. \quad (\text{O719C}_1)$$

Then  $D(a) = B$ , so  $f_a$  is the minimal polynomial of  $a$  over  $D$ . Put

$$W = v^d f_a(v^{-1}) \in H.$$

The fixed marked row has the exact compact test

$$\boxed{m_{719} = 1 \iff W^{(2^{rs}-1)/p} \neq 1.} \quad (\text{O719C}_2)$$

Thus an explicit pair  $(a, f_a)$  satisfying  $(\text{O719C}_1)$  and the nonidentity in  $(\text{O719C}_2)$  is an exact certificate. In the fixed polynomial bases, omitting the known monic coefficient, each of  $a$  and  $f_a$  occupies exactly  $rd = 3504820$  bits, or 438103 bytes when separately byte-aligned; two such files total 876206 bytes. A single continuously packed concatenation instead occupies 876205 bytes. An optional stored  $H$ -element occupies  $rs = 64261$  bits, or 8033 bytes. All four payloads  $a, f_a, W$ , and the final 719-phase are instantiated below.

*Proof.* The direct checks modulo  $s^2$  give  $v_s(2^r - 1) = 1$ . Hence a nontrivial element of  $\mu_s(D)$  is not an  $s$ -th power, and the prime Kummer criterion gives the field  $H$ . The two displayed congruences give  $c^M = \eta = \beta^M$ , hence  $(\beta/c)^M = 1$ . Since  $\gcd(s, M) = 1$ , the  $s$ -th power image in  $B^\times$  is exactly the subgroup of order  $M$ , so an  $a$  in  $(\text{O719C}_1)$  exists.

From  $x = ca^s + 1$  and  $A = x^r$  one obtains

$$D(A) \subseteq D(a) \subseteq B.$$

The polynomial  $f$  remains irreducible over  $D$ , because its degree  $d$  is coprime to  $r$ . Thus  $[B : D] = d$ , the inclusions are equalities, and the asserted  $f_a$  is the minimal polynomial.

Put  $E = BH = B(DK) = BK$ , the same field  $E$  as in Proposition 5.37, and let  $y = av$ . Then  $y^s = \beta$ , so  $y$  is one of the marked equation's Frobenius-conjugate roots. Since  $\gcd(d, s) = 1$ ,  $f_a$  remains irreducible over  $H$ , and in characteristic two

$$N_{E/H}(1 + y) = \prod_{\sigma \in \text{Gal}(B/D)} (1 + v\sigma(a)) = v^d f_a(v^{-1}) = W. \quad (\text{O719C}_3)$$

Norm transitivity and Proposition 5.37 give

$$N_{E/K}(1 + y) = N_{H/K}(W) = W^{(2^{rs}-1)/(2^s-1)}.$$

Raising to  $(2^s - 1)/p$  proves  $(\text{O719C}_2)$ . Frobenius conjugation preserves identity or nonidentity, so the allowed choice of  $a$  does not change the test.

Lean kernel-checks the scaled finite-product algebra underlying the last equality in  $(\text{O719C}_3)$  and conditional nested exponent multiplication. The field identifications, Kummer construction, minimal-polynomial and norm specializations, numerical exponent factorization, and the deduction from the finite certificate remain paper-level.  $\square$

**Proposition 5.39** (checked first crossed payload for the fixed  $p = 719$  mark). *Retain Proposition 5.38. In its fixed polynomial bases there is an explicitly stored element  $a \in B$  satisfying*

$$a^{359} = \frac{1+x}{c}, \quad ca^{359} + 1 = x, \quad (ca^{359} + 1)^{179} = A.$$

*Its continuously packed little-endian coefficient payload has 3504820 bits, hence 438103 bytes, and SHA-256*

*c7385164fedcbb971ff2bb239ef9e0a0d1ccb81b16edd81eee2a139030ba8758.*

*Consequently  $D(a) = B$ , so the minimal polynomial of  $a$  over  $D$  has degree 19580. The maintained exact verifier reconstructs  $x$  and  $c$  from the checked  $p = 359$  artifact and verifies the displayed equations. Its full-upstream mode additionally recomputes  $N_{B/D}(1+x)$ .*

*This constructs the first payload required by (O719C<sub>1</sub>). Proposition 5.40 supplies the remaining payloads and evaluates the final phase.*

*Proof.* The finite computation is exactly the maintained artifact verifier described above. From  $x = ca^{359} + 1$  and  $A = x^{179}$  one gets  $D(A) \subseteq D(a) \subseteq B$ . Since  $B = D(A)$ , equality follows. The finite computation is not kernel-checked in Lean.  $\square$

**Proposition 5.40** (exact crossed evaluation at the fixed  $p = 719$  mark). *Retain Propositions 5.38 and 5.39. There is an explicitly stored monic polynomial  $f_a \in D[T]$  of degree 19580, represented by its 19580 non-leading coefficients, such that  $f_a(a) = 0$ . Its separately byte-aligned coefficient payload has 438103 bytes and SHA-256*

*66a4aaa3a406d67bc9aba4ae56c9bc218f65d296d7c68725da6b1417854b92de.*

*The exact crossed norm*

$$W = v^{19580} f_a(v^{-1}) \in H$$

*has a 8033-byte payload with SHA-256*

*fc00d40eabdba738d950bf17317e68418d0b8813ba3325c1ce35b1975f9569c6.*

*Finally*

$$\Xi = W^{(2^{64261}-1)/719} \neq 1, \quad \Xi^{719} = 1, \quad (\text{O719E}_1)$$

*and the 8033-byte payload for  $\Xi$  has SHA-256*

*da5858b2e53bfce7e83944a41e264a0fa01a3f0a1b030b889239afd383708a81.*

*Consequently*

$$\boxed{m_{719} = 1.} \quad (\text{O719E}_2)$$

*Proof.* The maintained exact verifier checks the hashes and canonical padding of all four payloads, reconstructs the certified  $p = 359$  root and phase, verifies  $a^{359} = (1+x)/c$ , and recomputes  $W$  from the displayed scaled minimal-polynomial evaluation. Its authoritative full mode also verifies  $f_a(a) = 0$  by dense modular composition. It evaluates the Euler exponent exactly in base  $2^{179}$ , using the diagonal Frobenius action in the Kummer basis, and obtains (O719E<sub>1</sub>). Proposition 5.38 then gives (O719E<sub>2</sub>).

This is a locally replayable exact finite computation plus the paper-level crossed-tower deduction. The payload evaluation and finite-field specialization are not kernel-checked in Lean; the Lean development checks only the generic scaled-product and nested-exponent algebra used above.  $\square$

**Proposition 5.41** (the exact ordinary carry at  $p = 727$ ). *One has*

$$\text{ord}_{727}(2) = 121 = 11^2, \quad Q(121) = \{121\}, \quad \alpha_{727} = \kappa_{121} + 1.$$

*More precisely, the actual selected full-conductor polynomial is*

$$\begin{aligned} P_{\alpha_{11}}(X) &= X^{20} + X^{16} + X^5 + X + 1, \\ P_{\kappa_{121}}(X) &= P_{\alpha_{11}}(X^{121}) = X^{2420} + X^{1936} + X^{605} + X^{121} + 1. \end{aligned} \quad (\text{O727}_1)$$

*The second polynomial is irreducible over  $\mathbb{F}_2$ . In its quotient field, with  $x$  the class of  $X$ , exact binary arithmetic gives*

$$x^{(2^{2420}-1)/727} = 1, \quad (x+1)^{(2^{2420}-1)/727} \neq 1, \quad \left((x+1)^{(2^{2420}-1)/727}\right)^{727} = 1. \quad (\text{O727}_2)$$

*Consequently*

$$\boxed{m_{727} = 1}. \quad (\text{O727}_3)$$

*The canonical fixed-width SHA-256 fingerprints of the modulus and the nontrivial phase in the polynomial basis are, respectively,*

$$\begin{aligned} &a10b35b0caf07945af6d71c0f3fccb7b3e67c3f5ae3124e19b5482e960cec852, \\ &1b95861258bcf45a90c284011c3337561996e5cc70ee6e05374b121515103e57. \end{aligned} \quad (\text{O727}_4)$$

*Proof.* Direct modular arithmetic gives  $2^{121} = 1 \pmod{727}$  but  $2^{11} \neq 1 \pmod{727}$ , proving the order assertion. The selected dependency  $\alpha_{11} = \kappa_5 + 1$  and  $P_{\alpha_5}(T) = T^4 + T + 1$  give

$$P_{\alpha_{11}}(X) = P_{\alpha_5}((X+1)^5) = X^{20} + X^{16} + X^5 + X + 1.$$

Proposition 5.10 at the  $11^2$ -Kummer step gives the second polynomial in  $(\text{O727}_1)$  and its irreducibility. Independently, the maintained checker verifies the complete Rabin criterion:  $X^{2^{2420}} = X$  modulo the displayed polynomial and

$$\gcd(P_{\kappa_{121}}, X^{2^{2420/r}} + X) = 1 \quad (r = 2, 5, 11).$$

It then evaluates the two exact Euler powers in  $(\text{O727}_2)$ . The first equality proves that translate zero still has a 727-th root; the nonidentity at translate one proves that translate one does not. Hence the least finite translate is one.

The stdlib verifier is `experiments/ordinary_727_certificate.py`. Lean kernel-checks that 727 is prime, the order arithmetic above, and the selected degree 2420. The irreducibility and Euler evaluations are exact finite computation, not a kernel theorem.  $\square$

**Proposition 5.42** (ordinary affine orbit and the marked scalar). *Let  $Q$  be a power of two, let  $r$  be an odd prime with  $r \mid Q - 1$ , and let*

$$k = \mathbb{F}_Q, \quad K = \mathbb{F}_{Q^r}, \quad x^r = a \in k^\times,$$

*where  $X^r - a$  is irreducible over  $k$ . Put*

$$\lambda = x^{Q-1} = a^{(Q-1)/r} \in \mu_r(k).$$

Then  $\lambda$  has order  $r$ ,  $x^Q = \lambda x$ , and the complete affine translate orbit has the exact polynomial

$$\prod_{c \in k} (Y + 1 + cx) = Y^Q + \lambda Y + (1 + \lambda). \quad (\text{OAM}_1)$$

For every  $u \in k^\times$ , the rescaled root  $x_u = ux$  has the same Frobenius multiplier  $\lambda$ , and

$$\{1 + cx_u : c \in k\} = \{1 + cx : c \in k\}. \quad (\text{OAM}_2)$$

Let  $p \mid (Q^r - 1)/(Q - 1)$ , assume  $p \nmid Q - 1$ , and let  $\chi : K^\times \rightarrow \mu_p$  have exact order  $p$ . Then  $\chi$  is trivial on  $k^\times$ , so the complete multiset of labels

$$\{\chi(1 + cx) : c \in k\}$$

and every symmetric invariant of it are unchanged by  $x \mapsto ux$ . Their product is one. But the distinguished value transforms as

$$H(x) = \chi(1 + x), \quad H(x_u) = \chi(1 + ux), \quad (\text{OAM}_3)$$

so rescaling merely moves which nonzero scalar is called 1.

For the actual  $(Q, r, p) = (2^{19580}, 179, 359)$ , the bound (KE) holds. The locally certified equality  $m_{359} = 1$  makes  $H(x) \neq 1$ , whereas Proposition 5.28 supplies a nonzero  $u \in k^\times$  with  $H(x_u) = 1$ . By (OAM<sub>1</sub>)–(OAM<sub>2</sub>), these two marked Kummer lines have the same affine-orbit polynomial, labelled histogram, and every symmetric translate statistic; by (SPH<sub>0</sub>), their trace bits are opposite. Hence those aggregate data cannot evaluate the bit in (SPH<sub>1</sub>). The actual lower multiplicative ancestry fixes the mark  $c = 1$ ; discarding it restores this rescaling freedom.

*Proof.* Irreducibility implies  $x \neq 0$ , and

$$x^{Q-1} = (x^r)^{(Q-1)/r} = a^{(Q-1)/r} \in \mu_r(k).$$

Because  $r$  is prime, a smaller order would make  $\lambda = 1$ , hence  $x^Q = x$ , contrary to  $[k(x) : k] = r$ . The standard additive-polynomial identity  $\prod_{c \in k} (Z + c) = Z^Q + Z$ , applied to  $Z = (Y + 1)/x$ , gives

$$\prod_{c \in k} (Y + 1 + cx) = (Y + 1)^Q + x^{Q-1}(Y + 1),$$

which is (OAM<sub>1</sub>). Multiplication by  $u$  permutes  $k$ , proving (OAM<sub>2</sub>); it also leaves  $\lambda$  unchanged because  $u^{Q-1} = 1$ .

The hypothesis  $p \nmid Q - 1$  makes  $\chi$  trivial on  $k^\times$ . Taking  $\chi$  of the constant term in (OAM<sub>1</sub>) gives product one because  $1 + \lambda \in k^\times$ . The reindexing statement and (OAM<sub>3</sub>) are immediate. Lean kernel-checks the denominator-cleared collapse from the two additive-polynomial terms to the right side of (OAM<sub>1</sub>); the finite-field product and Kummer specialization remain paper-level. In the 359-instance,  $\sqrt{Q} = 2^{9790} > (358)(178) + 359$ , so (KE) applies; the certified nontrivial line and the trivial rescaled line then give the claimed opposite trace bits.  $\square$

*Certified datum.* The value  $m_{359} = 1$  is present in A380496, and Proposition 5.36 now independently evaluates the trace in (SPH<sub>1</sub>) by exact finite computation. Quadratic half-orbit norms do not decide it:  $\sum_{h \in H} h = 0$  in  $\mathbb{F}_{359}$ , so their symbol product is identically one. Proposition 5.28 likewise shows why the safe-prime eigenline data alone cannot select the Conway phase.

## 5.6 The ordinary-spine conjecture ( $O$ )

For each odd component prime  $r \neq 3$ , prove that the selected relative unit of Proposition 5.7 has full primary order at every divisor of  $\mathcal{L}_{r,a}$ . The signed absolute-Frobenius signed-ball theorem proves the larger primary factors; the remaining smaller factors, including the independent  $p = 359$  trace evaluation, are the exact unresolved core.

## 6 The cubic $3^k$ spine

Write  $\zeta := \kappa_{3^k}$  and  $h := 3^k$  throughout this section.

**Conjecture 6.1** ( $C_k$ : Gaussian-period primitivity). *The period*

$$\gamma_k = \zeta + \zeta^{-1} \in \mathbb{F}_{2^{3^k}}^\times$$

*is primitive for every  $k \geq 1$ .*

### 6.1 Structure

**Proposition 6.2** (cyclotomic recognition).  *$\zeta^{3^k} = \kappa_2$ , which has order 3; hence  $\zeta$  is a primitive  $3^{k+1}$ -st root of unity. Since 2 is a primitive root modulo  $3^{k+1}$ , the cyclotomic polynomial  $\Phi_{3^{k+1}}(x) = x^{2^h} + x^h + 1$  is irreducible over  $\mathbb{F}_2$ , the component field is  $\mathbb{F}_2(\zeta) = \mathbb{F}_{2^{2h}}$ , and it carries the index-2 subfield  $L = \mathbb{F}_{2^h}$ .*

**Proposition 6.3** (half-angle splitting). *With  $s = (3^{k+1} + 1)/2$  (the inverse of 2 modulo  $3^{k+1}$ ),*

$$\kappa_{3^k} + 1 = \zeta^s (\zeta^s + \zeta^{-s}),$$

*where  $\zeta^s$  lies in the norm-one circle  $U$  (of order exactly  $3^{k+1}$  within the subgroup of order  $2^h + 1$ ) and  $(\zeta^s + \zeta^{-s})^2 = \zeta + \zeta^{-1} =: \gamma_k \in L^\times$ . Since  $\mathbb{F}_{2^{2h}}^\times = L^\times \times U$  with coprime orders,*

$$\text{ord}(\kappa_{3^k} + 1) = 3^{k+1} \cdot \text{ord}(\gamma_k), \quad \text{ord}(\gamma_k) \mid 2^{3^k} - 1.$$

*Machine-checked corollaries:  $(\kappa + 1)^{2^h - 1} = \zeta^{-1}$  and  $N_{F/L}(\kappa + 1) = \gamma_k$ .*

**Proposition 6.4** (translates  $m = 2, 3$ ).  *$\kappa_{3^k} + 2$  and  $\kappa_{3^k} + 3$  split as in Proposition 6.3 with  $\gamma_k$  replaced by a Galois conjugate. Therefore  $\text{ord}(\kappa_{3^k} + m) = 3^{k+1} \cdot \text{ord}(\gamma_k)$  for  $m = 1, 2, 3$ , and  $\text{ord}(\kappa_{3^k}) = 3^{k+1}$ .*

**Proposition 6.5** (the  $2 \cdot 3^k$  exception, unconditionally). *Any prime  $p$  with  $f(p) = 2 \cdot 3^k$  satisfies  $p \mid 2^{3^k} + 1$ , hence  $p$  divides neither  $3^{k+1}$  nor  $2^{3^k} - 1$ , hence  $p$  never divides  $\text{ord}(\kappa_{3^k} + m)$  for  $m \in \{0, 1, 2, 3\}$ . By the order criterion,  $m_p \geq 4$ . The full primitivity conjecture  $C_k$  below is not needed; the splitting alone forces the bound. The argument applies beyond the current tables, e.g.  $p = 87211$ ,  $f = 54$ .*

**Proposition 6.6** (the 3-power column). *For a prime  $r$  with  $f(r) = 3^k$ , the values  $m \in \{0, 2, 3\}$  are impossible:  $m_r = 0$  would require  $r \mid \text{ord}(\kappa_{3^k}) = 3^{k+1}$ , a power of 3, while  $r \equiv 1 \pmod{3^k}$  is a prime distinct from 3; and  $m = 2, 3$  share  $m = 1$ 's order class by Proposition 6.4. Any failure of  $C_k$  therefore gives  $m_r \geq 4$  directly.*



**Proposition 6.7** (norm tower).  $\gamma_{k-1} = \gamma_k^3 + \gamma_k = N_{L_k/L_{k-1}}(\gamma_k)$  (minimal polynomial  $X^3 + X + \gamma_{k-1}$ ), so  $\text{ord}(\gamma_{k-1}) \mid \text{ord}(\gamma_k)$ . With lifting-the-exponent ( $v_r(2^{3^k} - 1) = v_r(2^{f(r)} - 1)$  for old primes  $r \neq 3$ ), full-order parts propagate upward, and conjecture  $C_k$  reduces to  $C_{k-1}$  plus primitivity at the new primes  $r \mid \Phi_{3^k}(2)$ .

**Proposition 6.8** (the norm-kernel seam). Put  $q = 2^{3^{k-1}}$ , so  $L_k = \mathbb{F}_{q^3}$  and  $L_{k-1} = \mathbb{F}_q$ . There is a canonical direct product

$$L_k^\times = L_{k-1}^\times \times U_k, \quad |U_k| = q^2 + q + 1 = \Phi_{3^k}(2),$$

where  $U_k = \ker N_{L_k/L_{k-1}}$ . If  $\gamma_k = a_k b_k$  in this product, then

$$\gamma_{k-1} = N(\gamma_k) = a_k^3,$$

and cubing is an automorphism of  $L_{k-1}^\times$ . Consequently

$$C_k \iff C_{k-1} \text{ and } b_k \text{ generates } U_k.$$

In particular, the norm recursion determines the old component  $a_k$  exactly but supplies no order information about the new component  $b_k$ .

*Proof.* The two displayed factor orders are coprime:

$$\gcd(q - 1, q^2 + q + 1) = \gcd(q - 1, 3) = 1,$$

because  $q = 2^{\text{odd}} \equiv 2 \pmod{3}$ . The cyclic group  $L_k^\times$  therefore splits as asserted. The relative norm is  $z \mapsto z^{1+q+q^2}$ ; it kills  $U_k$  and restricts on  $L_{k-1}^\times$  to  $a \mapsto a^3$ , which is bijective because  $3 \nmid q - 1$ . The final equivalence follows from coprimality of the two factor orders.  $\square$

## 6.2 Equivalent algebraic and cyclotomic forms

**Proposition 6.9** (primitive Dickson iterate). Let  $D_n(X, 1)$  be the Dickson polynomial characterized by  $D_n(u + u^{-1}, 1) = u^n + u^{-n}$ , and put  $f(X) = X^3 + X$ . Then

$$P_k(X) := D_{3^k}(X, 1) + 1 = f^{\circ k}(X) + 1$$

is the minimal polynomial of  $\gamma_k$  over  $\mathbb{F}_2$ . In particular it is irreducible of degree  $3^k$ , and

$$C_k \iff P_k(X) \text{ is a primitive polynomial over } \mathbb{F}_2.$$

The polynomials also obey the coefficient recursion

$$P_{k+1} = P_k^3 + P_k^2 + 1, \quad P_1 = X^3 + X + 1.$$

*Proof.* The Dickson composition law gives  $D_{3^k} = D_3^{\circ k} = f^{\circ k}$  in characteristic two. Since  $\zeta^{3^k}$  is a primitive cube root of unity,

$$D_{3^k}(\gamma_k, 1) = \zeta^{3^k} + \zeta^{-3^k} = 1.$$

On the other hand, the Frobenius orbit of  $\zeta + \zeta^{-1}$  has length  $3^k$ : the least  $j > 0$  with  $2^j \equiv \pm 1 \pmod{3^{k+1}}$  is  $3^k$ . Thus the displayed monic polynomial, which has degree  $3^k$ , is the minimal polynomial. Its root is primitive precisely when the polynomial is primitive. Finally, writing  $D_{3^k} = P_k + 1$  and using  $D_3(Y) = Y^3 + Y$  gives  $P_{k+1} = (P_k + 1)^3 + (P_k + 1) + 1 = P_k^3 + P_k^2 + 1$ .  $\square$

**Proposition 6.10** (the coefficient recursion does not preserve primitivity). *Over  $\mathbb{F}_2$ , put*

$$F = X^{10} + X^3 + 1, \quad H = F^3 + F^2 + 1 = X^{30} + X^{23} + X^{16} + X^{10} + X^9 + X^3 + 1.$$

*Then  $F$  is primitive and  $H$  is irreducible, but a root of  $H$  has order*

$$7161 = 3 \cdot 7 \cdot 11 \cdot 31 < 2^{30} - 1.$$

*Thus neither irreducibility of the next polynomial nor primitivity of the preceding polynomial turns the recursion in Proposition 6.9 into a formal induction. Additional structure of the actual  $P_k$  sequence—for example its power-of-three degrees and cyclotomic ancestry—is required.*

*Proof.* Rabin's criterion is certified by

$$X^{2^{10}} \equiv X \pmod{F}, \quad \gcd(F, X^{2^5} + X) = \gcd(F, X^{2^2} + X) = 1.$$

Moreover  $2^{10} - 1 = 3 \cdot 11 \cdot 31$ , and direct reduction gives

$$X^{1023} \equiv 1 \pmod{F},$$

while

$$\begin{aligned} X^{341} &\equiv X^7 + X^6 + X^5 + X^3 + X^2 + 1, \\ X^{93} &\equiv X^9 + X^7 + X^6 + X^3 + 1, \\ X^{33} &\equiv X^9 + X^6 + X^5 + X^3 + X^2 \end{aligned} \pmod{F}.$$

The last three residues are not one, so  $F$  is primitive.

For  $H$ , the exact Rabin reductions are

$$X^{2^{30}} \equiv X \pmod{H},$$

$$\gcd(H, X^{2^{15}} + X) = \gcd(H, X^{2^{10}} + X) = \gcd(H, X^{2^6} + X) = 1,$$

which prove irreducibility. Finally

$$X^{7161} \equiv 1 \pmod{H},$$

whereas

$$\begin{aligned} X^{2387} &\equiv X^{26} + X^{23} + X^{22} + X^{16} + X^{15} + X^7 + X^6 + X^3 + 1, \\ X^{1023} &\equiv X^{20} + X^{10} + X^6 + X^3 + 1, \\ X^{651} &\equiv X^{23} + X^{21} + X^9 + X^3, \\ X^{231} &\equiv X^{23} + X^{22} + X^{21} + X^{12} + X^9 + X^8 + X^7 + X^5 + X^3 + 1 \end{aligned} \pmod{H}.$$

These residues are not one and correspond to division of 7161 by its four prime factors, so the displayed order is exact.  $\square$

**Proposition 6.11** (shifted cyclotomic-unit form). *Put  $x_k = 1 + \zeta \in \mathbb{F}_{2^{2h}}^\times$ , where  $h = 3^k$  and  $N = 3^{k+1}$ . Then*

$$x_k^{2^h-1} = \zeta^{-1}, \quad x_k^{2^h+1} = \gamma_k, \quad \text{ord}(x_k) = N \text{ ord}(\gamma_k).$$

*Consequently*

$$C_k \iff \text{ord}(1 + \zeta) = 3^{k+1}(2^{3^k} - 1).$$

*Thus the cubic arm is equivalently maximal order for the reduction at 2 of the cyclotomic unit  $1 + \zeta_{3^{k+1}}$ .*

*Proof.* Because  $2^h \equiv -1 \pmod{N}$ ,  $x_k^{2^h} = 1 + \zeta^{-1} = \zeta^{-1}x_k$ . Dividing and multiplying by  $x_k$  gives the first two identities. The factors  $2^h - 1$  and  $2^h + 1$  are coprime. In the corresponding direct-product decomposition of  $\mathbb{F}_{2^{2h}}^\times$ , the first displayed power has exact order  $N$  and the second has order  $\text{ord}(\gamma_k)$ , so the order of  $x_k$  is their product.  $\square$

**Proposition 6.12** (norm-coherent cyclotomic lift). *Let  $\xi_k = \zeta_{3^{k+1}}$  in characteristic zero and  $u_k = 1 + \xi_k$ . Then 2 is inert in  $\mathbb{Q}(\xi_k)$  and*

$$N_{\mathbb{Q}(\xi_k)/\mathbb{Q}(\xi_{k-1})}(u_k) = u_{k-1}.$$

*The norm to the maximal real subfield is*

$$N_{\mathbb{Q}(\xi_k)/\mathbb{Q}(\xi_k + \xi_k^{-1})}(u_k) = 2 + \xi_k + \xi_k^{-1},$$

*whose reduction modulo the unique prime above 2 is  $\gamma_k$ . Thus Proposition 6.11 is the residue-field shadow of a norm-coherent cyclotomic-unit system in the cyclotomic  $\mathbb{Z}_3$ -tower. Conditional on  $C_{k-1}$ ,  $C_k$  is exactly the assertion that, for every prime  $\ell \mid \Phi_{3^k}(2)$ , the reduction of  $u_k$  is not an  $\ell$ -th power.*

*Proof.* The residue degree of 2 in  $\mathbb{Q}(\xi_k)$  is  $\text{ord}_{3^{k+1}}(2) = 2 \cdot 3^k = \varphi(3^{k+1})$ , proving inertness. The three relative conjugates of  $\xi_k$  are  $\xi_k, \xi_k\omega$ , and  $\xi_k\omega^2$ , so

$$\prod_{j=0}^2 (1 + \xi_k\omega^j) = 1 + \xi_k^3 = 1 + \xi_{k-1}.$$

Multiplication by the complex-conjugate factor gives the real norm. The final assertion follows from Proposition 6.11: in a cyclic group an element has the full  $\ell$ -primary part of the group order precisely when it is not an  $\ell$ -th power.  $\square$

### 6.3 The selected Kummer class and reciprocity boundary

**Proposition 6.13** (circular-unit and ray-class form). *Let*

$$F_k = \mathbb{Q}(\xi_k + \xi_k^{-1}), \quad \lambda_k = (1 - \xi_k)(1 - \xi_k^{-1}),$$

*let  $\mathfrak{p}_k = (2)$  be the unique prime of  $F_k$  above 2, and let  $\mathcal{C}_k \subset \mathcal{O}_{F_k}^\times$  be the real circular-unit group. If  $\sigma(\xi_k) = \xi_k^2$ , then*

$$c_k := \frac{\sigma(\lambda_k)}{\lambda_k} = (1 + \xi_k)(1 + \xi_k^{-1}) = 2 + \xi_k + \xi_k^{-1}$$

*is a circular unit, its Galois conjugates generate a subgroup of 2-power index in  $\mathcal{C}_k$ , and its reduction modulo  $\mathfrak{p}_k$  is  $\gamma_k$ . Since the residue-field unit group has odd order, that 2-power index disappears on reduction. Consequently*

$$\overline{\mathcal{C}_k} = \langle \gamma_k \rangle \subset (\mathcal{O}_{F_k}/\mathfrak{p}_k)^\times \simeq \mathbb{F}_{2^{3^k}}^\times, \quad (\clubsuit)$$

*and hence*

$$[\mathbb{F}_{2^{3^k}}^\times : \langle \gamma_k \rangle] = [(\mathcal{O}_{F_k}/\mathfrak{p}_k)^\times : \overline{\mathcal{C}_k}]. \quad (\spadesuit)$$

*In particular,  $C_k$  is exactly surjectivity modulo 2 of the real circular units of conductor  $3^{k+1}$ .*

*More precisely, put  $E_k = \mathcal{O}_{F_k}^\times$  and let  $\text{Cl}_{\mathfrak{p}_k}(F_k)$  be the ray class group of finite modulus  $\mathfrak{p}_k$ . The ray class exact sequence gives*

$$[\mathbb{F}_{2^{3^k}}^\times : \langle \gamma_k \rangle] = \frac{|\text{Cl}_{\mathfrak{p}_k}(F_k)|}{|\text{Cl}(F_k)|} [\overline{E_k} : \overline{\mathcal{C}_k}], \quad (\heartsuit)$$

where  $[\overline{E_k} : \overline{C_k}] \mid [E_k : C_k]$ . The circular-unit index theorem identifies the odd part of the latter index with the odd part of  $|\mathrm{Cl}(F_k)|$ . Thus, for an odd prime  $\ell \nmid |\mathrm{Cl}(F_k)|$ ,

$$v_\ell([\mathbb{F}_{2^{3^k}}^\times : \langle \gamma_k \rangle]) = v_\ell\left(\frac{|\mathrm{Cl}_{\mathfrak{p}_k}(F_k)|}{|\mathrm{Cl}(F_k)|}\right). \quad (\diamond)$$

Equation  $(\heartsuit)$  names the obstruction but does not reduce its difficulty: the ray class group is defined through this residue-unit quotient, and the class-number exclusion in  $(\diamond)$  is itself not known uniformly in  $k$ . Finally, with  $h = 3^k$ ,  $q = 2^{h/3}$ , and  $\tau = \sigma^{h/3}$ , the relative circular unit

$$\varepsilon_k = \frac{\tau(c_k)}{c_k}, \quad N_{F_k/F_{k-1}}(\varepsilon_k) = 1,$$

reduces to  $\eta_k$ . Thus the exact character obstruction of Proposition 6.37 is the new relative part of the same ray-class obstruction.

*Proof.* The class of 2 generates  $(\mathbb{Z}/3^{k+1}\mathbb{Z})^\times/\{\pm 1\}$ , so  $\sigma$  generates  $\mathrm{Gal}(F_k/\mathbb{Q})$ . The squares of the standard real circular units are generated, up to  $-1$ , by the quotients  $\sigma^j(\lambda_k)/\lambda_k$ . Telescoping gives

$$\frac{\sigma^j(\lambda_k)}{\lambda_k} = \prod_{i=0}^{j-1} \sigma^i(c_k),$$

so the conjugates of  $c_k$  generate this square subgroup. Squaring is an automorphism of the odd-order residue-field unit group, so the full circular-unit group and this subgroup have the same reduction image. Reduction commutes with  $\sigma$ , which becomes Frobenius at the inert prime 2, and

$$\overline{c_k} = (1 + \zeta)(1 + \zeta^{-1}) = \zeta + \zeta^{-1} = \gamma_k.$$

The reductions of the conjugates are therefore  $\gamma_k, \gamma_k^2, \gamma_k^{2^2}, \dots$ , and they generate the single multiplicative subgroup  $\langle \gamma_k \rangle$ . This proves  $(\clubsuit)$  and  $(\spadesuit)$ .

The exact sequence

$$E_k \longrightarrow (\mathcal{O}_{F_k}/\mathfrak{p}_k)^\times \longrightarrow \mathrm{Cl}_{\mathfrak{p}_k}(F_k) \longrightarrow \mathrm{Cl}(F_k) \longrightarrow 1$$

proves  $(\heartsuit)$ . The divisibility of reduction indices is immediate from  $C_k \subset E_k$ , and the circular-unit index theorem gives  $(\diamond)$ ; see Washington [43, Ch. 8]. Finally  $\tau$  generates  $\mathrm{Gal}(F_k/F_{k-1})$ , so the norm of its coboundary  $\tau(c_k)/c_k$  telescopes to 1. Modulo 2 it is  $\gamma_k^q/\gamma_k = \eta_k$ .  $\square$

**Proposition 6.14** (cubic pure-radical reflection descent). *Put  $h = 3^k$ , let  $\xi = \zeta_{3^{k+1}}$ , and retain*

$$K = F_k = \mathbb{Q}(\xi + \xi^{-1}), \quad c_k = (1 + \xi)(1 + \xi^{-1}).$$

For a prime  $\ell \mid \Phi_h(2)$ , put

$$B = K(\zeta_\ell + \zeta_\ell^{-1}), \quad M = K(\zeta_\ell).$$

The conductors are coprime, so  $K \cap \mathbb{Q}(\zeta_\ell) = \mathbb{Q}$  and  $M/B$  is quadratic. The element  $c_k$  is a global unit fixed by the nontrivial involution  $j \in \mathrm{Gal}(M/B)$ , and its reduction at the unique prime of  $K$  above two is  $\gamma_k$ .

Assume first that  $c_k \notin (M^\times)^\ell$ . Choose  $\theta^\ell = c_k$ , extend  $j$  by  $j(\theta) = \theta$ , and put

$$E_C = M(\theta), \quad L_C = E_C^{(j)} = B(\theta). \quad (\mathrm{CRD}_1)$$

Then  $E_C/B$  is dihedral of order  $2\ell$ , while  $L_C/B$  has degree  $\ell$ . The selected condition has the exact norm form

$$\boxed{\text{the } \ell\text{-primary condition in } C_k \text{ fails} \iff 2 \in N_{L_C/B}(L_C^\times)}. \quad (\text{CRD}_2)$$

Moreover  $L_C/B$  satisfies the Hasse norm principle, and its norm-one torus satisfies weak approximation. At every prime of  $B$  above two,

$$\begin{array}{l|l} \ell\text{-primary failure} & L_C \otimes_B B_{\mathfrak{r}} \simeq B_{\mathfrak{r}}^\ell, \\ \ell\text{-primary success} & L_C \otimes_B B_{\mathfrak{r}} \text{ is the unramified degree-}\ell \text{ field.} \end{array} \quad (\text{CRD}_3)$$

Every local norm condition for the element two away from the primes above two is automatic. Thus the reflection norm equation has no hidden class-group, Tate–Shafarevich, or Brauer–Manin obstruction: it is exactly the original selected residue condition.

If  $c_k \in (M^\times)^\ell$ , then  $c_k \in (B^\times)^\ell$  as well, the selected condition fails, and the construction degenerates to  $L_C = B$ ; in this convention  $(\text{CRD}_2)$  remains true.

*Proof.* Since  $\text{ord}_\ell(2) = h$ , one has  $\ell \nmid 3^{k+1}$ . Coprimality of cyclotomic conductors gives the field-intersection assertion. Also  $N_{\mathbb{Q}(\xi)/\mathbb{Q}}(1 + \xi) = \Phi_{3^{k+1}}(-1) = 1$ , so  $c_k$  is a global unit. The identity and reduction claim are

$$c_k = 2 + \xi + \xi^{-1}, \quad \overline{c_k} = \zeta + \zeta^{-1} = \gamma_k;$$

the first algebraic identity is kernel-checked in `Ogdoad.Excess`.

Let  $\sigma(\theta) = \zeta_\ell \theta$ . Since  $j$  fixes  $\theta$  and inverts  $\zeta_\ell$ , one has  $j\sigma j^{-1} = \sigma^{-1}$ , proving the dihedral and fixed-field assertions. The completionwise reflection-norm equivalence  $(\text{HD}_5)$  applies to this dihedral extension.

The unique completion  $K_{\mathfrak{p}}/\mathbb{Q}_2$  is unramified of degree  $h$  and already contains  $\mu_\ell$ , because  $\text{ord}_\ell(2) = h$ . Hence  $\mathfrak{p}$  splits completely in both  $B/K$  and  $M/K$ , and  $M/B$  splits at every prime above two. At each such completion, local Kummer theory says that adjoining an  $\ell$ -th root of  $c_k$  is split exactly when  $\gamma_k$  is an  $\ell$ -th power; otherwise it is the unramified degree- $\ell$  field. Its norm valuations are then divisible by  $\ell$ , whereas  $v_{\mathfrak{r}}(2) = 1$ . This proves  $(\text{CRD}_3)$  and the two-adic part of  $(\text{CRD}_2)$ .

At every finite place away from  $2\ell$ , the entries  $c_k$  and two are tame units. At a place above  $\ell$ , the totally ramified group  $\text{Gal}(M/K)$  fixes  $c_k$ , two, and the place while acting faithfully on  $\mu_\ell$ ; Hilbert-symbol covariance therefore forces  $(c_k, 2)_{\lambda, \ell} = 1$ . The archimedean places of  $M$  are complex. Cyclic Hasse for  $E_C/M$ , followed by the completionwise reflection equivalence, proves  $(\text{CRD}_2)$  and the Hasse norm principle for  $L_C/B$ . Independently, the metacyclic norm-torus theorem of Hoshi–Yamasaki gives both the Hasse norm principle and weak approximation for this odd-dihedral reflection field [18, Theorem 3.2 and Example 4.6].

Finally, if  $c_k = y^\ell$  in  $M$ , then  $c_k^2 = N_{M/B}(y)^\ell$ . Since 2 is invertible modulo the odd prime  $\ell$ , this implies  $c_k \in (B^\times)^\ell$ . Its residue is an  $\ell$ -th power and the degenerate convention follows.  $\square$

**Proposition 6.15** (the exceptional residue character is the Kummer regulator). *Retain Proposition 6.13, and put*

$$n = 3^k, \quad M = 2^n - 1, \quad A = (\mathcal{O}_{F_k}/\mathfrak{p}_k)^\times, \quad \Delta = \langle \sigma \rangle.$$

Let  $\ell \mid \Phi_n(2)$ , let  $\mathcal{O}$  be the ring of integers of an unramified extension of  $\mathbb{Q}_\ell$  containing  $\mu_n$ , and let  $\lambda$  be its maximal ideal. There is a unique  $\vartheta \in \mu_n(\mathcal{O})$  such that  $\vartheta \equiv 2 \pmod{\lambda}$ . In additive exponent notation, the corresponding character quotient of the residue module is

$$A_\vartheta := \frac{A \otimes_{\mathbb{Z}} \mathcal{O}}{(\sigma - \vartheta)(A \otimes_{\mathbb{Z}} \mathcal{O})} \simeq \frac{\mathcal{O}}{(2 - \vartheta)}. \quad (\text{RC}_1)$$

Moreover

$$v_\lambda(2 - \vartheta) = v_\ell(2^n - 1) = v_\ell(\Phi_n(2)). \quad (\text{RC}_2)$$

Thus the local ray determinant has a forced zero at every current prime; it can never prove the cubic arm by being an  $\ell$ -adic unit.

Choose a generator  $g_0$  of  $A$ , write  $\gamma_k = g_0^{e_k}$ , and let  $R_\vartheta$  be the class of  $e_k$  in the right side of (RC<sub>1</sub>). Then

$$(A/\overline{\mathcal{C}_k})_\vartheta \simeq \frac{\mathcal{O}}{(2 - \vartheta, R_\vartheta)}, \quad (\text{RC}_3)$$

and, with the compatible generator used in Proposition 6.41,

$$R_\vartheta \equiv L(1) \pmod{\lambda}. \quad (\text{RC}_4)$$

Consequently

$$\begin{aligned} \ell \nmid [A : \overline{\mathcal{C}_k}] &\iff R_\vartheta \not\equiv 0 \pmod{\lambda} \\ &\iff L(1) \neq 0 \iff \gamma_k \notin A^\ell. \end{aligned} \quad (\text{RC}_5)$$

The exceptional ray-class regulator is therefore exactly the selected Kummer coordinate, rather than an independent class-number factor.

*Proof.* The cyclotomic-divisor criterion gives  $\text{ord}_\ell(2) = n$ ; in particular  $\ell \nmid n$ . The residue class of two is consequently a simple root of  $X^n - 1$  modulo  $\ell$ , and Hensel lifting gives the unique  $\vartheta$ . The exponent map identifies the  $\mathbb{Z}[\Delta]$ -module  $A$  with  $\mathbb{Z}/M\mathbb{Z}$ , on which  $\sigma$  acts as multiplication by two. It follows immediately that

$$A_\vartheta \simeq \mathcal{O}/(M, 2 - \vartheta).$$

In  $\mathcal{O}$  one has

$$M = 2^n - 1 = \prod_{\zeta \in \mu_n} (2 - \zeta).$$

Every factor except  $2 - \vartheta$  is a unit modulo  $\lambda$ . Hence  $(M) = (2 - \vartheta)$ , which proves (RC<sub>1</sub>)–(RC<sub>2</sub>); the last equality in (RC<sub>2</sub>) again uses  $\text{ord}_\ell(2) = n$ .

Proposition 6.13 gives  $\overline{\mathcal{C}_k} = \langle \gamma_k \rangle$ . Passing to the  $\vartheta$ -quotient therefore adds the single relation  $e_k = 0$ , proving (RC<sub>3</sub>). For the final identification, use the generator  $g$  of  $\mathbb{F}_{2^{2n}}^\times$  from Proposition 6.41 and take  $g_0 = g^{2^n+1}$ . If  $z = g^{(2^{2n}-1)/(3n)}$ , then

$$\gamma_k = z^{-1}(1 + z)^2.$$

The  $3n$ -torsion factor has zero  $\ell$ -index, while  $2^n + 1 \equiv 2 \pmod{\ell}$ . Comparing exponents relative to  $g$  and  $g_0$  gives  $e_k \equiv L(1) \pmod{\ell}$ , which is (RC<sub>4</sub>). The equivalences in (RC<sub>5</sub>) now follow from cyclicity of  $A$ .  $\square$

**Remark 6.16** (what the ray-class Gras theorem does and does not say). All's local presentation [3, Cor. 2.4], specialized to the inert rational prime two, is the group-ring form of (RC<sub>1</sub>). His ray-class Gras theorem [3, Thm. 1.2] applies here because  $\ell \nmid 2|\Delta|$ , but it compares the Schmidt group of 2-cyclotomic units congruent to one modulo two with the corresponding nontrivial ray-class component. It does not identify that group with the reduction of the ordinary circular units in Proposition 6.13, and it supplies no nonvanishing of  $R_\vartheta$ . The elementary calculation above is the required specialization: the known Euler factor vanishes, and its remaining circular-unit regulator is precisely  $L(1)$ .

**Proposition 6.17** (the circular-unit circulant is residue-invisible). *Retain Proposition 6.13, put*

$$n = 3^k, \quad M = 2^n - 1, \quad R = \mathbb{Z}[\Delta], \quad \nu = 1 + \sigma + \cdots + \sigma^{n-1},$$

*and let  $\mathcal{L}_k$  be the multiplicative lattice generated by the conjugates of  $c_k$ . The exponent map identifies*

$$\mathcal{L}_k \simeq R/\mathbb{Z}\nu. \quad (\text{CL}_1)$$

*On this rank- $n - 1$  lattice the circulant difference operator  $\delta = \sigma - 1$  has*

$$|\det(\delta \mid R/\mathbb{Z}\nu)| = \prod_{\substack{\chi^n=1 \\ \chi \neq 1}} |\chi - 1| = n = 3^k. \quad (\text{CL}_2)$$

*This three-primary determinant does not imply that the index of the reduction of  $\mathcal{L}_k$  is three-primary. Indeed, if  $s = \text{ord}(\gamma_k)$ , reduction is exactly*

$$\rho_s : R/\mathbb{Z}\nu \longrightarrow \mathbb{Z}/s\mathbb{Z}, \quad [f] \longmapsto f(2) \pmod{s}, \quad (\text{CL}_3)$$

*where the target records exponents of  $\gamma_k$ . Hence*

$$[\mathbb{F}_{2^n}^\times : \overline{\mathcal{L}_k}] = M/s,$$

*while  $\delta$  acts on the target as multiplication by  $2 - 1 = 1$ . Thus the determinant in (CL<sub>2</sub>) is invisible to every residue quotient relevant to the cubic arm.*

*More strongly, this failure is formal rather than merely a gap in the determinant estimate. For every divisor  $d \mid M$ , let  $G = \langle g \rangle$  be cyclic of order  $M$ , with the  $\Delta$ -action  $\sigma(g) = g^2$ . Then*

$$\rho_d : R/\mathbb{Z}\nu \longrightarrow G, \quad [f] \longmapsto g^{\text{df}(2)} \quad (\text{CL}_4)$$

*is a well-defined equivariant reduction model whose image has index exactly  $d$ . It preserves the integral orbit lattice, the sole characteristic-zero norm relation, Frobenius covariance, and the circulant determinant. If  $n = 3h$ ,  $q = 2^h$ , and  $d \mid q^2 + q + 1$ , then  $\gcd(d, q - 1) = 1$  and the relative norm of  $g^d$  has order  $q - 1$ ; thus the model preserves maximal lower-field norm as well. If, in addition,  $M/d$  has a divisor on which two has order  $n$ , the Frobenius orbit of  $g^d$  still has length  $n$ . Consequently even the orbit degree and old-factor norm propagation can coexist with a non-three-primary current index. The models (CL<sub>4</sub>) are not finite-field counterexamples to  $C_k$ : they prove that an argument using only the conjugate exponent lattice and its circulant regulator cannot evaluate the selected reduction of the circular unit.*

*Proof.* Proposition 6.13 shows that the conjugates of  $c_k$  generate a finite-index subgroup of the real circular units, so  $\mathcal{L}_k$  has rank  $n - 1$ . They are totally positive and hence have no nontrivial torsion. Their norm relation is  $c_k' = 1$ . The kernel of the exponent map  $R \rightarrow \mathcal{L}_k$  therefore has rank one and contains the primitive vector  $\nu$ , so it is exactly  $\mathbb{Z}\nu$ . This proves (CL<sub>1</sub>).

The characteristic polynomial of  $\sigma$  on  $R/\mathbb{Z}\nu$  is  $(X^n - 1)/(X - 1)$ . Evaluating it at one gives (CL<sub>2</sub>). At the inert prime above two,  $\sigma$  reduces to Frobenius and  $\sigma^i(c_k)$  reduces to  $\gamma_k^{2^i}$ , proving (CL<sub>3</sub>). In particular the image has order  $s$ , and  $\sigma - 1$  evaluates to one.

For (CL<sub>4</sub>), well-definedness follows from  $\nu(2) = M$ , and equivariance follows from  $2^n \equiv 1 \pmod{M}$ . The image is  $\langle g^d \rangle$ , of index  $d$ . Finally put  $A = 1 + q + q^2 = M/(q - 1)$ . The relative norm is

$$N(g^d) = (g^d)^A.$$

Because  $q \equiv -1 \pmod{3}$ , one has  $\gcd(A, q - 1) = 1$ ; hence a divisor  $d \mid A$  is coprime to  $q - 1$ , and the displayed element has order  $q - 1$ . The Frobenius-orbit assertion follows because the orbit length of an element of order  $M/d$  is  $\text{ord}_{M/d}(2)$ .  $\square$

**Proposition 6.18** (exceptional eigenspace and Hilbert-reciprocity no-go). Put  $n = 3^k$ ,  $m = 3n$ , let

$$F = \mathbb{Q}(\xi_m + \xi_m^{-1}), \quad \mathfrak{p} = (2),$$

and write  $\Delta = \text{Gal}(F/\mathbb{Q}) = \langle \sigma \rangle$ , where  $\sigma(\xi_m) = \xi_m^2$ . Let  $\ell \mid \Phi_n(2)$ , and set

$$V_\ell = (\mathcal{O}_F/\mathfrak{p})^\times / ((\mathcal{O}_F/\mathfrak{p})^\times)^\ell.$$

Then  $V_\ell$  is one-dimensional over  $\mathbb{F}_\ell$ , and  $\sigma$  acts on it as the scalar 2. Let

$$\text{ev}_2 : \mathbb{F}_\ell[\Delta] \longrightarrow \mathbb{F}_\ell, \quad \sum a_j \sigma^j \longmapsto \sum a_j 2^j.$$

The following reciprocity operators vanish on  $V_\ell$ :

$$\text{ev}_2(1 + \sigma + \cdots + \sigma^{n-1}) = 0, \quad \text{ev}_2(2 - \sigma) = 0. \quad (\text{ER}_1)$$

If  $q = 2^{n/3}$  and  $\tau = \sigma^{n/3}$ , then

$$\text{ev}_2(1 + \tau + \tau^2) = 1 + q + q^2 = 0, \quad \text{ev}_2(\tau - 1) = q - 1 \neq 0. \quad (\text{ER}_2)$$

Moreover the plus projection of the classical Stickelberger element of  $\mathbb{Q}(\xi_m)$  acts as zero on  $V_\ell$ . Its zero is simple: for

$$\nu_n(X) = 1 + X + \cdots + X^{n-1}$$

one has

$$\nu_n(2) = 0, \quad \nu'_n(2) = n2^{n-1} \neq 0 \pmod{\ell}. \quad (\text{ER}_3)$$

Let

$$c = (1 + \xi_m)(1 + \xi_m^{-1}), \quad \gamma = \bar{c} \in \mathbb{F}_{2^n}^\times, \quad \mathcal{A}_{k,\ell} = \gamma^{(2^n-1)/\ell} \in \mu_\ell. \quad (\text{ER}_4)$$

Then cubic failure at  $\ell$  is exactly  $\mathcal{A}_{k,\ell} = 1$ . Although  $\mathcal{A}_{k,\ell}$  is a local Kummer/Artin symbol, global Hilbert reciprocity imposes no condition on this one distinguished value. More precisely, put  $E = F(\mu_\ell)$ . The prime  $\mathfrak{p}$  splits into  $\ell - 1$  primes in  $E/F$ . If  $\mathfrak{P}_0 \mid \mathfrak{p}$  is chosen so that

$$(c, 2)_{\mathfrak{P}_0, \ell} = \mathcal{A}_{k,\ell}$$

for the reciprocity convention sending the uniformizer 2 to arithmetic Frobenius, then the  $\ell - 1$  symbols above two are

$$(c, 2)_{a\mathfrak{P}_0, \ell} = \mathcal{A}_{k,\ell}^a, \quad a \in \mathbb{F}_\ell^\times. \quad (\text{ER}_5)$$

Every local symbol away from two is one, and Hilbert reciprocity reduces to

$$\prod_{a \in \mathbb{F}_\ell^\times} \mathcal{A}_{k,\ell}^a = \mathcal{A}_{k,\ell}^{\ell(\ell-1)/2} = 1, \quad (\text{ER}_6)$$

which holds for every possible value of  $\mathcal{A}_{k,\ell}$ . Thus norm coherence, the classical Stickelberger ideal, and global Kummer reciprocity cannot distinguish failure from nonfailure. A completion must compute a first-order unit regulator in the simple exceptional eigenspace, rather than another ideal factorization or product formula.



*Proof.* The standard cyclotomic-divisor criterion gives  $\text{ord}_\ell(2) = n$ . Since  $(\mathcal{O}_F/\mathfrak{p})^\times \simeq \mathbb{F}_{2^n}^\times$ , its quotient by  $\ell$ -th powers has order  $\ell$ , and Frobenius acts by  $[x] \mapsto [x^2] = 2[x]$ . This proves the module assertion. Equations (ER<sub>1</sub>) follow from  $2^n = 1 \pmod{\ell}$ . The element  $q = 2^{n/3}$  has exact order three modulo  $\ell$ , proving (ER<sub>2</sub>).

Let  $G = \text{Gal}(\mathbb{Q}(\xi_m)/\mathbb{Q})$ , let  $j$  be complex conjugation, and write

$$\theta_m = \frac{1}{m} \sum_{\substack{1 \leq a < m \\ (a,m)=1}} a \sigma_a^{-1}$$

for the Stickelberger element. Pairing  $a$  with  $m - a$  gives

$$(1 + j)\theta_m = \sum_{g \in G} g.$$

On an even representation, the right side is twice the norm element of  $\Delta$ , whose evaluation at two is zero by (ER<sub>1</sub>). Since  $\ell \neq 2, 3$ , division by two and by  $m$  is valid modulo  $\ell$ . Hence  $\theta_m$ , and therefore every classical Stickelberger-ideal multiple controlling Gauss–Jacobi-sum ideals, acts trivially on  $V_\ell$ . The particularly relevant integrality factor  $2 - \sigma$  vanishes separately. Differentiating

$$(X - 1)\nu_n(X) = X^n - 1$$

at  $X = 2$  gives (ER<sub>3</sub>), since  $\ell \nmid 2n$ .

The reduction of  $c$  is  $\gamma$ . In the local field  $F_{\mathfrak{p}}$ , which contains  $\mu_\ell$ , adjoining an  $\ell$ -th root  $z$  of  $c$  is the unramified Kummer extension selected by the residue of  $c$ . Arithmetic Frobenius satisfies

$$\frac{\text{Frob}(z)}{z} = \bar{z}^{2^n-1} = \gamma^{(2^n-1)/\ell},$$

proving (ER<sub>4</sub>) and the failure criterion. Also, for the relative circular unit  $\varepsilon = \tau(c)/c$ ,

$$(\varepsilon, 2)_{\mathfrak{p}, \ell} = \mathcal{A}_{k, \ell}^{q-1};$$

because  $q - 1 \not\equiv 0 \pmod{\ell}$ , the relative coboundary carries exactly the same unknown symbol and supplies no new relation.

It remains to audit global reciprocity. The completion  $F_{\mathfrak{p}}$  is the unramified degree- $n$  extension of  $\mathbb{Q}_2$ . Since  $\mathbb{Q}_2(\mu_\ell)/\mathbb{Q}_2$  is also unramified of degree  $\text{ord}_\ell(2) = n$ , it is already contained in  $F_{\mathfrak{p}}$ . The conductors  $m$  and  $\ell$  are coprime, so  $F \cap \mathbb{Q}(\mu_\ell) = \mathbb{Q}$ ; consequently  $[E : F] = \ell - 1$ , and  $\mathfrak{p}$  splits completely in  $E/F$ . Galois functoriality of the Hilbert symbol gives (ER<sub>5</sub>).

We now check every place above  $\ell$ , the only point at which wild local reciprocity might have supplied extra information. Let  $K_0$  be the maximal unramified subfield of an  $\ell$ -adic completion of  $E$ . That completion is  $K = K_0(\mu_\ell)$ , with absolute ramification index  $e = \ell - 1$ , and both  $c$  and  $2$  belong to  $K_0^\times$ . For any unit  $a \in K_0^\times$ , write

$$a = [\bar{a}] \langle a \rangle, \quad \langle a \rangle \in 1 + \ell \mathcal{O}_{K_0}.$$

The Teichmüller factor has order prime to  $\ell$ , hence is an  $\ell$ -th power. If  $U_i = 1 + \mathfrak{m}_K^i$ , then

$$1 + \ell \mathcal{O}_{K_0} \subset U_{\ell-1}.$$

The standard Hilbert-filtration orthogonality for a field containing  $\mu_\ell$  is [19]

$$(U_i, U_j)_\ell = 1 \quad \text{when} \quad i + j > \frac{\ell e}{\ell - 1} = \ell. \quad (\text{ER}_7)$$

Taking  $i = j = \ell - 1$  proves  $(c, 2)_{\lambda, \ell} = 1$  at every  $\lambda \mid \ell$ . At every finite place away from  $2\ell$ , both entries are units and the tame Hilbert-symbol formula is one; archimedean places contribute nothing because  $\ell$  is odd. The global product formula is therefore precisely (ER<sub>6</sub>). Since  $\sum_{a=1}^{\ell-1} a \equiv 0 \pmod{\ell}$ , this relation is tautological and leaves the single symbol at  $\mathfrak{P}_0$  arbitrary.  $\square$

**Proposition 6.19** (the simple exceptional zero does not evaluate its Bockstein). *Retain Proposition 6.18. Put*

$$\mathcal{A} = \mathbb{Z}_\ell[\Delta], \quad \mathfrak{m} = (\ell, \sigma - 2), \quad \nu = 1 + \sigma + \cdots + \sigma^{n-1},$$

and let  $u = (2^n - 1)/\ell \in \mathbb{Z}_\ell$ . Then

$$\nu \equiv \ell u + n2^{n-1}(\sigma - 2) \pmod{\mathfrak{m}^2}. \quad (\text{BZ}_1)$$

The coefficient  $n2^{n-1}$  is a unit modulo  $\ell$ , but this says only that the first-order exceptional coordinate is one-dimensional; it does not determine the cyclotomic unit's value in that coordinate. Indeed  $(\sigma - 1)\nu = \sigma^n - 1 = 0$ , while  $\sigma - 1$  is a unit modulo  $\mathfrak{m}^2$ . Thus (BZ<sub>1</sub>) is the cotangent relation

$$0 = \ell u + n2^{n-1}(\sigma - 2) \text{ in } \mathcal{A}/\mathfrak{m}^2, \quad (\text{BZ}'_1)$$

not a nonzero value of the sought unit class.

More sharply, let  $K/\mathbb{Q}_2$  be the unramified extension of degree  $n$ , and let

$$\lambda = (1 - \xi_m)(1 - \xi_m^{-1}) = tv, \quad t \in \mu_{2^n-1}, \quad v \in 1 + 2\mathcal{O}_K$$

be its Teichmüller/principal-unit decomposition at two. One has  $N_{K/\mathbb{Q}_2}(t) = 1$  and  $N_{K/\mathbb{Q}_2}(v) = 3$ . Conversely, for every  $t_0 \in \mu_{2^n-1}$  there is  $v_0 \in 1 + 2\mathcal{O}_K$  such that

$$\lambda_0 = t_0 v_0, \quad N_{K/\mathbb{Q}_2}(\lambda_0) = 3, \quad \overline{\lambda_0} = \overline{t_0}. \quad (\text{BZ}_2)$$

The residue on the right may have any  $\ell$ -power class. Hence the norm-three identity, Frobenius action, and the nonzero derivative in (BZ<sub>1</sub>) cannot by themselves prove cubic nonfailure. The missing datum is the distinguished cyclotomic unit's actual coordinate — for example through a derived local-unit regulator — not the formal existence of the Bockstein coordinate. This is a local countermodel to those data only; it is not a global cyclotomic counterexample.

*Proof.* Write  $\epsilon = \sigma - 2$ . Modulo  $(\ell^2, \ell\epsilon, \epsilon^2) = \mathfrak{m}^2$ , binomial expansion gives

$$\sum_{j=0}^{n-1} (2 + \epsilon)^j \equiv (2^n - 1) + \nu'_n(2)\epsilon \equiv \ell u + n2^{n-1}\epsilon.$$

Here the discarded part of  $\nu'_n(2) = n2^{n-1} - (2^n - 1)$  lies in  $\ell\epsilon$ . Since  $\text{ord}_\ell(2) = n$ , one has  $n \mid \ell - 1$ , so  $n2^{n-1} \not\equiv 0 \pmod{\ell}$ . This proves (BZ<sub>1</sub>). The identity  $(\sigma - 1)\nu = 0$  and  $\sigma - 1 = 1 + \epsilon \in (\mathcal{A}/\mathfrak{m}^2)^\times$  prove (BZ'<sub>1</sub>); its one nonzero linear coefficient leaves a one-dimensional cotangent space.

The real cyclotomic norm of  $\lambda$  is three. The norm of every Teichmüller unit is its residue-field norm, so  $N(t) = t^{2^n-1} = 1$ , leaving  $N(v) = 3$ . Conversely the norm map from the principal units of an unramified local extension onto the base principal units is surjective. Given  $t_0$ , choose  $v_0$  with norm three; then (BZ<sub>2</sub>) follows. In addition  $\sigma(t_0 v_0)/(t_0 v_0)$  reduces to  $t_0^2/t_0 = t_0$ , so even the relative-coboundary form carries the same arbitrary residue class. Choosing  $t_0$  inside or outside the subgroup of  $\ell$ -th powers proves the asserted neutrality.  $\square$

**Proposition 6.20** (the derivative regulator is the original Kummer class). *Retain the notation of Proposition 6.13, put  $n = 3^k$ , and write*

$$\nu_n(X) = 1 + X + \cdots + X^{n-1}, \quad \mathcal{R}_k^\partial := \prod_{j=0}^{n-1} \sigma^j(c_k)^j = c_k^{\sigma \nu'_n(\sigma)}.$$

*Then the first-order circular-unit regulator singled out by the simple exceptional zero is explicit:*

$$\mathcal{R}_k^\partial = \frac{\lambda_k^n}{3}, \quad \overline{\mathcal{R}_k^\partial} = \gamma_k^n. \quad (\text{DR}_1)$$

*Consequently, for every prime  $\ell \mid \Phi_n(2)$ , its reduction is an  $\ell$ -th power if and only if  $\gamma_k$  is. Thus the derivative regulator carries exactly the unknown Kummer class in  $C_k$ : the nonzero derivative identifies the coordinate but does not evaluate it.*

*For  $k \geq 2$  this identification is norm coherent in the sharp form*

$$N_{F_k/F_{k-1}}(\mathcal{R}_k^\partial) = (\mathcal{R}_{k-1}^\partial)^3. \quad (\text{DR}_2)$$

*Hence*

$$\mathcal{E}_k^\partial := \frac{\mathcal{R}_k^\partial}{\mathcal{R}_{k-1}^\partial} = \left( \frac{\lambda_k^3}{\lambda_{k-1}} \right)^{n/3}$$

*has relative norm one. At every current  $\ell \mid \Phi_n(2)$ , the lower-field factor is automatically an  $\ell$ -th power, so  $\overline{\mathcal{E}_k^\partial}$  again carries precisely the Kummer class of  $\gamma_k$ . In particular, passing to a derived relative unit does not create an inductive relation capable of proving  $C_k$ .*

*Proof.* Since  $c_k = \sigma(\lambda_k)/\lambda_k$ , the weighted product telescopes cyclically:

$$\prod_{j=0}^{n-1} \left( \frac{\sigma^{j+1}(\lambda_k)}{\sigma^j(\lambda_k)} \right)^j = \frac{\lambda_k^{n-1}}{\prod_{i=1}^{n-1} \sigma^i(\lambda_k)} = \frac{\lambda_k^n}{N_{F_k/\mathbb{Q}}(\lambda_k)}.$$

The real cyclotomic norm of  $\lambda_k$  is 3, proving the first identity in (DR<sub>1</sub>). Modulo the prime above two, both  $\lambda_k$  and  $c_k$  reduce to  $\gamma_k$ , while 3 reduces to one; hence the second identity follows. Because  $\text{ord}_\ell(2) = n$ , one has  $\ell \nmid n$ , so raising the one-dimensional  $\ell$ -power quotient to the  $n$ -th power is an automorphism. This proves the equivalence of Kummer classes.

Finally,  $N_{F_k/F_{k-1}}(\lambda_k) = \lambda_{k-1}$ . Taking the relative norm of  $\lambda_k^n/3$ , and remembering that the relative degree is three, gives

$$N_{F_k/F_{k-1}}(\mathcal{R}_k^\partial) = \frac{\lambda_{k-1}^n}{3^3} = \left( \frac{\lambda_{k-1}^{n/3}}{3} \right)^3,$$

which is (DR<sub>2</sub>). The displayed formula and norm-one property of  $\mathcal{E}_k^\partial$  follow immediately. Since  $\ell \nmid 2^{n/3} - 1$ , exponentiation by  $\ell$  is an automorphism of  $\mathbb{F}_{2^{n/3}}^\times$ ; every lower-field factor is therefore an  $\ell$ -th power. The relative unit has the same current Kummer class, as claimed.  $\square$

**Proposition 6.21** (weighted Frobenius words only erase or reparameterize). *Let  $x \in \mathbb{F}_{2^n}^\times$ , let the prime  $\ell \mid 2^n - 1$ , and put*

$$A = x^{(2^n-1)/\ell} \in \mu_\ell, \quad \sigma(x) = x^2.$$

*For a finite group-ring word  $w = \sum_i a_i \sigma^i$ , with  $a_i \in \mathbb{Z}$ , define*

$$x^w = \prod_i \sigma^i(x)^{a_i}, \quad \text{ev}_2(w) = \sum_i a_i 2^i \pmod{\ell}.$$

Then

$$(x^w)^{(2^n-1)/\ell} = A^{\text{ev}_2(w)}. \quad (\text{WO}_1)$$

Consequently a weighted Frobenius-orbit construction has only two possibilities. If  $\text{ev}_2(w) = 0$ , it is identically  $\ell$ -power flat. If  $\text{ev}_2(w) \neq 0$ , and in particular if that value is a unit modulo  $\ell$ , its nontriviality is exactly the nontriviality of the original selected class  $A$ , merely transported by an automorphism of  $\mu_\ell$ .

For the cubic arm, take  $n = 3^k$ , a current prime  $\ell \mid \Phi_n(2)$ , and  $\nu_n = 1 + \sigma + \cdots + \sigma^{n-1}$ . Then

$$\text{ev}_2(\nu_n) = 0, \quad \text{ev}_2(\sigma \nu'_n(\sigma)) = 2\nu'_n(2) \equiv n \pmod{\ell}. \quad (\text{WO}_2)$$

Thus the norm word is automatically flat, whereas the derivative regulator of Proposition 6.20 has Euler phase  $A^n$ . Its nonzero derivative does not evaluate the selected class; it is an invertible reparameterization of it.

Nor can the discrete coefficient  $L(1)$  itself have a normalization-independent formula in terms of a fixed integer Euler or Fermat quotient. In the notation of Proposition 6.41, put  $Q = 2^{2^n}$ ,  $m = 3n$ , and  $z = g^{(Q-1)/m}$ . Replacing the compatible primitive generator  $g$  by  $g^a$ , where

$$a \equiv 1 \pmod{m}, \quad \gcd(a, Q-1) = 1,$$

keeps  $z$  fixed but changes

$$L(1) \mapsto a^{-1}L(1) \pmod{\ell}. \quad (\text{WO}_3)$$

The Chinese remainder theorem permits every nonzero value of  $a \pmod{\ell}$  while retaining these conditions. Hence only the vanishing or nonvanishing of  $L(1)$  is intrinsic; a literal equality with a fixed scalar Euler quotient is not even well normalized.

*Proof.* Put  $E = (2^n - 1)/\ell$ . Frobenius commutes with taking the  $E$ -th power, so

$$(x^w)^E = \prod_i (x^{2^i})^{a_i E} = \prod_i (x^E)^{a_i 2^i} = A^{\sum_i a_i 2^i},$$

which proves (WO<sub>1</sub>). The two alternatives follow because  $A \in \mu_\ell$ .

The first congruence in (WO<sub>2</sub>) is  $\nu_n(2) = 2^n - 1 \equiv 0 \pmod{\ell}$ . Differentiating  $(X-1)\nu_n(X) = X^n - 1$  and evaluating at  $X = 2$  gives

$$\nu_n(2) + \nu'_n(2) = n2^{n-1}.$$

Therefore  $2\nu'_n(2) \equiv n2^n \equiv n \pmod{\ell}$ . Since  $\text{ord}_\ell(2) = n$ , one has  $n \mid \ell - 1$ , so  $\ell \nmid n$ .

Finally,  $a \equiv 1 \pmod{m}$  gives

$$(g^a)^{(Q-1)/m} = g^{(Q-1)/m} = z.$$

If  $1 - z = g^v = (g^a)^{v'}$ , then  $v' \equiv a^{-1}v \pmod{\ell}$ , proving (WO<sub>3</sub>). To realize a prescribed nonzero residue modulo  $\ell$ , impose it together with  $a \equiv 1 \pmod{m}$ , and impose  $a \equiv 1$  modulo every other prime divisor of  $Q - 1$ ; the Chinese remainder theorem applies because  $\ell \nmid m$ . The resulting  $a$  is coprime to  $Q - 1$ , as required.  $\square$

**Proposition 6.22** (the Coleman specialization has a forced exceptional zero). *Retain Proposition 6.13, put  $n = 3^k$  and  $m = 3n$ , and let  $\ell \mid \Phi_n(2)$  be prime. The norm-coherent real circular unit  $\mathbf{c} = (c_k)_k$  has, at the prime three, the Coleman power series*

$$f_{\mathbf{c}}(T) = \frac{(2+T)^2}{1+T}, \quad (1+T) \frac{d}{dT} \log f_{\mathbf{c}}(T) = \frac{T}{2+T}. \quad (\text{CO}_1)$$

Write  $\zeta_3^{\text{KL}}$  for the Kubota–Leopoldt three-adic zeta pseudo-measure. With the Coleman-map normalization of [36, Sections 7.2–7.4], its image on the even Iwasawa quotient is

$$\text{Col}(\mathbf{c}) = 2([2] - [1])\zeta_3^{\text{KL}}. \quad (\text{CO}_2)$$

Let  $\tilde{\lambda}$  be the characteristic-zero lift of the primitive even character in (MJ<sub>2</sub>): choose a primitive  $n$ -th root  $\vartheta$  and set

$$\tilde{\lambda}(\pm 2^i) = \vartheta^i \quad (0 \leq i < n).$$

There is a prime  $\mathfrak{L} \mid \ell$  of its coefficient field for which  $\vartheta \equiv 2 \pmod{\mathfrak{L}}$ . For every positive even integer  $r$ , Coleman interpolation gives the algebraic value

$$\text{ev}_{\tilde{\lambda}x^r} \text{Col}(\mathbf{c}) = 2(\tilde{\lambda}(2)2^r - 1)L(\tilde{\lambda}, 1-r) = -\frac{2}{r}(\tilde{\lambda}(2)2^r - 1)B_{r,\tilde{\lambda}}. \quad (\text{CO}_3)$$

At the least positive even weight for which the smoothing factor vanishes after reduction, namely  $r = n - 1$ , this value is forced to vanish modulo  $\mathfrak{L}$ :

$$\text{ev}_{\tilde{\lambda}x^{n-1}} \text{Col}(\mathbf{c}) \equiv 0 \pmod{\mathfrak{L}}. \quad (\text{CO}_4)$$

Indeed its smoothing factor reduces to  $2^n - 1$ . Dividing that forced factor before reduction leaves only

$$2L(\tilde{\lambda}, 2-n) = -\frac{2}{n-1}B_{n-1,\tilde{\lambda}}. \quad (\text{CO}_5)$$

The Coleman identity controls this quotient three-adically; it gives no  $\mathfrak{L}$ -unit theorem and no identification of (CO<sub>5</sub>) with the selected residue index  $L(1) = \text{ind}_{\ell}(1-z)$ . At weight zero the corresponding specialization is a noncritical three-adic  $L$ -value and has no canonical reduction at  $\mathfrak{L}$ .

Thus the standard Coleman/ $p$ -adic- $L$ -function specialization cannot prove  $\mathcal{R}_{\lambda} \not\equiv 1 \pmod{(\zeta_{\ell} - 1)^2}$ : its algebraic specialization meets a forced exceptional zero, while its noncritical specialization lives at the wrong prime. This is a route-exclusion theorem, not a failure of  $C_k$ . The separation is already exact at  $k = 2$ : Proposition 6.73 proves  $C_2$ , but for  $n = 9$ ,  $\ell = 73$ , and  $r = 8$ , the right side of (CO<sub>3</sub>) is zero modulo  $\mathfrak{L}$  because  $2^9 - 1 = 7 \cdot 73$ .

*Proof.* Write

$$c(2)_k = \frac{\xi_k^2 - 1}{\xi_k - 1} = 1 + \xi_k.$$

The standard cyclotomic-unit Coleman series is  $((1+T)^2 - 1)/T = 2 + T$ . Complex conjugation substitutes  $(1+T)^{-1} - 1$  for  $T$ , so its conjugate has series  $(2+T)/(1+T)$ . Their product is  $f_{\mathbf{c}}$ , and direct logarithmic differentiation proves (CO<sub>1</sub>).

The cyclotomic-unit formula  $\text{Col}(c(2)) = ([2] - [1])\zeta_3^{\text{KL}}$  and equivariance under complex conjugation give (CO<sub>2</sub>) after passage to the even quotient. Evaluating at the even character  $\tilde{\lambda}x^r$  and using the Kubota–Leopoldt interpolation formula gives the first equality in (CO<sub>3</sub>); the standard identity  $L(\chi, 1-r) = -B_{r,\chi}/r$  gives the second.

Since  $\text{ord}_\ell(2) = n$ , reduction at a suitable prime above  $\ell$  sends  $\vartheta$  to 2. Also  $n \mid \ell - 1$  and in fact  $n < \ell - 1$ , so  $r = n - 1$  is positive, even, and smaller than  $\ell - 1$ . The generalized Bernoulli quotient in  $(\text{CO}_3)$  is therefore  $\mathfrak{L}$ -integral: its conductor and  $r$  are prime to  $\ell$ , and no ordinary Bernoulli denominator occurring below weight  $\ell - 1$  contains  $\ell$ . Finally,

$$\tilde{\lambda}(2)2^{n-1} - 1 \equiv 2^n - 1 \equiv 0 \pmod{\mathfrak{L}},$$

which proves  $(\text{CO}_4)$ . Removing this displayed factor gives  $(\text{CO}_5)$ . The final assertions follow from the places at which the two interpolation values live and from the exact  $k = 2$  prime-step theorem.  $\square$

**Proposition 6.23** (semidirect ray extension forced by cubic failure). *Retain Proposition 6.13, put  $n = 3^k$ , and let  $\ell \mid \Phi_{3^k}(2)$ . Assume  $\ell \nmid |\text{Cl}(F_k)|$ . Then*

$$\gamma_k \in (\mathbb{F}_{2^n}^\times)^\ell \tag{RF}$$

*if and only if there exists a cyclic extension  $M/F_k$  of degree  $\ell$ , conductor exactly  $\mathfrak{p}_k = (2)$ , totally tamely ramified there and unramified at every other finite prime. This extension is Galois over  $\mathbb{Q}$ , with*

$$\text{Gal}(M/\mathbb{Q}) \cong C_\ell \rtimes_2 C_n, \quad \sigma\tau\sigma^{-1} = \tau^2. \tag{RF_1}$$

*Moreover, the unique prime of  $F_k$  above three splits completely in  $M/F_k$ . Both prescribed local behaviors are admissible: at two this is the standard tame extension with inertia  $C_\ell$ , residue degree  $n$ , and Frobenius action  $\tau \mapsto \tau^2$ ; at three the  $C_\ell$ -layer is split. Thus secant saturation and circular-unit reciprocity give no local contradiction. Without the class-number hypothesis, cubic failure yields the sharp alternative*

$$\ell \mid |\text{Cl}(F_k)| \quad \text{or the extension in (RF}_1\text{) exists.} \tag{RF_2}$$

*Proof.* Write  $E_k = \mathcal{O}_{F_k}^\times$  and  $\mathcal{C}_k$  for the circular units. Proposition 6.13 identifies the reduction of  $\mathcal{C}_k$  with  $\langle \gamma_k \rangle$ . If (RF) holds, then  $\mathcal{C}_k$  maps trivially to

$$V := \mathbb{F}_{2^n}^\times / (\mathbb{F}_{2^n}^\times)^\ell \cong C_\ell.$$

The odd part of  $[E_k : \mathcal{C}_k]$  is the odd part of the class number. Under the stated hypothesis its  $\ell$ -part vanishes, so  $E_k$  also maps trivially to  $V$ . The ray sequence then supplies an order- $\ell$  quotient of  $\text{Cl}_{\mathfrak{p}_k}(F_k)$ . Because the ordinary class group has no  $\ell$ -part, the quotient lies entirely in tame inertia at  $\mathfrak{p}_k$ . Class field theory gives the asserted  $M/F_k$ .

The quotient is Galois-stable. The generator  $\sigma$  of  $\text{Gal}(F_k/\mathbb{Q})$  acts on the residue field as  $x \mapsto x^2$ , hence on  $V$  by exponent two. Since  $\text{ord}_\ell(2) = n$ , this action is faithful; Schur–Zassenhaus, or the coprimality of  $n$  and  $\ell$ , gives  $(\text{RF}_1)$ . Conversely, tame inertia in any such extension gives the unique quotient  $\mathbb{F}_{2^n}^\times \twoheadrightarrow C_\ell$ . Global units lie in its kernel, so in particular  $\gamma_k$  is an  $\ell$ -th power.

Finally let

$$\lambda_k = (1 - \xi_k)(1 - \xi_k^{-1}), \quad c_k = (1 + \xi_k)(1 + \xi_k^{-1}).$$

The element  $\lambda_k$  generates the unique prime  $\mathfrak{r}_k$  above three, and  $\lambda_k/c_k \equiv 1 \pmod{\mathfrak{p}_k}$ , because both reduce to  $(1 + \zeta)(1 + \zeta^{-1}) = \gamma_k$ . Hence  $\mathfrak{r}_k = (\lambda_k/c_k)$  is trivial in the full ray class group and splits in  $M/F_k$ . The local statements now follow from  $\ell \mid 2^n - 1$  and the displayed semidirect action. If the class-number hypothesis fails, the first alternative in  $(\text{RF}_2)$  holds; otherwise the preceding construction proves the second.  $\square$

## 6.4 Kernel tower and quantitative order bounds

**Proposition 6.24** (explicit cubic kernel tower). *Define  $\eta_0 = 0$ . For  $k \geq 1$  put*

$$q_k = 2^{3^{k-1}}, \quad \eta_k = \gamma_k^{q_k-1}, \quad U_k = \ker \left( N_{\mathbb{F}_{q_k^3}/\mathbb{F}_{q_k}} \right).$$

*Then  $\eta_k \in U_k$ ,  $|U_k| = q_k^2 + q_k + 1 = \Phi_{3^k}(2)$ , and, with  $\omega = \zeta^{3^k} \in \mathbb{F}_4$ ,*

$$q_k \equiv 3^k - 1 \pmod{3^{k+1}}, \quad \eta_k = \frac{\omega + \omega^2 \zeta^2}{1 + \zeta^2}, \quad \zeta^2 = \frac{\eta_k + \omega}{\eta_k + \omega^2}.$$

*Moreover  $\eta_k$  has the irreducible cubic equation over  $\mathbb{F}_{q_k} = \mathbb{F}_{2^{3^{k-1}}}$*

$$X^3 + \eta_{k-1}X^2 + (\eta_{k-1} + 1)X + 1 = 0. \quad (*)$$

*Equivalently,*

$$N(\eta_k) = 1, \quad \text{Tr}(\eta_k) = \eta_{k-1}, \quad e_2(\eta_k) = \eta_{k-1} + 1.$$

*It also satisfies*

$$\eta_k^{q_k+1} + \eta_k + 1 = 0, \quad \eta_k + 1 = \eta_k^{q_k+1}, \quad \text{ord}(\eta_k + 1) = \text{ord}(\eta_k). \quad (\dagger)$$

*Thus  $\eta_k$  lies in the planar Singer difference set*

$$\mathcal{H}_k = \{y \in U_k : y^{q_k+1} + y + 1 = 0\}$$

*of size  $q_k + 1$  inside  $U_k$  [27]. Finally,*

$$C_k \iff C_{k-1} \text{ and } \text{ord}(\eta_k) = q_k^2 + q_k + 1. \quad (**)$$

*(Here  $C_0$  is the tautological assertion that 1 generates  $\mathbb{F}_2^\times$ .) Thus the new-prime obstruction is the single assertion that the distinguished root of  $(*)$  is a generator of the cubic norm-one torus (equivalently, multiplication by it is a Singer cycle).*

*Proof.* The congruence starts with  $2 \equiv 3-1 \pmod{9}$  and lifts by cubing: if  $a \equiv -1+3^k \pmod{3^{k+1}}$ , then  $a^3 \equiv -1+3^{k+1} \pmod{3^{k+2}}$ . Hence  $\zeta^{q_k} = \omega\zeta^{-1}$ . Dividing  $\gamma_k^{q_k}$  by  $\gamma_k$  now gives the stated Möbius formula; solving it gives the inverse formula.

Put  $t = \zeta^2$  and  $M(T) = (\omega + \omega^2 T)/(1+T)$ . The same formula one level down says  $\eta_{k-1} = M(t^3)$  (also for  $k = 1$ , when it gives  $\eta_0 = 0$ ). If  $a = \eta_{k-1}$  and  $X = M(t)$ , then

$$\left( \frac{X + \omega}{X + \omega^2} \right)^3 = \frac{a + \omega}{a + \omega^2}.$$

Cross-multiplication and  $\omega^2 + \omega + 1 = 0$  reduce exactly to  $(*)$ . Also  $\eta_k^{1+q_k+q_k^2} = \gamma_k^{q_k^3-1} = 1$ , so  $\eta_k \in U_k$ . It cannot lie in  $\mathbb{F}_{q_k}$ : otherwise its norm-one condition would give  $\eta_k^3 = 1$ , hence  $\eta_k = 1$  because  $3 \nmid q_k - 1$ ; the inverse Möbius formula would then give  $t = \omega$ , contradicting  $\text{ord}(t) = 3^{k+1} > 3$ . Thus  $(*)$  is irreducible, and its coefficients give the three symmetric identities.

The earlier minimal polynomial  $X^3 + X + \gamma_{k-1}$  gives  $\text{Tr}_{\mathbb{F}_{q_k^3}/\mathbb{F}_{q_k}}(\gamma_k) = 0$ . Dividing  $\gamma_k + \gamma_k^{q_k} + \gamma_k^{q_k^2} = 0$  by  $\gamma_k$  yields  $1 + \eta_k + \eta_k^{q_k+1} = 0$ , proving the first two parts of  $(\dagger)$ . Since  $\gcd(q_k + 1, q_k^2 + q_k + 1) = 1$ , exponentiation by  $q_k + 1$  is an automorphism of  $U_k$ , proving the equality of orders. The cited difference-set identification is the norm-one realization of the planar Singer set.

In the direct product of Proposition 6.8, raising to  $q_k - 1$  kills the old-field component and is an automorphism on  $U_k$ , since  $\gcd(q_k - 1, q_k^2 + q_k + 1) = 1$ . Therefore the kernel component of  $\gamma_k$  generates  $U_k$  exactly when  $\eta_k$  does; Proposition 6.8 gives  $(**)$ .  $\square$

**Proposition 6.25** (the normal Singer translate). *Retain Proposition 6.24, put*

$$\varepsilon_0 = 1, \quad \varepsilon_k = \eta_k + 1 \quad (k \geq 1),$$

and write  $q = q_k$ ,  $N = q^2 + q + 1$ . Then  $\varepsilon_k \in U_k$  has order  $\text{ord}(\eta_k)$  and has the irreducible relative cubic

$$X^3 + \varepsilon_{k-1}X^2 + (\varepsilon_{k-1} + 1)X + 1. \quad (\text{SN}_1)$$

Consequently

$$\text{Tr}_{L_k/L_{k-1}}(\varepsilon_k) = \varepsilon_{k-1}, \quad e_2(\varepsilon_k) = \varepsilon_{k-1} + 1, \quad N_{L_k/L_{k-1}}(\varepsilon_k) = 1. \quad (\text{SN}_2)$$

Moreover  $\varepsilon_k$  is normal over  $\mathbb{F}_2$ . More exactly, if  $\sigma(x) = x^2$  and

$$T_j = \sum_{r=0}^{3^{k-j}-1} \sigma^{r3^j} \quad (0 \leq j \leq k), \quad E_0 = T_0, \quad E_j = T_j + T_{j-1} \quad (j \geq 1),$$

then the  $E_j$  are the cyclotomic Frobenius projectors and

$$T_j(\varepsilon_k) = \varepsilon_j, \quad E_0(\varepsilon_k) = 1, \quad E_j(\varepsilon_k) = \eta_j + \eta_{j-1} \neq 0 \quad (j \geq 1). \quad (\text{SN}_3)$$

Thus the absolute Frobenius orbit of  $\varepsilon_k$  is an  $\mathbb{F}_2$ -basis of  $L_k$ . In the same decomposition

$$E_0(\eta_k) = 0, \quad E_j(\eta_k) = E_j(\varepsilon_k) \neq 0 \quad (j \geq 1),$$

so the orbit of  $\eta_k$  spans exactly the absolute trace-zero hyperplane and has rank  $3^k - 1$ .

For the immediate extension put  $x = \varepsilon_k$ ,  $y = x^q$ ,  $z = x^{q^2}$ , and  $b = \varepsilon_{k-1}$ . The trace Gram and cyclic Moore determinants are

$$G = (\text{Tr}(x^{q^i} x^{q^j}))_{0 \leq i, j < 3} = \begin{pmatrix} b^2 & b+1 & b+1 \\ b+1 & b^2 & b+1 \\ b+1 & b+1 & b^2 \end{pmatrix},$$

$$\det G = b^2(b^2 + b + 1)^2, \quad \det \begin{pmatrix} x & y & z \\ y & z & x \\ z & x & y \end{pmatrix} = b(b^2 + b + 1). \quad (\text{SN}_4)$$

Both determinants are nonzero. Finally

$$C_k \iff C_{k-1} \text{ and } \varepsilon_k \text{ generates } U_k. \quad (\text{SN}_5)$$

Thus the current Singer coordinate has a normal translate of exactly the same multiplicative order. This proves absolute and immediate normality; it does not assert that  $\varepsilon_k$  is completely normal. It is also distinct from the later trace-zero coordinate  $h_k = 1 + \gamma_k^{-1}$ .

*Proof.* In characteristic two, translating the cubic for  $\eta_k$  by  $X \mapsto X + 1$  gives  $(\text{SN}_1)$ ; translation preserves irreducibility, and its symmetric coefficients give  $(\text{SN}_2)$ . The Singer identity gives

$$\varepsilon_k = \eta_k^{q+1}, \quad \varepsilon_k^q = \eta_k^{-1}.$$

Since  $N = q(q+1) + 1$ , one has  $\gcd(q+1, N) = 1$ ; exponentiation by  $q+1$  is an automorphism of  $U_k$ , proving equality of orders and  $(\text{SN}_5)$  from Proposition 6.24.



The operator  $T_j$  is  $\text{Tr}_{L_k/L_j}$  and, since all relative degrees are odd, is the idempotent projector onto  $L_j$ . Trace transitivity and  $(\text{SN}_2)$  give  $T_j(\varepsilon_k) = \varepsilon_j$ , hence  $(\text{SN}_3)$ . Its last value is nonzero: equality  $\eta_j = \eta_{j-1}$  would put  $\eta_j$  in  $L_{j-1}$ , contrary to the irreducibility of its relative cubic. Now  $\text{ord}_{3^j}(2) = 2 \cdot 3^{j-1} = \varphi(3^j)$ , so  $\Phi_{3^j}$  is irreducible over  $\mathbb{F}_2$ . Each nonzero  $E_j$ -component therefore generates its whole simple Frobenius block. All blocks occur in  $\varepsilon_k$ , proving normality; omitting only the  $E_0$ -block gives the assertion for  $\eta_k$ .

Equation  $(\text{SN}_1)$  gives  $e_1 = b$ ,  $e_2 = b+1$ ,  $e_3 = 1$ . Hence the diagonal Gram entries are  $\text{Tr}(x^2) = b^2$  and the off-diagonal entries are  $\text{Tr}(xy) = xy + yz + zx = b+1$ , giving the first determinant in  $(\text{SN}_4)$ . In characteristic two the cyclic Moore determinant is

$$x^3 + y^3 + z^3 + xyz = e_1^3 + e_1 e_2 = b(b^2 + b + 1),$$

whose square is  $\det G$ . Here  $b \neq 0$ , and  $b^2 + b + 1 = 0$  would give an element of order three in  $L_{k-1}^\times$ ; this is impossible because  $3 \nmid 2^{3^{k-1}} - 1$ . The translated-cubic, Moore, Gram, and coprime-order algebra used here is kernel-checked in `Ogdoad.Excess`; the Frobenius-projector argument is paper-level.  $\square$

**Proposition 6.26** (reciprocal cubic Kummer descent). *Retain Proposition 6.24, write*

$$K = \mathbb{F}_q, \quad L = \mathbb{F}_{q^3}, \quad U = \ker(N_{L/K}), \quad a = \eta_{k-1},$$

and let the prime  $\ell \mid q^2 + q + 1$ . Define  $S_r(C, D) \in \mathbb{F}_2[C, D]$  by

$$S_0 = 1, \quad S_1 = C, \quad S_2 = C^2, \quad S_r = CS_{r-1} + DS_{r-2} + S_{r-3} \quad (r \geq 3). \quad (\text{CK}_1)$$

Then failure at  $\ell$ , namely  $\eta_k \in U^\ell$ , is equivalent to the existence of  $c, d \in K$  satisfying  $(\text{CK}_3)$  below. Every such pair automatically has

$$X^3 + cX^2 + dX + 1 \quad \text{irreducible over } K, \quad (\text{CK}_2)$$

and the two equations are

$$S_\ell(c, d) = a, \quad S_\ell(d, c) = a + 1. \quad (\text{CK}_3)$$

Given a witness  $(c, d)$ , choose a root  $b$  of  $(\text{CK}_2)$  and Frobenius-conjugate it so that  $b^\ell = \eta_k$ . Then  $c = \text{Tr}(b)$  and  $d = \text{Tr}(b^{-1})$ . For a fixed primitive  $\rho \in \mu_\ell \subset L$  and  $j \in \mathbb{Z}/\ell\mathbb{Z}$ , put

$$\begin{aligned} c_j &= \rho^j b + \rho^{jq} b^q + \rho^{jq^2} b^{q^2}, \\ d_j &= \rho^{j(1+q)} b^{1+q} + \rho^{j(q+q^2)} b^{q+q^2} + \rho^{j(q^2+1)} b^{q^2+1}. \end{aligned} \quad (\text{CK}_4)$$

Every  $c_j, d_j$  lies in  $K$ , every displayed cubic below is irreducible over  $K$ , and one has the exact factorization

$$X^{3\ell} + aX^{2\ell} + (a+1)X^\ell + 1 = \prod_{j \bmod \ell} (X^3 + c_j X^2 + d_j X + 1). \quad (\text{CK}_5)$$

The reciprocal cubic  $X^3 + dX^2 + cX + 1$  contains the conjugates of  $b^{-1}$ ; equivalently,  $b^{q+1} = b^{-q^2}$  is an  $\ell$ -th root of  $\eta_k + 1$ . Finally

$$\mathbb{F}_2(c, d) = K, \quad \sum_j c_j = \sum_j d_j = 0, \quad \prod_j (c_j + d_j) = 1. \quad (\text{CK}_6)$$

Thus cubic failure is a selected pair of reciprocal degree-three Dickson equations over the preceding field, rather than merely a factor of the degree- $3\ell$  Capelli composition.

*Proof.* Because  $\ell \mid q^2 + q + 1$  and  $3 \nmid q^2 + q + 1$ , one has  $\text{ord}_\ell(q) = 3$ . Suppose first that  $b \in U$  and  $b^\ell = \eta_k$ . The element  $b$  cannot lie in  $K$ , since its  $\ell$ -th power does not. Hence its conjugates have the irreducible minimal polynomial

$$X^3 + cX^2 + dX + 1, \quad c = \text{Tr}_{L/K}(b), \quad d = \text{Tr}_{L/K}(b^{-1}).$$

The recurrence (CK<sub>1</sub>) is the power-sum recurrence for these three roots. It gives

$$\text{Tr}(b^\ell) = S_\ell(c, d) = a, \quad \text{Tr}(b^{-\ell}) = S_\ell(d, c) = a + 1,$$

where the second equality uses the reciprocal cubic and the symmetric coefficients of the polynomial in Proposition 6.24.

Conversely, let  $c, d \in K$  satisfy (CK<sub>3</sub>), and let  $b_1, b_2, b_3$  be the roots of the corresponding cubic in an algebraic closure. Their product is one. The first elementary symmetric function of their  $\ell$ -th powers is  $S_\ell(c, d) = a$ , and the second is

$$b_1^{-\ell} + b_2^{-\ell} + b_3^{-\ell} = S_\ell(d, c) = a + 1,$$

and the third is one. Those powers are therefore the three distinct roots of the separable polynomial  $X^3 + aX^2 + (a + 1)X + 1$ . A reducible cubic over the finite field  $K$  would have a root  $b_i \in K$ , making  $b_i^\ell$  a  $K$ -root of that irreducible target polynomial. This is impossible, so (CK<sub>2</sub>) is automatic. We may therefore write the roots as  $b, b^q, b^{q^2}$ ; their product puts  $b$  in  $U$ . A Frobenius conjugate of  $b$  is consequently an  $\ell$ -th root of  $\eta_k$  in  $U$ ; choose this conjugate for the rest of the proof.

For the factorization, Frobenius cycles

$$\rho^j b, \quad \rho^{jq} b^q, \quad \rho^{jq^2} b^{q^2}$$

because  $q^3 \equiv 1 \pmod{\ell}$ . Their product is one because  $1 + q + q^2 \equiv 0 \pmod{\ell}$ , and their first two symmetric coefficients are (CK<sub>4</sub>). No orbit has size one, since its  $\ell$ -th power would put  $\eta_k$  in  $K$ . If a member of the triple indexed by  $j$  equalled a member indexed by  $j'$ , then raising to the  $\ell$ -th power and using irreducibility of the polynomial for  $\eta_k$  would first force the same Frobenius position; primitivity of  $\rho$  would then force  $j = j'$ . Thus the triples are disjoint. These are all roots of the left side of (CK<sub>5</sub>), proving the factorization and irreducibility. The reciprocal assertion follows from  $b^{1+q+q^2} = 1$  and  $(b^{q+1})^\ell = \eta_k^{q+1} = \eta_k + 1$ .

Since  $a = S_\ell(c, d)$  generates  $K/\mathbb{F}_2$ , one has  $\mathbb{F}_2(c, d) = K$ . Comparing the coefficients of  $X^{3\ell-1}$  and  $X$  in (CK<sub>5</sub>) gives the two sums in (CK<sub>6</sub>); evaluation at  $X = 1$  gives the product.  $\square$

**Proposition 6.27** (Dickson Jacobian and the selected cubic fibre). *Let  $F$  have characteristic two and let  $r \geq 1$  be odd. With the recurrence (CK<sub>1</sub>), put*

$$A_r = S_r(C, D), \quad B_r = S_r(D, C), \quad J_r = \det \frac{\partial(A_r, B_r)}{\partial(C, D)}.$$

*Then the following polynomial identity holds in  $F[C, D]$ :*

$$\boxed{(CD + 1)J_r = A_r B_r + 1.} \tag{CJ<sub>1</sub>}$$

*If  $X^3 + AX^2 + BX + 1$  is separable, the geometric fibre  $(A_r, B_r) = (A, B)$  consists of exactly  $r^2$  reduced points and is everywhere étale.*

Now specialize to Proposition 6.26. The geometric  $(\text{CK}_3)$ -fibre has  $\ell^2$  reduced points, and its  $K$ -rational locus has cardinality

$$\#\{(c, d) \in K^2 : (\text{CK}_3)\} = \begin{cases} \ell, & \eta_k \in U^\ell, \\ 0, & \eta_k \notin U^\ell. \end{cases} \quad (\text{CJ}_2)$$

At every rational point,

$$J_\ell(c, d) = \frac{a^2 + a + 1}{cd + 1} \neq 0. \quad (\text{CJ}_3)$$

Since  $\gcd(\ell, q - 1) = 1$ , all nonzero Jacobian values, the source and target discriminants

$$(cd + 1)^2, \quad (a^2 + a + 1)^2,$$

and their ratios are automatically  $\ell$ -th powers in  $K$ .

There is likewise only one Kummer obstruction behind the reciprocal pair. For a geometric  $b$  with  $b^\ell = \eta_k$ , set

$$\theta = b^{q^2+q+1} \in \mu_\ell. \quad (\text{CJ}_4)$$

It is independent of the choice  $b \mapsto \rho^j b$ , and

$$\eta_k \in U^\ell \iff \theta = 1.$$

The companion root  $b^{q+1}$  satisfies  $(b^{q+1})^\ell = \eta_k + 1$ , and its norm obstruction is  $\theta^{q+1} = \theta^{-q^2}$ . Since  $\gcd(q+1, \ell) = 1$ , it vanishes exactly when  $\theta$  does. Thus the ordered second equation in  $(\text{CK}_3)$  selects the target instead of its reciprocal, but supplies no independent residue class.

Under failure, put  $z_{k-1} := a^2 + a + 1$ . The  $\ell$  pairs in  $(\text{CK}_4)$  also obey the selected norm product

$$\boxed{\prod_{j \bmod \ell} (c_j^2 + c_j d_j + d_j^2) = a^2 + a + 1 = z_{k-1}.} \quad (\text{CJ}_5)$$

Every factor is nonzero. This recursion again has no  $\ell$ -residue content, because raising to the  $\ell$ -th power is an automorphism of  $K^\times$ .

Finally, the symmetric Berlekamp representative of a separable cubic  $X^3 + CX^2 + DX + 1$  is

$$\begin{aligned} \mathcal{B}(C, D) &= \frac{C^3 + D^3 + CD + 1}{(CD + 1)^2} \\ &= \frac{(C + D + 1)(C^2 + CD + D^2 + C + D + 1)}{(CD + 1)^2}. \end{aligned} \quad (\text{CJ}_6)$$

Its absolute trace is zero for every irreducible cubic, while the selected target has the displayed representative  $\mathcal{B}(a, a + 1) = 0$ . Therefore ramification, discriminant, inversion, and Berlekamp-trace invariants all leave precisely the original selected class  $\theta$ ; none excludes the rational fibre in  $(\text{CJ}_2)$ .

*Proof.* Work first in the Laurent ring with independent  $x, y$  and  $z = (xy)^{-1}$ . Put

$$C = x + y + z, \quad D = xy + yz + zx.$$

The Newton recurrence gives

$$(A_r, B_r) = (x^r + y^r + z^r, x^{-r} + y^{-r} + z^{-r}).$$

For logarithmic differentials,

$$\det \frac{\partial(C, D)}{\partial(\log x, \log y)} = (x + y)(y + z)(z + x) = CD + 1.$$

At the powered point, the corresponding determinant is  $A_r B_r + 1$ , while the power map  $(x, y) \mapsto (x^r, y^r)$  has logarithmic determinant  $r^2 = 1$  in characteristic two. The chain rule therefore gives explicitly

$$J_r(CD + 1) = r^2(A_r B_r + 1) = A_r B_r + 1.$$

Since  $C, D$  are algebraically independent, it is an identity in  $F[C, D]$ .

Label the three distinct target roots. Choose independently an  $r$ -th root of each of the first two and make the third the inverse of their product. This gives  $r^2$  source cubics. Two choices cannot give the same unordered source roots: raising to the  $r$ -th power would permute distinct labeled target roots, forcing the permutation and then both choices to agree. This accounts for the full geometric fibre. On it  $A_r B_r + 1 \neq 0$ ; (CJ<sub>1</sub>) forces both  $CD + 1$  and  $J_r$  to be nonzero, proving reducedness and étaleness.

The automatic irreducibility proved in Proposition 6.26 identifies every rational point with a root  $b$  whose  $\ell$ -th power is one of the three selected target roots. After fixing  $b^\ell = \eta_k$ , the  $\ell$  possible roots are exactly  $\rho^j b$ , and each cubic has exactly one root over  $\eta_k$ . Thus (CK<sub>5</sub>) gives all rational points under failure, while the equivalence in that proposition gives none under success. This proves (CJ<sub>2</sub>), and (CJ<sub>1</sub>) proves (CJ<sub>3</sub>). The power-residue and discriminant statements follow from  $\gcd(\ell, q-1) = 1$  and the characteristic-two discriminant formula  $\text{disc}(X^3 + CX^2 + DX + 1) = (CD + 1)^2$ .

For (CJ<sub>4</sub>), raising  $\theta$  to the  $\ell$ -th power uses  $N(\eta_k) = 1$ , while changing  $b$  by  $\rho^j$  changes  $\theta$  by  $\rho^{j(q^2+q+1)} = 1$ . Membership of  $b$  in  $U$  is exactly  $\theta = 1$ , and

$$N_{L/K}(b^{q+1}) = \theta^{q+1} = \theta^{-q^2}$$

proves the reciprocal claim.

Because  $\text{ord}_\ell(q) = 3$ , one has  $\ell \equiv 1 \pmod{3}$ . Evaluate (CK<sub>5</sub>) at  $\omega$  and  $\omega^2$ , where  $\omega^2 + \omega + 1 = 0$ . The two values on the left are respectively  $a + \omega$  and  $a + \omega^2$ , while the product of the corresponding two values of the  $j$ -th cubic is  $c_j^2 + c_j d_j + d_j^2$ . This proves (CJ<sub>5</sub>). A zero factor would either give  $c_j = d_j = 0$ , contradicting irreducibility, or put  $c_j/d_j \in \mathbb{F}_4 \setminus \mathbb{F}_2$ . But  $K$  has odd degree over  $\mathbb{F}_2$ , so  $K \cap \mathbb{F}_4 = \mathbb{F}_2$ .

For ordered roots  $x, y, z$ , direct simplification gives

$$\beta = \frac{x}{x+y} + \frac{y}{y+z} + \frac{z}{z+x}, \quad \beta^2 + \beta = \mathcal{B}(C, D).$$

If the cubic is irreducible and the cyclic ordering is Frobenius ordering, then  $\beta \in K$ , so the absolute trace of  $\mathcal{B}(C, D)$  is zero. Reversing the cyclic order changes  $\beta$  by one:  $\beta$  itself depends on the orientation, while  $\beta^2 + \beta$  is invariant. Substitution  $C = a, D = a + 1$  makes the numerator in (CJ<sub>6</sub>) zero. This proves the final audit.  $\square$

**Proposition 6.28** (absolute-trace obstruction to normal-period arguments). *For every  $k \geq 1$  one has*

$$\text{Tr}_{L_k/\mathbb{F}_2}(\gamma_k) = 0, \quad \text{Tr}_{L_k/\mathbb{F}_2}(\eta_k) = 0.$$

*Consequently neither  $\gamma_k$  nor  $\eta_k$  is normal over  $\mathbb{F}_2$ . In particular, a theorem whose hypothesis is that the selected Gaussian period is normal cannot prove the cubic arm for this family.*

*Proof.* The relative minimal polynomial of  $\gamma_k$  over  $L_{k-1}$  is  $X^3 + X + \gamma_{k-1}$ . Its quadratic coefficient is zero, so

$$\text{Tr}_{L_k/L_{k-1}}(\gamma_k) = 0.$$

Trace transitivity gives the first equality. By Proposition 6.24, the relative minimal polynomial of  $\eta_k$  has quadratic coefficient  $\eta_{k-1}$ ; in characteristic two,

$$\text{Tr}_{L_k/L_{k-1}}(\eta_k) = \eta_{k-1}.$$

Iterating trace transitivity reduces the absolute trace to that of  $\eta_0 = 0$ , proving the second equality.

If either element were normal, its full absolute-Frobenius orbit would be an  $\mathbb{F}_2$ -basis of  $L_k$ . The sum of that orbit is the absolute trace, so trace zero would give the nontrivial all-ones linear relation among the basis vectors. This is impossible.  $\square$

**Proposition 6.29** (the inverse selector is completely normal). *Put  $\gamma_0 = \beta_0 = 1$  and, for  $k \geq 1$ ,*

$$\beta_k = \gamma_k^{-1} \in L_k = \mathbb{F}_{2^{3^k}}.$$

*Then  $\beta_k$  is completely normal over  $\mathbb{F}_2$ : for every  $0 \leq s < k$ , its  $2^{3^s}$ -Frobenius orbit is an  $L_s$ -basis of  $L_k$ . In particular,*

$$\dim_{\mathbb{F}_2} \langle \beta_k^{2^i} : 0 \leq i < 3^k \rangle = 3^k. \quad (\text{CN}_1)$$

*By contrast, the exact Frobenius-orbit ranks of the original coordinate and its translate are*

$$\begin{aligned} \dim_{\mathbb{F}_2} \langle \gamma_k^{2^i} : 0 \leq i < 3^k \rangle &= 2 \cdot 3^{k-1}, \\ \dim_{\mathbb{F}_2} \langle (\gamma_k + 1)^{2^i} : 0 \leq i < 3^k \rangle &= 2 \cdot 3^{k-1} + 1. \end{aligned} \quad (\text{CN}_2)$$

*The inverse selector is also the derivative coordinate:*

$$XP'_k(X) = P_k(X) + 1, \quad P'_k(\gamma_k) = \beta_k. \quad (\text{CN}_{2a})$$

*Thus complete additive normality is unconditional after inversion. It supplies no criterion for multiplicative primitivity in  $C_k$ .*

*Proof.* The relation  $\gamma_k^3 + \gamma_k + \gamma_{k-1} = 0$  gives, on inversion,

$$\beta_k^3 + \beta_{k-1}\beta_k^2 + \beta_{k-1} = 0. \quad (\text{CN}_3)$$

Consequently

$$\text{Tr}_{L_k/L_{k-1}}(\beta_k) = \beta_{k-1},$$

and transitivity gives  $\text{Tr}_{L_k/L_j}(\beta_k) = \beta_j$  for every  $j < k$ . The denominator-cleared algebra in  $(\text{CN}_3)$  is kernel-checked in `Ogdoad.Excess`.

The base case of  $(\text{CN}_{2a})$  is immediate from  $P_1 = X^3 + X + 1$ . Differentiating  $P_{k+1} = P_k^3 + P_k^2 + 1$  in characteristic two shows that  $XP'_{k+1} = P_{k+1} + 1$  whenever  $XP'_k = P_k + 1$ . Evaluation at  $\gamma_k$  gives the second identity. The generic derivative induction step is also kernel-checked in `Ogdoad.Excess`.

For the absolute rank calculation, let  $\sigma(x) = x^2$ , and define the trace projectors

$$T_j = \sum_{r=0}^{3^{k-j}-1} \sigma^{r3^j} = \text{Tr}_{L_k/L_j}, \quad E_0 = T_0, \quad E_j = T_j + T_{j-1} \quad (1 \leq j \leq k).$$

Because

$$X^{3^k} - 1 = \prod_{j=0}^k \Phi_{3^j}(X)$$

and  $\text{ord}_{3^j}(2) = \varphi(3^j)$ , every factor  $\Phi_{3^j}$  is irreducible over  $\mathbb{F}_2$ . Hence  $E_j$  is the projector onto one simple Frobenius block. Now

$$E_0(\beta_k) = 1, \quad E_j(\beta_k) = \beta_j + \beta_{j-1} \neq 0.$$

Thus every simple block occurs, proving (CN<sub>1</sub>) and absolute normality.

For  $\gamma_k$ , the missing quadratic coefficient in  $X^3 + X + \gamma_{k-1}$  gives  $T_j(\gamma_k) = 0$  for every  $j < k$ . It therefore occupies exactly the top  $\Phi_{3^k}$ -block, of dimension  $\varphi(3^k) = 2 \cdot 3^{k-1}$ . Adding one supplies exactly the one-dimensional invariant block as well, proving (CN<sub>2</sub>).

Finally, Scheerhorn's completely-normal Dickson criterion applies directly to every intermediate extension:  $3^k$  is a power of the odd prime dividing  $2 + 1$ , and

$$P_k(X) = D_{3^k}(X, 1) + 1$$

is irreducible by Proposition 6.9. The reciprocal polynomial, whose root is  $\beta_k$ , is therefore completely normal [37, Corollary 1].  $\square$

**Proposition 6.30** (canonical half-circulant collapse of the two normal coordinates). *Retain Propositions 6.25 and 6.29. Put  $n = 3^k$ ,  $\sigma(x) = x^2$ , and regard  $\mathbb{F}_2[C_n]$  as  $\mathbb{F}_2[X]/(X^n - 1)$  acting through  $X = \sigma$ . Define*

$$D(X) = X + X^2 + \cdots + X^{n-2}$$

and

$$C_k(X) = \begin{cases} 1 + X + X^3 + \cdots + X^{n-2}, & k \text{ even}, \\ X^2 + X^4 + \cdots + X^{n-1}, & k \text{ odd}. \end{cases}$$

Then

$$\boxed{\beta_k = D(\sigma)\varepsilon_k, \quad \varepsilon_k = C_k(\sigma)\beta_k, \quad C_k D = 1} \quad (\text{NB}_1)$$

in  $\mathbb{F}_2[C_n]$ . At an  $n$ -th root  $\xi$ , the two circulants have spectral values

$$D(1) = C_k(1) = 1, \quad D(\xi) = \frac{\xi + 1}{\xi}, \quad C_k(\xi) = \frac{\xi}{\xi + 1} \quad (\xi \neq 1). \quad (\text{NB}_2)$$

Consequently their regular-representation determinants are one. The absolute normal bases generated by  $\beta_k$  and  $\varepsilon_k$  differ by this determinant-one Frobenius circulant and have the same full-field subspace polynomial  $Z^{2^n} + Z$ .

Let  $\ell \mid q^2 + q + 1$ , let  $\chi : U_k \rightarrow \mu_\ell$  have exact order  $\ell$ , and let  $\tilde{\chi} : L_k^\times \rightarrow \mu_\ell$  be its unique extension trivial on  $L_{k-1}^\times$ . Then

$$\eta_k \in U_k^\ell \iff \tilde{\chi}(\beta_k) = 1 \iff \tilde{\chi}(\varepsilon_k) = 1. \quad (\text{NB}_3)$$

Under failure both full absolute Frobenius orbits in (NB<sub>1</sub>) lie in  $\ker \tilde{\chi}$ ; conversely, membership of either orbit already is the original failure. Thus the coordinate axes, the cyclic half-block sums supplied by  $C_k$ , their  $(n - 2)$ -term complements supplied by  $D$ , and every determinant or subspace-polynomial invariant of this change of normal basis provide no second Kummer condition.

*Proof.* Write  $\varepsilon = \eta + 1$  and  $\varepsilon^{1/2} = \sigma^{-1}(\varepsilon)$ . The Singer identity and the selected identity  $h_k = 1 + \beta_k = \eta_k^{q/2+1}$  give

$$(\varepsilon + \varepsilon^{1/2})^2 = \varepsilon^2 + \varepsilon = \eta^2 + \eta = \eta^{q+2} = h_k^2.$$

Squaring is injective, so

$$\beta_k = 1 + \varepsilon + \sigma^{-1}(\varepsilon).$$

Since  $\text{Tr}_{L_k/\mathbb{F}_2}(\varepsilon) = 1$ , this is  $\beta_k = \sum_{i=1}^{n-2} \sigma^i(\varepsilon) = D(\sigma)\varepsilon$ .

It remains to invert  $D$ . Its support in  $\mathbf{Z}/n\mathbf{Z}$  is every residue except 0,  $-1$ . If  $S$  is the support of  $C_k$ , the coefficient of  $X^r$  in  $C_k D$  is

$$|S| + \mathbf{1}_S(r) + \mathbf{1}_S(r+1) \pmod{2}.$$

For  $n \equiv 1 \pmod{4}$ ,  $S = \{0\} \cup \{1, 3, \dots, n-2\}$ ; for  $n \equiv 3 \pmod{4}$ ,  $S = \{2, 4, \dots, n-1\}$ . In both cases  $|S|$  is odd, and the displayed coefficient is one precisely at  $r = 0$ . This proves (NB<sub>1</sub>); summing the geometric progression at  $\xi$  proves (NB<sub>2</sub>). An invertible matrix over  $\mathbb{F}_2$  has determinant one.

Finally,

$$\beta_k^{q-1} = \eta_k^{-1}, \quad \varepsilon_k = \eta_k^{q+1},$$

and both  $q-1$  and  $q+1$  are units modulo  $\ell$ . Applying  $\tilde{\chi}$  proves (NB<sub>3</sub>), and Frobenius stability of its kernel proves the orbit formulation. Lean kernel-checks the three-term square-root bridge and the Singer exponent identity; the universal cyclic support count and finite-field character interpretation remain paper-level.

This exhausts linear Frobenius-equivariant normal-basis invariants. It does not rule out a non-linear additive–multiplicative correlation tailored to the selected half-block incidence, although current character bounds allow the  $n$  prescribed orbit points.  $\square$

**Proposition 6.31** (an actual-degree two-normal-basis countermodel). *The complete incidence package in Proposition 6.30 is compatible with current Kummer failure, even at an actual cubic-tower degree and an actual current prime.*

Use the polynomial-bit convention in which bit  $i$  is the coefficient of  $X^i$ , put

$$p(X) = 0x329341f8b47dd7938af63, \quad E = \mathbb{F}_2[X]/(p), \quad q = 2^{27},$$

$$N = q^2 + q + 1 = 18014398643699713, \quad \ell = 2593,$$

and in the polynomial basis set

$$\begin{aligned} \eta &= 1629469875507523981620540, \\ \varepsilon &= 1629469875507523981620541, \\ \beta &= 2293671573472151973449566. \end{aligned}$$

Then  $p$  is irreducible of degree 81,  $\ell$  is a prime divisor of  $N$ , and

$$\eta^{q+1} = \eta + 1, \quad \eta^N = 1, \quad \eta^{N/\ell} = 1. \quad (\text{NBC}_1)$$

Moreover,

$$\varepsilon = \eta + 1 = \eta^{q+1}, \quad \beta^{q-1} = \eta^{-1}, \quad \beta = 1 + \eta^{q/2+1} = \sum_{i=1}^{79} \varepsilon^{2^i}. \quad (\text{NBC}_2)$$

Thus the exact Singer, normalized inverse, and half-circulant identities all hold, while  $\eta, \varepsilon, \beta$  lie in the index-2593 Kummer kernel of  $E^\times$ .

The normality data are equally strong. The absolute Moore determinant of  $\varepsilon$  is one. The relative Moore determinants of  $\beta$  over the subfields of degrees 1, 3, 9, 27 are respectively

$$1, \quad 996154457819232351242539, \quad 1913091018644430508445588, \quad 1836908266524337706399502, \quad (\text{NBC}_3)$$

all nonzero. Hence  $\varepsilon$  is normal over  $\mathbb{F}_2$  and  $\beta$  is completely normal.

The missing datum is exactly the selected lower recursion. If

$$a = \text{Tr}_{E/\mathbb{F}_{2^7}}(\eta) = \eta + \eta^q + \eta^{q^2},$$

then

$$a^{2^9+1} + a + 1 \neq 0. \quad (\text{NBC}_4)$$

Thus the model exits the Conway-selected trace ancestry immediately below the top step. It proves that the two full Frobenius orbits, their nonlinear kernel incidence, all Moore/subspace determinants, and the canonical half-circulant relation add no obstruction independent of that ancestry.

*Proof.* The exact maintained verifier in `experiments/, cubic_two_normal_countermodel.py`, performs carryless polynomial arithmetic modulo  $p$ . Rabin's criterion is certified by

$$X^{2^{81}} \equiv X \pmod{p}, \quad \gcd(p, X^{2^{27}} - X) = 1,$$

which suffices because the only prime divisor of 81 is three. It then checks (NBC<sub>1</sub>), (NBC<sub>2</sub>), the three full-field 2593-power identities, the Moore determinants (NBC<sub>3</sub>), and the strict inequality (NBC<sub>4</sub>).

Lean independently kernel-checks the Rabin residues, the Singer, norm, failure, normalized- $\beta$ , and half-circulant identities, all three full Kummer-kernel memberships, and the selected-ancestry break in `Ogdoad.Excess`. The standard interpretation of the script's nonzero Moore determinants as relative normality remains paper-level linear algebra.  $\square$

**Proposition 6.32** (self-dual inverse basis and determinant collapse). *Retain Proposition 6.29. For  $k \geq 1$ , put*

$$q = 2^{3^{k-1}}, \quad E = L_k, \quad F = L_{k-1},$$

and write

$$x = \beta_k, \quad y = x^q, \quad z = x^{q^2}, \quad a = \beta_{k-1}.$$

Then

$$x + y + z = a, \quad xy + yz + zx = 0, \quad xyz = a. \quad (\text{CN}_4)$$

Consequently the relative trace Gram matrix is diagonal:

$$(\text{Tr}_{E/F}(x^{q^i} x^{q^j}))_{0 \leq i, j < 3} = a^2 I_3. \quad (\text{CN}_5)$$

Thus  $(x/a, y/a, z/a)$  is a self-dual normal basis of  $E/F$ , and the immediate Moore determinant contains only the lower selector:

$$\det \begin{pmatrix} x & y & z \\ y & z & x \\ z & x & y \end{pmatrix} = x^3 + y^3 + z^3 + xyz = a^3. \quad (\text{CN}_6)$$



There is likewise no intermediate absolute determinant invariant. Put  $n = 3^k$ ,  $b_i = \beta_k^{2^i}$ , let  $\mathcal{C} = (b_{i+j})_{i,j \in \mathbb{Z}/n\mathbb{Z}}$ , and let  $\mathcal{M}$  be the  $\mathbb{F}_2$ -matrix of multiplication by  $\beta_k$  in the normal basis  $(b_j)$ . Then

$$\det \mathcal{C} = 1, \quad \mathcal{C}\mathcal{M} = \text{diag}(b_0, \dots, b_{n-1})\mathcal{C}. \quad (\text{CN}_7)$$

If  $Q_k$  is the reciprocal minimal polynomial of  $\beta_k$ , then for every  $f \in \mathbb{F}_2[X]$ ,

$$\det f(\mathcal{M}) = \prod_{i=0}^{n-1} f(b_i) = N_{L_k/\mathbb{F}_2}(f(\beta_k)) = \text{Res}(Q_k, f). \quad (\text{CN}_8)$$

In particular, for a current prime  $\ell \mid \Phi_n(2)$  and  $t = (2^n - 1)/\ell$ ,

$$\det(\mathcal{M}^t - I) = 0 \iff \beta_k^t = 1 \iff \text{the } \ell\text{-primary condition in } C_k \text{ fails.} \quad (\text{CN}_9)$$

On failure  $\mathcal{M}^t - I = 0$ ; on success it is invertible and its determinant is one. Hence rank, determinant, resultant, and dual-basis coordinates provide no state strictly between the original selected Euler test's zero and nonzero outcomes.

Finally this collapse persists for crossed Frobenius determinants. Let  $S(u) = u^q$ , put  $\vartheta = \beta_k^t$ , and let  $A$  be multiplication by  $\vartheta$ . Since  $N_{E/F}(\vartheta) = 1$ , Hilbert 90 gives  $\vartheta = v^{q-1}$ , and therefore

$$M_v^{-1} S M_v = A S. \quad (\text{CN}_{10})$$

Thus every characteristic-polynomial or determinant invariant of the twisted Frobenius  $AS$  is the corresponding invariant of  $S$ , whether or not  $\vartheta = 1$ . The only untwisted discriminator  $\det(A - I)$  is precisely  $(\text{CN}_9)$ .

*Proof.* Equation  $(\text{CN}_3)$  is the relative minimal polynomial

$$X^3 + aX^2 + a$$

of  $x$ , so its elementary symmetric coefficients give  $(\text{CN}_4)$ . The diagonal Gram entry is

$$\text{Tr}_{E/F}(x^2) = x^2 + y^2 + z^2 = a^2,$$

whereas every off-diagonal entry is  $xy + yz + zx = 0$ . Complete normality supplies the basis assertion. In characteristic two,

$$x^3 + y^3 + z^3 + xyz = (x + y + z)^3 + (x + y + z)(xy + yz + zx) = a^3,$$

which proves  $(\text{CN}_5)$ – $(\text{CN}_6)$ . These two polynomial calculations are independently kernel-checked in `Ogload.Excess`.

The matrix  $\mathcal{C}$  is invertible because  $(b_i)$  is a normal basis. Squaring every entry cyclically permutes its rows, so  $(\det \mathcal{C})^2 = \det \mathcal{C}$ ; nonvanishing over  $\mathbb{F}_2$  gives  $\det \mathcal{C} = 1$ . Evaluating multiplication by  $\beta_k$  in each Frobenius embedding gives the second identity in  $(\text{CN}_7)$ , and taking determinants proves  $(\text{CN}_8)$ . Since multiplication by a nonzero field element is invertible,  $\mathcal{M}^t - I$  is either the zero matrix when  $\beta_k^t = 1$ , or invertible otherwise. Its nonzero determinant lies in  $\mathbb{F}_2^\times = \{1\}$ , proving  $(\text{CN}_9)$ .

For the final assertion,  $\ell \mid q^2 + q + 1$ , so  $N_{E/F}(\vartheta) = \vartheta^{q^2+q+1} = 1$ . Hilbert 90 and the direct calculation

$$v^{-1} S(vu) = v^{q-1} S(u) = \vartheta S(u)$$

give  $(\text{CN}_{10})$ . Lean checks this generic conjugacy and the corresponding norm-one order-three twist identity. The normal-basis, finite-field norm, Hilbert–90, and resultant interpretations remain paper-level.  $\square$

**Proposition 6.33** (Singer-orbit sieve). *Let  $k \geq 1$ . Put  $n = 3^k$ ,  $q = 2^{n/3}$ , and  $N = q^2 + q + 1 = |U_k|$ . If a prime  $\ell \mid N$  satisfies*

$$\ell > \frac{N}{n(n-1)+1},$$

*then  $\eta_k \notin U_k^\ell$ . Consequently, if  $\ell^a \parallel N$ , then*

$$v_\ell(\text{ord } \eta_k) = a.$$

*Unconditionally,*

$$\text{ord}(\eta_k) \geq n(n-1)+1, \quad [U_k : \langle \eta_k \rangle] \leq \frac{N}{n(n-1)+1}.$$

*Proof.* Let

$$D = \{y \in U_k : y^{q+1} + y + 1 = 0\}.$$

This is the cyclic  $(N, q+1, 1)$  Singer difference set: every nonidentity element of  $U_k$  has a unique expression  $xy^{-1}$  with distinct  $x, y \in D$ . Since the minimal polynomial of  $\eta_k$  over  $\mathbb{F}_2$  has degree  $n$ , its Frobenius orbit

$$A = \{\eta_k^{2^i} : 0 \leq i < n\}$$

has  $n$  elements. The equation defining  $D$  is Frobenius-stable, so  $A \subset D$ .

Suppose  $\eta_k \in H := U_k^\ell$ . The subgroup  $H$  is characteristic in the cyclic group  $U_k$ , hence Frobenius-stable, and  $A \subset H$ . The  $n(n-1)$  quotients  $xy^{-1}$  for ordered distinct pairs  $(x, y) \in A^2$  are nonidentity elements of  $H$  and are all distinct by the Singer difference-set property. Therefore

$$|H| - 1 \geq n(n-1).$$

Since  $|H| = N/\ell$ , this contradicts the displayed strict inequality. In a cyclic group of order  $\ell^a m$ ,  $\ell \nmid m$ , lying outside the subgroup of  $\ell$ -th powers is equivalent to having the full  $\ell^a$  part in one's order, proving the first assertion.

Finally the same distinct quotients all lie in  $\langle \eta_k \rangle$ , whether or not an  $\ell$  is fixed. Thus  $\text{ord}(\eta_k) - 1 \geq n(n-1)$ , which gives the last two inequalities.  $\square$

**Remark 6.34** (why the translated orbit does not double the sieve). The identity  $y + 1 = y^{q+1}$  puts the translated Frobenius orbit  $A + 1 = A^{q+1}$  in the same subgroup as  $A$ . It supplies no second quotient family: modulo  $N$  one has  $(q+1)^2 \equiv q$  and  $q(q+1) \equiv -1$ , while the quotient set of  $A$  is stable under both  $q$ -th powers and inversion. Hence the internal quotient sets of  $A$  and  $A + 1$  coincide. Cross quotients already collide, for example  $(y+1)/y = y^q$ . Improving Proposition 6.33 therefore requires cross-correlation or character information beyond the  $(N, q+1, 1)$  difference-set axiom.

**Proposition 6.35** (partition-order bound). *Retain Proposition 6.33. Let  $B(n)$  be the number of partitions into distinct positive parts not divisible by 3 whose total is at most  $n = 3^k$ . Then*

$$\text{ord}(\eta_k) \geq B(n).$$

*In particular, if  $p_{\mathfrak{F}}^{\text{dist}}(n)$  counts such partitions of  $n$  itself, then*

$$\text{ord}(\eta_k) \geq p_{\mathfrak{F}}^{\text{dist}}(n) = \exp\left(\left(\frac{\sqrt{2}\pi}{3} + o(1)\right)\sqrt{n}\right). \quad (\text{B})$$

If a prime  $\ell \mid N$  satisfies

$$\ell > \frac{N}{B(n)},$$

then  $\eta_k \notin U_k^\ell$ , so the order of  $\eta_k$  contains the full  $\ell$ -primary part of  $N$ .

*Proof.* Put  $r = 3n$ ,  $t = \zeta^2$ , and retain  $\omega = \zeta^n$ , so  $t^n = \omega^2$  and

$$\eta_k = M(t), \quad M(X) = \frac{\omega + \omega^2 X}{1 + X}.$$

The element 2 generates  $(\mathbb{Z}/r\mathbb{Z})^\times$ , of order  $2n$ . For every positive  $j$  not divisible by 3, choose  $z_j$  with  $2^{z_j} \equiv j \pmod{r}$ . Frobenius gives

$$\eta_k^{2^{z_j}} = \omega^j \frac{1 + \omega^j t^j}{1 + t^j}. \quad (6)$$

For a finite set  $I$  of such  $j$ , put

$$s_I = \sum_{j \in I} j, \quad E_I = \sum_{j \in I} 2^{z_j}, \quad A_I(X) = \prod_{j \in I} (1 + \omega^j X^j), \quad B_I(X) = \prod_{j \in I} (1 + X^j).$$

Thus

$$\eta_k^{E_I} = \omega^{s_I} \frac{A_I(t)}{B_I(t)}. \quad (7)$$

All denominators are nonzero when  $s_I \leq n < r$ .

Suppose  $s_I, s_J \leq n$  and  $\eta_k^{E_I} = \eta_k^{E_J}$ . Cross-multiplication gives

$$F(X) := \omega^{s_I} A_I(X) B_J(X) + \omega^{s_J} A_J(X) B_I(X), \quad F(t) = 0. \quad (8)$$

Now  $M(t^{-1}) = M(t) + 1 = \eta_k + 1$ , and the Singer equation  $\eta_k + 1 = \eta_k^{q+1}$  shows that the assumed equality also holds after replacing  $\eta_k$  by  $M(t^{-1})$ . Hence  $F(t^{-1}) = 0$ . Over  $\mathbb{F}_4$ , the two elements  $t, t^{-1}$  have the distinct degree- $n$  minimal polynomials

$$X^n + \omega^2, \quad X^n + \omega,$$

because  $\text{ord}_r(4) = n$ . Therefore

$$X^{2n} + X^n + 1 \mid F(X). \quad (9)$$

But  $\deg F \leq s_I + s_J \leq 2n$ . In the only possible equality case  $s_I = s_J = n$ , the two leading coefficients are both  $\omega^{2n}$  and cancel in characteristic two. Thus  $\deg F < 2n$ , and (9) forces  $F = 0$ .

The constant coefficient first gives  $\omega^{s_I} = \omega^{s_J}$ . Cancel that scalar and all factors belonging to  $I \cap J$ . If  $j_0$  is the least remaining part, comparison of the  $X^{j_0}$ -coefficients gives  $\omega^{j_0} = 1$ : on one side  $j_0$  occurs in an  $A$ -factor, with coefficient  $\omega^{j_0}$ , and on the other in a  $B$ -factor, with coefficient 1; larger parts cannot contribute. This is impossible because  $3 \nmid j_0$ . Hence  $I = J$ .

The  $B(n)$  powers in (7) are therefore pairwise distinct, proving the order bound. Meinardus's partition theorem [28], applied to the Euler product

$$G(x) := \prod_{3 \nmid j} (1 + x^j) = \frac{\prod_{j \geq 1} (1 + x^j)}{\prod_{j \geq 1} (1 + x^{3j})}, \quad \log G(e^{-t}) \sim \frac{\pi^2}{18t},$$

gives (B). Finally, if  $\eta_k \in U_k^\ell$ , all  $B(n)$  displayed powers lie in the subgroup of order  $N/\ell$ . The strict inequality rules this out; the cyclic-group argument of Proposition 6.33 then gives the full  $\ell$ -primary assertion.  $\square$

**Remark 6.36** (ceiling of the height-budgeted amplification). The preceding construction is scale-optimal within the following precise class of degree arguments. For  $J = (\mathbb{Z}/3n\mathbb{Z})^\times / \{\pm 1\}$ , assign an atom indexed by  $j \in J$  the height

$$h(j) := \min\{|b| : b \equiv \pm j \pmod{3n}\}.$$

Suppose a family of integer exponent vectors is certified pairwise distinct only by bounding the Laurent span after cross-multiplication with the triangle estimate

$$w(e - e') := \sum_{j \in J} h(j) |e_j - e'_j| \leq 2n$$

and then invoking the two degree- $n$  minimal polynomials as above (including the verified leading-term cancellation at the boundary). Then the family has cardinality  $\exp(O(\sqrt{n}))$ .

Indeed, fix one vector  $e_0$ . Every difference  $e - e_0$  lies in the displayed weighted  $\ell^1$ -ball. Each positive height occurs only  $O(1)$  times, so the number of such lattice points is bounded by the coefficients through degree  $2n$  of a fixed power of

$$\prod_{m \geq 1} (1 + 2x^m + 2x^{2m} + \cdots) \preccurlyeq \prod_{m \geq 1} (1 - x^m)^{-2}.$$

Colored-partition bounds give  $\exp(O(\sqrt{n}))$ . Polynomially many Frobenius or cyclic shifts do not change that scale, whereas  $|U_k| = \exp((2 \log 2/3 + o(1))n)$ . Thus more triangle-budgeted subset encodings cannot prove primitivity. This does *not* rule out systematic factor cancellations or exact reduction modulo  $\Phi_{3n}$ ; those require new arithmetic input rather than the generic degree budget.

## 6.5 The exact remaining character test

**Proposition 6.37** (no transfer and the exact character test). *The kernel orders are pairwise coprime: for  $1 \leq i < k$ ,*

$$\gcd(\Phi_{3^i}(2), \Phi_{3^k}(2)) = 1.$$

*For each prime  $\ell \mid |U_k|$ , let  $\chi_{k,\ell} : U_k \rightarrow \mu_\ell$  be a character of exact order  $\ell$ . Then*

$$\mathrm{Hom}\left(\mathbb{F}_{2^{3^{k-1}}}^\times, \mu_\ell\right) = 1; \tag{\ddagger_0}$$

*no multiplicative character or power-residue datum from a previous level can transfer an  $\ell$ -part to level  $k$ . Moreover,*

$$C_k \iff C_{k-1} \text{ and } \chi_{k,\ell}(\eta_k) \neq 1 \text{ for every prime } \ell \mid |U_k|. \tag{\ddagger}$$

*Equivalently, with*

$$\rho_{k,\ell}(A) = \chi_{k,\ell}(A^{q_k-1}) \quad (A \in \mathbb{F}_{q_k}^\times),$$

*where  $\rho_{k,\ell}$  is trivial on  $\mathbb{F}_{q_k}^\times$ , the missing assertion is  $\rho_{k,\ell}(\gamma_k) \neq 1$  for every such  $\ell$ . This is the smallest power-residue form of the cubic obstruction.*

*Proof.* If a prime  $r$  divided two displayed cyclotomic values, then  $r \neq 3$  and the standard cyclotomic-divisor criterion would give both  $\mathrm{ord}_r(2) = 3^i$  and  $\mathrm{ord}_r(2) = 3^k$ , a contradiction. Thus every  $\ell \mid |U_k|$  is coprime to  $|\mathbb{F}_{2^{3^{k-1}}}^\times| = \prod_{j < k} \Phi_{3^j}(2)$ , proving  $(\ddagger_0)$ . In the cyclic group  $U_k$ , the kernel of an exact order- $\ell$  character is  $U_k^\ell$ . If  $|U_k| = \ell^a m$  with  $\ell \nmid m$ , then an element has the full  $\ell^a$  in its order exactly when it is not in  $U_k^\ell$ . Applying this for every prime divisor and using  $(**)$  proves  $(\ddagger)$ . The formula for  $\rho_{k,\ell}$  and  $\eta_k = \gamma_k^{q_k-1}$  give the final equivalence.  $\square$

**Proposition 6.38** (short cyclotomic-period form of the cubic obstruction). *Let  $k \geq 1$ , put  $n = 3^k$ ,  $q = 2^{n/3}$ , and let  $\ell$  be a prime divisor of  $\Phi_{3^k}(2)$ . Let  $\rho = \rho_{k,\ell}$  be the exact  $\ell$ -character from Proposition 6.37, realized with values in the complex roots of unity. In  $\mathbb{F}_{2^{2n}}$  let  $\mu_{3n}$  denote the roots of unity of order dividing  $3n = 3^{k+1}$ , and set*

$$\mathcal{S}_{k,\ell} := \sum_{\substack{t \in \mu_{3n} \\ t \neq 1}} \rho(t + t^{-1}). \quad (\text{S})$$

*Every argument of  $\rho$  in this sum is a nonzero element of  $\mathbb{F}_{2^n}$ . If  $\rho(\gamma_k) = \zeta_\ell^a$ , then*

$$\mathcal{S}_{k,\ell} = (n-1) + 2 \sum_{i=0}^{n-1} \zeta_\ell^{a2^i}. \quad (\text{P})$$

*Consequently*

$$\chi_{k,\ell}(\eta_k) = 1 \iff \mathcal{S}_{k,\ell} = 3n - 1. \quad (\text{C})$$

*Thus the surviving datum is exactly a short Gauss period over the order- $n$  subgroup generated by 2 in  $(\mathbb{Z}/\ell\mathbb{Z})^\times$ .*

*Proof.* Since  $2^n \equiv -1 \pmod{3n}$ , every  $t + t^{-1}$  in (S) is fixed by  $2^n$  and lies in  $\mathbb{F}_{2^n}$ ; it is nonzero unless  $t = 1$ . The  $n-1$  nonidentity elements of  $\mu_n$  give real cyclotomic elements in the lower field  $\mathbb{F}_{2^{n/3}}^\times$ , on which  $\rho$  is trivial.

The remaining  $2n$  elements are the primitive  $3n$ -th roots. Pairing them by inversion leaves the  $n$  Frobenius conjugates  $\gamma_k^{2^i}$ , each twice. This proves (P). Since  $q-1$  is invertible modulo  $\ell$ ,  $\chi_{k,\ell}(\eta_k) = 1$  exactly when  $\rho(\gamma_k) = 1$ , which gives the forward implication in (C). Conversely every summand in (S) has complex absolute value one. Equality with the number of summands  $3n-1$  forces equality in the triangle inequality and hence forces every summand to be 1, in particular  $\rho(\gamma_k) = 1$ .  $\square$

**Remark 6.39** (why a full Singer character sum is too coarse). The Singer difference-set character sum controls a set of size  $q+1$  and has the usual square-root magnitude. Equation (P) instead localizes to only  $3n-1$  cyclotomic terms, with the unknown part supported on the order- $n$  subgroup  $\langle 2 \rangle$  modulo  $\ell$ . Generic Weil or difference-set bounds do not evaluate this localized scalar; an all-level proof must rule out its extremal value in (C).

**Proposition 6.40** (cyclotomic-secant saturation). *Retain  $n = 3^k$ , put  $m = 3n$ , and let  $K = \mathbb{F}_{2^{2n}}$  contain a primitive  $m$ -th root  $\zeta$ . For every prime  $\ell \mid \Phi_{3^k}(2)$ , the following are equivalent:*

- (i)  $\eta_k \in U_k^\ell$ ;
- (ii)  $\gamma_k \in (\mathbb{F}_{2^n}^\times)^\ell$ ;
- (iii)  $1 + \zeta \in (K^\times)^\ell$ ;
- (iv) for every  $a \not\equiv b \pmod{m}$ ,  $\zeta^a + \zeta^b \in (K^\times)^\ell$ .

*Thus, for an exact order- $\ell$  character  $\chi$  of  $K^\times$ , failure is also equivalent to the extremal identity*

$$\sum_{\substack{t \in \mu_m \\ t \neq 1}} \chi(1-t) = m-1. \quad (\text{CS})$$

*It saturates every cyclotomic secant, not only the Frobenius orbit of the distinguished Gaussian period.*

*Proof.* The first equivalence follows from

$$\eta_k = \gamma_k^{2^{n/3}-1},$$

because  $\ell \nmid 2^{n/3} - 1$ . Next,

$$\gamma_k = \zeta^{-1}(1 + \zeta)^2.$$

The order of  $\zeta$  is coprime to  $\ell$ , and squaring is invertible on the  $\ell$ -primary quotient. Moreover the inclusion  $\mathbb{F}_{2^n}^\times \subset K^\times$  preserves that quotient, since  $\ell \nmid 2^n + 1$ . This proves the equivalence with (iii).

Write  $d = b - a$ . The factor  $\zeta^a$  is automatically an  $\ell$ -th power. If  $3 \nmid d$ , then (2) generates  $(\mathbb{Z}/m\mathbb{Z})^\times$ , so  $d \equiv 2^j \pmod{m}$  and

$$1 + \zeta^d = (1 + \zeta)^{2^j}.$$

If  $3 \mid d$ , then  $1 + \zeta^d \in \mathbb{F}_{2^{2n/3}}^\times$ . This group has order coprime to  $\ell$ , because  $\text{ord}_\ell(2) = n$ , so all its elements are  $\ell$ -th powers in  $K$ . This proves (iii) $\Rightarrow$ (iv); the converse is the case  $a = 0, b = 1$ .

Finally, all terms in (CS) have complex absolute value one. Part (iv) makes them all one; conversely equality with the positive real number  $m - 1$  forces every term to be one by equality in the triangle inequality.  $\square$

**Proposition 6.41** (the mixed-Jacobi top line of the cubic obstruction). *Let  $k \geq 1$ , put  $n = 3^k$ ,  $m = 3n$ , and let  $\ell \mid \Phi_n(2)$  be prime. Put  $Q = 2^{2n}$ , choose a generator  $g \in \mathbb{F}_Q^\times$ , and set*

$$z = g^{(Q-1)/m}.$$

*Thus  $z$  is a primitive  $m$ -th root of unity. Define  $\text{ind}_\ell(g^v) = v \pmod{\ell}$ , and let  $\chi_m, \chi_\ell$  be the multiplicative characters, extended by zero at zero, determined by*

$$\chi_m(g) = \zeta_m, \quad \chi_\ell(g) = \zeta_\ell.$$

*For  $1 \leq j < m$ , form the mixed Jacobi sums*

$$J_j = \sum_{x \in \mathbb{F}_Q} \chi_m(x)^j \chi_\ell(1 - x).$$

*Fix a prime above  $\ell$ , put  $\pi = \zeta_\ell - 1$ , and write  $\omega$  for the reduction of  $\zeta_m$ . Then*

$$J_j \equiv -1 + \pi C_j \pmod{\pi^2}, \quad C_j = \sum_{a=1}^{m-1} \omega^{ja} L(a), \quad L(a) = \text{ind}_\ell(1 - z^a). \quad (\text{MJ}_1)$$

*There is a primitive even character*

$$\lambda : (\mathbb{Z}/m\mathbb{Z})^\times \longrightarrow \mathbb{F}_\ell^\times, \quad \lambda(\pm 2^i) = 2^i \quad (0 \leq i < n), \quad (\text{MJ}_2)$$

*and*

$$L(a) = \begin{cases} 0, & 3 \mid a, \\ \lambda(a)L(1), & (a, m) = 1. \end{cases} \quad (\text{MJ}_3)$$

*Consequently, if*

$$G_j = \sum_{a \in (\mathbb{Z}/m\mathbb{Z})^\times} \lambda(a) \omega^{ja},$$

then  $C_j = L(1)G_j$ , and

$$G_j = \begin{cases} 0, & 3 \mid j, \\ \lambda(j)^{-1}G_1, & (j, m) = 1, \end{cases} \quad G_1G(\lambda^{-1}) = \lambda(-1)m = m \neq 0. \quad (\text{MJ}_4)$$

Here  $G(\lambda^{-1}) = \sum_{a \in (\mathbb{Z}/m\mathbb{Z})^\times} \lambda(a)^{-1} \omega^a$ . In particular  $G_1 \neq 0$ , and

$$\begin{aligned} \gamma_k \notin (\mathbb{F}_{2^n}^\times)^\ell &\iff 1 - z \notin (\mathbb{F}_Q^\times)^\ell \\ &\iff L(1) \neq 0 \\ &\iff C_1 \neq 0 \\ &\iff J_1 \not\equiv -1 \pmod{\pi^2}. \end{aligned} \quad (\text{MJ}_5)$$

Thus the entire first-order Jacobi vector is supported on the primitive top- $\Phi_m$  Fourier coordinates, and every primitive coordinate is a known nonzero multiple of the one selected coordinate  $L(1)$ . The remaining cubic theorem is precisely the all-level assertion

$$v_\pi(J_1 + 1) = 1. \quad (\text{MJ}_6)$$

*Proof.* The cyclotomic-divisor criterion gives  $\text{ord}_\ell(2) = n$ , while  $\text{ord}_m(2) = 2n$ . Hence  $m\ell \mid Q - 1$ .

Expand  $\zeta_\ell^v = (1 + \pi)^v \equiv 1 + v\pi \pmod{\pi^2}$  in the defining Jacobi sum. Its constant term is  $-1$ . To obtain the linear term, partition  $\mathbb{F}_Q^\times$  by the value of the discrete logarithm modulo  $m$ . For a nonzero class  $a$ ,

$$\prod_{v \equiv a \pmod{m}} (1 - g^v) = 1 - z^a;$$

this is the polynomial identity for the coset of the subgroup  $\langle g^m \rangle$ , evaluated at one. Taking  $\ell$ -indices gives the displayed coefficient. In the zero class, the term  $x = 1$  is omitted by the zero-extension of  $\chi_\ell$ , and

$$\prod_{h \in \langle g^m \rangle \setminus \{1\}} (1 - h) = \frac{Q - 1}{m} = 1 \quad \text{in characteristic two;}$$

its index is therefore zero. This proves (MJ<sub>1</sub>).

The group  $(\mathbb{Z}/m\mathbb{Z})^\times$  is generated by 2, has order  $2n$ , and satisfies  $2^n \equiv -1 \pmod{m}$ . Since 2 has order  $n$  modulo  $\ell$ , the rule in (MJ<sub>2</sub>) is well-defined and multiplicative. It is even, and its kernel is exactly  $\{\pm 1\}$ . It cannot factor through modulus  $m/3$ : the kernel of reduction to  $(\mathbb{Z}/(m/3)\mathbb{Z})^\times$  has order three, whereas  $\ker \lambda$  has order two. Thus its conductor is  $m$ .

If  $3 \mid a$ , then  $1 - z^a \in \mathbb{F}_{2^{2n/3}}^\times$ , whose order is coprime to  $\ell$ , because  $\text{ord}_\ell(2) = n \nmid 2n/3$ . Hence  $L(a) = 0$ . If  $(a, m) = 1$ , write  $a \equiv \pm 2^i \pmod{m}$ . Frobenius gives

$$\text{ind}_\ell(1 - z^{2^i}) = 2^i \text{ind}_\ell(1 - z).$$

For the negative sign use  $1 - z^{-2^i} = z^{-2^i}(1 - z^{2^i})$ ; the  $m$ -torsion factor has zero  $\ell$ -index. This proves (MJ<sub>3</sub>), and hence  $C_j = L(1)G_j$ .

For  $(j, m) = 1$ , changing variables gives  $G_j = \lambda(j)^{-1}G_1$ . If  $3 \mid j$ , the additive factor is constant on the three lifts of a unit modulo  $m/3$ , while  $\lambda$  is nontrivial on that order-three kernel, so each fibre sums to zero. Finally the standard Gauss identity gives

$$G(\lambda)G(\lambda^{-1}) = \lambda(-1)m = m.$$

Since  $\ell \nmid m$ , this proves (MJ<sub>4</sub>) and  $G_1 \neq 0$ .

The equivalence between  $L(1) = 0$  and  $1 - z$  being an  $\ell$ -th power is the definition of the index. Up to a Frobenius conjugate, the selected period satisfies

$$\gamma_k = z + z^{-1} = z^{-1}(1 + z)^2.$$

The  $m$ -torsion factor is an  $\ell$ -th power, squaring is invertible on the odd  $\ell$ -Kummer quotient, and extension from  $\mathbb{F}_{2^n}$  to  $\mathbb{F}_Q$  preserves that quotient because  $\ell \nmid 2^n + 1$ . This proves (MJ<sub>5</sub>). Finally, the indices coprime to  $m$  are exactly the primitive additive characters of the cyclic group of order  $m$ ; the other indices are divisible by three. Thus (MJ<sub>4</sub>) is precisely the claimed top-component statement, and (MJ<sub>6</sub>) is the surviving nonvanishing condition.  $\square$

**Remark 6.42** (higher Jacobi congruences do not bootstrap the first line). Arul's higher congruence theorem [1, Theorem 1.5], stated for a second prime conductor, expands

$$J + 1 = \sum_{i=1}^{\ell-1} \pi^i A_i \pmod{\pi^\ell}.$$

The coefficient field is unramified at  $\ell$ , so distinct nonzero summands have distinct valuations  $i/(\ell - 1)$  modulo the integers and cannot cancel. In particular

$$v_\pi(J + 1) = 1 \iff A_1 \neq 0.$$

Higher congruences impose additional power-residue conditions only after  $A_1 = 0$ ; they cannot force  $A_1 \neq 0$  from  $A_2, A_3, \dots$ . Proposition 6.41 proves the required composite-conductor first-order analogue directly, with  $A_1 = C_1 = G_1 L(1)$  and  $G_1 \neq 0$ . Hence this route again stops at the selected Kummer index  $L(1)$ , rather than providing an induction from a deeper Jacobi congruence.

**Corollary 6.43** (the selected weighted Jacobi resolvent). *Retain Proposition 6.41, put  $U = (\mathbb{Z}/m\mathbb{Z})^\times$ , and choose integer lifts  $w_j \equiv \lambda(j) \pmod{\ell}$ . Then*

$$\mathcal{R}_\lambda := \prod_{j \in U} (-J_j)^{w_j} \equiv 1 - 2n\pi C_1 \pmod{\pi^2}. \quad (\text{WR}_1)$$

*Consequently the cubic arm at  $(k, \ell)$  is equivalent to the one weighted congruence*

$$\mathcal{R}_\lambda \not\equiv 1 \pmod{\pi^2}. \quad (\text{WR}_2)$$

*By contrast, the full unweighted primitive product is identically flat:*

$$\prod_{j \in U} (-J_j) \equiv 1 \pmod{\pi^2}. \quad (\text{WR}_3)$$

*Thus an unweighted Hasse–Davenport relation necessarily loses the selected first-order phase; a proof must evaluate a genuinely weighted resolvent such as  $\mathcal{R}_\lambda$ .*

*Proof.* Since  $-J_j \equiv 1 - \pi C_j \pmod{\pi^2}$ , first-order expansion and Proposition 6.41 give

$$\mathcal{R}_\lambda \equiv 1 - \pi \sum_{j \in U} w_j C_j = 1 - \pi \sum_{j \in U} \lambda(j) \lambda(j)^{-1} C_1 = 1 - 2n\pi C_1.$$

Now  $\ell \nmid 2n$ , and  $C_1 = G_1 L(1)$  with  $G_1 \neq 0$ , proving (WR<sub>2</sub>). With every weight equal to one, the linear term instead contains  $C_1 \sum_{j \in U} \lambda(j)^{-1} = 0$ , which proves (WR<sub>3</sub>).  $\square$



**Corollary 6.44** (the first Jacobi phase is the selected current projection). *Retain Proposition 6.41. Put*

$$t = \frac{2^n - 1}{\ell}, \quad s = \frac{m + 1}{2}, \quad r \equiv st \pmod{m},$$

write  $I$  for the binary support of  $t$ , and set

$$\rho = g^{(Q-1)/\ell}, \quad \Theta = z^{-r} \prod_{i \in I} (1 + z^{2^i}) = z^{-r} (1 + z)^t. \quad (\text{CJ}_1)$$

Then

$$\Theta \in \mu_\ell, \quad \Theta^2 = \rho^{L(1)}. \quad (\text{CJ}_2)$$

Consequently, if  $\log_\rho : \mu_\ell \rightarrow \mathbb{F}_\ell$  is the discrete logarithm with base  $\rho$ , then

$$C_1 = 2G_1 \log_\rho(\Theta), \quad (\text{CJ}_3)$$

and

$$v_{\zeta_{\ell-1}}(J_1 + 1) = 1 \iff \Theta \neq 1. \quad (\text{CJ}_4)$$

Thus the first Jacobi derivative is literally the discrete logarithm of the selected current  $\ell$ -projection. Its Fourier and binary-support forms locate the missing scalar but do not replace its evaluation.

*Proof.* Since  $2^n \equiv -1 \pmod{m}$ ,

$$(1 + z)^{2^n - 1} = z^{-1}.$$

Also  $\ell t = 2^n - 1$  and  $2r \equiv t \pmod{m}$ , whence  $r\ell \equiv -1 \pmod{m}$ . Therefore

$$\Theta^\ell = z^{-r\ell} (1 + z)^{2^n - 1} = 1.$$

On the other hand,

$$\Theta^{2^n + 1} = z^{-r(2^n + 1)} (1 + z)^{t(2^n + 1)} = (1 + z)^{(Q-1)/\ell} = \rho^{L(1)}.$$

The omitted torsion factor is one because  $m \mid r(2^n + 1)$ . Since  $2^n + 1 \equiv 2 \pmod{\ell}$ , this proves (CJ<sub>2</sub>). Taking the  $\rho$ -logarithm gives  $L(1) = 2\log_\rho(\Theta)$ ; now Proposition 6.41 gives  $C_1 = G_1 L(1)$  with  $G_1 \neq 0$ , proving the remaining assertions.  $\square$

**Proposition 6.45** (real square resultant and the failed size strategy). *Retain Corollary 6.44, and put*

$$K = \mathbb{Q}(\zeta_m), \quad K^+ = \mathbb{Q}(\zeta_m + \zeta_m^{-1}), \quad s = \frac{m + 1}{2},$$

$$\beta = \zeta_m^{-s} (1 + \zeta_m), \quad x = \zeta_m^{-r} (1 + \zeta_m)^t = \beta^t.$$

Then  $\beta, x \in K^+$ , and

$$N_{K^+/\mathbb{Q}}(\beta) = N_{K^+/\mathbb{Q}}(x) = -1. \quad (\text{RN}_1)$$

Let  $F_m(Y)$  be the degree- $n$  minimal polynomial of  $\beta$ , and set

$$A_{k,\ell} = N_{K^+/\mathbb{Q}}(x - 1).$$

Up to the standard harmless sign convention for the first resultant,

$$A_{k,\ell} = \text{Res}(F_m(Y), Y^t - 1), \quad (\text{RN}_2)$$

while the full cyclotomic resultant is the exact square

$$\text{Res}(\Phi_m(X), (1+X)^t - X^r) = A_{k,\ell}^2. \quad (\text{RN}_3)$$

The prime two is inert in  $K^+$ , with residue degree  $n$ . If  $\Theta$  is the selected phase in  $(\text{CJ}_1)$ , then

$$\begin{aligned} \Theta = 1 &\iff x \equiv 1 \pmod{(2)} \\ &\iff 2^n \mid A_{k,\ell} \\ &\iff 2^{2n} \mid \text{Res}(\Phi_m(X), (1+X)^t - X^r). \end{aligned} \quad (\text{RN}_4)$$

Thus a bound  $0 < |A_{k,\ell}| < 2^n$  would prove the cubic arm. That proposed mechanism is false already in the second row. For

$$k = 2, \quad n = 9, \quad m = 27, \quad \ell = 73, \quad t = 7, \quad r = 17,$$

one has

$$F_{27}(Y) = Y^9 - 9Y^7 + 27Y^5 - 30Y^3 + 9Y + 1$$

and the exact resultant

$$|A_{2,73}| = 567780193 > 2^9, \quad (\text{RN}_5)$$

so the full resultant is  $567780193^2$ . Real descent and unit norm therefore sharpen failure to an inert-prime divisibility statement, but do not yield an archimedean contradiction; proving that  $A_{k,\ell}$  is odd is exactly the original selected residue test.

*Proof.* Since  $2s \equiv 1 \pmod{m}$ ,

$$\bar{\beta} = \zeta_m^s(1 + \zeta_m^{-1}) = \zeta_m^{s-1}(1 + \zeta_m) = \beta.$$

For representatives  $1 \leq a < m/2$ ,  $3 \nmid a$ , of the real embeddings,

$$\sigma_a(\beta) = (-1)^a 2 \cos(\pi a/m). \quad (\text{RN}_6)$$

These values are distinct, so  $\beta$  generates  $K^+$ . The number of odd representatives is  $(n + (-1)^k)/2$ , which is odd because  $3^k \equiv (-1)^k \pmod{4}$ . Equation  $(\text{RN}_6)$  proves the first equality in  $(\text{RN}_1)$ . The integer  $t$  is odd, proving the second.

Taking the product of  $\beta^t - 1$  over the real embeddings proves  $(\text{RN}_2)$ . Since  $x - 1 \in K^+$ , its norm from  $K$  is the square of its real norm. Multiplication by  $\zeta_m^r$  does not change the full rational norm, and this gives  $(\text{RN}_3)$ .

The order identities

$$\text{ord}_m(2) = 2n, \quad 2^n \equiv -1 \pmod{m}$$

show that two is inert in  $K$  and in  $K^+$ , of residue degrees  $2n$  and  $n$ , respectively. Thus for the unique prime  $\mathfrak{p} \mid 2$  in  $K^+$ ,

$$v_2(A_{k,\ell}) = n v_{\mathfrak{p}}(x - 1).$$

The reduction of  $x$  is  $\Theta$ , proving  $(\text{RN}_4)$ .

For the final exact obstruction, the displayed polynomial for  $F_{27}$  follows by separating the primitive- twenty-seven factor in the Dickson identity for

$$D_{27}(Y) - 2 = (Y - 2)(Y + 1)^2(Y^3 - 3Y + 1)^2 F_{27}(Y)^2.$$

Reducing  $F_{27}$  modulo  $Y^7 - 1$  gives

$$27Y^5 - 30Y^3 + Y^2 + 9Y - 8.$$

The determinant of multiplication by this element on  $\mathbb{Z}[Y]/(Y^7 - 1)$  is

$$\det \begin{pmatrix} -8 & 0 & 27 & 0 & -30 & 1 & 9 \\ 9 & -8 & 0 & 27 & 0 & -30 & 1 \\ 1 & 9 & -8 & 0 & 27 & 0 & -30 \\ -30 & 1 & 9 & -8 & 0 & 27 & 0 \\ 0 & -30 & 1 & 9 & -8 & 0 & 27 \\ 27 & 0 & -30 & 1 & 9 & -8 & 0 \\ 0 & 27 & 0 & -30 & 1 & 9 & -8 \end{pmatrix} = -567780193.$$

This proves (RN<sub>5</sub>). Squaring and using (RN<sub>3</sub>) gives the stated full resultant.  $\square$

**Proposition 6.46** (cyclic parity form of the cubic obstruction). *Put  $n = 3^k$ ,  $m = 3n$ , with  $k \geq 1$ , and let the prime  $\ell$  divide  $\Phi_n(2)$ . Define*

$$t = \frac{2^n - 1}{\ell}, \quad s = \frac{m + 1}{2}, \quad r \equiv st \pmod{m}, \quad 0 \leq r < m,$$

and let  $R_{k,\ell}(X)$  be the degree- $< m$  reduction of  $(1 + X)^t$  modulo  $X^m - 1$ . Then, independently of cubic failure,

$$\frac{X^n - 1}{X - 1} \mid R_{k,\ell}(X) + X^r. \quad (\text{CP}_1)$$

Moreover, for a primitive  $m$ -th root  $\zeta$  and  $\gamma_k = \zeta + \zeta^{-1}$ ,

$$\gamma_k \in (\mathbb{F}_{2^n}^\times)^\ell \iff R_{k,\ell}(X) + X^r = 1 + X + \cdots + X^{m-1}. \quad (\text{CP}_2)$$

Equivalently, write the binary expansion

$$t = \sum_{i \in I} 2^i.$$

Then cubic failure at  $\ell$  is exactly the group-algebra identity

$$\prod_{i \in I} (1 + X^{2^i}) = \sum_{\substack{0 \leq j < m \\ j \neq r}} X^j \quad \text{in } \mathbb{F}_2[X]/(X^m - 1). \quad (\text{CP}_3)$$

Thus, if

$$c_j = \# \left\{ J \subseteq I : \sum_{i \in J} 2^i \equiv j \pmod{m} \right\} \pmod{2},$$

failure says

$$c_r = 0, \quad c_j = 1 \quad (j \neq r). \quad (\text{CP}_4)$$

The remaining cubic theorem is therefore the universal impossibility of this complement-of-one parity pattern for the particular binary set  $I$  coming from  $(2^n - 1)/\ell$ .

*Proof.* Let  $\xi \neq 1$  be an  $n$ -th root of unity, of order  $e = 3^{a+1}$ , and put  $d = 3^a < n$ . The cyclotomic-divisor criterion gives  $\text{ord}_\ell(2) = n$ , so  $\gcd(\ell, 2^d - 1) = 1$ . Since  $d \mid n$ , it follows that

$$T := \frac{t}{2^d - 1} \in \mathbb{Z}.$$

Now  $2^d \equiv -1 \pmod{e}$ , and hence

$$(1 + \xi)^{2^d - 1} = \xi^{-1}.$$

Also  $t = T(2^d - 1) \equiv -2T \pmod{e}$ . Because  $2s \equiv 1 \pmod{m}$ , this gives  $-T \equiv st \equiv r \pmod{e}$ , and therefore

$$R_{k,\ell}(\xi) = (1 + \xi)^t = \xi^{-T} = \xi^r.$$

The polynomial  $(X^n - 1)/(X - 1)$  is squarefree, since  $n$  is odd, and its roots are exactly these nonidentity  $n$ -th roots. This proves (CP<sub>1</sub>).

At the primitive  $m$ -th root  $\zeta$ , one has

$$\gamma_k = \zeta^{-1}(1 + \zeta)^2.$$

The element  $\gamma_k$  is an  $\ell$ -th power exactly when  $\gamma_k^t = 1$ . Since squaring is bijective in characteristic two, this is equivalent to

$$(1 + \zeta)^t = \zeta^{st} = \zeta^r,$$

and hence to

$$\Phi_m(X) \mid R_{k,\ell}(X) + X^r. \quad (\text{CP}_5)$$

The two factors in (CP<sub>1</sub>) and (CP<sub>5</sub>) are coprime and satisfy

$$\frac{X^n - 1}{X - 1} \Phi_m(X) = \frac{X^m - 1}{X - 1} = 1 + X + \cdots + X^{m-1}.$$

The polynomial  $R_{k,\ell} + X^r$  has degree  $< m$ . Under failure it is therefore a scalar multiple of the displayed degree- $m - 1$  polynomial. Evaluation at  $X = 1$ , which is preserved by reduction modulo  $X^m - 1$ , gives

$$(R_{k,\ell} + X^r)(1) = (1 + 1)^t + 1 = 1;$$

the all-ones polynomial also has value  $m = 1$  in characteristic two. The scalar is consequently one, proving (CP<sub>2</sub>); the converse is immediate.

Finally, in the group algebra,

$$(1 + X)^t = \prod_{i \in I} (1 + X^{2^i}).$$

The coefficient of  $X^j$  after cyclic reduction is the parity of the subsets with exponent sum congruent to  $j$ . Rewriting (CP<sub>2</sub>) as  $R_{k,\ell} = \sum_{j \neq r} X^j$  proves (CP<sub>3</sub>)–(CP<sub>4</sub>).  $\square$

**Corollary 6.47** (exceptional residue and primitive CRT isolation). *Retain Proposition 6.46, write  $\ell = 1 + an$ , and set*

$$S_{k,\ell}(X) = R_{k,\ell}(X) + X^r.$$

*Then  $a$  is even and the exceptional residue is one of exactly three lifts of  $-1 \pmod{n}$ :*

$$r \equiv an - 1 \pmod{m}, \quad r = \begin{cases} m - 1, & a \equiv 0 \pmod{3}, \\ n - 1, & a \equiv 1 \pmod{3}, \\ 2n - 1, & a \equiv 2 \pmod{3}. \end{cases} \quad (\text{CP}_6)$$

Moreover, in the squarefree group algebra

$$\mathcal{A}_m = \mathbb{F}_2[X]/(X^m - 1) \cong \mathbb{F}_2 \times \prod_{j=1}^{k+1} \mathbb{F}_{2^{2 \cdot 3^j - 1}},$$

the CRT coordinates of  $S_{k,\ell}$  have the form

$$S_{k,\ell} = (1, 0, \dots, 0, \sigma_{\text{top}}). \quad (\text{CP}_7)$$

Cubic failure is exactly  $\sigma_{\text{top}} = 0$ . Thus every invariant which factors through a proper cyclic projection of  $C_m$ , or otherwise discards the top  $\Phi_m$ -component, is identical on success and failure. At a primitive  $m$ -th root  $\zeta$ , the surviving coordinate is precisely the original selected Kummer test, since

$$\zeta^{-r} S_{k,\ell}(\zeta) = (1 + \zeta)^t \zeta^{-r} + 1. \quad (\text{CP}_8)$$

Finally let  $w$  be the binary weight of  $t$ , and let  $C_j$  be the integer number of binary submasks of  $t$  congruent to  $j \pmod{m}$ . Failure necessarily implies

$$2^w \geq m - 1, \quad C_r \equiv 1 - m \pmod{4}. \quad (\text{CP}_9)$$

In particular,  $w < \lceil \log_2(m - 1) \rceil$  is an unconditional nonfailure criterion. These are exact screens, not an evaluation of the remaining top component.

*Proof.* The equality  $\text{ord}_m(2) = 2n$  gives  $2^n \equiv -1 \pmod{m}$ . Since  $\ell$  and  $n$  are odd,  $a$  is even. Also

$$(1 + an)^{-1} \equiv 1 - an \pmod{3n},$$

because  $3n \mid a^2 n^2$ . Hence

$$t \equiv -2\ell^{-1} \equiv 2an - 2 \pmod{m}.$$

As  $2r \equiv t \pmod{m}$ , this proves  $(\text{CP}_6)$ .

The standard factorization

$$X^m - 1 = (X - 1) \prod_{j=1}^{k+1} \Phi_{3^j}(X)$$

is squarefree, and every displayed cyclotomic factor is irreducible over  $\mathbb{F}_2$ , because  $\text{ord}_{3^j}(2) = 2 \cdot 3^{j-1}$ . Equation  $(\text{CP}_1)$  kills the coordinates for  $1 \leq j \leq k$ , while  $S_{k,\ell}(1) = 1$ . This proves  $(\text{CP}_7)$ ; equation  $(\text{CP}_5)$  says exactly that its last coordinate vanishes. Equation  $(\text{CP}_8)$  is immediate from the definition of  $S_{k,\ell}$ .

Under failure, the  $2^w$  submasks must occupy all  $m - 1$  nonexceptional residues, proving the first inequality in  $(\text{CP}_9)$ . Complementation gives  $C_j = C_{t-j}$ . The only fixed residue of this involution is  $r$ ; every other residue occurs in one of  $(m - 1)/2$  pairs, and failure makes the common count in each pair odd. Since  $2^w \equiv 0 \pmod{4}$  under the first inequality, summing all counts modulo four gives

$$0 \equiv C_r + m - 1 \pmod{4},$$

which completes the proof.  $\square$

**Corollary 6.48** (augmentation-ideal normalization of cubic failure). *Retain Proposition 6.46, and put*

$$A_m = \mathbb{F}_2[X]/(X^m - 1), \quad J_m = \sum_{j=0}^{m-1} X^j, \quad e_m = 1 + J_m.$$

The element  $e_m$  is the identity of the augmentation ideal  $e_m A_m$ . For  $0 \leq i < n$ , define

$$h_i = X^{-s2^i}(1 + X^{2^i}) = X^{s2^i} + X^{-s2^i} \in e_m A_m.$$

If  $I$  is the binary support of  $t = (2^n - 1)/\ell$ , then cubic failure at  $\ell$  is exactly

$$\prod_{i \in I} h_i = e_m. \quad (\text{CA}_1)$$

On the other hand, the full-product relation

$$\prod_{i=0}^{n-1} h_i = e_m \quad (\text{CA}_2)$$

holds unconditionally. Thus failure makes both the proper digit subset  $I$  and its complement multiply to the identity in  $e_m A_m$ .

Likewise, raising  $(\text{CP}_3)$  to the  $\ell$ -th power yields only the unconditional identity

$$(1 + X)^{2^n - 1} = J_m + X^{-1} \quad \text{in } A_m, \quad (\text{CA}_3)$$

because  $r\ell \equiv -1 \pmod{m}$ . Hence neither the full product nor the strongest immediate exponent check can exclude failure. The remaining cubic theorem is precisely the absence of the extra proper-subproduct relation  $(\text{CA}_1)$  for this selected binary support.

*Proof.* For  $y \in A_m$ , one has  $J_m y = \text{aug}(y) J_m$ . Since  $m$  is odd,  $e_m$  has augmentation zero and acts as the identity on the augmentation ideal.

Write  $P_I = \prod_{i \in I} (1 + X^{2^i})$ . Under  $(\text{CP}_3)$ ,  $P_I = J_m + X^r$ , and therefore

$$\prod_{i \in I} h_i = X^{-st} P_I = X^{-r} (J_m + X^r) = e_m.$$

Every step is reversible, proving  $(\text{CA}_1)$ .

For the full product, let  $\xi$  be an  $m$ -th root of unity in an algebraic closure. If  $\xi \neq 1$ , then

$$\prod_{i=0}^{n-1} (1 + \xi^{2^i}) = (1 + \xi)^{2^n - 1} = \frac{1 + \xi^{2^n}}{1 + \xi} = \xi^{-1},$$

because  $2^n \equiv -1 \pmod{m}$ . Also  $s(2^n - 1) \equiv -1 \pmod{m}$ , so the monomial factor in the product of the  $h_i$  contributes  $\xi$ . At  $\xi = 1$  the product vanishes. These are exactly the values of  $e_m$ ; since  $m$  is odd,  $X^m - 1$  is squarefree, proving  $(\text{CA}_2)$ . It follows that every  $h_i$  is a unit of  $e_m A_m$ , so  $(\text{CA}_1)$  also holds for the complementary subset.

Finally  $J_m + X^r = X^r e_m$ , and  $e_m$  is idempotent. Raising  $(1 + X)^t = X^r e_m$  to the  $\ell$ -th power gives

$$(1 + X)^{2^n - 1} = X^{r\ell} e_m = X^{-1} e_m = J_m + X^{-1},$$

which proves  $(\text{CA}_3)$ . □

**Proposition 6.49** (cubic block compression and forced midpoint). *Retain Proposition 6.46. Put*

$$d = \frac{n}{3}, \quad q = 2^d, \quad N = q^2 + q + 1, \quad u = \frac{N}{\ell},$$

so that  $t = (q - 1)u$ , and set

$$B(X) = 1 + X + \cdots + X^{n-2} \in A_m, \quad b = X^{-s(q-1)}B.$$

Then

$$(1 + X)^{q-1} = B(X) \quad \text{in } A_m, \tag{CB_1}$$

and cubic failure at  $\ell$  is exactly

$$B^u = X^{st}e_m, \quad \text{equivalently} \quad b^u = e_m. \tag{CB_2}$$

The normalized element satisfies, unconditionally,

$$b(X^{-1}) = b(X), \quad b^N = e_m. \tag{CB_3}$$

At a primitive  $m$ -th root  $\zeta$ ,

$$b(\zeta)^2 = \eta_k. \tag{CB_4}$$

Thus (CB<sub>2</sub>) is the original selected current-factor order question in compressed form, rather than a weaker consequence.

Moreover, if  $C_j$  is the number of binary submasks of  $t$  whose sum is  $j \pmod{m}$ , then

$$C_j = C_{t-j}. \tag{CB_5}$$

The unique fixed residue of  $j \mapsto t - j$  is  $r = st$ , and  $C_r$  is even. Hence complementation forces the exceptional zero required by (CP<sub>4</sub>) into exactly the right position; it cannot contradict cubic failure.

*Proof.* The lifted congruence in Proposition 6.24 gives

$$q \equiv n - 1 \pmod{3n}, \quad \frac{q - (n - 1)}{3n} \equiv 0 \pmod{2}.$$

Since  $q$  is a power of two,  $(1 + X)^{q-1} = \sum_{a=0}^{q-1} X^a$ . Cyclic reduction modulo  $X^m - 1$  cancels the even number of complete residue blocks and leaves exactly  $B$ , proving (CB<sub>1</sub>). Substitution in (CP<sub>2</sub>), followed by multiplication by  $X^{-st}$ , proves (CB<sub>2</sub>).

Because  $2s \equiv 1 \pmod{m}$  and  $q - 1 \equiv n - 2 \pmod{m}$ , reversing  $B$  proves  $b(X^{-1}) = b(X)$ . Also  $(q - 1)N = 2^n - 1$ . At every nonidentity  $m$ -th root  $\xi$ ,

$$(1 + \xi)^{2^n - 1} = \xi^{-1}, \quad s(2^n - 1) \equiv -1 \pmod{m},$$

so  $b(\xi)^N = 1$ ; at  $\xi = 1$ ,  $B(1) = n - 1 = 0$ . Squarefree interpolation proves  $b^N = e_m$ . Finally  $(1 + \zeta)^2 = \zeta \gamma_k$ , whence

$$b(\zeta)^2 = \zeta^{-(q-1)}(1 + \zeta)^{2(q-1)} = \gamma_k^{q-1} = \eta_k.$$

The binary integer  $t$  is odd. Complementing a submask is therefore fixed-point-free and changes its sum from  $j$  to  $t - j \pmod{m}$ , which proves (CB<sub>5</sub>). Since  $m$  is odd, the involution on residues has the unique fixed point  $r = st$ ; all submasks above that point occur in complementary pairs.  $\square$

**Proposition 6.50** (distinguished Wendt-factor obstruction). *Retain Proposition 6.49, and let  $P_{\eta,k}$  be the degree- $n$  minimal polynomial of  $\eta_k$  over  $\mathbb{F}_2$ . Define*

$$G_u(X) = \gcd(X^u + 1, (X + 1)^u + 1) \in \mathbb{F}_2[X].$$

*Then cubic failure at  $\ell$  is equivalent to*

$$P_{\eta,k} \mid G_u. \quad (\text{CW}_1)$$

*Under failure, the two distinct factors*

$$P_{\eta,k}(X), \quad P_{\eta,k}(X + 1) \quad (\text{CW}_2)$$

*both divide  $G_u$ , and the second is the reciprocal partner of the first. Every irreducible factor of  $G_u$  has degree exactly  $n$ ; translation pairs those factors without fixed points. Consequently*

$$G_u = 1 \quad \text{or} \quad 2n \mid \deg G_u. \quad (\text{CW}_3)$$

*The usual six-element group generated by  $X \mapsto X + 1$  and  $X \mapsto X^{-1}$  acts freely on the roots of  $G_u$ . Thus the surviving assertion is specifically that the distinguished pair (CW<sub>2</sub>), not merely some Wendt pair, never occurs.*

*Proof.* Failure says  $\eta_k^u = 1$ . Since  $\eta_k + 1 = \eta_k^{q+1}$ , it also gives  $(\eta_k + 1)^u = 1$ , and Frobenius gives the same two identities for every conjugate. This proves (CW<sub>1</sub>) and (CW<sub>2</sub>); the converse uses the first factor.

The relative trace recursion in Proposition 6.24 gives

$$\text{Tr}_{\mathbb{F}_{2^n}/\mathbb{F}_2}(\eta_k) = 0.$$

As  $n$  is odd, the trace of  $\eta_k + 1$  is one, so the two factors in (CW<sub>2</sub>) are distinct. Also  $\eta_k^{-1} = \eta_k^q + 1$ ; hence the translate is the minimal polynomial of  $\eta_k^{-1}$ , namely the reciprocal partner.

Let  $x$  be a root of  $G_u$ . The nontrivial orders of both  $x$  and  $x + 1$  divide  $u \mid \Phi_n(2)$ . Every prime  $p \mid u$  satisfies  $\text{ord}_p(2) = n$ : the only prime dividing  $n$  is three, and  $3 \nmid \Phi_n(2)$ . Thus  $x$  has degree at least  $n$ , while its order divides  $2^n - 1$ , so its degree is exactly  $n$ . Translation changes the trace by one and therefore pairs the degree- $n$  irreducible factors without fixed points, proving (CW<sub>3</sub>).

Finally none of a root  $x$ , its inverse, or its translate is fixed by an involution: the only possible values would be zero or one. A fixed point of the order-three map satisfies  $x^2 + x + 1 = 0$ , hence has order three, but  $3 \nmid u$ . The six-element action is therefore free.  $\square$

**Proposition 6.51** (Singer factorization of the Wendt gcd). *Retain Proposition 6.50. In  $L = \mathbb{F}_{q^3}$ , put*

$$U = \ker(N_{L/\mathbb{F}_q}) = \mu_N, \quad H = U^\ell = \mu_u,$$

*and let*

$$D = \{x \in U : x^{q+1} + x + 1 = 0\}.$$

*Thus  $D$  is the planar Singer  $(N, q + 1, 1)$  difference set from Proposition 6.24. Then*

$$\text{Roots}(G_u) = (D \cap H) \dot{\cup} ((D \cap H) + 1). \quad (\text{SW}_1)$$

*If  $\mathcal{O}$  runs over the absolute-Frobenius orbits in  $D \cap H$ , and  $P_{\mathcal{O}}$  is the corresponding minimal polynomial, then*

$$G_u(X) = \prod_{\mathcal{O} \subset D \cap H} P_{\mathcal{O}}(X)P_{\mathcal{O}}(X + 1), \quad \deg G_u = 2|D \cap H|. \quad (\text{SW}_2)$$



Every orbit  $\mathcal{O}$  has size  $n$ , and  $P_{\mathcal{O}}(X+1)$  is its reciprocal partner.

More precisely, choose a generator  $g$  of  $U$ , and put

$$a_j = |D \cap g^j H| \quad (j \in \mathbb{Z}/\ell\mathbb{Z}).$$

The quotient intersection numbers obey the exact moment identities

$$\begin{aligned} \sum_j a_j &= q+1, & \sum_j a_j^2 &= q+u, \\ \sum_j a_j a_{j+r} &= u \quad (r \neq 0), & a_{2j} &= a_j. \end{aligned} \tag{SW_3}$$

In particular, if

$$\ell - 1 < \sqrt{q} + \frac{1}{\sqrt{q}}, \tag{SW_4}$$

then every  $a_j$  is positive. In that range  $a_0 \geq n$  and  $\deg G_u \geq 2n$ : Wendt factors of the correct degree and symmetry are forced to exist even though the distinguished factor may still be absent.

*Proof.* A root  $x$  of  $G_u$  satisfies  $x, x+1 \in H \subset U$ . Put  $y = x^{q+1}$ . Since

$$x^q = \frac{y}{x}, \quad x^{q^2} = \frac{1}{y},$$

the condition  $N(x+1) = 1$ , after clearing  $xy$ , factors as

$$(y+x+1)((x+1)y+x) = 0. \tag{SW_5}$$

The first factor says  $x \in D$ . In the second case  $x+1 \in D$ . Conversely, if  $x \in D \cap H$ , then  $x+1 = x^{q+1} \in H$ , so both  $x$  and  $x+1$  are roots of  $G_u$ . The two sets in (SW<sub>1</sub>) are disjoint: if both  $x$  and  $x+1$  lay in  $D$ , their two Frobenius equations would give  $x^2 + x + 1 = 0$ , impossible because  $3 \nmid N$ .

Every nonidentity element of  $H$  has absolute degree  $n$ , by the cyclotomic-divisor argument in Proposition 6.50. The polynomial  $G_u$  is squarefree, and (SW<sub>1</sub>) therefore gives (SW<sub>2</sub>). For  $x \in D$ ,

$$(x+1)^q = x^{-1},$$

so translation of its absolute orbit is the inverse orbit, proving the reciprocal assertion.

The first identity in (SW<sub>3</sub>) is  $|D| = q+1$ . For the second,  $\sum a_j^2$  counts ordered pairs of points of  $D$  whose quotient lies in  $H$ . The identity quotient contributes  $q+1$ , and each of the other  $u-1$  elements occurs once because  $D$  has difference parameter one. If  $r \neq 0$ , the corresponding coset has  $u$  nonidentity quotients, proving the third identity. Squaring stabilizes  $D$  and sends the  $j$ -th coset to the  $2j$ -th, which proves the last.

If some  $a_j$  vanished, Cauchy's inequality applied to the remaining  $\ell-1$  entries would give

$$(q+1)^2 \leq (\ell-1)(q+u). \tag{SW_6}$$

But a direct simplification using  $u = N/\ell$  gives

$$\ell((q+1)^2 - (\ell-1)(q+u)) = q \left[ \left( \sqrt{q} + \frac{1}{\sqrt{q}} \right)^2 - (\ell-1)^2 \right],$$

which is positive under (SW<sub>4</sub>), a contradiction. Finally  $D \cap H$  is a union of size- $n$  orbits, so positivity of  $a_0$  implies  $a_0 \geq n$ .  $\square$

**Proposition 6.52** (the selected  $S_3$ -quotient of the Wendt pair). *For  $x \in D$ , put*

$$J(x) = \frac{(x^2 + x + 1)^3}{x^2(x + 1)^2}, \quad a = \text{Tr}_{\mathbb{F}_{q^3}/\mathbb{F}_q}(x).$$

*Then*

$$J(x) = a^2 + a + 1. \tag{SJ_1}$$

*Let  $d = 3^{k-1}$ , let  $W_{k-1}$  be the degree- $d$  minimal polynomial of*

$$z_{k-1} = \eta_{k-1}^2 + \eta_{k-1} + 1,$$

*and retain the selected polynomial  $P_{\eta,k}$ . Then*

$$P_{\eta,k}(X)P_{\eta,k}(X + 1) = [X(X + 1)]^{2d} \cdot W_{k-1}\left(\frac{(X^2 + X + 1)^3}{X^2(X + 1)^2}\right). \tag{SJ_2}$$

*Thus quotienting by the free  $S_3$ -action lowers the selected degree- $2n$  pair to one degree- $d$  trace factor. Cubic failure is still the assertion that this particular factor, rather than an unselected Singer intersection with the same orbit structure, occurs in the Wendt quotient.*

*Proof.* The cubic for  $x \in D$  is

$$T^3 + aT^2 + (a + 1)T + 1.$$

Substitution of  $x$  gives

$$a = \frac{x^3 + x + 1}{x(x + 1)}.$$

Putting this expression over a common denominator proves

$$a^2 + a + 1 = \frac{(x^2 + x + 1)^3}{x^2(x + 1)^2},$$

which is (SJ<sub>1</sub>).

The element  $\eta_{k-1}$  has odd degree  $d$ , while it is at most quadratic over  $z_{k-1}$ . Hence the two fields are equal and  $W_{k-1}$  has degree  $d$ . The invariant  $J$  is constant on the six-element orbit generated by  $X \mapsto X + 1$  and  $X \mapsto X^{-1}$ . On  $D$ , its order-three subgroup is relative Frobenius, while translation gives the reciprocal absolute orbit. Consequently the roots of the two sides of (SJ<sub>2</sub>) are the same. Both sides are monic of degree  $6d = 2n$ , which proves the identity.  $\square$

**Corollary 6.53** (reciprocal Conway selector and the exact order seam). *For  $j \geq 1$ , put*

$$z_j = \eta_j^2 + \eta_j + 1, \quad z_0 = 1.$$

*Then*

$$z_j = \gamma_j^{-2}. \tag{RS_1}$$

*If  $q = 2^{3^{j-1}}$ , then  $z_j$  has the irreducible cubic*

$$Z^3 + z_{j-1}Z^2 + z_{j-1} = 0 \tag{RS_2}$$

over  $\mathbb{F}_q$ . In particular

$$\mathrm{Tr}_{\mathbb{F}_{q^3}/\mathbb{F}_q}(z_j) = \mathrm{N}_{\mathbb{F}_{q^3}/\mathbb{F}_q}(z_j) = z_{j-1}, \quad z_{j-1} = \frac{z_j^3}{z_j^2 + 1}. \quad (\mathrm{RS}_3)$$

If  $P_j$  is the minimal polynomial of  $\gamma_j$ , the minimal polynomial of  $z_j$  is its reciprocal

$$W_j(X) = X^{3^j} P_j(X^{-1}). \quad (\mathrm{RS}_4)$$

Most importantly,

$$z_j^{q-1} = \eta_j^{-2}, \quad \mathrm{ord}(z_j) = \mathrm{ord}(z_{j-1}) \mathrm{ord}(\eta_j). \quad (\mathrm{RS}_5)$$

Thus  $W_j$  is primitive exactly when  $P_j$  is. For every current prime  $\ell \mid q^2 + q + 1$ ,

$$z_j \in (\mathbb{F}_{q^3}^\times)^\ell \iff \eta_j \in U_j^\ell. \quad (\mathrm{RS}_6)$$

The selected  $S_3$ -quotient is therefore a lossless reciprocal formulation of  $C_j$ , not an independent induction from  $C_{j-1}$ .

*Proof.* Put  $t = \zeta_j^2$  and  $M(t) = (\omega + \omega^2 t)/(1 + t) = \eta_j$ . Direct calculation in  $\mathbb{F}_4(t)$  gives

$$M(t)^2 + M(t) + 1 = \frac{t}{(1+t)^2}, \quad \gamma_j^2 = t + t^{-1} = \frac{(1+t)^2}{t},$$

which proves  $(\mathrm{RS}_1)$ .

Write  $b = \eta_{j-1}$ ,  $c = b^2 + b + 1 = z_{j-1}$ , and let  $a_i$  be the three relative conjugates of  $\eta_j$ . By Proposition 6.24,

$$e_1(a_i) = b, \quad e_2(a_i) = b + 1, \quad e_3(a_i) = 1.$$

For  $w_i = a_i^2 + a_i + 1$ , these identities give

$$e_1(w_i) = c, \quad e_2(w_i) = 0.$$

Moreover, over  $\mathbb{F}_4$ ,

$$\prod_i w_i = \prod_i (a_i + \omega) \prod_i (a_i + \omega^2) = (b + \omega)(b + \omega^2) = c.$$

This proves  $(\mathrm{RS}_2)$ . Since  $\eta_j$  is at most quadratic over  $z_j$ , while its absolute degree  $3^j$  is odd, the two elements generate the same field. Thus the cubic is irreducible, and its coefficients give  $(\mathrm{RS}_3)$ .

Squaring permutes the absolute conjugates of  $\gamma_j$ , while inversion sends them to the conjugates of  $z_j = \gamma_j^{-2}$ . The constant term of  $P_j$  is one, proving  $(\mathrm{RS}_4)$ . The first equality in  $(\mathrm{RS}_5)$  follows from  $\eta_j = \gamma_j^{q-1}$ .

Now

$$\mathbb{F}_{q^3}^\times \simeq \mathbb{F}_q^\times \times U_j, \quad \gcd(q-1, q^2 + q + 1) = 1.$$

Relative norm detects the  $\mathbb{F}_q^\times$ -component of  $z_j$  without changing its order, because it cubes that component and  $3 \nmid q-1$ . Raising to  $q-1$  detects the  $U_j$ -component without changing its order. Equations  $(\mathrm{RS}_3)$  and  $(\mathrm{RS}_5)$ , together with the oddness of  $|U_j|$ , therefore give the order product. Finally  $\ell$ -powering is an automorphism on  $\mathbb{F}_q^\times$ , while both  $q-1$  and 2 are invertible modulo  $\ell$ ; this proves  $(\mathrm{RS}_6)$ .  $\square$

**Proposition 6.54** (fixed-norm iterated cubic fibre). *Put*

$$L_j = \mathbb{F}_{2^{3j}}, \quad z_0 = 1, \quad z_j = \eta_j^2 + \eta_j + 1 = \gamma_j^{-2}.$$

*Fix  $k \geq 1$  and a prime  $\ell \mid \Phi_{3^k}(2)$ . For  $0 \leq j < k$ , let  $r_j \in L_j^\times$  be the unique element such that*

$$r_j^\ell = z_j. \quad (\text{IF}_0)$$

*Then  $r_0 = 1$ , and the roots are norm coherent:*

$$N_{L_j/L_{j-1}}(r_j) = r_{j-1} \quad (1 \leq j < k). \quad (\text{IF}_1)$$

*For indeterminates  $C, D, E$ , define generalized power sums by*

$$\begin{aligned} T_0(C, D, E) &= 1, & T_1(C, D, E) &= C, & T_2(C, D, E) &= C^2, \\ T_n(C, D, E) &= CT_{n-1} + DT_{n-2} + ET_{n-3} & (n \geq 3), \end{aligned} \quad (\text{IF}_2)$$

*and put*

$$(A, B) = \Psi_{\ell, E}(C, D) = (T_\ell(C, D, E), T_\ell(D, EC, E^2)).$$

*If  $J = \det(\partial(A, B)/\partial(C, D))$ , then in  $\mathbb{F}_2[C, D, E]$ ,*

$$\boxed{(CD + E)J = AB + E^\ell.} \quad (\text{IF}_3)$$

*The same identity holds with  $\ell$  replaced throughout by every odd integer  $n$ , defining  $\Psi_{n, E}$  from the same recurrences.*

*Now specialize  $E$  to a nonzero field element. For  $1 \leq j \leq k$ , take  $K = L_{j-1}$ ,  $E = r_{j-1}$ , and the fixed- $E$  selected fibre*

$$T_\ell(C, D, E) = z_{j-1}, \quad T_\ell(D, EC, E^2) = 0. \quad (\text{IF}_4)$$

*It has exactly  $\ell^2$  reduced geometric points and is étale. Every  $K$ -rational point gives an irreducible cubic  $X^3 + CX^2 + DX + E$ . For  $j < k$ , the  $K$ -rational fibre is a singleton, the minimal polynomial of  $r_j$ . At  $j = k$ , its cardinality is*

$$\#\Psi_{\ell, r_{k-1}}^{-1}(z_{k-1}, 0)(K) = \begin{cases} 0, & z_k \notin (L_k^\times)^\ell, \\ \ell, & z_k \in (L_k^\times)^\ell. \end{cases} \quad (\text{IF}_5)$$

*In the failure case the  $\ell$  points correspond to the  $\ell$  roots  $b^\ell = z_k$ , and all of them have  $N_{L_k/K}(b) = r_{k-1}$ . Thus every bad top lift collapses to the same complete selected lower norm ancestry.*

*Proof.* Put  $q = 2^{3^{k-1}}$ . Since  $\Phi_{3^k}(2) \equiv 1 \pmod{3}$ , one has  $\ell \neq 3$ . Also  $q^2 + q + 1 \equiv 0 \pmod{\ell}$ , while  $q \not\equiv 1 \pmod{\ell}$ ; hence  $\text{ord}_\ell(q) = 3$  and  $\text{ord}_\ell(2) = 3^k$ . Therefore  $\ell \nmid |L_j^\times|$  for  $j < k$ , which proves existence and uniqueness in (IF<sub>0</sub>). Taking norms and using Corollary 6.53 gives

$$N(r_j)^\ell = z_{j-1} = r_{j-1}^\ell.$$

Powering by  $\ell$  is injective on  $L_{j-1}^\times$ , proving (IF<sub>1</sub>).

For (IF<sub>3</sub>), work first over  $\mathbb{F}_2(E)$ , take independent  $x, y$ , and put  $z = E/(xy)$ . Then

$$C = x + y + z, \quad D = xy + yz + zx,$$

while  $A = x^\ell + y^\ell + z^\ell$ . The three pair-products  $xy, yz, zx$  have elementary coefficients  $D, EC, E^2$ , so the second recurrence gives  $B = (xy)^\ell + (yz)^\ell + (zx)^\ell$ . Logarithmic differentiation gives

$$\det \frac{\partial(C, D)}{\partial(\log x, \log y)} = CD + E, \quad \det \frac{\partial(A, B)}{\partial(\log x^\ell, \log y^\ell)} = AB + E^\ell.$$

The power map has logarithmic determinant  $\ell^2 = 1$  in characteristic two. The chain rule proves  $(\text{IF}_3)$  over  $\mathbb{F}_2(E)$ , hence as a polynomial identity.

Label the three distinct roots  $y_1, y_2, y_3$  of the target cubic  $X^3 + z_{j-1}X^2 + z_{j-1}$ . Choose  $\ell$ -th roots of  $y_1, y_2$  independently; among the roots of  $y_3$ , exactly one makes the product equal to  $E$ . This gives  $\ell^2$  source triples. The target labels are recovered after powering, so distinct choices do not coalesce on passing to coefficients. On the selected fibre,  $(\text{IF}_3)$  reads

$$(CD + E)J = E^\ell = z_{j-1} \neq 0,$$

which proves separability, reducedness, and étaleness.

At a rational point, the powered source roots are the roots of the irreducible selected target cubic. A reducible source cubic over the finite field  $K$  would have a  $K$ -root whose  $\ell$ -th power was a  $K$ -root of that target, a contradiction. Hence its roots lie in the unique cubic extension  $L_j$ , and the coefficient points are in bijection with roots  $b^\ell = z_j$  having norm  $E$ . For  $j < k$ , powering is bijective on  $L_j^\times$ , giving the singleton  $r_j$ . At the top there are no roots on success and exactly  $\ell$  on failure. Their norms all equal  $r_{k-1}$  by the same injectivity argument in  $K^\times$ , proving  $(\text{IF}_5)$ .  $\square$

**Proposition 6.55** (the full selected trace-coefficient flag). *Retain Proposition 6.24, fix  $k \geq 1$ , and let  $\ell \mid \Phi_{3^k}(2)$  be prime. For the generalized cubic power map of Proposition 6.54, define, for  $1 \leq j \leq k$ ,*

$$\mathcal{Y}_j = \Psi_{\ell,1}^{-1}(\eta_{j-1}, \eta_{j-1} + 1) \subset \mathbb{A}_{L_{j-1}}^2. \quad (\text{TA}_1)$$

Thus  $(C, D) \in \mathcal{Y}_j$  means that the  $\ell$ -th powers of the roots of

$$X^3 + CX^2 + DX + 1 \quad (\text{TA}_2)$$

have the selected reciprocal cubic

$$X^3 + \eta_{j-1}X^2 + (\eta_{j-1} + 1)X + 1. \quad (\text{TA}_3)$$

Each  $\mathcal{Y}_j$  is finite reduced étale of geometric degree  $\ell^2$ , and every  $L_{j-1}$ -rational point gives an irreducible reciprocal cubic. If  $J$  is the Jacobian determinant of  $\Psi_{\ell,1}$ , then at every geometric point,

$$\boxed{(CD + 1)J = \eta_{j-1}(\eta_{j-1} + 1) + 1 = z_{j-1} \neq 0.} \quad (\text{TA}_4)$$

In particular the source cubic has nonzero discriminant

$$\text{disc}(X^3 + CX^2 + DX + 1) = (CD + 1)^2. \quad (\text{TA}_5)$$

For  $j < k$ , let  $b_j \in L_j^\times$  be the unique element satisfying  $b_j^\ell = \eta_j$ . Then

$$\mathbb{N}_{L_j/L_{j-1}}(b_j) = 1, \quad \#\mathcal{Y}_j(L_{j-1}) = 1 \quad (1 \leq j < k), \quad (\text{TA}_6)$$

and the unique point is the coefficient pair of the minimal polynomial of  $b_j$ . At the top,

$$\#\mathcal{Y}_k(L_{k-1}) = \begin{cases} 0, & \eta_k \notin (L_k^\times)^\ell, \\ \ell, & \eta_k \in (L_k^\times)^\ell. \end{cases} \quad (\text{TA}_7)$$

Consequently the complete edgewise flag  $\mathcal{Y}_{\leq k} = \prod_{j=1}^k \mathcal{Y}_j(L_{j-1})$  has

$$\boxed{\# \mathcal{Y}_{\leq k} = \begin{cases} 0, & \eta_k^{(2^{3^k}-1)/\ell} \neq 1, \\ \ell, & \eta_k^{(2^{3^k}-1)/\ell} = 1. \end{cases}} \quad (\text{TA}_8)$$

Thus the complete recursively selected trace ancestry, together with all symmetric source coefficients, Jacobians, cubic discriminants, and their ordinary lower-field norms, is compatible with failure. Its sole rationality obstruction is the original top Euler class. This does not exclude a nonsymmetric cross-edge construction using a chosen top root or a mixed additive-multiplicative character correlation not determined by the edge coefficient fibres.

*Proof.* Let  $y_1, y_2, y_3$  be the roots of  $(\text{TA}_3)$ . Choose  $\ell$ -th roots  $x_1, x_2$  independently and put  $x_3 = (x_1 x_2)^{-1}$ . Since  $y_1 y_2 y_3 = 1$ , one has  $x_3^\ell = y_3$ . The target roots recover the labels after powering, so this gives exactly  $\ell^2$  geometric coefficient points.

The universal identity

$$(CD + E)J = AB + E^\ell$$

specializes at  $E = 1$ ,  $A = \eta_{j-1}$ , and  $B = \eta_{j-1} + 1$  to  $(\text{TA}_4)$ . Here  $z_{j-1} = \eta_{j-1}^2 + \eta_{j-1} + 1 \neq 0$ . Hence both  $CD + 1$  and  $J$  are nonzero, proving reducedness, étaleness, and  $(\text{TA}_5)$ .

A reducible source cubic over  $L_{j-1}$  would have a base-field root whose  $\ell$ -th power was a root of the irreducible target cubic, an impossibility. For  $j < k$ , one has  $\ell \nmid |L_j^\times|$ , so  $\ell$ -powering is bijective on  $L_j^\times$ . The target roots therefore have one Frobenius orbit of unique preimages, giving one coefficient pair. Moreover

$$N(b_j)^\ell = N(\eta_j) = 1,$$

and injectivity of  $\ell$ -powering downstairs gives  $N(b_j) = 1$ .

At  $j = k$ , a rational point supplies an  $\ell$ -th root in  $L_k$  of a conjugate of  $\eta_k$ , hence one of  $\eta_k$  after Frobenius conjugation. Conversely, under failure the  $\ell$  roots of  $\eta_k$  give  $\ell$  distinct coefficient pairs, as in Proposition 6.26. This proves  $(\text{TA}_7)$ ; all lower factors are singletons, so  $(\text{TA}_8)$  follows. Lean kernel-checks the selected Jacobian numerator, the characteristic-two reciprocal-cubic discriminant, and the norm-one uniqueness step. The finite-field fibre count and étale interpretation remain paper-level.  $\square$

**Proposition 6.56** (the selected cubic radicals span one Kummer-class line). *Retain Proposition 6.24. Fix  $k \geq 1$ , put*

$$L = L_k = \mathbb{F}_{q^3}, \quad q = 2^{3^{k-1}}, \quad N = q^2 + q + 1,$$

*and let  $\ell \mid N$  be prime. Choose integers  $1 \leq r < \ell$  and  $s \geq 0$  such that*

$$r(q-1) = 1 + \ell s. \quad (\text{RK}_1)$$

*For a chosen root  $t^\ell = \eta_k$  in  $\bar{L}$ , define*

$$e = t^{q+1}, \quad b = t^{-r} \beta_k^{-s}.$$

*Then*

$$\boxed{e^\ell = \varepsilon_k, \quad b^\ell = \beta_k.} \quad (\text{RK}_2)$$

*Moreover, for every  $\rho \in \mu_\ell$ ,*

$$t \mapsto \rho t, \quad e \mapsto \rho^{q+1} e, \quad b \mapsto \rho^{-r} b. \quad (\text{RK}_3)$$

All three weights  $1, q+1, -r$  are nonzero modulo  $\ell$ .

Writing the Kummer quotient additively,

$$[\varepsilon_k] = (q+1)[\eta_k], \quad [\beta_k] = -r[\eta_k] \quad \text{in } L^\times / (L^\times)^\ell. \quad (\text{RK}_4)$$

Consequently every choice of the three radicals generates the same field:

$$L(\eta_k^{1/\ell}) = L(\varepsilon_k^{1/\ell}) = L(\beta_k^{1/\ell}). \quad (\text{RK}_5)$$

The same holds for  $\gamma_k = \beta_k^{-1}$ .

For every  $1 \leq j < k$ ,  $\ell$ -powering is bijective on  $L_j^\times$ . Hence all lower selected  $\eta_j, \varepsilon_j, \beta_j, \gamma_j$  have unique  $\ell$ -th roots already in  $L_j$ , and the full selected radical compositum is

$$\boxed{L_k(\eta_j^{1/\ell}, \varepsilon_j^{1/\ell}, \beta_j^{1/\ell}, \gamma_j^{1/\ell} : 1 \leq j \leq k) = L_k(t).} \quad (\text{RK}_6)$$

Its degree over  $L_k$  is  $\ell$  on  $C$ -success at this prime  $\ell$ , and one on  $C$ -failure at  $\ell$ .

More intrinsically, the  $L$ -subalgebra generated by these chosen roots, their Frobenius conjugates, and the full lower ancestry is the single Kummer algebra

$$\mathcal{A}_{k,\ell} = L[T]/(T^\ell - \eta_k) = \bigoplus_{m=0}^{\ell-1} LT^m. \quad (\text{RK}_7)$$

The action  $T \mapsto \rho T$  preserves every selected coefficient and every lower canonical root, and

$$\mathcal{A}_{k,\ell}^{\mu_\ell} = L. \quad (\text{RK}_8)$$

Thus the compatible chosen roots, their Frobenius conjugates, and all cross-edge monomials supply no second radical torsor or canonical orientation. A remaining proof must establish a genuinely nonzero-weight correlation uniformly on this one torsor, or evaluate a weight-zero scalar in  $L$ . The proposition does not itself prove the required nonvanishing and does not exclude external character-sum or counting arguments.

*Proof.* Since  $q \equiv -1 \pmod{3}$ ,

$$\gcd(q-1, N) = \gcd(q-1, 3) = 1.$$

Thus  $q$  has order three modulo  $\ell$ ; in particular  $\mu_\ell \subset L$ , and both  $q-1$  and  $q+1$  are units modulo  $\ell$ . This gives  $r, s$  in (RK<sub>1</sub>).

The Singer and normalized-inverse identities give

$$\varepsilon_k = \eta_k^{q+1}, \quad \beta_k^{q-1} = \eta_k^{-1}.$$

Therefore

$$e^\ell = (t^\ell)^{q+1} = \eta_k^{q+1} = \varepsilon_k,$$

while

$$\eta_k^{-r} = \beta_k^{r(q-1)} = \beta_k^{1+\ell s},$$

and hence

$$b^\ell = \eta_k^{-r} \beta_k^{-\ell s} = \beta_k.$$

Replacing  $t$  by  $\rho t$  gives (RK<sub>3</sub>), and (RK<sub>4</sub>) follows. Since the displayed weights are units modulo  $\ell$ , Bézout recovers  $t$  from either  $e$  or  $b$ , proving (RK<sub>5</sub>); arbitrary choices differ from these compatible roots by elements of  $\mu_\ell \subset L$ .

Because  $q = 2^{3^{k-1}}$  has order three modulo  $\ell$ , 2 has order  $3^k$  modulo  $\ell$ . Thus  $\ell \nmid 2^{3^j} - 1 = |L_j^\times|$  for  $j < k$ , proving the lower-root assertion and (RK<sub>6</sub>). Standard prime-degree Kummer theory gives its stated degree.

Finally,  $T$  is a unit in  $\mathcal{A}_{k,\ell}$ , and every Laurent monomial reduces uniquely by

$$T^{\ell a+m} = \eta_k^a T^m, \quad 0 \leq m < \ell.$$

In particular products of Frobenius conjugates reduce by

$$\prod_i (T^{2^i})^{w_i} = \eta_k^a T^m \quad \text{when} \quad \sum_i w_i 2^i = \ell a + m.$$

All lower selected roots are coefficients in  $L$ , while  $e$  and  $b$  have the weights in (RK<sub>3</sub>). Since  $\mu_\ell \subset L$ , the distinct characters of  $\mu_\ell$  on the summands  $LT^m$  prove (RK<sub>8</sub>).  $\square$

**Proposition 6.57** (every nonzero cubic weight carries the Euler monodromy). *Retain Propositions 6.56 and 6.59. Put*

$$L = L_k = \mathbb{F}_{q^3}, \quad N = q^2 + q + 1 = \ell u, \quad \tau = \tau_k, \quad \Omega = \tau^u,$$

and choose  $v \in \bar{L}$  with  $v^\ell = \tau$ . Then

$$\boxed{v^{q^3} = \Omega^{q-1} v.} \tag{CWG_1}$$

For  $a \in L^\times$  and  $0 < m < \ell$ , the evaluated pure weight  $F = av^m$  satisfies

$$\boxed{F^{q^3} = \Omega^{m(q-1)} F.} \tag{CWG_2}$$

Since  $\text{ord}_\ell(q) = 3$ , both  $m$  and  $q-1$  are units modulo  $\ell$ , and therefore

$$\boxed{F \in L \iff \Omega = 1 \iff C\text{-failure at } \ell.} \tag{CWG_3}$$

The nonvanishing of  $F$  itself is automatic. Equivalently, on the radical torsor a nonzero homogeneous regular function  $aV^m$  vanishes nowhere.

The universal Kummer algebra is

$$\mathcal{B} = L[V]/(V^\ell - \tau) = \bigoplus_{m \bmod \ell} LV^m.$$

For  $P(X) \in L[X]$  with  $\deg P < \ell$ , separability gives the exact orientation-free branch product

$$\boxed{\text{Res}_X(X^\ell - \tau, P(X)) = \prod_{\rho \in \mu_\ell} P(\rho v).} \tag{CWG_4}$$

It is independent of the chosen root  $v$ , and is nonzero exactly when  $P$  vanishes on no root branch. In particular, for  $a, b \in L$  and  $0 < m < \ell$ ,

$$\boxed{\text{Res}_X(X^\ell - \tau, a + bX^m) = a^\ell + b^\ell \tau^m.} \tag{CWG_5}$$



The coefficients may be any everywhere-defined ancestry expressions in  $L$ . Pure nonzero weights therefore provide only an invertible power of the original Euler bit, while root-choice-invariant regular functions reduce to weight-zero arithmetic; the orbit product of one mixed evaluation is the resultant above and forgets which branch could vanish. This does not evaluate a special selected weight-zero resultant and therefore does not prove  $C_k$ . External character sums, inequalities, and direct weight-zero evaluations remain outside the conclusion.

*Proof.* Since  $q^3 - 1 = (q - 1)N = \ell(q - 1)u$ ,

$$v^{q^3} = v^{1+\ell(q-1)u} = v \tau^{(q-1)u} = \Omega^{q-1}v,$$

proving (CWG<sub>1</sub>); coefficients in  $L$  are fixed by  $q^3$ -powering, so (CWG<sub>2</sub>) follows. If  $F \in L$ , then (CWG<sub>2</sub>) and  $F \neq 0$  give  $\Omega^{m(q-1)} = 1$ . The exponent is a unit modulo  $\ell$ , while  $\Omega \in \mu_\ell$ , so  $\Omega = 1$ . Conversely (CWG<sub>1</sub>) shows that  $\Omega = 1$  makes  $v$  fixed by the  $q^3$ -power Frobenius, hence  $v \in L$ . This proves (CWG<sub>3</sub>).

The power basis gives the weight decomposition of  $\mathcal{B}$ . Its geometric roots are the distinct  $\rho v$ , and the first polynomial in the resultant is monic; the product formula for the resultant proves (CWG<sub>4</sub>). Finally  $\rho \mapsto \rho^m$  permutes  $\mu_\ell$ , and characteristic two gives

$$\prod_{\rho \in \mu_\ell} (a + b\rho^m v^m) = a^\ell + b^\ell v^{m\ell} = a^\ell + b^\ell \tau^m,$$

which proves (CWG<sub>5</sub>).

Lean kernel-checks the full-exponent factorization and both monodromy identities. The finite-field fixed-point and Kummer/Euler criteria, together with separability, the resultant formula, and the invariant-ring interpretation, remain paper-level.  $\square$

**Proposition 6.58** (the top cubic fibre is one Kummer coordinate). *Retain Proposition 6.54, and put*

$$K = L_{k-1} = \mathbb{F}_q, \quad L = L_k = \mathbb{F}_{q^3}, \quad N = q^2 + q + 1, \quad T = R_{L/K}^1 \mathbb{G}_m.$$

Let  $r = r_{k-1}$ , let  $c \in K^\times$  be the unique cube root of  $r$ , and set

$$\alpha = \frac{z_k}{c^\ell} \in T(K). \tag{KT_1}$$

Then

$$z_k \in (L^\times)^\ell \iff \alpha \in \ell T(K). \tag{KT_2}$$

The fixed-norm coefficient fibre is naturally isomorphic to the finite étale torsor

$$X = [\ell]^{-1}(\alpha)$$

under  $T[\ell]$ , of degree  $\ell^2$ . Its class

$$\kappa = \delta(\alpha) \in H^1(K, T[\ell]) \simeq T(K)/\ell T(K) \simeq \mathbb{F}_\ell \tag{KT_3}$$

vanishes exactly on failure; the desired  $C$ -conclusion is  $\kappa \neq 0$ .

There is an exact Euler representative containing the full selected lower ancestry:

$$\boxed{\Omega = \alpha^{N/\ell} = \frac{z_k^{N/\ell}}{r} \in T[\ell](K),} \quad \kappa = 0 \iff \Omega = 1. \tag{KT_4}$$

Moreover

$$\Omega^{q-1} = \eta_k^{-2N/\ell}, \quad (\text{KT}_5)$$

so this is an invertible power of the original selected obstruction, not a new lower invariant.

More precisely, identify

$$A = T[\ell](\overline{K}) \simeq \{(u_0, u_1, u_2) \in \mathbb{F}_\ell^3 : u_0 + u_1 + u_2 = 0\}.$$

Arithmetic  $q$ -Frobenius is  $F = qP$ , with  $P$  a three-cycle; its eigenvalues on  $A$  are  $1, q^2$ . Choosing a geometric origin makes Frobenius on  $X$  the affine map  $x \mapsto Fx + v$ , where the class of  $v$  on the fixed line is  $\kappa$ . For every  $s \geq 1$ ,

$$\#X(\mathbb{F}_{q^s}) = \begin{cases} \ell^{1+\mathbf{1}_{3|s}}, & \kappa = 0, \\ 0, & \kappa \neq 0 \text{ and } \ell \nmid s, \\ \ell^{1+\mathbf{1}_{3|s}}, & \kappa \neq 0 \text{ and } \ell \mid s. \end{cases} \quad (\text{KT}_6)$$

Equivalently, the Frobenius cycle types and zeta functions are

|                 | cycles                                 | $Z(X, t)$                                    |                    |
|-----------------|----------------------------------------|----------------------------------------------|--------------------|
| $\kappa = 0$    | $\ell[1] + \frac{\ell^2 - \ell}{3}[3]$ | $(1-t)^{-\ell}(1-t^3)^{-(\ell^2-\ell)/3}$    | (KT <sub>7</sub> ) |
| $\kappa \neq 0$ | $[\ell] + \frac{\ell-1}{3}[3\ell]$     | $(1-t^\ell)^{-1}(1-t^{3\ell})^{-(\ell-1)/3}$ |                    |

Thus the entire fibre geometry differs only by translation along one fixed line. The torus  $T$  has no nontrivial  $K$ -rational character. More precisely, let  $B$  be a finite-dimensional  $\mathbb{F}_\ell$ -representation of  $F$ , let  $f : A \rightarrow B$  be an  $F$ -equivariant linear homomorphism, and let  $v \in A^F$  represent  $\kappa$ . The induced quotient map sends

$$\kappa \longmapsto [f(v)] \in B/(F-1)B.$$

It kills  $\kappa$  exactly when  $f(v) \in (F-1)B$ , in particular if  $f$  kills  $A^F$ . When  $B = B^F \oplus (F-1)B$  is semisimple, nonvanishing of  $f(v)$  is sufficient to retain only an image of the same unresolved coordinate. Thus scalar characters and the ordinary norm/determinant routes are blind; a transfer to another anisotropic torus must be checked through its particular module map.

*Proof.* Since  $3 \nmid q-1$ , the cube root  $c$  is unique. The norm of  $z_k$  is  $z_{k-1} = r^\ell = c^{3\ell}$ , proving  $\alpha \in T(K)$ . If  $b^\ell = z_k$ , norm coherence gives  $N(b) = r = c^3$ , so  $b/c \in T(K)$ ; the converse is immediate. This proves (KT<sub>2</sub>). To identify  $X$  with the coefficient fibre, let  $\sigma_0, \sigma_1, \sigma_2$  be the three embeddings of  $L/K$ . Over  $\overline{K}$ , a point  $t = (t_0, t_1, t_2) \in X$  satisfies

$$t_i^\ell = \sigma_i(\alpha), \quad t_0 t_1 t_2 = 1.$$

Send it to

$$(C, D) = \left( c \sum_i t_i, c^2 \sum_{i < j} t_i t_j \right). \quad (\text{KT}_8)$$

The three source roots are  $ct_i$ ; their  $\ell$ -th powers are the three distinct conjugates of  $z_k$ , and their product is  $c^3 = r$ . Thus (KT<sub>8</sub>) lands in the selected fixed-norm coefficient fibre. Conversely those distinct powered targets recover the labeled  $t_i$ , so the map is a bijection on geometric points. It is  $K$ -defined, and both sides are finite étale; hence it is an isomorphism. The Kummer sequence, together with Lang's theorem  $H^1(K, T) = 0$ , gives (KT<sub>3</sub>).

Modulo  $q-1$ , one has  $N \equiv 3$ , hence  $c^N = c^3 = r$ . The cyclic Euler criterion in the group  $T(K)$ , of order  $N$ , now proves (KT<sub>4</sub>). Equation (KT<sub>5</sub>) follows from  $z_k^{q-1} = \eta_k^{-2}$ , since the base scalar  $c$  is killed by raising to  $q-1$ .

On the augmentation plane  $A$ , the three-cycle has eigenvalues  $q, q^2$ , because  $q$  has order three modulo  $\ell$ ; therefore  $qP$  has eigenvalues  $1, q^2$ . The cohomology quotient  $A/(F-1)A$  is exactly its fixed line. If  $v \in A^F$  represents  $\kappa$ , then over  $\mathbb{F}_{q^s}$  the exact restriction formula is

$$\text{res}_{\mathbb{F}_{q^s}/\mathbb{F}_q}(\kappa) = [(1 + F + \cdots + F^{s-1})v] = [sv] \in A/(F^s - 1)A. \quad (\text{KT}_9)$$

Moreover,

$$|A^{F^s}| = \ell^{1+1_{3|s}}.$$

This proves (KT<sub>6</sub>), and orbit enumeration gives (KT<sub>7</sub>). Finally the character lattice of the norm-one cubic torus is the  $C_3$ -augmentation lattice, whose invariant subgroup is zero. Functoriality of the quotient  $H^1 = A/(F-1)A$  sends  $[v]$  to  $[f(v)]$ ; the final assertions follow, with the stated semisimple direct-sum condition when it is invoked.  $\square$

**Proposition 6.59** (one normalized Singer element carries every current class). *For  $k \geq 1$ , put*

$$q = 2^{3^{k-1}}, \quad K = L_{k-1} = \mathbb{F}_q, \quad L = L_k = \mathbb{F}_{q^3}, \quad N = q^2 + q + 1.$$

*Let  $d_k \in K^\times$  be the unique cube root of  $z_{k-1}$ , and define*

$$d_k^3 = z_{k-1}, \quad s_k = d_k^2, \quad \tau_k = \frac{z_k}{d_k}. \quad (\text{NS}_1)$$

*Then  $\tau_k \in R_{L/K}^1 \mathbb{G}_m(K)$ , and its irreducible cubic over  $K$  is*

$$\boxed{X^3 + s_k X^2 + 1}. \quad (\text{NS}_2)$$

*In particular*

$$\text{Tr}_{L/K}(\tau_k) = s_k, \quad \text{Tr}_{L/K}(\tau_k^{-1}) = 0, \quad N_{L/K}(\tau_k) = 1. \quad (\text{NS}_3)$$

*Moreover*

$$\tau_k^{q-1} = \eta_k^{-2}, \quad \text{ord}(\tau_k) = \text{ord}(\eta_k). \quad (\text{NS}_4)$$

*The coefficients retain the complete selected ancestry. For  $k \geq 2$ ,*

$$N(d_k) = d_{k-1}, \quad N(s_k) = s_{k-1}, \quad \boxed{s_k^3 = s_{k-1} \tau_{k-1}^2}. \quad (\text{NS}_5)$$

*Also  $\mathbb{F}_2(s_k) = L_{k-1}$ ; conditional on  $C_{k-1}$ , the coefficient  $s_k$  is primitive in  $K^\times$ .*

*Conversely, the normalized recursion is Frobenius-rigid. Starting from  $s_1 = 1$ , every finite sequence in the prescribed fields satisfying (NS<sub>2</sub>) and the boxed recursion in (NS<sub>5</sub>) is a compatible absolute-Frobenius conjugate of the selected finite sequence. An infinite compatible sequence is similarly a conjugate under the procyclic group  $\text{Gal}(\bigcup_j L_j/\mathbb{F}_2) \simeq \mathbb{Z}_3$ . Thus retaining the complete recursion does not replace the primitivity problem: it reconstructs the same selected elements up to an order-preserving Galois action.*

*Now retain a current prime  $\ell \mid N$ , the lower root  $r = r_{k-1}$  with  $r^\ell = z_{k-1}$ , and the cube root  $c^3 = r$  from Proposition 6.58. Then*

$$c^\ell = d_k, \quad \boxed{\alpha_{k,\ell} = \tau_k, \quad \Omega_{k,\ell} = \tau_k^{N/\ell}}. \quad (\text{NS}_6)$$

Thus the current torsors are the preimages  $[\ell]^{-1}(\tau_k)$  of one selected norm-one element, while  $\Omega_{k,\ell} \in T[\ell](K)$  is their mod- $\ell$  Euler obstruction. At this level,

$$C_k \iff C_{k-1} \text{ and } \text{ord}(\tau_k) = N.$$

Consequently the universal cubic arm is the conjunction of these maximal-order statements over all levels, not a separate prime-dependent family of torsors.

There is a corresponding selected critical-value criterion. For odd  $u$ , use the generalized sums of  $(\mathbb{F}_2)$  and put

$$A_u(S) = T_u(S, 0, 1), \quad B_u(S) = T_u(0, S, 1), \quad \Delta_u(S) = A_u(S)B_u(S) + 1. \quad (\text{NS}_7)$$

If  $u = N/\ell$ , then

$$\boxed{\Omega_{k,\ell} = 1 \iff \Delta_u(s_k) = 0,} \quad (\text{NS}_8)$$

and more explicitly

$$\boxed{\Delta_u(s_k) = N_{L/K}(1 + \eta_k^{-2u}).} \quad (\text{NS}_9)$$

If  $S_k$  is the minimal polynomial of  $s_k$  over  $\mathbb{F}_2$ , the desired current- $\ell$  conclusion is therefore

$$\text{Res}_X(S_k(X), \Delta_u(X)) = 1. \quad (\text{NS}_{10})$$

For  $u = 1$ , one has  $\Delta_1 = 1$ , recovering the prime-step closure. For composite  $N$ , the displayed resultant remains a selected nonvanishing assertion. Primitivity of  $s_k$  alone cannot force it: primitive base coefficients with nonprimitive norm-one roots of  $X^3 + sX^2 + 1$  already occur in finite fields. Any induction must use the full recursion  $(\text{NS}_5)$ , not only the sparse cubic.

*Proof.* Because  $3 \nmid q-1$ , cube roots are unique in  $K^\times$ . Substitute  $z_k = d_k \tau_k$  and  $z_{k-1} = d_k^3$  into  $(\text{RS}_2)$ , then divide by  $d_k^3$ . This gives  $(\text{NS}_2)$ ; irreducibility follows because  $z_k$  and  $\tau_k$  differ by a base scalar. Its coefficients prove  $(\text{NS}_3)$ . A base scalar is killed by  $q-1$ , so  $(\text{RS}_5)$  gives the first identity in  $(\text{NS}_4)$ . On the norm-one group, both  $q-1$  and two are coprime to  $N$ ; hence this power preserves order.

Taking norms in  $d_k^3 = z_{k-1}$ , using  $N(z_{k-1}) = z_{k-2}$ , and invoking uniqueness of cube roots proves the first identity in  $(\text{NS}_5)$ ; squaring proves the second. Since  $z_{k-1} = d_{k-1} \tau_{k-1}$ ,

$$s_k^3 = z_{k-1}^2 = s_{k-1} \tau_{k-1}^2,$$

which proves the recursion. The equality  $s_k^3 = z_{k-1}^2$  also shows that  $s_k$  and  $z_{k-1}$  generate the same field: inverse Frobenius recovers  $z_{k-1}$  from  $z_{k-1}^2$ , while the unique cube root in  $K^\times$  is a power and recovers  $s_k$  from  $z_{k-1}^2$ . Under  $C_{k-1}$ , Corollary 6.53 makes  $z_{k-1}$  primitive. Squaring and cubing are automorphisms of the multiplicative group  $K^\times$ , so  $s_k$  is primitive as well.

For the rigidity assertion, the cubic at level one is irreducible, so its three roots are one absolute-Frobenius orbit. Inductively suppose the chosen pair at level  $j-1$  is a compatible Frobenius conjugate of the selected pair. The recursion determines  $s_j$  uniquely, because cubing is an automorphism of  $L_{j-1}^\times$ , and gives the corresponding conjugate of the selected coefficient. The three roots of  $X^3 + s_j X^2 + 1$  are its relative-Frobenius orbit, hence exactly the three extensions of that lower embedding to  $L_j$ . Choosing any of them extends the Frobenius-compatible chain and proves the induction.

The two identities  $c^3 = r$  and  $r^\ell = z_{k-1} = d_k^3$  give  $(c^\ell)^3 = d_k^3$ . Cube-root uniqueness proves  $c^\ell = d_k$ , and substitution in  $(\text{KT}_1)$ – $(\text{KT}_4)$  proves  $(\text{NS}_6)$ . Finally  $(\text{RS}_5)$ , together with  $\text{ord}(\tau_k) = \text{ord}(\eta_k)$ , proves the displayed equivalence between  $C_k$  and the lower-level assertion plus  $\text{ord}(\tau_k) = N$ .

The conjugates of  $\tau_k^u$  have polynomial

$$X^3 + A_u(s_k)X^2 + B_u(s_k)X + 1.$$

Its characteristic-two discriminant is  $\Delta_u(s_k)^2$ , and the odd-index form of (IF<sub>3</sub>) identifies  $\Delta_u(s_k)$  with the Jacobian of  $\Psi_{u,1}$  at  $(s_k, 0)$ . It vanishes exactly when two conjugates collide. In that event  $\tau_k^u \in K^\times$ , while also  $\tau_k^u$  has norm one; since  $K^\times \cap R_{L/K}^1 \mathbb{G}_m = \{1\}$ , this is precisely  $\Omega_{k,\ell} = 1$ . The converse is immediate, proving (NS<sub>8</sub>).

Put  $\delta = \tau_k^{q-1} = \eta_k^{-2}$ . The product of the three pairwise differences of the conjugates of  $\tau_k^u$  is

$$N_{L/K}(1 + \delta^u).$$

This is (NS<sub>9</sub>). Equivalently its denominator relative to  $u = 1$  is one, since

$$N(1 + \eta_k^{-2}) = N(1 + \eta_k)^2 = 1;$$

the last equality follows by evaluating the selected reciprocal cubic of  $\eta_k$  at one. Finally a nonzero element of the finite field generated by  $s_k$  has absolute norm one, while zero has norm zero. The product formula for the resultant proves (NS<sub>10</sub>).  $\square$

**Corollary 6.60** (trace-recursive reparameterization of the Singer class). *Set  $\gamma_0 = 1$ ,  $h_0 = 0$ , and, for  $k \geq 1$ ,*

$$h_k = 1 + \gamma_k^{-1}.$$

*Then  $h_k$  has the irreducible relative cubic*

$$\boxed{X^3 + h_{k-1}X^2 + X + 1} \tag{HTR<sub>1</sub>}$$

*over  $L_{k-1}$ . Consequently*

$$\text{Tr}(h_k) = h_{k-1}, \quad \text{Tr}(h_k^{-1}) = 1, \quad N(h_k) = 1. \tag{HTR<sub>2</sub>}$$

*In the notation of Proposition 6.59,*

$$\boxed{\tau_k^3 = h_k^4, \quad h_k = \frac{1 + \zeta_k^3}{(1 + \zeta_k)^3}, \quad h_k^2 = \eta_k^{q+2}, \quad h_k = \eta_k^{q/2+1}.} \tag{HTR<sub>3</sub>}$$

*All exponents in the last line are invertible on the norm-one group:*

$$\gcd(3, N) = \gcd(4, N) = \gcd(q/2 + 1, N) = 1.$$

*Hence*

$$\boxed{\text{ord}(\tau_k) = \text{ord}(h_k) = \text{ord}(\eta_k).} \tag{HTR<sub>4</sub>}$$

*Thus  $h_k$  gives a particularly sparse trace-recursive tower, but it is an invertible-power reparameterization of the same selected order coordinate, not an inductive order estimate.*

*Proof.* Substitute  $\gamma_k = (h_k + 1)^{-1}$  into  $\gamma_k^3 + \gamma_k + \gamma_{k-1} = 0$ , clear  $(h_k + 1)^3$ , and use  $\gamma_{k-1} = (h_{k-1} + 1)^{-1}$ . This gives (HTR<sub>1</sub>). The fractional-linear substitution preserves the relative degree, so the cubic is irreducible; its coefficients give (HTR<sub>2</sub>).

The identities  $\gamma_{k-1} = \gamma_k(\gamma_k + 1)^2$  and  $z_j = \gamma_j^{-2}$  give

$$\tau_k^3 = \frac{z_k^3}{z_{k-1}} = (1 + \gamma_k^{-1})^4 = h_k^4.$$

The displayed formula in  $\zeta_k$  is a direct simplification of  $1 + \gamma_k^{-1}$ . Using  $\eta_k = \gamma_k^{q-1}$  and  $\gamma_k^N = \gamma_{k-1}$  gives  $h_k^2 = \eta_k^{q+2}$ ; injectivity of squaring gives the last formula. The gcds are immediate except possibly the last: a common divisor of  $q/2 + 1$  and  $N = q^2 + q + 1$  divides three, while  $q/2 + 1 \not\equiv 0 \pmod{3}$ . The order equalities now follow from (HTR<sub>3</sub>).  $\square$

**Proposition 6.61** (one-branch norm descent of the cubic critical value). *Retain Proposition 6.59. For every positive odd  $u$ , the critical polynomial in  $(\mathbf{NS}_7)$  has a unique compression*

$$\boxed{\Delta_u(S) = \widehat{\Delta}_u(S^3)}, \quad \widehat{\Delta}_u(V) \in \mathbb{F}_2[V]. \quad (\text{CND}_1)$$

For independent indeterminates  $V, Y$ , define an operator  $\mathcal{C} : \mathbb{F}_2[V] \rightarrow \mathbb{F}_2[V]$  by

$$\boxed{\mathcal{C}(F)(V) = \text{Res}_Y(Y^3 + VY^2 + V, F(Y))}. \quad (\text{CND}_2)$$

At every actual normalized edge  $j \geq 2$ ,

$$\boxed{\mathcal{C}(F)(s_{j-1}^3) = N_{L_{j-1}/L_{j-2}} F(s_j^3)}. \quad (\text{CND}_3)$$

Now let  $u = N/\ell$  be a current value at level  $k$ . Then

$$\boxed{\mathcal{C}^{\circ(k-1)}(\widehat{\Delta}_u)(1) = N_{L_{k-1}/\mathbb{F}_2} \Delta_u(s_k)}. \quad (\text{CND}_4)$$

For  $k = 1$ , the left side means zero iterations and  $s_1 = 1$ . Consequently

$$\boxed{\Omega_{k,\ell} = 1 \iff \mathcal{C}^{\circ(k-1)}(\widehat{\Delta}_u)(1) = 0}, \quad (\text{CND}_5)$$

and desired nonfailure is exactly the terminal value one.

This is stricter than the unstructured resultant in  $(\mathbf{NS}_{10})$ : one fixed cubic norm operator carries the selected critical value through every actual normalized ancestry edge. It is still a lossless descent, not a universal nonvanishing theorem. The terminal bit is the same selected order coordinate, and its degree is not forced below the minimal polynomial by the recursion.

*Proof.* Give  $C, D, E$  the weights 1, 2, 3. The recurrence (IF<sub>2</sub>) makes  $T_n(C, D, E)$  homogeneous of weight  $n$ . Therefore every exponent of  $S$  in  $A_u(S)$  is congruent to  $u$  modulo three, while every exponent in  $B_u(S)$  is congruent to  $2u$ . Every exponent in  $A_u B_u + 1$  is consequently divisible by three, proving the existence and uniqueness in (CND<sub>1</sub>).

Put  $U_j = s_j^3 = z_{j-1}^2$ . Squaring the selected irreducible cubic (RS<sub>2</sub>) for  $z_{j-1}$  shows that, over  $L_{j-2}$ , the element  $U_j$  has the irreducible polynomial

$$Y^3 + U_{j-1}Y^2 + U_{j-1}.$$

The monic root-product formula for (CND<sub>2</sub>) therefore gives (CND<sub>3</sub>) with no sign or leading factor. Iteration and transitivity of norms give (CND<sub>4</sub>), using  $s_1 = 1$ . Finally (NS<sub>8</sub>) says that  $\Omega_{k,\ell} = 1$  exactly when  $\Delta_u(s_k) = 0$ ; a finite-field norm vanishes exactly with its argument, and every nonzero element of  $\mathbb{F}_2$  is one. This proves (CND<sub>5</sub>).  $\square$

**Proposition 6.62** (cyclotomic factorization of the cubic critical value). *Let  $u \geq 1$  be odd and  $d = (u - 1)/2$ . There is a unique monic  $H_u \in \mathbb{F}_2[X]$  such that*

$$\mathcal{D}_u(X, 1) = XH_u(X)^2, \quad \deg H_u = d, \quad H_u(0) = 1. \quad (\text{CCF}_1)$$

Its roots are the distinct nonzero elements

$$x_\rho = \rho + \rho^{-1}, \quad \{\rho, \rho^{-1}\} \subset \mu_u \setminus \{1\}, \quad (\text{CCF}_2)$$

where all nonidentity  $u$ -th roots, not only primitive roots, occur. The critical compression in Proposition 6.61 has the exact factorization, with multiplicity,

$$\begin{aligned} \widehat{\Delta}_u(V) &= \text{Res}_X(H_u(X), VX + (X+1)^3) \\ &= \prod_{\{\rho, \rho^{-1}\}} (x_\rho V + (x_\rho + 1)^3) \\ &= \prod_{\{\rho, \rho^{-1}\}} \left( V + \frac{(x_\rho + 1)^3}{x_\rho} \right). \end{aligned} \quad (\text{CCF}_3)$$

The last equality uses  $\prod x_\rho = H_u(0) = 1$ . The displayed critical values need not be distinct; for example  $\widehat{\Delta}_7(V) = (V+1)^3$ .

The selected specialization remembers the actual Conway cubic exactly. For arbitrary  $z, Y$  in characteristic two, put

$$C_V(X) = X^3 + X^2 + (V+1)X + 1.$$

Then

$$C_{z^2}(1 + zY^2) = z^3(Y^3 + Y)^2 + z^2. \quad (\text{CCF}_4)$$

At level  $k$ , take  $z = z_{k-1} = \gamma_{k-1}^{-2}$  and  $V = s_k^3 = z_{k-1}^2$ . Because  $z\gamma_{k-1}^2 = 1$ , this becomes

$$C_V(1 + zY^2) = z^3(Y^3 + Y + \gamma_{k-1})^2. \quad (\text{CCF}_5)$$

Consequently

$$\widehat{\Delta}_u(s_k^3) = N_{L_k/L_{k-1}} H_u(1 + z_{k-1}\gamma_k^2). \quad (\text{CCF}_6)$$

For  $u = N/\ell$ , failure is equivalently that the selected cubic  $C_{s_k^3}$  shares a root with  $H_u$ . This reconstructs the exact cyclotomic collision behind the terminal bit; it does not force the resultant nonzero when  $u$  is composite.

*Proof.* The standard Dickson factorization  $\mathcal{D}_u(\rho + \rho^{-1}, 1) = \rho^u + \rho^{-u}$ , with reciprocal pairs grouped together, gives (CCF<sub>1</sub>)–(CCF<sub>2</sub>). Let  $t_1, t_2, t_3$  be the roots of  $T^3 + ST^2 + 1$ , so  $e_1 = S, e_2 = 0, e_3 = 1$ . By the definitions in (NS<sub>7</sub>),

$$\Delta_u(S) = \prod_{i < j} (t_i^u + t_j^u).$$

Factor each power difference over  $\mu_u$ , remove the  $\rho = 1$  layer, and pair  $\rho$  with  $\rho^{-1}$ . For  $x = \rho + \rho^{-1}$ , the resulting symmetric product is

$$\prod_{i < j} (t_i^2 + xt_i t_j + t_j^2) = xS^3 + (x+1)^3.$$

Indeed  $e_2 = 0$  gives  $t_i t_j = t_k(t_i + t_j)$ ; each factor is then  $(S + t_k)(S + (x+1)t_k)$ , and multiplication over  $k$  gives the displayed identity. Taking the product over the roots of the monic  $H_u$  proves the resultant and both products in (CCF<sub>3</sub>), including multiplicity.

Expanding  $C_{z^2}(1 + zY^2)$  proves (CCF<sub>4</sub>). The selected relation  $z\gamma_{k-1}^2 = 1$  gives (CCF<sub>5</sub>). The roots of the right-hand Conway cubic are the three relative conjugates of  $\gamma_k$ , so its transported roots are precisely the roots of the monic  $C_V$ . The root-product formula for the resultant now yields (CCF<sub>6</sub>). Finally (NS<sub>8</sub>) identifies its vanishing with the original Euler obstruction.  $\square$

**Proposition 6.63** (the full ancestral Jacobian line collapses). *Retain Proposition 6.9, put  $f(X) = X^3 + X$ , and extend its notation by  $P_0(X) = X + 1$ , so that  $P_i(X) = f^{\circ i}(X) + 1$  for every  $i \geq 0$ . For  $m \geq 0$ , the Dickson square root from Proposition 6.62 has the exact factorization*

$$\boxed{\begin{aligned} f^{\circ m}(X) &= X H_{3^m}(X)^2, & H_{3^m}(X) &= \prod_{i=0}^{m-1} P_i(X), \\ (f^{\circ m})'(X) &= H_{3^m}(X)^2. \end{aligned}} \quad (\text{AJ}_1)$$

Put  $\gamma_0 = 1$ . For  $1 \leq m \leq k$ , define

$$R_{k,m} := H_{3^m}(\gamma_k), \quad J_{k,m} := (f^{\circ m})'(\gamma_k).$$

Then the complete selected ancestry gives

$$\boxed{\begin{aligned} R_{k,m} &= \prod_{i=0}^{m-1} (\gamma_{k-i} + 1) = (\gamma_k + 1) c_{k,m}, \\ c_{k,m} &:= \prod_{i=1}^{m-1} (\gamma_{k-i} + 1) \in L_{k-1}^\times, \\ J_{k,m} &= R_{k,m}^2 = \frac{\gamma_{k-m}}{\gamma_k}. \end{aligned}} \quad (\text{AJ}_2)$$

In particular, all nonempty ancestral square roots span the one-dimensional  $L_{k-1}$ -space  $(\gamma_k + 1)L_{k-1}$ , and all partial Jacobians span  $(\gamma_k + 1)^2 L_{k-1}$ . At the full depth,

$$R_{k,k} = \prod_{j=1}^k (\gamma_j + 1), \quad J_{k,k} = \gamma_k^{-1} = \beta_k. \quad (\text{AJ}_3)$$

Now let  $\ell \mid \Phi_{3^k}(2)$  be prime, put  $q = 2^{3^k-1}$ , and define

$$\mathcal{E}_\ell(x) := x^{(2^{3^k}-1)/\ell} \in \mu_\ell.$$

For every  $1 \leq m \leq k$ ,

$$\boxed{\mathcal{E}_\ell(J_{k,m}) = \mathcal{E}_\ell(\gamma_k)^{-1}, \quad \mathcal{E}_\ell(R_{k,m})^2 = \mathcal{E}_\ell(\gamma_k)^{-1}.} \quad (\text{AJ}_4)$$

Since  $\ell$  is odd and  $\eta_k = \gamma_k^{q-1}$ , with  $\gcd(q-1, \ell) = 1$ , this gives the exact equivalence

$$\boxed{\mathcal{E}_\ell(R_{k,m}) = 1 \iff \mathcal{E}_\ell(J_{k,m}) = 1 \iff \mathcal{E}_\ell(\eta_k) = 1.} \quad (\text{AJ}_5)$$

More generally, let  $F \in L_{k-1}[T_1, \dots, T_k]$  be homogeneous of total degree  $d$ . Whenever the displayed evaluations are nonzero, homogeneity and (AJ<sub>2</sub>) give

$$\boxed{\begin{aligned} \mathcal{E}_\ell(F(R_{k,1}, \dots, R_{k,k})) &= \mathcal{E}_\ell(\gamma_k + 1)^d, \\ \mathcal{E}_\ell(F(J_{k,1}, \dots, J_{k,k})) &= \mathcal{E}_\ell(\gamma_k)^{-d}. \end{aligned}} \quad (\text{AJ}_6)$$

Thus if  $\ell \nmid d$ , either residue test is one exactly on the original current failure; if  $\ell \mid d$ , it is identically one. The coefficients of  $F$  may depend nonsymmetrically on every separate lower selected ancestor.



Therefore partial iterated Jacobians, their canonical Dickson square roots, lower-ancestry linear combinations, and homogeneous determinants supply no second current Kummer coordinate.

This is a homogeneous residue-class collapse only. It does not evaluate whether a selected homogeneous expression is zero, does not cover sums of different total degrees, and does not exclude a correlation using other top-point data. In particular it does not prove the required  $\mathcal{E}_\ell(\eta_k) \neq 1$ .

*Proof.* In characteristic two,

$$f(Y) = Y^3 + Y = Y(Y+1)^2, \quad f'(Y) = Y^2 + 1 = (Y+1)^2.$$

Starting with  $f^{\circ 0}(X) = X$ , induction therefore gives

$$f^{\circ(m+1)}(X) = f^{\circ m}(X)(f^{\circ m}(X) + 1)^2 = X \left( \prod_{i=0}^m P_i(X) \right)^2.$$

The product is monic with constant term one, so uniqueness in  $(\text{CCF}_1)$  identifies it with  $H_{3^{m+1}}$ . Differentiating  $XH_{3^m}^2$  in characteristic two proves the last identity in  $(\text{AJ}_1)$ .

The selected recursion says  $f^{\circ i}(\gamma_k) = \gamma_{k-i}$  for  $0 \leq i \leq k$ . Evaluation of  $(\text{AJ}_1)$  gives the product formulas in  $(\text{AJ}_2)$ ; division of  $\gamma_{k-m} = \gamma_k R_{k,m}^2$  gives its last equality. All factors are nonzero. Separating the  $i = 0$  factor puts every remaining factor in  $L_{k-1}^\times$ , proving the line assertions and  $(\text{AJ}_3)$ .

For a current prime,  $\text{ord}_\ell(2) = 3^k$ . Hence  $\ell \nmid |L_{k-1}^\times|$ , so every nonzero lower-field coefficient is an  $\ell$ -th power in  $L_k$  and has  $\mathcal{E}_\ell$ -value one. Applying  $\mathcal{E}_\ell$  to  $(\text{AJ}_2)$  proves  $(\text{AJ}_4)$ . The square map on  $\mu_\ell$  is injective, and

$$\mathcal{E}_\ell(\eta_k) = \mathcal{E}_\ell(\gamma_k)^{q-1}$$

with  $q - 1$  invertible modulo  $\ell$ , proving  $(\text{AJ}_5)$ .

Finally,  $(\text{AJ}_2)$  and homogeneity give the exact factorizations

$$\begin{aligned} F(R_{k,1}, \dots, R_{k,k}) &= (\gamma_k + 1)^d F(1, c_{k,2}, \dots, c_{k,k}), \\ F(J_{k,1}, \dots, J_{k,k}) &= (\gamma_k + 1)^{2d} F(1, c_{k,2}^2, \dots, c_{k,k}^2). \end{aligned}$$

The remaining factors lie in  $L_{k-1}$ . Applying  $\mathcal{E}_\ell$ , and using  $\mathcal{E}_\ell(\gamma_k + 1)^2 = \mathcal{E}_\ell(\gamma_k)^{-1}$ , proves  $(\text{AJ}_6)$  and the stated dichotomy.  $\square$

**Proposition 6.64** (the mixed ancestral algebra is exactly the selected cubic field). *Retain Proposition 6.63, put*

$$K = L_{k-1}, \quad z = R_{k,1} = \gamma_k + 1,$$

and extend the notation there by  $c_{k,1} = 1$ . Then every  $c_{k,m}$  lies in  $K^\times$ , and

$$R_{k,m} = c_{k,m} z, \quad J_{k,m} = R_{k,m}^2 \quad (1 \leq m \leq k). \quad (\text{AMA}_1)$$

Consequently

$$K[R_{k,1}, \dots, R_{k,k}, J_{k,1}, \dots, J_{k,k}] = K[z] = L_k. \quad (\text{AMA}_2)$$

This equality has the following exact formal presentation. Let

$$\text{ev} : K[T_1, \dots, T_k, U_1, \dots, U_k] \longrightarrow L_k, \quad T_m \longmapsto R_{k,m}, \quad U_m \longmapsto J_{k,m}.$$

Then

$$\ker(\text{ev}) = (T_m - c_{k,m} T_1 \ (2 \leq m \leq k), U_m - T_m^2 \ (1 \leq m \leq k), T_1^3 + T_1^2 + \gamma_{k-1}). \quad (\text{AMA}_3)$$

Thus every formal mixed-degree ancestral polynomial has, modulo this kernel, a unique representative

$$A + BT_1 + CT_1^2, \quad A, B, C \in K, \quad (\text{AMA}_4)$$

and its selected evaluation in  $L_k$  vanishes exactly when  $A = B = C = 0$ . The restriction to the selected evaluation is essential: the nonzero formal polynomial

$$T_1^3 + T_1^2 + \gamma_{k-1}$$

already vanishes there.

Moreover the first ancestral root recovers the original selected endpoint:

$$\gamma_k = z + 1, \quad \mathcal{E}_\ell(\eta_k) = ((z + 1)^{q-1})^{(q^3-1)/\ell}, \quad q = 2^{3^{k-1}}. \quad (\text{AMA}_5)$$

Hence allowing zeros and arbitrary mixtures of ancestral degrees supplies neither a larger coordinate algebra nor an independent Kummer coordinate: it recovers the whole selected cubic field, including the original Euler endpoint itself. This is a presentation theorem, not a nonvanishing theorem. A mixed-degree proof could still succeed by proving that a specific quadratic normal remainder in  $(\text{AMA}_4)$  is nonzero; the proposition says precisely that such a step is new selected arithmetic, not a consequence of the ancestral generators or their formal relations.

*Proof.* Equation  $(\text{AMA}_1)$  is  $(\text{AJ}_2)$ , with  $c_{k,1} = 1$ ; its displayed product also proves  $c_{k,m} \in K^\times$ . Since  $T_1$  maps to  $z$ , this proves  $(\text{AMA}_2)$ .

The minimal polynomial  $X^3 + X + \gamma_{k-1}$  of  $\gamma_k$  over  $K$ , translated by  $X \mapsto X + 1$ , is

$$X^3 + X^2 + \gamma_{k-1},$$

which is again irreducible. Therefore every listed generator in  $(\text{AMA}_3)$  lies in  $\ker(\text{ev})$ . Conversely, after quotienting by the listed ideal, the linear relations eliminate  $T_2, \dots, T_k$ , the quadratic relations eliminate every  $U_m$ , and the last relation leaves exactly

$$K[T_1]/(T_1^3 + T_1^2 + \gamma_{k-1}).$$

The induced map from this quotient to  $K[z] = L_k$  is the standard minimal-polynomial isomorphism. This proves the reverse ideal containment, hence  $(\text{AMA}_3)$ , and Euclidean division by the monic cubic gives  $(\text{AMA}_4)$ . Its uniqueness follows equally from the  $K$ -linear independence of  $1, z, z^2$ . Finally  $z = \gamma_k + 1$  and  $q^3 = 2^{3^k}$ , together with  $\eta_k = \gamma_k^{q-1}$ , give  $(\text{AMA}_5)$ .  $\square$

**Proposition 6.65** (the selected Singer trace is the whole remaining phase). *Retain Proposition 6.51. For  $a \in \mathbb{F}_q$  put*

$$D_a = \{x \in D : \text{Tr}_{\mathbb{F}_{q^3}/\mathbb{F}_q}(x) = a\}.$$

*Each  $D_a$  is empty or one three-element relative-Frobenius orbit. Let  $\chi : U \rightarrow \mu_\ell$  have exact order  $\ell$ , where  $\text{ord}_\ell(q) = 3$ , and put  $H = \ker \chi$ . If  $D_a \neq \emptyset$ , define*

$$\Theta_\chi(a) = \sum_{x \in D_a} \chi(x).$$

*Then*

$$D_a \subset H \iff \Theta_\chi(a) = 3. \quad (\text{ST}_1)$$

In particular, current failure is exactly

$$\Theta_\chi(\eta_{j-1}) = 3. \quad (\text{ST}_2)$$

For any nontrivial additive character  $\psi$  of  $\mathbb{F}_q$ , Fourier inversion gives

$$\Theta_\chi(a) = \frac{1}{q} \sum_{b \in \mathbb{F}_q} \psi(-ba) \sum_{x \in D} \chi(x) \psi\left(b \operatorname{Tr}_{\mathbb{F}_{q^3}/\mathbb{F}_q}(x)\right). \quad (\text{ST}_3)$$

Whenever the moment criterion (SW<sub>4</sub>) forces  $D \cap H \neq \emptyset$ , some trace fibre actually attains the extremal value 3. Hence a uniform character bound cannot exclude (ST<sub>2</sub>); one must distinguish the selected Conway fibre.

*Proof.* The trace- $a$  elements of  $D$  are the roots of

$$T^3 + aT^2 + (a+1)T + 1.$$

Relative Frobenius has no fixed point on  $D$ : an element of  $D \cap \mathbb{F}_q$  would lie in  $\mathbb{F}_q^\times \cap U = \{1\}$ , while  $1 \notin D$ . Thus every nonempty fibre is one orbit of length three.

For  $x \in D_a$ , put  $\theta = \chi(x)$ . Then

$$\Theta_\chi(a) = \theta + \theta^q + \theta^{q^2}.$$

The fibre lies in  $H$  exactly when  $\theta = 1$ . Conversely, equality to 3 in the displayed sum forces all three unit complex numbers to equal one, by equality in the triangle inequality. This proves (ST<sub>1</sub>). The distinguished fibre is the relative orbit of  $\eta_j$ , whose trace is  $\eta_{j-1}$ , proving (ST<sub>2</sub>). Character orthogonality on the additive group of  $\mathbb{F}_q$  gives (ST<sub>3</sub>). Finally, if  $x \in D \cap H$ , its whole relative orbit lies in  $H$ , so its trace fibre has  $\Theta_\chi = 3$ .  $\square$

**Proposition 6.66** (the recursive trace–Fourier flag collapses at the top edge). *Retain Propositions 6.24, 6.50, and 6.65. Put*

$$L_j = \mathbb{F}_{2^{3j}}, \quad q = 2^{3^{k-1}}, \quad N = q^2 + q + 1,$$

and write

$$D_k = \{x \in L_k^\times : x^{1+q+q^2} = 1, x^{q+1} + x + 1 = 0\}.$$

For  $0 \leq j < k$ , let  $\operatorname{Tr}_{k/j} = \operatorname{Tr}_{L_k/L_j}$  and define the complete lower trace flag

$$\mathfrak{T}_k(x) = (\operatorname{Tr}_{k/j}(x))_{0 \leq j < k}.$$

Trace transitivity gives the exact factorization

$$\boxed{\mathfrak{T}_k(x) = \left( \operatorname{Tr}_{L_{k-1}/L_j}(\operatorname{Tr}_{k/(k-1)}(x)) \right)_{0 \leq j < k}.} \quad (\text{RTF}_1)$$

Consequently the literal full-ancestry fibre is already the immediate trace fibre:

$$\boxed{\begin{aligned} \mathcal{F}_k &:= \{x \in D_k : \operatorname{Tr}_{k/j}(x) = \eta_j \text{ for every } 0 \leq j < k\} \\ &= \{x \in D_k : \operatorname{Tr}_{k/(k-1)}(x) = \eta_{k-1}\} \\ &= \{\eta_k, \eta_k^q, \eta_k^{q^2}\}. \end{aligned}} \quad (\text{RTF}_2)$$

This collapse holds before choosing a multiplicative character and is exact for every weighting which depends only on the iterated trace tuple  $\mathfrak{T}_k(x)$ . Its additive Fourier form is especially sharp. Put

$$\psi_j(y) = (-1)^{\text{Tr}_{L_j/\mathbb{F}_2}(y)}.$$

For arbitrary  $b_j \in L_j$ , embedded in  $L_{k-1}$ , one has

$$\prod_{j=0}^{k-1} \psi_j(b_j \text{Tr}_{k/j}(x)) = \psi_{k-1} \left( \left( \sum_{j=0}^{k-1} b_j \right) \text{Tr}_{k/(k-1)}(x) \right). \quad (\text{RTF}_3)$$

Thus stacking additive Fourier modes from every lower trace level creates no mode beyond the single immediate-edge additive character in  $(\mathbf{ST}_3)$ .

Let  $\ell \mid N$  be prime, put  $u = N/\ell$ , let  $\chi : U_k \rightarrow \mu_\ell$  have exact order  $\ell$ , and write  $\theta = \chi(\eta_k)$ . Then

$$\sum_{x \in \mathcal{F}_k} \chi(x) = \theta + \theta^q + \theta^{q^2} = \Theta_\chi(\eta_{k-1}). \quad (\text{RTF}_4)$$

Moreover

$$\begin{aligned} C\text{-failure at } (k, \ell) &\iff \theta = 1 \iff \mathcal{F}_k \subset \ker \chi \\ &\iff \sum_{x \in \mathcal{F}_k} \chi(x) = 3 \iff P_{\eta, k} \mid G_u. \end{aligned} \quad (\text{RTF}_5)$$

Hence the complete literal lower trace ancestry, its arbitrary additive multi-Fourier projectors, and the distinguished Wendt incidence all return to the same selected three-point phase orbit. This does not cover a weight which depends on the top point through data other than its iterated traces, a nonsymmetric chosen-root expression, or a genuinely mixed additive–multiplicative correlation between the top point and separate lower ancestors. Nor does it evaluate  $\Theta_\chi(\eta_{k-1})$ ; the selected nonvanishing remains open.

*Proof.* For  $j < k$ , trace transitivity is

$$\text{Tr}_{k/j} = \text{Tr}_{L_{k-1}/L_j} \circ \text{Tr}_{k/(k-1)},$$

which proves (RTF<sub>1</sub>). The selected tower satisfies  $\text{Tr}_{L_i/L_{i-1}}(\eta_i) = \eta_{i-1}$  at every edge. Therefore the immediate equality  $\text{Tr}_{k/(k-1)}(x) = \eta_{k-1}$  forces every lower equality in the first line of (RTF<sub>2</sub>), and the converse is immediate.

For  $x \in D_k$  with relative trace  $a$ , its relative orbit has cubic

$$X^3 + aX^2 + (a+1)X + 1.$$

At  $a = \eta_{k-1}$ , this is the selected irreducible cubic of Proposition 6.24; its roots are  $\eta_k, \eta_k^q, \eta_k^{q^2}$ . This proves (RTF<sub>2</sub>).

For the Fourier assertion, trace transitivity and  $L_j$ -linearity give

$$\text{Tr}_{L_j/\mathbb{F}_2}(b_j \text{Tr}_{k/j}(x)) = \text{Tr}_{L_k/\mathbb{F}_2}(b_j x) = \text{Tr}_{L_{k-1}/\mathbb{F}_2}(b_j \text{Tr}_{k/(k-1)}(x)).$$

Summing over  $j$  and exponentiating by  $-1$  proves (RTF<sub>3</sub>).

Finally multiplicativity gives  $\chi(\eta_k^{q^i}) = \theta^{q^i}$ , proving (RTF<sub>4</sub>). Since  $\ker \chi = U_k^\ell$ , failure is  $\theta = 1$ . Equality of the three-term sum with three forces the three complex unit summands to be one,

by equality in the triangle inequality. Proposition 6.50 identifies the same failure with  $P_{\eta,k} \mid G_u$ , proving (RTF<sub>5</sub>).

Lean kernel-checks the abstract compatible-fibre equality, factorization of a full flag weight through its immediate coordinate, and summation of linear Fourier functionals after composition with one top map. Finite-field trace transitivity, the Singer-fibre specialization, and the character/Wendt identifications remain paper-level.  $\square$

**Proposition 6.67** (the selected trace hyperplane is self-polar). *Retain Proposition 6.24. Put*

$$F = L_{k-1} = \mathbb{F}_q, \quad E = L_k = \mathbb{F}_{q^3}, \quad q = 2^{3^{k-1}}, \quad \gamma = \gamma_k,$$

and let  $\rho : E^\times \rightarrow \mu_\ell \subset \mathbb{C}^\times$  be the exact current character of Proposition 6.37. Thus  $\rho$  is nontrivial and  $\rho|_{F^\times} = 1$ . For  $a \in E^\times$ , put

$$H_a = \{x \in E : \text{Tr}_{E/F}(ax) = 0\}, \quad \Sigma_a(\rho) = \sum_{[x] \in H_a^\times / F^\times} \rho(x). \quad (\text{CPH}_1)$$

The summand is independent of the representative because  $\rho$  is trivial on  $F^\times$ . If  $\psi : F \rightarrow \mathbb{C}^\times$  is any nontrivial additive character and

$$G_\psi(\rho) = \sum_{x \in E^\times} \rho(x) \psi(\text{Tr}_{E/F}(x)),$$

then

$$\boxed{\Sigma_a(\rho) = q^{-1} \rho(a)^{-1} G_\psi(\rho), \quad |\Sigma_a(\rho)| = \sqrt{q}.} \quad (\text{CPH}_2)$$

In particular the orientation is inverse in  $a$ .

The selected depressed cubic implies

$$\text{Tr}_{E/F}(\gamma) = \text{Tr}_{E/F}(\gamma^2) = 0, \quad \boxed{H_\gamma = F + F\gamma.} \quad (\text{CPH}_3)$$

Consequently

$$\boxed{\frac{\Sigma_\gamma(\rho)}{\Sigma_1(\rho)} = \rho(\gamma)^{-1}, \quad C\text{-failure at } (k, \ell) \iff \Sigma_\gamma(\rho) = \Sigma_1(\rho).} \quad (\text{CPH}_4)$$

Moreover  $F + F(\gamma + c) = F + F\gamma$  for every  $c \in F$ . Hence every projective-line correlation obtained by translating the selected top coordinate by an arbitrary lower-ancestry scalar gives the same  $\Sigma_\gamma$ . The mixed additive–multiplicative hyperplane method therefore recovers the selected Euler phase exactly, but does not evaluate it.

*Proof.* Since every  $F^\times$ -orbit in  $H_a^\times$  has size  $q - 1$ ,

$$\Sigma_a(\rho) = \frac{1}{q-1} \sum_{x \in E^\times} \rho(x) \mathbf{1}_{\text{Tr}(ax)=0}.$$

Additive orthogonality gives

$$\mathbf{1}_{\text{Tr}(ax)=0} = \frac{1}{q} \sum_{u \in F} \psi(u \text{Tr}(ax)).$$

The  $u = 0$  term vanishes because  $\rho$  is nontrivial. For  $u \in F^\times$ , substituting  $y = uax$  and using  $\rho(u) = 1$  gives

$$\sum_{x \in E^\times} \rho(x) \psi(\text{Tr}(uax)) = \rho(a)^{-1} G_\psi(\rho).$$

There are  $q - 1$  such values of  $u$ , proving the first identity in (CPH<sub>2</sub>). The lifted additive character  $x \mapsto \psi(\text{Tr}(x))$  is nontrivial; the standard finite-field Gauss-sum identity therefore gives  $|G_\psi(\rho)| = |E|^{1/2} = q^{3/2}$ , proving the absolute value.

The relative minimal polynomial of  $\gamma$  is  $X^3 + X + \gamma_{k-1}$ . Its zero quadratic coefficient gives  $\text{Tr}(\gamma) = 0$ , and  $\text{Tr}(\gamma^2) = \text{Tr}(\gamma)^2 = 0$ . Thus  $F + F\gamma \subseteq H_\gamma$ . The left side has  $F$ -dimension two because  $\gamma \notin F$ ; the right side has dimension two because the finite-field trace pairing is nondegenerate and  $\gamma \neq 0$ . This proves (CPH<sub>3</sub>). Dividing the two nonzero instances of (CPH<sub>2</sub>), with  $a = \gamma, 1$ , gives (CPH<sub>4</sub>). Proposition 6.37 identifies  $\rho(\gamma) = 1$  with current failure. Finally  $F + F(\gamma + c) = F + F\gamma$  is immediate.  $\square$

**Proposition 6.68** (exact one-step lifting of the previous Singer set). *Let  $j \geq 2$ , put  $r = 2^{3^{j-2}}$ ,  $q = r^3$ , and fix  $\omega \in \mathbb{F}_4 \setminus \mathbb{F}_2$ . For  $s \in \{r, q\}$ , write*

$$U_s = \ker(N_{\mathbb{F}_{s^3}/\mathbb{F}_s}), \quad D_s = \{x \in U_s : x^{s+1} + x + 1 = 0\}.$$

*In a common algebraic closure put*

$$M(T) = \frac{\omega + \omega^2 T}{1 + T}.$$

*Then  $M$  gives a bijection*

$$\{t : t^{s+1} = \omega^2\} \xrightarrow{\sim} D_s,$$

*and, for  $t^{q+1} = \omega^2$ ,*

$$\text{Tr}_{\mathbb{F}_{q^3}/\mathbb{F}_q}(M(t)) = M(t^3). \quad (\text{SL}_1)$$

*Consequently the restricted trace map*

$$\text{Tr} : \text{Tr}^{-1}(D_r) \cap D_q \longrightarrow D_r$$

*is surjective with every fibre of cardinality exactly three, and*

$$|\text{Tr}^{-1}(D_r) \cap D_q| = 3(r + 1). \quad (\text{SL}_2)$$

*Thus precisely  $r + 1$  of the  $(q + 1)/3$  nonempty current Singer trace fibres lie over the previous Singer set. In particular, the selected coefficient's membership in  $D_r$  does not distinguish its current fibre: every previous Singer point has a current three-point fibre satisfying the same norm-one, Singer, and cubic-trace identities.*

*Proof.* For  $x \in D_s$ , put

$$t = \frac{x + \omega}{x + \omega^2}.$$

Since  $s$  is a power of two with odd exponent,  $\omega^s = \omega^2$ . Using  $x^s = (x + 1)/x$  gives  $t^s = \omega^2/t$ , and hence  $t^{s+1} = \omega^2$  and  $x = M(t)$ . Conversely,  $t^{s+1} = \omega^2$  gives

$$M(t)^s = \frac{M(t) + 1}{M(t)}.$$

Iterating this order-three Möbius map shows that  $M(t)$  belongs to  $\mathbb{F}_{s^3}$ , has norm one, and lies in  $D_s$ . This proves the bijection.

For  $x = M(t) \in D_q$ , put  $a = \text{Tr}_{\mathbb{F}_{q^3}/\mathbb{F}_q}(x)$ . The relative orbit has cubic

$$X^3 + aX^2 + (a + 1)X + 1.$$

Substitution and cross-multiplication give

$$\left(\frac{x+\omega}{x+\omega^2}\right)^3 = \frac{a+\omega}{a+\omega^2}.$$

Thus  $t^3 = M^{-1}(a)$ , proving  $(\mathbf{SL}_1)$ .

Now let  $a = M(u) \in D_r$ , so  $u^{r+1} = \omega^2$ , and choose any of the three roots  $t^3 = u$ . Write

$$C = \frac{r^2 - r + 1}{3}, \quad q + 1 = 3(r + 1)C.$$

For  $j = 2$ , one has  $r = 2$  and  $C = 1$ . For  $j \geq 3$ ,  $r \equiv -1 \pmod{9}$ , so  $r^2 - r + 1 \equiv 3 \pmod{9}$ . In either case  $C \equiv 1 \pmod{3}$ , and

$$t^{q+1} = (t^3)^{(r+1)C} = (\omega^2)^C = \omega^2.$$

All three roots therefore give distinct points of  $D_q$  with trace  $a$ . Conversely,  $(\mathbf{SL}_1)$  shows that every point above  $D_r$  arises this way, proving  $(\mathbf{SL}_2)$ .  $\square$

**Remark 6.69** (the complete cubic recursion is Frobenius-rigid). Suppose  $\xi_0 = 0$  and, for every  $i \geq 1$ , the element  $\xi_i$  is a root of

$$X^3 + \xi_{i-1}X^2 + (\xi_{i-1} + 1)X + 1.$$

Then  $\xi_i = \eta_i^{2^{e_i}}$  for some  $e_i$ , with

$$e_i \equiv e_{i-1} \pmod{3^{i-1}}.$$

Indeed, after  $\xi_{i-1} = \eta_{i-1}^{2^{e_{i-1}}}$ , the stage- $i$  cubic is the  $2^e$ -Frobenius twist of the selected cubic. Its three roots are

$$\eta_i^{2^{e+t3^{i-1}}}, \quad t = 0, 1, 2.$$

Induction proves the assertion. Thus imposing the complete recursive chain determines one absolute Frobenius orbit, whose members all have the same multiplicative order. Proposition 6.68 therefore rules out a one-step membership argument; it does not create a false recursive tower.

**Remark 6.70** (the natural secant Kummer cover is rational). Over  $\mathbb{F}_4$ , the Möbius function

$$M(T) = \frac{\omega + \omega^2 T}{1 + T}$$

which expresses the cyclotomic secant has Kummer cover

$$Z^\ell = M(T), \quad T = \frac{Z^\ell + \omega}{Z^\ell + \omega^2}.$$

Its function field is therefore  $\mathbb{F}_4(Z)$ , of genus zero. A positive-genus Weil estimate on this cover cannot rule out the distinguished torsion fibre; the remaining information is its actual Artin value.

**Proposition 6.71** (Capelli form of the surviving lemma). *Put*

$$A(X) = X^3 + X + 1, \quad B(X) = X^2 + X, \quad R(X) = \frac{A(X)}{B(X)}.$$

*Let  $Q_0(X) = X$  and, for  $k \geq 1$ , define the monic binary polynomial*

$$Q_k(X) = B(X)^{3^{k-1}} Q_{k-1}(R(X)). \quad (\star)$$

Then  $Q_k$  is the minimal polynomial of  $\eta_k$  over  $\mathbb{F}_2$  and is irreducible of degree  $3^k$ . Its first four values are

$$\begin{aligned} Q_1(X) &= X^3 + X + 1, \\ Q_2(X) &= X^9 + X + 1, \\ Q_3(X) &= X^{27} + X^{25} + X^{24} + X^{19} + X^{18} + X^{16} + X^{10} + X^9 + X^3 + X + 1, \\ Q_4(X) &= X^{81} + X^{64} + X^{16} + X + 1. \end{aligned}$$

For every prime  $\ell \mid \Phi_{3^k}(2)$ , the following are equivalent:

- (i)  $\chi_{k,\ell}(\eta_k) \neq 1$ ;
- (ii)  $\eta_k$  is not an  $\ell$ -th power in  $\mathbb{F}_{2^{3^k}}^\times$ ;
- (iii)  $X^\ell - \eta_k$  is irreducible over  $\mathbb{F}_{2^{3^k}}$ ;
- (iv)  $Q_k(X^\ell)$  is irreducible over  $\mathbb{F}_2$ .

Thus the surviving character lemma is equivalently the assertion that all the explicitly recursive compositions  $Q_k(X^\ell)$ , for  $\ell \mid \Phi_{3^k}(2)$ , are irreducible. At the first composite level  $k = 4$ , the three targets are the monomial compositions of the displayed pentanomial  $Q_4$  by  $\ell \in \{2593, 71119, 97685839\}$ .

*Proof.* Proposition 6.24 gives  $R(\eta_k) = \eta_{k-1}$  and an irreducible cubic for  $\eta_k$  over  $\mathbb{F}_{2^{3^{k-1}}}$ . Since  $\eta_{k-1} \in \mathbb{F}_2(\eta_k)$ , induction gives  $[\mathbb{F}_2(\eta_k) : \mathbb{F}_2] = 3^k$ . Clearing the denominator in the minimal polynomial of  $\eta_{k-1}$  gives  $(\star)$ , a monic polynomial of degree  $3^k$  having  $\eta_k$  as a root: its leading term is  $A(X)^{3^{k-1}} = X^{3^k} + \dots$ . It is therefore the minimal polynomial. Direct substitution in  $(\star)$  gives the four displayed polynomials.

Here  $\Phi_{3^k}(2) = q_k^2 + q_k + 1 \equiv 1 \pmod{3}$ , so a prime  $\ell \mid \Phi_{3^k}(2)$  is not 3. The cyclotomic-divisor criterion therefore gives  $\text{ord}_\ell(2) = 3^k$ , and hence  $\mu_\ell \subset \mathbb{F}_{2^{3^k}}$ . Since  $\ell$  is prime, Kummer theory says that  $X^\ell - \eta_k$  either splits off a linear factor or is irreducible, according as  $\eta_k$  is or is not an  $\ell$ -th power. This proves the equivalence of (ii) and (iii), while Capelli's lemma gives the equivalence of (iii) and (iv). The direct product  $\mathbb{F}_{2^{3^k}}^\times \simeq \mathbb{F}_{q_k}^\times \times U_k$  and  $\ell \nmid (q_k - 1)$  show that being an  $\ell$ -th power in the full field is equivalent, for  $\eta_k \in U_k$ , to lying in  $U_k^\ell$ . This is the equivalence with (i).  $\square$

**Remark 6.72** (Singer membership is not primitivity). The difference-set identification cannot replace the character test. Indeed, exponentiation

$$s_k : U_k \longrightarrow U_k, \quad s_k(y) = y^{q_k+1},$$

is an automorphism of order 6: one has  $\gcd(q_k + 1, |U_k|) = 1$  and  $(q_k + 1)^3 \equiv -1 \pmod{|U_k|}$ , while  $(q_k + 1)^2 \equiv q_k \not\equiv 1$ . The Singer equation is simply  $s_k(y) = y + 1$ . Thus its consequence  $\text{ord}(y + 1) = \text{ord}(y)$  is automatic order preservation by an automorphism and supplies no order divisor by itself. For example, take  $q = 32$  and let  $y$  be a root of  $X^3 + X^2 + 1$  over  $\mathbb{F}_2$ . Then  $y$  has order 7 and  $q + 1 \equiv 5 \pmod{7}$ ; the defining cubic gives  $y^5 + y + 1 = 0$ . Hence  $y \in \{x : x^{q+1} + x + 1 = 0\}$ , but  $y$  does not generate the norm-one group of order  $q^2 + q + 1 = 7 \cdot 151$ . What must be used in the Conway tower is the recursive coefficient  $\eta_{k-1}$  in  $(\star)$ , not Singer membership alone.

**Proposition 6.73** (prime-step closure). *If  $\Phi_{3^k}(2)$  is prime, then*

$$C_k \iff C_{k-1}.$$



Consequently  $C_1, C_2, C_3$  hold analytically: the three new kernel orders are respectively

$$7, \quad 73, \quad 262657,$$

and all are prime.

*Proof.* Proposition 6.24 proves that  $\eta_k$  has degree 3 over  $\mathbb{F}_{q_k}$ , so in particular  $\eta_k \neq 1$ . A nonidentity element of a group of prime order generates it, and  $(**)$  gives the first claim. The primality of 7 and 73 is immediate. For completeness,  $262657 - 1 = 2^9 \cdot 3^3 \cdot 19$ , and base 5 gives the Pocklington certificate

$$5^{262656} \equiv 1 \pmod{262657}, \quad \gcd\left(5^{262656/r} - 1, 262657\right) = 1 \quad (r = 2, 3, 19),$$

which proves the third primality assertion.  $\square$

**Proposition 6.74** (cubic-arm equivalence). *For primes  $r$  with  $f(r) \mid 3^k$  a 3-power:*

$$C_k \iff m_r = 1 \text{ for every prime } r \text{ with } f(r) \in \{3, 9, \dots, 3^k\}.$$

*The family formula is the 0/1/4 candidate of Section 1 restricted to the 3-power column.*

**Proposition 6.75** (cyclotomic-unit residue reduction). *Let  $h = 3^k$  and let  $\ell$  be any prime divisor of  $2^h - 1$ . Then*

$$\gamma_k = \zeta^{-1}(1 + \zeta)^2$$

*is an  $\ell$ -th power in  $\mathbb{F}_{2^h}$  if and only if  $1 + \zeta$  is an  $\ell$ -th power in  $\mathbb{F}_{2^{2h}}$ . The same equivalence holds at every power  $\ell^a \mid 2^h - 1$ ; non- $\ell$ -power already implies that  $\gamma_k$  has the full  $\ell$ -primary part of  $2^h - 1$  in its order.*

*Proof.* Because  $\ell \mid 2^h - 1$ , one has  $\ell \nmid 2^h + 1$ , so extension from  $\mathbb{F}_{2^h}^\times$  to  $\mathbb{F}_{2^{2h}}^\times$  preserves the  $\ell$ -primary quotient. Also  $\ell \neq 3$ , hence  $\zeta^{-1}$  is an  $\ell$ -th power in the  $3^{k+1}$ -group generated by  $\zeta$ ; squaring is an automorphism of every odd  $\ell$ -primary quotient. The displayed factorization proves both assertions.  $\square$

General multiplicative-order bounds for Gaussian periods do not imply maximal order [2]. Here the norm recursion  $\gamma_k^3 + \gamma_k = \gamma_{k-1}$  propagates the old factors, but a proof must still show that each new prime-power factor of  $\Phi_{3^k}(2)$  divides  $\text{ord}(\gamma_k)$ . Irreducibility of the corresponding Dickson polynomial is likewise weaker than primitivity [26]. Proposition 6.24 supplies the appropriate cubic replacement and places the root on a Singer difference set, but Remark 6.72 shows that Singer membership alone cannot prove generation.

## 6.6 The cubic-spine conjecture (C)

The universal cubic arm is Conjecture 6.1. The prime-step argument proves  $C_k$  analytically for  $k \leq 3$ , and Proposition 6.35 proves the required primary order above an explicit partition threshold at every level. The remaining task is the selected Kummer-class nonvanishing at the smaller new primary factors. Propositions 6.15 and 6.45 show respectively that the current ray Euler factor has a forced simple zero whose circular-unit regulator is exactly  $L(1)$ , and that the real norm criterion has no viable size bound. Proposition 8.32 rules out a generic cyclic-code or difference-set contradiction. Finite order checks, their factorization dependencies, and the distinction between local certificates and external primality records are isolated in Appendix A.3. Proposition 6.22 also closes the natural three-adic Coleman route: the circular unit has the explicit series  $(2+T)^2/(1+T)$ ,

but its algebraic  $\tilde{\lambda}x^{3^k-1}$ -specialization is forced to vanish modulo every current  $\ell$ , even at the proved level  $(k, \ell) = (2, 73)$ . After division by that exceptional factor one is left with a generalized Bernoulli value for which Coleman theory supplies no  $\ell$ -adic unit statement or cross-prime identification with the selected Kummer index.

## 7 The singleton-even Conway–Fermat arm

The candidate contains strong explicit maximal-order assertions even before the non-cyclotomic singleton chains are reached. This prevents a proof from following from field degree or irreducibility alone.

### 7.1 Maximal projective order

**Proposition 7.1** (the singleton-even arm is Conway–Fermat maximality). *For  $n \geq 0$ , put*

$$h_n = 2^{n+1}, \quad c_n = \kappa_{h_n} = 2^{2^n}, \quad E_n = \mathbb{F}_{2^{h_n}}, \quad F_n = 2^{2^n} + 1.$$

*With  $E_{-1} = \mathbb{F}_2$ , let*

$$\delta_n = \text{ord}(c_n E_{n-1}^\times) \quad \text{in} \quad E_n^\times / E_{n-1}^\times.$$

*Then  $1 < \delta_n \mid F_n$ , and the  $m = 0$  arm of Conjecture 1.1 for every prime  $p \mid F_n$  is equivalent to*

$$\delta_n = F_n.$$

*Proof.* The Conway finite-nimber tower satisfies

$$c_n^2 + c_n = c_{n-1}c_{n-2} \cdots c_0$$

(with the empty product 1). Thus  $E_n/E_{n-1}$  is quadratic and  $c_n \notin E_{n-1}$ , so the coset order is nontrivial. Its ambient quotient has order

$$\frac{2^{2^{n+1}} - 1}{2^{2^n} - 1} = F_n,$$

which proves  $1 < \delta_n \mid F_n$ ; equivalently the relative norm is  $c_n^{F_n} = c_{n-1} \cdots c_0$ .

If  $p \mid F_n$ , then  $2^{2^n} \equiv -1 \pmod{p}$ , hence  $f(p) = 2^{n+1} = h_n$ . Fermat numbers are pairwise coprime, so the entire  $p$ -primary part of  $|E_n^\times|$  lies in the quotient by  $E_{n-1}^\times$ . The full power criterion therefore gives  $m_p = 0$  iff  $v_p(\delta_n) = v_p(F_n)$ . Requiring this for every prime divisor of  $F_n$  is exactly  $\delta_n = F_n$ .  $\square$

**Corollary 7.2** (exact primitive-product equivalence). *Put  $a_n = \prod_{j=0}^n c_j$ . For every  $n \geq 2$ ,*

$$\text{ord}(c_n) = \text{ord}(a_n) = \prod_{j=1}^n \delta_j. \tag{PP_1}$$

*Consequently*

$$c_0 c_n \text{ is primitive in } E_n \iff \delta_j = F_j \quad (1 \leq j \leq n). \tag{PP_2}$$

*Thus the all-level Conway–Fermat arm is exactly the all-level primitive-product problem studied by Popovych [33], in both directions, rather than merely an analogous maximal-order question.*

*Proof.* Write  $q = 2^{2^n}$ . Since  $\gcd(q-1, q+1) = 1$ , the cyclic group  $E_n^\times$  is the direct product of  $E_{n-1}^\times$ , of order  $q-1$ , and its subgroup of order  $F_n = q+1$ . Write accordingly  $c_n = g_n u_n$ , where  $u_n \in E_{n-1}^\times$  and  $\text{ord}(g_n) = \delta_n$ . The tower norm gives

$$a_{n-1} = c_n^{q+1} = u_n^2,$$

so  $\text{ord}(u_n) = \text{ord}(a_{n-1})$ . The two factors have coprime orders, and therefore

$$\text{ord}(c_n) = \delta_n \text{ord}(a_{n-1}), \quad a_n = a_{n-1} c_n = g_n u_n^3.$$

At  $n = 1$ , the quotient has prime order  $F_1 = 5$ , so  $\delta_1 = 5$ ; the same decomposition gives  $u_1^3 = 1$  and therefore  $a_1 = g_1$ , of order 5. Inductively  $\text{ord}(u_n) = \prod_{j=1}^{n-1} \delta_j$ , which is prime to three because every  $\delta_j \mid F_j$  for  $j \geq 1$ . Cubing therefore preserves the order of  $u_n$ , proving (PP<sub>1</sub>).

The element  $c_0$  has order three, coprime to  $\text{ord}(c_n)$ , while

$$|E_n^\times| = 3 \prod_{j=1}^n F_j.$$

Hence  $c_0 c_n$  is primitive exactly when  $\prod_{j=1}^n \delta_j = \prod_{j=1}^n F_j$ . Since the Fermat numbers are pairwise coprime and  $\delta_j \mid F_j$ , this is equivalent to all the displayed equalities in (PP<sub>2</sub>).  $\square$

**Proposition 7.3** (the Conway and Wiedemann norm-one drivers are not Frobenius-identical). *Put  $q = |E_{n-1}| = 2^{2^n}$  and*

$$w_n = c_n^{q-1} = \frac{c_n + 1}{c_n}.$$

*The canonical isomorphism*

$$E_n^\times / E_{n-1}^\times \longrightarrow \ker(N_{E_n/E_{n-1}}), \quad z E_{n-1}^\times \longmapsto z^{q-1},$$

*preserves order, and therefore*

$$\text{ord}(w_n) = \delta_n, \quad w_n + w_n^{-1} = a_{n-1}^{-1}. \quad (\text{WW}_1)$$

*If  $d_n = a_{n-1}^{-1}$ , then the Conway drivers themselves obey*

$$d_{n+1} = d_n(w_n + 1) = \frac{(w_n + 1)^3}{w_n}. \quad (\text{WW}_2)$$

*This has the same norm-one shape as Wiedemann's iterated quadratic tower [41], whose selected elements satisfy*

$$x_n + x_n^{-1} = x_{n-1}, \quad \text{ord}(x_n) \mid F_n,$$

*but it is not the same recursion. For every  $n \geq 3$ , no  $\mathbb{F}_2$ -automorphism of an algebraic closure can send  $x_n$  to  $w_n$ . Thus Wiedemann maximality cannot be transferred to the Conway–Fermat arm by a Frobenius identification of the selected elements.*

*Proof.* The displayed quotient map is injective because its kernel is  $E_{n-1}^\times$ , and both groups have order  $q+1 = F_n$ , proving the order assertion. The Conway relation gives  $c_n(c_n + 1) = a_{n-1}$ , whence

$$w_n + w_n^{-1} = \frac{(c_n + 1)^2 + c_n^2}{c_n(c_n + 1)} = a_{n-1}^{-1}$$

in characteristic two. Also  $a_n = a_{n-1}c_n$  and  $c_n = (w_n + 1)^{-1}$ , so

$$d_{n+1} = d_n(w_n + 1) = (w_n + w_n^{-1})(w_n + 1) = \frac{(w_n + 1)^3}{w_n},$$

which proves (WW<sub>2</sub>).

If an  $\mathbb{F}_2$ -automorphism sent  $x_n$  to  $w_n$ , it would send  $x_{n-1} = x_n + x_n^{-1}$  to  $a_{n-1}^{-1}$ , and hence preserve their multiplicative orders. But  $\text{ord}(x_{n-1}) \mid F_{n-1}$ , whereas

$$\text{ord}(a_{n-1}^{-1}) = \prod_{j=1}^{n-1} \delta_j$$

by (PP<sub>1</sub>), and this order is divisible by  $\delta_1 = F_1 = 5$ . Fermat numbers are pairwise coprime, so  $5 \nmid F_{n-1}$  for  $n \geq 3$ , a contradiction.  $\square$

**Proposition 7.4** (the classical  $Q$ -transform stops at irreducibility). *Retain Proposition 7.1, put  $m = 2^n$ , let  $A_{n-1}$  be the minimal polynomial of  $a = a_{n-1}$ , and set*

$$f(X) = X^m A_{n-1}(X^{-1}), \quad u = \frac{c_n + 1}{c_n}.$$

*Then  $f$  is the minimal polynomial of  $a^{-1}$ , and the minimal polynomial of  $u$  is exactly the classical binary  $Q$ -transform*

$$f^Q(U) = U^m f(U + U^{-1}). \quad (\text{QT}_1)$$

*It is irreducible of degree  $2m$ . If  $e = \text{ord}(u)$ , the complete self-reciprocal order information supplied by this transformation is*

$$e \mid F_n, \quad \min\{j > 0 : 2^j \equiv \pm 1 \pmod{e}\} = m. \quad (\text{QT}_2)$$

*Every nontrivial divisor  $e \mid F_n$  satisfies (QT<sub>2</sub>). Thus the full  $Q$ -transform and self-reciprocal irreducibility theorems do not distinguish  $e = F_n$  from any proper fibotomic stratum.*

*Nor is the Conway tower obtained by iterating the  $Q$ -transform. After the quadratic lift one has*

$$a_n = \frac{u}{(u+1)^3}, \quad a_n^{-1} = \frac{(u+1)^3}{u}, \quad (\text{QT}_3)$$

*so the next step is a cubic pushforward followed by another quadratic preimage, not a second application of  $U \mapsto U + U^{-1}$ .*

*Proof.* The identity (WW<sub>1</sub>) gives

$$u + u^{-1} = a^{-1},$$

so  $u$  is a root of (QT<sub>1</sub>). The constant and linear coefficients of  $A_{n-1}$  are one by Proposition 7.26. More specifically, Proposition 7.29 gives  $\text{Tr}(a) = 1$ , so the coefficient of  $X^{m-1}$  in  $A_{n-1}$ , equivalently  $f'(0)$ , is one. The Varshamov–Garakov criterion in the form of Meyn and Götz [29] therefore proves that  $f^Q$  is irreducible of degree  $2m$ .

The order classification of self-reciprocal irreducibles [42] gives (QT<sub>2</sub>). Conversely, if  $1 < e \mid 2^m + 1$ , then  $2^m \equiv -1 \pmod{e}$ . The order of two modulo  $e$  divides  $2m$ , does not divide  $m$ , and  $m$  is a power of two; hence it is  $2m$ , and the first exponent giving either sign is  $m$ . This proves the exact limitation. Finally  $a = u/(u+1)^2$  and  $c_n = (u+1)^{-1}$ , which gives (QT<sub>3</sub>).  $\square$

**Proposition 7.5** (two-step quartic and exact Kummer dichotomy). *Let  $n \geq 2$ , put  $B = a_{n-2}$ ,  $K_0 = E_{n-2}$ , and  $c = c_n$ . Then the minimal polynomial of  $c$  over  $K_0$  is*

$$f_n(X) = X^4 + (B+1)X^2 + BX + B^3. \quad (\text{TQ}_1)$$

*For every prime  $\ell \mid F_n$ , the following are equivalent:*

- (i) *the order  $\delta_n$  contains the full  $\ell$ -primary part of  $F_n$ ;*
- (ii)  *$c$  is not an  $\ell$ -th power in  $E_n^\times$ ;*
- (iii)  *$f_n(X^\ell)$  is irreducible over  $K_0$ .*

*If these conditions fail, then  $f_n(X^\ell)$  is a product of exactly  $\ell$  distinct irreducible quartics over  $K_0$ . Thus a minimal-counterexample argument may be descended across two quadratic levels without discarding the selected coefficient  $B$ , but failure remains a complete quartic splitting pattern rather than a degree contradiction.*

*Proof.* Write  $a = a_{n-1} = Bc_{n-1}$ . The two consecutive Conway relations give

$$c_{n-1}^2 + c_{n-1} = B, \quad c^2 + c = a.$$

Multiplying the first by  $B^2$  and using  $a = Bc_{n-1}$  gives

$$a^2 + Ba + B^3 = 0.$$

Eliminating  $a = c^2 + c$  now gives  $f_n(c) = 0$ . Moreover  $B \neq 0$ , so  $c_{n-1} = a/B$  belongs to  $K_0(c)$ . Consequently

$$K_0(c) \supset K_0(c_{n-1}) = E_{n-1},$$

and adjoining  $c$  gives  $E_n$ . Since  $[E_n : K_0] = 4$ , the displayed quartic is the minimal polynomial.

Fix  $\ell \mid F_n$ . The entire  $\ell$ -primary part of  $E_n^\times$  lies in  $E_n^\times/E_{n-1}^\times$ , so (i) and (ii) are the exact cyclic power criterion. Also  $\mu_\ell \subset E_n$ . Capelli's lemma gives

$$f_n(X^\ell) \text{ irreducible over } K_0 \iff X^\ell - c \text{ irreducible over } E_n,$$

and Kummer theory over  $E_n$  identifies the latter condition with  $c \notin E_n^{\times\ell}$ . This proves the equivalence.

Under failure write  $c = t^\ell$  with  $t \in E_n$ . Every root of  $f_n(X^\ell)$  lies in  $E_n$ , since  $\mu_\ell \subset E_n$ . No such root lies in the unique intermediate field  $E_{n-1}$ : otherwise its  $\ell$ -th power, one of the four  $K_0$ -conjugates of  $c$ , would lie there, forcing  $c$  itself into  $E_{n-1}$ . Hence every root has degree four over  $K_0$ . The polynomial is separable and has  $4\ell$  roots, so it factors into exactly  $\ell$  distinct irreducible quartics.  $\square$

**Proposition 7.6** (coefficient conic of the bad quartics). *Retain Proposition 7.5, put  $Q = |K_0|$ ,  $L = E_{n-1}$ , and suppose that  $c = t^\ell$  in  $E_n$ . Set*

$$r = N_{E_n/L}(t), \quad e = r + r^Q, \quad d = rr^Q.$$

*Then*

$$r^\ell = a_{n-1}, \quad e \neq 0, \quad d^\ell = B^3. \quad (\text{TQ}_2)$$

*For  $\zeta \in \mu_\ell$ , put*

$$y_\zeta = t\zeta, \quad s_\zeta = y_\zeta + y_\zeta^{Q^2}.$$

Writing  $\mathcal{D}_\ell(S, P)$  for the Dickson power sum of Proposition 7.31, one has

$$\mathcal{D}_\ell(s_\zeta, r) = 1. \quad (\text{TQ}_3)$$

The  $\ell$  irreducible quartic factors of  $f_n(X^\ell)$  are exactly

$$\begin{aligned} g_{s_\zeta}(X) &= N_{L/K_0}(X^2 + s_\zeta X + r) \\ &= X^4 + \alpha_\zeta X^3 + \beta_\zeta X^2 + \gamma_\zeta X + d, \end{aligned}$$

where

$$\alpha_\zeta = s_\zeta + s_\zeta^Q, \quad \beta_\zeta = e + s_\zeta s_\zeta^Q, \quad \gamma_\zeta = s_\zeta r^Q + s_\zeta^Q r.$$

Every coefficient triple lies on the conic

$$\gamma_\zeta^2 + \alpha_\zeta e \gamma_\zeta + \alpha_\zeta^2 d = e^2(\beta_\zeta + e). \quad (\text{TQ}_4)$$

Conversely,  $\alpha_\zeta$  and  $\gamma_\zeta$  recover

$$s_\zeta = \frac{\gamma_\zeta + \alpha_\zeta r}{e},$$

and  $\beta_\zeta$  is then forced. Thus the coefficient conic is a lossless change of coordinates on the same Dickson fibre  $\mathcal{D}_\ell(S, r) = 1$ ; it supplies no independent obstruction to the bad factorization.

*Proof.* Taking the norm of  $t^\ell = c$  from  $E_n$  to  $L$  gives  $r^\ell = a_{n-1}$ . If  $r$  belonged to  $K_0$ , then so would  $a_{n-1} = Bc_{n-1}$ ; since  $B \neq 0$  and  $c_{n-1} \notin K_0$ , this is impossible. Hence  $e \neq 0$ . A second norm and the quadratic equation for  $a_{n-1}$  give

$$d^\ell = N_{L/K_0}(a_{n-1}) = B^3.$$

Because  $\ell \mid Q^2 + 1$ , every  $\zeta \in \mu_\ell$  has norm one from  $E_n$  to  $L$ . Thus  $y_\zeta$  has relative trace  $s_\zeta$  and norm  $r$ . The Dickson identity gives

$$\mathcal{D}_\ell(s_\zeta, r) = y_\zeta^\ell + (y_\zeta^{Q^2})^\ell = c + c^{Q^2} = 1.$$

The traces  $s_\zeta$  are distinct: equality would identify the two unordered root pairs of the corresponding quadratic; the crossed identification has  $\ell$ -th powers  $c$  and  $c + 1$ , which is impossible. Taking the  $Q$ -conjugate norm of these quadratics gives all  $\ell$  quartics and the displayed coefficient formulas; their irreducibility was proved in Proposition 7.5.

Finally

$$\gamma_\zeta + \alpha_\zeta r = e s_\zeta.$$

Taking the norm from  $L$  to  $K_0$  and expanding both sides gives  $(\text{TQ}_4)$ . Since  $e \neq 0$ , the same identity recovers  $s_\zeta$ , after which  $\beta_\zeta = e + s_\zeta s_\zeta^Q$  is forced. This proves the lossless assertion.  $\square$

**Proposition 7.7** (Fibonacci-polynomial criterion). *Fix  $n \geq 0$ , put  $m = 2^n$ ,  $q = 2^m$ , and write  $K = E_{n-1} = \mathbb{F}_q$ ,  $L = E_n = \mathbb{F}_{q^2}$  and  $a = a_{n-1} = \prod_{j < n} c_j$ . If*

$$S_0(X) = 0, \quad S_1(X) = 1, \quad S_{r+2}(X) = S_{r+1}(X) + X S_r(X),$$

then

$$\delta_n = \min\{r > 0 : S_r(a) = 0\},$$

and hence

$$\delta_n = F_n \iff S_{F_n/\ell}(a) \neq 0 \quad \text{for every prime } \ell \mid F_n. \quad (\text{F}_n)$$

More precisely, for odd  $d > 1$  define the fibotomic polynomial

$$\mathfrak{F}_d(X) = \prod_{\substack{\{v, v^{-1}\} \\ \text{ord}(v)=d}} \left( X - \frac{v}{(v+1)^2} \right),$$

where the product takes one representative of each inverse-pair of primitive  $d$ -th roots. Then  $\mathfrak{F}_d \in \mathbb{F}_2[X]$ ,

$$S_r(X) = \prod_{\substack{d|r \\ d>1}} \mathfrak{F}_d(X) \quad (r \text{ odd}), \quad (\text{B}_r)$$

and, if  $A_{n-1}$  is the minimal polynomial of  $a_{n-1}$  over  $\mathbb{F}_2$ ,

$$\delta_n = d \iff A_{n-1} \mid \mathfrak{F}_d.$$

Every irreducible factor of every  $\mathfrak{F}_d$  with  $1 < d \mid F_n$  has the same degree  $m$ . Thus the Conway–Fermat conjecture is exactly the assertion  $A_{n-1} \mid \mathfrak{F}_{F_n}$ ; field degree cannot distinguish that stratum from any proper-divisor stratum.

*Proof.* Put  $c = c_n$  and  $u = (c+1)/c$ . Since  $c^2 + c = a$  and  $c^q = c+1$ , the element  $u$  lies in the norm-one group of order  $q+1 = F_n$ , and  $\delta_n = \text{ord}(u)$ . The two roots  $c, c+1$  of  $T^2 + T + a$  give the Binet identity

$$S_r(a) = c^r + (c+1)^r = \frac{1+u^r}{(1+u)^r}. \quad (\text{B})$$

This proves the first criterion. In a cyclic group of order  $F_n$  an element has full order precisely when its  $F_n/\ell$ -th power is nontrivial for every prime  $\ell \mid F_n$ , proving  $(\text{F}_n)$ .

For odd  $r$ , the nontrivial  $r$ -th roots  $v$  occur in inverse pairs, and the common value  $v/(v+1)^2$  of a pair is, by (B), a root of  $S_r$ . Frobenius permutes the pairs, so the displayed factors have coefficients in  $\mathbb{F}_2$ . Both sides of  $(\text{B}_r)$  are monic of degree  $(r-1)/2$ , which proves the factorization and the divisibility criterion.

Finally  $2^m \equiv -1 \pmod{d}$  for every nontrivial  $d \mid F_n$ . Because  $m$  is a power of two,  $\text{ord}_d(2) = 2m$ , and the least  $j > 0$  with  $2^j \equiv \pm 1 \pmod{d}$  is  $m$ . This is exactly the Frobenius orbit length of  $v/(v+1)^2$ : indeed

$$\frac{v}{(v+1)^2} = \frac{w}{(w+1)^2} \iff (v+w)(vw+1) = 0,$$

so only the inverse pair  $\{v, v^{-1}\}$  has the same image. This proves the equal-degree assertion.  $\square$

**Proposition 7.8** (trailing-zero compression and its exact circularity). *Let  $K$  be a characteristic-two field, let  $a \in K$ , and put*

$$T_t(a) = \sum_{j=0}^{t-1} a^{2^j}.$$

*For the Fibonacci values of Proposition 7.7, one has, for all  $t, h \geq 0$ ,*

$$\boxed{S_{2^t h+1}(a) = S_{h+1}(a)^{2^t} + T_t(a) S_h(a)^{2^t}.} \quad (\text{TZ}_1)$$

*If  $S_h(a) \neq 0$ , this is equivalently*

$$\boxed{S_{2^t h+1}(a) = 0 \iff \left( \frac{S_{h+1}(a)}{S_h(a)} \right)^{2^t} = T_t(a).} \quad (\text{TZ}_2)$$

Now retain the Conway–Fermat notation, assume  $n \geq 2$ , and put  $m = 2^n$ . Let  $d > 1$  divide  $F_n$ , write

$$d = 1 + 2^t h, \quad 0 < t \leq m,$$

where  $t$  is a chosen power-of-two divisor of  $d - 1$ , not necessarily its exact 2-adic valuation, so  $h$  need not be odd. Suppose  $\delta_n \mid d$  and  $\delta_n > 2^t$ . Then

$$S_h(a) \neq 0, \quad S_{h+1}(a) \neq 0. \quad (\text{TZ}_3)$$

In particular, for a failed prime-factor test  $d = F_n/\ell > 1$ , one may take  $t = n + 2$ : every nontrivial divisor of  $F_n$  is  $1 \pmod{2^{n+2}}$ , and so is larger than  $2^{n+2}$ . Writing

$$R_h = \frac{S_{h+1}(a)}{S_h(a)},$$

the hypothetical failure then has the choice-free form

$$\boxed{R_h = 1 + T_{m-t}(a)}. \quad (\text{TZ}_4)$$

This sharper formula is nevertheless automatic on every relevant proper-order stratum. Indeed, if  $u = (c_n + 1)/c_n$  has order  $\delta_n \mid d$ , and  $k = 2^{m-t}$ , then

$$2^t h \equiv -1 \pmod{\delta_n}, \quad 2^m \equiv -1 \pmod{\delta_n} \implies h \equiv k \pmod{\delta_n}. \quad (\text{TZ}_5)$$

Thus the continued-fraction, Cassini, trace, and Frobenius identities obtained functorially from  $(\text{TZ}_4)$  hold on every corresponding order- $\delta_n$  stratum. By themselves they do not distinguish the selected Conway factor from a proper fibotomic factor; new full-ancestry information could still make such an identity useful.

*Proof.* In characteristic two the doubling identities are

$$S_{2r} = S_r^2, \quad S_{2r+1} = S_{r+1}^2 + aS_r^2.$$

Induction on  $t$ , using  $T_{t+1} = T_t^2 + a$ , proves  $(\text{TZ}_1)$ ; division by  $S_h^{2^t}$  proves  $(\text{TZ}_2)$ .

By Proposition 7.7, the zero indices are exactly the multiples of  $\delta_n$ . If  $\delta_n \mid h$ , then  $\delta_n \mid 2^t h$  and  $\delta_n \mid 2^t h + 1 = d$ , forcing  $\delta_n \mid 1$ . If  $\delta_n \mid h + 1$ , then

$$\delta_n \mid 2^t(h + 1) - d = 2^t - 1,$$

contrary to  $\delta_n > 2^t$ . This proves  $(\text{TZ}_3)$ . The standard prime-divisor congruence for  $F_n$  gives the stated choice  $t = n + 2$ , and  $n + 2 \leq 2^n = m$  for  $n \geq 2$ .

The absolute trace of  $a = a_{n-1}$  is one, so

$$1 = T_m(a) = T_t(a) + T_{m-t}(a)^{2^t}.$$

Under failure,  $(\text{TZ}_2)$  therefore gives

$$R_h^{2^t} = (1 + T_{m-t}(a))^{2^t}.$$

Frobenius is injective, proving  $(\text{TZ}_4)$ .



Finally multiply  $2^t h \equiv -1 \pmod{\delta_n}$  by  $2^{m-t}$  and use  $2^m \equiv -1 \pmod{\delta_n}$  to obtain (TZ<sub>5</sub>). Hence  $u^h = u^k$ . The Binet identity and  $S_k(a) = c_n^k + (c_n + 1)^k = 1$  give the explicit calculation

$$T_{m-t}(a) = c_n^k + c_n,$$

$$R_h = c_n \frac{1 + u^{h+1}}{1 + u^h} = c_n \frac{1 + u^{k+1}}{1 + u^k} = c_n^k + c_n + 1.$$

Thus (TZ<sub>4</sub>) is automatic on the whole order- $\delta_n$  stratum; the identities derived from it alone cannot select the Conway ancestry. Conversely (TZ<sub>2</sub>) returns the original zero  $S_d(a) = 0$ . The compression is exact but supplies no independent nonvanishing invariant.  $\square$

**Proposition 7.9** (full-ancestry trace descent is the lower tautology). *For  $i \geq 0$ , put  $d = 2^i$ ,  $K_0 = E_{i-1} = \mathbb{F}_{2^d}$ , and  $K = E_i = \mathbb{F}_{2^{2d}}$ . Write  $A = a_{i-1} \in K_0$ ,  $a = a_i \in K$ , and suppose  $0 \leq t \leq d$ . Thus*

$$a^{2^d} = a + A, \quad T_d(A) = 1.$$

*For every integer  $D \geq t$ , put*

$$C_{D,t}(x) = 1 + T_{D-t}(x).$$

*Then*

$$\boxed{\text{Tr}_{K/K_0}(C_{2d,t}(a)) = C_{d,t}(A)}. \quad (\text{CT}_1)$$

*The value on the right is exactly the lower full-Fermat continued fraction:*

$$C_{d,t}(A) = \frac{S_{2^{d-t+1}}(A)}{S_{2^{d-t}}(A)}. \quad (\text{CT}_2)$$

*Consequently, when the hypothetical ratio in (TZ<sub>4</sub>) is descended through the actual quadratic ancestry, every eligible relative trace lands on the lower level's unconditional full-Fermat zero identity. Relative trace therefore preserves all ancestry but produces no contradiction to a proper top order.*

*Proof.* Partial Frobenius traces are additive and commute with Frobenius powering. Hence  $a^{2^d} = a + A$  gives

$$\begin{aligned} \text{Tr}_{K/K_0}(1 + T_{2d-t}(a)) &= T_{2d-t}(A) \\ &= T_d(A) + T_{d-t}(A) = 1 + T_{d-t}(A), \end{aligned}$$

which is (CT<sub>1</sub>). The doubling formulas used in Proposition 7.8 give

$$S_{2^r}(A) = 1, \quad S_{2^{r+1}}(A) = 1 + T_r(A).$$

Taking  $r = d - t$  proves (CT<sub>2</sub>). The formal identities—additivity, Frobenius commutation, index splitting, and the first displayed relative-trace calculation—are independently kernel-checked in `Ogdoad.Excess`.  $\square$

## 7.2 The selected factor and the limits of one-step induction

**Proposition 7.10** (coarse top-step data cannot force maximality). *For every divisor  $d > 1$  of  $F_n$  there are elements  $a_d \in \mathbb{F}_q$  and  $c_d \in \mathbb{F}_{q^2}$  such that*

$$[\mathbb{F}_2(a_d) : \mathbb{F}_2] = m, \quad [\mathbb{F}_2(c_d) : \mathbb{F}_2] = 2m, \quad c_d^q = c_d + 1, \quad c_d^2 + c_d = a_d,$$

$$\text{Tr}_{\mathbb{F}_q/\mathbb{F}_2}(a_d) = 1, \quad \text{N}_{\mathbb{F}_{q^2}/\mathbb{F}_q}(c_d) = a_d,$$

*but the order of  $c_d \mathbb{F}_q^\times$  is exactly  $d$ . In particular all proper-divisor strata obey the same degree, trace, Artin–Schreier and relative-norm data as the Conway top step.*

*Proof.* Choose  $v$  of order  $d$  in the norm-one subgroup of  $\mathbb{F}_q^\times$  and put

$$c_d = \frac{1}{v+1}, \quad a_d = \frac{v}{(v+1)^2}.$$

Then  $v^q = v^{-1}$ , so  $c_d^q = c_d + 1$ ; the remaining displayed identities follow directly. The orbit calculation in the proof of Proposition 7.7 gives the two field degrees. Consequently  $T^2 + T + a_d$  is irreducible over  $\mathbb{F}_q$ , which is equivalent to the stated absolute trace, and  $(c_d + 1)/c_d = v$  gives quotient order  $d$ .  $\square$

**Proposition 7.11** (the exact recursively selected factor). *Set  $a_{-1} = 1$ ,  $a_i = \prod_{j \leq i} c_j$ , and let  $A_i$  be its minimal polynomial over  $\mathbb{F}_2$ , with  $A_{-1}(Y) = Y + 1$ . Then*

$$a_i^2 + a_{i-1}a_i + a_{i-1}^3 = 0$$

and

$$A_i(X) = \text{Res}_Y(A_{i-1}(Y), X^2 + YX + Y^3). \quad (\text{R}_i)$$

Hence the unresolved Conway–Fermat lemma is the precise recursive claim that the factor selected by  $(\text{R}_i)$  lies in the full-order fibotomic stratum, rather than in any equal-degree  $\mathfrak{F}_d$  with  $d \mid F_{i+1}$  and  $d < F_{i+1}$ .

*Proof.* Multiplying  $c_i^2 + c_i = a_{i-1}$  by  $a_{i-1}^2$  gives the first identity. We prove the degree assertion and the resultant formula inductively from  $a_{-1} = 1$ . The resultant in  $(\text{R}_i)$  is monic of degree  $2 \deg A_{i-1}$  and vanishes at  $a_i$ . Since  $c_i = a_i/a_{i-1} \notin E_{i-1}$ , one has  $a_i \notin E_{i-1}$ . But  $E_i/\mathbb{F}_2$  has degree  $2^{i+1}$ , and every proper subfield of  $E_i$  is contained in its unique index-two subfield  $E_{i-1}$ . Therefore  $a_i$  has full degree  $2^{i+1} = 2 \deg A_{i-1}$ , so the resultant is its minimal polynomial.  $\square$

**Proposition 7.12** (rational normalization, unique parents, and finite-field collapse). *For a monic polynomial  $A \in \mathbb{F}_2[Y]$ , write*

$$\mathcal{T}(A)(X) = \text{Res}_Y(A(Y), X^2 + YX + Y^3).$$

The correspondence  $X^2 + YX + Y^3 = 0$  has rational normalization  $\mathbf{P}_c^1$ , with

$$Y = \alpha(c) := c^2 + c, \quad X = \beta(c) := c^3 + c^2. \quad (\text{RDy}_1)$$

Consequently, after base change to  $\overline{\mathbb{F}}_2$ , its transform on effective zero-cycles on the  $X$ -line is

$$\text{div}_0 \mathcal{T}(A) = \beta_* \alpha^* \text{div}_0(A). \quad (\text{RDy}_2)$$

If  $A$  is irreducible of degree  $m$  and a root of  $A$  has absolute trace one, then  $\mathcal{T}(A)$  is irreducible of degree  $2m$ , and its roots again have trace one. On this locus  $\mathcal{T}$  is injective. More precisely, if  $B = \mathcal{T}(A)$ ,  $x$  is a root of  $B$ , and  $q = 2^m$ , then the parent root and hence its minimal polynomial are recovered by

$$y = x^q + x. \quad (\text{RDy}_3)$$

Conversely, let  $B$  be an irreducible trace-one polynomial of degree  $2m$ , let  $x$  be one of its roots, and put  $y = x^{2^m} + x$ . Then  $B$  has a parent under  $\mathcal{T}$  if and only if

$$\boxed{x^{2^m+1} = y^3}. \quad (\text{RDy}_4)$$

When this holds, the parent is unique.

The rational dynamics adds no shorter selector. If  $H \in \overline{\mathbb{F}}_2(t)$  satisfies

$$H \circ \alpha = H \circ \beta,$$

then  $H$  is constant; neither  $\alpha$  nor  $\beta$  factors rationally through the other. Their critical portrait is likewise universal:  $\alpha$  branches only at infinity,  $\beta$  has critical values  $0, \infty$ , and that two-point set is invariant. Every trace-one orbit avoids it. Thus all trace-one seeds of the same degree  $m$  have iterates irreducible of degrees  $2^r m$ , with common splitting fields  $\mathbb{F}_{2^{2^r m}}$ .

In the norm-one coordinates

$$\Phi(v) = \frac{v}{(v+1)^2}, \quad \Psi(v) = \frac{v}{(v+1)^3},$$

the transition is  $\Phi(w) = \Psi(v)$ . There is no rational selection  $w = R(v)$ , since it would force  $2 \deg R = 3$ . Hence critical-orbit, splitting-field, degree, and rational-first-integral data do not select the full Fermat stratum. The actual  $A_i$  is instead the unique node whose repeated parent chain ends at  $A_{-1} = X + 1$ ; applying  $(\text{RDy}_4)$  along that chain is exactly the existing full trace–norm ancestry, not a new nonvanishing invariant.

*Proof.* In the function field of the correspondence put  $c = X/Y$ . Then  $c^2 + c + Y = 0$ , which gives  $(\text{RDy}_1)$ ; the defining property of the resultant gives  $(\text{RDy}_2)$ .

Let  $y$  be a trace-one root of  $A$ , choose  $c^2 + c = y$ , and put  $x = yc$ . The Artin–Schreier criterion makes  $Z^2 + Z + y$  irreducible over  $\mathbb{F}_{2^m}$ , so

$$c^{2^m} = c + 1, \quad x^{2^m} = x + y.$$

Therefore  $y = x^{2^m} + x$  and  $c = x/y$  both belong to  $\mathbb{F}_2(x)$ , whence  $x$  has degree  $2m$ . The monic degree- $2m$  resultant is its minimal polynomial, and trace transitivity gives  $\text{Tr}_{2m/1}(x) = \text{Tr}_{m/1}(y) = 1$ . This also proves parent recovery and injectivity.

For a child, the two relative conjugates are  $x, x + y$ , so a parent necessarily gives

$$x^{2^m+1} = x(x + y) = y^3.$$

Conversely assume  $(\text{RDy}_4)$  and put  $c = x/y$ . Then

$$c^2 + c = \frac{x(x + y)}{y^2} = y.$$

The element  $y$  cannot lie in a proper subfield of  $\mathbb{F}_{2^m}$ : otherwise  $\mathbb{F}_2(y, c)$ , and hence  $\mathbb{F}_2(x)$ , would have degree strictly less than  $2m$ . Thus its minimal polynomial has degree  $m$ . Moreover trace transitivity and the trace-one hypothesis on  $B$  give  $\text{Tr}_{m/1}(y) = \text{Tr}_{2m/1}(x) = 1$ . The preceding construction therefore shows that its transform is  $B$ . This proves  $(\text{RDy}_4)$  and uniqueness.

Degrees of nonconstant rational maps multiply. Hence  $H \circ \alpha = H \circ \beta$  would give  $2 \deg H = 3 \deg H$ , while a factorization of either projection through the other would give  $2 = 3 \deg R$  or  $3 = 2 \deg R$ . The derivative and pole calculations for  $c^2 + c$  and  $c^2(c + 1)$  give the stated critical portrait. Iterating the first part proves the common finite-field tower. Finally  $\Phi(R(v)) = \Psi(v)$  has the same impossible degree equation, and the unique-parent conclusion follows from  $(\text{RDy}_3)$ . Lean independently checks  $\beta = c\alpha$ ,  $\alpha(c + 1) = \alpha(c)$ ,  $\beta(c + 1) = \beta(c) + \alpha(c)$ , the curve equation, and  $\beta(c)\beta(c + 1) = \alpha(c)^3$  in `Ogdoad.Excess`.  $\square$

**Proposition 7.13** (the full ancestry chain is an étale zero–unit dichotomy). *For  $n \geq 1$ , put*

$$I_n = (X_0^2 + X_0 + 1, \\ X_i^2 + X_{i-1}X_i + X_{i-1}^3 : 1 \leq i < n) \subseteq \mathbb{F}_2[X_0, \dots, X_{n-1}].$$

*Evaluation at the complete Conway ancestry is an isomorphism*

$$\mathbb{F}_2[X_0, \dots, X_{n-1}]/I_n \xrightarrow{\sim} E_{n-1}, \quad X_i \mapsto a_i. \quad (\text{FE}_1)$$

*Consequently*

$$I_n \cap \mathbb{F}_2[X_{n-1}] = (A_{n-1}(X_{n-1})),$$

*and, for every  $R \in \mathbb{F}_2[X]$ ,*

$$I_n + (R(X_{n-1})) = \begin{cases} I_n, & A_{n-1} \mid R, \\ (1), & A_{n-1} \nmid R. \end{cases} \quad (\text{FE}_2)$$

*The quotient in (FE<sub>1</sub>) is finite étale of degree  $2^n$ ; geometrically its points form one free Frobenius orbit. Moreover*

$$\det(m_{R(a_{n-1})} : E_{n-1} \rightarrow E_{n-1}) = N_{E_{n-1}/\mathbb{F}_2}(R(a_{n-1})) = \text{Res}(A_{n-1}, R). \quad (\text{FE}_3)$$

*Thus for  $R = S_d$ , adjoining the complete ancestry equations and the hypothetical proper-order equation produces no intermediate intersection: it gives the whole selected orbit or the empty scheme. Proving the second case is exactly  $A_{n-1} \nmid S_d$ , the original Conway–Fermat target.*

*Proof.* Successively adjoining the variables in their displayed monic quadratic equations makes the source of (FE<sub>1</sub>) a vector space of dimension at most  $2^n$ . Evaluation is onto: its image contains all the  $a_i$ , and the top one generates  $E_{n-1}$  by Proposition 7.11. Since  $[E_{n-1} : \mathbb{F}_2] = 2^n$ , evaluation is an isomorphism. The elimination and zero–unit assertions are then just divisibility by the minimal polynomial and the fact that the quotient is a field.

The Jacobian of the displayed generators is lower triangular with diagonal  $1, X_0, \dots, X_{n-2}$ . Every  $a_i$  is nonzero, so it is nonsingular at every geometric point; Frobenius transitivity follows from the full degree of  $a_{n-1}$ . Finally, the determinant of multiplication by an element is its field norm, and the norm of  $R(a_{n-1})$  is the displayed monic resultant.  $\square$

**Proposition 7.14** (odd fibotomic intersections are reduced). *For every odd  $d > 1$ , the Fibonacci polynomial  $S_d(X)$  is squarefree over  $\mathbb{F}_2$ . More explicitly, if  $u^d = 1$ ,  $u \neq 1$ , and  $\Phi(u) = u/(1+u)^2$ , then*

$$S_d(\Phi(u)) = 0, \quad S'_d(\Phi(u)) = \frac{1}{u(1+u)^{d-2}} \neq 0. \quad (\text{FS}_1)$$

*Equivalently, in characteristic two,*

$$S'_{2r}(X) = 0, \quad S'_{2r+1}(X) = S_r(X)^2. \quad (\text{FS}_2)$$

*Hence if a selected factor  $A_{n-1}$  divides  $S_d$ , it occurs with multiplicity one. Writing  $S_d = A_{n-1}Q$ , its higher Hasse derivatives at  $a = a_{n-1}$  merely give*

$$S_d^{[j]}(a) = \sum_{s=1}^j A_{n-1}^{[s]}(a) Q^{[j-s]}(a), \quad (\text{FS}_3)$$

*which recursively determines the quotient jets because  $A'_{n-1}(a) \neq 0$ . Tangent multiplicity and local jets therefore do not strengthen the selected nondivisibility problem by themselves.*

*Proof.* The Binet parametrization gives

$$S_d(\Phi(u)) = \frac{1 + u^d}{(1 + u)^d}.$$

Differentiate at a nontrivial  $d$ -th root  $u$ . Since  $d$  is odd and the characteristic is two, the derivative of the right side is  $u^{-1}(1 + u)^{-d}$ , while  $\Phi'(u) = (1 + u)^{-2}$ . This proves (FS<sub>1</sub>). The inverse-pair parametrization  $u \sim u^{-1}$  supplies all  $(d - 1)/2 = \deg S_d$  roots, so  $S_d$  is squarefree.

Alternatively, differentiating the Fibonacci recurrence and using the doubling identities proves (FS<sub>2</sub>) by induction. This derivative identity and its nonvanishing consequence are independently kernel-checked in `Ogdoad.Excess`. Formula (FS<sub>3</sub>) is the Hasse product rule after evaluating at a root of  $A_{n-1}$ .  $\square$

**Proposition 7.15** (the natural ancestry filtration loses Frobenius). *Retain the full-chain ideal  $I_n$  of Proposition 7.13 and give  $X_i$  weight  $2^i$ . For every monomial order refining this weight, the displayed generators of  $I_n$  are a Gröbner basis with initial ideal*

$$(X_0^2, \dots, X_{n-1}^2).$$

Consequently the associated graded algebra is

$$\text{gr } E_{n-1} \simeq \mathbb{F}_2[x_0, \dots, x_{n-1}]/(x_0^2, \dots, x_{n-1}^2), \quad (\text{GF}_1)$$

and Frobenius annihilates every positive-degree element of this graded algebra. Thus a max-weight, Newton-polygon, or unique-leading-monomial argument cannot propagate a nonzero leading class through the squaring steps in Fibonacci fast doubling; each such step necessarily falls to uncontrolled lower filtration.

The exact loss at one Conway edge can be written without division. If

$$x^2 = Ax + A^3, \quad S_r(x) = U_r + xV_r,$$

then

$$\begin{aligned} U_{2r} &= U_r^2 + A^3V_r^2, & V_{2r} &= AV_r^2, \\ U_{2r+1} &= U_{r+1}^2 + A^3V_{r+1}^2 + A^4V_r^2, & V_{2r+1} &= AV_{r+1}^2 + U_r^2 + (A^3 + A^2)V_r^2. \end{aligned} \quad (\text{GF}_2)$$

The three summands in either odd update may cancel in every leading grade. Moreover every ordered multiplicative valuation on the finite field  $E_{n-1}$  is trivial. Hence replacing the weight filtration by a genuine field valuation cannot repair the loss. This rules out a leading-term proof by itself; it does not rule out an induction that controls the complete lower normal form by new ancestry-sensitive information.

*Proof.* The leading term of the first chain equation is  $X_0^2$ . For  $i > 0$ , the three terms have weights

$$2^{i+1}, \quad 3 \cdot 2^{i-1}, \quad 3 \cdot 2^{i-1},$$

so the unique leading term is  $X_i^2$ . These leading monomials are pairwise coprime; Buchberger's coprime criterion proves the Gröbner-basis claim and (GF<sub>1</sub>). In characteristic two, the square of a sum of squarefree positive-degree monomials is the sum of their squares, each of which contains some  $x_i^2$ , so it vanishes in the associated graded.

The characteristic-two doubling identities are

$$S_{2r} = S_r^2, \quad S_{2r+1} = S_{r+1}^2 + xS_r^2.$$

Substitute  $S_r = U_r + xV_r$ , reduce every  $x^2$  by  $Ax + A^3$ , and collect the constant and  $x$ -coefficients to obtain  $(\mathbf{GF}_2)$ . Both algebraic reductions are independently kernel-checked in the Lean companion. Finally, if  $v$  is an ordered multiplicative valuation and  $z \neq 0$ , then  $z^{|E_{n-1}^\times|} = 1$ , so  $|E_{n-1}^\times|v(z) = 0$  and hence  $v(z) = 0$ .  $\square$

**Proposition 7.16** (the top-bit block is the same projective-order problem). *Fix one Conway edge, put*

$$A = a_{i-1}, \quad c = c_i, \quad a_i = Ac, \quad c^2 = c + A,$$

*and let  $M$  be the regular multiplication matrix of  $A$  on  $E_{i-1}$ . In the  $E_{i-1}$ -decomposition  $E_i = E_{i-1} \oplus cE_{i-1}$ , multiplication by the literal top Conway bit  $a_i$  has block matrix*

$$M_i = \begin{pmatrix} 0 & M^2 \\ M & M \end{pmatrix}. \quad (\text{CB}_1)$$

*Put*

$$D = \text{diag}(M, M), \quad C_M = \begin{pmatrix} 0 & M \\ I & I \end{pmatrix}.$$

*Then  $M_i = DC_M = C_M D$ , and for  $r \geq 1$ ,*

$$C_M^r = \begin{pmatrix} MS_{r-1}(M) & MS_r(M) \\ S_r(M) & S_{r+1}(M) \end{pmatrix}. \quad (\text{CB}_2)$$

*Consequently*

$$C_M^r \text{ is a lower-field scalar} \iff S_r(A) = 0,$$

*where a lower-field scalar means  $\text{diag}(N_b, N_b)$ , with  $N_b$  regular multiplication by some  $b \in E_{i-1}$ . Equivalently, this is a scalar in  $\text{GL}_2(E_{i-1})$  after restriction of scalars. The first return of either  $C_M$  or  $M_i$  in  $\text{PGL}_2(E_{i-1})$  is exactly  $\delta_i$ . Moreover*

$$\begin{aligned} \det(XI - M_i) &= \det(X^2I + XM + M^3) \\ &= \text{Res}_Y(A_{i-1}(Y), X^2 + YX + Y^3). \end{aligned} \quad (\text{CB}_3)$$

*Thus the nested multiplication matrix retains the complete Conway ancestry, but its characteristic polynomial is the existing selected resultant and its Singer/projective order is the existing Fibonacci first zero. A conjugacy to a Singer generator would already prove the Conway–Fermat conjecture; it is not an independent route to it.*

*Proof.* For  $x_0, x_1 \in E_{i-1}$ ,

$$(Ac)(x_0 + cx_1) = A^2x_1 + c(Ax_0 + Ax_1),$$

which proves  $(\text{CB}_1)$ . The factorization by  $D$  is immediate, and its factors commute because every block is a polynomial in  $M$ . Induction using  $S_{r+2} = S_{r+1} + XS_r$  proves  $(\text{CB}_2)$ . Its off-diagonal blocks vanish exactly when  $S_r(M) = 0$ ; evaluating regular multiplication at  $A$  identifies this with  $S_r(A) = 0$ . The Fibonacci criterion now gives the projective order.

Finally the commuting-block determinant formula gives the first line of  $(\text{CB}_3)$ , and regular multiplication turns that determinant into the displayed field norm and resultant. The coordinate identity, companion-power formula, and scalar equivalence are independently kernel-checked in `Ogdoad.Excess`.  $\square$

**Proposition 7.17** (the first-failure cube character is automatic). *Assume  $n \geq 2$ , put*

$$q = 2^{2^n}, \quad Q = \sqrt{q},$$

*and let  $v = u_{n-1}$  be the previous norm-one driver. Exact Conway ancestry gives*

$$a_{n-1} = \frac{v}{(v+1)^3}. \quad (\text{CU}_1)$$

*Since  $v$  lies in the torus of order  $Q+1$  and  $3 \nmid Q+1$ , choose  $\rho$  in that torus with  $\rho^3 = v$ . Then*

$$a_{n-1} = \left( \frac{\rho}{v+1} \right)^3. \quad (\text{CU}_2)$$

*Thus the old coefficient is automatically a cube before its multiplicative order is used. If  $n$  is a first failed level, the earlier maximalities give*

$$\text{ord}(a_{n-1}) = \frac{q-1}{3};$$

*this says only that the already cubic element generates the cube subgroup.*

*The fibotomic coordinate has an exact compatible transport. Put  $\Phi(x) = x/(x+1)^2$ . Whenever the denominators are nonzero,*

$$\Phi(x^3) = \frac{\Phi(x)^3}{(1+\Phi(x))^2} = (1+\Phi(x)) \left( \frac{\Phi(x)}{1+\Phi(x)} \right)^3. \quad (\text{CU}_3)$$

*For every  $e > 1$  dividing  $q+1 = F_n$ ,  $3 \nmid e$ , so cubing permutes the exact-order- $e$  norm-one elements. Consequently, for every cubic character  $\chi : \mathbb{F}_q^\times \rightarrow \mu_3$ ,*

$$\sum_{\text{ord}(u)=e} \chi(\Phi(u)) = \sum_{\text{ord}(x)=e} \chi(1+\Phi(x)). \quad (\text{CU}_4)$$

*The cube character therefore transports bijectively to a translate-one character on the same fibotomic stratum. By itself this excludes no proper stratum: it supplies no nonvanishing or emptiness theorem for the translate-one character. Old maximal order and cubic-subgroup membership supply no first-failure contradiction by themselves beyond the selected prime-factor divisibilities  $A_{n-1} \mid S_{F_n/\ell}$ , for primes  $\ell \mid F_n$ .*

*Proof.* Equation (CU<sub>1</sub>) is the lower expression for  $a_{n-1}$  in (E). The group of order  $Q+1$  has no three-torsion, so cubing is bijective and (CU<sub>2</sub>) follows directly. Corollary 7.2 gives the displayed first-failure order, and a cyclic group of order  $q-1$  has a cube subgroup of order  $(q-1)/3$ .

In characteristic two,

$$x^3 + 1 = (x+1)(x^2 + x + 1), \quad 1 + \Phi(x) = \frac{x^2 + x + 1}{(x+1)^2}.$$

Substitution proves both forms in (CU<sub>3</sub>). If  $x$  has exact order  $e$ , the excluded denominators would force order one or three, which cannot divide a Fermat number. Substitute  $u = x^3$  in the left side of (CU<sub>4</sub>); the last factor in (CU<sub>3</sub>) is a cube and disappears under  $\chi$ . The three rational identities used here are independently kernel-checked in `Ogdoad.Excess`.  $\square$

**Proposition 7.18** (exact quadratic remainder descent). *Retain Proposition 7.11. For  $R(X) \in \mathbb{F}_2[X]$ , divide  $R(X)$ , now regarded as an element of  $\mathbb{F}_2[Y][X]$ , by the monic quadratic*

$$Q_Y(X) = X^2 + YX + Y^3.$$

*There are unique  $U_R, V_R \in \mathbb{F}_2[Y]$  such that*

$$R(X) \equiv U_R(Y) + XV_R(Y) \pmod{Q_Y(X)}. \quad (\text{QR}_1)$$

*Then, for every  $i \geq 0$ ,*

$$\boxed{A_i(X) \mid R(X) \iff A_{i-1}(Y) \mid U_R(Y) \quad \text{and} \quad A_{i-1}(Y) \mid V_R(Y).} \quad (\text{QR}_2)$$

*Consequently the selected divisibility  $A_{n-1} \mid S_{F_n/\ell}$  can be pushed, without loss, through every preceding Conway quadratic: iterating  $(\text{QR}_1)$  gives an exact binary dependency tree of divisibilities at  $A_{-1} = Y + 1$ . This criterion uses the complete deterministic ancestry, so a one-step countermodel with the wrong predecessor does not certify the selected divisibility. Its leaf conditions are nevertheless an exact re-expression of the original selected Fibonacci value; degree alone does not make any leaf coprimality automatic.*

*Proof.* Euclidean division by the monic polynomial  $Q_Y$  proves existence and uniqueness in  $(\text{QR}_1)$ . Substitute  $Y = a_{i-1}$  and  $X = a_i$ . Proposition 7.11 gives  $Q_{a_{i-1}}(a_i) = 0$ , so

$$R(a_i) = U_R(a_{i-1}) + a_i V_R(a_{i-1}). \quad (\text{QR}_3)$$

The extension  $\mathbb{F}_2(a_i)/\mathbb{F}_2(a_{i-1})$  has degree two; hence  $1, a_i$  is a basis over the lower field. Thus the right side of  $(\text{QR}_3)$  vanishes exactly when both lower coefficients vanish. The minimal-polynomial criterion gives  $(\text{QR}_2)$ , and iteration proves the final assertion.  $\square$

**Proposition 7.19** (complete remainder coordinates and Euclidean-tail exhaustion). *Retain Proposition 7.18. For  $i \geq -1$ , write*

$$\mathcal{R}_i = \{R \in \mathbb{F}_2[X] : \deg R < 2^{i+1}\}, \quad \mathcal{R}_{-1} = \mathbb{F}_2.$$

*For  $i \geq 0$ , let*

$$R(X) \equiv U_R(Y) + XV_R(Y) \pmod{X^2 + YX + Y^3}$$

*and define*

$$\sigma_i : \mathcal{R}_i \longrightarrow \mathcal{R}_{i-1} \times \mathcal{R}_{i-1}, \quad \sigma_i(R) = (U_R \bmod A_{i-1}, V_R \bmod A_{i-1}), \quad (\text{RCE}_1)$$

*using the unique reduced representatives. Then every  $\sigma_i$  is an  $\mathbb{F}_2$ -linear isomorphism. Iteration gives a canonical isomorphism*

$$\boxed{\mathbf{C}_n : \mathcal{R}_{n-1} \xrightarrow{\sim} \mathbb{F}_2^{2^n}.} \quad (\text{RCE}_2)$$

*For every  $R \in \mathbb{F}_2[X]$ , the vector  $\mathbf{C}_n(R \bmod A_{n-1})$  is exactly the ordered leaf vector of the complete binary quadratic-remainder descent. Consequently*

$$\boxed{A_{n-1} \mid R \iff R \bmod A_{n-1} = 0 \iff \mathbf{C}_n(R \bmod A_{n-1}) = 0.} \quad (\text{RCE}_3)$$

*Every binary leaf pattern occurs for exactly one reduced polynomial.*

*Write the first monic division as*

$$R = Q_R A_{n-1} + \rho_R, \quad \deg \rho_R < 2^n.$$



Then

$$\boxed{R \mapsto (Q_R, \mathbf{C}_n(\rho_R))} \quad (\text{RCE}_4)$$

is a bijection

$$\mathbb{F}_2[X] \xrightarrow{\sim} \mathbb{F}_2[X] \times \mathbb{F}_2^{2^n}, \quad (Q, \mathbf{c}) \mapsto A_{n-1}Q + \mathbf{C}_n^{-1}(\mathbf{c}). \quad (\text{RCE}_5)$$

Thus the raw first quotient and the complete ancestry-coordinate vector are independent for unrestricted input polynomials, even with the literal Conway chain held fixed.

After the first division, the monic Euclidean remainder sequence for  $(R, A_{n-1})$  is the sequence for  $(A_{n-1}, \rho_R)$ . Its entire tail, and every subresultant polynomial remainder sequence derived from that tail, is a deterministic function of  $\rho_R$ ; conversely the first tail entry recovers  $\rho_R$ . For  $R = S_d$ ,

$$\gcd(A_{n-1}, S_d) = \begin{cases} A_{n-1}, & \rho_{S_d} = 0, \\ 1, & \rho_{S_d} \neq 0, \end{cases} \quad \text{Res}(A_{n-1}, S_d) = \begin{cases} 0, & \rho_{S_d} = 0, \\ 1, & \rho_{S_d} \neq 0. \end{cases} \quad (\text{RCE}_6)$$

Therefore the complete intermediate remainder tree and the complete Euclidean/subresultant tail are information-equivalent to the original selected residue  $S_d \bmod A_{n-1}$ ; they are not an additional ancestry invariant. The independence in  $(\text{RCE}_4)$  does not require its free input to be a Fibonacci polynomial. Hence it does not exclude a special coefficient identity relating the actual quotient  $Q_{S_d}$  to  $\rho_{S_d}$ , nor a direct proof that one coordinate of  $\rho_{S_d}$  is nonzero. Those are the surviving coefficient-sensitive routes.

*Proof.* Evaluation identifies  $\mathcal{R}_i$  with  $E_i$ , since  $A_i$  is the degree- $2^{i+1}$  minimal polynomial of  $a_i$ . Under these identifications,  $\sigma_i$  is the coordinate map

$$E_i \xrightarrow{\sim} E_{i-1} \times E_{i-1}, \quad u + a_i v \mapsto (u, v),$$

by  $(\text{QR}_3)$  and the basis  $1, a_i$ . This proves  $(\text{RCE}_2)$ – $(\text{RCE}_3)$ , including unique realization of every leaf vector. Monic division gives unique  $Q_R, \rho_R$ , and combining that bijection with  $\mathbf{C}_n$  proves  $(\text{RCE}_4)$ – $(\text{RCE}_5)$ .

The Euclidean algorithm discards  $Q_R$  after its first step and continues from  $(A_{n-1}, \rho_R)$ . Finally  $A_{n-1}$  is irreducible. A nonzero  $\rho_R$  has smaller degree and is therefore coprime to  $A_{n-1}$ ; the resultant lies in  $\mathbb{F}_2$ , so it is one in that case and zero otherwise. This proves  $(\text{RCE}_6)$ . Lean kernel-checks the monic quotient–remainder uniqueness and free-realization core, together with zero reflection and product independence for an abstract additive coordinate equivalence. Its identification with the actual Conway tree and the Euclidean/subresultant interpretation remain paper-level.  $\square$

**Proposition 7.20** (Fibonacci quotient defects recover the endpoint). *Fix  $n \geq 1$ , put*

$$A = A_{n-1}, \quad m = \deg A = 2^n,$$

*and let  $d > 1$  be odd. Write the actual monic Euclidean division*

$$S_d = AQ_d + \rho_d, \quad \deg \rho_d < m. \quad (\text{FQD}_1)$$

*Define the two special quotient defects*

$$\mathcal{O}_d = S'_d + A'Q_d, \quad \mathcal{E}_d = S_{d+1} + XA'Q_d. \quad (\text{FQD}_2)$$

Then

$$\boxed{\mathcal{E}_d + X\mathcal{O}_d = S_d.} \quad (\text{FQD}_3)$$

More precisely,

$$\begin{aligned} \mathcal{O}_d &= A Q'_d + \rho'_d, \\ \mathcal{E}_d &= A(Q_d + X Q'_d) + \rho_d + X \rho'_d. \end{aligned} \tag{FQD_4}$$

Consequently the reduced defects are

$$\mathcal{O}_d \bmod A = \rho'_d, \quad \mathcal{E}_d \bmod A = \rho_d + X \rho'_d. \tag{FQD_5}$$

If

$$\rho_d(X) = \sum_{j=0}^{m-1} r_j X^j,$$

then

$$\rho'_d(X) = \sum_j r_{2j+1} X^{2j}, \quad \rho_d(X) + X \rho'_d(X) = \sum_j r_{2j} X^{2j}. \tag{FQD_6}$$

Thus the two defects are exactly the shifted odd-coordinate block and the even-coordinate block of the selected remainder. In particular

$$\rho_d = (\mathcal{E}_d \bmod A) + X(\mathcal{O}_d \bmod A), \tag{FQD_7}$$

and hence

$$A \mid S_d \iff A \mid \mathcal{O}_d \text{ and } A \mid \mathcal{E}_d. \tag{FQD_8}$$

Thus the reduced defect pair is information-equivalent to the original selected endpoint, rather than an independent obstruction. This derivative/next-value coupling does not control higher  $A$ -adic data or global coefficient relations in the unreduced quotient  $Q_d$ . A surviving coefficient-sensitive proof must use such data or directly prove that one of the two parity blocks in (FQD<sub>5</sub>) is nonzero.

*Proof.* Write  $d = 2h + 1$ . The characteristic-two derivative and doubling identities give

$$S'_d = S_h^2 = S_{2h} = S_{d-1}. \tag{FQD_{11}}$$

The Fibonacci recurrence therefore becomes

$$S_{d+1} = S_d + X S_{d-1} = S_d + X S'_d. \tag{FQD_{12}}$$

Substituting this in the definitions of the defects makes both copies of  $X A' Q_d$ , and both copies of  $X S'_d$ , cancel in characteristic two, proving (FQD<sub>3</sub>).

Differentiate (FQD<sub>1</sub>):

$$S'_d = A' Q_d + A Q'_d + \rho'_d.$$

Adding  $A' Q_d$  proves the first identity in (FQD<sub>4</sub>). Substituting (FQD<sub>1</sub>) and this differentiated identity into

$$\mathcal{E}_d = S_d + X S'_d + X A' Q_d$$

proves the second. Reduction modulo  $A$  gives (FQD<sub>5</sub>). Since differentiation kills even powers and takes  $X^{2j+1}$  to  $X^{2j}$ , formula (FQD<sub>6</sub>) follows. The two parity supports are disjoint, and

$$(\rho_d + X \rho'_d) + X \rho'_d = \rho_d,$$

which proves (FQD<sub>7</sub>). The forward implication in (FQD<sub>8</sub>) follows from (FQD<sub>4</sub>); the reverse implication follows from (FQD<sub>3</sub>).

The abstract defect reconstruction, odd/even remainder decomposition, and exact-factor divisibility formulas are kernel-checked in `Ugdoad.Excess`. Their specialization to formal polynomial derivatives, the selected quotient, and the finite-field unit statements remain paper-level.  $\square$

**Proposition 7.21** (one-branch norm descent for the Fermat arm). *For  $R(X) \in \mathbb{F}_2[X]$ , retain the remainder*

$$R(X) \equiv U_R(Y) + XV_R(Y) \pmod{X^2 + YX + Y^3}$$

*from Proposition 7.18, and define*

$$\mathcal{N}(R)(Y) = \text{Res}_X(X^2 + YX + Y^3, R(X)).$$

*Then*

$$\boxed{\mathcal{N}(R) = U_R^2 + YU_RV_R + Y^3V_R^2} \quad (\text{QN}_1)$$

*and, for every  $i \geq 0$ , every actual Conway quadratic edge satisfies*

$$\boxed{A_i \mid R \iff A_{i-1} \mid \mathcal{N}(R).} \quad (\text{QN}_2)$$

*After identifying  $\mathbb{F}_2[Y]$  with  $\mathbb{F}_2[X]$  by  $Y \mapsto X$ , regard  $\mathcal{N}$  as an endomorphism before iterating it. Consequently, for every  $n \geq 0$ ,*

$$\boxed{\mathcal{N}^{\circ n}(R)(1) = N_{E_{n-1}/\mathbb{F}_2}(R(a_{n-1}))}. \quad (\text{QN}_3)$$

*The right side is one exactly when  $R(a_{n-1}) \neq 0$ .*

*The Fibonacci family itself has a closed first norm layer. Put  $W_d = \mathcal{N}(S_d)$ . Then, for every  $d \geq 0$ ,*

$$W_0 = 0, \quad W_1 = W_2 = 1, \quad W_3 = 1 + Y + Y^3,$$

*and*

$$\boxed{W_{d+4} = W_{d+3} + YW_{d+2} + Y^3W_{d+1} + Y^6W_d}. \quad (\text{QN}_4)$$

*For every  $n \geq 0$ , prime  $\ell \mid F_n$ , and current test  $d = F_n/\ell$ , however,*

$$\mathcal{N}^{\circ n}(S_d)(1) = \begin{cases} 0, & \delta_n \mid d, \\ 1, & \delta_n \nmid d. \end{cases} \quad (\text{QN}_5)$$

*Thus the single terminal bit compresses the complete binary remainder tree, but its required value one is exactly the original selected Conway–Fermat assertion. The recurrence exposes rather than removes the remaining obstruction.*

*Proof.* Let  $x, x'$  be the two roots of  $X^2 + YX + Y^3$ . Their sum is  $Y$  and their product is  $Y^3$ , so

$$(U_R + xV_R)(U_R + x'V_R) = U_R^2 + YU_RV_R + Y^3V_R^2.$$

This is the product formula for the resultant and proves (QN<sub>1</sub>). At  $Y = a_{i-1}$ , the two roots are the relative conjugates of  $a_i$ ; hence

$$\mathcal{N}(R)(a_{i-1}) = N_{E_i/E_{i-1}} R(a_i).$$

The field norm and its argument vanish together, proving (QN<sub>2</sub>). Iterate, use transitivity of norms, descend to  $A_{-1} = Y + 1$ , and use  $a_{-1} = 1$  to obtain (QN<sub>3</sub>).

For (QN<sub>4</sub>), let  $\lambda_1, \lambda_2$  and  $\mu_1, \mu_2$  be the roots of

$$T^2 + T + x, \quad T^2 + T + x'.$$

The four products  $\lambda_i \mu_j$  have characteristic polynomial

$$T^4 + T^3 + YT^2 + Y^3T + Y^6.$$

The Binet formula gives

$$W_d = S_d(x)S_d(x') = \sum_{i,j} (\lambda_i \mu_j)^d,$$

so their power-sum recurrence is exactly  $(\mathbf{QN}_4)$ , with the displayed initial values obtained directly. Finally  $S_d(a_{n-1}) = 0$  exactly when  $\delta_n \mid d$ , by Proposition 7.7;  $(\mathbf{QN}_3)$  now proves  $(\mathbf{QN}_5)$ .  $\square$

**Corollary 7.22** (the Fermat terminal word is the order word). *For every  $n \geq 0$  and  $d \geq 0$ , put*

$$T_n(d) = \mathcal{N}^{\circ n}(S_d)(1).$$

*Then*

$$\boxed{T_n(d) = \epsilon_{\delta_n}(d)}. \quad (\mathbf{ZOW}_1)$$

*Consequently  $T_n$  has least period  $\delta_n$ , exact linear complexity  $\delta_n - 1$ , and exactly  $\delta_n$  states in both its 2-kernel and minimal binary DFAO. Since  $\delta_n \mid F_n$ , for every prime  $\ell \mid F_n$ ,*

$$\boxed{T_n(F_n/\ell) = 1 \iff v_\ell(\delta_n) = v_\ell(F_n)}. \quad (\mathbf{ZOW}_2)$$

*Thus exact quadratic ancestry compression does not reduce the state complexity below the unknown Conway–Fermat quotient order. It presents that order itself as the terminal binary word.*

*Proof.* Proposition 7.21 gives  $T_n(d) = 0$  exactly when  $S_d(a_{n-1}) = 0$ , while Proposition 7.7 identifies the latter condition with  $\delta_n \mid d$ . This proves  $(\mathbf{ZOW}_1)$ . Proposition 5.13 supplies the period, complexity, and automaton assertions. Finally, for divisors  $\delta_n \mid F_n$ , the failure of  $\delta_n \mid F_n/\ell$  is equivalent to equality of the two  $\ell$ -adic valuations, proving  $(\mathbf{ZOW}_2)$ .  $\square$

**Proposition 7.23** (the complementary selected resultant is the opposite bit). *Fix  $n \geq 1$ , put*

$$m = 2^n, \quad q = 2^m, \quad N = q + 1 = F_n, \quad a = a_{n-1}, \quad A = A_{n-1},$$

*and let  $d \mid N$ . Fibonacci divisibility defines the cofactor*

$$J_{N,d}(X) = \frac{S_N(X)}{S_d(X)} \in \mathbb{F}_2[X].$$

*Then*

$$S_N(X) = 1 + \sum_{r=0}^{m-1} X^{2^r}, \quad S'_N(X) = 1,$$

*and hence*

$$1 = S'_d(X)J_{N,d}(X) + S_d(X)J'_{N,d}(X). \quad (\mathbf{QNC}_1)$$

*At the complete actual Conway ancestry one has*

$$S_d(a)J_{N,d}(a) = 0, \quad S'_d(a)J_{N,d}(a) + S_d(a)J'_{N,d}(a) = 1. \quad (\mathbf{QNC}_2)$$

*Consequently exactly one of  $S_d(a)$  and  $J_{N,d}(a)$  vanishes. More precisely,*

$$\begin{aligned} S_d(a) = 0 &\implies S'_d(a)J_{N,d}(a) = 1, \\ J_{N,d}(a) = 0 &\implies S_d(a)J'_{N,d}(a) = 1. \end{aligned} \quad (\mathbf{QNC}_3)$$

The two selected resultants are therefore complementary bits:

$$\boxed{\text{Res}(A, S_d) + \text{Res}(A, J_{N,d}) = 1 \quad \text{in } \mathbb{F}_2.} \quad (\text{QNC}_4)$$

Equivalently, for the exact one-branch Conway norm operator of Proposition 7.21,

$$\boxed{\mathcal{N}^{\text{on}}(S_d)(1) + \mathcal{N}^{\text{on}}(J_{N,d})(1) = 1.} \quad (\text{QNC}_5)$$

For  $d = N/\ell$ , the first summand is the desired primary success bit; the cofactor, its derivative, and its complete ancestry descent supply exactly the opposite bit, not a second obstruction. More generally, in the complete fibotomic factorization of  $S_N$ , exactly one selected resultant vanishes, and its position is the unknown order  $\delta_n$ . Thus factor-cofactor splitting determines no preferred position. This proposition does not rule out a new ancestry-sensitive identity involving a polynomial outside this factorization.

*Proof.* The standard strong divisibility of Fibonacci polynomials gives  $S_d \mid S_N$ . Fast doubling gives

$$S_N(X) = S_{2^m+1}(X) = 1 + \sum_{r=0}^{m-1} X^{2^r},$$

whose formal derivative is one. Differentiating  $S_N = S_d J_{N,d}$  proves (QNC<sub>1</sub>).

The selected element has absolute trace one, so  $S_N(a) = 1 + \text{Tr}_{E_{n-1}/\mathbb{F}_2}(a) = 0$ . Evaluation of the factorization and of (QNC<sub>1</sub>) gives (QNC<sub>2</sub>). The product identity says that at least one factor vanishes, while the differentiated identity says they cannot both vanish; it also gives the inverse identities (QNC<sub>3</sub>).

Because  $A$  is monic,

$$\text{Res}(A, R) = N_{E_{n-1}/\mathbb{F}_2}(R(a)) \in \mathbb{F}_2.$$

A finite-field norm vanishes exactly with its argument, and the only nonzero element of  $\mathbb{F}_2$  is one. Thus the two resultants are 0, 1 in some order, proving (QNC<sub>4</sub>). Proposition 7.21 identifies the two resultants with the two iterated terminal norms in (QNC<sub>5</sub>). The fibotomic statement follows from the squarefree factorization of  $S_N$ .

Lean kernel-checks the power-of-two-plus-one Fibonacci derivative and the abstract product-zero/differentiated-Bézout-one implication, including the complementary-bit conclusion over  $\mathbb{F}_2$ , in `Ugdoad.Excess`. The Fibonacci divisibility, selected finite-field specialization, and resultant identification remain paper-level.  $\square$

**Proposition 7.24** (selected CRT projectors and ancestry trace annihilation). *Retain the notation of Proposition 7.23, and put*

$$\mathcal{T}_N = \mathbb{F}_2[X]/(S_N), \quad r = \deg S_d = \frac{d-1}{2}, \quad s = \deg J_{N,d} = \frac{N-d}{2}.$$

In  $\mathcal{T}_N$ , define

$$e_{\text{fail}} = [S'_d J_{N,d}], \quad e_{\text{succ}} = [S_d J'_{N,d}].$$

Then

$$e_{\text{fail}} + e_{\text{succ}} = 1, \quad e_{\text{fail}} e_{\text{succ}} = 0, \quad e_{\text{fail}}^2 = e_{\text{fail}}, \quad e_{\text{succ}}^2 = e_{\text{succ}}. \quad (\text{QNP}_1)$$

Under the canonical Chinese-remainder isomorphism

$$\mathcal{T}_N \simeq \mathbb{F}_2[X]/(S_d) \times \mathbb{F}_2[X]/(J_{N,d}),$$

their orientations are

$$e_{\text{fail}} = (1, 0), \quad e_{\text{succ}} = (0, 1). \quad (\text{QNP}_2)$$

Moreover, every Bézout certificate

$$US_d + VJ_{N,d} = 1$$

produces the same projectors:

$$[VJ_{N,d}] = e_{\text{fail}}, \quad [US_d] = e_{\text{succ}}. \quad (\text{QNP}_3)$$

Thus the terminal support information returned by the Euclidean algorithm, or by any normalized final subresultant certificate, is canonical.

Evaluation at the actual selected Conway element  $a = a_{n-1}$  gives

$$(e_{\text{fail}}(a), e_{\text{succ}}(a)) = \begin{cases} (1, 0), & S_d(a) = 0, \\ (0, 1), & J_{N,d}(a) = 0. \end{cases} \quad (\text{QNP}_4)$$

In particular the orientation relative to the desired resultants is

$$\boxed{e_{\text{succ}}(a) = \text{Res}(A, S_d), \quad e_{\text{fail}}(a) = \text{Res}(A, J_{N,d}).} \quad (\text{QNP}_5)$$

This orientation is destroyed by every proper additive descent through the literal Conway ancestry. For  $-1 \leq j \leq n-2$ ,

$$\boxed{\text{Tr}_{E_{n-1}/E_j}(e_{\text{fail}}(a)) = \text{Tr}_{E_{n-1}/E_j}(e_{\text{succ}}(a)) = 0,} \quad (\text{QNP}_6)$$

whereas

$$\boxed{\text{Norm}_{E_{n-1}/E_j}(e_{\star}(a)) = e_{\star}(a) \quad (\star \in \{\text{fail}, \text{succ}\}).} \quad (\text{QNP}_7)$$

Thus norm descent retains exactly the original endpoint bit and trace descent retains none of it.

The ambient multiplication characteristic polynomials are

$$\chi_{\mu_{e_{\text{fail}}}}(T) = T^s(T+1)^r, \quad \chi_{\mu_{e_{\text{succ}}}}(T) = T^r(T+1)^s. \quad (\text{QNP}_8)$$

For  $d = N/\ell$ , with  $\ell \mid N$  prime, one has  $N \equiv d \equiv 1 \pmod{4}$ , so  $r$  and  $s$  are even and both ambient traces vanish. On the selected quotient  $E_{n-1} = \mathbb{F}_2(a)$ , whose degree is  $m = 2^n$ , multiplication by either selected projector has

$$\chi_{\mu_{e_{\star}(a)}}(T) = (T + e_{\star}(a))^m = T^m + e_{\star}(a). \quad (\text{QNP}_9)$$

Hence for  $n \geq 1$  its trace and multiplication-characteristic-polynomial discriminant vanish in both branches, while its determinant is  $e_{\star}(a)$ , again exactly the original selected resultant.

Consequently terminal Euclidean quotients only through their final support projector, final Bézout or subresultant certificates, proper additive ancestry traces, and multiplication trace or discriminant cannot decide which fibotomic factor vanishes. This does not rule out an identity involving intermediate Euclidean quotient polynomials before support collapse, a polynomial outside the factorization of  $S_N$ , or any identity that directly evaluates  $e_{\text{succ}}(a)$ .

*Proof.* Proposition 7.23 gives

$$S_d J_{N,d} = S_N = 0 \quad \text{in } \mathcal{T}_N$$

and

$$S'_d J_{N,d} + S_d J'_{N,d} = S'_N = 1.$$

Their two summands therefore add to one and have product  $S'_d S_d J_{N,d} J'_{N,d} = 0$ . Multiplying their sum by either summand proves idempotence, giving (QNP<sub>1</sub>). Reduction modulo the two coprime factors proves (QNP<sub>2</sub>). The same reductions send  $VJ_{N,d}$  to  $(1, 0)$  and  $US_d$  to  $(0, 1)$ , proving the uniqueness assertion (QNP<sub>3</sub>).

Since  $S_N(a) = 0$ , evaluation defines a map  $\mathcal{T}_N \rightarrow E_{n-1}$ . The differentiated identity and the fact that exactly one of  $S_d(a)$  and  $J_{N,d}(a)$  vanishes give (QNP<sub>4</sub>). Both projectors evaluate in  $\mathbb{F}_2$ . A selected resultant is the norm of the corresponding selected value to  $\mathbb{F}_2$ , hence is zero exactly when that value is zero and otherwise equals one. This proves the orientation in (QNP<sub>5</sub>).

For  $-1 \leq j \leq n-2$ ,

$$[E_{n-1} : E_j] = 2^{n-1-j}$$

is even. The relative trace of either scalar bit is therefore zero in characteristic two, while its relative norm is that bit raised to a positive power and hence is unchanged. This proves (QNP<sub>6</sub>)–(QNP<sub>7</sub>).

The CRT decomposition makes multiplication by  $e_{\text{fail}}$  the identity on the degree- $r$  factor and zero on the degree- $s$  factor; the roles reverse for  $e_{\text{succ}}$ . This gives (QNP<sub>8</sub>). A prime divisor  $\ell$  of  $2^{2^n} + 1$  has  $\text{ord}_\ell(2) = 2^{n+1}$ , so for  $n \geq 1$  one has  $\ell \equiv 1 \pmod{4}$ , and consequently  $d = N/\ell \equiv 1 \pmod{4}$ . Finally, on the degree- $m$  selected field, multiplication by  $e_\star(a) \in \mathbb{F}_2$  is scalar. Since  $m$  is a power of two,

$$(T + e_\star(a))^m = T^m + e_\star(a).$$

For  $m > 1$  this has zero trace and zero discriminant, while its constant term, equivalently its determinant in characteristic two, is  $e_\star(a)$ .

The abstract complementary-idempotent identities, selected field branches, uniqueness of Bézout support projectors, and the quadratic trace/norm core are kernel-checked in `0gdoad.Excess`. The polynomial CRT specialization, resultant orientation, and characteristic-polynomial interpretation remain paper-level.  $\square$

**Proposition 7.25** (one-step maximality is not an inductive invariant). *Let*

$$C(Y) = Y^{16} + Y^{15} + Y^{12} + Y^{11} + Y^{10} + Y^8 + 1$$

*and apply the exact Conway resultant operator of Proposition 7.11:*

$$\begin{aligned} B(X) &= \text{Res}_Y(C(Y), X^2 + YX + Y^3) \\ &= X^{32} + X^{31} + X^{30} + X^{27} + X^{26} + X^{24} + X^{23} + X^{22} \\ &\quad + X^{19} + X^{15} + X^{14} + X^{11} + X^8 + X^7 + 1. \end{aligned} \tag{NI<sub>1</sub>}$$

*Then  $C$  is an irreducible factor of the full-order fibotomic stratum  $\mathfrak{F}_{F_4}$ , where  $F_4 = 65537$ , whereas  $B$  is an irreducible factor of the proper-order stratum*

$$\mathfrak{F}_{6700417}, \quad 6700417 = F_5/641. \tag{NI<sub>2</sub>}$$

*Moreover, if  $y$  is a root of  $C$ , then*

$$\text{ord}(y) = 21845 = \frac{2^{16} - 1}{3} = 5 \cdot 17 \cdot 257. \tag{NI_{2a}}$$

*If  $x$  is a root of  $B$ , then in fact*

$$\text{ord}(x) = \frac{2^{32} - 1}{3} = F_1 F_2 F_3 F_4. \tag{NI_{2b}}$$

Thus the countermodel also satisfies the exact multiplicative-order consequence that all preceding Conway–Fermat maximalities impose on both its degree-sixteen element and its successor. In particular, the successor contains every preceding Fermat factor with full multiplicity even though its next norm-one component loses the factor 641. Even that strengthened predecessor/successor order data, together with the exact resultant transition, does not force target maximality.

Thus the exact one-step resultant can send a full-order predecessor factor to a proper-order target factor, already at the first composite Fermat level. In particular, predecessor maximality plus the transition formula alone is not an induction theorem. This is not a counterexample to the distinguished Conway path:  $C$  is an alternative factor at the  $A_3$  level and need not have the earlier ancestry of the actual  $A_3$ .

*Proof.* First, exact modular squaring gives

$$Y^{2^{16}} \equiv Y \pmod{C},$$

and

$$Y^{2^{56}} + Y \equiv R := Y^{15} + Y^{14} + Y^{13} + Y^9 + Y^8 + Y^2 + Y \pmod{C}.$$

The Bézout identity  $UC + VR = 1$  holds for

$$\begin{aligned} U &= Y^{14} + Y^{13} + Y^{12} + Y^{11} + Y^7 + Y^6 + Y^5 + Y^4 \\ &\quad + Y^3 + Y + 1, \\ V &= Y^{15} + Y^{14} + Y^{12} + Y^{11} + Y^9 + Y^8 + Y^5 + Y^3 + Y + 1. \end{aligned}$$

Rabin’s criterion, for which two is the only prime divisor of sixteen, therefore proves that  $C$  is irreducible.

The Fibonacci polynomials satisfy, in characteristic two,

$$S_{2r} = S_r^2, \quad S_{2r+1} = S_{r+1}^2 + Y S_r^2,$$

and hence

$$S_{2^{m+1}}(Y) = 1 + \sum_{j=0}^{m-1} Y^{2^j}. \quad (\text{NI}_3)$$

For a root  $y$  of  $C$ , the coefficient of  $Y^{15}$  gives  $\text{Tr}_{\mathbb{F}_{2^{16}}/\mathbb{F}_2}(y) = 1$ . Equation  $(\text{NI}_3)$  therefore gives  $S_{65537}(y) = 0$ . Since 65537 is prime,  $y$  belongs to the full-order stratum  $\mathfrak{F}_{65537}$ .

The predecessor’s own multiplicative order is certified independently by the exact reductions

$$\begin{aligned} Y^{2^{1845}} &\equiv 1, \\ Y^{4^{369}} &\equiv 7576, \quad Y^{1^{285}} \equiv 1653, \quad Y^{8^5} \equiv 2464 \pmod{C}, \end{aligned}$$

where hexadecimal bit  $j$  is the coefficient of  $Y^j$ . The last three residues are not one. Since  $21845 = 5 \cdot 17 \cdot 257$ , these identities prove  $(\text{NI}_{2a})$ . Under  $\delta_j = F_j$  for  $1 \leq j \leq 3$ , Corollary 7.2 gives the same order for  $a_3$ .

Now choose  $z$  with  $z^2 + z = y$  and put  $x = yz$ . The trace just computed makes  $Z^2 + Z + y$  irreducible over  $\mathbb{F}_{2^{16}}$ . Thus  $x \notin \mathbb{F}_{2^{16}}$ ; because every proper subfield of  $\mathbb{F}_{2^{32}}$  lies in  $\mathbb{F}_{2^{16}}$ ,  $x$  has degree thirty-two over  $\mathbb{F}_2$ . Moreover



$$x^2 + yx + y^3 = y^2(z^2 + z + y) = 0. \quad (\text{NI}_4)$$

Modulo the relation in  $(\text{NI}_4)$ , direct reduction gives

$$B(X) = C(Y)(Q_1(Y)X + Q_0(Y)), \quad (\text{NI}_5)$$

where

$$\begin{aligned} Q_1 &= Y^{29} + Y^{28} + Y^{25} + Y^{24} + Y^{21} + Y^{18} + Y^{17} + Y^{16} \\ &\quad + Y^{15} + Y^{14} + Y^{11} + Y^{10} + Y^9 + Y^6, \\ Q_0 &= Y^{32} + Y^{31} + Y^{29} + Y^{28} + Y^{27} + Y^{26} + Y^{25} + Y^{23} \\ &\quad + Y^{20} + Y^{14} + Y^{12} + Y^{11} + Y^{10} + Y^9 + 1. \end{aligned}$$

Thus  $B(x) = 0$ . Its displayed degree is thirty-two, so it is the minimal polynomial of  $x$ , and consequently the stated resultant.

The multiplicative order of this successor can also be read exactly, without sampling. Put  $q = 2^{16}$ . Since  $y^q = y$  and the two roots of  $Z^2 + Z + y$  are  $z, z + 1$ , one has  $z^q = z + 1$ . Therefore, with  $v = (z + 1)/z$ ,

$$x^{q+1} = y^3, \quad x^{q-1} = v. \quad (\text{NI}_{5a})$$

The element  $v$  is a nonidentity element of the norm-one group of order  $q + 1 = 65537$ ; that order is prime, so  $\text{ord}(v) = q + 1$ . On the other hand,  $(\text{NI}_{2a})$  and  $\gcd(3, 21845) = 1$  give  $\text{ord}(y^3) = (q - 1)/3$ .

Now

$$\mathbb{F}_{q^2}^\times = \mathbb{F}_q^\times \times \ker\left(N_{\mathbb{F}_{q^2}/\mathbb{F}_q}\right),$$

because the two factor orders  $q - 1$  and  $q + 1$  are coprime. Write the corresponding components of  $x$  as  $x_-$  and  $x_+$ . Equation  $(\text{NI}_{5a})$  has orders  $\text{ord}(x_-^2) = \text{ord}(x_-)$  and  $\text{ord}(x_+^{-2}) = \text{ord}(x_+)$ , since both factor groups have odd order. It follows that

$$\text{ord}(x_-) = \frac{q - 1}{3}, \quad \text{ord}(x_+) = q + 1,$$

and their coprime product proves  $(\text{NI}_{2b})$ . Notice that the loss of 641 below concerns the following quotient of order  $2^{32} + 1 = F_5$ , not either component in this decomposition; keeping these two adjacent quadratic levels separate is essential.

It remains to identify the target stratum. From a pair  $(f, g) = (S_r, S_{r+1})$ , put

$$A = f^2, \quad D = g^2 + Xf^2.$$

Reading one further binary digit sends the pair to  $(A, D)$  for a zero and to  $(D, g^2)$  for a one. Repeated reduction modulo  $B$  along the binary word

$$6700417 = (11001100011110110000001)_2$$

ends at the exact residue pair

$$(S_{6700417}, S_{6700418}) \equiv (00000000, 1875\text{a}343) \pmod{B}, \quad (\text{NI}_6)$$

where a hexadecimal bit in position  $j$  is the coefficient of  $X^j$ . This is an  $O(\log 6700417)$  polynomial-identity certificate, not field sampling. Hence  $S_{6700417}(x) = 0$ . The integer 6700417 is prime, so  $x$  belongs to the exact order-6700417 fibotomic stratum. Finally, if  $v = (z + 1)/z$ , then

$$y = \frac{v}{(v+1)^2}, \quad x = yz = \frac{v}{(v+1)^3},$$

showing directly that this is the same fibotomic/elliptic transition as the Conway resultant.  $\square$

### 7.3 Ancestry refinements and exact no-go results

**Proposition 7.26** (reciprocal-trace ancestry and a trace-refined no-go). *Every distinguished Conway factor satisfies*

$$\mathrm{Tr}_{E_i/\mathbb{F}_2}(a_i^{-1}) = 1. \quad (\mathrm{RT}_1)$$

*Equivalently, the constant and linear coefficients of every  $A_i$  are both one. More generally, this reciprocal-trace bit is preserved by every full-degree transition of Proposition 7.11.*

*This extra ancestry invariant still does not make one-step maximality inductive. Put*

$$C_1(Y) = Y^{16} + Y^{15} + Y^{12} + Y^{11} + Y^6 + Y + 1.$$

*Then  $C_1$  is an irreducible full-order factor of  $\mathfrak{F}_{65537}$ , has constant and linear coefficients one, and its exact Conway resultant is*

$$\begin{aligned} B_1(X) = & X^{32} + X^{31} + X^{30} + X^{28} + X^{27} + X^{26} + X^{24} + X^{23} + X^{22} \\ & + X^{20} + X^{19} + X^{18} + X^{16} + X^{15} + X^{14} + X^{13} + X^{12} + X^{11} \\ & + X^9 + X^8 + X^7 + X^5 + X^2 + X + 1. \end{aligned} \quad (\mathrm{RT}_2)$$

*The polynomial  $B_1$  is irreducible but belongs to the proper-order stratum*

$$\mathfrak{F}_{6700417}, \quad 6700417 = F_5/641. \quad (\mathrm{RT}_3)$$

*Thus predecessor maximality, the exact transition, and the first reciprocal coefficient together still do not force target maximality.*

*Proof.* The two  $E_i/E_{i-1}$ -conjugates of  $a_i^{-1}$  have sum

$$\mathrm{Tr}_{E_i/E_{i-1}}(a_i^{-1}) = \frac{a_{i-1}}{a_{i-1}^3} = a_{i-1}^{-2}.$$

Taking absolute traces and using compatibility with squaring gives

$$\mathrm{Tr}(a_i^{-1}) = \mathrm{Tr}(a_{i-1}^{-1})^2.$$

Starting with  $a_{-1} = 1$  proves  $(\mathrm{RT}_1)$ . The absolute norms are also one, so the constant coefficient is one and the linear coefficient is norm times reciprocal trace.

For the more general preservation statement, let  $A$  be a monic irreducible polynomial with constant coefficient one and root  $t$ . Its transition satisfies

$$\mathrm{Res}_Y(A(Y), X^2 + YX + Y^3) = \mathrm{Norm}_{\mathbb{F}_2(t)/\mathbb{F}_2}(1 + t^{-2}X + t^{-3}X^2).$$

The coefficient of  $X$  is therefore  $\mathrm{Tr}(t^{-2}) = \mathrm{Tr}(t^{-1})^2$ , the same bit as the linear coefficient of  $A$ .

We now certify the countermodel. Modulo  $C_1$ ,

$$Y^{2^{16}} = Y, \quad Y^{2^{56}} + Y = R := Y^{13} + Y^9 + Y^7 + Y^4 + Y,$$

and the exact Bézout identity  $UC_1 + VR = 1$  holds for

$$\begin{aligned} U &= Y^{10} + Y^9 + Y^7 + Y^6 + Y^5 + Y^4 + Y^3 + 1, \\ V &= Y^{13} + Y^{11} + Y^{10} + Y^7 + Y^3 + Y^2 + 1. \end{aligned}$$

Rabin's criterion proves  $C_1$  irreducible. If  $y$  is a root, its trace is one, and the identity

$$S_{2m+1}(Y) = 1 + \sum_{j=0}^{m-1} Y^{2^j}$$

gives  $S_{65537}(y) = 0$ . Since 65537 is prime, this is the full-order stratum.

Direct expansion of the resultant gives  $(RT_2)$ . If  $z^2 + z = y$  and  $x = yz$ , then  $x^2 + yx + y^3 = 0$ . The trace-one condition makes  $Z^2 + Z + y$  irreducible over  $\mathbb{F}_{2^{16}}$ , so  $x$  has degree thirty-two. Hence the displayed degree-thirty-two resultant is irreducible.

Finally, repeated use of

$$S_{2r} = S_r^2, \quad S_{2r+1} = S_{r+1}^2 + XS_r^2$$

along  $6700417 = (11001100011110110000001)_2$  gives the exact residues

$$(S_{6700417}, S_{6700418}) \equiv (0, 3407b7e4) \pmod{B_1},$$

where hexadecimal bit  $j$  is the coefficient of  $X^j$ . The second residue is nonzero, and 6700417 is prime. The fibotomic criterion proves  $(RT_3)$ .  $\square$

**Proposition 7.27** (nine absolute moments do not force one-step maximality). *The distinguished degree-sixteen factor is*

$$A_3 = X^{16} + X^{15} + X^{11} + X^{10} + X^8 + X^7 + X^6 + X^3 + X^2 + X + 1.$$

Put

$$C = X^{16} + X^{15} + X^{11} + X^8 + X^5 + X^3 + X^2 + X + 1.$$

Then  $A_3 + C = X^{10} + X^7 + X^6 + X^5$ . If  $a$  and  $y$  are roots of  $A_3$  and  $C$ , respectively, then

$$(\text{Tr}(a^k))_{k=1}^5 = (\text{Tr}(y^k))_{k=1}^5 = (1, 1, 1, 1, 0),$$

and

$$\text{Tr}(a^{-k}) = \text{Tr}(y^{-k}) = 1 \quad (1 \leq k \leq 4).$$

The factor  $C$  is irreducible and lies in the full-order stratum  $\mathfrak{S}_{65537}$ . Nevertheless its exact Conway resultant is

$$\begin{aligned} B &= X^{32} + X^{31} + X^{30} + X^{20} + X^{19} + X^{16} + X^{15} + X^{14} \\ &\quad + X^{11} + X^8 + X^5 + X + 1, \end{aligned}$$

which is irreducible and lies in the proper stratum  $\mathfrak{S}_{6700417}$ , where  $6700417 = F_5/641$ . Thus predecessor maximality, the exact transition, the first five forward power traces, and the first four inverse-power traces still do not force target maximality. These are absolute  $\mathbb{F}_2$ -moment data only:  $C$  does not have the distinguished relative trace-norm ancestry of  $A_3$ .

*Proof.* For a monic polynomial  $P = X^{16} + e_1X^{15} + \cdots + e_{16}$ , Newton's identities in characteristic two read

$$p_k + e_1p_{k-1} + \cdots + e_{k-1}p_1 + (k \bmod 2)e_k = 0, \quad p_k = \sum_{P(r)=0} r^k.$$

Although the final term vanishes for even  $k$ , equality of  $e_1, \dots, e_s$  still inductively forces equality of  $p_1, \dots, p_s$ . The leading five coefficient tuples of  $A_3$  and  $C$  are both  $(1, 0, 0, 0, 1)$ , giving  $(p_1, \dots, p_5) = (1, 1, 1, 1, 0)$ . Apply the same identities to  $X^{16}P(X^{-1})/P(0)$ : the first four coefficient tuples are both  $(1, 1, 1, 0)$ , giving the asserted inverse traces.

We give exact polynomial certificates for the remaining claims. In the following hexadecimal strings, bit  $j$  is the coefficient of degree  $j$ . Modulo  $C = 1892\mathbf{f}$ ,

$$Y^{2^{16}} = Y, \quad R_C := Y^{2^8} + Y = 3877,$$

and

$$U_C = \mathbf{cca}, \quad V_C = 73\mathbf{b1}, \quad U_C C + V_C R_C = 1.$$

Rabin's criterion proves that  $C$  is irreducible. Its  $Y^{15}$ -coefficient gives  $\text{Tr}(y) = 1$ , and hence

$$S_{65537}(y) = 1 + \sum_{j=0}^{15} y^{2^j} = 0.$$

Since 65537 is prime,  $C$  lies in the full-order stratum.

In the quotient by  $X^2 + YX + Y^3$ , exact reduction gives

$$B(X) = C(Y)(Q_1(Y)X + Q_0(Y)), \quad Q_1 = 31180083, \quad Q_0 = 1\mathbf{a928cc53}.$$

If  $z^2 + z = y$  and  $x = yz$ , then  $x^2 + yx + y^3 = 0$ . The trace-one condition makes  $Z^2 + Z + y$  irreducible over  $\mathbb{F}_{2^{16}}$ , so  $x$  has degree thirty-two. Hence the monic degree-thirty-two polynomial  $B$  is the exact resultant and is irreducible. Independently, its Rabin certificate is

$$X^{2^{32}} = X \pmod{B}, \quad R_B := X^{2^{16}} + X = \mathbf{e55f1c7a},$$

$$U_B = 3295\mathbf{d0c3}, \quad V_B = 6748328\mathbf{a}, \quad U_B B + V_B R_B = 1.$$

Finally the exact Fibonacci residues are

$$(S_{6700417}, S_{6700418}) \equiv (0, \mathbf{bda50b33}) \pmod{B}, \quad S_{641} \equiv 7\mathbf{acf6362} \pmod{B}.$$

As 6700417 is prime, the fibotomic criterion places  $B$  in the claimed exact proper-order stratum. If  $C$  had the distinguished relative trace-norm ancestry, Proposition 7.29 would force it to be a Frobenius conjugate of  $a_3$ , contrary to  $C \neq A_3$ ; this proves the final scope statement.  $\square$

**Remark 7.28** (higher reciprocal moments). The full inverse-power trace sequence propagates exactly, but quickly becomes deeper ancestry rather than a short invariant. If  $t = a_{i-1}$  and

$$D_0 = 0, \quad D_1 = t^{-2}, \quad D_j = t^{-2}D_{j-1} + t^{-3}D_{j-2},$$

then

$$\text{Tr}_{E_i/\mathbb{F}_2}(a_i^{-j}) = \text{Tr}_{E_{i-1}/\mathbb{F}_2}(D_j(a_{i-1})).$$

For any fixed factor, sufficiently many moments recover its minimal polynomial. What is not known is any uniformly bounded initial segment which forces the full fibotomic stratum throughout the tower.

**Proposition 7.29** (complete ancestry and exact trace-fibre exhaustion). *For  $i \geq 0$  and  $-1 \leq j < i$ , the distinguished elements satisfy*

$$\mathrm{Tr}_{E_i/E_j}(a_i) = a_j, \quad \mathrm{Norm}_{E_i/E_j}(a_i) = a_j^{3^{i-j}}. \quad (\mathrm{AN}_1)$$

*Conversely, fix  $i \geq 0$  and  $x_i \in E_i$ . Put*

$$x_j = \mathrm{Tr}_{E_i/E_j}(x_i) \quad (-1 \leq j < i).$$

*If  $x_{-1} = 1$  and*

$$\mathrm{Norm}_{E_j/E_{j-1}}(x_j) = x_{j-1}^3 \quad (0 \leq j \leq i), \quad (\mathrm{AN}_2)$$

*then  $x_i$  is an  $\mathbb{F}_2$ -Frobenius conjugate of  $a_i$ . Thus  $(\mathrm{AN}_1)$ – $(\mathrm{AN}_2)$  are a necessary-and-sufficient ancestry invariant for the actual resultant factor.*

*Now fix  $n \geq 1$ , put*

$$F = E_{n-1} = \mathbb{F}_q, \quad K = E_n = \mathbb{F}_{q^2},$$

*and let  $\ell \mid q+1$ . Set*

$$H = (K^\times)^\ell, \quad U = \ker \mathrm{Norm}_{K/F}.$$

*For every  $t \in F^\times$ ,*

$$\#\{h \in H : \mathrm{Tr}_{K/F}(h) = t\} = \frac{q+1}{\ell} - 1. \quad (\mathrm{AN}_3)$$

*If  $t = a_{n-1}$ , every element counted in  $(\mathrm{AN}_3)$  has absolute trace one and is a normal, hence full-degree, element of  $K/\mathbb{F}_2$ . Thus, whenever  $q+1$  is composite, full degree, normality, the complete Frobenius-orbit length, and the actual relative-trace value all occur among  $\ell$ -th powers.*

*The normalized relative norms of the elements in  $(\mathrm{AN}_3)$  form exactly*

$$\mathcal{B}_\ell = \left\{ \frac{u}{(u+1)^2} : u \in U^\ell, u \neq 1 \right\}, \quad |\mathcal{B}_\ell| = \frac{1}{2} \left( \frac{q+1}{\ell} - 1 \right). \quad (\mathrm{AN}_4)$$

*More precisely, an element  $h$  of relative trace  $t$  is an  $\ell$ -th power and has relative norm  $t^3$  if and only if  $t \in \mathcal{B}_\ell$ . For  $t = a_{n-1}$  this is equivalent to*

$$S_{(q+1)/\ell}(a_{n-1}) = 0. \quad (\mathrm{AN}_5)$$

*Finally  $[a_n] = [c_n]$  in  $K^\times/K^{\times\ell}$ . Consequently the entire earlier ancestry is recoverable exactly, but after the last quadratic step its power-residue content is precisely the original fibotomic membership  $(\mathrm{AN}_5)$ . Trace, separate norm, normality, and Frobenius-orbit invariants give no additional obstruction.*

*Proof.* The relation

$$a_i^2 + a_{i-1}a_i + a_{i-1}^3 = 0$$

is the minimal polynomial of  $a_i$  over  $E_{i-1}$ . It gives immediate trace  $a_{i-1}$  and norm  $a_{i-1}^3$ ; transitivity gives  $(\mathrm{AN}_1)$ .

For the converse, transitivity of trace gives  $\mathrm{Tr}_{E_j/E_{j-1}}(x_j) = x_{j-1}$ . Together with  $(\mathrm{AN}_2)$  this says that  $x_j$  is a root of

$$X^2 + x_{j-1}X + x_{j-1}^3.$$

Starting from  $x_{-1} = a_{-1} = 1$ , induct on  $j$ . If  $x_{j-1}$  is a Frobenius conjugate of  $a_{j-1}$ , the displayed polynomial is the corresponding Frobenius conjugate of the polynomial for  $a_j$ ; its two roots are

the two lifts of that conjugacy to  $E_j$ . Hence  $x_j$ , and finally  $x_i$ , is a Frobenius conjugate of the distinguished element.

Since  $\gcd(\ell, q-1) = 1$ , one has  $F^\times \subset H$ . Multiplication by any  $t \in F^\times$  bijects the trace-one and trace- $t$  fibres in  $H$ . In characteristic two,  $\text{Tr}_{K/F}(h) = 0$  is equivalent to  $h^q = h$ , so the nonzero trace-zero elements are exactly  $F^\times$ , all of which lie in  $H$ . If the common nonzero-fibre size is  $M$ , then

$$\frac{q^2 - 1}{\ell} = |H| = (q-1) + (q-1)M,$$

which gives (AN<sub>3</sub>).

The distinguished  $a_{n-1}$  has absolute trace one by (AN<sub>1</sub>). Thus every  $h$  in its relative-trace fibre also has absolute trace one. To justify normality, let  $N = [K : \mathbb{F}_2]$ , a power of two, and let  $\rho$  generate its cyclic Galois group. After choosing one normal generator,  $K$  is the regular module over

$$\mathbb{F}_2[\langle \rho \rangle] \simeq \mathbb{F}_2[X]/((X+1)^N),$$

a local ring. An element is a normal generator exactly when its group-ring coordinate is a unit, equivalently when multiplication by  $(\rho+1)^{N-1}$  does not kill it. Since

$$(\rho+1)^{N-1} = 1 + \rho + \cdots + \rho^{N-1},$$

this last value is its absolute trace. Hence absolute trace one is equivalent to normality.

Write  $h = tz$  in a nonzero trace fibre. Then  $\text{Tr}_{K/F}(z) = 1$ , so  $z^q = z + 1$  and

$$\text{Norm}_{K/F}(h) = t^2 z(z+1).$$

Put  $u = (z+1)/z \in U$ ; then  $z = 1/(u+1)$  and

$$z(z+1) = \frac{u}{(u+1)^2}. \quad (\text{AN}_6)$$

We claim  $z \in H$  if and only if  $u \in U^\ell$ . The forward implication follows from  $u = z^{q-1}$ . Conversely, if  $u \in U^\ell$ , then  $u \in H$ . The right side of (AN<sub>6</sub>) lies in  $F^\times \subset H$ , so in  $K^\times/H$  one has

$$[u] = 2[u+1].$$

Because  $\ell$  is odd, multiplication by two is invertible; hence  $[u] = 0$  forces  $[u+1] = 0$ , and  $z = (u+1)^{-1} \in H$ . The map  $u \mapsto u/(u+1)^2$  identifies precisely the inverse pair  $\{u, u^{-1}\}$ , proving (AN<sub>4</sub>).

The equality  $\text{Norm}(h) = t^3$  is, after division by  $t^2$ , exactly  $t = z(z+1)$ . This proves the first assertion following (AN<sub>4</sub>). The set  $U^\ell$  has order  $(q+1)/\ell$ , and the Fibonacci-polynomial factorization identifies its nontrivial inverse pairs with the roots of  $S_{(q+1)/\ell}$ , proving (AN<sub>5</sub>). Finally  $a_n = a_{n-1}c_n$  and  $a_{n-1} \in F^\times \subset H$ , so  $[a_n] = [c_n]$ .  $\square$

**Corollary 7.30** (a completely normal primitive candidate). *For every  $n \geq 0$ , the elements  $a_n$  and  $c_0 a_n$  are normal over every intermediate field  $E_j$ ,  $-1 \leq j < n$ . In particular they are completely normal over  $\mathbb{F}_2$ . Moreover*

$$\text{ord}(c_0 a_n) = 3 \prod_{j=1}^n \delta_j, \quad (\text{CN}_1)$$

so

$$c_0 a_n \text{ is primitive in } E_n \iff \delta_j = F_j \quad (1 \leq j \leq n). \quad (\text{CN}_2)$$

Thus the singleton-even conjecture may equivalently be stated as primitivity of one explicit completely normal element.

By contrast, the Popovych product  $c_0 c_n$  is normal only over the immediately preceding field when  $n \geq 2$ :

$$\text{Tr}_{E_n/E_{n-1}}(c_0 c_n) = c_0, \quad \text{Tr}_{E_n/E_j}(c_0 c_n) = 0 \quad (-1 \leq j \leq n-2). \quad (\text{CN}_3)$$

Hence complete normality does not supply the missing primary order, and no absolute normal-basis theorem applies directly to  $c_0 c_n$ .

*Proof.* For a cyclic extension  $L/K$  of characteristic two and degree a power of two, an element  $x \in L$  is normal over  $K$  if and only if  $\text{Tr}_{L/K}(x) \neq 0$ . Indeed, after choosing a normal generator, the regular module is

$$K[C_{2^r}] \simeq K[X]/((X+1)^{2^r}),$$

a local ring. The coordinate of  $x$  is a unit exactly when its augmentation is nonzero, and that augmentation is nonzero exactly when the relative trace of  $x$  is nonzero.

Proposition 7.29 gives

$$\text{Tr}_{E_n/E_j}(a_n) = a_j \neq 0 \quad (-1 \leq j < n),$$

so  $a_n$  is normal over every  $E_j$ . For  $j \geq 0$ , the scalar  $c_0$  lies in  $E_j$ , and hence

$$\text{Tr}_{E_n/E_j}(c_0 a_n) = c_0 a_j \neq 0.$$

At the bottom,

$$\text{Tr}_{E_n/\mathbb{F}_2}(c_0 a_n) = \text{Tr}_{E_0/\mathbb{F}_2}(c_0 a_0) = \text{Tr}_{E_0/\mathbb{F}_2}(c_0^2) = 1.$$

This proves complete normality. Corollary 7.2 gives  $\text{ord}(a_n) = \prod_{j=1}^n \delta_j$ , which is prime to three; since  $c_0$  has order three,  $(\text{CN}_1)$  follows. The identity

$$|E_n^\times| = 3 \prod_{j=1}^n F_j$$

and the pairwise coprimality of the Fermat numbers prove  $(\text{CN}_2)$ .

Finally the quadratic equation for  $c_n$  has trace one, so the first trace in  $(\text{CN}_3)$  is  $c_0$ . Every further relative trace takes the lower-field element  $c_0$  through a positive even extension degree and is therefore zero in characteristic two. The normality criterion above completes the proof.  $\square$

**Proposition 7.31** ( $\ell$ -th-root ancestry and the exact Dickson obstruction). *For variables  $S, P$ , define the characteristic-two Dickson power sums by*

$$\mathcal{D}_0(S, P) = 0, \quad \mathcal{D}_1(S, P) = S, \quad \mathcal{D}_{r+2}(S, P) = S\mathcal{D}_{r+1}(S, P) + P\mathcal{D}_r(S, P).$$

*Thus, whenever  $x + y = S$  and  $xy = P$ ,*

$$\mathcal{D}_r(S, P) = x^r + y^r.$$

Fix  $n \geq 2$  and a prime  $\ell \mid F_n$ . Put  $r_{-1} = 1$ , and for  $0 \leq j < n$  let  $r_j \in E_j^\times$  be the unique element such that

$$r_j^\ell = a_j.$$

Then, for every  $0 \leq j < n$ ,

$$N_{E_j/E_{j-1}}(r_j) = r_{j-1}^3, \quad \mathcal{D}_\ell(\text{Tr}_{E_j/E_{j-1}}(r_j), r_{j-1}^3) = r_{j-1}^\ell. \quad (\text{RA})$$

Set  $b = r_{n-1}$ . Failure of the  $\ell$ -primary Conway–Fermat condition at the top step has the exact descended form

$$c_n \in (E_n^\times)^\ell \iff \text{there exists } s \in E_{n-1} \text{ such that } \mathcal{D}_\ell(s, b) = 1. \quad (\text{RD})$$

If additionally  $\delta_j = F_j$  for every  $1 \leq j < n$ , then, with  $q = |E_{n-1}| = 2^{2^n}$ ,

$$\text{ord}(b) = \text{ord}(a_{n-1}) = \frac{q-1}{3}.$$

Consequently all exact lower trace, norm, and multiplicative-order data are automatically compatible with a hypothetical top failure; the single selected equation (RD) is the remaining obstruction.

*Proof.* Because  $\ell \mid F_n$ , it is coprime to  $|E_j^\times|$  for every  $j < n$ . Hence the  $\ell$ -power map is an automorphism on every lower multiplicative group and the elements  $r_j$  are unique. Applying the  $\ell$ -power map to the norm and using Proposition 7.29 gives

$$N(r_j)^\ell = N(a_j) = a_{j-1}^3 = (r_{j-1}^3)^\ell.$$

Injectivity of the  $\ell$ -power map on  $E_{j-1}^\times$  proves the first identity in (RA). If  $r'_j$  is the relative conjugate of  $r_j$ , then

$$\begin{aligned} \mathcal{D}_\ell(r_j + r'_j, r_j r'_j) &= r_j^\ell + (r'_j)^\ell \\ &= \text{Tr}_{E_j/E_{j-1}}(a_j) = a_{j-1} = r_{j-1}^\ell, \end{aligned}$$

which proves the second.

The group  $E_{n-1}^\times$  is uniquely  $\ell$ -divisible, while  $E_n^\times/E_{n-1}^\times$  is cyclic. Hence the coset of  $c_n$  misses the full  $\ell$ -primary part of the quotient if and only if it is an  $\ell$ -th power there, equivalently if and only if  $c_n \in (E_n^\times)^\ell$ .

Suppose first that  $c_n = t^\ell$ . Since  $\text{Tr}_{E_n/E_{n-1}}(c_n) = 1$  and  $N_{E_n/E_{n-1}}(c_n) = a_{n-1}$ , injectivity below gives

$$N(t) = b.$$

For  $s = \text{Tr}(t)$ , the power-sum identity therefore gives

$$\mathcal{D}_\ell(s, b) = \text{Tr}(t^\ell) = 1.$$

Conversely, suppose  $\mathcal{D}_\ell(s, b) = 1$ , and let  $t, t'$  be the roots of

$$X^2 + sX + b.$$

They lie in the unique extension of  $E_{n-1}$  of degree at most two, hence in  $E_n$ . Their  $\ell$ -th powers satisfy

$$t^\ell + (t')^\ell = 1, \quad t^\ell (t')^\ell = b^\ell = a_{n-1}.$$



Thus  $t^\ell, (t')^\ell$  are the two roots of

$$X^2 + X + a_{n-1},$$

namely  $c_n, c_n + 1$ . In particular  $c_n$  is an  $\ell$ -th power, proving (RD).

Finally, prior maximality, Corollary 7.2, and the  $n = 1$  base case established in its proof give

$$\text{ord}(a_{n-1}) = \prod_{j=1}^{n-1} F_j = \frac{q-1}{3}.$$

Taking the unique  $\ell$ -th root preserves this order because  $\ell$  is coprime to  $q-1$ . □

**Corollary 7.32** (choice-free lower root). *Retain Proposition 7.31, and put*

$$q = |E_{n-1}|, \quad d = \frac{q+1}{\ell}, \quad Q = \frac{q}{2}, \quad a = a_{n-1}, \quad b = r_{n-1}.$$

*Then the unique lower  $\ell$ -th root occurring in the Dickson obstruction is determined without any root choice:*

$$b^2 = a^d, \quad b = (a^d)^Q. \tag{RE}$$

*Consequently the exact failure criterion (RD) is*

$$c_n \in (E_n^\times)^\ell \iff \text{there exists } s \in E_{n-1} \text{ such that } \mathcal{D}_\ell(s, (a^d)^Q) = 1. \tag{RD'}$$

*Thus the descent contains no auxiliary choice of a compatible lower root. It still asks whether one selected Dickson value vanishes, so (RE) is a lossless normalization rather than a nonvanishing theorem.*

*Proof.* Since  $b \in \mathbb{F}_q$ , one has  $b^q = b$ . The identities  $b^\ell = a$  and  $\ell d = q+1$  give

$$b^2 = b^{q+1} = b^{\ell d} = a^d.$$

As  $q = 2Q$ , Frobenius fixity then gives

$$b = b^q = (b^2)^Q = (a^d)^Q.$$

Substitution in (RD) proves (RD'). The two displayed identities are formalized in the companion Lean file. □

**Proposition 7.33** (hypothetical-failure Fibonacci blocks collapse). *Retain Corollary 7.32. Thus*

$$q = 2^{2^n}, \quad N = q+1, \quad d = \frac{N}{\ell}, \quad Q = \frac{q}{2}, \quad a = a_{n-1}.$$

*Assume the hypothetical selected failure*

$$S_d(a) = 0,$$

*and put  $b_F = S_{d+1}(a)$ . Then every shifted Fibonacci value is already determined by the initial length- $d$  block and this one scalar:*

$$\boxed{S_{kd+r}(a) = b_F^k S_r(a) \quad (k, r \geq 0).} \tag{FBC_1}$$

*Moreover*

$$\boxed{b_F^2 = a^d, \quad b_F^\ell = a, \quad b_F = (a^d)^Q.} \tag{FBC_2}$$

Consequently  $b_F$  is exactly the choice-free lower  $\ell$ -th root  $b = r_{n-1}$  of Corollary 7.32; the adjacent Fibonacci value does not supply a second Kummer coordinate.

There is a universal polynomial form. In  $\mathbb{F}_2[X]$ ,

$$\boxed{\begin{aligned} S_{kd+r}(X) &\equiv S_{d+1}(X)^k S_r(X) \pmod{S_d(X)}, \\ S_{d+1}(X)^2 &\equiv X^d \pmod{S_d(X)}, \\ S_{d+1}(X)^\ell &\equiv X \pmod{S_d(X)}. \end{aligned}} \quad (\text{FBC}_3)$$

Thus every selected consequence depending only on the residues of consecutive or  $d$ -shifted Fibonacci values modulo the hypothetical failure factor—including ring identities obtained from their continued fractions, Cassini relations, or Euclidean reductions after passage to that quotient—reduces to the initial block together with the already-known root  $(a^d)^Q$ . It cannot create a new nonvanishing constraint.

This does not determine whether  $A_{n-1} \mid S_d$ . In particular it does not eliminate coefficient- or degree-sensitive information in the intermediate Euclidean quotient polynomials before reduction modulo  $S_d$ , nor arbitrary subresultants of  $A_{n-1}$  and  $S_d$ . Those remain possible ancestry-sensitive inputs.

*Proof.* For fixed  $d$ , both sequences

$$r \mapsto S_{d+r}(a), \quad r \mapsto S_{d+1}(a)S_r(a)$$

satisfy  $T_{r+2} = T_{r+1} + aT_r$ . Under  $S_d(a) = 0$ , their initial values are respectively  $(0, S_{d+1}(a))$  and  $(0, S_{d+1}(a))$ . They are therefore equal. Iterating this one-block identity proves (FBC<sub>1</sub>).

Cassini's identity in characteristic two is

$$S_{r+1}(X)^2 + S_r(X)S_{r+2}(X) = X^r.$$

At  $r = d$ , the failure hypothesis gives  $b_F^2 = a^d$ . Taking  $k = \ell, r = 0$  in (FBC<sub>1</sub>) gives  $S_N(a) = 0$ , while  $S_q(X) = 1$  because  $q$  is a power of two. The Fibonacci recurrence therefore gives

$$S_{N+1}(a) = S_{q+2}(a) = S_{q+1}(a) + aS_q(a) = a.$$

Taking  $k = \ell, r = 1$  in (FBC<sub>1</sub>) now yields

$$b_F^\ell = S_{\ell d+1}(a) = S_{N+1}(a) = a.$$

Finally  $b_F \in \mathbb{F}_q$ , so  $b_F^q = b_F$ . Since  $q = 2Q$ ,

$$b_F = (b_F^2)^Q = (a^d)^Q.$$

This proves (FBC<sub>2</sub>) and identifies  $b_F$  with the unique lower  $\ell$ -th root.

Apply the same recurrence argument in the quotient ring  $\mathbb{F}_2[X]/(S_d)$ . There  $S_d = 0$ , so the block identity and Cassini give the first two congruences of (FBC<sub>3</sub>). Taking  $k = \ell, r = 1$ , using  $\ell d = q + 1$ ,  $S_q = 1$ , and

$$S_{q+2} = S_{q+1} + XS_q,$$

gives the third congruence.

Lean kernel-checks the recurrence block identity, its iteration, Cassini in characteristic two, and the forced  $\ell$ -th-root identity. Their specialization to the selected Conway field and the polynomial-quotient interpretation remain paper-level.  $\square$

**Corollary 7.34** (the first Fibonacci quotient digit under failure). *Retain Proposition 7.33, and write*

$$S_d = AQ_d + \rho_d, \quad A = A_{n-1}.$$

*Under the hypothetical failure  $A \mid S_d$ , the first  $A$ -adic quotient digit is*

$$\boxed{Q_d \equiv (A')^{-1} S'_d \equiv (XA')^{-1} S_{d+1} \pmod{A}.} \quad (\text{FQD}_9)$$

*Here  $X$  and  $A'$  are units modulo  $A$ : one has  $A(0) = 1$ , while the irreducible finite-field polynomial  $A$  is separable. With*

$$a = a_{n-1}, \quad b_F = S_{d+1}(a) = (a^d)^{q/2},$$

*one has*

$$\boxed{Q_d(a) = \frac{b_F}{aA'(a)}.} \quad (\text{FQD}_{10})$$

*Thus the class of  $Q_d$  in  $\mathbb{F}_2[X]/(A) \simeq \mathbb{F}_q$ , and hence every Conway-coordinate descent, norm, resultant, or trace of that first digit, is already determined by the selected minimal polynomial and the existing choice-free Kummer root. This does not control higher  $A$ -adic digits, equivalently  $S_d \bmod A^j$  for  $j \geq 3$ , or global coefficient relations in the unreduced quotient.*

*Proof.* Under failure,  $\rho_d = 0$ . The first identity in (FQD<sub>4</sub>) gives

$$S'_d + A'Q_d \equiv 0 \pmod{A}.$$

Invertibility of  $A'$  proves the first congruence in (FQD<sub>9</sub>). The relation  $S_{d+1} \equiv XS'_d \pmod{A}$  proves the second.

Evaluation at  $a$  gives

$$S_{d+1}(a) = aS'_d(a),$$

because  $S_d(a) = 0$ . Hence  $S'_d(a) = b_F/a$ , and evaluation of (FQD<sub>9</sub>) gives (FQD<sub>10</sub>).  $\square$

**Proposition 7.35** (local uniformizers and quotient endpoint windows collapse). *Fix  $n \geq 2$ , and put*

$$M = 2^n, \quad q = 2^M, \quad N = q + 1, \quad A = A_{n-1}.$$

*Let  $\ell$  be a proper prime divisor of  $N$ , put  $d = N/\ell > 1$ , and assume the hypothetical failure  $A \mid S_d$ . Write*

$$Q_d = \frac{S_d}{A}, \quad Q_N^{\text{full}} = \frac{S_N}{A}.$$

*The second quotient is unconditional, since  $S_N(a_{n-1}) = 1 + \text{Tr}(a_{n-1}) = 0$ .*

*For a nonzero polynomial  $P$ , write*

$$P^\vee(X) = X^{\deg P} P(X^{-1}).$$

*Then the two endpoint coefficient windows satisfy*

$$\boxed{Q_d \equiv Q_N^{\text{full}} \pmod{X^{4M}}, \quad Q_d^\vee \equiv (Q_N^{\text{full}})^\vee \pmod{X^M}.} \quad (\text{QW}_1)$$

*Thus the first  $4M$  low coefficients and the first  $M$  leading coefficients of the hypothetical quotient are exactly those of the unconditional full-Fermat cofactor; neither window depends on  $\ell$ .*

There is also an all-order local statement. Let

$$\widehat{\mathcal{O}}_A = \widehat{\mathbb{F}_2[X]_{(A)}}$$

be the  $A$ -adic completion. The factorization  $S_d = AQ_d$  has  $Q_d \in \widehat{\mathcal{O}}_A^\times$ , and for every  $r \geq 1$ ,

$$\boxed{(S_d^r) = (A^r) \text{ in } \widehat{\mathcal{O}}_A.} \quad (\text{QW}_2)$$

Choose a coefficient-field identification  $\widehat{\mathcal{O}}_A \simeq \mathbb{F}_q[[T]]$ , with  $A \mapsto T$ . Then

$$S_d \mapsto T U_d(T), \quad U_d(0) = Q_d(a_{n-1}) = \frac{b_F}{a_{n-1}A'(a_{n-1})} \neq 0, \quad (\text{QW}_3)$$

and substitution  $T \mapsto T U_d(T)$  is a continuous  $\mathbb{F}_q$ -algebra automorphism.

Consequently no argument depending only on the unmarked completed local ring, its principal-ideal filtration, or the two endpoint windows in (QW<sub>1</sub>) can exclude failure. This does not erase the higher digits in the fixed  $A$ -coordinate: a relation coupling them to the global coordinate  $X$ , to middle coefficients of  $Q_d$ , or to lower Conway generators is not invariant under this reparametrization and remains a possible ancestry-sensitive input.

*Proof.* The Euler–Lucas congruence gives

$$\ell \equiv 1 \pmod{2^{n+2}}.$$

Since  $N \equiv 1 \pmod{2^{n+2}}$ , also

$$d = 1 + 4Mh$$

for an integer  $h \geq 1$ . The trailing-zero identity gives

$$S_d = S_{h+1}^{4M} + T_{n+2}(X)S_h^{4M}.$$

Every positive-index Fibonacci polynomial has constant coefficient one. Hence, modulo  $X^{4M}$ , both displayed Frobenius powers are one, and

$$S_d \equiv 1 + T_{n+2}(X) \pmod{X^{4M}}.$$

On the other hand

$$S_N = 1 + T_M(X) \equiv 1 + T_{n+2}(X) \pmod{X^{4M}},$$

because  $M \geq n + 2$ . Therefore  $A(Q_d - Q_N^{\text{full}})$  is divisible by  $X^{4M}$ . Since  $A(0) = 1$ ,  $A$  is a unit modulo  $X^{4M}$ , proving the first congruence.

Put  $D = \deg S_d = (d - 1)/2 = 2Mh$ . The standard coefficient formula is

$$[X^{D-j}]S_d = \binom{D+j}{2j} \pmod{2}.$$

For  $0 < j < M$ , the low  $n + 1$  binary digits of  $D + j$  are those of  $j$ , while the highest nonzero digit of  $2j$  is one position above the highest nonzero digit of  $j$ . Lucas's theorem therefore makes the displayed coefficient zero. Hence  $S_d^\vee \equiv 1 \pmod{X^M}$ . The sparse formula for  $S_N$  likewise gives  $S_N^\vee \equiv 1 \pmod{X^M}$ . Reciprocal factorization yields

$$S_d^\vee = A^\vee Q_d^\vee, \quad S_N^\vee = A^\vee (Q_N^{\text{full}})^\vee.$$

Since  $A^\vee(0) = 1$ , cancellation modulo  $X^M$  proves the second congruence.

Finally  $S_d$  is squarefree, so  $A \nmid Q_d$ ; hence  $Q_d$  is a unit in the  $A$ -adic completion, proving (QW<sub>2</sub>). The residue in (QW<sub>3</sub>) is Corollary 7.34. A formal series  $TU_d(T)$  with nonzero linear coefficient has a unique compositional inverse, which proves the automorphism assertion.

Lean kernel-checks the unit-rescaling statement for every ideal power, quotient-window cancellation by a common unit, and the nilpotent trailing-zero identity. The Fibonacci coefficient formula, Lucas window, completed-local-ring identification, and selected specialization remain paper-level.  $\square$

**Corollary 7.36** (the factor-sensitive endpoint boundary is exact). *Retain Proposition 7.35, and put*

$$s = v_2(\ell - 1), \quad R = 2^s.$$

Then

$$\boxed{v_2(d - 1) = s, \quad R \geq 4M, \quad s < M.} \quad (\text{QW}_4)$$

Under the hypothetical failure  $A \mid S_d$ , the first coefficients beyond the factor-sensitive endpoint windows are exact:

$$\boxed{Q_d + Q_N^{\text{full}} \equiv X^R \pmod{X^{R+1}}, \quad Q_d^\vee + (Q_N^{\text{full}})^\vee \equiv X^{R/4} \pmod{X^{R/4+1}}.} \quad (\text{QW}_5)$$

Thus the quotients agree in their first  $R$  low coefficients and their first  $R/4$  leading coefficients, and differ by one at the next coefficient on each side. Since  $R \geq 4M$ , this strictly sharpens (QW<sub>1</sub>).

Consequently even the first coefficient beyond either uniform window is not a new selected obstruction: its location is determined by  $v_2(\ell - 1)$ , and its defect is always one. Relations involving later coefficients, the middle of the quotient, both endpoints simultaneously, or the fixed  $A$ -adic coordinate remain outside the conclusion.

*Proof.* Since  $2^M \equiv -1 \pmod{\ell}$  and  $M = 2^n$ , one has  $\text{ord}_\ell(2) = 2M$ . For  $n \geq 2$ , this first gives  $\ell \equiv 1 \pmod{8}$ , so two is a quadratic residue modulo  $\ell$ . Euler's criterion then shows that  $2M \mid (\ell - 1)/2$ , whence

$$4M \mid \ell - 1.$$

Therefore  $R \geq 4M$ .

Properness gives  $\ell < N = 2^M + 1$ , hence

$$2^s = R \leq \ell - 1 < 2^M,$$

so  $s < M$ . Write  $\ell - 1 = Ra$ , with  $a$  odd. From  $\ell d = N$ ,

$$\ell(d - 1) = N - \ell = 2^M - (\ell - 1) = R(2^{M-s} - a).$$

The factor in parentheses is positive and odd, while  $\ell$  is odd. Thus  $v_2(d - 1) = s$ . Write

$$d = 1 + Rg, \quad g \text{ odd.}$$

The trailing-zero identity at the exact extracted power  $R = 2^s$  gives

$$S_d = S_{g+1}^R + T_s(X)S_g^R.$$

Both Fibonacci polynomials have constant coefficient one, so

$$S_d \equiv 1 + T_s(X) \pmod{X^R}.$$

Moreover the second summand contributes nothing at degree  $R$ , while

$$[X^R]S_{g+1}^R = [X]S_{g+1} = \binom{g-1}{1} = 0 \quad \text{in } \mathbb{F}_2.$$

On the other hand,

$$S_N = 1 + T_M(X) = 1 + T_s(X) + T_{M-s}(X)^R.$$

Since  $s < M$ , the last term has coefficient one at  $X^R$ . Hence

$$S_d + S_N \equiv X^R \pmod{X^{R+1}}. \quad (\text{QW}_6)$$

For the leading boundary put

$$D = \deg S_d = \frac{d-1}{2} = \frac{R}{2}g.$$

The Fibonacci coefficient formula gives

$$[X^{D-j}]S_d = \binom{D+j}{2j} \pmod{2}.$$

If  $0 < j < R/4$ , the low  $s-1$  binary digits of  $D+j$  are those of  $j$ , whereas the highest nonzero digit of  $2j$  lies one place above the highest digit of  $j$  and still below the  $R/2$ -place. Lucas's theorem therefore makes the coefficient zero. At  $j = R/4$ , removing the common  $2^{s-2}$ -shift in Lucas's product gives

$$\binom{D+R/4}{R/2} \equiv \binom{2g+1}{2} = g(2g+1) \equiv 1 \pmod{2}.$$

Thus

$$S_d^\vee \equiv 1 + X^{R/4} \pmod{X^{R/4+1}}.$$

The leading exponent of  $S_N$  is  $2^{M-1}$ , and its next exponent is  $2^{M-2}$ ; hence the first reciprocal gap is  $2^{M-2}$ . Since  $s < M$ ,

$$R/4 = 2^{s-2} < 2^{M-2},$$

and therefore

$$S_N^\vee \equiv 1 \pmod{X^{R/4+1}}. \quad (\text{QW}_7)$$

Finally,

$$S_d = AQ_d, \quad S_N = AQ_N^{\text{full}},$$

with  $A(0) = 1$ . In  $\mathbb{F}_2[X]/(X^{R+1})$ , multiplication by  $A$  fixes  $X^R$ , so cancellation in  $(\text{QW}_6)$  gives the first congruence of  $(\text{QW}_5)$ . Reciprocal factorization uses the common factor  $A^\vee$ , whose constant coefficient is one because  $A$  is monic. The same argument applied to  $(\text{QW}_7)$  gives the second congruence.

Lean kernel-checks the abstract common-factor boundary-defect transport. The Fermat-divisor valuation, Fibonacci boundary coefficients, and Lucas calculation remain paper-level.  $\square$

**Proposition 7.37** (two fixed-coordinate quotient-jet blocks collapse). *Retain Corollary 7.36. Thus*

$$d = 1 + Rg, \quad R = 2^s, \quad g \text{ odd}, \quad T = T_s(X) = \sum_{i=0}^{s-1} X^{2^i}.$$

For a polynomial  $P$ , write  $P^{[j]}$  for its  $j$ -th Hasse derivative. Put

$$r = \frac{g-1}{2}, \quad B = S_g, \quad C = S_{g+1}, \quad H = S_r^{2R}.$$

Then, as identities in  $\mathbb{F}_2[X]$ ,

$$S_d^{[j]} = \begin{cases} S'_d, & j = 2^i, \quad 0 \leq i < s, \\ TH, & j = R, \\ H, & j = R + 2^i, \quad 0 \leq i < s, \\ 0, & 1 \leq j < 2R \text{ otherwise,} \end{cases} \quad (\text{QJ}_1)$$

and the apparent new scalar  $H$  satisfies the exact global identity

$$X^R H = S_d + (1 + T)S'_d. \quad (\text{QJ}_2)$$

Assume now the hypothetical failure  $A \mid S_d$ , and write the unique finite  $A$ -adic expansion, extended by zero digits, as

$$Q_d = \frac{S_d}{A} = \sum_{\nu \geq 0} A^\nu q_\nu, \quad \deg q_\nu < M.$$

Then every fixed-coordinate digit

$$q_0, q_1, \dots, q_{2R-2} \quad (\text{QJ}_3)$$

is determined recursively by  $A$ , the fixed arithmetic parameter  $s$ , and the already-known first digit  $q_0 \equiv (A')^{-1}S'_d \pmod{A}$ . The second digit is explicitly

$$(A')^2 q_1 \equiv (A' + A^{[2]})q_0 + A'q'_0 \pmod{A}. \quad (\text{QJ}_4)$$

The first Hasse jet not eliminated by these identities occurs at order  $2R$ . With  $U = S_{r+1}$  and  $V = S_r$ , it is

$$S_d^{[2R]} = (1 + T)(U')^{2R} + TX^R(V')^{2R}. \quad (\text{QJ}_5)$$

Equivalently, if  $r = 2t$  this is  $(1 + T)S_t^{4R}$ , while if  $r = 2t + 1$  it is  $TX^R S_t^{4R}$ . At the selected root  $a = a_{n-1}$ , both displayed prefactors are nonzero, and therefore

$$S_d^{[2R]}(a) = 0 \iff S_{[r/2]}(a) = 0. \quad (\text{QJ}_6)$$

If  $r = 2t$  is even, this value is automatically nonzero:

$$r \equiv 0 \pmod{2} \implies S_d^{[2R]}(a) \neq 0. \quad (\text{QJ}_7)$$

Thus the first term not reduced to the old quotient residue is exactly a smaller selected Fibonacci value, and only the odd- $r$  branch can retain a zero there. The proposition does not prove that remaining value nonzero, and hence does not prove the Conway–Fermat maximal-order assertion.

*Proof.* For  $R = 2^s$ , the Hasse–Frobenius rule is

$$(F^R)^{[j]} = 0 \quad (R \nmid j), \quad (F^R)^{[Rm]} = (F^{[m]})^R. \quad (*)$$

Moreover  $T^{[j]} = 1$  when  $j = 2^i$ ,  $0 \leq i < s$ , and is zero for every other  $0 < j < R$ . The trailing-zero compression and the characteristic-two doubling identities give

$$S_d = C^R + TB^R, \quad C = S_{2r+2} = S_{r+1}^2, \quad B = S_{2r+1} = S_{r+1}^2 + XS_r^2.$$

In particular  $S'_d = B^R$ ,  $C' = 0$ , and  $B' = S_r^2$ . Applying  $(*)$  and the Hasse product rule gives  $(QJ_1)$ . Also

$$X^R H = (XS_r^2)^R = (B + C)^R = B^R + C^R = S_d + (1 + T)S'_d,$$

which proves  $(QJ_2)$ .

Let  $a = a_{n-1}$ . Since  $A$  is separable,  $A'(a) \neq 0$ . For every  $m \geq 0$ , repeated Hasse product expansion gives a triangular identity

$$Q_d^{[m]}(a) = (A'(a))^m q_m(a) + \Phi_m(A; q_0, \dots, q_{m-1})(a), \quad (\dagger)$$

where  $\Phi_m$  is explicit: a term from  $A^\nu q_\nu$  can survive at  $a$  only if  $\nu \leq m$ , and for  $\nu = m$  every one of the  $m$  copies of  $A$  must receive Hasse order one. Likewise  $S_d = AQ_d$  gives

$$S_d^{[m+1]}(a) = A'(a)Q_d^{[m]}(a) + \sum_{i=2}^{m+1} A^{[i]}(a)Q_d^{[m+1-i]}(a). \quad (\ddagger)$$

Thus the coefficient of  $q_m(a)$  in  $(\ddagger)$  is  $(A'(a))^{m+1} \neq 0$ . Reduction of  $(QJ_2)$  modulo  $A$ , together with  $A(0) = 1$ , determines  $H \bmod A$  from  $S'_d \bmod A$ . Hence  $(QJ_1)$ ,  $(\dagger)$ , and  $(\ddagger)$  determine  $q_m(a)$ , and therefore the unique degree- $< M$  polynomial  $q_m$ , successively for  $0 \leq m \leq 2R - 2$ .

For  $m = 1$ ,  $S_d^{[2]} = S'_d$  because  $2 < R$ , and reduction modulo  $A$  gives

$$A^{[2]}q_0 + A'(q'_0 + A'q_1) \equiv A'q_0,$$

which is  $(QJ_4)$  in characteristic two.

Finally,  $(*)$  at order  $2R$  gives

$$S_d^{[2R]} = (C^{[2]})^R + T(B^{[2]})^R.$$

Writing  $U = S_{r+1}$ ,  $V = S_r$ , the doubling formulas give  $C^{[2]} = (U')^2$  and  $B^{[2]} = (U')^2 + X(V')^2$ , proving  $(QJ_5)$ . The two parity specializations follow from  $S'_{2t} = 0$  and  $S'_{2t+1} = S_t^2$ . Finally

$$T_s(a)^2 + T_s(a) = a^{2^s} + a.$$

If  $T_s(a)$  were zero or one, then  $a^{2^s} = a$ . The exact degree  $[\mathbb{F}_2(a) : \mathbb{F}_2] = M$  would force  $M \mid s$ , contrary to  $0 < s < M$  in  $(QW_4)$ . Thus both  $T_s(a)$  and  $1 + T_s(a)$  are nonzero; also  $a \neq 0$  because  $A(0) = 1$ . This proves  $(QJ_6)$ .

For  $(QJ_7)$ , suppose  $r = 2t$  and the displayed value vanishes. Proposition 7.7 then gives  $\delta_n \mid t$ . The hypothetical failure gives  $\delta_n \mid d$ , while  $g = 4t + 1$ , so

$$\delta_n \mid d - 4Rt = R + 1.$$

Choose a prime  $p \mid \delta_n$ . Since also  $p \mid N = 2^M + 1$ , and  $M$  is a power of two, one has  $\text{ord}_p(2) = 2M$ . But  $p \mid R + 1 = 2^s + 1$  also gives  $\text{ord}_p(2) \mid 2s$ , hence  $M \mid s$ , contrary to  $0 < s < M$  in  $(QW_4)$ . This proves  $(QJ_7)$ .

Lean kernel-checks the second-digit rearrangement, the exact identity  $(QJ_2)$  and its quotient specialization, and the ring normalization in  $(QJ_5)$ . The Hasse block bookkeeping and the specialization to the selected Conway polynomial remain paper-level.  $\square$



**Proposition 7.38** (the odd fixed jet is an exact cofactor–Pierpont collision). *Retain Proposition 7.37, assume the hypothetical failure  $A \mid S_d$ , and suppose*

$$r = 2t + 1.$$

*Put*

$$\Gamma_{n,\ell} := \gcd(d, 3R + 1) = \gcd(d, 2^{M-s} - 3).$$

*Then*

$$\boxed{S_d^{[2R]}(a) = 0 \iff S_t(a) = 0 \iff \delta_n \mid t} \quad (\text{QJ}_8)$$

$$\iff \delta_n \mid \Gamma_{n,\ell}.$$

*In particular,  $\Gamma_{n,\ell} = 1$  is a sufficient arithmetic certificate that the odd order- $2R$  jet is nonzero.*

*Every prime  $p \mid \Gamma_{n,\ell}$  satisfies*

$$\boxed{\text{ord}_p(2) = 2M, \quad 2^{M-s} \equiv 3 \pmod{p}.} \quad (\text{QJ}_9)$$

*If  $v = v_2(s)$ , then  $v < n$  and*

$$\boxed{\text{ord}_p(3) = 2^{n+1-v}, \quad p \mid 3^{2^{n-v}} + 1, \quad p \equiv 1 \pmod{12}.} \quad (\text{QJ}_{10})$$

*Thus a surviving prime is simultaneously a primitive Fermat divisor for base two and a primitive generalized-Fermat divisor for base three, with the additional oriented discrete-log condition in (QJ<sub>9</sub>).*

*There is a sharp small-window dichotomy. Every prime in (QJ<sub>10</sub>) is one modulo both  $4M$  and  $12$ , hence one modulo  $12M$ . If*

$$3R + 1 < (12M + 1)^2,$$

*then*

$$\boxed{\Gamma_{n,\ell} \in \{1, 3R + 1\}.} \quad (\text{QJ}_{11})$$

*In the nontrivial case  $3R + 1$  is prime. Consequently, in this window,*

$$S_d^{[2R]}(a) = 0 \implies \Gamma_{n,\ell} = \delta_n = 3R + 1.$$

*In particular this applies whenever  $s \leq 2n + 5$ .*

*Using the cofactor  $d = F_n/\ell$  gives the sharp factor-sensitive screen. The weaker full-Fermat gcd can retain the selected factor  $\ell$  itself and therefore fail to certify nonvanishing when  $\Gamma_{n,\ell} = 1$ . The proposition does not prove  $\Gamma_{n,\ell} = 1$ , nor does a nontrivial  $\Gamma_{n,\ell}$  determine the selected order  $\delta_n$ .*

*Proof.* The first equivalence in (QJ<sub>8</sub>) is (QJ<sub>6</sub>), and the Fibonacci zero-set criterion gives

$$S_t(a) = 0 \iff \delta_n \mid t.$$

Under hypothetical failure one also has  $\delta_n \mid d$ . Since  $g = 4t + 3$ ,

$$d = 1 + R(4t + 3) = (3R + 1) + 4Rt.$$

Thus  $\delta_n \mid t$  implies  $\delta_n \mid 3R + 1$ . Conversely, if  $\delta_n \mid 3R + 1$ , then  $\delta_n \mid 4Rt$ ; and  $\gcd(\delta_n, 4R) = 1$ , because  $\delta_n \mid F_n$  is odd. Hence  $\delta_n \mid t$ . This proves (QJ<sub>8</sub>).

Since  $d \mid N = 2^M + 1$ , the identities

$$2^{M-s}(3R + 1) - 3N = 2^{M-s} - 3,$$

$$N - R(2^{M-s} - 3) = 3R + 1$$

prove the two gcd presentations of  $\Gamma_{n,\ell}$ .

Let  $p \mid \Gamma_{n,\ell}$  be prime. Then  $p \mid F_n$ , so  $2^M \equiv -1 \pmod{p}$ . Because  $M = 2^n$ , this forces  $\text{ord}_p(2) = 2M$ . The second gcd presentation gives  $2^{M-s} \equiv 3 \pmod{p}$ . Since  $0 < s < M$ ,

$$\gcd(2M, M-s) = 2^{v_2(s)}.$$

Therefore

$$\text{ord}_p(3) = \frac{2M}{\gcd(2M, M-s)} = 2^{n+1-v_2(s)},$$

which proves the primitive base-three assertion. Also  $p \equiv 1 \pmod{8}$ , so two is a quadratic residue modulo  $p$ ; hence so is three. Quadratic reciprocity then gives  $p \equiv 1 \pmod{12}$ .

Finally every prime divisor of  $\Gamma_{n,\ell}$  is congruent to one modulo both  $4M$  and  $12$ , hence modulo  $12M$ . Therefore every divisor of  $\Gamma_{n,\ell}$  is one modulo  $12M$ , while  $R \geq 4M$  gives  $3R+1 \equiv 1 \pmod{12M}$ . If  $1 < \Gamma_{n,\ell} < 3R+1$ , then  $\Gamma_{n,\ell} \equiv 1 \pmod{12M}$ ; from  $3R+1 = \Gamma_{n,\ell}((3R+1)/\Gamma_{n,\ell})$ , the complementary quotient is also congruent to one modulo  $12M$ . They are therefore at least  $12M+1$ , contradicting  $3R+1 < (12M+1)^2$ . Hence  $\Gamma_{n,\ell}$  is either one or  $3R+1$ . If the latter were composite, the same argument applied to a proper factor and its complement would give the same contradiction. This proves (QJ<sub>11</sub>). Finally  $s \leq 2n+5$  gives  $3R+1 \leq 96M^2+1 < (12M+1)^2$ .

Lean kernel-checks the quotient-ring collision equivalence, the linear combination relating the two collision forms, and the cancellation by the power-of-two unit. The Fibonacci zero-set, finite-field order calculations, and small-window specialization remain paper-level.  $\square$

**Proposition 7.39** (the cofactor collision has an isolated recovered jet). *Retain Proposition 7.38, and put  $B = 3R+1$ . At the opposite end of the factor range one has the unconditional exclusion*

$$\boxed{M-s \leq n+3 \implies \Gamma_{n,\ell} = 1.} \quad (\text{QJ}_{12})$$

Combined with (QJ<sub>11</sub>), a proper nontrivial collision  $1 < \Gamma_{n,\ell} < B$  can occur only when

$$2n+6 \leq s \leq M-n-4. \quad (\text{QJ}_{13})$$

There is also an exact next-jet statement. Suppose the odd branch  $r = 2t+1$  has  $S_d^{[2R]}(a) = 0$ , and first let  $t > 0$ . Write

$$t = 2^u h, \quad h \text{ odd}, \quad W = 2^{u+2}R.$$

Then

$$\boxed{S_d^{[2R+W]}(a) = T_s(a)a^R(S'_h(a))^W = T_s(a)a^R S_{(h-1)/2}(a)^{2W} \neq 0.} \quad (\text{QJ}_{14})$$

The omitted case is exact: if  $t = 0$ , then  $d = B$  and

$$S_d = 1 + T_s(1 + X^R), \quad \deg S_d = 3R/2 < 2R, \quad (\text{QJ}_{15})$$

so every Hasse jet of order at least  $2R$  vanishes. Hence a vanished odd order- $2R$  jet cannot remain Hasse-flat unless the selected cofactor is exactly  $d = 3R+1$ .

This is not a maximal-order proof: the recovered nonzero jet is compatible with  $A \mid S_d$  and determines a later fixed- $A$  quotient digit rather than contradicting hypothetical failure. Nor do (QJ<sub>11</sub>)-(QJ<sub>13</sub>) exclude the endpoint possibility  $\Gamma_{n,\ell} = B$ .

*Proof.* For (QJ<sub>12</sub>), suppose that  $p \mid \Gamma_{n,\ell}$  is prime and put  $x = M - s$ . Then (QJ<sub>9</sub>)–(QJ<sub>10</sub>) give  $p \geq 12M + 1$  and  $p \mid 2^x - 3$ . If  $x = 1$ , this says  $p \mid -1$ , which is impossible. If  $2 \leq x \leq n + 3$ , then

$$0 < 2^x - 3 \leq 2^{n+3} - 3 = 8M - 3 < 12M + 1 \leq p,$$

again impossible. This proves the exclusion. The strict-factor part of (QJ<sub>11</sub>) proves (QJ<sub>13</sub>).

For the recovered jet, (QJ<sub>5</sub>) and repeated doubling give the polynomial identity

$$S_d^{[2R]} = T_s X^R S_t^{4R} = T_s X^R S_h^W. \quad (*)$$

Lucas's theorem gives

$$\binom{2R+W}{2R} \equiv \binom{2+2^{u+2}}{2} \equiv 1 \pmod{2}.$$

Apply the order- $W$  Hasse derivative to (\*). The Hasse composition rule and the displayed parity identify its left side with  $S_d^{[2R+W]}$ . Since  $S_h^W$  is a  $W$ -th Frobenius power, the Hasse–Frobenius and product rules, evaluated at  $a$  with  $S_h(a) = 0$ , leave exactly

$$T_s(a) a^R (S'_h(a))^W.$$

The vanishing hypothesis and (QJ<sub>8</sub>) give  $\delta_n \mid t$ . Since  $\delta_n$  is odd,  $\delta_n \mid h$ . Writing  $h = 2m + 1$ , differentiation of  $S_h = S_{m+1}^2 + X S_m^2$  gives  $S'_h = S_m^2$ . If this vanished at  $a$ , the Fibonacci zero-set criterion would also give  $\delta_n \mid m$ ; then  $\delta_n \mid h - 2m = 1$ , impossible because  $\delta_n > 1$ . The other factors are nonzero by the proof of (QJ<sub>6</sub>), proving (QJ<sub>14</sub>).

If  $t = 0$ , then  $g = 3$  and  $d = 1 + 3R = B$ . Trailing-zero compression with  $S_3 = 1 + X$  and  $S_4 = 1$  gives  $S_d = 1 + T_s(1 + X^R)$ , whose degree is  $3R/2$ . This proves (QJ<sub>15</sub>).

Lean kernel-checks the strengthened short-factor window, the upper-end size contradiction, the odd-index divisibility contradiction, the Lucas parity and Hasse-composition coefficient, and non-vanishing of the recovered product. The selected Fibonacci identities and their Hasse specialization remain paper-level, as in Propositions 7.37 and 7.38.  $\square$

**Proposition 7.40** (a failed Fermat factor has a lower semiconjugacy flag). *Retain the notation of Proposition 7.39. Thus*

$$d = 1 + Rg, \quad R = 2^s, \quad g \text{ odd},$$

*and assume the hypothetical failure  $A_{n-1} \mid S_d$ . Put*

$$v = v_2(s), \quad \rho = n - v, \quad e = (g - 1)/2,$$

*and let  $\mathcal{N}$  be the quadratic resultant operator of Proposition 7.21. Then*

$$\boxed{N_{E_{n-1}/E_{v-1}}(S_g(a_{n-1})) = N_{E_{n-1}/E_{v-1}}(a_{n-1})^e.} \quad (\text{QJ}_{16})$$

*Equivalently, the actual lower selected polynomial obeys*

$$\boxed{A_{v-1} \mid \mathcal{F}_{n,\ell}(X) := \mathcal{N}^{\circ\rho}(S_g)(X) + X^{3^e}.} \quad (\text{QJ}_{17})$$

*Thus a single nonzero remainder  $\mathcal{F}_{n,\ell} \bmod A_{v-1}$  excludes the original failure; the test lives at degree  $2^v$ , not at the top degree  $2^n$ .*

*The odd-jet collision has a factor-independent specialization. Define*

$$P_0 = X + 1, \quad P_{j+1} = \mathcal{N}(P_j), \quad D_j = P_j + X^{3^j}.$$

If  $S_d^{[2R]}(a_{n-1}) = 0$ , then

$$\boxed{A_{v-1} \mid D_\rho.} \quad (\text{QJ}_{18})$$

Here  $P_j$  is monic of degree  $3^j$  with constant coefficient one. Consequently  $D_j \neq 0$ ,  $\deg D_j < 3^j$ , and

$$\boxed{3^{n-v} \leq 2^v \implies S_d^{[2R]}(a_{n-1}) \neq 0.} \quad (\text{QJ}_{19})$$

The implication in (QJ<sub>19</sub>) is an unconditional exclusion of the surviving odd collision, under the standing hypothetical-failure context. It is not a proof that  $A_{n-1} \nmid S_d$ : a nonzero Hasse jet is compatible with divisibility. If  $v = 0$ , then  $E_{v-1} = E_{-1} = \mathbb{F}_2$  and  $A_{v-1} = A_{-1} = X + 1$ ; both nonzero relative norms are one, so (QJ<sub>18</sub>) is automatic and carries no new information.

*Proof.* Write

$$w = w_n = \frac{c_n + 1}{c_n}, \quad \phi(z) = \frac{z}{(z + 1)^2},$$

so that  $\text{ord}(w) = \delta_n$  and  $a_{n-1} = \phi(w)$ . For every odd  $m$ , the Binet formula gives

$$S_m(\phi(z)) = \frac{1 + z^m}{(1 + z)^m}, \quad \phi(z^m) = \frac{\phi(z)^m}{S_m(\phi(z))^2}. \quad (*)$$

Failure gives  $\delta_n \mid d$ , hence  $Rg \equiv -1 \pmod{\delta_n}$ . Let  $L = 2^\rho$  and

$$x_j = a_{n-1}^{R^j} = \phi(w^{R^j}) \quad (j \in \mathbf{Z}/L\mathbf{Z}).$$

Because  $s = 2^v s_0$  with  $s_0$  odd, Frobenius  $z \mapsto z^R$  has order  $L$  on  $E_{n-1}/E_{v-1}$ ; the  $x_j$  are exactly the relative conjugates. Since  $\gcd(R, \delta_n) = 1$  and  $\delta_n > 1$ , the denominators below are nonzero. Moreover  $\phi(z^{-1}) = \phi(z)$ , so  $Rg \equiv -1 \pmod{\delta_n}$  gives

$$\phi((w^{R^{j+1}})^g) = \phi(w^{-R^j}) = x_j.$$

Applying (\*) to  $z = w^{R^{j+1}}$  gives

$$x_j S_g(x_{j+1})^2 = x_{j+1}^g.$$

Multiplication around the cycle yields

$$N(a_{n-1}) N(S_g(a_{n-1}))^2 = N(a_{n-1})^g,$$

where all three norms are from  $E_{n-1}$  to  $E_{v-1}$ . The first norm is nonzero. Cancelling it and using injectivity of the square map in a finite field proves (QJ<sub>16</sub>).

Propositions 7.21 and 7.29 give

$$\mathcal{N}^{\circ\rho}(S_g)(a_{v-1}) = N(S_g(a_{n-1})), \quad N(a_{n-1}) = a_{v-1}^{3^\rho},$$

which proves (QJ<sub>17</sub>).

If the odd order- $2R$  jet vanishes, (QJ<sub>8</sub>) gives  $\delta_n \mid 3R + 1$ . Repeating the cyclic argument with  $m = 3$ , rather than  $m = g$ , uses  $3R \equiv -1 \pmod{\delta_n}$  and  $S_3 = X + 1$ ; it gives

$$N(a_{n-1} + 1) = N(a_{n-1}).$$

The same selected norm descent is exactly  $P_\rho(a_{v-1}) = a_{v-1}^{3^\rho}$ , proving (QJ<sub>18</sub>).

Finally, if  $P$  is monic of degree  $h$  and  $P(0) = 1$ , then  $\mathcal{N}(P)$  is monic of degree  $3h$ , while  $\mathcal{N}(P)(0) = P(0)^2 = 1$ . For the degree assertion, if  $y, y'$  are the roots of  $Y^2 + XY + X^3$ , then

$yy' = X^3$ . The leading product  $(yy')^h = X^{3h}$  is unique: every paired cross-term  $y^i y'^j + y^j y'^i$ , with  $i \leq h, j < h$ , has degree at most  $2i + j \leq 3h - 1$ . At  $X = 0$ , the resultant is  $\text{Res}_Y(Y^2, P(Y)) = P(0)^2$ . Induction gives the assertions about  $P_j, D_j$ . If  $3^\rho \leq 2^v = \deg A_{v-1}$ , the nonzero polynomial  $D_\rho$  has degree strictly below  $A_{v-1}$ , contradicting (QJ<sub>18</sub>). This proves (QJ<sub>19</sub>).

Lean kernel-checks the denominator-cleared cubic identity, the general cyclic-product reindexing, cancellation to the exact half power in characteristic two, the selected quadratic-edge norm formula, and the final degree-divisibility exclusion. The Binet specialization, finite-field orbit identification, and resultant degree bookkeeping remain paper-level.  $\square$

The fast `--jet-only` mode of `experiments/fermat_selected_screen.py` evaluates  $\Gamma_{n,\ell}$  and the branch-appropriate gcd without constructing the selected field. Both are one on every published prime-factor row in the pinned  $F_{12}, \dots, F_{18}$  table, so the first uncollapsed Hasse jet is nonzero for all those coordinates even under hypothetical failure. This is an exact integer screen of a higher jet, not a replacement for the selected residue computation and not an all-level proof.

**Proposition 7.41** (Dickson all-or-nothing factorization). *Retain Proposition 7.31. Thus  $n \geq 2$ ,*

$$K = E_{n-1} = \mathbb{F}_q, \quad L = E_n = \mathbb{F}_{q^2}, \quad a = a_{n-1}, \quad b = r_{n-1}, \quad b^\ell = a,$$

where the prime  $\ell$  divides  $F_n = q + 1$ . Put  $c = c_n$ , so that

$$c^2 + c = a, \quad c^q = c + 1,$$

and define

$$f_{\ell,n}(X) = \mathcal{D}_\ell(X, b) + 1 \in K[X].$$

Then  $f_{\ell,n}$  is separable of degree  $\ell$ , and exactly one of the following occurs:

- (i)  $c \in L^{\times\ell}$ , in which case  $f_{\ell,n}$  splits into  $\ell$  distinct linear factors over  $K$ ;
- (ii)  $c \notin L^{\times\ell}$ , in which case  $f_{\ell,n}$  is irreducible over  $K$ .

Consequently the Dickson equation in (RD) never has an intermediate number of roots: it has either  $\ell$  roots in  $K$  or none.

More explicitly, choose  $t \in \overline{K}$  with  $t^\ell = c$ , and put

$$\vartheta = \frac{t^{q+1}}{b} \in \mu_\ell.$$

This element is independent of the chosen  $\ell$ -th root, while the standard finite-field Kummer symbol is

$$c^{(q^2-1)/\ell} = \vartheta^{-2}.$$

With the convention  $\text{Res}(f, g) = \prod_{f(s)=0} g(s)$ , one has

$$\text{Res}_X(f_{\ell,n}(X), X^q - X) = (1 + \vartheta)^\ell. \quad (\text{DF})$$

Thus the resultant vanishes precisely in the Conway–Fermat failure case.

Finally, if  $A_{n-1}$  is the minimal polynomial of  $a_{n-1}$  over  $\mathbb{F}_2$ , then

$$C_n(Y) := A_{n-1}(Y^2 + Y)$$

is the degree- $2^{n+1}$  minimal polynomial of  $c_n$ . The absolute composition

$$C_n(X^\ell) = A_{n-1}(X^{2\ell} + X^\ell) \quad (\text{AC})$$

is irreducible of degree  $2^{n+1}\ell$  in case (ii), while in case (i) it is the product of exactly  $\ell$  distinct irreducible polynomials, each of degree  $2^{n+1}$ .

*Proof.* Because  $\ell \mid q + 1$ , one has  $\mu_\ell \subset L$  and  $q \equiv -1 \pmod{\ell}$ . Fix a primitive  $\ell$ -th root  $\zeta$ , choose  $t^\ell = c$ , and put  $\tilde{t} = b/t$ . Since

$$\tilde{t}^\ell = \frac{b^\ell}{c} = \frac{a}{c} = c + 1,$$

the  $\ell$  elements, indexed by  $\mathbb{Z}/\ell\mathbb{Z}$ ,

$$s_i = t\zeta^i + \tilde{t}\zeta^{-i}$$

satisfy

$$\mathcal{D}_\ell(s_i, b) = (t\zeta^i)^\ell + (\tilde{t}\zeta^{-i})^\ell = c + (c + 1) = 1.$$

They are distinct. Indeed, equality  $s_i = s_j$  identifies the unordered root pairs of the corresponding polynomials  $X^2 + s_iX + b$ . Besides  $i = j$ , the only possible pairing would identify  $t\zeta^i$  with  $\tilde{t}\zeta^{-j}$ ; taking  $\ell$ -th powers would give  $c = c + 1$ . Hence the  $s_i$  are all the roots of the monic degree- $\ell$  polynomial  $f_{\ell,n}$ .

There is a unique  $k \in \mathbb{Z}/\ell\mathbb{Z}$  such that

$$t^q = \tilde{t}\zeta^k.$$

Since  $\zeta^q = \zeta^{-1}$ , direct substitution gives

$$s_i^q = s_{i-k}, \tag{DF_1}$$

and applying  $q$ -Frobenius once more gives

$$t^{q^2} = t\zeta^{-2k}. \tag{DF_2}$$

As  $\ell$  is odd,  $k = 0$  if and only if  $t \in L$ . Because all  $\ell$ -th roots of  $c$  differ by an element of  $\mu_\ell \subset L$ , this is equivalent to  $c \in L^{\times\ell}$ . If  $k = 0$ , every  $s_i$  is fixed by  $q$ -Frobenius. If  $k \neq 0$ , translation by  $-k$  is one  $\ell$ -cycle because  $\ell$  is prime. Equation (DF<sub>1</sub>) therefore proves the split/irreducible dichotomy.

The quotient  $t^{q+1}/b$  is  $\zeta^k$ , and replacing  $t$  by another  $\ell$ -th root does not change it because  $\ell \mid q + 1$ . Equation (DF<sub>2</sub>) yields

$$c^{(q^2-1)/\ell} = t^{q^2-1} = \vartheta^{-2}.$$

For the resultant, characteristic two and (DF<sub>1</sub>) give

$$\text{Res}(f_{\ell,n}, X^q - X) = \prod_i (s_{i-k} + s_i).$$

Moreover

$$s_{i-k} + s_i = (1 + \vartheta) (\vartheta^{-1}t\zeta^i + \tilde{t}\zeta^{-i}).$$

After pulling out the factors  $\zeta^{-i}$ , whose product is one, the fact that multiplication by two permutes  $\mathbb{Z}/\ell\mathbb{Z}$  gives

$$\prod_i (\vartheta^{-1}t\zeta^i + \tilde{t}\zeta^{-i}) = (\vartheta^{-1}t)^\ell + \tilde{t}^\ell = c + (c + 1) = 1.$$

This proves (DF).

The element  $a$  has absolute trace one, so  $Y^2 + Y + a$  is irreducible over  $K$ . Since  $a = c^2 + c$  and  $[K : \mathbb{F}_2] = 2^n$ , the element  $c$  has degree  $2^{n+1}$  over  $\mathbb{F}_2$ ; hence  $\mathcal{C}_n$  is its minimal polynomial. Capelli's lemma now says that  $\mathcal{C}_n(X^\ell)$  is irreducible over  $\mathbb{F}_2$  if and only if  $X^\ell - c$  is irreducible over  $L$ . Since  $\mu_\ell \subset L$  and  $\ell$  is prime, this holds exactly when  $c \notin L^{\times\ell}$ .

In the other case every root  $x$  of  $\mathcal{C}_n(X^\ell)$  lies in  $L$ , because its  $\ell$ -th power is a Frobenius conjugate of  $c$  and all its  $\ell$ -th roots lie in  $L$ . That conjugate has degree  $2^{n+1}$ , so

$$2^{n+1} \leq [\mathbb{F}_2(x) : \mathbb{F}_2] \leq [L : \mathbb{F}_2] = 2^{n+1}.$$

Separability then partitions the  $2^{n+1}\ell$  roots into exactly  $\ell$  distinct Frobenius orbits of length  $2^{n+1}$ , proving the final assertion.  $\square$

**Proposition 7.42** (reverse-Dickson trace divisor). *Retain Proposition 7.41, and put*

$$m = 2^n, \quad q = 2^m, \quad K = \mathbb{F}_q.$$

Let  $A_{n-1} \in \mathbb{F}_2[T]$  be the degree- $m$  minimal polynomial of  $a = a_{n-1}$ , and define

$$\begin{aligned} R_\ell(Z) &= Z^\ell \mathcal{D}_\ell(Z^{-1}, 1) = \mathcal{D}_\ell(1, Z)^2, \\ P_{\ell,a}(Z) &= Z^\ell + a R_\ell(Z), \\ H_m(Z) &= 1 + \sum_{j=0}^{m-1} Z^{2^j}, \\ \mathcal{P}_{\ell,n}(Z) &= \prod_{r=0}^{m-1} P_{\ell,a^{2^r}}(Z) = R_\ell(Z)^m A_{n-1}\left(\frac{Z^\ell}{R_\ell(Z)}\right). \end{aligned} \tag{TD}$$

The displayed rational expression is a polynomial in  $\mathbb{F}_2[Z]$ . Moreover

$$R_\ell \in \mathbb{F}_2[Z^2], \quad R_\ell(0) = 1, \quad [Z^{\ell-1}]R_\ell = 1,$$

and the following are equivalent:

- (i)  $c_n \in E_n^{\times \ell}$ , so the singleton-even arm fails at  $\ell$ ;
- (ii)  $P_{\ell,a}$  splits completely over  $K$ ;
- (iii)

$$\mathcal{P}_{\ell,n}(Z) \mid H_m(Z) \quad \text{in } \mathbb{F}_2[Z]. \tag{TD_1}$$

In the failure case every root of  $P_{\ell,a}$  has absolute trace one, and

$$\sum_{P_{\ell,a}(z)=0} z = \prod_{P_{\ell,a}(z)=0} z = a. \tag{TD_2}$$

There is also an absolute all-or-nothing alternative:  $\mathcal{P}_{\ell,n}$  is irreducible over  $\mathbb{F}_2$  of degree  $m\ell$  on success, while on failure it is the product of exactly  $\ell$  distinct irreducible polynomials of degree  $m$ .

Consequently failure forces

$$m\ell \leq \frac{q}{2}. \tag{TD_3}$$

Equivalently, for the complementary divisor  $d = (q+1)/\ell$ , one must have  $d \geq 2m+1$ .

*Proof.* The Dickson polynomial  $\mathcal{D}_\ell(X, 1)$  contains only odd powers because  $\ell$  is odd. Termwise comparison gives the first identity in (TD) and the three assertions about  $R_\ell$ . In particular

$$P'_{\ell,a}(Z) = Z^{\ell-1};$$

since  $P_{\ell,a}(0) = a \neq 0$ , this polynomial is separable.

For  $s \neq 0$ , set  $z = b/s^2$ . Dickson homogeneity gives

$$\mathcal{D}_\ell(1, z) = s^{-\ell} \mathcal{D}_\ell(s, b),$$

and hence

$$P_{\ell,a}(z) = \frac{a}{s^{2\ell}} (1 + \mathcal{D}_\ell(s, b)^2). \quad (\text{TD}_4)$$

Thus  $s \mapsto b/s^2$  is a Frobenius-equivariant bijection between the roots of  $\mathcal{D}_\ell(X, b) + 1$  and the roots of  $P_{\ell,a}$ ; its inverse sends  $z$  to the unique square root of  $b/z$ . The split/irreducible dichotomy in Proposition 7.41 therefore proves the equivalence of (i) and (ii).

Suppose failure holds, and let  $s \in K$  be one of the Dickson roots. If  $t, t'$  are the roots of  $X^2 + sX + b$ , then

$$\{t^\ell, (t')^\ell\} = \{c_n, c_n + 1\}.$$

Neither right-hand element belongs to  $K$ , so the quadratic is irreducible over  $K$ . The Artin–Schreier criterion now gives

$$\text{Tr}_{K/\mathbb{F}_2}(b/s^2) = 1.$$

Thus the  $\ell$  roots of  $P_{\ell,a}$  are distinct trace-one elements of  $K$ .

The Frobenius conjugates of  $a$  are distinct. The root sets of two different conjugate polynomials  $P_{\ell,a^{2^r}}$  are disjoint: a common root  $z$  would force  $R_\ell(z) = 0$ , then  $z^\ell = 0$ , contrary to  $R_\ell(0) = 1$ . Therefore failure supplies  $m\ell$  distinct trace-one roots of  $\mathcal{P}_{\ell,n}$ . The roots of  $H_m$  are exactly the trace-one elements of  $K$ , proving (iii). Conversely, (iii) puts every root of  $P_{\ell,a}$  in  $K$ ; equation (TD<sub>4</sub>) then puts every Dickson root in  $K$ , so Proposition 7.41 gives failure. The coefficient and constant terms of  $P_{\ell,a}$  give (TD<sub>2</sub>).

On success  $P_{\ell,a}$  is irreducible over  $K$ . If  $z$  is one of its roots, then

$$a = \frac{z^\ell}{R_\ell(z)} \in \mathbb{F}_2(z),$$

so  $K = \mathbb{F}_2(a) \subseteq \mathbb{F}_2(z)$ , and  $z$  has absolute degree  $m\ell$ . Hence  $\mathcal{P}_{\ell,n}$  is its irreducible polynomial. On failure every root  $z \in K$  still has absolute degree  $m$ , by the same displayed identity, and the disjoint conjugate fibres form exactly  $\ell$  Frobenius orbits. This proves the factorization alternative. Finally  $\deg H_m = 2^{m-1} = q/2$ , giving (TD<sub>3</sub>); the equivalent form follows from  $q \geq 2m$ .  $\square$

**Proposition 7.43** (the Conway trace is the top-bit functional). *For every  $n \geq 0$ , put  $m = 2^n$ ,  $q = 2^m$ ,  $K = E_{n-1} = \mathbb{F}_q$ , and  $a = a_{n-1} = \prod_{j < n} c_j$ , with the empty product  $a_{-1} = 1$ . In the ordinary binary representatives of the numbers in  $K$ , one has*

$$a = \prod_{j=0}^{n-1} c_j = 2^{m-1} = q/2. \quad (\text{TB}_1)$$

For the Conway bit basis  $e_i = 2^i$ ,  $0 \leq i < m$ ,

$$\text{Tr}_{K/\mathbb{F}_2}(e_i) = \begin{cases} 0, & i < m-1, \\ 1, & i = m-1. \end{cases} \quad (\text{TB}_2)$$

Consequently absolute trace is literally the top-bit coordinate:

$$\ker \text{Tr}_{K/\mathbb{F}_2} = \{0, \dots, q/2 - 1\}, \quad \text{Tr}_{K/\mathbb{F}_2}^{-1}(1) = \{q/2, \dots, q-1\}. \quad (\text{TB}_3)$$



In particular  $a$  is the least Conway ordinal of absolute trace one.

Now assume  $n \geq 2$ , let the prime  $\ell \mid F_n$ , and retain the polynomial  $P_{\ell,a}$  of Proposition 7.42. Let

$$L_m(Z) = \sum_{i=0}^{m-1} Z^{2^i}.$$

Then the singleton-even arm fails at  $\ell$  exactly when

$$\boxed{P_{\ell,a}(Z + a) \mid L_m(Z) \text{ in } K[Z]}. \quad (\text{TB}_4)$$

Thus the bad reverse-Dickson fibre is centered, at the literal least trace-one nimber, inside the trace-zero linearized polynomial.

*Proof.* Products of distinct Conway–Fermat generators require no quadratic reduction, and therefore agree with their ordinary binary product. Hence

$$a = \prod_{j < n} 2^{2^j} = 2^{\sum_{j < n} 2^j} = 2^{2^n - 1} = 2^{m-1},$$

which proves (TB<sub>1</sub>).

We prove (TB<sub>2</sub>) by induction on  $n$ . The assertion for  $K = \mathbb{F}_2$  is immediate. For  $n \geq 1$ , put  $d = m/2$  and  $K_0 = E_{n-2} = \mathbb{F}_{2^d}$ . If  $i < d$ , then  $e_i \in K_0$ , so its relative trace through the quadratic extension  $K/K_0$  is zero. If  $i \geq d$ , disjoint Fermat support gives

$$e_i = c_{n-1} e_{i-d}.$$

The two roots of the Conway equation for  $c_{n-1}$  differ by one; hence  $\text{Tr}_{K/K_0}(c_{n-1}) = 1$ , and

$$\text{Tr}_{K/K_0}(e_i) = e_{i-d}.$$

Trace transitivity and the induction hypothesis leave the value one only at  $i - d = d - 1$ , namely  $i = m - 1$ . Linearity now proves (TB<sub>3</sub>).

By Proposition 7.42, failure is equivalent to complete splitting of the separable polynomial  $P_{\ell,a}$  over  $K$ , and every one of its roots has trace one. Since  $a$  also has trace one, translation by  $a$  puts all those roots in the trace-zero space. Its vanishing polynomial is the separable linearized polynomial  $L_m$ , proving the forward implication in (TB<sub>4</sub>). Conversely, the divisibility puts every root of  $P_{\ell,a}(Z + a)$ , and hence every root of  $P_{\ell,a}$ , in  $K$ . The same proposition then gives failure.  $\square$

**Proposition 7.44** (additive top-bit countermodels). *Let  $m = 2^n \geq 2$ , put  $q = 2^m$ , and write*

$$K = \mathbb{F}_q, \quad L = \mathbb{F}_{q^2}, \quad U = \ker(N_{L/K}) = \mu_{q+1}.$$

*Let the odd prime  $\ell \mid q + 1$  satisfy  $(q + 1)/\ell > 1$ . Choose*

$$w \in U^\ell \setminus \{1\}, \quad u_0 \in U, \quad u_0^\ell = w, \quad \Phi(u) = \frac{u}{(u+1)^2}, \quad a = \Phi(w). \quad (\text{AT}_1)$$

*Then  $a \in K$  has absolute trace one and exact degree  $m$  over  $\mathbb{F}_2$ . With*

$$T_\ell(Z) = \mathcal{D}_\ell(1, Z), \quad P_{\ell,a}(Z) = Z^\ell + aT_\ell(Z)^2,$$

one has

$$G(Z) := P_{\ell,a}(Z + a) = \prod_{v \in \mu_\ell} (Z + \Phi(u_0 v) + a),$$

$$G(Z) \mid L_m(Z) := \sum_{i=0}^{m-1} Z^{2^i}. \quad (\text{AT}_2)$$

More precisely, if

$$\mathcal{T} = \{\Phi(u_0 v) + a : v \in \mu_\ell\}, \quad W = \text{span}_{\mathbb{F}_2}(\mathcal{T}),$$

and  $M_W(Z) = \prod_{w_0 \in W} (Z + w_0)$  is the subspace polynomial, then

$$\sum_{t \in \mathcal{T}} t = 0, \quad \lceil \log_2 \ell \rceil \leq \dim_{\mathbb{F}_2} W \leq \min(m-1, \ell-1), \quad (\text{AT}_3)$$

and

$$G \mid M_W \mid L_m. \quad (\text{AT}_4)$$

The trace operator is self-adjoint for  $\langle x, y \rangle = \text{Tr}_{K/\mathbb{F}_2}(xy)$ , with

$$(\ker \text{Tr})^\perp = \mathbb{F}_2, \quad (\text{AT}_5)$$

and the coefficient and derivative constraints are compatible as well:

$$[Z^{\ell-1}]G = 0, \quad G'(Z) = (Z + a)^{\ell-1}. \quad (\text{AT}_6)$$

Finally, choose any basis of  $\ker \text{Tr}$  and append  $a$ . In the induced binary labeling of  $K$ , the element  $a$  is literally the top bit and the least trace-one label. Hence the finite-field structure, top-bit trace rule, centered linearized divisibility, Moore bounds, and adjoint identities are jointly compatible with failure. This is not a Conway counterexample: the adapted basis need not be the multiplicative Fermat basis. It proves that an exclusion must use that special multiplicative ancestry, not only additive trace or a bare choice of binary labeling.

*Proof.* For  $u \in U \setminus \{1\}$ , put  $c = (1 + u)^{-1}$ . Since  $u^q = u^{-1}$ , one has

$$c^q = c + 1, \quad \Phi(u) = c^2 + c.$$

The Artin–Schreier criterion therefore gives  $\text{Tr}_{K/\mathbb{F}_2}(\Phi(u)) = 1$ . In particular  $a$  has trace one. If  $a$  lay in a proper subfield  $\mathbb{F}_{2^d} \subsetneq K$ , then  $m/d$  would be an even power of two and trace transitivity would give

$$\text{Tr}_{K/\mathbb{F}_2}(a) = (m/d) \text{Tr}_{\mathbb{F}_{2^d}/\mathbb{F}_2}(a) = 0,$$

a contradiction. Thus  $a$  has degree  $m$ .

The Dickson semiconjugacy calculation of Proposition 7.48 applies to every  $w = u_0^\ell$  and gives the product in (AT<sub>2</sub>). All values  $\Phi(u_0 v)$  have trace one, so their translates by  $a$  lie in the trace kernel, whose vanishing polynomial is  $L_m$ . The roots are distinct by the inverse-pair calculation in (KC<sub>1</sub>): for distinct  $v, v'$ , equality could only pair  $u_0 v$  with its inverse, which after raising to  $\ell$  would force  $w = w^{-1}$ , and the odd-order group  $U$  would then force  $w = 1$ . This proves the divisibility.

The original roots have sum  $a$ , while translating  $\ell$  of them adds  $\ell a = a$ ; hence  $\sum_{t \in \mathcal{T}} t = 0$ . Since  $W$  contains  $\ell$  distinct points, its dimension is at least  $\lceil \log_2 \ell \rceil$ . The displayed nontrivial dependence

and  $W \subseteq \ker \text{Tr}$  give the two upper bounds. Root-set inclusion among separable split polynomials proves (AT<sub>4</sub>).

For the trace pairing,

$$\langle \text{Tr}(x), y \rangle = \text{Tr}(x) \text{Tr}(y) = \langle x, \text{Tr}(y) \rangle.$$

The one-dimensional space  $\mathbb{F}_2$  is orthogonal to the trace kernel, so nondegeneracy gives (AT<sub>5</sub>). Finally  $P'_{\ell,a}(Z) = Z^{\ell-1}$ , and translation cancels its next-to-leading coefficient because  $\ell$  is odd. This proves (AT<sub>6</sub>). Extending a basis of the trace kernel by  $a$  gives the final binary labeling.  $\square$

**Proposition 7.45** (half-field additive saturation of a failure fibre). *Retain Proposition 7.42. Write*

$$q = Q^2 = 2^m, \quad K = \mathbb{F}_q, \quad K_0 = \mathbb{F}_Q,$$

*and suppose that the Conway–Fermat condition fails at  $\ell$ . Let  $S \subset K$  be the  $\ell$ -element root set of  $P_{\ell,a}$ . Then, for every  $t \in K_0$ ,*

$$P_{\ell,a}(Z+t) \mid H_m(Z). \quad (\text{AS}_1)$$

*These translated polynomials are pairwise distinct; indeed,*

$$[Z^{\ell-1}]P_{\ell,a}(Z+t) = a+t. \quad (\text{AS}_2)$$

*Define the relative-trace polynomial*

$$\Theta_{\ell,a}(Y) = \prod_{z \in S} (Y + z^Q + z) \in K_0[Y].$$

*Then*

$$\prod_{t \in K_0} P_{\ell,a}(Z+t) = \Theta_{\ell,a}(Z^Q + Z). \quad (\text{AS}_3)$$

*If  $\Theta_{\ell,a}$  is separable, then*

$$\ell \leq \frac{Q}{2}. \quad (\text{AS}_4)$$

*Equivalently, failure at a prime  $\ell > Q/2$  forces distinct roots  $z, z' \in S$  with*

$$z^Q + z = (z')^Q + z', \quad \text{or equivalently} \quad z + z' \in K_0. \quad (\text{AS}_5)$$

*Thus separability of this one selected relative-trace fibre would give a square-root-scale sieve strictly sharper than the unconditional degree bound (TD<sub>3</sub>). It is an additional selected assertion, not a consequence of one-step trace and norm data.*

*Proof.* Every  $z \in S$  has absolute trace one by Proposition 7.42. For  $t \in K_0$ ,

$$\text{Tr}_{K/\mathbb{F}_2}(t) = \text{Tr}_{K_0/\mathbb{F}_2}(t+t) = 0.$$

Hence all roots  $z+t$  of  $P_{\ell,a}(Z+t)$  are again roots of the trace-one polynomial  $H_m$ , proving (AS<sub>1</sub>). Because  $R_\ell$  has degree  $\ell-1$  and leading coefficient one, while  $\ell$  is odd, translation gives (AS<sub>2</sub>).

Interchanging the two products gives

$$\begin{aligned} \prod_{t \in K_0} P_{\ell,a}(Z+t) &= \prod_{z \in S} \prod_{t \in K_0} (Z+t+z) \\ &= \prod_{z \in S} (Z^Q + Z + z^Q + z), \end{aligned}$$

which is (AS<sub>3</sub>). If  $\Theta_{\ell,a}$  is separable, its  $\ell$  relative-trace values are distinct. The  $Q$  translates of each root then give  $Q\ell$  distinct trace-one elements of  $K$ . There are exactly  $q/2 = Q^2/2$  such elements, proving (AS<sub>4</sub>). Negating this injectivity condition gives (AS<sub>5</sub>).  $\square$

**Corollary 7.46** (full trace-kernel translation is exactly saturated). *Retain Proposition 7.45, and put*

$$\mathcal{T}_0 = \ker(\mathrm{Tr}_{K/\mathbb{F}_2}), \quad \mathcal{H} = \{z \in K : \mathrm{Tr}_{K/\mathbb{F}_2}(z) = 1\}.$$

*Under the hypothetical failure,*

$$\prod_{t \in \mathcal{T}_0} P_{\ell,a}(Z+t) = H_m(Z)^\ell. \quad (\text{AS}_6)$$

*The  $|\mathcal{T}_0|$  translated polynomials in this product are pairwise distinct. Thus the full additive translation attack is exactly saturated: each point of the trace-one hyperplane occurs with multiplicity  $\ell$ , so averaging over the whole trace kernel cannot yield a counting inequality or any Fourier coefficient beyond the two characters forced by the affine trace hyperplane.*

*Proof.* Let  $S \subset \mathcal{H}$  be the  $\ell$ -element root set of  $P_{\ell,a}$ . Interchanging finite products gives

$$\prod_{t \in \mathcal{T}_0} P_{\ell,a}(Z+t) = \prod_{s \in S} \prod_{t \in \mathcal{T}_0} (Z+t+s).$$

For each fixed  $s \in \mathcal{H}$ , translation by  $s$  is a bijection from  $\mathcal{T}_0$  to  $\mathcal{H}$ , so the inner product is  $H_m(Z)$ . This proves  $\text{AS}_6$ . Distinctness follows from  $\text{AS}_2$ . Lean checks the underlying generic finite-group product identity; the trace-hyperplane specialization is paper-level.  $\square$

**Proposition 7.47** (relative-trace collision factorization). *Retain Proposition 7.45, and put*

$$U = \ker(\mathrm{N}_{\mathbb{F}_{q^2}/K}), \quad U_0 = \ker(\mathrm{N}_{K/K_0}), \quad \Phi(u) = \frac{u}{(u+1)^2}.$$

*Under the hypothetical failure, choose  $u_0 \in U$  with  $u_0^\ell = w = (c_n + 1)/c_n$ , so that, as in Proposition 7.48,*

$$S = \{\Phi(u_0 v) : v \in \mu_\ell\}.$$

*For  $x^\ell = w$  and  $v \in \mu_\ell \setminus \{1\}$ , set*

$$A = x^Q, \quad B = v^Q, \quad D = (1+x)(1+xv), \quad D_Q = (1+A)(1+AB)$$

*and*

$$N_0 = A(1+B)D + x(1+v)D_Q, \\ N_1 = N_0 + DD_Q.$$

*All four factors in  $DD_Q$  are nonzero, and the collision equation factors exactly as*

$$\Phi(A) + \Phi(AB) + \Phi(x) + \Phi(xv) = \frac{N_0 N_1}{(DD_Q)^2}. \quad (\text{AX}_1)$$

*There is a more intrinsic form. Let  $s \in \mu_\ell$  be the unique square root of  $v$ , put  $y = xs$ , and set*

$$\alpha = \Phi(y), \quad \beta = \Phi(s).$$

*Then  $y^\ell = w$ ,  $s^\ell = 1$ , and  $\alpha \neq \beta$ . The two factors in  $(\text{AX}_1)$  give the disjoint alternatives*

$$N_0 = 0 \iff \alpha/\beta \in K_0^\times, \\ N_1 = 0 \iff \alpha/\beta \in U_0. \quad (\text{AX}_2)$$

Consequently the selected polynomial  $\Theta_{\ell,a}$  is separable if and only if

$$\frac{\Phi(y)}{\Phi(s)} \notin K_0^\times \cup U_0 \quad (\text{AX}_3)$$

for every  $y^\ell = w$  and every  $s \in \mu_\ell \setminus \{1\}$ . Thus its separability is exactly a selected two-axis transversality assertion, not merely a generic property of a transported multiplicative coset.

Equivalently, put  $\sigma = s + s^{-1}$  and use the Dickson power sum  $\mathcal{D}_\ell$  of Proposition 7.31. A collision exists if and only if there are

$$s \in \mu_\ell \setminus \{1\}, \quad \tau \in K_0^\times \cup U_0$$

such that

$$\mathcal{D}_\ell(\sigma, 1) = 0, \quad \mathcal{D}_\ell(\tau\sigma, 1) = a^{-1}. \quad (\text{AX}_4)$$

Thus the Conway value is not discarded: the remaining assertion is that  $a^{-1}$  avoids the two axis-restrictions of the Dickson map through every fibotomic zero. In particular, for  $z \in K^\times \setminus K_0^\times$  put

$$J(z) = \frac{N_{K/K_0}(z)}{\text{Tr}_{K/K_0}(z)^2} = \Phi(z^{Q-1}),$$

the selected ancestry  $J(a) = a_{n-2}$  makes every collision satisfy

$$J(\mathcal{D}_\ell(\tau\sigma, 1)) = a_{n-2}. \quad (\text{AX}_5)$$

This last equality is a necessary shadow of (AX<sub>4</sub>), not an exclusion by itself.

Indeed, generic separability is false even at square-root scale. For every fixed  $s \in \mu_\ell \setminus \{1\}$ , there are exactly  $Q/2$  elements  $\lambda \in K_0^\times$  for which  $\lambda\Phi(s)$  has absolute trace one, and hence exactly  $Q$  elements  $y \in U \setminus \{1\}$  such that

$$\frac{\Phi(y)}{\Phi(s)} \in K_0^\times. \quad (\text{AX}_6)$$

If  $\ell \leq Q$ , at least one of these  $y$ 's lies outside  $\mu_\ell$ . For  $w' = y^\ell \neq 1$ , the nonselected transported coset over  $w'$  then has a type- $N_0$  relative-trace collision. This construction includes the range  $Q/2 < \ell \leq Q$ ; therefore the square-root inequality alone cannot exclude the collision of the actual Conway coset.

*Proof.* For  $t \in U \setminus \{1\}$ , put  $c_t = (1+t)^{-1}$ . Then  $\Phi(t) = c_t^2 + c_t$ . Hence, with

$$h = c_x + c_{xv} = \frac{x(1+v)}{(1+x)(1+xv)},$$

one has

$$\Phi(x) + \Phi(xv) = h^2 + h. \quad (\text{AX}_7)$$

The assumptions  $x^\ell = w \neq 1$  and  $v^\ell = 1$  show that neither  $x$  nor  $xv$  is one; the same holds after  $Q$ -Frobenius, so the displayed denominators are nonzero. Moreover  $x, v \in U$ , and inversion fixes  $h$ , so  $h^q = h$  and  $h \in K$ .

The two relative traces collide precisely when

$$0 = (h^2 + h)^Q + (h^2 + h) = (h^Q + h)^2 + (h^Q + h).$$

Now  $h^Q + h \in K_0$ , and the roots of  $T^2 + T$  are 0, 1. Thus a collision is exactly  $h^Q = h$  or  $h^Q = h + 1$ . Directly,

$$h^Q + h = \frac{N_0}{DD_Q},$$

so multiplying  $(h^Q + h)(h^Q + h + 1)$  by  $(DD_Q)^2$  proves  $(\text{AX}_1)$  and identifies its two factors.

Because  $\ell$  is odd, squaring permutes  $\mu_\ell$ , so  $s$  is unique. Substitution of  $x = y/s$  and  $v = s^2$  gives

$$h = \frac{y(1+s)^2}{(y+s)(1+ys)} = \frac{\Phi(y)}{\Phi(y) + \Phi(s)} = \frac{\rho}{\rho + 1}, \quad \rho = \frac{\alpha}{\beta}. \quad (\text{AX}_8)$$

Here  $y^\ell = w$  and  $s^\ell = 1$ . The standard identity

$$\Phi(r) = \Phi(t) \iff (r+t)(1+rt) = 0$$

shows that  $\alpha = \beta$  would force  $y = s^{\pm 1}$ , contrary to  $w \neq 1$ . Thus every denominator in  $(\text{AX}_8)$  is nonzero. It follows that

$$h^Q = h \iff \rho^Q = \rho \iff \rho \in K_0^\times,$$

whereas

$$h^Q = h + 1 \iff \frac{\rho^Q}{\rho^Q + 1} = \frac{1}{\rho + 1} \iff \rho^{Q+1} = 1 \iff \rho \in U_0.$$

Since  $K_0^\times \cap U_0$  has order  $\gcd(Q-1, Q+1) = 1$ , and  $\rho \neq 1$ , these alternatives are disjoint. Centering an unordered pair of distinct elements of  $u_0\mu_\ell$  reverses the construction uniquely up to replacing  $s$  by  $s^{-1}$ . This proves  $(\text{AX}_2)$ – $(\text{AX}_3)$ .

Since  $\Phi(t)^{-1} = t + t^{-1}$ , take  $\tau = \beta/\alpha = \rho^{-1}$ . Both axes are closed under inversion, and

$$y + y^{-1} = \tau(s + s^{-1}) = \tau\sigma.$$

The Dickson power-sum identity and  $y^\ell = w$ ,  $s^\ell = 1$  give  $(\text{AX}_4)$ . Conversely, suppose  $(\text{AX}_4)$  holds and let  $y, y^{-1}$  be the roots of

$$X^2 + \tau\sigma X + 1.$$

Then

$$y^\ell + y^{-\ell} = a^{-1} = w + w^{-1},$$

so  $y^\ell \in \{w, w^{-1}\}$ . The element  $y$  cannot lie in  $K$ , because then  $y^\ell \in K$ , whereas  $w \in U \setminus \{1\}$  and  $U \cap K^\times = \{1\}$ . As the quadratic has coefficients in  $K$ , it follows that  $y^q = y^{-1}$ , so  $y \in U$ ; replace  $y$  by  $y^{-1}$  if necessary to arrange  $y^\ell = w$ . Now

$$\frac{\Phi(y)}{\Phi(s)} = \tau^{-1} \in K_0^\times \cup U_0,$$

and  $(\text{AX}_2)$  gives a collision. Finally,  $J(z^{-1}) = J(z)$ , while the Conway trace and norm recursion gives  $J(a) = a_{n-2}$ , proving  $(\text{AX}_5)$ .

For the final assertion fix  $s \neq 1$  and write  $\beta = \Phi(s)$ . The element  $\beta$  has absolute trace one. Therefore  $b_0 = \beta^Q + \beta$  is nonzero: if  $\beta \in K_0$ , its absolute trace from the quadratic extension  $K/K_0$  would vanish. For  $\lambda \in K_0$ , transitivity of trace gives

$$\text{Tr}_{K/\mathbb{F}_2}(\lambda\beta) = \text{Tr}_{K_0/\mathbb{F}_2}(\lambda b_0).$$

This nonzero  $\mathbb{F}_2$ -linear functional takes the value one on exactly  $Q/2$  elements of  $K_0$ , none of them zero. Every trace-one element of  $K$  has exactly the inverse pair of lifts under  $\Phi : U \setminus \{1\} \rightarrow K$ , so these scalars produce exactly  $Q$  values of  $y$  satisfying  $(\text{AX}_6)$ . At most  $\ell - 1$  of them lie in  $\mu_\ell$ . If  $\ell \leq Q$ , choose one outside, put  $w' = y^\ell$ , and center the pair  $y/s, ys$ . It consists of distinct members of the same  $\mu_\ell$ -coset and has a collision by  $(\text{AX}_2)$ , completing the construction.  $\square$

**Proposition 7.48** (the reverse-Dickson divisor is a transported Kummer coset). *Retain Proposition 7.42, put  $N = q + 1$ , and write*

$$U = \ker\left(N_{\mathbb{F}_{q^2}/\mathbb{F}_q}\right) = \mu_N.$$

For odd  $r$ , define

$$T_r(Z) = \mathcal{D}_r(1, Z), \quad F_r(Z) = \frac{Z^r}{T_r(Z)^2}, \quad \Phi(u) = \frac{u}{(u+1)^2}.$$

Then  $\Phi$  induces a bijection

$$(U \setminus \{1\})/(u \sim u^{-1}) \xrightarrow{\sim} \{z \in \mathbb{F}_q : \text{Tr}_{\mathbb{F}_q/\mathbb{F}_2}(z) = 1\}, \quad (\text{KC}_1)$$

and one has the semiconjugacy

$$T_r(\Phi(u)) = \frac{1+u^r}{(1+u)^r}, \quad F_r(\Phi(u)) = \Phi(u^r). \quad (\text{KC}_2)$$

Consequently

$$H_m(Z) = T_N(Z) = \prod_{\substack{\{u, u^{-1}\} \\ u \in U \setminus \{1\}}} (Z + \Phi(u)). \quad (\text{KC}_3)$$

For the selected Conway point put

$$w = \frac{c_n + 1}{c_n} \in U, \quad a = \Phi(w) = a_{n-1}.$$

Failure at  $\ell$  is equivalent to  $w \in U^\ell$ . If  $u_0^\ell = w$ , then

$$P_{\ell,a}(Z) = Z^\ell + aT_\ell(Z)^2 = \prod_{v \in \mu_\ell} (Z + \Phi(u_0 v)). \quad (\text{KC}_4)$$

Thus the reverse-Dickson fibre is exactly one multiplicative  $\mu_\ell$ -coset transported through  $\Phi$ . Furthermore

$$T_\ell \mid H_m, \quad \gcd(\mathcal{P}_{\ell,n}, T_\ell) = 1,$$

so failure gives the sharper factor statement

$$\mathcal{P}_{\ell,n}(Z)T_\ell(Z) \mid H_m(Z). \quad (\text{KC}_5)$$

There are exactly

$$\frac{N/\ell - 1}{2}$$

distinct trace-one values  $\Phi(w)$  with  $w \in U^\ell \setminus \{1\}$ , each realizing the same transported coset factorization.

*Proof.* For  $u \in U \setminus \{1\}$ , put  $c = (u+1)^{-1}$ . Since  $u^q = u^{-1}$ ,

$$c^q = c + 1, \quad c^2 + c = \Phi(u).$$

The Artin-Schreier trace criterion gives  $\text{Tr}_{\mathbb{F}_q/\mathbb{F}_2}(\Phi(u)) = 1$ . Direct simplification also gives

$$\Phi(u) = \Phi(v) \iff (u+v)(1+uv) = 0,$$

so the fibres are precisely inverse pairs. Both sides of (KC<sub>1</sub>) have  $q/2$  elements.

The roots of  $X^2 + X + \Phi(u)$  are  $(1+u)^{-1}$  and  $u(1+u)^{-1}$ . The Dickson power-sum identity therefore proves the first formula in (KC<sub>2</sub>), and the second follows on squaring the denominator. Taking  $r = N$  shows that the monic degree- $q/2$  polynomials  $T_N$  and  $H_m$  have the same roots, proving (KC<sub>3</sub>).

One has  $w = c_n^{q-1}$ . In the coprime decomposition  $\mathbb{F}_{q^2}^\times \simeq \mathbb{F}_q^\times \times U$ , the  $\ell$ -power map is an automorphism on  $\mathbb{F}_q^\times$ . Hence  $c_n$  is an  $\ell$ -th power exactly when  $w \in U^\ell$ , by Proposition 7.31. For  $v \in \mu_\ell$ , semiconjugacy gives

$$F_\ell(\Phi(u_0v)) = \Phi((u_0v)^\ell) = \Phi(w) = a.$$

These  $\ell$  values are distinct. Indeed, an inverse collision inside  $u_0\mu_\ell$  would imply  $w^2 = 1$ , hence  $w = 1$  because  $|U|$  is odd; but  $w = 1$  is impossible. The monic degree- $\ell$  polynomial  $P_{\ell,a}$  therefore has precisely the roots in (KC<sub>4</sub>).

The same argument on  $\mu_\ell \setminus \{1\}$  gives  $T_\ell \mid T_N$ . Also

$$P_{\ell,a} \equiv Z^\ell \pmod{T_\ell}, \quad T_\ell(0) = 1,$$

and the same congruence holds for every Frobenius conjugate. This proves coprimality and (KC<sub>5</sub>). Finally the image  $U^\ell$  has order  $N/\ell$ ; its nonidentity elements form  $(N/\ell - 1)/2$  inverse pairs, and (KC<sub>1</sub>) proves the count.  $\square$

**Remark 7.49** (exact fibotomic support of a hypothetical failure). Use the fibotomic polynomials  $\mathfrak{F}_s$  of Proposition 7.7. Put

$$e = v_\ell(N), \quad \delta = \text{ord}(w) = \ell^f \delta_0, \quad (\ell, \delta_0) = 1.$$

Under failure  $0 \leq f \leq e - 1$ . If  $f = 0$ , the coset  $u_0\mu_\ell$  contains exactly one element of order  $\delta$  and  $\ell - 1$  elements of order  $\ell\delta$ . Accordingly  $\mathcal{P}_{\ell,n}$  has exactly one degree- $m$  irreducible factor in  $\mathfrak{F}_\delta$  and exactly  $\ell - 1$  in  $\mathfrak{F}_{\ell\delta}$ . If  $f \geq 1$ , every element of the coset has order  $\ell\delta$ , and all  $\ell$  factors lie in  $\mathfrak{F}_{\ell\delta}$ . This includes the cases  $v_\ell(N) > 1$ .

These support constraints do not close the arm. Every trace-one element of  $\mathbb{F}_q$  has absolute degree  $m$ : if it lay in a proper subfield  $\mathbb{F}_{2^d}$ , then  $m/d$  would be even and its absolute trace would vanish. Hence the only resulting capacity inequality is

$$\varphi(\delta) \geq 2m.$$

It is automatic, because  $\delta > 1$  divides  $2^m + 1$ , while  $m$  is a power of two; therefore  $\text{ord}_\delta(2) = 2m$ . The full divisor, derivative, Hasse-derivative, Newton-sum, coefficient, and fibotomic-support packages are thus realized by transported nonselected cosets. Excluding the selected one still requires the recursive Conway ancestry.

**Remark 7.50** (the reverse-Dickson gap is selected but not yet excluded). Put  $e = v_2(\ell - 1)$ . For  $e \geq 2$ , substitution  $X = W + W^{-1}$  gives

$$\mathcal{D}_\ell(X, 1) + X = (W^{\ell-1} + 1)(W + W^{-\ell}).$$

At  $W = 1$  the two factors vanish to orders  $2^e$  and 2, while  $X$  has order 2. Therefore

$$\mathcal{D}_\ell(X, 1) \equiv X \pmod{X^{2^{e-1}+1}}. \tag{TG}$$



Every prime divisor of the Fermat number  $F_n = q + 1$  has  $e \geq n + 2$ . Hence

$$\begin{aligned} P_{\ell,a}(Z) &= Z^\ell + aZ^{\ell-1} + \text{terms of degree at most } \ell - (2m + 1), \\ \mathcal{P}_{\ell,n}(Z) &= Z^{m(\ell-1)} A_{n-1}(Z) + \text{terms of degree at most } m\ell - (2m + 1). \end{aligned} \quad (\text{TG}_1)$$

This long leading gap still does not contradict (TD<sub>1</sub>). For  $n > 2$ , every nontrivial complementary divisor  $d \mid F_n$  satisfies  $d \equiv 1 \pmod{4m}$ , hence  $d \geq 4m + 1$  and  $m\ell < q/4$ . A hypothetical quotient  $H_m/\mathcal{P}_{\ell,n}$  is therefore long enough to absorb the leading gap. The divisor criterion retains the actual selected polynomial  $A_{n-1}$ ; its elementary degree and coefficient consequences alone do not decide it.

## 7.4 Elliptic and generalized-Jacobian reformulations

**Proposition 7.51** (supersingular elliptic form of the Conway transition). *For  $i \geq 0$  put*

$$u_i := \frac{c_i + 1}{c_i}.$$

*Fix  $n \geq 1$ , set  $Q = 2^{2^{n-1}}$ ,  $v = u_{n-1}$  and  $u = u_n$ . Then the exact Conway recursion is*

$$\frac{u}{(u+1)^2} = \frac{v}{(v+1)^3}, \quad (\text{E})$$

*or equivalently*

$$u(v+1)^3 = v(u+1)^2, \quad u + u^{-1} = v^2 + v + 1 + v^{-1}.$$

*The normalization of this correspondence is the fixed supersingular elliptic curve over  $\mathbb{F}_2$*

$$\mathcal{E} : \quad y^2 + y = x^3 + x^2. \quad (\text{W})$$

*The maps to the  $v$ - and  $u$ -lines have degrees two and three, respectively; in particular the transition has no rational section recovering  $u$  from  $v$ .*

*On the original  $(u, v)$  correspondence, the rational map*

$$\mathcal{A}(u, v) = \left( \frac{u+v}{uv+1}, v^{-1} \right)$$

*extends to an automorphism of order four, with*

$$\mathcal{A}^2(u, v) = (u^{-1}, v).$$

*For the distinguished Conway point  $P_n = (u_n, u_{n-1})$  one has*

$$\text{Frob}_Q(P_n) = \mathcal{A}(P_n) \quad \text{or} \quad \mathcal{A}^{-1}(P_n). \quad (\text{Q})$$

*Thus, at  $P_n$ , relative Frobenius is one of the two quarter-turns  $\mathcal{A}^{\pm 1}$  on the elliptic correspondence.*

*Proof.* Write  $a_i = \prod_{j \leq i} c_j$ . The identities

$$a_{n-1} = \frac{u}{(u+1)^2}, \quad a_{n-1} = a_{n-2}c_{n-1} = \frac{v}{(v+1)^2} \frac{1}{v+1}$$

give (E); inversion gives the second displayed form.

Put  $U = u + 1$ ,  $V = v + 1$ , and blow up the affine singularity  $(U, V) = (0, 0)$  by  $X = U/V$ ,  $Y = V$ . After division by  $V^2$  the equation becomes

$$XY^2 + (X^2 + 1)Y + X^2 = 0.$$

Its homogenization

$$H = XY^2 + (X^2 + Z^2)Y + X^2Z$$

has partial derivatives  $(Y^2, X^2 + Z^2, X^2)$  and is therefore smooth. The point  $[1 : 1 : 0]$  is a flex. The linear change

$$(x, y, z) = (Z, X, X + Y + Z)$$

takes  $H = 0$  to  $y^2z + yz^2 = x^3 + x^2z$ , which is (W). This curve has five  $\mathbb{F}_2$ -points, hence trace  $-2$ , and is supersingular. The original equation has degrees two in  $u$  and three in  $v$ , proving the degrees of the two projections. A rational section of the nontrivial degree-two finite cover would induce a retraction of its degree-two function-field extension, impossible for a proper finite field extension.

Direct substitution in (E) gives

$$\frac{\mathcal{A}_u}{(\mathcal{A}_u + 1)^2} = v \frac{u}{(u + 1)^2} = \frac{v^{-1}}{(v^{-1} + 1)^3},$$

so  $\mathcal{A}$  preserves the correspondence. A second substitution gives  $\mathcal{A}^2(u, v) = (u^{-1}, v)$ , proving the order assertion. Thus this birational self-map and its birational inverse extend over the smooth projective normalization.

Finally  $v^Q = v^{-1}$  and

$$\left( \frac{u}{(u + 1)^2} \right)^Q = \left( \frac{v}{(v + 1)^3} \right)^Q = v \frac{u}{(u + 1)^2}.$$

The fibres of  $z \mapsto z/(z + 1)^2$  are the inverse-pairs  $\{z, z^{-1}\}$ . The first coordinate of  $\text{Frob}_Q(P_n)$  is therefore either  $\mathcal{A}_u(P_n)$  or its inverse, while the second coordinate is  $v^{-1}$ , proving (Q).  $\square$

**Remark 7.52** ( $X_0(11)$  reduction, but not a modular tower). The elliptic curve  $\mathcal{E}$  is the good reduction modulo two of

$$y^2 + y = x^3 - x^2 - 10x - 20,$$

the standard  $X_0(11)$  model (Cremona 11a1, LMFDB 11.a2) [25]. Its discriminant is  $-11^5$ , and reduction gives (W). This observation does not turn the Conway recursion into a modular-curve tower. The two Conway coordinate maps have degrees two and three, whereas  $X_0(11) \rightarrow X(1)$  has degree twelve and the standard degeneracy maps  $X_0(11^{r+1}) \rightarrow X_0(11^r)$ ,  $r \geq 1$ , have degree eleven. Moreover

$$j = -\frac{2^{12}3^{13}}{11^5}$$

is nonintegral, so the characteristic-zero curve has no complex multiplication. The order-four endomorphism  $i = \pi + 1$  in Proposition 7.78 is extra supersingular structure created after reduction; it does not lift to an endomorphism of this characteristic-zero  $X_0(11)$  model.

**Proposition 7.53** (the successive-point overlap has zero CM component). *On  $\mathcal{E}$ , the two Conway coordinates are*

$$u = \frac{x^2 + y}{x}, \quad v = \frac{y + 1}{x},$$

of degrees three and two respectively. Let

$$\mathcal{C} = \mathcal{E} \times_{v, \mathbf{P}^1, u} \mathcal{E}$$

be the normalization of the fibre product, and write  $p, q : \mathcal{C} \rightarrow \mathcal{E}$  for the first and second projections. Then  $\mathcal{C}$  is integral,

$$\deg p = 3, \quad \deg q = 2, \quad (\text{OV}_1)$$

and the correspondence induced on  $\text{Pic}^0(\mathcal{E})$  vanishes in both directions:

$$q_* p^* = 0, \quad p_* q^* = 0. \quad (\text{OV}_2)$$

Equivalently, its numerical class has bidegree  $(3, 2)$  and zero  $\text{End}(\mathcal{E})$ -component. In particular the overlap, when specialized to  $n \geq 2$ ,

$$v(P_n) = u(P_{n-1})$$

is not the graph of an isogeny and has no nonzero isogeny or CM component; in particular it is not a Hecke correspondence acting nontrivially on  $\text{Pic}^0(\mathcal{E})$ .

This failure persists at the Kummer level. Put

$$\mathfrak{c} = \frac{1}{u+1}, \quad \mathfrak{d} = \frac{1}{v+1}.$$

On  $\mathcal{C}$ ,

$$q^* \mathfrak{c} = p^* \mathfrak{d}, \quad (\text{OV}_3)$$

For  $n \geq 2$ , put  $\tilde{P}_n = (P_n, P_{n-1}) \in \mathcal{C}$ . Then

$$(p^* \mathfrak{c})(\tilde{P}_n) = c_n, \quad (q^* \mathfrak{c})(\tilde{P}_n) = (p^* \mathfrak{d})(\tilde{P}_n) = c_{n-1}.$$

For every prime  $\ell > 3$ , the class of  $p^* \mathfrak{c}$  is not in the span of  $p^* \mathfrak{d}$  in  $\overline{\mathbb{F}}_2(\mathcal{C})^\times / \overline{\mathbb{F}}_2(\mathcal{C})^{\times \ell}$ . Thus the correspondence yields no monomial Kummer relation between the new function  $p^* \mathfrak{c}$  and the transported old function  $p^* \mathfrak{d} = q^* \mathfrak{c}$ . This global divisor statement does not decide the specialized class  $[c_n]$ , including when  $\ell \mid F_n$ .

*Proof.* The  $v$ -map is the separable Galois double cover whose deck involution is  $P \mapsto T - P$ ; the  $u$ -map is separable of odd degree three. Their function fields over  $\overline{\mathbb{F}}_2(\mathbf{P}^1)$  have coprime degrees and hence are linearly disjoint. This proves integrality and  $(\text{OV}_1)$ . Norm-conorm compatibility in the compositum of these function fields gives

$$q_* p^* = u^* v_*$$

on degree-zero divisor classes. But  $\text{Pic}^0(\mathbf{P}^1) = 0$ , so this map vanishes; the transpose argument proves the other equality in  $(\text{OV}_2)$ .

Equation  $(\text{OV}_3)$  is the defining equality  $v \circ p = u \circ q$ . The divisors on  $\mathcal{E}$  are

$$\begin{aligned} \text{div}(\mathfrak{c}) &= [-T] + 2[O] - 3[-2T], \\ \text{div}(\mathfrak{d}) &= [O] + [T] - 2[-2T]. \end{aligned}$$

At a point of  $\mathcal{C}$  above  $-T$  under  $p$ , the valuation of  $p^* \mathfrak{c}$  is the ramification index, hence one, two, or three, while the valuation of  $p^* \mathfrak{d}$  is zero. Since every prime divisor of  $F_n$ ,  $n \geq 1$ , is larger than three, no relation

$$p^* \mathfrak{c} = (p^* \mathfrak{d})^r h^\ell$$

is possible. This proves the Kummer assertion.  $\square$

**Proposition 7.54** (new-prime character neutrality and the exact Kummer target). *Retain Proposition 7.51 and let  $\ell$  be a prime divisor of  $Q^2 + 1 = F_n$ . If  $\chi : E_n^\times = \mathbb{F}_{Q^4}^\times \rightarrow \mu_\ell$  has exact order  $\ell$ , and*

$$\alpha = \chi(u), \quad \beta = \chi(u+1),$$

*then*

$$\alpha = \beta^2, \quad \beta^{Q^2} = \beta^{-1}. \quad (\text{K})$$

*The second identity is tautological because  $Q^2 \equiv -1 \pmod{\ell}$ ; in particular the exact lower transition and the quarter-turn Frobenius symmetry do not by themselves exclude  $\beta = 1$ .*

*The following are equivalent:*

- (i) *the order  $\delta_n$  contains the full  $\ell$ -primary part of  $F_n$ ;*
- (ii)  *$u_n \notin U_n^\ell$ , where  $U_n = \ker N_{E_n/E_{n-1}}$ ;*
- (iii)  *$c_n \notin (E_n^\times)^\ell$ ;*
- (iv)  *$X^\ell - c_n$  is irreducible over  $E_n$ .*

*Consequently the all-level Conway–Fermat theorem is exactly the nontriviality, for every  $\ell \mid F_n$ , of this new Kummer symbol at the distinguished point of (W). No lower-field multiplicative character can supply its value.*

*Proof.* The lower field has order  $Q^2$ , and  $\gcd(\ell, Q^2 - 1) = 1$ , so  $\chi$  is trivial on all of  $E_{n-1}^\times$ , in particular on every expression in  $v$ . Applying  $\chi$  to (E) gives  $\alpha = \beta^2$ . Since  $u^{Q^2} = u^{-1}$ ,

$$(u+1)^{Q^2} = u^{-1} + 1 = \frac{u+1}{u},$$

which gives the other identity in (K). It reduces to  $\beta^{-1} = \beta^{-1}$  modulo  $\ell$ .

In the cyclic group  $U_n$  of order  $F_n$ , nonmembership in  $U_n^\ell$  is equivalent to having the full  $\ell$ -primary order. Here

$$U_n^\ell = U_n \cap (E_n^\times)^\ell. \quad (\text{K}_0)$$

Indeed, if  $u = x^\ell \in U_n$ , then  $N_{E_n/E_{n-1}}(x)^\ell = 1$ ; because  $\gcd(\ell, Q^2 - 1) = 1$ , this forces  $N(x) = 1$ , so  $x \in U_n$ . The reverse inclusion is immediate. Moreover (E) lies in the lower field, hence is an  $\ell$ -th power in  $E_n$ . In  $E_n^\times / (E_n^\times)^\ell$  it therefore says

$$[u] = 2[u+1].$$

Since  $\ell$  is odd,  $u$  is an  $\ell$ -th power exactly when  $u+1$  is. But  $u+1 = c_n^{-1}$ , proving the equivalence of the first three assertions. Finally  $\mu_\ell \subset E_n$ , so the prime-degree Kummer criterion proves the equivalence with irreducibility of  $X^\ell - c_n$ .  $\square$

**Proposition 7.55** (a Fermat Kummer coordinate is born once and transported by 3/2). *Fix  $n \geq 1$ , let  $M = \ell^e \parallel F_n$ , and, in additive notation, put*

$$\mathcal{H}_j = E_j^\times / (E_j^\times)^M.$$

*Then*

$$\mathcal{H}_j = 0 \quad (j < n), \quad \mathcal{H}_j \cong \mathbf{Z}/M\mathbf{Z} \quad (j \geq n). \quad (\text{KT}_1)$$

For  $j > n$ , inclusion induces an isomorphism  $\iota_j : \mathcal{H}_{j-1} \xrightarrow{\sim} \mathcal{H}_j$ . If

$$\kappa_j = [a_j]_M,$$

then

$$\kappa_n = [c_n]_M = -\frac{1}{2}[w_n]_M, \quad 2[c_j]_M = \iota_j(\kappa_{j-1}), \quad \boxed{\kappa_j = \frac{3}{2}\iota_j(\kappa_{j-1})} \quad (j > n). \quad (\text{KT}_2)$$

Thus, under the canonical identifications along the tail,

$$\boxed{\kappa_j = (3/2)^{j-n}\kappa_n \quad (j \geq n).} \quad (\text{KT}_3)$$

The multiplier  $3/2$  is a unit modulo  $M$ . Consequently the exact  $\ell$ -adic divisibility of the selected class is invariant along the whole future tower, and for every  $j \geq n$ ,

$$c_n \in (E_n^\times)^\ell \iff a_j \in (E_j^\times)^\ell. \quad (\text{KT}_4)$$

For  $j > n$ , the current driver  $w_j$  has order dividing  $F_j$ , which is coprime to  $M$ , and hence

$$[w_j]_M = 0, \quad [w_j + 1]_M = -\frac{1}{2}\iota_j(\kappa_{j-1}). \quad (\text{KT}_5)$$

In particular a failure born at  $(n, \ell)$  is equivalent to every—or any—future selected translate  $w_j + 1$  being an  $\ell$ -th power. These are not independent future conditions.

Let  $A_j$  be the minimal polynomial of  $a_j$  over  $\mathbb{F}_2$ . The same transport has the exact arboreal form

$$\begin{aligned} \kappa_n \notin \ell\mathcal{H}_n &\iff A_j(X^\ell) \text{ is irreducible over } \mathbb{F}_2, \\ \kappa_n \in \ell\mathcal{H}_n &\iff A_j(X^\ell) \text{ is a product of exactly } \ell \text{ distinct irreducibles of degree } [E_j : \mathbb{F}_2] \end{aligned} \quad (\text{KT}_6)$$

for one, equivalently every,  $j \geq n$ . Hence the infinite future resultant tree supplies no accumulating obstruction: its irreducible or completely split behavior is an exact copy of the single Kummer coordinate born at level  $n$ .

The transport is equally explicit on the selected Euler symbols. Put

$$\mathfrak{h}_j = a_j^{(|E_j|-1)/\ell} \in \mu_\ell \quad (j \geq n).$$

Then

$$\mathfrak{h}_n = w_n^{F_n/\ell}, \quad \boxed{\mathfrak{h}_j = \mathfrak{h}_{j-1}^3 = \mathfrak{h}_n^{3^{j-n}} \quad (j > n).} \quad (\text{KT}_7)$$

Since  $\ell \neq 3$ , cubing is an automorphism of  $\mu_\ell$ . Thus the selected Euler residue remains nontrivial or trivial along the entire tail. Motivated by the birth Hilbert-symbol normalization, put  $\hat{\mathfrak{h}}_j = \mathfrak{h}_j^{-2}$ . Then

$$\hat{\mathfrak{h}}_n = H \quad \text{in } (\text{GL}_6), \quad \hat{\mathfrak{h}}_j = \hat{\mathfrak{h}}_{j-1}^3.$$

There is also a simultaneous direct-product diagonalization. At the edge  $E_{j-1} \subset E_j$ , put  $q = |E_{j-1}|$ , let  $s_j \in E_{j-1}$  be the unique square root of  $a_{j-1}$ , and put  $g_j = c_j/s_j$ . Then

$$N(g_j) = 1, \quad c_j = s_j g_j, \quad a_j = s_j^3 g_j, \quad w_j = g_j^{-2}. \quad (\text{KT}_8)$$

Because  $q-1$  and  $q+1$  are coprime, multiplication gives the canonical split

$$E_j^\times \cong E_{j-1}^\times \times U_j, \quad a_j \longleftrightarrow (a_{j-1}^{3/2}, w_j^{-1/2}). \quad (\text{KT}_9)$$

Consequently, for  $1 \leq m \leq j$ , the  $U_m$ -coordinate of  $a_j$  is

$$w_m^{-\frac{1}{2}(\frac{3}{2})^{j-m}}, \quad (\text{KT}_{10})$$

where the exponent is read in  $\mathbf{Z}/F_m\mathbf{Z}$ . Every displayed exponent is a unit. Thus multiplicative CRT diagonalizes the complete ancestry into independently born Fermat coordinates; only the additive Conway equation can couple them, and evaluating that coupling is the original selected birth-symbol problem.

*Proof.* The cyclic group  $E_j^\times$  has order  $\prod_{r=0}^j F_r$ . Pairwise coprimality of the Fermat numbers gives (KT<sub>1</sub>). For  $j > n$ , suppose that a lower element  $x$  becomes an  $M$ -th power in  $E_j$ . Taking the quadratic norm shows that  $x^2$  is an  $M$ -th power in  $E_{j-1}$ . Since  $M$  is odd, multiplication by two is invertible on  $\mathcal{H}_{j-1}$ , so  $x$  was already an  $M$ -th power. Thus  $\iota_j$  is injective; its source and target both have order  $M$ , so it is an isomorphism.

At the birth edge every lower element is uniquely  $M$ -divisible, whence  $\kappa_n = [c_n]_M$ . The relation  $c_n(c_n + 1) = a_{n-1}$  gives  $[w_n]_M = [c_n + 1]_M - [c_n]_M = -2[c_n]_M$ . For  $j > n$ , the norm identity

$$N_{E_j/E_{j-1}}(c_j) = a_{j-1}$$

and the fact that norm of an included lower element is its square imply  $2[c_j]_M = \iota_j(\kappa_{j-1})$ . Since  $a_j = a_{j-1}c_j$ , multiplying its class by two gives

$$2\kappa_j = 2\iota_j(\kappa_{j-1}) + 2[c_j]_M = 3\iota_j(\kappa_{j-1}),$$

which proves (KT<sub>2</sub>)–(KT<sub>3</sub>). No Fermat number  $F_n$ ,  $n \geq 1$ , is divisible by two or three, so both denominators and the multiplier are units modulo  $M$ . This proves (KT<sub>4</sub>). For  $j > n$ ,  $w_j$  lies in the norm-one group of order  $F_j$ , coprime to  $M$ , while  $w_j + 1 = c_j^{-1}$ ; this proves (KT<sub>5</sub>).

Finally  $\mu_\ell \subset E_j$  for  $j \geq n$ . Prime-degree Kummer theory and Capelli's lemma say that  $A_j(X^\ell)$  is irreducible exactly when  $a_j$  is not an  $\ell$ -th power in  $E_j$ . In the power case  $X^\ell - a_j$  splits into  $\ell$  distinct linear factors over  $E_j$ . Every root retains full absolute degree, because its  $\ell$ -th power is the full-degree element  $a_j$ . This gives exactly  $\ell$  distinct irreducible factors of degree  $[E_j : \mathbb{F}_2]$ , proving (KT<sub>6</sub>).

For the birth identity in (KT<sub>7</sub>), put  $q = |E_{n-1}|$ . The lower factor in  $a_n = a_{n-1}c_n$  disappears at exponent  $(q^2 - 1)/\ell$ , and the Conway-unit Euler identity gives

$$\mathfrak{h}_n = c_n^{(q-1)(q+1)/\ell} = w_n^{(q+1)/\ell}.$$

For  $j > n$ , put  $Q = |E_{j-1}|$  and  $h = (Q - 1)/\ell$ . Frobenius fixity and  $c_j^{Q+1} = a_{j-1}$  give

$$\mathfrak{h}_j = (a_{j-1}c_j)^{(Q+1)h} = (a_{j-1}^h)^2 a_{j-1}^h = \mathfrak{h}_{j-1}^3.$$

For (KT<sub>8</sub>), the norm of both  $c_j$  and the included lower element  $s_j$  is  $a_{j-1} = s_j^2$ , so  $g_j$  has norm one. Since  $g_j^q = g_j^{-1}$  and  $c_j^q = c_j + 1$ , one has

$$w_j = \frac{c_j^q}{c_j} = \frac{g_j^q}{g_j} = g_j^{-2}.$$

The remaining identities are immediate. Coprimality of  $q - 1$  and  $q + 1$  proves (KT<sub>9</sub>), and iteration multiplies every old coordinate exponent by  $3/2$ , proving (KT<sub>10</sub>). Lean independently checks this generic norm/Frobenius cube identity and the denominator-free transport identities  $2[c_j] = \iota_j(\kappa_{j-1})$  and  $2\kappa_j = 3\iota_j(\kappa_{j-1})$ .  $\square$

**Proposition 7.56** (Dickson–Conway resultant and parallel rays). *For odd  $\ell$ , let  $\mathcal{D}_0(S, R) = 0$ ,  $\mathcal{D}_1(S, R) = S$ , and*

$$\mathcal{D}_{r+2}(S, R) = S\mathcal{D}_{r+1}(S, R) + R\mathcal{D}_r(S, R)$$

*in characteristic two. With the monic convention  $\text{Res}_S(f, g) = \prod_{f(s)=0} g(s)$ , one has the universal identity*

$$\boxed{\text{Res}_S(\mathcal{D}_\ell(S, R) + 1, Z^2 + SRZ + R^3) = Z^{2\ell} + R^\ell Z^\ell + R^{3\ell}.} \quad (\text{DCR}_1)$$

*Now let  $\ell$  be prime and fix  $M = \ell^e \parallel F_n$ . At the birth edge put  $a = a_{n-1}$  and  $Q_a(T) = T^2 + aT + a^3$ . If the coordinate born at  $(n, \ell)$  fails, the unique lower root  $R^\ell = a$  lies in  $E_{n-1}$ , and*

$$Q_a(Z^\ell) = \prod_{\mathcal{D}_\ell(s, R)=1} (Z^2 + sRZ + R^3), \quad (\text{DCR}_2)$$

*where the  $\ell$  factors are distinct irreducible quadratics over  $E_{n-1}$ . This is the coefficient-exact birth-edge refinement of Proposition 7.41.*

*At a later edge  $j > n$ , use the normalized roots below: put  $R = r_{j-1}$ ,  $T = r_j$ ,  $q = |E_{j-1}|$ , and  $d = T + T^q \in E_{j-1}$ . Since now  $\mu_\ell \subset E_{j-1}$ , one instead has*

$$Q_{a_{j-1}}(Z^\ell) = \prod_{i \bmod \ell} (Z^2 + \zeta^i dZ + \zeta^{2i} R^3). \quad (\text{DCR}_{2a})$$

*These are again  $\ell$  distinct irreducible quadratics. If the born coordinate succeeds,  $Q_{a_{j-1}}(Z^\ell)$  is irreducible of degree  $2\ell$  at every  $j \geq n$ . Thus relative Frobenius is respectively  $\ell$  disjoint transpositions or one  $2\ell$ -cycle. Both patterns are closed under every later Conway transition; factor counts, symmetric coefficients, and relative arboreal monodromy add no new obstruction.*

*More precisely, in the failure case choose the unique lower root  $r_{n-1}^\ell = a_{n-1}$ . Every one of the  $\ell$  choices  $r_n^\ell = a_n$  automatically satisfies*

$$N_{E_n/E_{n-1}}(r_n) = r_{n-1}^3. \quad (\text{DCR}_3)$$

*Given one such  $r_n$ , there is for each  $j > n$  a unique continuation*

$$r_j^\ell = a_j, \quad N_{E_j/E_{j-1}}(r_j) = r_{j-1}^3. \quad (\text{DCR}_4)$$

*All normalized infinite rays therefore form one  $\mu_\ell$ -torsor, not an  $\ell$ -branching tree. If two rays differ by  $r'_j = \zeta_j r_j$ , then only after the birth edge,*

$$\zeta_j^2 = \zeta_{j-1}^3, \quad \zeta_j = \zeta_n^{(3/2)^{j-n}} \quad (j > n). \quad (\text{DCR}_5)$$

*Hence the full compatible root tower is precisely the root-level form of the 3/2-transport in (KT<sub>3</sub>).*

*Proof.* The Dickson polynomial is monic of  $S$ -degree  $\ell$ . Work over an algebraic closure of  $\mathbb{F}_2(R)$ . Choose  $\alpha + \beta = 1$ ,  $\alpha\beta = R^\ell$ , then choose  $x^\ell = \alpha$ , put  $y = R/x$ , and let  $\zeta$  be a primitive  $\ell$ -th root of unity. The  $\ell$  elements

$$s_i = x\zeta^i + y\zeta^{-i} \quad (i \bmod \ell)$$

satisfy  $\mathcal{D}_\ell(s_i, R) = x^\ell + y^\ell = 1$ . They are distinct: an equality identifies the unordered root pairs of  $T^2 + s_i T + R$ ; the crossed identification would give  $\alpha = \beta$ , impossible because  $\alpha + \beta = 1$ . Moreover

$$Z^2 + s_i RZ + R^3 = (Z + Rx\zeta^i)(Z + Ry\zeta^{-i}).$$

Multiplying over  $i$  gives

$$(Z^\ell + R^\ell \alpha)(Z^\ell + R^\ell \beta) = Z^{2\ell} + R^\ell Z^\ell + R^{3\ell},$$

which proves (DCR<sub>1</sub>). At the birth edge, substituting  $R^\ell = a$  gives the right side  $Q_a(Z^\ell)$ , while Proposition 7.41 puts every  $s_i$  in  $E_{n-1}$ . A displayed quadratic cannot have a root there, since its  $\ell$ -th power would be one of the two roots of  $Q_a$ , both outside  $E_{n-1}$ . This proves (DCR<sub>2</sub>).

At a future edge Frobenius fixes  $\zeta$  and pairs  $T\zeta^i$  with  $T^q\zeta^i$ , which gives (DCR<sub>2a</sub>) directly from  $TT^q = R^3$ . Their constants are distinct because squaring permutes  $\mu_\ell$ , and the same root argument proves that the quadratics are irreducible. Under success, an  $\ell$ -th root of  $a_j$  has relative degree  $2\ell$ , proving the asserted irreducibility at every edge.

For (DCR<sub>3</sub>), the quotient of  $N(r_n)$  by  $r_{n-1}^3$  lies in  $\mu_\ell \cap E_{n-1}^\times = \{1\}$ . At a later edge an arbitrary  $\ell$ -th root has norm  $\xi r_{j-1}^3$  for some  $\xi \in \mu_\ell$ ; multiplying it by the unique  $\eta \in \mu_\ell$  with  $\eta^2 = \xi^{-1}$  gives (DCR<sub>4</sub>), and uniqueness follows because squaring is bijective on odd-order  $\mu_\ell$ . Comparing two continuations proves (DCR<sub>5</sub>). The single quadratic factorization underlying (DCR<sub>1</sub>) is independently kernel-checked in `0gdoad.Excess`; the root enumeration and resultant are paper-level. Base-rationality of the constant-term Dickson factors uses the birth congruence  $|E_{n-1}| \equiv -1 \pmod{\ell}$ ; later edges require (DCR<sub>2a</sub>). These identities prove coefficient and monodromy compatibility under hypothetical failure, not the value of the initial selected Kummer coordinate.  $\square$

**Proposition 7.57** (exact cyclotomic lift of the selected Fermat symbol). *Put  $m = 2^n$ ,  $q = 2^m$ , and  $N = q + 1 = F_n$ . Choose a generator  $\omega$  of the norm-one group*

$$U_n = \ker\left(N_{\mathbb{F}_{q^2}/\mathbb{F}_q}\right)$$

*independently of the Conway element, and write*

$$w_n = \frac{c_n + 1}{c_n} = \omega^{r_n}, \quad r_n \in \mathbf{Z}/N\mathbf{Z}. \quad (\text{CS}_1)$$

*Let  $\mathfrak{P}_\omega \mid 2$  be the prime of  $\mathbf{Q}(\zeta_N)$  specified by  $\zeta_N \bmod \mathfrak{P}_\omega = \omega$ , and define the cyclotomic unit*

$$\varepsilon_{r_n} = (1 + \zeta_N^{r_n})^{-1}.$$

*Its residue is  $c_n$ . For every prime  $\ell \mid N$ , the  $\ell$ -th power-residue symbol, identified with its reduction in  $\mu_\ell \subset \mathbb{F}_{q^2}^\times$ , is*

$$\boxed{\left(\frac{\varepsilon_{r_n}}{\mathfrak{P}_\omega}\right)_\ell = \omega^{r_n N/\ell} = w_n^{N/\ell}.} \quad (\text{CS}_2)$$

*Consequently*

$$\left(\frac{\varepsilon_{r_n}}{\mathfrak{P}_\omega}\right)_\ell \neq 1 \iff \ell \nmid r_n \iff v_\ell(\delta_n) = v_\ell(N) \iff A_{n-1} \nmid S_{N/\ell}. \quad (\text{CS}_3)$$

*Thus the all-level Popovych problem is exactly the assertion that the Conway-selected discrete logarithm  $r_n$  is a unit modulo  $F_n$  at every level.*

*This description is noncircular only with the independent generator  $\omega$ . Defining the prime above two by  $\zeta_N \bmod \mathfrak{P} = w_n$  would already assume that  $w_n$  has order  $N$ . With an honest primitive  $\omega$ , the undetermined coordinate  $r_n \bmod \ell$  remains exactly the original obstruction. The ambient*



presentation in  $\mathbf{Q}(\zeta_N)$ , the local logarithm, and the principal ideal of the circular unit do not determine it. The minimal odd cyclotomic conductor is not blind: since

$$\zeta_N^{r_n} = \varepsilon_{r_n}^{-1} - 1, \quad \mathbf{Q}(\varepsilon_{r_n}) = \mathbf{Q}(\zeta_{\delta_n}), \quad \delta_n = \frac{N}{\gcd(N, r_n)},$$

proving that this conductor is  $N$  is equivalent to the original maximal-order assertion. Under  $\ell$ -failure it drops to a divisor of  $N/\ell$ , but every nontrivial divisor  $d \mid F_n$  still has  $\text{ord}_d(2) = 2m$ , so residue-field degree does not exclude that drop. The principal-unit part at two is uniquely  $\ell$ -divisible and the logarithm kills the Teichmüller torsion where  $(\text{CS}_2)$  lives; every  $\varepsilon_{r_n}$  also has unit principal ideal. Circular-unit distribution and reciprocity can transport the value in  $(\text{CS}_2)$ , but without an additional ancestry-sensitive congruence they do not determine  $r_n \bmod \ell$ .

*Proof.* The elementary Fermat congruence gives  $\text{ord}_N(2) = 2m$ , so  $\mathfrak{P}_\omega$  has residue field  $\mathbb{F}_{q^2}$ . The element  $w_n \neq 1$ . Put  $d = \text{ord}(w_n) = N/\gcd(N, r_n)$ . Then  $d > 1$  is odd,  $\zeta_N^{r_n}$  is a primitive  $d$ -th root, and  $\Phi_d(-1) = 1$ . Hence  $1 + \zeta_N^{r_n}$  is a cyclotomic unit of absolute norm one. Reduction gives

$$\varepsilon_{r_n} \bmod \mathfrak{P}_\omega = (1 + w_n)^{-1} = c_n.$$

Put  $h = N/\ell$ . In the residue field  $w_n^q = w_n^{-1}$ , and hence

$$(1 + w_n)^{q-1} = \frac{1 + w_n^q}{1 + w_n} = w_n^{-1}.$$

Euler's definition of the power-residue symbol now gives

$$\varepsilon_{r_n}^{(q^2-1)/\ell} = (1 + w_n)^{-(q-1)h} = w_n^h,$$

which is  $(\text{CS}_2)$ . In the cyclic group of order  $N$ ,  $\omega^{r_n N/\ell} \neq 1$  exactly when  $\ell \nmid r_n$ , and this is equivalent to retaining the full  $\ell$ -primary part of the order. Proposition 7.7 supplies the last equivalence in  $(\text{CS}_3)$ . The Euler-exponent identity used here is independently kernel-checked in `Ogdoad.Excess`.  $\square$

**Proposition 7.58** (the canonical global lift leaves a free cyclotomic torsor). *Define a characteristic-zero lift of the complete Conway chain by*

$$\mathcal{K}_{-1} = \mathbf{Q}, \quad \mathcal{A}_{-1} = 1,$$

and, for  $i \geq 0$ ,

$$C_i^2 + C_i + \mathcal{A}_{i-1} = 0, \quad \mathcal{A}_i = \mathcal{A}_{i-1}C_i, \quad \mathcal{K}_i = \mathcal{K}_{i-1}(C_i). \quad (\text{GL}_1)$$

At two, choose the Hensel branch reducing successively to the Conway elements  $c_i$ . Then 2 is inert and unramified through  $\mathcal{K}_n/\mathbf{Q}$ . On the other hand, 3 is totally ramified, with a unique prime at every level. More precisely, if

$$D_i = 1 - 4\mathcal{A}_i, \quad Y_i = 2C_i + 1,$$

where  $D_{-1} = -3$ , then

$$Y_i^2 = D_{i-1}, \quad 2D_i = Y_i^3 - Y_i^2 - Y_i + 3, \quad (\text{GL}_2)$$

and  $D_i$  is a uniformizer at the prime above three.

Put

$$W_i = -\frac{C_i + 1}{C_i}.$$

Then, characteristic-freely,

$$\mathcal{A}_{i-1} = \frac{W_i}{(W_i + 1)^2}, \quad \mathcal{A}_i = -\frac{W_i}{(W_i + 1)^3}.$$

Every  $C_i$ ,  $C_i + 1$ ,  $\mathcal{A}_i$ , and  $W_i$  is an algebraic unit; in particular

$$\operatorname{div}(W_i) = 0, \quad N_{\mathcal{K}_i/\mathbf{Q}}(W_i) = 1. \quad (\text{GL}_3)$$

At the prime above three,  $C_i \equiv W_i \equiv 1$ . Hence, for every prime  $\ell \mid F_n$  with  $n \geq 1$ ,  $W_n$  is an  $\ell$ -th power in the full 3-adic completion.

At two, all lower-field units are  $\ell$ -th powers. The first reparametrization therefore gives

$$[W_n] = -2[C_n] \quad \text{in } \mathcal{K}_{n,\mathfrak{p}}^\times / \mathcal{K}_{n,\mathfrak{p}}^{\times \ell}. \quad (\text{GL}_{3a})$$

For the cyclotomic unit  $\varepsilon_{r_n}$  of Proposition 7.57, the compatible Hensel lifts satisfy  $C_n/\varepsilon_{r_n} \in 1 + 2\mathcal{O}$ . This principal-unit group is uniquely  $\ell$ -divisible, so

$$[W_n] = -2[\varepsilon_{r_n}]. \quad (\text{GL}_{3b})$$

Thus higher two-adic precision retains exactly the same selected Kummer class, up to an invertible scalar.

Nevertheless global reciprocity does not determine its selected symbol at two. Put  $M = \mathcal{K}_n(\zeta_\ell)$ . Then

$$\mathcal{K}_n \cap \mathbf{Q}(\zeta_\ell) = \mathbf{Q}, \quad (\text{GL}_4)$$

and the unique prime  $\mathfrak{p} \mid 2$  of  $\mathcal{K}_n$  splits completely in  $M/\mathcal{K}_n$ . Label the primes above it by

$$\mathfrak{q}_a = \tau_a \mathfrak{q}_1, \quad \tau_a(\zeta_\ell) = \zeta_\ell^a, \quad a \in (\mathbf{Z}/\ell\mathbf{Z})^\times.$$

For the tame Hilbert-symbol orientation

$$(x, y)_{\mathfrak{q}, \ell} = \left( (-1)^{v(x)v(y)} \frac{x^{v(y)}}{y^{v(x)}} \bmod \mathfrak{q} \right)^{(|k(\mathfrak{q})|-1)/\ell},$$

write  $H = (W_n, 2)_{\mathfrak{q}_1, \ell}$ . Equivariance gives

$$(W_n, 2)_{\mathfrak{q}_a, \ell} = H^a, \quad \prod_{a \in (\mathbf{Z}/\ell\mathbf{Z})^\times} H^a = 1 \quad \text{for every } H \in \mu_\ell. \quad (\text{GL}_5)$$

If  $q = 2^{2^n}$  and  $N = q + 1 = F_n$ , reduction at  $\mathfrak{q}_1$  gives

$$H = \left( w_n^{N/\ell} \right)^{-2}. \quad (\text{GL}_6)$$

Thus  $H \neq 1$  is exactly the desired full  $\ell$ -primary order condition. All finite places away from  $2\ell$  are tame-trivial, the place above three is already power-trivial, and there are no real places. Global reciprocity therefore says only

$$\prod_{\lambda \mid \ell} (W_n, 2)_{\lambda, \ell} = 1; \quad (\text{GL}_7)$$

the product over the complete orbit above two is identically one for every possible  $H$ . In fact, for  $n \geq 2$  the stronger pointwise statement is

$$(u, b)_{\lambda, \ell} = 1 \quad (u, b \in \mathcal{K}_n^\times, \lambda \mid \ell). \quad (\text{GL}_{7a})$$

Consequently the canonical lift, its principal divisor, its total 3-adic ramification, and unweighted global reciprocity do not select the Conway residue symbol. Any reciprocity proof must add a genuinely weighted  $\ell$ -adic ray or Jacobi coefficient that breaks the torsor in  $(\text{GL}_5)$ ; proving that coefficient nonzero is a new arithmetic input, not a consequence of the lifted ancestry alone.

*Proof.* The reductions of the quadratics in  $(\text{GL}_1)$  are precisely the successive irreducible Conway quadratics. Hensel lifting therefore makes each local extension at two the unramified quadratic extension and proves the first assertion.

The identities  $(\text{GL}_2)$  follow directly from  $C_i^2 + C_i = -\mathcal{A}_{i-1}$ . Starting from the uniformizer  $D_{-1} = -3$ , the first identity is Eisenstein at the unique prime over three. In the resulting ramified quadratic extension, the linear term  $-Y_i$  in the second identity has strictly smallest valuation, so  $D_i$  is again a uniformizer. This proves total ramification inductively.

The relative norm of each of  $C_i$  and  $C_i + 1$  is  $\mathcal{A}_{i-1}$ . Induction therefore makes them, and hence  $\mathcal{A}_i$  and  $W_i$ , units. The nontrivial relative conjugation sends  $W_i$  to  $W_i^{-1}$ , proving  $(\text{GL}_3)$ . At three,  $Y_i \equiv 0$  gives  $C_i \equiv 1$  and then  $W_i \equiv 1$ . Since  $\ell$  is odd and different from three, powering by  $\ell$  is an automorphism of both the order-two residue-unit group and the pro-3 principal units.

Solving  $W_i = -(C_i + 1)/C_i$  for  $C_i$  and using  $C_i^2 + C_i = -\mathcal{A}_{i-1}$  gives the two displayed rational reparametrizations. At two,  $\ell \mid q + 1$  is coprime to the lower residue-group order  $q - 1$ , proving  $(\text{GL}_{3a})$ . The two characteristic-zero units  $C_n$  and  $\varepsilon_{r_n}$  have the same reduction  $c_n$ ; their ratio is therefore a principal unit, on which odd  $\ell$ -powering is an automorphism. This proves  $(\text{GL}_{3b})$ .

Every nontrivial subfield of  $\mathcal{K}_n$  is ramified at three, whereas  $\mathbf{Q}(\zeta_\ell)$  is not; since the latter has abelian Galois group, this proves  $(\text{GL}_4)$ . The residue field at  $\mathfrak{p}$  is  $\mathbb{F}_{q^2}$  and contains  $\mu_\ell$ , so  $\mathfrak{p}$  splits completely in  $M$ . Hilbert-symbol equivariance proves the first equality in  $(\text{GL}_5)$ , and the second follows because  $\sum_{a=1}^{\ell-1} a$  is divisible by  $\ell$ .

Finally  $W_n \bmod \mathfrak{p} = w_n$ , and the displayed tame convention gives

$$H = w_n^{(q^2-1)/\ell} = \left(w_n^{N/\ell}\right)^{q-1} = \left(w_n^{N/\ell}\right)^{-2},$$

since  $w_n^N = 1$ . The standard tame local formula makes all other places outside  $\ell$  trivial as stated. The product formula now yields  $(\text{GL}_7)$ , while  $(\text{GL}_5)$  proves that it imposes no condition on  $H$ . For  $(\text{GL}_{7a})$ , let  $\mathfrak{l} = \lambda \cap \mathcal{K}_n$ . Every prime divisor  $\ell \mid F_n$ ,  $n \geq 2$ , satisfies  $\ell \equiv 1 \pmod{2^{n+2}}$ , whereas  $[\mathcal{K}_{n,\mathfrak{l}} : \mathbf{Q}_\ell] \leq 2^{n+1}$ . Thus  $\mathcal{K}_{n,\mathfrak{l}}$  cannot contain  $\mathbf{Q}_\ell(\zeta_\ell)$ , and the decomposition stabilizer of  $\lambda$  in  $\Delta$  contains some  $\tau_a \neq 1$ . Since  $\tau_a$  fixes  $u$ ,  $b$ , and  $\lambda$ , equivariance gives

$$(u, b)_{\lambda, \ell} = \tau_a((u, b)_{\lambda, \ell}) = (u, b)_{\lambda, \ell}^a.$$

The symbol has prime order dividing  $\ell$ , so it is one.

Lean independently checks the characteristic-free reparametrization, the two identities  $(\text{GL}_2)$ , and the corresponding odd-Kummer square-class equivalence in `Ogdoad.Excess`.  $\square$

**Proposition 7.59** (the semiprimitive Gauss table remembers only the selected label). *Retain Proposition 7.58. Put*

$$m = 2^n, \quad q = 2^m, \quad k_0 = \mathbb{F}_q, \quad k = \mathbb{F}_{q^2},$$

and identify the residue copy of  $\mu_\ell \subset k^\times$  with the fixed global copy by inverse reduction at  $\mathfrak{q}_1$ . Define

$$\chi(x) = x^{(q^2-1)/\ell} \in \mu_\ell, \quad \psi(x) = (-1)^{\text{Tr}_{k/\mathbb{F}_2}(x)}.$$

Thus  $\chi$  has order  $\ell$  and is trivial on  $k_0^\times$ . For  $a \in (\mathbf{Z}/\ell\mathbf{Z})^\times$  and  $b \in k^\times$ , put

$$G_a(b) = \sum_{x \in k^\times} \chi(x)^a \psi(bx).$$

Then

$$\boxed{G_a(1) = q, \quad G_a(b) = q\chi(b)^{-a}, \quad G_a(w_n) = qH^{-a}.} \quad (\text{SG}_1)$$

For  $\xi \in \mu_\ell$ , define the labelled signed period

$$P_\xi(b) = \sum_{\substack{x \in k^\times \\ \chi(x) = \xi}} \psi(bx) \in \mathbf{Z}.$$

Its complete table is

$$\boxed{P_\xi(b) = \begin{cases} \frac{(\ell-1)q-1}{\ell}, & \xi = \chi(b)^{-1}, \\ -\frac{q+1}{\ell}, & \xi \neq \chi(b)^{-1}. \end{cases}} \quad (\text{SG}_2)$$

Both values are integers because  $\ell \mid q+1$ . In particular,

$$H = 1 \iff P_1(w_n) = \frac{(\ell-1)q-1}{\ell}, \quad (\text{SG}_3)$$

whereas for  $H \neq 1$ , one has  $P_1(w_n) = -(q+1)/\ell$ .

Consequently the unordered period multiset, every symmetric function of that multiset, all Gauss absolute values, and all principal-ideal or Stickelberger data are independent of  $H$ . Explicitly,

$$|G_a(w_n)| = q, \quad (G_a(w_n)) = (q), \quad G_a(w_n)G_{-a}(w_n) = q^2,$$

and

$$\prod_{a=1}^{\ell-1} G_a(w_n) = q^{\ell-1}. \quad (\text{SG}_4)$$

The datum which remains is exactly the label of the unique exceptional period, equivalently the root-of-unity phase

$$\frac{G_a(w_n)}{q} = H^{-a}.$$

Thus the semiprimitive Gauss table gives a two-valued integral reformulation of the selected problem, but not a nonvanishing theorem.

*Proof.* For  $y \in k^\times$ , consider the  $k_0^\times$ -line  $yk_0^\times$ . For  $t \in k_0$ ,

$$\psi(ty) = (-1)^{\text{Tr}_{k_0/\mathbb{F}_2}(t \text{Tr}_{k/k_0}(y))}.$$

Because  $k/k_0$  is quadratic of characteristic two,

$$\text{Tr}_{k/k_0}(y) = 0 \iff y^q = y \iff y \in k_0.$$

Hence

$$\sum_{t \in k_0^\times} \psi(ty) = \begin{cases} q-1, & yk_0^\times = k_0^\times, \\ -1, & yk_0^\times \neq k_0^\times. \end{cases} \quad (\text{SG}_5)$$

The second row follows because the corresponding nontrivial additive character sums to zero on  $k_0$ , and deleting  $t = 0$  removes one.

The character  $\chi^a$  is constant on these lines and is a nontrivial character of the cyclic quotient  $k^\times/k_0^\times$ . Its sum over that quotient is zero. The exceptional line has character value one, so

$$G_a(1) = (q-1) - \sum_{\substack{C \in k^\times/k_0^\times \\ C \neq k_0^\times}} \chi(C)^a = (q-1) - (-1) = q.$$

Changing variables  $y = bx$  gives  $G_a(b) = \chi(b)^{-a} G_a(1)$ . Moreover

$$\chi(w_n) = w_n^{(q^2-1)/\ell} = \left(w_n^{(q+1)/\ell}\right)^{q-1} = \left(w_n^{(q+1)/\ell}\right)^{-2} = H,$$

because  $q \equiv -1 \pmod{\ell}$ . This proves (SG<sub>1</sub>).

Each fibre of  $\chi$  consists of  $(q+1)/\ell$   $k_0^\times$ -lines. In  $P_\xi(b)$ , the unique line on which the additive character is trivial is  $b^{-1}k_0^\times$ , whose  $\chi$ -label is  $\chi(b)^{-1}$ . If that line belongs to the fibre, its contribution is  $q-1$  and the remaining  $(q+1)/\ell - 1$  lines contribute  $-1$ ; otherwise all  $(q+1)/\ell$  lines contribute  $-1$ . This proves (SG<sub>2</sub>). Its two values differ by  $q$ , proving (SG<sub>3</sub>).

Finally  $H$  is a root of unity, so multiplication by  $H^{-a}$  changes neither absolute values nor principal ideals. Pairing  $a$  with  $-a$ , or using  $\sum_{a=1}^{\ell-1} a \equiv 0 \pmod{\ell}$ , proves (SG<sub>4</sub>).

More generally, let  $\rho_0$  be a multiplicative character of  $k_0^\times$  and put  $\rho = \rho_0 \circ N_{k/k_0}$ . Since  $N(w_n) = 1$ , the same change of variables gives

$$G_{\chi^a \rho, w_n} = H^{-a} G_{\chi^a \rho, 1}. \quad (\text{SG}_6)$$

Thus arbitrary lower multiplicative characters do not repair the loss: Gauss–Jacobi quotients of total  $\chi$ -weight zero erase the phase, whereas nonzero total weight retains exactly the unknown  $H$ . This statement covers lower characters factoring through the norm, not arbitrary nonlinear ancestry-dependent sums. Lean kernel-checks the abstract nontrivial line sum, multiplicative change of variables, exceptional-line aggregation, denominator-cleared period values, and label-blind symmetric sums in `Ogdoad.Excess`; the finite-extension trace-line identification remains paper-level.  $\square$

**Proposition 7.60** (the complete Conway bit basis has one Kummer phase). *Retain Proposition 7.59. Thus  $m = 2^n$ ,  $q = 2^m$ ,  $E_{n-1} = \mathbb{F}_q$ , and  $E_n = \mathbb{F}_{q^2}$ . Write*

$$c = c_n = 2^m, \quad e_i = 2^i \quad (0 \leq i < 2m), \quad C = \chi(c),$$

where the  $e_i$  are the literal Conway bit basis. Then

$$\boxed{\chi(e_i) = 1 \quad (0 \leq i < m), \quad \chi(e_{m+i}) = C \quad (0 \leq i < m), \quad H = C^{-2}.} \quad (\text{CBK}_1)$$

In particular,

$$H = 1 \iff C = 1 \iff e_i \in (E_n^\times)^\ell \quad (0 \leq i < 2m). \quad (\text{CBK}_2)$$

More generally, for finite subsets  $S, T \subset \{0, \dots, m-1\}$ ,

$$\chi\left(\prod_{i \in S} e_i \prod_{j \in T} e_{m+j}\right) = C^{|T|}. \quad (\text{CBK}_3)$$

Thus every multiplicative-character identity formed from the complete actual bit ancestry has only one current coordinate: upper-block weight zero is tautologically invisible, while nonzero weight

modulo  $\ell$  is an invertible power of the original selected phase. In the Gauss table of Proposition 7.59, all lower-bit twists equal  $q$ , and all upper-bit twists equal the one value  $qC^{-a}$ . Evaluating every bit therefore supplies no phase independent of  $H$ .

*Proof.* For  $i < m$ , one has  $e_i \in E_{n-1}^\times$ , and

$$\frac{q^2 - 1}{\ell} = (q - 1) \frac{q + 1}{\ell},$$

so  $\chi(e_i) = 1$ . The disjoint-Fermat-support rule for nim multiplication is literal on these basis elements:

$$e_{m+i} = ce_i \quad (0 \leq i < m).$$

Multiplicativity proves the second assertion and (CBK<sub>3</sub>). The Conway quadratic gives  $c^q = c + 1$ . Since  $q \equiv -1 \pmod{\ell}$ ,

$$H = \chi\left(\frac{c+1}{c}\right) = \frac{\chi(c)^q}{\chi(c)} = C^{-2}.$$

Squaring is an automorphism of the odd-order group  $\mu_\ell$ , and  $\ker \chi = (E_n^\times)^\ell$ , proving (CBK<sub>2</sub>). The Gauss-sum statement follows by substituting these values into (SG<sub>1</sub>).

Lean kernel-checks the abstract multiplicative phase, monomial-weight, and odd-torsion inverse-square cores in `0gdoad.Excess`. The literal Conway bit multiplication and finite-field Kummer interpretation remain paper-level. The proposition rules out multiplicative monomial arguments; it does not rule out nonlinear additive combinations or a new identity coupling  $\chi(x + y)$  to the selected ancestry.  $\square$

**Corollary 7.61** (the labelled first upper block). *Retain Proposition 7.60, put*

$$a = a_{n-1} = c(c + 1), \quad d = \frac{q + 1}{\ell},$$

and, for  $\xi \in \mu_\ell$ , define

$$B_\xi(X) = (X + c + 1)^d + \xi(X + c)^d \in E_n[X].$$

Then

$$\boxed{X^q + X = \prod_{\xi \in \mu_\ell} B_\xi(X).} \quad (\text{CBK}_4)$$

For  $x \in E_{n-1}$ , one has

$$B_\xi(x) = 0 \iff \chi(c + x) = \xi. \quad (\text{CBK}_5)$$

Consequently

$$\#\{x \in E_{n-1} : \chi(c + x) = \xi\} = \begin{cases} d - 1, & \xi = 1, \\ d, & \xi \neq 1, \end{cases} \quad (\text{CBK}_6)$$

and, for every nonzero  $r \in \mathbf{Z}/\ell\mathbf{Z}$ ,

$$\sum_{x \in E_{n-1}} \chi(c + x)^r = -1.$$

The trivial-label factor is already defined over the lower field and is exactly

$$B_1(X) = S_d(X^2 + X + a), \quad B_1(0) = S_d(a). \quad (\text{CBK}_7)$$

In particular,

$$H = 1 \iff \chi(c) = 1 \iff B_1(0) = 0 \iff S_d(a) = 0. \quad (\text{CBK}_8)$$

Thus the product factorization, factor degrees, and every unmarked histogram or label power moment of the literal ordinal interval  $c + E_{n-1} = \{q, \dots, 2q - 1\}$  are fixed independently of the target. The individual labelled factor retains one missing incidence: whether its marked first point  $c$  lies on the trivial-labelled factor. This is precisely the original selected Fibonacci zero.

*Proof.* Put  $A = (X + c + 1)^d$  and  $B = (X + c)^d$ . The cyclotomic product in characteristic two gives

$$\prod_{\xi \in \mu_\ell} (A + \xi B) = A^\ell + B^\ell.$$

Since  $c^q = c + 1$ , expanding the right-hand side gives

$$(X + c + 1)^{q+1} + (X + c)^{q+1} = X^q + X,$$

which proves (CBK<sub>4</sub>). For  $x \in E_{n-1}$ ,

$$\chi(c + x) = \left( \frac{c + x + 1}{c + x} \right)^d,$$

so (CBK<sub>5</sub>) follows. Projectivizing identifies  $x \mapsto (c + x)E_{n-1}^\times$  with all lines in  $E_n^\times/E_{n-1}^\times$  except the base line. Every character label occurs  $d$  times on the full quotient, and the deleted base line has label one, proving (CBK<sub>6</sub>) and the moment formula.

Finally, the characteristic-two Binet identity applied to the roots  $X + c$  and  $X + c + 1$ , whose product is  $X^2 + X + a$ , gives (CBK<sub>7</sub>). Its specialization at zero and Proposition 7.60 give (CBK<sub>8</sub>). Lean kernel-checks the deleted-uniform-fibre count and the characteristic-two norm collapse in **Ogdoad.Excess**; the cyclotomic product, projective-line identification, and Binet specialization remain paper-level.  $\square$

**Proposition 7.62** (universal translation and marked-endpoint exhaustion). *Retain Corollary 7.61 and put*

$$F_\xi(Y) = (Y + 1)^d + \xi Y^d.$$

Then

$$B_\xi(X) = F_\xi(X + c), \quad \prod_{\xi \in \mu_\ell} F_\xi(Y) = Y^q + Y + 1. \quad (\text{CBK}_9)$$

Thus the complete labelled factor family modulo common translation is universal: the Conway ancestry occurs only in the marked evaluation point  $Y = c$ . More precisely, if  $C = \chi(c)$ , then

$$F_\xi(c) = c^d(C + \xi), \quad F_1(c) = c^d(C + 1) = S_d(a). \quad (\text{CBK}_{10})$$

Hence the unique factor vanishing at the marked endpoint has label  $C$ , the failure assertion is that this label is one, and the desired maximal-order assertion is that it is not one.

All simultaneous-translation invariants are selector-blind. If  $d = 1$ , then  $B_1 = 1$  and the desired nonvanishing is automatic. Assume  $d > 1$ . In the raw factor normalization above one has

$$\text{disc}(B_1) = 1, \quad (\text{CBK}_{11})$$

and, for distinct labels,

$$\text{Res}(B_1, B_\eta) = (1 + \eta)^{d-1} \quad (\eta \neq 1), \quad \text{Res}(B_\xi, B_\eta) = (\xi + \eta)^d \quad (\xi, \eta \neq 1). \quad (\text{CBK}_{12})$$

The factor symmetries likewise contain no marked information:

$$B_\xi(X+1) = \xi B_{\xi^{-1}}(X), \quad \text{Frob}_q^{\text{coeff}}(B_\xi) = \xi^{-1} B_\xi. \quad (\text{CBK}_{13})$$

The individual coefficients may retain the translation, but exhaust to the same Fibonacci word:

$$[X^j]B_1 = \binom{d}{j} S_{d-j}(a) \quad (0 \leq j < d). \quad (\text{CBK}_{14})$$

If  $\binom{d}{j}$  is odd and  $A_{n-1}$  is the selected minimal polynomial of  $a$ , then

$$N_{E_{n-1}/\mathbb{F}_2}([X^j]B_1) = \text{Res}_T(A_{n-1}(T), S_{d-j}(T)). \quad (\text{CBK}_{15})$$

When  $\binom{d}{j}$  is odd, equivalently, coefficient vanishing says  $\text{ord}(w) \mid d-j$ . The norm of the constant coefficient  $j=0$  is the original selected resultant. Thus these data supply no selector-blind second invariant: every single coefficient or Hasse jet is a shifted selected divisibility problem. Any gain must add genuinely ancestry-sensitive input, possibly through a relation involving one or more shifted values.

*Proof.* The first identity is the definition after the common translation. The cyclotomic product gives

$$\prod_{\xi \in \mu_\ell} ((Y+1)^d + \xi Y^d) = (Y+1)^{q+1} + Y^{q+1} = Y^q + Y + 1,$$

proving (CBK<sub>9</sub>). Since  $(c+1)/c = w$  and  $w^d = C$ , evaluation at  $c$  proves (CBK<sub>10</sub>).

Translation does not change discriminants or resultants, so compute with the  $F_\xi$ . At a root  $y$  of  $F_1$ ,

$$F'_1(y) = \frac{y^{d-1}}{y+1}, \quad \prod y = F_1(0) = 1, \quad \prod (y+1) = F_1(1) = 1.$$

This proves (CBK<sub>11</sub>). At a root of  $F_1$ , the first resultant in (CBK<sub>12</sub>) follows directly from  $F_\eta(y) = (1+\eta)y^d$  and  $\prod y = 1$ . For  $\xi \neq 1$ , the raw leading coefficient is  $1+\xi$  and  $\prod y = (1+\xi)^{-1}$ ; the same root evaluation gives the second resultant exactly. Direct substitution and  $c^q = c+1$  give (CBK<sub>13</sub>), where the Frobenius acts coefficientwise.

Finally the binomial theorem gives

$$[X^j]B_1 = \binom{d}{j} ((c+1)^{d-j} + c^{d-j}) = \binom{d}{j} S_{d-j}(a),$$

the last equality being the same characteristic-two Binet identity as in (CBK<sub>7</sub>). Taking the norm over the selected Frobenius orbit is the monic resultant in (CBK<sub>15</sub>). Lean kernel-checks the universal trace-one collapse and the marked phase-defect identity; the cyclotomic product, resultant products, and selected finite-field specialization remain paper-level.  $\square$

**Proposition 7.63** (weighted reciprocity and a circular cyclotomic-difference selector). *Retain Proposition 7.58, put  $\Delta = (\mathbf{Z}/\ell\mathbf{Z})^\times$ , and label the primes above two by  $\mathbf{q}_a = \tau_a \mathbf{q}_1$ . For  $x \in M^\times$  define its weighted two-adic first moment*

$$\mathcal{E}_2(x) = \sum_{a \in \Delta} a v_{\mathbf{q}_a}(x) \pmod{\ell}. \quad (\text{FWS}_1)$$

Then

$$\prod_{\mathbf{q}|2} (W_n, x)_{\mathbf{q}, \ell} = H^{\mathcal{E}_2(x)}. \quad (\text{FWS}_2)$$



Consequently, if the principal divisor  $(x)$  is supported only above  $2\ell$ , global reciprocity gives the exact single-orbit identity

$$H^{\mathcal{E}_2(x)} = \prod_{\lambda|\ell} (W_n, x)_{\lambda, \ell}^{-1}. \quad (\text{FWS}_3)$$

A proof of  $H \neq 1$  could therefore come from a mixed element with  $\mathcal{E}_2(x) \neq 0$ , together with an independently evaluated right-hand side.

Neither unmixed factor can provide such an element. One has

$$\mathcal{E}_2(x) = 0 \quad \text{for every } x \in \mathcal{K}_n^\times \text{ and every } x \in \mathbf{Q}(\zeta_\ell)^\times. \quad (\text{FWS}_4)$$

In particular this excludes classical Gauss, Jacobi, and Stickelberger principalizers lying in the cyclotomic factor. Indeed, if  $D = \langle 2 \rangle \subset \Delta$  is the decomposition group at two, then  $-1 \in D$ , because  $2^{2^n} \equiv -1 \pmod{\ell}$ . Hence a divisor from the cyclotomic factor has

$$v_{\mathfrak{q}_a}(x) = v_{\mathfrak{q}_{-a}}(x),$$

and the  $a, -a$  contributions to  $(\text{FWS}_1)$  cancel. A usable selector must break this inverse-pair symmetry with genuinely  $\mathcal{K}_n$ -specific data.

The lifted recursion also contributes no hidden compensator. It gives

$$W_n = \mathcal{A}_{n-1} C_n^{-2} = \left( \prod_{j < n} C_j \right) C_n^{-2}. \quad (\text{FWS}_{4a})$$

For  $j < n$ , the two-adic residue of  $C_j$  has order prime to  $\ell$ , so  $C_j$  is a local  $\ell$ -th power above two. Applying reciprocity to each lower  $C_j$  therefore gives, for every  $x$  as above,

$$\prod_{\lambda|\ell} (C_j, x)_{\lambda, \ell} = 1, \quad \prod_{\lambda|\ell} (W_n, x)_{\lambda, \ell} = \left( \prod_{\lambda|\ell} (C_n, x)_{\lambda, \ell} \right)^{-2}. \quad (\text{FWS}_{4b})$$

Thus complete lower ancestry cancels from the  $\ell$ -adic compensator: the right side of  $(\text{FWS}_3)$  is again the current selected  $C_n$ -character, up to the invertible scalar  $-2$ .

The most natural mixed principal element is already exactly circular. Put

$$h = F_n/\ell, \quad T = W_n^h, \quad x = T - \zeta_\ell.$$

Then

$$N_{M/\mathcal{K}_n}(x) = \Phi_\ell(T), \quad (\text{FWS}_5)$$

and, after reduction at the prime above two,

$$\Phi_\ell(w_n^h) = 0 \iff w_n^h \neq 1 \iff H \neq 1. \quad (\text{FWS}_6)$$

Thus  $(x)$  has any divisor support above two exactly when the desired maximal-order condition is already true; in the failure case  $w_n^h = 1$  and  $\Phi_\ell(1) = \ell \equiv 1 \pmod{2}$ . The pure idèle with component two at  $\mathfrak{q}_1$  and one at the other primes does have first moment one and evaluates to  $H$ , but it is not principal. Making it into a principal selector whose divisor is supported above  $2\ell$  would require an  $\ell$ -adic ray-class compensator. Its existence and the evaluation of the right side of  $(\text{FWS}_3)$  are precisely the remaining selected relative ray-class problem. The lifted Conway norm recursion neither forces that coefficient to vanish nor proves it nonzero.

*Proof.* At  $\mathfrak{q}_a$ , tame bilinearity and Proposition 7.58 give

$$(W_n, x)_{\mathfrak{q}_a, \ell} = H^{av_{\mathfrak{q}_a}(x)}.$$

Multiplication proves (FWS<sub>2</sub>), and the Hilbert product formula proves (FWS<sub>3</sub>). Valuations of an element of  $\mathcal{K}_n$  are constant on the full orbit. For an element of  $\mathbf{Q}(\zeta_\ell)$ , they are constant on the decomposition cosets of  $D$ ; pairing  $a$  with  $-a$  proves (FWS<sub>4</sub>). The first identity in (FWS<sub>4a</sub>) is  $W_n = \mathcal{A}_{n-1}C_n^{-2}$ , and the second follows by expanding  $\mathcal{A}_{n-1}$ . At two, every lower residue has order dividing  $\prod_{r < n} F_r$ , which is coprime to  $\ell$ ; the principal-unit part is also uniquely  $\ell$ -divisible. Its local symbols are therefore trivial. At places away from  $2\ell$ , both entries are units and the tame symbol is trivial. The product formula applied to each  $C_j$ , followed by multiplicativity, proves (FWS<sub>4b</sub>).

Equation (FWS<sub>5</sub>) is the defining product for  $\Phi_\ell$ . The reduction  $t = w_n^h$  satisfies  $t^\ell = 1$ . In characteristic two and for odd  $\ell$ , the geometric sum  $\Phi_\ell(t) = 1 + t + \cdots + t^{\ell-1}$  vanishes exactly when  $t \neq 1$ , which proves (FWS<sub>6</sub>). Lean independently checks the inverse-pair/coset first-moment cancellation and this odd geometric-sum dichotomy.  $\square$

**Proposition 7.64** (an affine mixed selector and its oriented tame residue). *Retain Proposition 7.63, assume  $n \geq 2$ , put*

$$q = 2^{2^n}, \quad M_0 = \mathcal{K}_{n-1}(\zeta_\ell), \quad \mathfrak{r} = \mathfrak{q}_1 \cap M_0, \quad \eta = \zeta_\ell \bmod \mathfrak{r} \in \mathbb{F}_{q^2}.$$

*There are algebraic integers  $U, V \in \mathcal{O}_{\mathcal{K}_{n-1}}$  such that, for*

$$R = U + V\zeta_\ell \in M_0, \quad x = C_n - R \in M,$$

*one has*

$$v_{\mathfrak{q}_1}(x) = 1, \quad v_{\mathfrak{q}_a}(x) = 0 \quad (a \neq 1), \quad (W_n, x)_{\lambda, \ell} = 1 \quad (\lambda \mid \ell). \quad (\text{AFS}_1)$$

*In particular  $\mathcal{E}_2(x) = 1$ . Define the finite oriented tame residue*

$$\mathcal{T}_R = \prod_{\mathfrak{v} \nmid 2\ell} \left( \overline{W_n}^{v_{\mathfrak{v}}(x)} \right)^{(|k(\mathfrak{v})|-1)/\ell} \in \mu_\ell. \quad (\text{AFS}_2)$$

*Here inverse reduction identifies the copy of  $\mu_\ell$  in each residue field with the fixed global  $\mu_\ell \subset M$ . Then global reciprocity evaluates the selected symbol exactly as*

$$\boxed{H = \mathcal{T}_R^{-1}}. \quad (\text{AFS}_3)$$

*Moreover*

$$N_{M/M_0}(x) = \mathcal{A}_{n-1} + R + R^2. \quad (\text{AFS}_4)$$

*At conjugate components of a divisor of the right side, the two values of  $W_n$  are reciprocal. Consequently the ordinary norm or resultant forgets  $\mathcal{T}_R$ : equation AFS<sub>3</sub> is an oriented half-resultant rather than an unweighted norm formula.*

*Proof.* Since  $\ell \mid q+1$  and  $\ell \nmid q-1$ , the reduction  $\eta$  does not lie in  $\mathbb{F}_q$ . Hence  $\{1, \eta\}$  is an  $\mathbb{F}_q$ -basis of  $\mathbb{F}_{q^2}$ , and there are unique  $u, v \in \mathbb{F}_q$ , with  $v \neq 0$ , such that

$$c_n = u + v\eta.$$

For every sufficiently large integer  $N$ , simultaneous Chinese remaindering in  $\mathcal{O}_{\mathcal{K}_{n-1}}$  chooses initial lifts  $U \equiv u$  and  $V \equiv v$  at the prime above two while putting both in  $\mathfrak{l}^N$  at every prime  $\mathfrak{l} \mid \ell$ . With the convention  $\mathfrak{q}_a = \tau_a \mathfrak{q}_1$ , comparison through  $\tau_a^{-1}$  gives

$$\zeta_\ell \bmod \mathfrak{q}_a = \eta^{a^{-1}}, \quad x \bmod \mathfrak{q}_a = v(\eta - \eta^{a^{-1}}).$$

Because  $\eta$  has order  $\ell$ , this vanishes exactly for  $a = 1$ . The extension at two is unramified, so two is a uniformizer. Replacing  $U$  by  $U + 2T$  changes  $x/2 \bmod \mathfrak{q}_1$  by  $-\overline{T} \in \mathbb{F}_q$ . Choose the residue of  $T$  to avoid the single forbidden value, and simultaneously impose  $T \in \mathfrak{l}^N$  at every  $\mathfrak{l} \mid \ell$ . This proves the first two assertions of **AFS**<sub>1</sub> without changing any reduction modulo two.

For  $\lambda \mid \ell$ , put  $y = C_n - U \in \mathcal{K}_n^\times$ . The chosen  $U$  is  $\ell$ -adically deep and  $C_n$  is a global unit, so  $y$  is a local unit. Taking  $N$  beyond the finitely many local  $\ell$ -th-power thresholds makes

$$\frac{x}{y} = 1 - \frac{V\zeta_\ell}{y} \in M_\lambda^{\times \ell}$$

at every such  $\lambda$ . The pointwise identity  $\mathbf{GL}_{7a}$  gives  $(W_n, y)_{\lambda, \ell} = 1$ , and bilinearity proves the last assertion of **AFS**<sub>1</sub>. Notice that this kills the pairing; it does not assert that  $x$  itself is an  $\ell$ -th power.

At  $\mathfrak{q}_1$ , the tame formula and  $v_{\mathfrak{q}_1}(x) = 1$  give  $(W_n, x)_{\mathfrak{q}_1, \ell} = H$ . Every other prime above two contributes one, every prime above  $\ell$  contributes one by **AFS**<sub>1</sub>, and the remaining finite factors are exactly **AFS**<sub>2</sub>. The Hilbert product formula proves **AFS**<sub>3</sub>.

The top conjugation fixes  $R$  and sends  $C_n$  to  $-1 - C_n$ . Hence

$$(C_n - R)(-1 - C_n - R) = \mathcal{A}_{n-1} + R + R^2,$$

which proves **AFS**<sub>4</sub>. If a split oriented component has  $C_n \equiv r$ , then  $\mathcal{A}_{n-1} + r + r^2 = 0$  and

$$W_n \equiv \frac{\mathcal{A}_{n-1}}{r^2} = -\frac{r+1}{r}.$$

The conjugate component has  $C_n \equiv -1 - r$  and sees the reciprocal value. At a ramified component  $r = -1/2$ , this value is one. Thus the full norm combines and cancels the two orientations, whereas the divisor of  $x$  retains only one. Lean independently checks the affine interpolation, quadratic norm, and reciprocal-orientation identities; the simultaneous approximation, valuation statements, Hilbert symbols, and global reciprocity remain paper-level.  $\square$

**Corollary 7.65** (the affine selector norm is genuinely mixed). *Retain Proposition 7.64, let  $\mathfrak{p}$  be the unique prime of  $\mathcal{K}_{n-1}$  above two, and put*

$$g = N_{M/M_0}(x) = \mathcal{A}_{n-1} + R + R^2.$$

*The primes of  $M_0$  above  $\mathfrak{p}$  are naturally indexed by  $\Delta/\{\pm 1\}$ ; write  $\mathfrak{r}_{\bar{a}} = \mathfrak{q}_a \cap M_0$ . Their normalized valuations satisfy*

$$v_{\mathfrak{r}_{\bar{a}}}(g) = \begin{cases} 1, & \bar{a} = \bar{1}, \\ 0, & \bar{a} \neq \bar{1}. \end{cases} \quad (\text{AFS}_{4c})$$

*Consequently*

$$\text{Stab}_\Delta(g) \subseteq \{\pm 1\}, \quad [\mathcal{K}_{n-1}(g) : \mathcal{K}_{n-1}] \geq \frac{\ell - 1}{2}. \quad (\text{AFS}_{4d})$$

*In particular  $g \notin \mathcal{K}_{n-1}$ . Thus the relative norm of an affine principalizer isolating the named prime cannot lie in the lower Conway field. The norm element, and its prime-above-two valuation divisor, has  $\Delta$ -orbit size at least  $(\ell - 1)/2$ .*

*Proof.* The residue size at  $\mathfrak{p}$  is  $q$ , and  $q \equiv -1 \pmod{\ell}$ . Hence the decomposition group of  $\mathfrak{p}$  in  $M_0/\mathcal{K}_{n-1}$  is  $\{\pm 1\}$ : the extension is unramified there, has residue degree two, and has  $(\ell - 1)/2 > 1$  primes above  $\mathfrak{p}$ . The top involution sends  $\mathfrak{q}_a$  to  $\mathfrak{q}_{-a}$ , so  $\mathfrak{r}_a$  splits in  $M/M_0$  as the pair  $\mathfrak{q}_a, \mathfrak{q}_{-a}$ , each with residue degree one. The element  $x = C_n - R$  is integral. Equation AFS<sub>1</sub> therefore gives

$$v_{\mathfrak{r}_a}(g) = v_{\mathfrak{q}_a}(x) + v_{\mathfrak{q}_{-a}}(x),$$

which is exactly AFS<sub>4c</sub>.

If  $\tau_b(g) = g$ , its unique prime-above-two support must be fixed, so  $\mathfrak{r}_b = \mathfrak{r}_{\bar{1}}$ , equivalently  $b = \pm 1$ . More precisely,

$$\text{Stab}_{\Delta}(g) = \text{Gal}(M_0/\mathcal{K}_{n-1}(g)),$$

so the Galois orbit–stabilizer equality proves AFS<sub>4d</sub>. Equivalently, every element descended from  $\mathcal{K}_{n-1}$  has the same valuation at all primes above  $\mathfrak{p}$ , whereas  $g$  has the nonconstant vector  $(1, 0, \dots, 0)$ .

The companion Lean file checks the two abstract group-action cores: an invariant function on a transitive orbit is constant, and a symmetry preserving a uniquely supported value fixes that support point. The cyclotomic decomposition, norm-valuation formula and Galois covariance, and the stabilizer-index/Galois-degree passage remain paper-level.  $\square$

**Corollary 7.66** (affine normalization loses no oriented ray data). *Let  $s$  be the top involution of  $M/M_0$  and put  $Y = 2C_n + 1$ , so  $s(Y) = -Y$ . For any  $x_0 \in M^\times$  with  $x_0 \neq s(x_0)$ , define*

$$z = \frac{Y}{x_0 - s(x_0)}, \quad x = zx_0.$$

*Then  $z \in M_0^\times$ ,  $x - s(x) = Y$ , and*

$$x = C_n - R \quad \text{for some } R \in M_0. \quad (\text{AFS}_{4a})$$

*The fractional divisor added by  $z$  is  $s$ -invariant. Its two oriented tame values are reciprocal, so it contributes trivially to the anti-invariant  $W_n$ -character. Hence every nonsymmetric oriented principal divisor has an affine representative carrying the same oriented Artin value.*

*This is a fractional-divisor normalization, not an integral principalizer theorem:  $z$  can have poles or nonunit components at two or  $\ell$ . It need not preserve the simple-zero conditions, the vanishing  $\ell$ -adic pairings, or  $S$ -unit support in Proposition 7.64.*

*Proof.* The denominator  $x_0 - s(x_0)$  is anti-invariant, as is  $Y$ ; therefore  $s(z) = z$ . It follows that

$$x - s(x) = z(x_0 - s(x_0)) = Y.$$

Since  $C_n - s(C_n) = 2C_n + 1 = Y$ , the element  $R = C_n - x$  is fixed by  $s$ , proving AFS<sub>4a</sub>. The divisor of an element of  $M_0^\times$  has equal multiplicities on each split pair. Top conjugation sends  $W_n$  to  $W_n^{-1}$ , so the two residue values cancel. Lean checks the fixed-scalar and anti-trace identities; the divisor-character interpretation is paper-level.  $\square$

**Corollary 7.67** (the local oriented residue cover is rational). *At a split tame support prime, write  $r = R \bmod \mathfrak{p}$  for the orientation selected by  $x = C_n - R$ . Then*

$$W_n = -\frac{r+1}{r}, \quad z^\ell = W_n \iff r = -\frac{1}{1+z^\ell}. \quad (\text{AFS}_{4b})$$

*Thus the local Kummer cover of the oriented residue is rational. In particular no positive-genus Weil bound applied only to this residue equation can force its value to be a nonpower; the remaining datum is which global oriented divisor the selected principal element carries.*

*Proof.* Substitution gives

$$-\frac{(-1/(1+z^\ell))+1}{-1/(1+z^\ell)} = z^\ell.$$

Conversely solving  $z^\ell = -(r+1)/r$  for  $r$  gives the displayed formula whenever the denominators are nonzero. Lean checks this rational parametrization over an arbitrary field.  $\square$

**Proposition 7.68** (Jacobi collapse of the affine half-resultant). *Retain Proposition 7.64, put*

$$A = \mathcal{A}_{n-1}, \quad g = A + R + R^2, \quad \mathfrak{a}_R = (g)\mathfrak{r}^{-1},$$

*and take the approximation depth there sufficiently large that  $g/A$  is an  $\ell$ -th power at every prime above  $\ell$ . For  $z \in M_0^\times$  and an integral ideal  $\mathfrak{b}$  prime to  $\ell$  such that  $z$  is a unit at every  $\mathfrak{p} \mid \mathfrak{b}$ , write*

$$\left(\frac{z}{\mathfrak{b}}\right)_\ell = \prod_{\mathfrak{p} \mid \mathfrak{b}} \bar{z}^{v_{\mathfrak{p}}(\mathfrak{b})(|k(\mathfrak{p})|-1)/\ell} \in \mu_\ell.$$

*Then  $\mathfrak{a}_R$  is integral and prime to  $2\ell$ , and*

$$\mathcal{T}_R = \left(\frac{AR^{-2}}{\mathfrak{a}_R}\right)_\ell = \left(\frac{R}{\mathfrak{a}_R}\right)_\ell^{-2} = \left(\frac{R}{\mathfrak{r}}\right)_\ell^2, \quad (\text{AJ}_1)$$

$$H = \left(\frac{R}{\mathfrak{r}}\right)_\ell^{-2}. \quad (\text{AJ}_2)$$

*More generally,*

$$\left(\frac{B}{\mathfrak{a}_R}\right)_\ell = 1 \quad (\text{AJ}_3)$$

*for every global unit  $B \in \mathcal{O}_{K_{n-1}}^\times$ , so every lower  $C_i$  and  $\mathcal{A}_i$  cancels separately. Since  $R \bmod \mathfrak{r} = c_n$ ,*

$$\mathcal{T}_R = c_n^{2(q^2-1)/\ell}, \quad H = c_n^{-2(q^2-1)/\ell}. \quad (\text{AJ}_4)$$

*Consequently the half-resultant is nontrivial exactly when the original selected Conway–Fermat symbol is nontrivial. The affine selector is a noncircular principal representative, but its Jacobi evaluation is an exact normal form for the original obstruction, not a proof of nonvanishing.*

*Proof.* Proposition 7.64 gives  $v_{\mathfrak{r}}(g) = 1$ , makes  $g$  a unit at every other prime above two, and makes it a unit at every prime above  $\ell$ . Hence  $\mathfrak{a}_R$  is integral and prime to  $2\ell$ . At a prime in its support, the selected top orientation has  $C_n \equiv R$ , and therefore  $W_n \equiv AR^{-2}$ ; at a ramified collision the same residue is one. The norm multiplicities agree, proving the first equality of  $\text{AJ}_1$ .

For any lower global unit  $B$ , reciprocity for  $(B, g)_\ell$  has no contribution away from the divisor of  $g$  and the primes above  $\ell$ . At the latter primes,  $g/A$  is an  $\ell$ -th power, so  $(B, g) = (B, A)$ . The product of these remaining symbols is one by applying reciprocity to the two global units  $B, A$ . Thus  $(B/(g))_\ell = 1$ . At  $\mathfrak{r}$ , the residue of  $B$  lies in  $\mathbb{F}_q^\times$ , while  $\ell \mid q+1$  and  $\ell \nmid q-1$ ; hence  $(B/\mathfrak{r})_\ell = 1$ . Since  $(g) = \mathfrak{r}\mathfrak{a}_R$ , this proves  $\text{AJ}_3$ , and the case  $B = A$  gives

$$\mathcal{T}_R = \left(\frac{R}{\mathfrak{a}_R}\right)_\ell^{-2}. \quad (\text{AJ}_5)$$

It remains to locate the  $R$ -symbol. Let  $\mathfrak{b}$  be the prime-to- $\ell$  part of the principal divisor  $(R)$ , and put

$$K_\ell = \prod_{\lambda \mid \ell} (A, R)_{\lambda, \ell}.$$

Reciprocity for  $(A, R)$ , with  $A$  a global unit, gives

$$1 = \left( \frac{A}{\mathfrak{b}} \right)_\ell K_\ell. \quad (\text{AJ}_6)$$

The ideals  $(R)$  and  $(g)$  are coprime, since a common prime would divide the global unit  $A = g - R - R^2$ . In reciprocity for  $(R, g)$ , the primes of  $g$  contribute  $(R/(g))_\ell$ ; the primes of  $\mathfrak{b}$  contribute  $(A/\mathfrak{b})_\ell^{-1}$ , because  $g \equiv A \pmod{R}$ ; and the primes above  $\ell$  contribute  $K_\ell^{-1}$ , because  $g/A$  is an  $\ell$ -th power and the Hilbert symbol is alternating. Equation  $\text{AJ}_6$  therefore yields

$$\left( \frac{R}{(g)} \right)_\ell = 1.$$

Using  $(g) = \mathfrak{r}\mathfrak{a}_R$  gives  $(R/\mathfrak{a}_R)_\ell = (R/\mathfrak{r})_\ell^{-1}$ , which with  $\text{AJ}_5$  proves  $\text{AJ}_1$ . Equation  $\text{AFS}_3$  proves  $\text{AJ}_2$ . Finally  $R \bmod \mathfrak{r} = c_n$  and  $|k(\mathfrak{r})| = q^2$ , proving  $\text{AJ}_4$ . The multiplicative reduction from  $\text{AJ}_3$  and the product-one relation to  $\text{AJ}_1$ – $\text{AJ}_2$  is independently checked in Lean; the Jacobi and Hilbert reciprocity argument is paper-level.  $\square$

**Corollary 7.69** (even the unrestricted norm-two equation has no solution). *Retain the fields of Proposition 7.64. Even without imposing the selector congruences on  $R$ , no  $R \in M_0$  satisfies*

$$\mathcal{A}_{n-1} + R + R^2 = 2. \quad (\text{AFS}_5)$$

*For the selected  $R$ , lower-field values are already excluded by Corollary 7.65; the present statement is stronger in a different direction because it rules out the equation for every  $R \in M_0$ . It does not exclude a general  $\{2, \ell\}$ -unit value on the left of  $\text{AFS}_5$ .*

*Proof.* Put  $D = 1 - 4\mathcal{A}_{n-1}$ . A solution of  $\text{AFS}_5$  would give

$$(2R + 1)^2 = D + 8. \quad (\text{AFS}_6)$$

At the unique two-adic completion of  $\mathcal{K}_{n-1}$ , the unit  $D$  defines the unramified quadratic extension  $\mathcal{K}_n/\mathcal{K}_{n-1}$ , and  $(D + 8)/D \in 1 + 8\mathcal{O}$  is a square. Thus  $D + 8$  has the same nontrivial unramified local square class as  $D$ . Consequently

$$L = \mathcal{K}_{n-1}(\sqrt{D + 8})$$

is a quadratic subfield of  $M_0$  whose base-two localization is the unramified quadratic field.

On the other hand,  $\text{GL}_4$  makes  $M_0/\mathcal{K}_{n-1}$  cyclic of degree  $\ell - 1$ . Its unique quadratic subfield is  $\mathcal{K}_{n-1}(\sqrt{\ell})$ , because  $\ell \equiv 1 \pmod{4}$ . In fact every  $\ell \mid F_n$ ,  $n \geq 2$ , satisfies  $\ell \equiv 1 \pmod{8}$ , so  $\ell$  is a square in  $\mathbf{Q}_2$ ; hence  $\mathcal{K}_{n-1}(\sqrt{\ell})/\mathcal{K}_{n-1}$  splits at the chosen prime above two. It therefore cannot equal  $L$ , contradicting uniqueness of the quadratic subfield of  $M_0/\mathcal{K}_{n-1}$ . Lean checks the algebraic equivalence between  $\text{AFS}_5$  and  $\text{AFS}_6$ ; the local and global square-class argument is paper-level.  $\square$

Classical power reciprocity does evaluate the oriented tame half-resultant: Proposition 7.68 collapses it to the single base-field symbol  $(R/\mathfrak{r})_\ell^2$ . No cited theorem proves that symbol nontrivial; because  $R \bmod \mathfrak{r} = c_n$ , doing so is exactly the original selected Conway–Fermat problem. Iwasawa’s formula [19] computes a local symbol from a critical unit coefficient but is an equality, not a nonvanishing theorem. Rohrlich’s Jacobi-sum characters [35] and Helou’s cyclotomic-unit formulas [17] remain in the pure cyclotomic factor annihilated by  $(\text{FWS}_4)$ ; the latter reduces orthogonality to additional Fermat-quotient congruences rather than excluding it universally. The integral Brumer–Stark theorem of Dasgupta–Kakde–Silliman–Wang [10] assumes a totally real base and an abelian

CM extension; the explicit class-field construction of Dasgupta–Kakde [11] has the same structural scope. Here  $M/\mathcal{K}_n$  has the required split prime but  $\mathcal{K}_n \supset \mathbf{Q}(\zeta_3)$  is not totally real, whereas the cyclotomic presentation over  $\mathbf{Q}$  is CM but two has residue degree  $2^{n+1}$ , not one. Thus none of these theorems proves the remaining named symbol nontrivial.

**Proposition 7.70** (the first selected  $\ell$ -ray coefficient is an explicit Witt polynomial). *Retain the lifted chain of Proposition 7.58, let  $\mathfrak{l}$  be a prime of  $\mathcal{K}_n$  above  $\ell$ , and put  $F = \mathcal{K}_{n,\mathfrak{l}}$ . Assume that the displayed Conway order is étale along this branch, equivalently*

$$2\bar{C}_i + 1 \neq 0 \quad (0 \leq i \leq n) \quad (\text{ER}_0)$$

*in the successive residue fields. This holds, in particular, if  $\ell$  divides none of the absolute norms of  $D_i = 1 - 4\mathcal{A}_i$  for  $i < n$ . Under this hypothesis  $F/\mathbf{Q}_\ell$  is unramified. Put  $k = \mathcal{O}_F/\mathfrak{l}\mathcal{O}_F$ ,  $|k| = \ell^f$ , and write  $[x] \in \mathcal{O}_F$  for the Teichmüller lift of  $x \in k$ . Define the second-Witt addition defect*

$$\mathfrak{d}_\ell(x) = \frac{[x] + 1 - [x+1]}{\ell} \in k.$$

*It is explicit:*

$$\mathfrak{d}_\ell(x)^\ell = \sum_{r=1}^{\ell-1} \frac{(-1)^r}{r} x^r. \quad (\text{ER}_1)$$

*Write, modulo  $\ell^2$ ,*

$$C_i = [\bar{C}_i](1 + \ell b_i), \quad \mathcal{A}_i = [\bar{\mathcal{A}}_i](1 + \ell s_i), \quad s_{-1} = 0.$$

*Then  $s_i = s_{i-1} + b_i$ , and the lifted Conway equation gives*

$$(2\bar{C}_i + 1)b_i = (\bar{C}_i + 1)s_{i-1} - \mathfrak{d}_\ell(\bar{C}_i). \quad (\text{ER}_2)$$

*For the standard Fermat quotient*

$$q_\ell(W_n) = \frac{\overline{W_n^{\ell^f-1} - 1}}{\ell},$$

*one therefore has*

$$q_\ell(W_n) = 2b_n - s_{n-1} = \frac{s_{n-1} - 2\mathfrak{d}_\ell(\bar{C}_n)}{2\bar{C}_n + 1}. \quad (\text{ER}_3)$$

*Equivalently, set  $P_{-1} = 0$ ,  $Q_{-1} = 1$ , and recursively*

$$\begin{aligned} P_i &= (3\bar{C}_i + 2)P_{i-1} - \mathfrak{d}_\ell(\bar{C}_i)Q_{i-1}, \\ Q_i &= (2\bar{C}_i + 1)Q_{i-1}. \end{aligned} \quad (\text{ER}_4)$$

*Then*

$$\boxed{q_\ell(W_n) \neq 0 \iff P_{n-1} - 2\mathfrak{d}_\ell(\bar{C}_n)Q_{n-1} \neq 0.} \quad (\text{ER}_5)$$

*Thus the first ramified local Kummer coefficient is an explicit full-ancestry truncated-log polynomial. Its nonvanishing makes the local character of  $W_n$  nontrivial at the unique prime  $\lambda$  of  $F(\zeta_\ell)$  above  $\mathfrak{l}$ , but it does not imply  $H \neq 1$ : it neither evaluates that character on the distinguished prime above two nor supplies a useful local second argument. Indeed  $(W_n, 2)_{\lambda, \ell} = 1$  pointwise by  $(\text{GL}_{7a})$ . If  $(\text{ER}_0)$  fails, the ramified unit filtration must replace this first-Witt expansion; the denominator-cleared identities remain valid, but  $(\text{ER}_5)$  is not available by cancellation.*

*Proof.* The binomial expansion modulo  $\ell^2$  first gives

$$\mathfrak{d}_\ell(x^\ell) = \sum_{r=1}^{\ell-1} \frac{(-1)^r}{r} x^r.$$

Teichmuller Frobenius gives  $\mathfrak{d}_\ell(x^\ell) = \mathfrak{d}_\ell(x)^\ell$ , proving (ER<sub>1</sub>). Teichmuller multiplicativity gives  $s_i = s_{i-1} + b_i$ . Expanding  $C_i^2 + C_i + \mathcal{A}_{i-1} = 0$  to first order and dividing by the nonzero residue  $\bar{C}_i$  gives (ER<sub>2</sub>). Since  $W_n = \mathcal{A}_{n-1} C_n^{-2}$ , the chosen Fermat-quotient convention gives  $q_l(W_n) = 2b_n - s_{n-1}$ ; eliminating  $b_n$  proves (ER<sub>3</sub>). Induction in (ER<sub>2</sub>) gives  $s_i = P_i/Q_i$  and hence (ER<sub>4</sub>)–(ER<sub>5</sub>). In  $F(\zeta_\ell)$ , one has  $\ell = u(\zeta_\ell - 1)^{\ell-1}$  for a unit  $u$ . Hence a nonzero Fermat quotient gives a nonzero class in  $U_{\ell-1}/U_\ell$ , so  $W_n$  is not a local  $\ell$ -th power. The relative discriminant identity

$$N_{\mathcal{K}_i/\mathcal{K}_{i-1}}(D_i) = 1 + 4\mathcal{A}_{i-1} + 16\mathcal{A}_{i-1}^3$$

makes the excluded derivative primes explicit. Lean kernel-checks this identity, the one-step cleared parent update, the cleared terminal identity, and their generic nonvanishing equivalence. It does not formalize the recursive  $P_i, Q_i$  induction, Teichmuller expansion, or Fermat-quotient interpretation, which remain paper-level.  $\square$

**Corollary 7.71** (the last simple ramified branch has a nonzero leading class). *Suppose that (ER<sub>0</sub>) fails, and let  $r \leq n$  be the last non-étale index on the chosen  $\ell$ -adic branch. Assume that its discriminant zero is simple:*

$$\mathfrak{l}_i = \mathfrak{l} \cap \mathcal{K}_i, \quad F_i = \mathcal{K}_{i, \mathfrak{l}_i}, \quad v_{r-1}(D_{r-1}) = 1, \quad D_{r-1} = 1 - 4\mathcal{A}_{r-1}. \quad (\text{RR}_0)$$

Here  $v_i$  denotes the normalized valuation on  $F_i$ , and every Teichmuller bracket below is taken in  $F_n$ , equivalently in its maximal unramified subfield. Put  $Y_r = 2C_r + 1$ . Then  $Y_r$  is a uniformizer of the ramified quadratic completion and

$$\mathcal{A}_r \equiv [-1/8](1 - Y_r) \pmod{Y_r^2}. \quad (\text{RR}_1)$$

In particular its leading parent coefficient is  $s_r = -1$ .

Assume for  $r < j \leq n$  that  $2\bar{C}_j + 1 \neq 0$ , and for  $r < j < n$  that the branch-sensitive factors  $3\bar{C}_j + 2 \neq 0$ . If  $r < n$ , the later first-order coefficients obey

$$(2\bar{C}_j + 1)s_j = (3\bar{C}_j + 2)s_{j-1}, \quad (\text{RR}_2)$$

and the terminal norm-one unit has

$$s_{n-1} = - \prod_{j=r+1}^{n-1} \frac{3\bar{C}_j + 2}{2\bar{C}_j + 1} \neq 0, \quad (\text{RR}_3)$$

$$\text{coeff}_{Y_r}(W_n) = -\frac{s_{n-1}}{2\bar{C}_n + 1} \neq 0.$$

If  $r = n$ , the Cayley identity

$$W_n = \frac{1 + Y_n}{1 - Y_n}$$

instead gives leading coefficient two directly. In either case  $W_n$  is not an  $\ell$ -th power in the corresponding completion after adjoining  $\zeta_\ell$ . Thus the local Kummer character is genuinely nontrivial through every last-simple-ramification branch satisfying the displayed noncritical conditions.

This still does not prove  $H \neq 1$ . A nonzero weight-one equivariant character may be nonzero on this ramified inertia line and vanish on the entire distinguished prime-above-two orbit. The companion Lean file checks the displayed equivariance, inertia nonvanishing, and selected-orbit vanishing in an explicit two-coordinate toy model.



*Proof.* The equation  $Y_r^2 = D_{r-1}$  and  $(\text{RR}_0)$  show that the quadratic completion is tamely ramified and that  $Y_r$  is a uniformizer. Rational units differ from their Teichmüller lifts first at  $\ell$ -depth, which has  $Y_r$ -depth at least two after this simple ramified quadratic step. Since

$$C_r = \frac{-1 + Y_r}{2}, \quad \mathcal{A}_{r-1} \equiv [1/4] \pmod{Y_r^2},$$

one obtains  $(\text{RR}_1)$ . This quadratic step is totally ramified, so lower-field principal perturbations acquire  $Y_r$ -depth at least two; the later steps are unramified because  $r$  was the last non-étale index. Their Teichmüller addition defects likewise have depth at least two. Writing

$$\mathcal{A}_j = [\bar{\mathcal{A}}_j](1 + Y_r s_j), \quad C_j = [\bar{C}_j](1 + Y_r b_j) \pmod{Y_r^2},$$

therefore gives

$$(2\bar{C}_j + 1)b_j = (\bar{C}_j + 1)s_{j-1}, \quad s_j = s_{j-1} + b_j.$$

Elimination proves  $(\text{RR}_2)$ . At the terminal step it gives

$$(2\bar{C}_n + 1)(s_{n-1} - 2b_n) = -s_{n-1},$$

which proves  $(\text{RR}_3)$ . Because  $r$  was the last non-étale stage,  $Y_r$  remains a uniformizer thereafter. Put  $L = F_n(\zeta_\ell)$  and  $d = e(L/F_n)$ . Then

$$d \mid \ell - 1, \quad v_L(Y_r) = d, \quad e_L := v_L(\ell) = e(F_n/\mathbf{Q}_\ell)d \geq 2d.$$

The Teichmüller factor of  $W_n$  is already an  $\ell$ -th power because  $\ell$  is coprime to the order of the residue-field unit group. If  $z \in U_m$ , the first possible depth of  $z^\ell - 1$  is at least  $\min\{\ell m, e_L + m\}$ , with cancellation only raising it. Neither term can equal  $d$ : one has  $\ell m > d$  and  $e_L + m > d$ . Hence a principal unit with the nonzero depth- $d$  leading class in  $(\text{RR}_3)$  cannot be an  $\ell$ -th power. The case  $r = n$  follows from the displayed Cayley identity and  $(1 + Y)/(1 - Y) - 1 = 2Y/(1 - Y)$ .

If the lower completion before the critical step is unramified and  $\mathcal{A}_{r-1} = [1/4](1 + \ell s)$ , then

$$\frac{D_{r-1}}{\ell} \equiv q_\ell(4) - s, \quad q_\ell(4) = 2\mathfrak{d}_\ell(-1/2).$$

Thus  $s \neq q_\ell(4)$  is exactly the simple-discriminant case; equality is a genuine second-Witt or higher-filtration branch, not a failure of the first-order algebra. Lean kernel-checks the Cayley identities, the homogeneous recurrence, the terminal nonvanishing, and these two critical coefficient identities. The local valuation interpretation remains paper-level.  $\square$

**Proposition 7.72** (the remaining value is one specified Artin class). *Put  $\mathcal{E} = M(\theta)$ ,  $\theta^\ell = W_n$ , and let  $\mathfrak{f}_W$  be the Artin conductor of  $\mathcal{E}/M$ , supported above  $\ell$ . Since  $W_n$  is fixed by  $\Delta$  and  $\mu_\ell \subset M$ , the extension  $\mathcal{E}/K_n$  is Galois; hence  $\mathfrak{f}_W$  is  $\Delta$ -stable. With arithmetic Frobenius, the Artin–Kummer character*

$$\chi_W : \text{Cl}_{\mathfrak{f}_W}(M)/\text{Cl}_{\mathfrak{f}_W}(M)^\ell \longrightarrow \mu_\ell, \quad \chi_W([a]) = \frac{\text{Art}_{\mathcal{E}/M}(a)(\theta)}{\theta}$$

satisfies

$$\boxed{\chi_W([q_1]) = H,} \tag{RCF_1}$$

and, for  $a \in \Delta$ ,

$$\chi_W(\tau_a x) = \chi_W(x)^a. \tag{RCF_2}$$

Consequently the desired  $\ell$ -primary Popovych condition is exactly

$$[\mathfrak{q}_1] \notin \ker \chi_W. \quad (\text{RCF}_3)$$

If  $[\mathcal{E} : M] = \ell$ , this says precisely that  $\mathfrak{q}_1$  is inert rather than split in  $\mathcal{E}/M$ .

This formulation also exhausts the standard group-theoretic information. Every  $\Delta$ -fixed ray class is killed by  $\chi_W$ , and  $\chi_W([\mathfrak{q}_a \mathfrak{q}_{-a}]) = 1$ . Conversely, the orbit relations alone admit base value one or any element of  $\mu_\ell$ . Reflection theorems identify the  $\Delta$ -trivial Kummer radical generated by  $[W_n]$  with the reflected cyclotomic ray eigenspace, but compare ranks; they do not decide whether the projection of  $[\mathfrak{q}_1]$  pairs nontrivially with  $[W_n]$ . After enlarging  $\mathfrak{f}_W$  to a sufficiently large  $\Delta$ -invariant modulus induced from  $\mathcal{K}_n$ , ambiguous ray-class formulas count fixed classes and norm indices, but still cannot evaluate  $(\text{RCF}_1)$ . The desired condition is stronger than existence: even after proving  $\chi_W$  nontrivial, one must evaluate it at one distinguished mixed class.

*Proof.* Local Artin reciprocity at  $\mathfrak{q}_1$  and the Hilbert-symbol orientation of Proposition 7.58 give  $(\text{RCF}_1)$ . Because  $W_n$  is fixed by  $\Delta$  while  $\tau_a$  acts on  $\mu_\ell$  by the  $a$ -th power, Artin equivariance gives  $(\text{RCF}_2)$ . A nonidentity Frobenius in a cyclic degree- $\ell$  extension has order  $\ell$ , proving the inert/split assertion.

If a source class is fixed by some  $a \neq 1$ , equivariance says its image equals its own  $a$ -th power and hence is one. The inverse-pair assertion is immediate from  $(\text{RCF}_2)$ . Finally an equivariant orbit map has the form  $a \mapsto ah$  in additive  $\mathbb{F}_\ell$ -coordinates, with arbitrary base scalar  $h$ ; the orbit relations permit both  $h = 0$  and  $h \neq 0$ . These abstract fixed-class and scalar-freedom statements are kernel-checked in `Ogdoad.Excess`. Gras's reflection theorem [15, Theorems 4.4 and 5.18] and the Chevalley–Gras formula of Li–Yu [24, Theorem 2.1], applied after the stated modulus enlargement, have exactly the rank/counting scope just described and do not evaluate the named Artin class.  $\square$

**Corollary 7.73** (Hasse norm and dihedral descent boundary). *Retain Proposition 7.72 with  $n \geq 2$ . Then*

$$H = 1 \iff 2 \in N_{\mathcal{E}/M}(\mathcal{E}^\times). \quad (\text{HD}_1)$$

Thus failure of the Popovych  $\ell$ -condition is exactly solvability of one global cyclic norm equation.

If  $H = 1$ , choose  $y \in \mathcal{E}^\times$  with  $N_{\mathcal{E}/M}(y) = 2$ , put  $G = \text{Gal}(\mathcal{E}/M)$ , and let  $P = \sum_{\gamma \in G} n_\gamma \gamma \in \mathbf{Z}[G]$ . Then

$$N_{\mathcal{E}/M}(y^P) = 2^{\text{aug}(P)}, \quad \text{aug}(P) = \sum_{\gamma \in G} n_\gamma. \quad (\text{HD}_{1a})$$

Hence every unweighted operation in the cyclic Kummer direction has the same norm valuation at every  $\mathfrak{q}_a \mid 2$ . In particular, if some  $y^P$  descends to  $M$ , then  $(y^P)^\ell = 2^{\text{aug}(P)}$ , so its own valuation vector is constant. It cannot equal the affine selector, supported only at  $\mathfrak{q}_1$ , or its mixed relative norm, supported only on the  $\{\pm 1\}$ -coset of that prime. A coupling would have to weight the  $\Delta$ -orbit, which is precisely the missing mixed ray-class operation.

Assume now that  $[\mathcal{E} : M] = \ell$ , put  $M' = \mathcal{K}_{n-1}(\zeta_\ell)$ , and let  $\iota$  be the top involution of  $M/M'$ . It extends to  $\mathcal{E}$  by  $\iota(\theta) = \theta^{-1}$ . If  $\sigma(\theta) = \zeta_\ell \theta$ , then

$$\iota \sigma \iota^{-1} = \sigma^{-1}, \quad (\text{HD}_2)$$

so  $\text{Gal}(\mathcal{E}/M')$  is dihedral. The involution exchanges  $\mathfrak{q}_1$  and  $\mathfrak{q}_{-1}$ . For their common prime  $\mathfrak{r}$  of  $M'$ , choosing a prime of  $\mathcal{E}$  above  $\mathfrak{q}_1$  gives the Frobenius representative  $\sigma^h$  determined by  $H = \zeta_\ell^h$ . Every homomorphism from this dihedral group to an abelian group kills  $\sigma$ . Consequently every abelian quotient or descent of this specific Kummer extension sees the same image for  $h = 0$  and  $h \neq 0$ ; a successful lower-field descent must retain its nonabelian dihedral Frobenius.

Nevertheless the norm equation itself has an exact nonabelian reflection descent. Put  $L = \mathcal{E}^{(\iota)}$ , so  $[L : M'] = \ell$ . More explicitly, if  $t = \theta + \theta^{-1}$ , and if the classical power sums are defined by

$$D_0^{\text{cl}}(T) = 2, \quad D_1^{\text{cl}}(T) = T, \quad D_{m+2}^{\text{cl}}(T) = TD_{m+1}^{\text{cl}}(T) - D_m^{\text{cl}}(T),$$

then

$$L = M'(t), \quad D_\ell^{\text{cl}}(t) = W_n + W_n^{-1}. \quad (\text{HD}_{2a})$$

For every  $a \in M'^\times$ ,

$$a \in N_{\mathcal{E}/M}(\mathcal{E}^\times) \iff a \in N_{L/M'}(L^\times). \quad (\text{HD}_3)$$

In particular, when  $[\mathcal{E} : M] = \ell$ ,

$$H = 1 \iff 2 \in N_{L/M'}(L^\times). \quad (\text{HD}_4)$$

The equivalence is stable under every completion. For a place  $v$  of  $M'$ , put

$$\mathcal{E}_v = \mathcal{E} \otimes_{M'} M'_v, \quad M_v = M \otimes_{M'} M'_v, \quad L_v = L \otimes_{M'} M'_v.$$

Then, for every  $a \in (M'_v)^\times$ ,

$$a \in N_{L_v/M'_v}(L_v^\times) \iff \text{diag}(a) \in N_{\mathcal{E}_v/M_v}(\mathcal{E}_v^\times). \quad (\text{HD}_5)$$

Consequently, for every  $a \in M'^\times$ , this non-Galois prime-degree extension satisfies the Hasse norm principle:

$$a \in N_{L/M'}(L^\times) \iff a \in N_{L_v/M'_v}(L_v^\times) \text{ for every place } v. \quad (\text{HD}_6)$$

At a prime  $\mathfrak{r}_{\bar{a}} \mid 2$ , one has the sharper dichotomy

$$\left. \begin{array}{l} H = 1 \\ H \neq 1 \end{array} \right| \begin{array}{l} L_v \simeq (M'_v)^\ell, \\ L_v/M'_v \text{ is the unramified extension of degree } \ell. \end{array} \quad (\text{HD}_7)$$

In the second row every norm has valuation divisible by  $\ell$ , whereas  $v(2) = 1$ . All local conditions away from the primes above two are automatic. Thus the reflection field retains the nonabelian marked Frobenius, but it creates no new relative class-group or global norm obstruction: its norm-two equation is exactly the original selected local splitting question at two.

The quadratic discriminant resolvent is blind as well. Put  $A = W_n + W_n^{-1} \in M'$ . Since  $[\mathcal{E} : M] = \ell$ , one has  $A^2 - 4 \neq 0$ : otherwise  $W_n = \pm 1$ , hence  $W_n$  is an  $\ell$ -th power because  $\ell$  is odd. In fact the coefficient and its discriminant already lie in the lower ancestry:

$$A = \mathcal{A}_{n-1}^{-1} - 2, \quad A^2 - 4 = \frac{D_{n-1}}{\mathcal{A}_{n-1}^2}.$$

Then

$$\text{disc}_T(D_\ell^{\text{cl}}(T) - A) = (-1)^{\ell(\ell-1)/2} \ell^\ell (A^2 - 4)^{(\ell-1)/2} \in (M'^\times)^2. \quad (\text{HD}_8)$$

Indeed  $\ell \equiv 1 \pmod{4}$ , so the sign is positive and  $(\ell-1)/2$  is even, while the quadratic Gauss subfield of  $\mathbf{Q}(\zeta_\ell) \subset M'$  contains  $\sqrt{\ell}$ . Equivalently, both the identity and the nontrivial  $\ell$ -cycle in  $\text{HD}_7$  are even permutations. Neither the polynomial nor field discriminant can distinguish success from failure.

*Proof.* For a cyclic Kummer extension, local norm theory says that 2 is a norm at a place  $v$  exactly when  $(W_n, 2)_{v, \ell} = 1$ . Away from  $2\ell$  both entries are tame units, and at three  $W_n$  is already an  $\ell$ -th power. Every place above  $\ell$  is pointwise trivial by  $(\mathrm{GL}_{7a})$ , and the archimedean places are complex. At  $\mathfrak{q}_a \mid 2$ , the symbol is  $H^a$ , so all two-adic local norm conditions hold exactly when  $H = 1$ . The cyclic Hasse norm theorem proves  $(\mathrm{HD}_1)$  [16].

For  $P \in \mathbf{Z}[G]$ , norm invariance under every  $\gamma \in G$  gives

$$N(y^P) = \prod_{\gamma \in G} N(y)^{n_\gamma} = 2^{\sum n_\gamma},$$

which proves  $(\mathrm{HD}_{1a})$ . In particular the common further norm is  $N_{\mathcal{E}/M'}(y) = 4$ , with constant prime-above-two valuations; this is fully compatible with the mixed affine-norm theorem rather than contradicting it.

The top involution sends  $W_n$  to  $W_n^{-1}$ , proving that the stated extension of  $\iota$  is well-defined; direct evaluation on  $\theta$  gives  $(\mathrm{HD}_2)$ . At the prime of  $\mathcal{K}_{n-1}$  above two, arithmetic Frobenius in  $\mathrm{Gal}(M/\mathcal{K}_{n-1}) = \langle \iota \rangle \times \Delta$  is the diagonal element  $\iota\tau_{-1}$ , because the residue size is  $q \equiv -1 \pmod{\ell}$ . Hence  $\iota\tau_{-1}$  stabilizes  $\mathfrak{q}_1$ , so  $\iota(\mathfrak{q}_1) = \mathfrak{q}_{-1}$ ; the chosen Frobenius above  $\mathfrak{r}$  is the stated translation. Finally, in any abelian quotient the images of  $\sigma$  and  $\sigma^{-1}$  are equal by  $(\mathrm{HD}_2)$ . The image of  $\sigma$  therefore has order dividing both two and the odd prime  $\ell$ , hence is trivial. The abstract augmentation identity underlying  $\mathrm{HD}_{1a}$  and this final dihedral group-theoretic step are independently kernel-checked in `Ogdoad.Excess`.

To prove  $\mathrm{HD}_3$ , first let  $z \in L$ . Its  $M'$ -embeddings are the restrictions of  $\sigma^j$ ,  $0 \leq j < \ell$ , so

$$N_{L/M'}(z) = \prod_{j=0}^{\ell-1} \sigma^j(z) = N_{\mathcal{E}/M}(z).$$

The identity  $\mathrm{HD}_{2a}$  is the classical Dickson–Chebyshev power-sum formula applied to  $t = \theta + \theta^{-1}$ . If two conjugates  $\zeta_\ell^j \theta + \zeta_\ell^{-j} \theta^{-1}$  and  $\zeta_\ell^k \theta + \zeta_\ell^{-k} \theta^{-1}$  coincide, then either  $j = k$  or  $\zeta_\ell^{j+k} \theta^2 = 1$ . The latter implies  $W_n^2 = 1$ , hence that  $W_n$  is an  $\ell$ -th power because  $\ell$  is odd, contrary to  $[\mathcal{E} : M] = \ell$ . Thus the  $\ell$  conjugates are distinct and  $t$  generates the reflection fixed field. Conversely, suppose  $N_{\mathcal{E}/M}(y) = a \in M'^\times$ , put  $r = (\ell + 1)/2$ , and define

$$z = \frac{(y\iota(y))^r}{a}.$$

Then  $\iota(z) = z$ , hence  $z \in L$ , while conjugating the cyclic norm by  $\iota$  gives  $N(\iota(y)) = a$ . Therefore

$$N_{L/M'}(z) = N_{\mathcal{E}/M}(z) = \frac{(a^2)^r}{a^\ell} = a^{2r-\ell} = a.$$

This proves  $\mathrm{HD}_3$ , and  $\mathrm{HD}_4$  follows from  $\mathrm{HD}_1$ .

The proof just given is an identity in finite etale algebras and therefore commutes with the base change  $M' \rightarrow M'_v$ . In particular the same formula proves  $\mathrm{HD}_5$ . If  $a$  is a local norm from  $L_v$  for every  $v$ , equation  $\mathrm{HD}_5$  makes its diagonal image a local norm from  $\mathcal{E}/M$  at every place of  $M$ . The cyclic Hasse norm theorem produces a global  $y \in \mathcal{E}^\times$ , and the symmetrization above produces the required global  $z \in L^\times$ . This proves  $\mathrm{HD}_6$ . As an independent cross-check, Hoshi–Yamasaki’s metacyclic norm-torus theorem gives both the Hasse norm principle and weak approximation here: the odd dihedral group has trivial Schur multiplier and the reflection stabilizer is cyclic [18, Theorem 3.2 and Example 4.6]. Thus, once the norm fibre is nonempty, finitely many additional local open conditions on a witness cannot supply a new obstruction.

At  $\mathfrak{r}_{\bar{a}}$ , the extension  $M/M'$  splits into the two primes  $\mathfrak{q}_a, \mathfrak{q}_{-a}$ . If  $H = 1$ , their Frobenius translations are trivial, so this prime splits completely in  $\mathcal{E}/M'$ , and hence in  $L/M'$ . If  $H \neq 1$ , the Frobenius is a nonidentity  $\sigma^h$ ; since  $\ell$  is prime it generates  $\langle \sigma \rangle$ , which acts transitively on the  $\ell$  cosets of the reflection subgroup. This makes  $L_v/M'_v$  unramified of degree  $\ell$  and proves HD<sub>7</sub>. The remaining local assertions are the place-by-place calculation already used for HD<sub>1</sub>.

For  $f_A(T) = D_\ell^{\text{cl}}(T) - A$ , the classical Chebyshev description gives

$$f_A(T) = 2T_\ell(T/2) - A, \quad f'_A(T) = \ell U_{\ell-1}(T/2).$$

The latter polynomial has leading coefficient  $\ell$  and roots  $x_j = 2 \cos(j\pi/\ell)$ ,  $1 \leq j < \ell$ . At these roots,  $D_\ell^{\text{cl}}(x_j) = 2(-1)^j$ . With  $m = (\ell - 1)/2$ , therefore,

$$\text{Res}(f'_A, f_A) = \ell^\ell \prod_{j=1}^{\ell-1} (2(-1)^j - A) = \ell^\ell ((2 - A)(-2 - A))^m = \ell^\ell (A^2 - 4)^m.$$

The monic discriminant formula supplies the sign in HD<sub>8</sub>, and the square-class argument in the statement proves its final assertion. The two lower-ancestry identities follow directly from  $W_n = -(C_n + 1)/C_n$ ,  $C_n(C_n + 1) = -\mathcal{A}_{n-1}$ , and  $D_{n-1} = 1 - 4\mathcal{A}_{n-1}$ . Thus even the reflection polynomial is lower-defined; the selected information is its factorization at the named prime above two, not a hidden top-field coefficient. Moreover  $\text{disc}(\mathcal{O}_{M'}[t]) = \mathfrak{d}_{L/M'}[\mathcal{O}_L : \mathcal{O}_{M'}[t]]^2$  as ideals. Hence neither the polynomial discriminant nor the relative field-discriminant ideal retains the selected Frobenius; the index changes the discriminant only by a square.

Lean checks the generic reflection symmetrization, its exponent identity, and the resulting fixed-witness equivalence in a commutative group. The finite-étale base-change interpretation, cyclic Hasse theorem, fixed-field degree, discriminant calculation, and prime-valuation statements remain paper-level. If  $W_n$  is already a global  $\ell$ -th power, then  $\mathcal{E} = M$  and the dihedral construction degenerates; that case is itself a stronger global failure and is not covered by HD<sub>3</sub>–HD<sub>7</sub>.  $\square$

**Proposition 7.74** (the unramified Euler factor is the selected datum). *Retain Corollary 7.73, assume  $[\mathcal{E} : M] = \ell$ , and put*

$$Q = |k(\mathfrak{q}_1)| = |k(\mathfrak{r})|, \quad T = Q^{-s}.$$

*Use arithmetic Frobenius and write  $\chi(\text{Frob}_{\mathfrak{q}_1}) = H$  for the local cyclic Kummer character. Then*

$$L_{\mathfrak{q}_1}(T, \chi^a) = (1 - H^a T)^{-1}. \quad (\text{EF}_1)$$

*Let*

$$G = D_{2\ell} = \langle \sigma, \iota : \sigma^\ell = \iota^2 = 1, \iota \sigma \iota^{-1} = \sigma^{-1} \rangle, \quad \Pi = \text{Ind}_{\langle \iota \rangle}^G \mathbf{1}.$$

*At  $\mathfrak{r}$ , the Frobenius  $\sigma^h$ , where  $H = \zeta_\ell^h$ , acts on the  $\ell$  reflection cosets by translation. Consequently*

$$L_{\mathfrak{r}}(T, \Pi) = \begin{cases} (1 - T)^{-\ell}, & H = 1, \\ (1 - T^\ell)^{-1}, & H \neq 1, \end{cases} \quad (\text{EF}_2)$$

*and, for the augmentation quotient  $\Pi_0 = \Pi/\mathbf{1}$ ,*

$$L_{\mathfrak{r}}(T, \Pi_0) = \prod_{a=1}^{\ell-1} L_{\mathfrak{q}_1}(T, \chi^a) = \begin{cases} (1 - T)^{-(\ell-1)}, & H = 1, \\ \frac{1 - T}{1 - T^\ell}, & H \neq 1. \end{cases} \quad (\text{EF}_3)$$

Thus the named Euler factor sees failure exactly, but only by containing the original unknown  $H$ .

All standard conductor and normalized local-root-number data at this prime are blind. If  $\psi$  has conductor zero and the Haar measure is self-dual, then

$$a(\chi^a) = a(\Pi) = a(\Pi_0) = 0, \quad \varepsilon_{\mathfrak{r}}(s, \chi^a, \psi) = \varepsilon_{\mathfrak{r}}(s, \Pi, \psi) = \varepsilon_{\mathfrak{r}}(s, \Pi_0, \psi) = 1. \quad (\text{EF}_4)$$

The same local statement applies to the ordinary, cubic, and conductor-five reflection fields above: at the selected prime they are likewise split or unramified of odd prime degree. Hence discriminant, Artin conductor, epsilon factor, and local root number cannot evaluate any of the selected  $Z/O/C/D$  Frobenius values. This is a local no-go only; a new global ancestry identity could in principle correlate ramified factors elsewhere with  $H$ .

*Proof.* Equation (EF<sub>1</sub>) is the definition of the unramified local factor. Translation by zero fixes all reflection cosets, whereas a nonzero translation of the prime-order group is one  $\ell$ -cycle. This gives (EF<sub>2</sub>). Removing the invariant line gives (EF<sub>3</sub>), equivalently

$$\prod_{a=0}^{\ell-1} (1 - H^a T) = 1 - T^\ell \quad (H \neq 1).$$

Every representation in the statement is unramified. Deligne's conductor-zero normalization gives epsilon factor one for an unramified one-dimensional character; multiplicativity over its complex eigencharacters proves (EF<sub>4</sub>) [12, §§3.4–3.6 and Theorem 4.1(1)]. With geometric Frobenius,  $H$  is replaced by  $H^{-1}$ , leaving every split/nontrivial dichotomy unchanged.  $\square$

**Proposition 7.75** (the full Artin package leaves the named factor free). *Retain Proposition 7.74, put*

$$F = M' = \mathcal{K}_{n-1}(\zeta_\ell), \quad K = M = \mathcal{K}_n(\zeta_\ell), \quad m = \frac{\ell-1}{2}, \quad d = [F : \mathbb{Q}].$$

*For a nontrivial character  $\chi$  of  $\text{Gal}(\mathcal{E}/K)$ , the augmentation representation has the exact decomposition*

$$\Pi_0 \simeq \bigoplus_{a \in \mathbb{F}_\ell^\times / \{\pm 1\}} \text{Ind}_{K/F} \chi^a. \quad (\text{GAF}_1)$$

*Consequently*

$$L_F(s, \Pi_0) = \frac{\zeta_L(s)}{\zeta_F(s)} = \prod_{a \in \mathbb{F}_\ell^\times / \{\pm 1\}} L_K(s, \chi^a). \quad (\text{GAF}_2)$$

*Both  $F$  and  $L$  are totally imaginary. With  $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s} \Gamma(s)$ , the completed quotient is*

$$\Lambda_F(s, \Pi_0) = \left( \frac{|D_L|}{|D_F|} \right)^{s/2} \Gamma_{\mathbb{C}}(s)^{md} L_F(s, \Pi_0), \quad (\text{GAF}_3)$$

*and satisfies*

$$\Lambda_F(s, \Pi_0) = \Lambda_F(1-s, \Pi_0), \quad W(\Pi_0) = +1. \quad (\text{GAF}_4)$$

*Thus the global sign is independent of  $H$ .*

*Let  $\mathfrak{f}_W$  be the common conductor of the nonzero powers  $\chi^a$ . It is supported above  $\ell$ , and*

$$\mathfrak{f}_F(\Pi_0) = \mathfrak{d}_{L/F} = (\mathfrak{d}_{K/F} N_{K/F} \mathfrak{f}_W)^m. \quad (\text{GAF}_5)$$

If  $v \nmid \ell$  is ramified in  $K/F$ , then for every  $a$  the local induced representation is  $\mathbf{1} \oplus \eta_v$ , where  $\eta_v$  is the ramified quadratic character of  $K_v/F_v$ . Hence

$$L_v(T_v, \Pi_0) = (1 - T_v)^{-m}, \quad W_v(\Pi_0) = 1. \quad (\text{GAF}_6)$$

The remaining  $\ell$ -adic Gauss factors have global product one by  $(\text{GAF}_4)$ ; none contains the unramified value  $H$ .

By contrast, there are  $m$  primes of  $F$  above two, and multiplying their augmentation factors gives

$$\prod_{\mathfrak{r}|2} L_{\mathfrak{r}}(T, \Pi_0) = \begin{cases} (1 - T)^{-m(\ell-1)}, & H = 1, \\ \left(\frac{1 - T}{1 - T^\ell}\right)^m, & H \neq 1. \end{cases} \quad (\text{GAF}_7)$$

This product distinguishes success from failure only because it already contains the named unknown Frobenius. Conductor, discriminant, gamma factor, ramified local constants, and the functional-equation sign do not select between its two rows.

*Proof.* The nontrivial characters of the rotation subgroup occur in inverse pairs under the reflection, which gives  $(\text{GAF}_1)$ . Artin formalism for the reflection permutation representation gives  $(\text{GAF}_2)$ . Since

$$r_2(L) - r_2(F) = \frac{\ell d - d}{2} = md,$$

the quotient of the completed Dedekind zeta functions is exactly  $(\text{GAF}_3)$ , and the two zeta functional equations prove  $(\text{GAF}_4)$ .

Every nonzero power of  $\chi$  has the same kernel and conductor. The conductor–discriminant formula, followed by induction from  $K$  to  $F$ , proves  $(\text{GAF}_5)$ . At a finite  $v \nmid \ell$  ramified in  $K/F$ , the Kummer extension is unramified. Quadratic inertia is a reflection in the odd dihedral group and is normal in its decomposition group. A reflection subgroup is self-normalizing, so the decomposition group is that reflection subgroup itself; the rotation extension therefore splits locally. This gives  $\mathbf{1} \oplus \eta_v$  and the Euler factor in  $(\text{GAF}_6)$ . A quadratic local root number has fourth power one, while Fermat divisors satisfy  $\ell \equiv 1 \pmod{8}$ , so  $4 \mid m$  and  $W(\eta_v)^m = 1$ . Deligne’s induction and epsilon-factor formalism supplies the standard local normalization [12].

Finally, the primes above two are indexed by  $a \in \mathbb{F}_\ell^\times / \{\pm 1\}$ . At each one the Frobenius is the translation represented by  $H^a$ . It is the identity for every  $a$  exactly when  $H = 1$ ; otherwise it is one  $\ell$ -cycle. Multiplying the two cases of  $(\text{EF}_3)$  over the  $m$  primes proves  $(\text{GAF}_7)$ . The representation-theoretic and number-field identities in this proposition remain paper-level; Lean checks only the underlying odd-dihedral abelianization and local split/inert permutation cores recorded earlier.  $\square$

**Corollary 7.76** (only the top generator carries the Brauer obstruction). *Retain Corollary 7.73, put  $M_0 = \mathcal{K}_{n-1}(\zeta_\ell)$ , and write the  $\ell$ -torsion norm-residue symbol additively. Then*

$$\{\mathcal{A}_{n-1}, 2\}_\ell = 0 \quad \text{in } \text{Br}(M_0)[\ell]. \quad (\text{TB}_1)$$

If  $s$  is the top involution and

$$\alpha = \{C_n, 2\}_\ell, \quad \beta = \{W_n, 2\}_\ell \quad \text{in } \text{Br}(M)[\ell],$$

then

$$\alpha + s\alpha = 0, \quad \boxed{\beta = -2\alpha}. \quad (\text{TB}_2)$$

Consequently

$$H \neq 1 \iff \beta \neq 0 \iff \alpha \neq 0. \quad (\text{TB}_3)$$

Thus every lower ancestral factor has zero symbol with two after adjoining  $\zeta_\ell$ ; the whole obstruction is the anti-invariant class of the single top generator. Corestriction kills the class at issue, so this reduction gives no inductive nonvanishing theorem by itself.

*Proof.* We prove (TB<sub>1</sub>) place by place. Away from  $2\ell$ , both entries are units and the tame symbol is trivial. At a place above two, the completion of  $M_0$  is unramified with residue field  $\mathbb{F}_{q^2}$ , where  $q = 2^{2^n}$ , while the reduction of  $\mathcal{A}_{n-1}$  lies in  $\mathbb{F}_q^\times$ . Since  $\gcd(\ell, q-1) = 1$ , its residue is an  $\ell$ -th power, and  $\ell$ -powering is an automorphism of the pro-two principal units. Thus  $\mathcal{A}_{n-1}$  is already a local  $\ell$ -th power. At a place above  $\ell$ , the decomposition-stabilizer argument of (GL<sub>7a</sub>), now with  $[\mathcal{K}_{n-1} : \mathbf{Q}] = 2^n$ , kills the symbol pointwise. All infinite places are complex. The identical argument applies to every lower unit  $C_i$  and  $\mathcal{A}_i$ ,  $i < n$ . Brauer–Hasse–Noether proves (TB<sub>1</sub>).

Projection for corestriction gives

$$\text{cor}_{M/M_0}(\alpha) = \{N(C_n), 2\}_\ell = \{\mathcal{A}_{n-1}, 2\}_\ell = 0.$$

Restricting back to  $M$  yields  $\alpha + s\alpha = 0$ . Finally  $W_n = \mathcal{A}_{n-1}C_n^{-2}$  and bilinearity give  $\beta = -2\alpha$ . Multiplication by two is invertible on  $\ell$ -torsion, proving (TB<sub>3</sub>). Lean independently checks the abstract anti-invariant and coboundary identities in `0gdoad.Excess`.  $\square$

**Proposition 7.77** (four-point saturation of a Fermat failure). *Fix  $n \geq 2$ , put*

$$c = c_n, \quad d = c_{n-1}, \quad b = a_{n-2}, \quad Q = |E_{n-2}| = 2^{2^{n-1}}.$$

*Then, for one  $\epsilon \in \mathbb{F}_2$ ,*

$$c^Q = c + d + \epsilon, \quad c^{Q^2} = c + 1. \tag{FP_1}$$

*Consequently, if  $\ell \mid F_n$  and  $c$  is an  $\ell$ -th power in  $E_n$ , the complete affine plane*

$$c + \mathbb{F}_2 + \mathbb{F}_2 d = \{c, c+1, c+d, c+d+1\} \tag{FP_2}$$

*consists of  $\ell$ -th powers. Its product is nevertheless the old-field element*

$$N_{E_n/E_{n-2}}(c) = b^3, \tag{FP_3}$$

*which is automatically an  $\ell$ -th power. Thus the exact four-point saturation and its norm are compatible with failure; they do not supply the missing nonresidue.*

*Proof.* The tower equations give

$$d^2 + d = b, \quad c^2 + c = bd = d^3 + d^2, \quad d^Q = d + 1.$$

For  $s = c^Q + c$ , subtraction of the equation for  $c$  from its  $Q$ -th power gives  $s^2 + s = b$ . Hence  $s \in \{d, d+1\}$ , proving the first identity in (FP<sub>1</sub>); applying  $Q$ -Frobenius once more gives the second. Frobenius preserves the subgroup of  $\ell$ -th powers, so one power value forces all four values in (FP<sub>2</sub>). Finally,

$$c(c+1) = bd, \quad (c+d)(c+d+1) = b(d+1),$$

and their product is  $b^2 d(d+1) = b^3$ . Since  $\ell \mid Q^2 + 1$  is coprime to  $|E_{n-2}^\times| = Q - 1$ , every old-field element is an  $\ell$ -th power in  $E_n$ .  $\square$



**Proposition 7.78** (CM form and the Kummer-cover obstruction). *On  $\mathcal{E} : y^2 + y = x^3 + x^2$ , let  $O$  be the origin and  $T = (0, 0)$ . Then  $\mathcal{E}(\mathbb{F}_2) = \langle T \rangle \cong \mathbb{Z}/5\mathbb{Z}$ , with*

$$-T = (0, 1), \quad 2T = (1, 1), \quad -2T = (1, 0).$$

*On the normalized Conway correspondence, the function  $\mathfrak{c} = 1/(u + 1)$  is*

$$\mathfrak{c} = \frac{x}{x^2 + x + y}, \quad \text{div}(\mathfrak{c}) = [-T] + 2[O] - 3[-2T]. \quad (\text{EC}_1)$$

*The automorphism  $\mathcal{A}$  satisfies*

$$\mathcal{A}^2(P) = T - P, \quad \mathfrak{c}(\mathcal{A}^2 P) = \mathfrak{c}(P) + 1. \quad (\text{EC}_2)$$

*If  $\pi$  is 2-power Frobenius and  $i = \pi + 1$ , then  $i^2 = -1$ ,  $i(T) = 2T$ , and, after possibly replacing  $\mathcal{A}$  by its inverse,*

$$\mathcal{A} = \tau_{2T} \circ i.$$

*Both orientations fix  $R_0 = -2T$ .*

*For  $n \geq 4$ , set*

$$Q = 2^{2^{n-1}}, \quad R = \sqrt{Q} = 2^{2^{n-2}},$$

*and let  $P_n$  be the distinguished Conway point. If  $X_n = P_n - R_0$ , then*

$$\text{Frob}_Q(X_n) = \pm i X_n, \quad X_n \in \ker([R] \mp i). \quad (\text{EC}_3)$$

*Each kernel is cyclic of order  $R^2 + 1 = Q + 1$ .*

*For every prime  $\ell \mid Q^2 + 1$ , the Kummer cover*

$$\mathcal{C}_\ell : \quad z^\ell = \mathfrak{c}$$

*is geometrically connected, ramified exactly above  $-T, O, -2T$ , and has genus*

$$g(\mathcal{C}_\ell) = \frac{3\ell - 1}{2}. \quad (\text{EC}_4)$$

*The automorphism  $\mathcal{A}$  does not lift to  $\mathcal{C}_\ell$ . Hence the quarter-turn on the base supplies no action on the Kummer fibre above  $P_n$ ; the surviving Conway condition is exactly*

$$\mathfrak{c}(P_n) = c_n \notin (\mathbb{F}_{Q^4}^\times)^\ell. \quad (\text{EC}_5)$$

*It is the Artin-symbol problem for one distinguished fibre of this ramified cover, not a consequence of the CM action on  $\mathcal{E}$ .*

*Proof.* In the blow-up coordinates of Proposition 7.51,  $U = u + 1$ ,  $V = v + 1$ ,  $X = U/V$ ,  $Y = V$ , and  $(x : y : z) = (Z : X : X + Y + Z)$ . On  $z = 1$ , homogeneous representatives give

$$V = \frac{1 + x + y}{x}, \quad U = \frac{y(1 + x + y)}{x^2}.$$

Since  $y(1 + x + y) = xy + x^3 + x^2 = x(x^2 + x + y)$ , this proves the formula for  $\mathfrak{c} = 1/U$ . With  $d = x^2 + x + y$ , direct intersection gives

$$\text{div}(x) = [T] + [-T] - 2[O], \quad \text{div}(d) = [T] + 3[-2T] - 4[O].$$

Indeed  $d = 0$  gives  $y = x^2 + x$ , whence the curve equation gives  $x(x+1)^3 = 0$ , and  $d$  has pole order four at  $O$ . Subtraction proves (EC<sub>1</sub>).

The function  $v = (y+1)/x$  has pole divisor  $[O] + [T]$ , so the deck involution of its degree-two map is  $J(P) = T - P$ . Since  $\mathcal{A}^2$  fixes  $v$  and is nontrivial, it equals  $J$ . Also  $\mathcal{A}^2$  sends  $u$  to  $u^{-1}$ , and therefore

$$\mathfrak{c}(\mathcal{A}^2 P) = \frac{1}{u(P)^{-1} + 1} = \frac{u(P)}{u(P) + 1} = \mathfrak{c}(P) + 1,$$

proving (EC<sub>2</sub>).

The Frobenius polynomial of  $\mathcal{E}/\mathbb{F}_2$  is  $\pi^2 + 2\pi + 2 = 0$ . Thus  $i = \pi + 1$  satisfies  $i^2 = -1$ , and it acts on  $\mathcal{E}(\mathbb{F}_2)$  as multiplication by two. Write  $\mathcal{A} = \tau_S \circ \alpha$ , with  $\alpha(O) = O$ . As  $\mathcal{A}$  is defined over  $\mathbb{F}_2$ ,  $\alpha$  commutes with  $\pi$ . The centralizer of  $\pi$  in the supersingular endomorphism algebra is  $\mathbb{Q}(i)$ , whose integral units are  $\{\pm 1, \pm i\}$ . The linear part of  $\mathcal{A}^2$  is  $[-1]$ , so  $\alpha = \pm i$ . Comparing translation parts gives

$$\mathcal{A} = \tau_{2T} \circ i \quad \text{or} \quad \mathcal{A} = \tau_{-T} \circ (-i);$$

these are inverse and both fix  $R_0 = -2T$ .

For  $n \geq 4$ ,  $2^{n-1}$  is divisible by eight. Since  $\pi^8 = [16]$ ,

$$\pi^{2^{n-1}} = [2^{2^{n-2}}] = [R].$$

Translation of  $R_0$  to the origin makes  $\mathcal{A}^{\pm 1}$  the linear maps  $\pm i$ , proving (EC<sub>3</sub>). The separable endomorphisms  $[R] \mp i$  have degree  $R^2 + 1$ . For every odd prime  $p$ , the Tate module  $T_p(\mathcal{E})$  is free of rank one over  $\mathbb{Z}_p[i]$  (the polynomial  $X^2 + 1$  is separable away from 2). Hence the  $p$ -primary kernel is the additive quotient  $\mathbb{Z}_p[i]/(R \mp i)$ . The matrix of  $R \mp i$  over  $\mathbb{Z}_p$  has entries with gcd one and determinant  $R^2 + 1$ , so its first Smith invariant is one. Combining the odd primary parts shows that the geometric kernel is cyclic of order  $R^2 + 1$ .

Every prime  $\ell \mid Q^2 + 1$  is coprime to 15. Thus the valuations 1, 2,  $-3$  in (EC<sub>1</sub>) are nonzero modulo  $\ell$ . Kummer theory makes  $z^\ell = \mathfrak{c}$  geometrically connected and totally ramified at exactly those three points. Riemann–Hurwitz gives

$$2g(\mathcal{C}_\ell) - 2 = 3(\ell - 1),$$

which proves (EC<sub>4</sub>).

It remains to prove non-lifting. Choose  $\mathcal{A} = \tau_{2T} \circ i$ , put  $B_j = \mathcal{A}^j(O)$ , and retain  $R_0 = -2T$ . Then

$$(B_0, B_1, B_2, B_3) = (O, 2T, T, -T).$$

For  $D_j = \text{div}(\mathfrak{c} \circ \mathcal{A}^{-j})$ , pullback gives

$$D_j = 2[B_j] + [B_{j+3}] - 3[R_0]. \tag{EC_6}$$

If  $\sum_{j=0}^3 r_j D_j \equiv 0 \pmod{\ell}$ , comparison at the four  $B_j$ 's gives  $2r_j + r_{j+1} = 0$  cyclically. Iteration gives  $15r_0 = 0$ , and  $\ell \nmid 15$  forces every  $r_j = 0$ . Hence

$$\mathfrak{c}, \quad \mathfrak{c} \circ \mathcal{A}, \quad \mathfrak{c} \circ \mathcal{A}^2, \quad \mathfrak{c} \circ \mathcal{A}^3$$

are linearly independent in

$$\overline{\mathbb{F}}_2(\mathcal{E})^\times / \overline{\mathbb{F}}_2(\mathcal{E})^{\times \ell}.$$

A lift of  $\mathcal{A}$  would normalize the cyclic deck group, hence send  $z$  to  $gz^r$  for  $r \in (\mathbb{Z}/\ell\mathbb{Z})^\times$ , and would require  $\mathfrak{c} \circ \mathcal{A} = g^\ell \mathfrak{c}^r$ , contradicting this independence.

Finally, set

$$\chi(t) = t^{(Q^4-1)/\ell}, \quad \tau = \chi(\mathfrak{c}(P_n)).$$

Since  $\text{Frob}_Q^2(P_n) = \mathcal{A}^2(P_n)$ , (EC<sub>2</sub>) gives  $\mathfrak{c}(P_n)^{Q^2} = \mathfrak{c}(P_n) + 1$ . Because  $Q^2 \equiv -1 \pmod{\ell}$ , the four CM/Frobenius character values are only

$$\tau, \quad \tau^{\pm Q}, \quad \tau^{-1}, \quad \tau^{\mp Q}.$$

All norm identities among them are tautological. This proves neither  $\tau = 1$  nor  $\tau \neq 1$ : the desired assertion remains (EC<sub>5</sub>).  $\square$

**Proposition 7.79** (Miller-unit and generalized-Jacobian location). *Retain Proposition 7.78, put  $S = -2T$ ,  $q = Q^2$ , and  $K_n = \mathbb{F}_{Q^4} = \mathbb{F}_{q^2}$ . Then*

$$\mathfrak{c}^{-1} = f_{3,S}$$

*up to a nonzero scalar, where  $f_{3,S}$  is the Miller tripling function with divisor*

$$\text{div}(f_{3,S}) = 3[S] - [3S] - 2[O]. \quad (\text{GJ}_1)$$

*For every prime  $\ell \mid q + 1$ ,*

$$\#\mathcal{E}(K_n) = (q - 1)^2, \quad \mathcal{E}(K_n)[\ell] = 0, \quad [\ell] : \mathcal{E}(K_n) \xrightarrow{\sim} \mathcal{E}(K_n). \quad (\text{GJ}_2)$$

*Let  $B = \{-T, O, -2T\}$  and  $U = \mathcal{E} \setminus B$ . The Kummer class of  $\mathfrak{c}$  on  $U$  has nonzero tame residue vector*

$$(1, 2, -3) \pmod{\ell}. \quad (\text{GJ}_3)$$

*It is therefore ramified and does not come from an unramified  $\mu_\ell$ -class on  $\mathcal{E}$ . In the generalized Jacobian*

$$1 \longrightarrow \mathbb{G}_m^2 \longrightarrow J_B \longrightarrow \mathcal{E} \longrightarrow 0, \quad (\text{GJ}_4)$$

*Hilbert 90 and (GJ<sub>2</sub>) give*

$$J_B(K_n)/\ell J_B(K_n) \cong (K_n^\times/K_n^{\times\ell})^2. \quad (\text{GJ}_5)$$

*Thus the residue vector has a genuinely ramified toric projection; the ordinary  $K_n$ -rational elliptic quotient cannot detect that projection. Ordinary  $K_n$ -rational Tate–Lichtenbaum and formal-group data cannot by themselves decide (EC<sub>5</sub>). This does not assert that the ramified class evaluates nontrivially at the distinguished point, nor does it exclude a calculation using non-rational  $\mathcal{E}[\ell]$  or the full extension structure of  $J_B$ . An elliptic completion must evaluate that generalized-Jacobian Miller-unit residue.*

*Proof.* The point  $T$  has order five, so  $3S = 3(-2T) = -T$ . Equation (EC<sub>1</sub>) therefore gives

$$\text{div}(\mathfrak{c}^{-1}) = 3[S] - [3S] - 2[O],$$

which proves (GJ<sub>1</sub>).

For  $n \geq 4$ , the Frobenius relation used in Proposition 7.78 gives

$$\pi^{2^{n+1}} = [q].$$

This is Frobenius on  $K_n = \mathbb{F}_{2^{2n+1}}$ , so

$$\#\mathcal{E}(K_n) = \deg([q] - [1]) = (q - 1)^2.$$

As  $\ell \mid q + 1$  is odd, it is coprime to  $q - 1$ . This proves (GJ<sub>2</sub>); multiplication by  $\ell$  is likewise an automorphism of the characteristic-two formal group because its linear coefficient is a unit.

The divisor (EC<sub>1</sub>) gives the residue vector (GJ<sub>3</sub>), which is nonzero because  $\ell \nmid 6$ . The standard generalized-Jacobian sequence for the three-point reduced rational modulus is (GJ<sub>4</sub>). Its torus is split; Hilbert 90 makes the map on  $K_n$ -points surjective, and the kernel created on passing modulo  $\ell$  vanishes because  $\mathcal{E}(K_n)[\ell] = 0$ . This proves (GJ<sub>5</sub>) and the stated scope of the toric obstruction.  $\square$

**Proposition 7.80** (exact extension character and cohomological exhaustion). *Retain Proposition 7.79 and put*

$$D_1 = [-T] - [-2T], \quad D_2 = [O] - [-2T]$$

*in  $\text{Div}^0(B)$ . The character lattice of the split torus in  $J_B$  is*

$$X^*(\mathbb{G}_m^2) = \text{Div}^0(B) = \mathbb{Z}D_1 \oplus \mathbb{Z}D_2,$$

*and the extension is classified, under  $\mathcal{E} \cong \text{Pic}^0(\mathcal{E})$ , by*

$$\lambda(aD_1 + bD_2) = (a + 2b)T. \tag{GJ_6}$$

*Thus its entire extension class is rational five-torsion. Moreover*

$$r := \text{div}(\mathfrak{c}) = D_1 + 2D_2, \quad \lambda(r) = 5T = O,$$

*so  $r$  defines a character  $\varphi_r : J_B \rightarrow \mathbb{G}_m$ . If  $j_T : U \rightarrow J_B$  is the generalized Abel map normalized by  $j_T(T) = 0$ , then*

$$\varphi_r(j_T(P)) = \mathfrak{c}(P). \tag{GJ_7}$$

*For every prime  $\ell \mid q + 1$ , even the non-rational geometric  $\mathcal{E}[\ell]$  supplies no further rational Kummer class:*

$$H^1(K_n, \mu_\ell^2) \xrightarrow{\sim} H^1(K_n, J_B[\ell]), \quad J_B(K_n)/\ell J_B(K_n) \cong (K_n^\times/K_n^{\times\ell})^2. \tag{GJ_8}$$

*In torus coordinates dual to  $(D_1, D_2)$ , the character is  $(x, y) \mapsto xy^2$ , hence the nonzero linear functional  $(x, y) \mapsto x + 2y$  modulo  $\ell$ . Naturality of Kummer theory gives*

$$\varphi_{r,*}(\delta_J(j_T(P_n))) = P_n^*[\mathfrak{c}] = [\mathfrak{c}(P_n)] \in K_n^\times/K_n^{\times\ell}. \tag{GJ_9}$$

*Consequently (EC<sub>5</sub>) is exactly the assertion that the normalized Abel class of  $P_n$  avoids the hyperplane  $x + 2y = 0$ . Torus translation above any fixed elliptic point realizes every value of this functional. Hence the CM point, the full extension class, and non-rational elliptic  $\ell$ -torsion do not force (EC<sub>5</sub>); the remaining datum is the explicit toric coordinate of this one distinguished normalized Abel lift.*

*Proof.* For a reduced three-point rational modulus, the generalized-Jacobian torus is  $(\mathbb{G}_m^3)/\mathbb{G}_m$ , with character lattice  $\text{Div}^0(B)$ . The standard extension-class map sends a boundary divisor to its class in  $\text{Pic}^0(\mathcal{E})$ . Here

$$(-T) - (-2T) = T, \quad O - (-2T) = 2T,$$

which proves (GJ<sub>6</sub>). Equation (EC<sub>1</sub>) gives  $r = D_1 + 2D_2$ , and its image is  $T + 4T = O$ . Characters of  $J_B$  are precisely the torus characters whose pushout extension class vanishes, so  $r$  yields  $\varphi_r$ . The

generalized-Albanese universal property gives (GJ<sub>7</sub>) after normalization at  $T$ . Indeed,  $T \notin B$  and  $\mathfrak{c}(T) \in \mathbb{F}_2^\times = \{1\}$ .

The  $K_n$ -Frobenius on  $\mathcal{E}$  is  $[q]$ , and  $q \equiv -1 \pmod{\ell}$ . It therefore acts as  $[-1]$  on  $\mathcal{E}[\ell]$ . Since  $\ell$  is odd,

$$\mathcal{E}[\ell](K_n) = 0, \quad H^1(K_n, \mathcal{E}[\ell]) = \mathcal{E}[\ell]/(\text{Frob} - 1)\mathcal{E}[\ell] = 0.$$

The cohomology sequence of

$$0 \longrightarrow \mu_\ell^2 \longrightarrow J_B[\ell] \longrightarrow \mathcal{E}[\ell] \longrightarrow 0$$

then proves the first isomorphism in (GJ<sub>8</sub>). Lang's theorem and Kummer theory prove the second. The natural Kummer square for  $\varphi_r$ , together with (GJ<sub>7</sub>), gives (GJ<sub>9</sub>). Finally  $\mathbb{G}_m^2(K_n)/\ell \rightarrow J_B(K_n)/\ell$  is an isomorphism: surjectivity uses bijectivity of  $[\ell]$  on  $\mathcal{E}(K_n)$ , and injectivity uses  $\mathcal{E}(K_n)[\ell] = 0$ . The character  $(x, y) \mapsto xy^2$  is surjective on the  $\ell$ -quotient, proving the fixed-fibre statement and the asserted exact boundary.  $\square$

**Proposition 7.81** (explicit Rosenlicht coordinate and its reciprocity boundary). *Retain Proposition 7.80, write  $S = -2T$ ,  $P = P_n$ , and put  $X = P - S$ . For  $\ell \mid q + 1$ , let  $Z \in \mathcal{E}(K_n)$  be the unique point with*

$$[\ell]Z = P - T, \quad W = T + Z. \quad (\text{RC}_1)$$

*If  $e\ell \equiv 1 \pmod{5(Q+1)}$ , then*

$$Z = [e]X + [2e]T, \quad W = [e]X + [2e+1]T, \quad (\text{RC}_2)$$

*so the division point remains an explicit translate of the same cyclic CM kernel as  $X$ . Choose a Miller function  $F = F_{\ell, Z}$  with*

$$\text{div}(F) = \ell[Z] - [[\ell]Z] - (\ell - 1)[O].$$

*In the Rosenlicht coordinates dual to  $D_1 = [-T] - [S]$ ,  $D_2 = [O] - [S]$ , the normalized Abel Kummer class of  $P$  is*

$$\rho_1 = \frac{F(2T)}{F(S)}, \quad \rho_2 = \frac{F(2T)}{F(-T)}, \quad (\text{RC}_3)$$

*and its Miller-unit character is exactly*

$$[\mathfrak{c}(P)] = [\rho_1 \rho_2^2] \quad \text{in } K_n^\times / K_n^{\times \ell}. \quad (\text{RC}_4)$$

*There is also a formula avoiding the  $\ell$ -division point. On  $\mathcal{E} : y^2 + y = x^3 + x^2$ , set*

$$A_5 = 1 + x + x^2 + y + xy, \quad B_5 = 1 + x + xy, \quad g_1 = \frac{A_5}{B_5}, \quad g_2 = \frac{1}{B_5}.$$

*Then*

$$\text{div}(g_1) = 5D_1, \quad \text{div}(g_2) = 5D_2, \quad \mathfrak{c}^5 = g_1 g_2^2. \quad (\text{RC}_5)$$

*If  $5\bar{5} \equiv 1 \pmod{\ell}$ , the full normalized Abel class in  $J_B(K_n)/\ell J_B(K_n)$  has coordinates*

$$([g_1(P)]^{\bar{5}}, [g_2(P)]^{\bar{5}}). \quad (\text{RC}_6)$$

More explicitly, for the consecutive Conway variables  $u = u_n$ ,  $v = u_{n-1}$ , put

$$U = u + 1, \quad V = v + 1, \quad D = V^2 + V + U.$$

Then the normalized point has  $x = V/D$ ,  $y = U/D$ , and

$$g_1(P) = \frac{D^2 + D(U + V) + V^2 + UV}{D^2 + VD + UV}, \quad g_2(P) = \frac{D^2}{D^2 + VD + UV}, \quad (\text{RC}_7)$$

with  $g_1(P)g_2(P)^2 = U^{-5}$ .

These formulas compute the missing toric coordinate exactly, but do not force it to be nonzero modulo  $\ell$ . Indeed, Weil reciprocity gives

$$\mathfrak{c}(P) = \mathfrak{c}(W)^\ell \frac{F(2T)^3}{F(S)F(-T)^2}. \quad (\text{RC}_8)$$

Applying reciprocity instead to  $G(Y) = \mathfrak{c}(Y + T)$ , whose divisor is  $[S] + 2[-T] - 3[2T]$ , returns precisely the inverse identity

$$\frac{F(S)F(-T)^2}{F(2T)^3} = \frac{\mathfrak{c}(W)^\ell}{\mathfrak{c}(P)}. \quad (\text{RC}_9)$$

Thus the boundary Miller values and the rational five-torsion splitting exhaust the Rosenlicht coordinate, but collapsing them by the evident reciprocity law is exactly equivalent to the original Kummer value  $[\mathfrak{c}(P)]$ ; it supplies no independent nonresidue constraint.

*Proof.* Since  $P = X + 3T$ , one has  $P - T = X + 2T$ . Multiplication by  $\ell$  is invertible both on the cyclic kernel containing  $X$  and on  $\langle T \rangle$ , which proves  $(\text{RC}_2)$ .

Put  $h(Y) = F(Y - T)$ . Translation of the displayed divisor of  $F$  gives

$$\text{div}(h) = \ell[W] - [P] - (\ell - 1)[T].$$

The Rosenlicht evaluation of this principal relation against  $D_1, D_2$  is

$$\frac{h(S)}{h(-T)} = \frac{F(2T)}{F(S)}, \quad \frac{h(S)}{h(O)} = \frac{F(2T)}{F(-T)},$$

which proves  $(\text{RC}_3)$ . Weil reciprocity between  $h$  and  $\mathfrak{c}$ , using

$$\text{div}(\mathfrak{c}) = [-T] + 2[O] - 3[S], \quad \mathfrak{c}(T) = 1,$$

gives  $(\text{RC}_8)$ , and hence  $(\text{RC}_4)$ .

Direct intersection with the Weierstrass equation, including the local orders at the indicated five-torsion points, gives

$$\text{div}(A_5) = 5[-T] - 5[O], \quad \text{div}(B_5) = 5[S] - 5[O].$$

Thus  $g_1, g_2$  have the divisors asserted in  $(\text{RC}_5)$ . Both sides of  $\mathfrak{c}^5 = g_1 g_2^2$  have the same divisor and value one at  $T$ , proving the exact identity. Since five is invertible modulo  $\ell$ , this proves  $(\text{RC}_6)$ . Substitution of  $x = V/D$ ,  $y = U/D$  gives  $(\text{RC}_7)$ ; here  $\mathfrak{c}(P) = 1/U$ , so its last identity is also immediate from  $(\text{RC}_5)$ .

Finally, the divisor of  $G$  follows by translating that of  $\mathfrak{c}$ . Weil reciprocity between  $F$  and  $G$  is  $(\text{RC}_9)$ , proving the exact neutrality statement.  $\square$

**Proposition 7.82** (five-point equivariant generalized-Jacobian exhaustion). *Retain the Conway point  $P = P_n$ , the quarter-turn  $\mathcal{A}$ , and  $S = -2T$ . Put*

$$B_0 = O, \quad B_1 = 2T, \quad B_2 = T, \quad B_3 = -T, \quad D = \mathcal{E}(\mathbb{F}_2) = \{S, B_0, B_1, B_2, B_3\}.$$

For  $j \in \mathbb{Z}/4\mathbb{Z}$ , set

$$e_j = [B_j] - [S], \quad f_j = \mathbf{c} \circ \mathcal{A}^{-j}, \quad r_j = \operatorname{div}(f_j) = 2e_j + e_{j+3}. \quad (\text{FE}_1)$$

Then  $D$  is  $\mathcal{A}$ -invariant, and the four classes  $r_j$  form a basis of  $\operatorname{Div}^0(D) \otimes \mathbb{F}_\ell$  for every prime  $\ell \mid F_n$ ,  $n \geq 2$ .

Let  $J_D$  be the generalized Jacobian of the five-point reduced modulus, let  $K_n = \mathbb{F}_{Q^4}$ , where  $Q = 2^{2^{n-1}}$ , and choose any  $R \in \mathcal{E}(\mathbb{F}_{16}) \setminus D$ . If  $j_R$  denotes the generalized Abel map normalized at  $R$ , then

$$\mathbf{c}(P) \in K_n^{\times \ell} \iff [j_R(P)] = 0 \text{ in } J_D(K_n)/\ell J_D(K_n). \quad (\text{FE}_2)$$

Thus even the simultaneous Rosenlicht data of the complete quarter-turn orbit contain no obstruction beyond the original Kummer value.

For  $n \geq 3$ , put  $z = [j_R(P)]$  and let  $\sigma = \operatorname{Frob}_Q$ . If  $\sigma(P) = \mathcal{A}^\epsilon(P)$ , with  $\epsilon \in \{1, -1\}$ , then

$$Qz = \mathcal{A}_*^\epsilon z. \quad (\text{FE}_3)$$

Inside the split rank-four torus quotient this eigenspace is one-dimensional over  $\mathbb{F}_\ell$ , and evaluation by  $\varphi_{r_0}$  is an isomorphism from it to  $K_n^\times / K_n^{\times \ell}$ . Under this isomorphism  $z$  maps to  $[\mathbf{c}(P)]$ . Consequently the full  $\mathcal{A}$ -invariant generalized Jacobian, its rational five-torsion extension class, and the exact Frobenius quarter-turn relation determine precisely one scalar—the scalar in (EC<sub>5</sub>)—but do not determine whether that scalar vanishes.

*Proof.* The automorphism  $\mathcal{A}$  fixes  $S$  and cycles the other four  $\mathbb{F}_2$ -points as

$$O \mapsto 2T \mapsto T \mapsto -T \mapsto O.$$

Equation (EC<sub>6</sub>) gives (FE<sub>1</sub>). In the basis  $(e_0, e_1, e_2, e_3)$ , the four rows  $r_j$  form the circulant matrix  $2I + C$ , where  $C$  is a four-cycle. Hence

$$\det(2I + C) = 2^4 - 1 = 15. \quad (\text{FE}_4)$$

No prime divisor of  $F_n$ ,  $n \geq 2$ , divides 15, proving the basis assertion. Notice also that the union of the supports of the four  $f_j$  is exactly  $D$ .

The Frobenius polynomial of  $\mathcal{E}/\mathbb{F}_2$  gives  $\#\mathcal{E}(\mathbb{F}_{16}) = 25$ , so a point  $R \in \mathcal{E}(\mathbb{F}_{16}) \setminus D$  exists. Every  $f_j(R)$  is a nonzero element of  $\mathbb{F}_{16}$ ; since  $\gcd(\ell, 15) = 1$ , it is an  $\ell$ -th power already in  $\mathbb{F}_{16}$ . The five-point generalized Jacobian fits into

$$1 \longrightarrow \mathbb{G}_m^4 \longrightarrow J_D \longrightarrow \mathcal{E} \longrightarrow 0.$$

As in (GJ<sub>2</sub>),  $\#\mathcal{E}(K_n) = (Q^2 - 1)^2$ , so multiplication by  $\ell$  is bijective on  $\mathcal{E}(K_n)$ . Hilbert 90 and the absence of rational  $\ell$ -torsion therefore give

$$J_D(K_n)/\ell J_D(K_n) \simeq (K_n^\times / K_n^{\times \ell})^4. \quad (\text{FE}_5)$$

For  $r_j = \operatorname{div}(f_j)$ , let  $\varphi_{r_j} : J_D \rightarrow \mathbb{G}_m$  be the corresponding principal boundary character. Generalized Abel evaluation is

$$\varphi_{r_j}(j_R(P)) = \frac{f_j(P)}{f_j(R)} \text{ in } K_n^\times / K_n^{\times \ell}. \quad (\text{FE}_6)$$

Writing  $\sigma = \text{Frob}_Q$ , if  $\mathfrak{c}(P)$  is an  $\ell$ -th power, then  $\sigma(P) = \mathcal{A}^{\pm 1}(P)$  and Frobenius stability of  $K_n^{\times \ell}$  make every  $f_j(P)$  an  $\ell$ -th power. Equations (FE<sub>4</sub>)–(FE<sub>6</sub>) then give  $j_R(P) = 0$  modulo  $\ell$ . The converse follows by evaluating  $\varphi_{r_0}$ , since  $f_0 = \mathfrak{c}$ . This proves (FE<sub>2</sub>).

Suppose now that  $n \geq 3$ . Then  $\sigma$  fixes  $\mathbb{F}_{16}$ , and hence  $R$ . Moreover

$$\#J_D(\mathbb{F}_{16}) = 15^4 \cdot 25$$

is prime to  $\ell$ . Therefore the affine base-point correction  $[j_R(\mathcal{A}^e R)]$  vanishes modulo  $\ell$ , and

$$\sigma j_R(P) = j_R(\sigma P) = \mathcal{A}_*^e j_R(P) + j_R(\mathcal{A}^e R)$$

reduces to (FE<sub>3</sub>). On the right side of (FE<sub>5</sub>),  $\sigma$  acts on the scalar Kummer quotient as multiplication by  $Q$ , while  $\mathcal{A}_*^e$  acts on the rank-four cocharacter lattice as a four-cycle. Since  $Q^2 \equiv -1 \pmod{\ell}$ ,  $Q$  has exact order four modulo  $\ell$ . The  $Q$ -eigenspace of that four-cycle is consequently one-dimensional.

Finally, if  $\varphi_{r_0}$  vanished on a vector in this eigenspace, its four  $\mathcal{A}^*$ -translates  $r_j$  would all vanish there. The basis assertion would force the vector to be zero. Thus  $\varphi_{r_0}$  is nonzero, and hence an isomorphism, on the one-dimensional eigenspace. Its value is  $[f_0(P)/f_0(R)] = [\mathfrak{c}(P)]$ , completing the proof.  $\square$

**Proposition 7.83** (equivariant Kummer countermodels at the actual ordinal degree). *Retain the elliptic curve  $\mathcal{E}$ , its order-four automorphism  $\mathcal{A}$ , the five-point boundary  $D = \mathcal{E}(\mathbb{F}_2)$ , and the four functions*

$$f_j = \mathfrak{c} \circ \mathcal{A}^{-j}, \quad j \in \mathbb{Z}/4\mathbb{Z},$$

*from Proposition 7.82. Let  $\ell \nmid 15$  be an odd prime, and let  $Y_\ell$  be the normalization of the Kummer cover*

$$z_j^\ell = f_j, \quad j = 0, 1, 2, 3. \quad (\text{EK}_1)$$

*Then  $Y_\ell/\mathcal{E}$  is geometrically connected of degree  $\ell^4$ , is ramified exactly over the five points of  $D$ , and has*

$$2g(Y_\ell) - 2 = 5\ell^4 - 5\ell^3. \quad (\text{EK}_2)$$

*The order-four automorphism lifts as*

$$\tilde{\mathcal{A}}(P, z_0, z_1, z_2, z_3) = (\mathcal{A}P, z_3, z_0, z_1, z_2). \quad (\text{EK}_3)$$

*Let  $Q$  be a power of two with  $\ell \mid Q^2 + 1$ . If*

$$Q + 1 - 2g(Y_\ell)\sqrt{Q} > 5\ell^3, \quad (\text{EK}_4)$$

*then there exists  $P \in \mathcal{E}(\mathbb{F}_{Q^4}) \setminus D$  such that*

$$\text{Frob}_Q(P) = \mathcal{A}(P), \quad f_j(P) \in \mathbb{F}_{Q^4}^{\times \ell} \quad (0 \leq j < 4). \quad (\text{EK}_5)$$

*In particular  $\mathfrak{c}(P)$  is an  $\ell$ -th power.*

*Condition (EK<sub>4</sub>) holds at an actual Conway–Fermat field size:*

$$Q = 2^{256}, \quad \ell = 2424833 \mid F_9 = 2^{512} + 1. \quad (\text{EK}_6)$$

*Thus the exact field degree, elliptic transition, quarter-turn Frobenius relation, and complete four-function Kummer orbit admit structural failure points. This is not a counterexample to  $\delta_9 = F_9$ : the constructed point is not asserted to be the one selected by the full Conway ancestry. It proves that an ancestry-insensitive CM, isogeny, generalized-Jacobian, or reciprocity argument cannot force the required nonresidue.*



*Proof.* The divisor vectors of the four functions are the rows of  $2I + C$ , where  $C$  is a four-cycle. Their determinant is 15, so they are independent modulo  $\ell$ . Kummer theory over the geometric function field therefore makes  $(\text{EK}_1)$  connected of degree  $\ell^4$ .

The union of the supports of the four divisors is exactly  $D$ . At each of its five points the valuation vector is nonzero modulo  $\ell$ , so inertia has order  $\ell$  and there are  $\ell^3$  geometric points above it. Since the base curve has genus one, Riemann–Hurwitz gives

$$2g(Y_\ell) - 2 = 5\ell^4 \left(1 - \frac{1}{\ell}\right),$$

which is  $(\text{EK}_2)$ . The identity

$$f_j(\mathcal{AP}) = f_{j-1}(P)$$

proves that  $(\text{EK}_3)$  preserves the four Kummer equations and has order four.

Twist  $Y_\ell$  over  $\mathbb{F}_Q$  by the cocycle sending arithmetic  $Q$ -Frobenius to  $\tilde{\mathcal{A}}$ . Its  $\mathbb{F}_Q$ -points correspond to geometric points  $y$  satisfying

$$\text{Frob}_Q(y) = \tilde{\mathcal{A}}(y).$$

Hasse–Weil and  $(\text{EK}_4)$  give more such points than the at most  $5\ell^3$  points above the branch boundary. Choose a nonbranch point. Four Frobenius iterations give  $\text{Frob}_Q^4(y) = y$ , so all its coordinates lie in  $\mathbb{F}_{Q^4}$ . Projecting to  $\mathcal{E}$  and reading the four  $z_j$  proves  $(\text{EK}_5)$ .

Finally, 2424833 is a prime divisor of  $F_9$  and is smaller than  $2^{22}$ . Its primality and the divisibility  $2424833 \mid 2^{512} + 1$  are independently kernel-checked in the companion Lean file. Hence

$$g(Y_\ell) < 3\ell^4 < 2^{90}, \quad 2g(Y_\ell)\sqrt{Q} < 2^{219}, \quad 5\ell^3 < 2^{69},$$

whereas  $Q = 2^{256}$ . This proves  $(\text{EK}_4)$  and  $(\text{EK}_6)$ . □

When  $F_n$  is prime, irreducibility and  $\delta_n > 1$  force  $\delta_n = F_n$ , recovering Lenstra’s Fermat-prime rule. When  $F_n$  is composite, irreducibility forces only one nontrivial divisor: it does not force every prime-power factor. The condition  $\delta_n = F_n$  is exactly the maximal-coset-order hypothesis studied for Conway’s quadratic tower by Popovych [33]. It is strictly weaker than primitivity of  $c_n$ , which already fails at  $n = 2$  since  $\text{ord}(c_2) = 5 \cdot 17 = 85 < 255$ ; Popovych uses the all-level hypothesis instead to prove that products such as  $c_0 c_n$  are primitive. Wiedemann’s related iterated quadratic tower poses the analogous full-Fermat-order problem [41], but Proposition 7.3 proves that its selected norm-one element is not Frobenius-identifiable with the Conway element after the first levels. Thus the apparently easy “singleton even implies zero” clause already contains a longstanding explicit maximal-coset-order problem. Popovych verified the Conway condition only through  $n = 11$ ; the recent analysis of Cagliero–Herman–Szechtman still retains the quotient order as an undetermined divisor [6]. Their Euler–Lucas argument does give the unconditional congruence

$$\delta_n \equiv 1 \pmod{2^{n+2}}, \quad \delta_n \geq 1 + 2^{n+2} \quad (n \geq 2),$$

but this remains far below  $F_n$  and does not force any prescribed prime factor of a composite  $F_n$ . In fact this congruence cannot distinguish the selected order from any proper candidate. For every prime  $\ell \mid F_n$ , one has  $\text{ord}_\ell(2) = 2^{n+1}$ . Since  $\ell \equiv 1 \pmod{8}$ , the element two is a square modulo  $\ell$ , and a square root of two has order  $2^{n+2}$ ; hence  $\ell \equiv 1 \pmod{2^{n+2}}$ . It follows that every nontrivial divisor  $d \mid F_n$  already satisfies

$$d \equiv 1 \pmod{2^{n+2}}, \quad \text{ord}_d(2) = 2^{n+1}.$$

**Exact selected-factor screen.** The companion script `experiments/fermat_selected_screen.py` reconstructs the selected polynomial without a precomputed Conway table: starting from  $A_{-1}(Y) = Y + 1$ , it iterates

$$A_i(X) = \text{Res}_Y(A_{i-1}(Y), X^2 + XY + Y^3)$$

and evaluates  $S_{F_n/\ell}(X) \bmod A_{n-1}(X)$  by exact binary-polynomial fast doubling. The companion Lean file kernel-checks the cubic Horner reduction and the multiplication-matrix determinant giving each resultant step; the complete bit-packed iteration remains script-level. For a published prime divisor  $\ell \mid F_n$ , the criterion above says that a zero remainder is a genuine failure of the  $\ell$ -coordinate, whereas a nonzero remainder certifies only that published coordinate. The current FermatSearch tables give the following exact screen [34]:

| $n$ | $\deg A_{n-1}$ | $\text{wt}(A_{n-1})$ | published primes | zero residues | residual cofactor |
|-----|----------------|----------------------|------------------|---------------|-------------------|
| 12  | 4096           | 2093                 | 6                | 0             | $C_{1133}$        |
| 13  | 8192           | 4119                 | 4                | 0             | $C_{2391}$        |
| 14  | 16384          | 8179                 | 1                | 0             | $C_{4880}$        |
| 15  | 32768          | 16369                | 3                | 0             | $C_{9808}$        |
| 16  | 65536          | 32809                | 2                | 0             | $C_{19694}$       |
| 17  | 131072         | 65521                | 2                | 0             | $C_{39395}$       |
| 18  | 262144         | 131087               | 2                | 0             | $C_{78884}$       |

Here every displayed database factor is a proved prime occurring to the first power, while the displayed  $C_d$  are exact cofactors proved composite but not factored. The script checks exact divisibility but imports primality from that source; its degrees and hashes are reproducibility fingerprints rather than standalone formal certificates. The optional FLINT backend agrees bit-for-bit with the dependency-free reducer at the orientation gate and records the complete per-factor fingerprints in `experiments/fermat_selected_screen_results.md`. Thus no selected counterexample occurs at any currently published prime from  $F_{12}$  through  $F_{18}$ , but none of these levels is proved maximal and the all-level problem remains open.

There is also an exact bridge to the current primitive-quadratic classification. Assume  $n \geq 2$  and the lower Fermat levels are maximal, so that  $a = a_{n-1}$  has order  $(q-1)/3$  in  $\mathbb{F}_q^\times$ , where  $q = 2^{2^n}$ . Since  $v_3(q-1) = 1$ , the element  $c_0^2 a$  is primitive, while  $x = c_0 c_n$  has irreducible polynomial

$$X^2 + c_0 X + c_0^2 a. \tag{VQ}$$

Vega's theorem [38] makes every irreducible quadratic with nonzero trace coefficient and primitive constant term primitive when  $q+1$  is prime, a power of two, or twice a prime. His converse classification [39] proves that these are exactly the universal field sizes. Here  $q+1 = F_n$  is odd, so the theorem closes precisely the already-easy Fermat-prime levels; every composite Fermat level lies outside the universal classification. Vega's general coefficient recurrence specializes in (VQ) to  $h_i = c_0^{i-1} S_i(a)$ , and therefore returns the same selected first zero rather than bounding it.

The newer first-zero theorem of Zhu–Wu [44, Theorem 4.1] identifies the binomial and multiplicative orders of an irreducible quadratic from its first vanishing Lucas coefficient. Applied here, it recovers exactly the reverse-Dickson/Lucas criterion already isolated above; it does not prove that the first zero is  $F_n$ . Their later primitive-coefficient criterion assumes both that the constant term is primitive and that  $q+1 = 2^m \pi$  with  $m > 0$  and  $\pi$  an odd prime. Although the rescaling (VQ) supplies the primitive constant,  $q+1 = F_n$  is odd and may be composite. Thus that theorem does not cover the selected composite-Fermat polynomial.

Thus the Euler–Lucas congruence is automatic from  $1 < \delta_n \mid F_n$ . Proposition 7.55 further shows that the missing  $\ell$ -primary class is born at one edge and merely propagated by  $3/2$  through every future edge, while Proposition 7.56 proves that even the complete compatible root tower is one torsor of parallel rays. Proposition 7.63 proves that all unmixed classical cyclotomic principalizers have zero selector weight and that the natural mixed principal element has two-adic support exactly when the target is already true. Proposition 7.70 computes the first local ramified coefficient as an ancestry-sensitive Witt polynomial at every étale contracted prime, but local nontriviality does not evaluate the selected prime above two. Proposition 7.72 identifies the remaining obligation exactly as  $\chi_W([q_1]) \neq 1$  in one mixed ray class group. Corollary 7.73 equivalently asks that two not be a global norm from the associated cyclic Kummer extension and proves that every abelian descent of its lower dihedral closure kills the selected translation. No all-level proof or counterexample is presently known.

## 7.5 Comparison with the cubic arm

These two embedded arms are instances of the same projective statement. The quadratic arm asks that the class of  $c_n$  generate  $E_n^\times/E_{n-1}^\times$ . In the cubic arm the natural map

$$U_k \longrightarrow \mathbb{F}_{q_k}^\times / \mathbb{F}_{q_k}^\times$$

is an isomorphism, and the increment from  $C_{k-1}$  to  $C_k$  asks that the class of  $\gamma_k$  generate this quotient. Equivalently, the projective class of the multiplication-by- $\gamma_k$  matrix generates a Singer cycle in  $\mathrm{PGL}_3(q_k)$ . The 0/1/4 rule therefore contains a quadratic and a cubic maximal-projective-order conjecture among its four exact arms, rather than two unrelated order coincidences.

## 7.6 The singleton-even conjecture ( $Z$ )

The singleton-even part of the zero arm is exactly the assertion  $\delta_n = F_n$  for all  $n$ . The results above identify the recursively selected fibotomic factor and exhaust the trace, norm, elliptic, and generalized-Jacobian data currently available without evaluating its Kummer class. Thus the remaining theorem is not another coarse invariant: it is the nonvanishing of the explicit distinguished Miller-unit value in Proposition 7.82. Equivalently, in the nontrivial Kummer case, failure is the assertion that two is a norm from the explicit degree- $\ell$  reflection field

$$M'(\theta + \theta^{-1})/M', \quad D_\ell^{\mathrm{cl}}(\theta + \theta^{-1}) = W_n + W_n^{-1}.$$

This non-Galois norm form is the sharpest lower-field target retained by the full dihedral ancestry; its abelian quotients erase the marked Frobenius. However, the explicit symmetrization is valid over every completion and proves the Hasse norm principle for this reflection field. The norm-two equation fails locally exactly when the named prime above two has nontrivial translation Frobenius, and has no additional global class-group obstruction. Thus even this nonabelian descent is an exact reformulation of the selected birth symbol, not an independent route to its nonvanishing.

# 8 The exceptional $2 \cdot 3^k$ arm

## 8.1 Why component data do not decide the excess

$\mathcal{Q}(f(p))$  alone does not determine  $m_p$ :

| $\mathcal{Q}$ | $m = 4$             | $m = 1$            |
|---------------|---------------------|--------------------|
| $\{9\}$       | $p = 19, f = 18$    | $p = 73, f = 9$    |
| $\{81\}$      | $p = 163, f = 162$  | $p = 2593, f = 81$ |
| $\{243\}$     | $p = 1459, f = 486$ | $p = 487, f = 243$ |

The split for  $\mathcal{Q} = \{9\}$  is explained by the order criterion:  $\text{ord}(\kappa_9 + 1) = 3^3 \cdot (2^9 - 1) = 27 \cdot 511 = 13797$  and  $511 = 7 \cdot 73$ , so  $73 \mid \text{ord}(\kappa_9 + 1)$  but  $19 \nmid \text{ord}(\kappa_9 + 1)$ ; adding  $m = 4$  changes the order and introduces the factor 19.

The lower bound of the exception column is unconditional:  $m_p \geq 4$  for all  $p$  with  $f(p) = 2 \cdot 3^k$  (Proposition 6.5).

*Conjecture.* The matching upper bound  $m_p \leq 4$  on that column; its exact current-level formulation is  $D'_k$  in Conjecture 8.4, not the Gaussian-period conjecture  $C_k$ .

## 8.2 The corrected norm and current-factor target

The  $m = 4$  translate  $\kappa_{3^k} + 4$  involves the number  $4 \in \mathbb{F}_{16}$ , which does not lie in  $\mathbb{F}_2(\zeta)$  (as  $4 \nmid 2 \cdot 3^k$ ); the translate therefore lies in the degree-4  $\cdot 3^k$  compositum  $\mathbb{F}_{2^{4h}}$ , where  $h = 3^k$ .

**Proposition 8.1** (the corrected norm). *Let  $\sigma = \text{Frob}^{2h}$  generate  $\text{Gal}(\mathbb{F}_{2^{4h}}/\mathbb{F}_{2^{2h}})$ . Then  $\sigma(4) = 5$ , and with  $\omega := \zeta^h = \kappa_2$ ,*

$$N := N_{\mathbb{F}_{2^{4h}}/\mathbb{F}_{2^{2h}}}(\kappa_{3^k} + 4) = (\kappa + 4)(\kappa + 5) = \kappa^2 + \kappa + \omega.$$

*Proof.*  $\mathbb{F}_{16} \cap \mathbb{F}_{2^{2h}} = \mathbb{F}_4$  because  $\gcd(4, 2 \cdot 3^k) = 2$ , so  $\sigma|_{\mathbb{F}_{16}}$  is the nontrivial element of  $\text{Gal}(\mathbb{F}_{16}/\mathbb{F}_4)$ , which swaps the two roots 4, 5 of the Artin–Schreier minimal polynomial  $y^2 + y + \omega$  of 4 over  $\mathbb{F}_4$ . (Equivalently:  $2h \equiv 2 \pmod{4}$  and the Frobenius orbit of 4 is  $4 \rightarrow 6 \rightarrow 5 \rightarrow 7$ , so  $\sigma(4) = 4^{2^2} = 5$ .) Then  $(\kappa + 4)(\kappa + 5) = \kappa^2 + \kappa + (4 \otimes 5)$  and  $4 \otimes 5 = 4^2 + 4 = \omega$  by the same Artin–Schreier relation.  $\square$

The corrected norm collapses the  $m = 4$  test into the *same* sparse trinomial model  $\mathbb{F}_2[x]/(x^{2h} + x^h + 1)$  as the rest of the 3-power analysis: no compositum arithmetic is needed.

**Corollary 8.2** (in-field criterion). *For a prime  $p$  with  $f(p) = 2 \cdot 3^k$  and  $v_p(2^{2h} - 1) = 1$ , write  $\overline{N} = N^{2^h}$  and  $M_k := \overline{N}/N = N^{2^h-1} \in U$ . Then*

$$m_p = 4 \iff p \mid \text{ord}(M_k) \iff M_k^{(2^h+1)/p} \neq 1.$$

*Proof.*  $m_p \geq 4$  is Proposition 6.5, so  $m_p = 4$  iff  $\kappa + 4$  has no  $p$ -th root. By Propositions 3.1 and 3.4 with  $E = 4h$ ,  $f = 2h$  (and  $v_p(2^{4h} - 1) = v_p(2^{2h} - 1) = 1$  by lifting the exponent, since  $p \nmid 2$ ), that holds iff  $N^{(2^{2h}-1)/p} \neq 1$ . Since  $\mathbb{F}_{2^{2h}}^\times = L^\times \times U$  with coprime orders,  $p \mid 2^h + 1$ , and  $(2^{2h} - 1)/p = (2^h - 1) \cdot ((2^h + 1)/p)$ , the test equals  $M_k^{(2^h+1)/p} \neq 1$ .  $\square$

**Proposition 8.3** (exact current-level form of the exception conjecture). *Let  $k \geq 1$ . Retain the notation above, put  $q = 2^h$ ,  $n = 3^{k+1} = 3h$ , and write*

$$\zeta := \kappa_{3^k} \in K := \mathbb{F}_{2^{2h}}, \quad \tilde{K} := \mathbb{F}_{2^{4h}}, \quad \beta := \zeta + 4.$$

*Thus  $\zeta$  is a primitive  $n$ -th root of unity. Define*

$$N := N_{\tilde{K}/K}(\beta) = \zeta^2 + \zeta + \zeta^h, \quad M_k := N^{q-1}, \quad \Psi_k := \frac{\Phi_{2 \cdot 3^k}(2)}{3},$$

and

$$R_k(Y) := \text{Res}_X \left( X^{2h} + X^h + 1, Y + X^h + X^2 + X \right) \in \mathbb{F}_2[Y].$$

Then  $R_k$  is the irreducible minimal polynomial of  $N$  over  $\mathbb{F}_2$  and has degree  $2h$ . The prime divisors of  $\Psi_k$  are exactly the primes  $\ell \neq 3$  with  $\text{ord}_\ell(2) = 2 \cdot 3^k$ , with full valuations

$$v_\ell(\Psi_k) = v_\ell(2^{2h} - 1) = v_\ell(2^h + 1).$$

Moreover the following are equivalent:

- (a)  $m_\ell = 4$  for every prime  $\ell$  with  $f(\ell) = 2 \cdot 3^k$ ;
- (b)  $\Psi_k \mid \text{ord}(M_k)$ ;
- (c)  $N \notin (K^\times)^\ell$  for every prime  $\ell \mid \Psi_k$ ;
- (d)  $R_k(X^\ell)$  is irreducible over  $\mathbb{F}_2$  for every prime  $\ell \mid \Psi_k$ .

*Proof.* The order identity  $\text{ord}_{3^{k+1}}(2) = 2h$  makes  $X^{2h} + X^h + 1$  the irreducible minimal polynomial of  $\zeta$ . We first prove that  $N = \zeta^h + \zeta^2 + \zeta$  also has degree  $2h$ . Suppose  $N^{2^j} = N$ , and put  $a \equiv 2^j \pmod{3h}$ . The resulting vanishing polynomial, of degree less than  $3h$ , has support equal to the symmetric difference of

$$\{ah, a, 2a\} \quad \text{and} \quad \{h, 1, 2\}. \quad (1)$$

Every multiple of  $X^{2h} + X^h + 1$  of degree less than  $3h$  has support a disjoint union of triples  $\{r, r+h, r+2h\}$ . If  $a \equiv 2 \pmod{3}$ , the support in (1) contains  $h$  and  $2h$  but not 0, a contradiction. If  $a \equiv 1 \pmod{3}$ , the  $h$ -terms cancel and each residue class modulo  $h$  occurs at most twice; a nonempty union of such triples is again impossible. Hence the support is empty, so  $\{a, 2a\} = \{1, 2\}$  modulo  $3h$ . Since  $h \geq 3$ , this forces  $a = 1$ . Thus the Frobenius orbit of  $N$  has length  $2h$ , and the displayed resultant is its irreducible minimal polynomial.

Writing  $A = 2^{3^{k-1}}$  gives  $\Phi_{2 \cdot 3^k}(2) = A^2 - A + 1 \equiv 3 \pmod{9}$ , so division by 3 removes exactly the prime dividing the cyclotomic index. The standard cyclotomic-divisor criterion now says that the prime divisors of  $\Psi_k$  are precisely the  $\ell$  with  $\text{ord}_\ell(2) = 2h$ . Their entire primary contribution to  $2^{2h} - 1$  lies in this cyclotomic factor and, since  $\ell \nmid 2^h - 1$ , in  $2^h + 1$ . This proves the valuation statement.

Fix  $\ell \mid \Psi_k$ . Propositions 6.4 and 6.5 make the translates  $\zeta, \zeta + 1, \zeta + 2, \zeta + 3$  into  $\ell$ -th powers, so  $m_\ell \geq 4$ . The corrected norm and the exact power criterion give

$$\beta^{(2^{4h}-1)/\ell} = N^{(2^{2h}-1)/\ell},$$

and therefore

$$m_\ell = 4 \iff N \notin (K^\times)^\ell. \quad (2)$$

This uses the full  $\ell$ -primary valuation and has no squarefreeness or non-Wieferich hypothesis. Since  $q - 1 \equiv -2 \pmod{\ell}$ , raising to  $q - 1$  is invertible on  $K^\times / (K^\times)^\ell$ , so

$$N \in (K^\times)^\ell \iff M_k \in (K^\times)^\ell. \quad (3)$$

The entire  $\ell$ -primary part of  $K^\times$  lies in its norm-one factor of order  $q + 1$ . In a cyclic group, nonmembership in the subgroup of  $\ell$ -th powers is equivalent to the order containing the full  $\ell$ -primary part. Equations (2)–(3) prove the equivalence of the first three assertions.

Finally  $\mu_\ell \subset K$ . Kummer theory says that  $X^\ell - N$  is irreducible over  $K$  exactly when  $N \notin (K^\times)^\ell$ . Since  $K = \mathbb{F}_2(N)$ , Capelli's lemma says in turn that this is equivalent to irreducibility of  $R_k(X^\ell)$  over  $\mathbb{F}_2$ , completing the proof.  $\square$

**Conjecture 8.4** ( $D'_k$ : the exact current exception factor). *Conjecture. One has*

$$D'_k : \quad \Psi_k \mid \text{ord}(M_k).$$

By Proposition 8.3, this is exactly the  $m = 4$  upper bound for the primes at level  $f = 2 \cdot 3^k$ .

### 8.3 Projection, fibotomic, and translated-binomial forms

**Proposition 8.5** (exact current-layer projection and dominant-factor sieve). *Let  $k \geq 2$ , put*

$$s = 3^{k-1}, \quad h = 3s, \quad a = 2^s, \quad q = a^3 = 2^h, \quad \Lambda_k = a^2 - a + 1 = 3\Psi_k,$$

*and retain  $N, M_k, \zeta, \omega = \zeta^h$  from Proposition 8.3. Define*

$$P_k := M_k^{a+1}.$$

*Then  $P_k^{\Lambda_k} = 1$ , and for every prime  $\ell \mid \Psi_k$ ,*

$$v_\ell(\text{ord } P_k) = v_\ell(\text{ord } M_k). \quad (\text{D}_1)$$

*Moreover*

$$P_k = \frac{\zeta^4 + \zeta + \omega}{\omega^2 \zeta^4 + \zeta^3 + 1}. \quad (\text{D}_2)$$

$$[\mathbb{F}_2(P_k) : \mathbb{F}_2] = 2h. \quad (\text{D}_3)$$

*and*

$$v_3(\text{ord } P_k) = \begin{cases} 1, & k \text{ odd}, \\ 0, & k \text{ even}. \end{cases} \quad (\text{D}_4)$$

*Set  $c_k = 3$  for odd  $k$  and  $c_k = 1$  for even  $k$ . Then*

$$\text{ord}(P_k) \geq c_k(2h + 1). \quad (\text{D}_5)$$

*Consequently, if  $\ell \mid \Psi_k$  and*

$$\ell > \frac{\Lambda_k}{c_k(2h + 1)}, \quad (\text{D}_6)$$

*then  $P_k$ , and hence  $M_k$ , contains the full  $\ell$ -primary part of  $\Lambda_k$  in its order. In particular  $D'_k$  follows whenever  $\Psi_k$  is prime; this proves  $D'_2$  and  $D'_3$  analytically, since  $\Psi_2 = 19$  and  $\Psi_3 = 87211$ .*

*Proof.* The factorization

$$q + 1 = (a + 1)(a^2 - a + 1) = (a + 1)\Lambda_k$$

and  $M_k^{q+1} = 1$  give  $P_k^{\Lambda_k} = 1$ . Also  $\gcd(\Lambda_k, a + 1) = 3$ . Since  $k \geq 2$ , one has  $a \equiv -1 \pmod{9}$ , so  $\Lambda_k \equiv 3 \pmod{9}$  and  $\gcd(\Psi_k, a + 1) = 1$ . Exponentiation by  $a + 1$  therefore preserves all the primary valuations belonging to  $\Psi_k$ , proving (D<sub>1</sub>).

The lifted congruence

$$2^{3^{k-1}} \equiv 3^k - 1 \pmod{3^{k+1}}$$

gives, modulo  $3h$ ,

$$a \equiv h - 1, \quad a^2 \equiv h + 1, \quad a^3 \equiv -1, \quad a^4 \equiv 2h + 1. \quad (\text{D}_7)$$

As  $M_k = N^{a^3-1}$ ,

$$P_k = \frac{N^{a^4} N^{a^3}}{N^a N}.$$

Using  $N = \zeta^2 + \zeta + \omega$  and (D<sub>7</sub>), direct Frobenius expansion gives

$$\begin{aligned} N^a &= \omega^2 \zeta^{-2} + \omega \zeta^{-1} + \omega^2, \\ N^{a^3} &= \zeta^{-2} + \zeta^{-1} + \omega^2, \\ N^{a^4} &= \omega \zeta^2 + \omega^2 \zeta + \omega. \end{aligned}$$

Multiplying  $N^{a^3}$  and  $N^a$  by the same cancelling factor  $\zeta^2$ , and using  $1 + \omega + \omega^2 = 0$ , yields

$$\begin{aligned} N^{a^4}(\zeta^2 N^{a^3}) &= \zeta^4 + \zeta + \omega, \\ (\zeta^2 N^a)N &= \omega^2 \zeta^4 + \zeta^3 + 1, \end{aligned}$$

which proves (D<sub>2</sub>).

The minimal polynomial of  $\zeta$  over  $\mathbb{F}_2$  is  $X^{2h} + X^h + 1$ . Write the numerator and denominator in (D<sub>2</sub>) as  $C, D$ . Since  $P_k^q = P_k^{-1}$ , membership in  $\mathbb{F}_q$  would force  $P_k = 1$ . But  $C = D$ , after reducing  $\zeta^{2h} = \zeta^h + 1$ , would give

$$\zeta^{h+4} + \zeta^h + \zeta^3 + \zeta + 1 = 0, \quad (\text{D}_8)$$

a nonzero polynomial of degree  $h + 4 < 2h$ .

If  $P_k \in \mathbb{F}_{a^2}$ , its order divides  $\gcd(\Lambda_k, a^2 - 1) = 3$ , so  $P_k \in \{1, \omega, \omega^2\}$ . The first case was excluded; the other two respectively give

$$\zeta^{h+3} + \zeta = 0$$

and

$$\zeta^{h+4} + \zeta^{h+3} + \zeta^4 + \zeta^3 + \zeta + 1 = 0.$$

Both are again nonzero polynomials of degree below  $2h$ . Every proper divisor of  $2h = 2 \cdot 3^k$  divides either  $h$  or  $2h/3$ ; hence every proper subfield of  $\mathbb{F}_{2^{2h}}$  lies in either  $\mathbb{F}_q$  or  $\mathbb{F}_{a^2}$ . This proves (D<sub>3</sub>).

For the 3-part, observe that

$$P_k^{\Lambda_k/3} = M_k^{(q+1)/3} = N^{(q^2-1)/3} = N_{\mathbb{F}_{2^{2h}}/\mathbb{F}_4}(N). \quad (\text{D}_9)$$

Choose  $u \in \mathbb{F}_{16}$  with  $u^2 + u = \omega$ . Then  $u^4 = u + 1$ ,  $u^5 = \omega$ , and  $u$  has order 15. The extensions  $\mathbb{F}_{2^{2h}}/\mathbb{F}_4$  and  $\mathbb{F}_{16}/\mathbb{F}_4$  are linearly disjoint, and over  $\mathbb{F}_{16}$  the minimal polynomial of  $\zeta$  is  $X^h + \omega$ , irreducible because  $\text{ord}_{3h}(16) = h$ . Since

$$N = (\zeta + u)(\zeta + u + 1),$$

base change in (D<sub>9</sub>) gives

$$N_{\mathbb{F}_{2^{2h}}/\mathbb{F}_4}(N) = (u^h + \omega)((u + 1)^h + \omega).$$

The residues of  $h = 3^k$  modulo 15 are 3, 9, 12, 6 as  $k \equiv 1, 2, 3, 0 \pmod{4}$ ; reduction with  $u^4 = u + 1$  gives

$$P_k^{\Lambda_k/3} = \begin{cases} \omega^2, & k \text{ odd}, \\ 1, & k \text{ even}. \end{cases}$$

Because  $v_3(\Lambda_k) = 1$ , this proves (D<sub>4</sub>).

Let  $e = \text{ord}(P_k)$ . From (D<sub>3</sub>),  $\text{ord}_e(2) = 2h$ . Since  $e \mid \Lambda_k$  and  $e \notin \{1, 3\}$ , some prime  $r \mid \Psi_k$  divides  $e$ ; every such  $r$  has  $\text{ord}_r(2) = 2h$ , so  $r \equiv 1 \pmod{2h}$  and  $r \geq 2h + 1$ . When  $k$  is odd, (D<sub>4</sub>) also gives  $3r \mid e$ , which proves (D<sub>5</sub>).

If the full  $\ell$ -primary part of  $\Lambda_k$  were missing from  $e$ , then  $e \mid \Lambda_k/\ell$ . This contradicts (D<sub>5</sub>) under (D<sub>6</sub>). Equation (D<sub>1</sub>) transfers the conclusion back to  $M_k$ , and the last assertions follow.  $\square$

**Proposition 8.6** (fibotomic form of the current power-residue test). *Retain Proposition 8.5, put*

$$U = \{x \in \mathbb{F}_{q^2}^\times : x^{q+1} = 1\}, \quad \pi(x) = \frac{x}{(x+1)^2} = \frac{1}{x+x^{-1}},$$

and set  $Z = \pi(M_k)$ . The map  $\pi$  is two-to-one from  $U \setminus \{1\}$  onto

$$\mathcal{T}_q = \{z \in \mathbb{F}_q : \text{Tr}_{\mathbb{F}_q/\mathbb{F}_2}(z) = 1\},$$

with fibres  $\{x, x^{-1}\}$ .

For an odd prime  $\ell$ , define

$$H_\ell(W) = \sum_{j=0}^{(\ell-1)/2} \binom{\ell-j-1}{j} W^j \in \mathbb{F}_2[W].$$

If  $\ell \mid \Psi_k$ , then

$$M_k \in U^\ell \iff \text{there is } W \in \mathcal{T}_q \text{ with } W^\ell + ZH_\ell(W)^2 = 0. \quad (\text{F})$$

The polynomial

$$\mathcal{F}_{\ell,Z}(W) = W^\ell + ZH_\ell(W)^2$$

either has no root in  $\mathbb{F}_q$ , or splits there into  $\ell$  distinct linear factors. Finally  $H_\ell$  is squarefree and is a product of exactly  $(\ell-1)/(2h)$  irreducible polynomials over  $\mathbb{F}_2$ , each of degree  $h$ . Thus  $D'_k$  is equivalently the assertion that  $\mathcal{F}_{\ell,Z}$  is rootless over  $\mathbb{F}_q$  for every prime  $\ell \mid \Psi_k$ .

*Proof.* For  $x \in U$ , one has  $x^q = x^{-1}$ , so  $\pi(x) \in \mathbb{F}_q$ . The equation  $\pi(x) = z$  is

$$X^2 + z^{-1}X + 1 = 0.$$

The characteristic-two quadratic criterion says that this polynomial is irreducible over  $\mathbb{F}_q$  exactly when  $\text{Tr}(z) = 1$ . Its roots are inverse and have norm one. Counting, or the same quadratic argument in reverse, proves the stated two-to-one description.

The defining polynomial identity is

$$\frac{X^\ell + 1}{(X+1)^\ell} = H_\ell\left(\frac{X}{(X+1)^2}\right). \quad (\text{H})$$

If  $M_k = T^\ell$  with  $T \in U$ , put  $W = \pi(T)$ . Identity (H) gives

$$Z = \pi(T^\ell) = \frac{W^\ell}{H_\ell(W)^2},$$

so  $W$  satisfies (F). Conversely, lift a trace-one solution  $W$  to  $T \in U$ . Then (H) gives  $\pi(T^\ell) = \pi(M_k)$ , hence  $T^\ell = M_k$  or  $M_k^{-1}$ . Replacing  $T$  by  $T^{-1}$  in the second case proves  $M_k \in U^\ell$ .

In fact every  $\mathbb{F}_q$ -root of  $\mathcal{F}_{\ell,Z}$  has trace one. Indeed,  $H_\ell(W) \neq 0$  at a root, since otherwise the equation would force  $W = 0$ , whereas  $H_\ell(0) = 1$ . Lift a nonzero root through  $\pi(T) = W$ . If  $\text{Tr}(W) = 0$ , the quadratic criterion puts  $T$  in  $\mathbb{F}_q$ . Identity (H) then gives  $\pi(T^\ell) = Z$ , but for  $y \in \mathbb{F}_q \setminus \{1\}$ ,

$$\pi(y) = \left(\frac{y}{y+1}\right)^2 + \frac{y}{y+1}$$

has absolute trace zero, whereas  $Z = \pi(M_k)$  has trace one. This is a contradiction.



If one root exists, therefore lift it to  $T \in U$  and orient  $T$  so that  $T^\ell = M_k$ . Then the  $\ell$  elements

$$\pi(T\xi), \quad \xi \in \mu_\ell,$$

are roots in  $\mathbb{F}_q$ . They are distinct: an inverse collision would give  $T^2 \in \mu_\ell$ , hence  $M_k^2 = 1$ , whereas  $M_k \neq 1$  because  $N \notin \mathbb{F}_q$ . Since  $\mathcal{F}_{\ell,Z}$  has degree  $\ell$ , it splits completely. Its derivative is  $W^{\ell-1}$ , and trace-one roots are nonzero, so they are simple.

Finally, the roots of  $H_\ell$  are

$$\pi(\xi), \quad \xi \in \mu_\ell \setminus \{1\},$$

with  $\xi$  and  $\xi^{-1}$  identified. Their Frobenius-orbit length is the least  $d > 0$  with  $2^d \equiv \pm 1 \pmod{\ell}$ . Since  $\text{ord}_\ell(2) = 2h$ , this least  $d$  is  $h$ . The number and degrees of the factors follow, and the roots are distinct, proving squarefreeness.  $\square$

**Proposition 8.7** (Conway  $C$ -to- $D$  selector bridge). *Retain Proposition 8.6; in particular  $k \geq 2$ . Put*

$$h = 3^k, \quad t = 2^{h/3}, \quad q = 2^h = t^3,$$

and write

$$\gamma = \gamma_k = \zeta + \zeta^{-1}, \quad g_k = \gamma^2 + \gamma + 1.$$

For the corrected exceptional norm  $N = \zeta^2 + \zeta + \omega$ , with  $\overline{N} = N^q$ , one has

$$\text{Tr}_{\mathbb{F}_{q^2}/\mathbb{F}_q}(N) = g_k, \quad \text{N}_{\mathbb{F}_{q^2}/\mathbb{F}_q}(N) = g_k^{t^2}. \quad (\text{CD}_1)$$

Consequently

$$N^2 + g_k N + g_k^{t^2} = 0. \quad (\text{CD}_2)$$

The auxiliary periods are norm coherent:

$$\text{N}_{\mathbb{F}_{2^{3k}}/\mathbb{F}_{2^{3k-1}}}(g_k) = g_{k-1}, \quad \text{Tr}_{\mathbb{F}_{2^{3k}}/\mathbb{F}_{2^{3k-1}}}(g_k) = 1. \quad (\text{CD}_3)$$

In particular

$$\mathbb{F}_2(g_k) = \mathbb{F}_2(\gamma_k) = \mathbb{F}_q, \quad \text{Tr}_{\mathbb{F}_q/\mathbb{F}_2}(g_k) = 1. \quad (\text{CD}_4)$$

The coefficient  $Z = \pi(M_k)$  in Proposition 8.6 is therefore the explicit ancestral power

$$\boxed{Z = g_k^{t^2-2}}. \quad (\text{CD}_5)$$

Moreover

$$\gcd(t^2 - 2, t^3 - 1) = 1, \quad (\text{CD}_6)$$

so  $Z$  and  $g_k$  have the same multiplicative order and generate the same field over  $\mathbb{F}_2$ . If

$$w = \frac{N}{g_k},$$

then

$$w^2 + w + Z = 0, \quad \overline{w} = w + 1, \quad M_k = \frac{w+1}{w} = w^{q-1}. \quad (\text{CD}_7)$$

For every current prime  $\ell \mid \Psi_k$ , failure at  $\ell$  is equivalent to each of

$$M_k \in U^\ell, \quad w \in (\mathbb{F}_{q^2}^\times)^\ell, \quad \mathcal{F}_{\ell, g_k^{t^2-2}}(W) \text{ having a root in } \mathbb{F}_q. \quad (\text{CD}_8)$$

The two roots  $w, w+1 = w^q$  have the same  $\ell$ -power status, while their Kummer classes are opposite. Thus the remaining datum is the unordered opposite pair  $\{[w], -[w]\}$ , equivalently the norm-one phase  $\{M_k, M_k^{-1}\}$ ; it is not a choice of orientation between the two roots.

*Proof.* The  $q$ -Frobenius sends  $\zeta$  to  $\zeta^{-1}$  and  $\omega$  to  $\omega^2$ . Hence

$$N + \overline{N} = \zeta^2 + \zeta^{-2} + \zeta + \zeta^{-1} + \omega + \omega^2 = \gamma^2 + \gamma + 1 = g_k.$$

The relation  $\zeta^t = \omega\zeta^{-1}$  from Proposition 6.24, together with direct expansion, gives

$$N\overline{N} = 1 + \gamma + \gamma^t + (\gamma^t + \gamma)^2.$$

The three conjugates of  $\gamma_k$  over the preceding cubic field sum to zero, so  $\gamma + \gamma^t + \gamma^{t^2} = 0$ . Therefore

$$N\overline{N} = 1 + \gamma^{t^2} + \gamma^{2t^2} = g_k^{t^2},$$

which proves (CD<sub>1</sub>)–(CD<sub>2</sub>).

Let  $x_1, x_2, x_3$  be the roots of  $P(X) = X^3 + X + \gamma_{k-1}$ . Since  $X^2 + X + 1 = (X + \omega)(X + \omega^2)$ ,

$$\begin{aligned} \prod_{i=1}^3 (x_i^2 + x_i + 1) &= P(\omega)P(\omega^2) \\ &= (\gamma_{k-1} + \omega^2)(\gamma_{k-1} + \omega) = g_{k-1}. \end{aligned}$$

This is the norm identity in (CD<sub>3</sub>). The relative trace is

$$\sum_i (x_i^2 + x_i + 1) = \left( \sum_i x_i \right)^2 + \sum_i x_i + 1 = 1.$$

The element  $\gamma$  is at most quadratic over  $\mathbb{F}_2(g_k)$ , while  $[\mathbb{F}_2(\gamma) : \mathbb{F}_2] = h$  is odd. Hence that possible quadratic index is one, proving the field equality in (CD<sub>4</sub>). Transitivity applied to the relative trace one proves the absolute trace assertion.

For  $M_k = \overline{N}/N$ , elementary cancellation gives

$$\pi(M_k) = \frac{\overline{N}/N}{(\overline{N}/N + 1)^2} = \frac{N\overline{N}}{(N + \overline{N})^2} = g_k^{t^2-2},$$

proving (CD<sub>5</sub>). If an integer divides both  $t^2 - 2$  and  $t^3 - 1$ , then it divides  $2t - 1$ , and hence divides seven. Since  $k \geq 2$ , one has  $t \equiv 1 \pmod{7}$ , and therefore seven does not divide  $t^2 - 2$ . This proves (CD<sub>6</sub>), and exponentiation by  $t^2 - 2$  is therefore an automorphism of the cyclic group  $\mathbb{F}_q^\times$ .

Dividing (CD<sub>2</sub>) by  $g_k^2$  gives the first identity in (CD<sub>7</sub>); the remaining two follow from relative trace one. For a current prime,  $\ell \mid q + 1$  and  $\ell \nmid q - 1$ . The direct product  $\mathbb{F}_{q^2}^\times = \mathbb{F}_q^\times \times U$  gives

$$U^\ell = U \cap (\mathbb{F}_{q^2}^\times)^\ell,$$

because  $\ell$ -powering is an automorphism of  $\mathbb{F}_q^\times$ . Multiplication by  $q - 1$  is consequently invertible on the one-dimensional  $\ell$ -power quotient of  $\mathbb{F}_{q^2}^\times$ , so

$$w^{q-1} \in U^\ell \iff w \in (\mathbb{F}_{q^2}^\times)^\ell.$$

Proposition 8.6 supplies the final equivalence in (CD<sub>8</sub>). Frobenius preserves  $\ell$ -th powers and sends  $w$  to  $w + 1$ , proving the last assertion.  $\square$

**Remark 8.8** (what the selector bridge does not transfer). The bridge transports the complete selected  $D$ -coefficient through the recursively selected  $C$ -period and the norm-coherent tower  $(g_k)$ , but it is not an implication  $C_k \Rightarrow D'_k$ . The current primes in the two arms satisfy respectively  $\text{ord}_\ell(2) = h$  and  $\text{ord}_\ell(2) = 2h$ , so they are disjoint. At a  $D$ -current prime every element of  $\mathbb{F}_q^\times$ , including  $g_k$  and  $Z$ , is automatically an  $\ell$ -th power. The missing class lives in the selected quadratic root  $w$ , not in its base-field trace or norm. There is no known uniform exponent, coprime to  $q - 1$ , which expresses  $g_k = \gamma_k^2 + \gamma_k + 1$  as a power of  $\gamma_k$ ; the map  $x \mapsto x^2 + x + 1$  need not preserve multiplicative order. Thus  $C$ -primitivity supplies no order transfer to  $g_k$ .

**Proposition 8.9** (actual-selector primitivity still does not transfer). *There is an exact finite-field model which retains a primitive degree-nine scalar, the literal rational selector of Proposition 8.7, and the trace-one Artin–Schreier coupling, but fails the current  $D$ -condition at 19. Namely, in*

$$E = \mathbb{F}_2[z]/(z^{18} + z^9 + 1) = \mathbb{F}_{2^{18}},$$

put

$$\gamma = (z + z^{-1})^{25}, \quad g = \gamma^2 + \gamma + 1, \quad Z = g^{62}.$$

Then  $\gamma \in \mathbb{F}_{2^9}$  has order 511. Nevertheless, with

$$M = z^{10}, \quad w = (z^{10} + 1)^{-1},$$

one has

$$Z = \frac{M}{(M+1)^2} = w^2 + w, \quad w^{512} = w + 1, \quad \frac{w+1}{w} = M, \quad \text{ord}(M) = 27, \quad (\text{ACD}_1)$$

and  $w$  is a nineteenth power in  $E^\times$ . Thus primitivity of  $\gamma$ , even after applying the exact selected rational map and retaining the correct trace-one quadratic root, does not force the new 19-primary coordinate.

This is not a counterexample to the selected implication  $C_2 \Rightarrow D'_2$ . Its lower norm is

$$N_{\mathbb{F}_{2^9}/\mathbb{F}_{2^3}}(g) = 0\mathbf{x}8048,$$

whereas the recursively selected lower coefficient is  $0\mathbf{x}9001$ . Complete cubic ancestry removes precisely the freedom used by the model: Frobenius conjugation preserves both the Artin–Schreier equation and nineteenth-power status. Hence any proof of the actual implication must use the selected norm ancestry, not merely primitivity, the rational formula, degree, trace, or the quadratic root.

The exact degree-nine census makes this boundary quantitative. Among the 432 primitive values of  $\gamma$ , 216 give a trace-one selector and exactly 9 of those fail at 19; the failures form one absolute Frobenius orbit. Imposing the selected lower norm leaves 60 primitive values and no failures. Imposing in addition the selected lower trace leaves 9 values and still no failures. This is finite evidence at  $k = 2$ , not a proof that norm ancestry forces success at every level.

*Proof.* In the polynomial basis  $1, z, \dots, z^{17}$ , exact binary arithmetic gives

$$\gamma = 0\mathbf{x}14831, \quad g = 0\mathbf{x}16d03, \quad Z = 0\mathbf{x}279cf, \quad w = 0\mathbf{x}1b7b6.$$

The checks

$$\gamma^{73} = 0\mathbf{x}9000 \neq 1, \quad \gamma^7 = 0\mathbf{x}154b \neq 1$$

prove  $\text{ord}(\gamma) = 511 = 7 \cdot 73$ . Direct multiplication gives  $(\text{ACD}_1)$ , and  $y = 0\text{x}203\text{ba}$  satisfies  $y^{19} = w$ . The same exact calculation gives  $g^{73} = 0\text{x}8048 \neq 0\text{x}9001$ . Exhausting all primitive exponents gives the count triples/pairs

$$(432, 216, 9), \quad (60, 0), \quad (9, 0),$$

for respectively primitive/trace-one/failing values, selected-norm fibre/failures, and selected-norm-and-trace fibre/failures. The failing exponents are

$$25, 50, 67, 100, 134, 200, 268, 289, 400,$$

one doubling orbit modulo 511. These identities are independently rerun by the maintained C/D experiment. Lean kernel-checks the generic fact that a ring map preserves simultaneously an Artin–Schreier equation and a chosen  $\ell$ -th root; the finite certificate and its selected-norm mismatch remain script-level.  $\square$

**Proposition 8.10** (selected trace and norm still do not transfer). *There is an exact  $k = 4$  finite-field model which retains  $C$ -primitivity, the literal rational selector of Proposition 8.7, and the selected lower trace and norm, but fails the current  $D$ -condition at  $\ell = 163$ .*

Put  $t = 2^{27}$ ,  $q = t^3 = 2^{81}$ , choose a primitive 243-rd root  $z \in \mathbb{F}_{q^2}$ , and define

$$\gamma_4 = z + z^{-1}, \quad a = \gamma_4^3 + \gamma_4 = \gamma_3, \quad C = a^2 + a + 1 = g_3.$$

There is an element  $b \in \mathbb{F}_t$  for which

$$K = \mathbb{F}_t[X]/(X^3 + X^2 + bX + C) \cong \mathbb{F}_q$$

is a field. Writing  $g = X \bmod (X^3 + X^2 + bX + C)$ , one can choose  $\gamma \in K$  such that

$$\gamma^2 + \gamma + 1 = g, \quad \text{ord}(\gamma) = q - 1. \quad (\text{STN}_1)$$

Moreover, with

$$Z = g^{t^2-2}, \quad L = K[w]/(w^2 + w + Z),$$

one has

$$\text{Tr}_{K/\mathbb{F}_t}(g) = 1, \quad \text{N}_{K/\mathbb{F}_t}(g) = C, \quad \text{Tr}_{K/\mathbb{F}_2}(Z) = 1, \quad w^{(q^2-1)/163} = 1. \quad (\text{STN}_2)$$

Thus the selected Artin–Schreier root  $w$  is a 163-rd power even though its  $C$ -antecedent is primitive and both selected lower coordinates in  $(\text{STN}_2)$  agree with the Conway tower.

The missing ancestry coordinate is exactly

$$e_2(g, g^t, g^{t^2}) = b, \quad b \neq a + 1. \quad (\text{STN}_3)$$

For the actual selected cubic, the three elementary symmetric coordinates are  $(1, a + 1, a^2 + a + 1)$ . Hence this is not a counterexample to  $C_4 \Rightarrow D'_4$ : it proves that trace and norm, even together with primitivity and the literal rational selector, do not suffice; the middle coefficient is the remaining full-ancestry datum.

*Proof.* The maintained exact verifier `experiments/cyclotomic_3k_family.py` uses the ambient polynomial basis  $\mathbb{F}_2[z]/(z^{162} + z^{81} + 1)$ . Its fixed coefficient is

$$b = 0\text{x}8240041049041000209009000249201249000241.$$

It checks  $g^{t^3} = g$ , while  $g^t \neq g$  and  $g^{t^2} \neq g$ , certifying irreducibility of the cubic; it then checks the three symmetric identities in  $(\text{STN}_2)$  and  $(\text{STN}_3)$ . The factorization

$$q - 1 = 7 \cdot 73 \cdot 2593 \cdot 71119 \cdot 262657 \cdot 97685839$$

and one nonidentity power for each prime factor certify  $\text{ord}(\gamma) = q - 1$ . Absolute trace one makes the quadratic for  $w$  irreducible, and the final displayed power certifies that  $w$  is a 163-rd power. The concatenated fixed witness has SHA-256 fingerprint

$$6d3366ea5cc7a3a0b42a1f0cb4a48adf19eef95892b3a21c32786e88cf7424b0.$$

Geometrically, after splitting the relative cubic, the trace-norm fibre is the plane cubic

$$xy(1 + x + y) = C.$$

For the selected  $C \neq 0, 1$  it is smooth of genus one. The function  $e_2$  is nonconstant on this fibre, and exact Conway ancestry cuts it by the additional equation  $e_2 = a + 1$ , leaving the selected Frobenius orbit. Consequently point counts or character sums on the whole genus-one fibre can prove generic intersections but cannot evaluate that fixed degree-three divisor.

Lean kernel-checks the symmetric identities forcing trace one, middle coefficient  $a + 1$ , and norm  $a^2 + a + 1$  from the selected cubic ancestry. The explicit finite-field witness and its primitivity and power tests remain script-level exact computation.  $\square$

**Proposition 8.11** (one selected cubic step still does not transfer). *Even the complete one-step cubic coefficient, together with primitivity on both sides, does not imply the exceptional D-condition without the full lower Conway ancestry.*

Put  $t = 2^{27}$ ,  $q = t^3 = 2^{81}$ , and use the same cyclotomic ambient field  $\mathbb{F}_2[z]/(z^{162} + z^{81} + 1)$  as in Proposition 8.10. There is a primitive element

$$a = 0x9209240041201241041008008040048008040040 \in \mathbb{F}_t$$

such that  $X^3 + X + a$  is irreducible over  $\mathbb{F}_t$ . In

$$K = \mathbb{F}_t[X]/(X^3 + X + a)$$

put  $\gamma = X$ . Then both  $a$  and  $\gamma$  are primitive in their respective multiplicative groups and

$$\boxed{\gamma^3 + \gamma = a.} \tag{OSC_1}$$

For  $g = \gamma^2 + \gamma + 1 = (1, 1, 1)$  in the basis  $(1, \gamma, \gamma^2)$ , its three relative conjugates satisfy the complete selected symmetric fingerprint

$$\text{Tr}_{K/\mathbb{F}_t}(g) = 1, \quad e_2(g, g^t, g^{t^2}) = a + 1, \quad N_{K/\mathbb{F}_t}(g) = a^2 + a + 1. \tag{OSC_2}$$

Nevertheless, with  $Z = g^{t^2-2}$  and

$$L = K[w]/(w^2 + w + Z),$$

one has

$$\text{Tr}_{K/\mathbb{F}_2}(Z) = 1, \quad w^{(q^2-1)/163} = 1. \tag{OSC_3}$$

Thus a primitive lower ancestor, a primitive selected cubic child, the literal rational D-selector, and all three exact one-step coefficients remain compatible with 163-failure.

This is not a counterexample to  $C_4 \Rightarrow D'_4$ . The element  $a$  is not the Conway-selected  $\gamma_3$ ; it does not continue the recursive ancestry below this step. Conversely, for a fixed actual lower  $a = \gamma_{k-1}$ , the three equations in  $(\text{OSC}_2)$  give the single cubic

$$T^3 + T^2 + (a+1)T + (a^2 + a + 1), \quad (\text{OSC}_4)$$

so the slice consists only of the selected Frobenius orbit. Any division-polynomial, Jacobi, or character evaluation on that zero-dimensional slice is therefore exactly the original  $D$ -evaluation.

*Proof.* The maintained verifier `experiments/cyclotomic_3k_family.py` checks the fixed witness. The factorizations

$$t - 1 = 7 \cdot 73 \cdot 262657,$$

$$q - 1 = 7 \cdot 73 \cdot 2593 \cdot 71119 \cdot 262657 \cdot 97685839$$

and the corresponding nonidentity powers certify primitivity of  $a$  and  $\gamma$ . The equalities  $\gamma^q = \gamma$  and  $\gamma^t \neq \gamma$  certify irreducibility of  $X^3 + X + a$ , while  $\text{Tr}_{K/\mathbb{F}_2}(Z) = 1$  makes the Artin–Schreier quadratic irreducible. It checks  $(\text{OSC}_1)$ – $(\text{OSC}_3)$  directly; the fixed witness has SHA–256 fingerprint

$$5c5b45506d7f0b94c9b58727bdc1d38112f9f3c8f9f61dc5000d17724360e032.$$

The generic symmetric calculations giving  $(\text{OSC}_2)$  are already kernel-checked in `Ugdoad.Excess`. Lean now also checks that  $x^2 + x + 1$  has only the pair  $x, x + 1$  in each fibre and that the three selected symmetric coordinates expand to  $(\text{OSC}_4)$ . The fixed finite-field primitivity and 163-power certificates remain exact script-level computation.  $\square$

**Proposition 8.12** (*C-tower-faithful exceptional countermodel*). *Let  $k \geq 1$ ,  $h = 3^k$ ,  $q = 2^h$ , and retain the compatible selected Conway roots  $\zeta_j = \kappa_{3^j}$ , so  $\zeta_{j-1} = \zeta_j^3$ . Thus  $\zeta_k \in \mathbb{F}_{q^2}$  has order  $3h$  and  $\zeta_k^q = \zeta_k^{-1}$ . Put*

$$\gamma_k = \zeta_k + \zeta_k^{-1}, \quad Z_k^\circ = \gamma_k^{-1} = \frac{\zeta_k}{(\zeta_k + 1)^2}, \quad w_k^\circ = \frac{1}{\zeta_k + 1}. \quad (\text{TC}_1)$$

Then

$$(w_k^\circ)^2 + w_k^\circ = Z_k^\circ, \quad (w_k^\circ)^q = w_k^\circ + 1, \quad \frac{w_k^\circ + 1}{w_k^\circ} = (w_k^\circ)^{q-1} = \zeta_k. \quad (\text{TC}_2)$$

The coefficient  $Z_k^\circ \in \mathbb{F}_q$  has exact degree  $h$  over  $\mathbb{F}_2$ , satisfies

$$\text{Tr}_{\mathbb{F}_q/\mathbb{F}_2}(Z_k^\circ) = 1, \quad (\text{TC}_3)$$

and, for  $k \geq 2$ , is norm coherent:

$$N_{\mathbb{F}_{2^{3k}}/\mathbb{F}_{2^{3k-1}}}(Z_k^\circ) = Z_{k-1}^\circ. \quad (\text{TC}_4)$$

Moreover

$$\text{ord}(Z_k^\circ) = \text{ord}(\gamma_k), \quad (\text{TC}_5)$$

so  $C_k$  makes  $Z_k^\circ$  primitive in  $\mathbb{F}_q^\times$ .

Now let  $k \geq 2$ , let  $\ell \mid \Psi_k$ , and put  $a = v_\ell(q+1) = v_\ell(\Psi_k)$ . Then this model misses the entire current primary factor:

$$\zeta_k \in U^{\ell^a}, \quad w_k^\circ \in (\mathbb{F}_{q^2}^\times)^{\ell^a}, \quad v_\ell\left(\text{ord} \frac{w_k^\circ + 1}{w_k^\circ}\right) = 0. \quad (\text{TC}_6)$$

This is not a counterexample to  $D'_k$ : its coefficient is  $Z_k^\circ = \gamma_k^{-1}$ , whereas the actual Conway coefficient is

$$Z_{D,k} = g_k^{t^2-2}, \quad g_k = \gamma_k^2 + \gamma_k + 1, \quad t = 2^{h/3}.$$

It proves that trace one, exact degree, norm-coherent actual  $C$ -period ancestry, and even  $C_k$ -primitivity of an Artin–Schreier constant give no direct current-primary order transfer. It does not rule out an implication  $C_k \Rightarrow D'_k$  for the actual selector; any such implication must use the special rational function selecting  $Z_{D,k}$ .

*Proof.* Since  $(\zeta_k + 1)^2 = \zeta_k(\zeta_k + \zeta_k^{-1})$ , the identities in (TC<sub>1</sub>)–(TC<sub>2</sub>) are direct. The element  $\zeta_k$  has irreducible quadratic

$$X^2 + \gamma_k X + 1$$

over  $\mathbb{F}_q$ . Dividing by  $\gamma_k^2$  gives an Artin–Schreier equation with coefficient  $\gamma_k^{-2}$ , so that coefficient has absolute trace one. Absolute trace is invariant under squaring, hence  $\text{Tr}(\gamma_k^{-1}) = 1$ . The field  $\mathbb{F}_2(\gamma_k)$  has degree  $h$ , and inversion preserves it, which proves the degree assertion. Proposition 6.7 gives  $N(\gamma_k) = \gamma_{k-1}$ ; inversion proves (TC<sub>4</sub>). Equation (TC<sub>5</sub>) is immediate.

For a current prime,  $\text{ord}_\ell(2) = 2h$ , so  $\ell \nmid 3h$ . The element  $\zeta_k$ , whose order is  $3h$ , is therefore an  $\ell^a$ -th power in the norm-one group  $U$ . Also  $\ell \nmid q - 1$ . In the direct product

$$\mathbb{F}_{q^2}^\times = \mathbb{F}_q^\times \times U,$$

$\ell^a$ -powering is an automorphism on the first factor, while raising the second factor to  $q - 1$  is invertible on its  $\ell$ -primary power quotient. The last identity in (TC<sub>2</sub>) therefore gives  $w_k^\circ \in (\mathbb{F}_{q^2}^\times)^{\ell^a}$ , and the order assertion follows.  $\square$

**Proposition 8.13** (the actual exceptional root's lower norm). *Retain  $k \geq 2$ ,  $h = 3^k$ ,  $t = 2^{h/3}$ , and the actual normalized root*

$$w_k = \frac{N_k}{g_k}, \quad N_k = \zeta_k^2 + \zeta_k + \omega$$

from Proposition 8.7. Put  $\eta = \zeta_k^3$ . Then

$$\boxed{N_{\mathbb{F}_{2^{2h}}/\mathbb{F}_{2^{2h/3}}}(w_k) = \frac{\eta^2 + \omega^2\eta + 1}{g_{k-1}} = \frac{\eta^2}{\eta^2 + \omega\eta + 1}.} \quad (\text{TN}_1)$$

For every current prime  $\ell \mid \Psi_k$ , even this exact lower norm is automatically an  $\ell^{v_\ell(\Psi_k)}$ -th power in  $\mathbb{F}_{2^{2h/3}}^\times$ . It therefore carries no current Kummer coordinate.

*Proof.* Proposition 8.63's twisted norm formula for  $N_k$ , divided by the norm-coherence formula for  $g_k$ , gives the first expression in (TN<sub>1</sub>). In characteristic two,

$$g_{k-1} = \eta^{-2}(\eta^4 + \eta^3 + \eta^2 + \eta + 1) = \eta^{-2}(\eta^2 + \omega\eta + 1)(\eta^2 + \omega^2\eta + 1),$$

which gives the second. Finally  $\text{ord}_\ell(2) = 2h$ , so  $\ell \nmid 2^{2h/3} - 1$ . Powering by the full current  $\ell$ -primary factor is therefore bijective in the lower multiplicative group.  $\square$

**Proposition 8.14** (coprime torus projections of the exceptional root). *Retain  $k \geq 2$ , and put*

$$\begin{aligned} t &= 2^{3^{k-1}}, & Q &= t^2, & A &= Q + t + 1, & B &= Q - t + 1 = 3\Psi_k, \\ \Gamma &= AB = Q^2 + Q + 1, & K &= \mathbb{F}_{t^6}, & E &= \mathbb{F}_{t^2}. \end{aligned}$$

For the actual normalized root  $w = w_k$ , set

$$H = N_{K/E}(w), \quad R = w^{Q-1}.$$

Then

$$R \in \ker N_{K/E} = \mu_\Gamma, \quad \gcd(A, B) = 1, \quad (\text{DCP}_1)$$

and its two coprime projections are exactly

$$\boxed{R^A = P_k = M_k^{t+1}}, \quad (\text{DCP}_2)$$

$$\boxed{R^B = Z_{D,k}^{t-1} = g_k^{(Q-2)(t-1)}}. \quad (\text{DCP}_3)$$

Thus  $P_k$  is literally the minus-torus coordinate of the relative coboundary  $R$ ; the complementary plus coordinate is the  $g_k$ -coboundary.

For every  $\ell^a \parallel \Psi_k$ ,

$$v_\ell(\text{ord } w) = v_\ell(\text{ord } R) = v_\ell(\text{ord } P_k), \quad \ell \nmid \text{ord}(R^B), \quad (\text{DCP}_4)$$

and the exact full-primary criterion is

$$v_\ell(\text{ord } P_k) = a \iff P_k^{B/\ell} \neq 1 \iff w \notin (K^\times)^\ell. \quad (\text{DCP}_5)$$

There is no squarefreeness assumption here.

The lower norm has the exact compatibility

$$\boxed{P_k^{B/3} = H^{(Q-1)/3}}. \quad (\text{DCP}_6)$$

By  $(D_9)$ , this common value is  $\omega^2$  for odd  $k$  and one for even  $k$ . Hence the lower norm determines precisely the already-known 3-component, and no current  $\ell$ -component.

More generally, the map

$$\Theta : K^\times \longrightarrow E^\times \times \mu_\Gamma, \quad x \longmapsto (N_{K/E}(x), x^{Q-1}) \quad (\text{DCP}_7)$$

has kernel  $\mu_3$ , and its image is exactly

$$\left\{ (H_0, R_0) : H_0^{(Q-1)/3} = R_0^{\Gamma/3} \right\}. \quad (\text{DCP}_8)$$

Consequently, after fixing  $H$  and  $R^B$ , multiplication of  $R$  by an arbitrary current Sylow element of  $\mu_B$  preserves both data while  $R^A$  ranges freely through that current coordinate. These compatible variations need not preserve the selected Artin–Schreier equation; that equation is the remaining coupling.

The coupling itself is lossless but supplies no order evaluation. If  $x^{t^3} = x + 1$ , put  $Z = x^2 + x$ , and form  $H, R$  from  $x$  as above. Then

$$N_{\mathbb{F}_{t^3}/\mathbb{F}_t}(Z) = H^{t+1}, \quad Z^{t-1} = R^B. \quad (\text{DCP}_9)$$

The map

$$\Xi : \mathbb{F}_{t^3}^\times \longrightarrow \mathbb{F}_t^\times \times \mu_A, \quad z \longmapsto (N_{\mathbb{F}_{t^3}/\mathbb{F}_t}(z), z^{t-1}) \quad (\text{DCP}_{10})$$



is a bijection. Thus  $H^{t+1}$  and  $R^B$  reconstruct a unique  $Z$ . Such a reconstructed scalar comes from an Artin–Schreier pair  $x, x+1 \in K \setminus \mathbb{F}_{t^3}$  satisfying  $x^{t^3} = x+1$  exactly when

$$\mathrm{Tr}_{\mathbb{F}_{t^3}/\mathbb{F}_2}(Z) = 1.$$

For the actual selected data, the two roots  $w, w+1$  replace  $P_k$  only by  $P_k^{-1}$ , leaving its order unchanged.

This reconstruction has an exact one-variable order form. With

$$M = \frac{w+1}{w}, \quad Z = w^2 + w, \quad P = M^{t+1},$$

one has

$$M + M^{-1} = Z^{-1}, \quad \boxed{P + P^{-1} = \mathcal{D}_{t+1}(Z^{-1}, 1)}. \quad (\mathrm{DCP}_{11})$$

Consequently, for every  $\ell^a \parallel \Psi_k$ ,

$$\boxed{P^{B/\ell} = 1 \iff \mathcal{D}_{(t^3+1)/\ell}(Z^{-1}, 1) = 0.} \quad (\mathrm{DCP}_{12})$$

Thus the selected trace-one scalar determines the inverse pair  $\{P, P^{-1}\}$  and rewrites failure as one Dickson critical value; it does not evaluate that value.

The aggregate reconstruction is rigid on the Artin–Schreier locus. For any root  $x^{t^3} = x+1$ , put

$$Z_x = x^2 + x, \quad H_x = N_{K/E}(x), \quad C_x = (x^{Q-1})^B, \quad P_x = \left(\frac{x+1}{x}\right)^{t+1}.$$

If two such roots  $x, y$  satisfy

$$H_x^{t+1} = H_y^{t+1}, \quad C_x = C_y,$$

then

$$y \in \{x, x+1\}, \quad P_y \in \{P_x, P_x^{-1}\}. \quad (\mathrm{DCP}_{13})$$

For the actual selected data, fixing the exact norm  $H$  sharpens this to

$$\left. \begin{array}{l} k = 2 \\ k \geq 3 \end{array} \right| \begin{array}{l} \{w, w+1\} \text{ have the same } (H, C_x), \\ w \text{ is the unique Artin–Schreier root with the fixed } (H, C_x). \end{array} \quad (\mathrm{DCP}_{14})$$

This is a rigidity theorem, not a nonprimitive countermodel: once the exact Artin–Schreier coupling is imposed, the projection data reconstruct the selected minus coordinate instead of bounding its order independently.

Finally the selected  $C$ -dependence of the plus projection is the explicit reciprocal-quartic identity

$$\eta_k = \gamma_k^{t-1} \implies \boxed{(g_k^{t-1})^2 = \frac{\eta_k^4 + \eta_k^3 + 1}{\eta_k^4 + \eta_k + 1}.} \quad (\mathrm{DCP}_{15})$$

The denominator in the reciprocal-quartic identity is nonzero. Indeed  $X^4 + X + 1$  is irreducible over  $\mathbb{F}_2$ , while  $\eta_k \in \mathbb{F}_{2^{3^k}}$  and  $\gcd(4, 3^k) = 1$ . Therefore any proof of  $C_k \Rightarrow D'_k$  that proceeds through these projections would have to control this special rational image and the selected Artin–Schreier coupling, then still evaluate the disjoint minus coordinate  $P_k$ ; primitive order does not transfer directly.

*Proof.* The identities

$$AB = Q^2 + Q + 1, \quad \gcd(A, B) = 1$$

are direct; the gcd uses that  $t \equiv -1 \pmod{3}$ . Since  $N_{K/E}(R) = R^{1+Q+Q^2} = 1$ , one has  $R \in \mu_\Gamma$ . Also

$$(Q-1)A = (t^3-1)(t+1),$$

and  $w^{t^3-1} = M_k$ , proving (DCP<sub>2</sub>). Using  $w^{t^3} = w + 1$ , one obtains

$$\begin{aligned} R^B &= w^{(Q-1)(Q-t+1)} = \frac{w^{t^4} w^t}{w^{t^3} w} \\ &= \frac{(w^t + 1)w^t}{(w + 1)w} = (w^2 + w)^{t-1} = Z_{D,k}^{t-1}, \end{aligned}$$

and (CD<sub>5</sub>) gives the last expression in (DCP<sub>3</sub>).

A current  $\ell$  divides  $B/3$  and is coprime to  $(Q-1)A$ . Raising to  $Q-1$  and then to  $A$  therefore preserves its full primary order, while  $R^B$  kills it. The cyclic Euler criterion proves (DCP<sub>4</sub>)–(DCP<sub>5</sub>). Furthermore

$$P_k^{B/3} = R^{\Gamma/3} = w^{(Q-1)\Gamma/3} = H^{(Q-1)/3},$$

which proves (DCP<sub>6</sub>).

The kernel of  $\Theta$  consists of the elements of  $E^\times$  whose cube is one, hence is  $\mu_3$ . Every image pair satisfies the compatibility in (DCP<sub>8</sub>). Conversely, both the image and the compatible-pair set have

$$\frac{(Q-1)\Gamma}{3}$$

elements: the latter condition is the equality of two surjective maps to  $\mu_3$ . This proves (DCP<sub>8</sub>). A current Sylow element  $c \in \mu_B$  satisfies  $c^B = c^{\Gamma/3} = 1$ , while  $c \mapsto c^A$  is an automorphism of its Sylow group, proving the variation assertion.

For an Artin–Schreier root,  $Z = x^{t^3+1}$ . Expanding exponents gives

$$(t^3 + 1)(t^2 + t + 1) = (t + 1)(t^4 + t^2 + 1),$$

which is the norm identity in (DCP<sub>9</sub>); the second is the same exponent calculation used in (DCP<sub>3</sub>). Since

$$|\mathbb{F}_{t^3}^\times| = (t-1)A, \quad \gcd(t-1, A) = 1,$$

$\Xi$  is injective and the two sides have the same cardinality, so it is bijective. The characteristic-two quadratic criterion gives the stated trace-one boundary and the two conjugate roots. When the trace is zero the quadratic instead has two roots in  $\mathbb{F}_{t^3}$ , so they do not satisfy the required nontrivial Frobenius translation.

The first identity in (DCP<sub>11</sub>) is a direct calculation, and the Dickson identity  $\mathcal{D}_n(u + u^{-1}, 1) = u^n + u^{-n}$  gives the second. Since  $P$  has odd order,

$$\mathcal{D}_{B/\ell}(P + P^{-1}, 1) = 0 \iff P^{B/\ell} = 1.$$

Dickson composition and  $(t+1)B = t^3 + 1$  prove (DCP<sub>12</sub>).

If  $x, y$  satisfy the hypotheses of (DCP<sub>13</sub>), then (DCP<sub>9</sub>) and injectivity of  $\Xi$  give  $Z_x = Z_y$ . Thus  $(x+y)^2 + (x+y) = 0$ , so  $y = x$  or  $x+1$ , and the displayed claim about  $P_y$  follows by inversion. Moreover

$$N_{K/E}(w+1) = N_{K/E}(w^{t^3}) = H^t.$$

For the actual norm, (TN<sub>1</sub>) gives, with  $\eta = \zeta_k^3$ ,

$$H = \frac{\eta^2}{\eta^2 + \omega\eta + 1}, \quad H^t = \frac{1}{\eta^2 + \omega^2\eta + 1}.$$

Their equality is equivalent to

$$(\eta^3 + \omega)(\eta + \omega^2) = \eta^4 + \omega^2\eta^3 + \omega\eta + 1 = 0.$$

Since  $\eta$  has order  $3^k$ , for  $k \geq 2$  this occurs exactly when  $k = 2$ , proving (DCP<sub>14</sub>). The statement concerns coordinates that actually arise from Artin–Schreier roots; an arbitrary ambient pair in (DCP<sub>8</sub>) need not do so.

For the reciprocal-quartic identity, put  $y = \zeta_k^2$ . Directly from the selected period formulas,

$$\eta_k = \frac{\omega + \omega^2 y}{1 + y}, \quad y = \frac{\eta_k + \omega}{\eta_k + \omega^2}, \quad g_k^2 = \frac{y^5 + 1}{y^2(y + 1)}, \quad y^t = \omega^2 y^{-1}.$$

Substitution, followed by  $1 + \omega + \omega^2 = 0$ , gives the displayed reciprocal-quartic quotient.  $\square$

**Proposition 8.15** (the actual exceptional terminal resolvent). *Retain  $k \geq 2$ , and write*

$$t = 2^{3^{k-1}}, \quad Q = t^2, \quad q = t^3, \quad a = \gamma_{k-1}, \quad C = a^2 + a + 1.$$

Put  $r_i = \gamma_k^{t^i}$ , with indices modulo three, and define the oriented cyclic resolvents

$$\mathcal{K} = \sum_i r_i^2 r_{i+1}, \quad \mathcal{K}' = \sum_i r_i r_{i+1}^2, \quad \chi = \mathcal{K}/a.$$

Then

$$\mathcal{K} + \mathcal{K}' = a, \quad \mathcal{K}\mathcal{K}' = a^2 + 1, \quad \boxed{\chi^2 + \chi = 1 + a^{-2}}. \quad (\text{DTR}_1)$$

Reversing cyclic orientation sends  $\chi$  to  $\chi + 1$ .

The actual selected scalar

$$Y = Z_{D,k}^{-1} = g_k^{2-Q} \in \mathbb{F}_q$$

has the irreducible monic polynomial over  $\mathbb{F}_t$

$$\boxed{m_\chi(T) = T^3 + \frac{a^2\chi + a^2 + 1}{C}T^2 + \frac{a^2(a+1)\chi + a^3 + a + 1}{C}T + C.} \quad (\text{DTR}_2)$$

Equivalently, its denominator-free equation is

$$CT^3 + (a\mathcal{K} + a^2 + 1)T^2 + (a(a+1)\mathcal{K} + a^3 + a + 1)T + C^2 = 0. \quad (\text{DTR}_3)$$

For every current prime  $\ell \mid \Psi_k$ , with  $n = (q+1)/\ell$ ,

$$\boxed{\text{failure at } \ell \iff m_\chi(T) \mid \mathcal{D}_n(T, 1).} \quad (\text{DTR}_4)$$

This divisor has no independent norm shadow. If  $V = \mathcal{D}_n(Y, 1)$ , then

$$\boxed{\text{Res}_T(m_\chi, \mathcal{D}_n) = N_{\mathbb{F}_q/\mathbb{F}_t}(V) = \text{Tr}_{\mathbb{F}_q/\mathbb{F}_t}(V)^2,} \quad (\text{DTR}_5)$$

and these three quantities vanish exactly under failure.

Finally, for  $Z = Y^{-1}$ , orientation reversal changes its relative trace by

$$\frac{a^2(a+1)}{C^2} = \left(\frac{a+1}{C}\right)^2 + \frac{a+1}{C}, \quad (\text{DTR}_6)$$

and leaves its relative norm  $C^{-1}$  fixed. Therefore absolute trace and lower norm cannot distinguish the two orientations. The actual  $t$ -Frobenius does select one of them; the comparison proves blindness of these scalar invariants, not ambiguity of the Conway datum.

*Proof.* The  $r_i$  are the roots of  $X^3 + X + a$ , so their elementary symmetric functions are  $e_1 = 0, e_2 = 1, e_3 = a$ . The cyclic-resolvent identities give

$$\mathcal{K} + \mathcal{K}' = e_1 e_2 + e_3 = a, \quad \mathcal{K} \mathcal{K}' = e_1^3 e_3 + e_2^3 + e_3^2 = 1 + a^2,$$

which proves (DTR<sub>1</sub>).

Put  $x_i = r_i^2 + r_i + 1$ . Direct symmetric reduction gives

$$e_1(x) = 1, \quad e_2(x) = a + 1, \quad e_3(x) = C, \quad \mathcal{K}_x := \sum_i x_i^2 x_{i+1} = a\mathcal{K} + C.$$

The three conjugates of  $Y$  are  $Y_i = x_i^2 / x_{i-1}$ . Reducing their elementary symmetric functions over the common denominator  $C = x_0 x_1 x_2$  gives

$$\sum_i Y_i = \frac{a\mathcal{K} + a^2 + 1}{C}, \quad \sum_{i < j} Y_i Y_j = \frac{a(a+1)\mathcal{K} + a^3 + a + 1}{C}, \quad \prod_i Y_i = C.$$

This proves (DTR<sub>2</sub>)–(DTR<sub>3</sub>). Here  $a, C \neq 0$ , and  $\gcd(Q-2, q-1) = 1$ . Exponentiation by  $2-Q$  therefore recovers  $g_k$  from  $Y$ , so  $Y$  has relative degree three and the polynomial is irreducible. Proposition 8.14, specifically (DCP<sub>12</sub>), and the minimal-polynomial criterion prove (DTR<sub>4</sub>).

Let  $M = M_k$  and  $\xi = M^n \in \mu_\ell$ . Since  $Y = M + M^{-1}$ , Dickson's identity gives  $V = \xi + \xi^{-1}$ . Moreover  $\text{ord}_\ell(t) = 6$ , hence  $t^2 - t + 1 \equiv 0 \pmod{\ell}$ . With  $x = \xi$  and  $y = \xi^t$ , direct multiplication of the three relative conjugates gives

$$N(V) = \text{Tr}(V)^2, \quad xy \text{Tr}(V) = (x+y)(x+1)(y+1).$$

The second expression vanishes exactly when  $\xi = 1$ ; thus both sides of (DTR<sub>5</sub>) vanish exactly under failure. The first equality in (DTR<sub>5</sub>) is the root-product formula for the irreducible  $m_\chi$ . Finally take the reciprocal of (DTR<sub>2</sub>) and replace  $\chi$  by  $\chi + 1$ . The relative norm stays  $C^{-1}$ , while the trace changes by the first term in (DTR<sub>6</sub>); the displayed Artin–Schreier identity proves equality of the absolute traces.  $\square$

**Proposition 8.16** (translated-binomial form of  $D'_k$ ). *Retain  $k \geq 2$ ,  $h = 3^k$ , and choose  $u \in \mathbb{F}_{16}$  with  $u^2 + u = \omega$ . Put*

$$L = \mathbb{F}_{16^h} = \mathbb{F}_{2^{4h}}, \quad \beta = \zeta + u.$$

*Then the minimal polynomial of  $\beta$  over  $\mathbb{F}_{16}$  is the irreducible degree- $h$  polynomial*

$$B_k(Y) = (Y + u)^h + \omega.$$

*For every prime  $\ell \mid \Psi_k$ , the following are equivalent:*

- (i) the full  $\ell$ -primary part occurs in  $\text{ord}(M_k)$ ;
- (ii)  $\beta \notin (L^\times)^\ell$ ;
- (iii)  $B_k(X^\ell) = (X^\ell + u)^h + \omega$  is irreducible over  $\mathbb{F}_{16}$ .

More precisely,  $B_k(X^\ell)$  is either irreducible of degree  $h\ell$ , or is a product of exactly  $\ell$  irreducibles of degree  $h$ .

*Proof.* Since

$$\text{ord}_{3h}(16) = \frac{\text{ord}_{3h}(2)}{\gcd(\text{ord}_{3h}(2), 4)} = h,$$

the element  $\zeta$  has degree  $h$  over  $\mathbb{F}_{16}$ . As  $\zeta^h = \omega$ , its minimal polynomial there is  $X^h + \omega$ ; translation by  $u$  gives  $B_k$ .

Let  $K = \mathbb{F}_{2^{2h}}$ . The corrected norm is

$$N = N_{L/K}(\beta) = \beta^{2^{2h}+1}.$$

Because  $2^{2h} \equiv 1 \pmod{\ell}$ , this norm induces multiplication by two on  $L^\times / (L^\times)^\ell$ . Hence it first gives

$$\beta \in (L^\times)^\ell \iff N \in (L^\times)^\ell.$$

The natural map

$$K^\times / (K^\times)^\ell \longrightarrow L^\times / (L^\times)^\ell$$

is injective: restriction followed by the degree-two norm is multiplication by two, which is invertible modulo the odd prime  $\ell$ . Since  $N \in K$ , this sharpens the equivalence to

$$\beta \in (L^\times)^\ell \iff N \in (K^\times)^\ell,$$

and Proposition 8.3 identifies the negation with the full  $\ell$ -primary assertion. Kummer theory and Capelli's lemma give the irreducibility equivalence.

If  $\beta = \alpha^\ell$  with  $\alpha \in L$ , every root of  $B_k(X^\ell)$  lies in  $L$ , because  $\mu_\ell \subset L$ . Its  $\ell$ -th power is a conjugate of  $\beta$ , of degree  $h$  over  $\mathbb{F}_{16}$ ; the root therefore has degree both at least and at most  $h$ . All irreducible factors consequently have degree  $h$ , and there are exactly  $\ell$  of them.  $\square$

**Corollary 8.17** (short translated cyclotomic period). *Let  $\chi : L^\times \rightarrow \mu_\ell$  have exact order  $\ell$ , realize its values as complex roots of unity, and write  $\chi(\beta) = \xi_\ell^r$ . Then*

$$\mathcal{J}_{k,\ell} := \sum_{\theta^h = \omega} \chi(\theta + u) = \sum_{j=0}^{h-1} \xi_\ell^{r 16^j},$$

and

$$\beta \in (L^\times)^\ell \iff \mathcal{J}_{k,\ell} = h.$$

*Proof.* The roots of  $X^h + \omega$  are  $\zeta^{16^j}$ ,  $0 \leq j < h$ , and  $u \in \mathbb{F}_{16}$ . Thus

$$\chi(\zeta^{16^j} + u) = \chi((\zeta + u)^{16^j}) = \xi_\ell^{r16^j}.$$

If  $r = 0$ , every summand is one. Conversely, equality with the positive real number  $h$  forces equality in the complex triangle inequality and hence forces every summand, in particular the first, to be one.  $\square$

**Proposition 8.18** (full  $\mathbb{F}_{16}$ -line saturation). *Retain Proposition 8.16, put  $s = 3^{k-1}$ , and choose  $u$  so that it has order fifteen and  $\omega = u^5$ . For a prime  $\ell \mid \Psi_k$ , write*

$$\mathcal{C} = L^\times / (L^\times)^\ell, \quad v(c) = [\zeta + c] \quad (c \in \mathbb{F}_{16}), \quad A = 2^s \pmod{\ell},$$

*using additive notation for  $\mathcal{C}$ . Then  $A$  has order six,  $A^2 - A + 1 = 0$ , and  $A^3 = -1$ . The four base-field classes vanish,*

$$v(c) = 0 \quad (c \in \mathbb{F}_4), \tag{L_{16,1}}$$

*while for  $c \in \mathbb{F}_{16}^\times$ ,*

$$v(\omega c^{-2^s}) = Av(c), \quad v(c^{-2^h}) = -v(c). \tag{L_{16,2}}$$

*For the twelve exponents not divisible by five one has*

| $j$ | $\{m : v(u^m) = A^j v(u)\}$ |            |
|-----|-----------------------------|------------|
| 0   | $\{1, 4\}$                  |            |
| 1   | $\{3, 12\}$                 |            |
| 2   | $\{11, 14\}$                |            |
| 3   | $\{7, 13\}$                 |            |
| 4   | $\{6, 9\}$                  |            |
| 5   | $\{2, 8\}$ .                | (L_{16,3}) |

*Consequently*

$$\zeta + u \in (L^\times)^\ell \iff \zeta + c \in (L^\times)^\ell \quad \text{for every } c \in \mathbb{F}_{16}. \tag{L_{16,4}}$$

*If  $D'_k$  holds for  $\ell$ , none of the twelve translates with  $c \notin \mathbb{F}_4$  is an  $\ell$ -th power. Equivalently, for an exact-order- $\ell$  complex character  $\chi$  of  $L^\times$  and  $t = \chi(\zeta + u)$ ,*

$$\sum_{c \in \mathbb{F}_{16}} \chi(\zeta + c) = 4 + 2 \sum_{j=0}^5 t^{A^j}, \tag{L_{16}}$$

*and failure of  $D'_k$  is exactly the extremal value sixteen. Thus Frobenius, conjugation, and all  $\mathbb{F}_4$ -translates generate only scalar multiples of the one unresolved Kummer class  $v(u)$ .*

*Proof.* One has  $\text{ord}_\ell(2) = 2h = 6s$ , proving the assertions about  $A$ . For  $c = 0$ , the element  $\zeta$  has  $\ell$ -coprime order  $3h$ . For  $c \in \{1, \omega, \omega^2\}$ , Propositions 6.4 and 6.5 give order dividing  $3h(2^h - 1)$ , again coprime to  $\ell$ . This proves (L\_{16,1}); constants in  $\mathbb{F}_{16}^\times$  and  $\zeta$  also have zero class.

The congruence  $2^s \equiv h - 1 \pmod{3h}$  gives  $\zeta^{2^s} = \omega \zeta^{-1}$ . Discarding scalar factors with zero class,

$$Av(c) = [(\zeta + c)^{2^s}] = [\omega \zeta^{-1} + c^{2^s}] = [\zeta + \omega c^{-2^s}] = v(\omega c^{-2^s}).$$

Likewise  $2^h \equiv -1 \pmod{3h}$  and modulo  $\ell$ , so

$$-v(c) = [(\zeta + c)^{2^h}] = [\zeta^{-1} + c^{2^h}] = [\zeta + c^{-2^h}].$$

On exponents modulo fifteen the two transformations are

$$R(m) = 5 - (2^s \bmod 15)m, \quad S(m) = -(2^h \bmod 15)m.$$

For odd  $k$  they are  $(5 - 2m, 7m)$ , and for even  $k$  they are  $(5 - 8m, 13m)$ . Iterating either pair on the twelve residues outside  $5\mathbb{Z}/15\mathbb{Z}$  gives the table  $(L_{16,3})$ . Since every  $A^j$  is nonzero modulo  $\ell$ , all twelve classes vanish exactly when  $v(u)$  vanishes, proving  $(L_{16,4})$  and its nonvanishing counterpart.

Applying  $\chi$  to the table gives each  $t^{A^j}$  twice, while the four  $\mathbb{F}_4$ -terms are one. If  $t = 1$ , the sum is sixteen; conversely equality with the positive real number sixteen forces every unit-modulus summand to be one by equality in the triangle inequality.  $\square$

**Proposition 8.19** (adjacent quartic Kummer reduction). *Retain Proposition 8.18. Thus*

$$h = 3^k, \quad s = h/3, \quad K = \mathbb{F}_{4^h} = \mathbb{F}_4(\zeta), \quad L = \mathbb{F}_{16^h}, \quad A = 2^s \pmod{\ell},$$

where  $\ell \mid \Psi_k$ , and choose  $u \in \mathbb{F}_{16}$  of order fifteen with  $u^5 = \omega = \zeta^h$ . Put

$$N = \zeta^2 + \zeta + \omega, \quad N' = \zeta^2 + \zeta + \omega^2, \quad D = \zeta^4 + \zeta, \quad W = D + 1.$$

Then  $D$  is unconditionally an  $\ell$ -th power in  $K$ , while in the cyclic Kummer quotient  $K^\times/(K^\times)^\ell$  one has

$$\boxed{[W] = (1 + A^5)[N].} \tag{AQ_1}$$

The scalar  $1 + A^5$  is nonzero modulo  $\ell$ . Consequently

$$\boxed{D'_k \text{ holds at } \ell \iff W \notin (K^\times)^\ell.} \tag{AQ_2}$$

Thus the selected exceptional assertion is equivalently the adjacent-value exclusion

$$D \in (K^\times)^\ell, \quad D + 1 \notin (K^\times)^\ell. \tag{AQ_3}$$

In fact every nonzero translate gives an equivalent detector:

$$D'_k \text{ holds at } \ell \iff D + c \notin (K^\times)^\ell \text{ for any (equivalently every) } c \in \mathbb{F}_4^\times. \tag{AQ_{3'}}$$

On failure the complete affine line  $D + \mathbb{F}_4$  lies in  $(K^\times)^\ell$ .

This criterion has an immediate cubic form. Put

$$K_- = \mathbb{F}_{4^s}, \quad a = \zeta^3 \in K_-.$$

The element  $W$  has irreducible minimal polynomial

$$\boxed{m_W(T) = T^3 + T^2 + T + \Phi_5(a)} \tag{AQ_4}$$

over  $K_-$ , and

$$N_{K/K_-}(W) = \Phi_5(a) \in (K_-^\times)^\ell. \tag{AQ_5}$$

Hence Capelli's criterion gives the sparse exact target

$$\boxed{D'_k \text{ holds at } \ell \iff X^{3\ell} + X^{2\ell} + X^\ell + \Phi_5(a) \text{ is irreducible over } K_-} \tag{AQ_6}$$

On failure the displayed polynomial is a product of exactly  $\ell$  distinct irreducible cubics. Equations (AQ<sub>1</sub>)–(AQ<sub>6</sub>) are exact reductions, not a nonvanishing proof: the norm in (AQ<sub>5</sub>) is automatically an  $\ell$ -th power, so the new form isolates rather than resolves the selected additive coupling between  $D$  and  $D + 1$ .

*Proof.* Use additive notation in  $\mathcal{C}_L = L^\times / (L^\times)^\ell$ , and retain  $v(c) = [\zeta + c]$  from Proposition 8.18. For every  $b \in \mathbb{F}_{16}$ ,

$$\prod_{c \in b + \mathbb{F}_4} (\zeta + c) = (\zeta + b)^4 + (\zeta + b) = D + b^4 + b. \quad (\text{AQ}_7)$$

The four affine  $\mathbb{F}_4$ -cosets, together with the orbit table ( $L_{16,3}$ ), therefore give

| $b + \mathbb{F}_4$   | $\prod_{c \in b + \mathbb{F}_4} (\zeta + c)$ | class in $\mathcal{C}_L$ |
|----------------------|----------------------------------------------|--------------------------|
| $\mathbb{F}_4$       | $D$                                          | 0                        |
| $u + \mathbb{F}_4$   | $D + 1$                                      | $2(1 + A^5)v(u)$         |
| $u^3 + \mathbb{F}_4$ | $D + \omega^2$                               | $2(A + A^2)v(u)$         |
| $u^6 + \mathbb{F}_4$ | $D + \omega$                                 | $2(A^3 + A^4)v(u)$ .     |

(AQ<sub>8</sub>)

Here, for example,  $u + \mathbb{F}_4 = \{u, u^2, u^4, u^8\}$ ; the other two rows follow from  $u^3 + \mathbb{F}_4 = \{u^3, u^{11}, u^{12}, u^{14}\}$  and  $u^6 + \mathbb{F}_4 = \{u^6, u^7, u^9, u^{13}\}$ .

The quadratic constant-field norm gives

$$N = (\zeta + u)(\zeta + u^4), \quad N' = (\zeta + u^2)(\zeta + u^8),$$

so the same table gives

$$[N]_L = 2v(u), \quad [N']_L = 2A^5v(u).$$

Direct multiplication in characteristic two gives

$$NN' = (\zeta^2 + \zeta + \omega)(\zeta^2 + \zeta + \omega^2) = \zeta^4 + \zeta + 1 = W. \quad (\text{AQ}_9)$$

Restriction

$$K^\times / (K^\times)^\ell \longrightarrow L^\times / (L^\times)^\ell$$

is injective: norm after restriction is multiplication by  $[L : K] = 2$ , which is invertible modulo the odd prime  $\ell$ . Thus (AQ<sub>8</sub>)–(AQ<sub>9</sub>) descend to (AQ<sub>1</sub>), and the first row of (AQ<sub>8</sub>) shows that  $D$  is an  $\ell$ -th power already in  $K$ . Since  $A$  has exact order six modulo  $\ell$ , the equality  $1 + A^5 = 0$  would give  $A^5 = A^3$ , hence  $A^2 = 1$ , a contradiction. Likewise  $1 + A \neq 0$ , so the other two coefficients  $2A(1 + A)$  and  $2A^3(1 + A)$  in (AQ<sub>8</sub>) are nonzero. Proposition 8.3 identifies  $[N] \neq 0$  with  $D'_k$ , proving (AQ<sub>2</sub>)–(AQ<sub>3</sub>). Together with the other two nonzero coefficients and injectivity, this also proves (AQ<sub>3'</sub>) and its failure statement.

The degree-three Frobenius of  $K/K_-$  sends  $\zeta$  to  $\zeta^{4^s} = \omega\zeta$ , hence sends  $D$  successively to  $\omega D$  and  $\omega^2 D$ . Since  $D \neq 0$ , the three conjugates

$$1 + D, \quad 1 + \omega D, \quad 1 + \omega^2 D$$

are distinct. Their product polynomial is

$$\prod_{i=0}^2 (T + (1 + \omega^i D)) = T^3 + T^2 + T + (1 + D^3).$$



With  $a = \zeta^3$ , one has  $D = \zeta(a + 1)$ , and therefore

$$1 + D^3 = 1 + a(a + 1)^3 = a^4 + a^3 + a^2 + a + 1 = \Phi_5(a). \quad (\text{AQ}_{10})$$

This proves (AQ<sub>4</sub>). It also proves the norm identity in (AQ<sub>5</sub>). The value is nonzero because  $a$  has 3-power order  $h$ , whereas a root of  $\Phi_5$  has order five; it is an  $\ell$ -th power because  $\text{ord}_\ell(4) = h$ , so  $\ell \nmid |K_-^\times| = 4^s - 1$ .

Finally  $\mu_\ell \subset K$ . For the prime  $\ell$ , Kummer theory says that  $X^\ell - W$  is irreducible over  $K$  exactly when  $W$  is not an  $\ell$ -th power. Capelli's lemma applied to the irreducible cubic  $m_W$  proves (AQ<sub>6</sub>). If  $W = y^\ell$  in  $K$ , every  $\rho^r y$ ,  $\rho \in \mu_\ell$ , has degree exactly three over  $K_-$ , because its  $\ell$ -th power has degree three. The composition therefore splits into exactly  $\ell$  distinct irreducible cubics. This proves the final assertion.  $\square$

**Proposition 8.20** (exceptional gap and Conway-block product blindness). *Retain Proposition 8.18, so  $k \geq 2$ , and put  $K = \mathbb{F}_{2^{2h}}$ . For every prime  $\ell \mid \Psi_k$ ,*

$$m_\ell = 4 \quad \text{or} \quad m_\ell \geq 16. \quad (\text{G}_{16})$$

*Thus a failure of  $D'_k$  cannot produce an excess in the interval  $5, \dots, 15$ .*

*There is also no contradiction obtained by multiplying a complete finite-Conway-subfield block. More precisely, for  $a \geq 1$ , put*

$$C_a = \mathbb{F}_{2^{2^a}}, \quad Q_a = |C_a| = 2^{2^a}, \quad L_a = \mathbb{F}_{2^{2^a h}} = KC_a.$$

*Then, in  $L_a$ ,*

$$\prod_{c \in C_a} (\zeta + c) = \zeta^{Q_a} + \zeta = \zeta(1 + \zeta^{Q_a - 1}) \in (L_a^\times)^\ell. \quad (\text{G}_{C_a})$$

*Consequently even an extension of the affine-line saturation from  $\mathbb{F}_{16}$  to any complete finite Conway subfield would have a product identity which is automatically invisible in the unresolved  $\ell$ -Kummer quotient.*

*Proof.* Choose the element  $u$  of Proposition 8.18 to be the finite number 4. If  $m_\ell \neq 4$ , then the lower bound  $m_\ell \geq 4$  and the exact power criterion say that  $\zeta + 4$  is an  $\ell$ -th power in  $L = \mathbb{F}_{2^{4h}}$ . Equation (L<sub>16,4</sub>) then makes every  $\zeta + c$ ,  $c \in \mathbb{F}_{16}$ , an  $\ell$ -th power. The finite numbers  $0, 1, \dots, 15$  are exactly the elements of  $\mathbb{F}_{16}$ , so none of these sixteen translates can be the excess. This proves (G<sub>16</sub>).

For the second assertion, the roots of  $X^{Q_a} + X$  are precisely the elements of  $C_a$ , which gives the displayed product formula. Since  $2^a$  is even, lifting the exponent gives

$$v_3(Q_a - 1) = v_3(2^{2^a} - 1) = 1.$$

The element  $\zeta$  has order  $3h = 3^{k+1}$ , so  $\zeta^{Q_a - 1}$  has exact order  $h$  and lies in

$$\mathbb{F}_{2^{\text{ord}_h(2)}} = \mathbb{F}_{2^{2 \cdot 3^{k-1}}}.$$

This lower field embeds in  $K$ , hence in every  $L_a$ . On the other hand  $\text{ord}_\ell(2) = 2h$ . Hence  $\ell$  divides neither  $3h$  nor the multiplicative order of that lower field. It follows that both  $\zeta$  and  $1 + \zeta^{Q_a - 1}$  are  $\ell$ -th powers in  $L_a$ , proving (G<sub>C<sub>a</sub></sub>).  $\square$

**Proposition 8.21** (constant-extension line criterion and an explicit exceptional bound). *Retain  $k \geq 2$ ,  $h = 3^k$ ,*

$$K = \mathbb{F}_{4^h} = \mathbb{F}_4(\zeta), \quad \ell \mid \Psi_k,$$

*and let  $\chi : K^\times \rightarrow \mu_\ell$  have exact order  $\ell$ . For  $a \geq 1$ , put*

$$t = 2^{a-1}, \quad C_a = \mathbb{F}_{2^{2^a}} = \mathbb{F}_{4^t}, \quad L_a = KC_a = \mathbb{F}_{4^{ht}},$$

*and let*

$$\chi_a = \chi \circ N_{L_a/K}, \quad S_{a,k,\ell} = \sum_{c \in C_a} \chi_a(\zeta + c). \quad (\text{CE}_1)$$

*Then*

$$\zeta + C_a \subset (L_a^\times)^\ell \iff S_{a,k,\ell} = 4^t. \quad (\text{CE}_2)$$

*More precisely, attach to  $\chi$  the primitive even Dirichlet character modulo  $X^h + \omega$  over  $\mathbb{F}_4[X]$ , and write its  $L$ -polynomial as*

$$L_\chi(T) = (1 - T) \prod_{j=1}^{h-2} (1 - \alpha_j T), \quad |\alpha_j| = 2.$$

*Then constant-field extension gives the exact selected moment*

$$S_{a,k,\ell} = - \left( 1 + \sum_{j=1}^{h-2} \alpha_j^t \right), \quad |S_{a,k,\ell}| \leq 1 + (h-2)2^t. \quad (\text{CE}_3)$$

*Consequently*

$$2^t \geq h - 1 \implies m_\ell \leq 4^t - 1. \quad (\text{CE}_4)$$

*If  $t$  is the least power of two at least  $\lceil \log_2(h-1) \rceil$ , then*

$$m_\ell < 4^t < 16h^4. \quad (\text{CE}_5)$$

*Thus this gives a direct polynomial upper bound on the exceptional column, but not an absolute one. For any fixed  $a$ , the Riemann–Weil bound in  $(\text{CE}_3)$  is compatible with the extremal value  $S_{a,k,\ell} = 4^t$  for all sufficiently large  $h$ . A fixed-window proof by this route must therefore prove the selected nonextremality*

$$S_{a,k,\ell} \neq 4^t$$

*uniformly in  $k$  and  $\ell$ . At  $a = 2$ , this is exactly the full- $\mathbb{F}_{16}$ -line formulation of  $D'_k$ ; larger fixed  $a$  give genuinely weaker selected least-nonresidue assertions rather than a consequence of the four maximal-order arms.*

*Proof.* Since  $h$  is odd and  $t$  is a power of two, the fields  $K$  and  $C_a$  are linearly disjoint over  $\mathbb{F}_4$ , and  $[L_a : K] = t$ . Norm followed by inclusion on the cyclic  $\ell$ -Kummer quotients is multiplication by  $t$ . As  $\ell \nmid t$ , the norm induces an isomorphism

$$L_a^\times / (L_a^\times)^\ell \xrightarrow{\sim} K^\times / (K^\times)^\ell. \quad (\text{CE}_6)$$

In particular  $\chi_a$  has exact order  $\ell$ , its kernel is  $(L_a^\times)^\ell$ , and no term  $\zeta + c$  vanishes because  $\zeta$  has odd degree  $h > 1$  over  $\mathbb{F}_4$ , whereas every element of  $C_a$  has degree dividing  $t$ . Equation  $(\text{CE}_2)$

now follows from equality in the complex triangle inequality: a sum of  $4^t$  roots of unity equals the positive real number  $4^t$  precisely when every summand is one.

The polynomial  $X^h + \omega$  remains irreducible after extension of constants to  $C_a$ , again because  $\gcd(h, t) = 1$ . Since  $\ell \neq 3$ , the order- $\ell$  character is trivial on  $\mathbb{F}_4^\times$ , so the attached Dirichlet character is even. The character  $\chi_a$  is exactly its constant-field norm lift. If the reciprocal roots of  $L_\chi$  are  $1, \alpha_1, \dots, \alpha_{h-2}$ , constant extension raises them to their  $t$ -th powers. The degree-one coefficient of the extended  $L$ -polynomial is on the one hand

$$\sum_{c \in C_a} \chi_a(\zeta + c) = S_{a,k,\ell},$$

and on the other hand the negative of the sum of those reciprocal roots. This proves the equality in (CE<sub>3</sub>); the function-field Riemann hypothesis gives  $|\alpha_j| = 2$ , proving its inequality. Notice that 1 is the reciprocal root of the trivial even factor  $1 - T$ , so neither the sign nor the additive constant is being suppressed.

If  $2^t \geq h - 1$ , then

$$1 + (h - 2)2^t \leq 1 + (2^t - 1)2^t = 4^t - 2^t + 1 < 4^t.$$

Equations (CE<sub>2</sub>)–(CE<sub>3</sub>) therefore supply some  $c \in C_a$  for which  $\zeta + c$  is not an  $\ell$ -th power in  $L_a$ , and hence is not one in its defining subfield. The finite numbers  $0, 1, \dots, 4^t - 1$  are exactly the elements of  $C_a$ , proving (CE<sub>4</sub>).

Finally put  $R = \lceil \log_2(h - 1) \rceil$ . The least power of two  $t \geq R$  satisfies

$$t < 2R < 2(\log_2 h + 1).$$

Hence

$$4^t < 4^{2(\log_2 h + 1)} = 16h^4,$$

which proves (CE<sub>5</sub>). For fixed  $t$ , the right side of the bound in (CE<sub>3</sub>) eventually exceeds  $4^t$ , proving the stated limitation. The case  $a = 2$  has  $t = 2$  and  $C_a = \mathbb{F}_{16}$ , so Proposition 8.18 gives the final identification with  $D'_k$ .  $\square$

**Corollary 8.22** (composite-conductor cyclotomic-unit form of  $D'_k$ ). *Retain Proposition 8.20, put  $N_0 = 15h$ , and choose  $\theta \in L^\times$  so that*

$$\theta^5 = \zeta, \quad \theta^h = u.$$

*Then  $\theta$  has exact order  $N_0$ , and*

$$\rho := \frac{\zeta}{u} = \theta^{5-h}$$

*also has exact order  $N_0$ . If  $\tilde{\rho}$  is a primitive complex  $N_0$ -th root of unity,*

$$\varepsilon = 1 - \tilde{\rho} \in \mathbb{Q}(\mu_{N_0})^\times$$

*is a global cyclotomic unit. There is a prime  $\mathfrak{p} \mid 2$  whose residue field is  $L = \mathbb{F}_{2^{4h}}$  and for which*

$$\bar{\varepsilon} = 1 + \rho, \quad \zeta + u = u(1 + \rho).$$

*For every prime  $\ell \mid \Psi_k$ ,*

$$D'_k \text{ holds at } \ell \iff \left( \frac{\varepsilon}{\mathfrak{p}} \right)_\ell \neq 1. \quad (\text{CU}_{15h})$$

*Equivalently, the chosen prime over two has nontrivial Frobenius in the Kummer extension obtained by adjoining an  $\ell$ -th root of  $\varepsilon$  after adjoining  $\mu_\ell$ .*

*Proof.* The elements  $\zeta$  and  $u$  have orders  $3h$  and  $15$ , and their  $h$ -th and fifth powers agree:  $\zeta^h = u^5 = \omega$ . The compatible pair therefore lifts in the cyclic group of order  $\text{lcm}(3h, 15) = 15h$  to the stated  $\theta$ . Its two displayed images have their full orders, so  $\theta$  has order  $15h$ . Moreover

$$\gcd(5 - h, 15h) = 1$$

because  $h$  is a power of three and is not divisible by five; hence  $\rho$  is primitive of the same order.

The conductor  $15h$  has two distinct prime divisors, so  $\Phi_{15h}(1) = 1$ , which makes  $1 - \tilde{\rho}$  a unit. Also

$$\text{ord}_{15h}(2) = \text{lcm}(\text{ord}_{3h}(2), \text{ord}_5(2)) = \text{lcm}(2h, 4) = 4h.$$

Thus a prime over two has residue field  $L$ , and it may be chosen to send  $\tilde{\rho}$  to  $\rho$ . Reduction gives the two displayed identities. Finally  $\ell \nmid 15$ , so  $u$  is an  $\ell$ -th power. Proposition 8.16 identifies  $D'_k$  at  $\ell$  with  $\zeta + u$  being a non- $\ell$ -th-power; the standard power-residue/Frobenius criterion now gives  $(\text{CU}_{15h})$ .  $\square$

**Proposition 8.23** (exceptional failure as simultaneous Jacobi flatness). *Let  $k \geq 2$ , put*

$$h = 3^k, \quad M = 15h, \quad Q = 2^{4h},$$

*and let the prime  $\ell$  divide  $\Psi_k$ . Choose the reduction  $z \in \mathbb{F}_Q^\times$  of the selected primitive  $M$ -th root in Corollary 8.22, choose a generator  $g$  with  $g^{(Q-1)/M} = z$ , and form the mixed Jacobi sums*

$$J_j = \sum_{x \in \mathbb{F}_Q} \chi_M(x)^j \chi_\ell(1 - x) \quad (1 \leq j < M)$$

*of Proposition 5.21. At a chosen  $\ell$ -adic place put  $\pi = \zeta_\ell - 1$ . Then failure of  $D'_k$  at  $\ell$  is equivalent to the simultaneous first-order flatness*

$$J_j \equiv -1 \pmod{\pi^2} \quad (1 \leq j < M). \quad (\text{DJ})$$

*Thus mixed Jacobi sums see the complete exceptional Kummer vector, but the missing theorem is exactly that this vector is not identically zero; Stickelberger ideal factorization alone does not evaluate its first local phase.*

*Proof.* Write the unique coprime-order decomposition

$$z = ab, \quad \text{ord}(a) = 3h, \quad \text{ord}(b) = 5,$$

and put  $q = 2^h$ ,  $L = \mathbb{F}_{q^4} = \mathbb{F}_Q$ , and  $K_0 = \mathbb{F}_{q^2}$ . By Corollary 8.22, failure at  $\ell$  says that  $1 + ab$  is an  $\ell$ -th power in  $L$ . The norm  $L^\times \rightarrow K_0^\times$  acts on the one-dimensional  $\ell$ -Kummer quotient by

$$q^2 + 1 \equiv 2 \pmod{\ell},$$

because  $\text{ord}_\ell(2) = 2h$ . It is therefore an isomorphism on that quotient, and

$$U = (1 + ab)(1 + ab^{-1})$$

is an  $\ell$ -th power in  $K_0$ .

Since  $a^q = a^{-1}$ ,

$$(1 + a)^{q-1} = a^{-1}, \quad (1 + a^5)^{q-1} = a^{-5}.$$

Here  $q - 1 \equiv -2 \pmod{\ell}$ , while  $a$  has order prime to  $\ell$ . Hence both  $1 + a$  and  $1 + a^5$  are  $\ell$ -th powers. If

$$U' = (1 + ab^2)(1 + ab^{-2}),$$

then

$$UU' = \Phi_5(a) = \frac{1 + a^5}{1 + a}$$

is an  $\ell$ -th power. Thus  $U'$  is an  $\ell$ -th power, and the same norm isomorphism implies that  $1 + ab^2$  is an  $\ell$ -th power in  $L$ .

Now

$$\text{ord}_M(2) = 4h, \quad \varphi(M) = 8h.$$

The primitive  $M$ -th roots consequently form two Frobenius orbits, represented by  $ab$  and  $ab^2$ . These representatives are not in the same orbit: a power  $2^j$  fixing the  $3h$ -component has  $j \equiv 0 \pmod{2h}$ , whereas no such exponent sends  $b$  to  $b^2$ . Hence  $1 + v$  is an  $\ell$ -th power for every primitive  $M$ -th root  $v$ .

It remains to treat proper-order powers of  $z$ . If  $d \mid M$  is proper, the divisibility  $2h \mid \text{ord}_d(2)$  can occur only for  $d = 3h$ . For an element  $v$  of that order,  $v^q = v^{-1}$ , and

$$(1 + v)^{q-1} = v^{-1}.$$

As above,  $q - 1 \equiv -2 \pmod{\ell}$  and  $\ell \nmid \text{ord}(v)$ , so  $1 + v$  is an  $\ell$ -th power. For every other proper  $d$ , the smallest finite field containing  $v$  has unit group of order prime to  $\ell$ ; its  $\ell$ -power map is an automorphism. We have therefore proved

$$1 - z^a \in (\mathbb{F}_Q^\times)^\ell \quad (1 \leq a < M).$$

Proposition 5.21 now gives (DJ). Conversely, (DJ) makes every Fourier coefficient  $C_j$  vanish, so (MJ<sub>2</sub>) at  $a = 1$  says that  $1 - z$  is an  $\ell$ -th power. This is precisely failure of  $D'_k$  at  $\ell$ .  $\square$

**Proposition 8.24** (the exceptional Jacobi vector has one selected coordinate). *Retain Proposition 8.23, and for*

$$U = (\mathbb{Z}/M\mathbb{Z})^\times$$

define

$$\lambda(\pm 2^i) = 2^i \pmod{\ell} \quad (0 \leq i < 4h). \quad (\text{DJ}_1)$$

Then  $\lambda : U \rightarrow \mathbb{F}_\ell^\times$  is a primitive even Dirichlet character modulo  $M$ . If

$$L(a) = \text{ind}_\ell(1 - z^a), \quad G_j = \sum_{a \in U} \lambda(a) \overline{\zeta_M}^{ja},$$

then

$$L(a) = \begin{cases} 0, & (a, M) > 1, \\ \lambda(a)L(1), & (a, M) = 1, \end{cases} \quad C_j = L(1)G_j. \quad (\text{DJ}_2)$$

Moreover

$$G_1 G(\lambda^{-1}) = \lambda(-1)M = M \not\equiv 0 \pmod{\ell}, \quad (\text{DJ}_3)$$

and consequently

$$D'_k \text{ holds at } \ell \iff L(1) \neq 0 \iff C_1 \neq 0 \iff J_1 \not\equiv -1 \pmod{(\zeta_\ell - 1)^2}. \quad (\text{DJ}_4)$$

Thus the simultaneous Jacobi flatness in Proposition 8.23 is a rank-one condition, not  $M - 1$  independent conditions.

This coordinate is exactly the first regulator of the quadratic-twist antiunit in Proposition 8.36. With

$$E_1 = (1 - ab)(1 - ab^{-1}), \quad E_2 = (1 - ab^2)(1 - ab^{-2}), \quad d_k = E_1/E_2,$$

one has

$$\text{ind}_\ell(E_1) = 2L(1), \quad \text{ind}_\ell(E_2) = -2L(1), \quad \text{ind}_\ell(d_k) = 4L(1), \quad (\text{DJ}_5)$$

and hence

$$C_1 = \frac{G_1}{4} \text{ind}_\ell(d_k). \quad (\text{DJ}_6)$$

In particular the exact remaining exceptional theorem is the single nonvanishing assertion

$$\text{ind}_\ell(d_k) \neq 0 \quad (\ell \mid \Psi_k). \quad (\text{DJ}_7)$$

*Proof.* The group  $U$  has order  $\varphi(M) = 8h$ , while  $\text{ord}_M(2) = 4h$ . Moreover  $-1 \notin \langle 2 \rangle$ , because a power of two congruent to  $-1$  modulo  $3h$  has exponent  $h \pmod{2h}$ , whereas a power of two congruent to  $-1$  modulo five has exponent  $2 \pmod{4}$ , and these conditions are incompatible for odd  $h$ . Hence

$$U = \{\pm 2^i : 0 \leq i < 4h\}.$$

Since  $\text{ord}_\ell(2) = 2h$ , the rule in  $(\text{DJ}_1)$  is well-defined, multiplicative, and even. It is primitive. Indeed it cannot factor modulo  $M/3$ : the kernel of  $U \rightarrow (\mathbb{Z}/(M/3)\mathbb{Z})^\times$  has order three, whereas the kernel of  $\lambda$  has order four. It cannot factor modulo  $M/5 = 3h$  either. The CRT class  $r$  with

$$r \equiv 1 \pmod{3h}, \quad r \equiv 2 \pmod{5}$$

belongs to the reduction kernel but satisfies  $\lambda(r) = -1$ : in the factorization

$$\lambda = \lambda_- \chi_5,$$

the first factor is trivial on  $r$ , while  $\chi_5(2) = -1$ . Removing either prime divisor of  $M$  therefore destroys the character, which proves primitivity.

If  $(a, M) > 1$ , then  $z^a$  has proper order. The proof of Proposition 8.23 shows unconditionally that every such  $1 - z^a$  is an  $\ell$ -th power, and hence  $L(a) = 0$ . If  $(a, M) = 1$ , write  $a = \pm 2^i$ . Frobenius covariance gives  $L(2^i) = 2^i L(1)$ ; replacing  $z^{2^i}$  by its inverse changes  $1 - z^{2^i}$  only by an  $M$ -torsion factor, whose  $\ell$ -index is zero. This proves the first identity in  $(\text{DJ}_2)$ . Substitution into the Fourier coefficient  $(\text{MJ}_1)$  of Proposition 5.21 gives  $C_j = L(1)G_j$ .

The standard Gauss identity for a primitive Dirichlet character gives  $(\text{DJ}_3)$ . Here  $\ell \nmid M$ , so  $G_1 \neq 0$ . Corollary 8.22 identifies  $D'_k$  with  $L(1) \neq 0$ , while the first-order Jacobi expansion identifies  $C_1 \neq 0$  with the final congruence in  $(\text{DJ}_4)$ .

Finally the two exponents carrying  $ab$  and  $ab^{-1}$  are trivial under the quadratic character modulo five, so both factors of  $E_1$  have index  $L(1)$ . The exponents carrying  $ab^2$  and  $ab^{-2}$  have quadratic

character  $-1$ , so both factors of  $E_2$  have index  $-L(1)$ . This proves (DJ<sub>5</sub>), and (DJ<sub>6</sub>)–(DJ<sub>7</sub>) follow because four and  $G_1$  are nonzero modulo  $\ell$ .  $\square$

**Proposition 8.25** (canonical four-Jacobi detector). *Retain Proposition 8.24. Write the primitive Dirichlet character as  $\lambda = \lambda_{-\chi_5}$ , where  $\chi_5 = (\frac{\cdot}{5})$ . Decompose the selected character  $\chi_M$  of Proposition 5.21 uniquely as  $AB$ , where  $A, B$  have orders  $3h, 5$ , and set  $X = \chi_\ell$  with the normalization  $X(g) = \zeta_\ell$  used there. For  $r = 1, 2, 3, 4$ , let  $j_r \bmod M$  be the CRT class*

$$j_r \equiv 1 \pmod{3h}, \quad j_r \equiv r \pmod{5},$$

and put  $J_r = J(AB^r, X)$ . In the fraction field of the local cyclotomic ring define

$$\mathcal{D}_{k,\ell} = \frac{J_1 J_4}{J_2 J_3}.$$

Then, with  $\pi = \zeta_\ell - 1$ ,

$$\boxed{\mathcal{D}_{k,\ell} \equiv 1 - 4\pi C_1 = 1 - \pi G_1 \operatorname{ind}_\ell(d_k) \pmod{\pi^2}.} \quad (\text{FJ}_1)$$

Consequently

$$D'_k \text{ fails at } \ell \iff \mathcal{D}_{k,\ell} \equiv 1 \pmod{\pi^2}, \quad (\text{FJ}_2)$$

and

$$\operatorname{ind}_\ell(d_k) = -G_1^{-1} \frac{\overline{\mathcal{D}_{k,\ell} - 1}}{\pi}. \quad (\text{FJ}_3)$$

This four-term quotient is canonical and minimal under the conductor-five symmetries. Indeed, an inversion-invariant integer weight vector on the four phases has the form  $(u, v, v, u)$ ; zero augmentation forces  $v = -u$ , and the resulting detector line over  $\mathbb{F}_\ell$  is spanned by  $(1, -1, -1, 1)$ . The primitive integral weights are its two signs.

There is also a single-antiunit form. For a multiplicative character  $C$ , fixed in the conductor-five coordinate, such that  $C^2$  and the displayed characters are nontrivial, put

$$\mathcal{H}_5(C) = \frac{J(CB, CB^{-1})}{J(CB^2, CB^{-2})} = \frac{G(CB)G(CB^{-1})}{G(CB^2)G(CB^{-2})}. \quad (\text{FJ}_4)$$

Here  $G(\cdot)$  denotes the finite-field Gauss sum; the common  $G(C^2)$  denominator has cancelled. Then

$$\boxed{\mathcal{D}_{k,\ell} = \frac{\mathcal{H}_5(A)}{\mathcal{H}_5(AX)}}. \quad (\text{FJ}_5)$$

For  $C = A$  or  $AX$ , let  $E$  be the field generated by the values of  $C$ . Conductor-five conjugation  $B \mapsto B^{-1}$  fixes  $\mathcal{H}_5(C)$ , so it lies in  $E(\zeta_5 + \zeta_5^{-1})$ . The nontrivial automorphism of this quadratic extension, induced by  $B \mapsto B^2$ , sends  $\mathcal{H}_5(C)$  to its inverse. Hence

$$N_{E(\sqrt{5})/E} \mathcal{H}_5(C) = 1. \quad (\text{FJ}_6)$$

If the forward  $X$ -tangent is normalized by

$$\partial_X \log \mathcal{H}_5(A) = \frac{\overline{\mathcal{H}_5(AX)/\mathcal{H}_5(A) - 1}}{\pi},$$

then (FJ<sub>3</sub>) reads  $\operatorname{ind}_\ell(d_k) = G_1^{-1} \partial_X \log \mathcal{H}_5(A)$ . Order-five product formulas evaluate the invariant norm and therefore erase, rather than evaluate, this anti-invariant tangent.

*Proof.* The four CRT classes have

$$C_{j_1} = C_{j_4} = C_1, \quad C_{j_2} = C_{j_3} = -C_1.$$

Indeed  $G_j = \lambda(j)^{-1}G_1$ , while  $\lambda(j_r) = \chi_5(r)$ . Since  $J_r \equiv -1 + \pi C_{j_r} \pmod{\pi^2}$ , and every denominator is a  $\pi$ -adic unit,

$$\frac{(1 - \pi C_1)^2}{(1 + \pi C_1)^2} \equiv 1 - 4\pi C_1 \pmod{\pi^2}.$$

Now use (DJ<sub>6</sub>) and the nonvanishing of  $G_1$  to obtain (FJ<sub>1</sub>)–(FJ<sub>3</sub>). The minimality assertion is the displayed two-dimensional linear calculation.

Finally the Gauss–Jacobi identity gives

$$\frac{J(AB, X)J(AB^{-1}, X)}{J(AB^2, X)J(AB^{-2}, X)} = \frac{\mathcal{H}_5(A)}{\mathcal{H}_5(AX)},$$

because the common factor  $G(X)^2$  cancels. Replacing  $B$  by  $B^2$  interchanges numerator and denominator in (FJ<sub>4</sub>), while replacing it by  $B^{-1}$  only swaps factors. This proves the last claims.  $\square$

**Proposition 8.26** (relative Eisenstein determinant). *Retain Proposition 8.25, and put*

$$S = 2^{2h}, \quad K = \mathbb{F}_S, \quad L = \mathbb{F}_{S^2} = \mathbb{F}_Q.$$

*The characters  $A$  and  $X$  are the norm lifts of unique characters  $a_0, x_0$  of  $K^\times$ , while  $B|_{K^\times} = 1$ :*

$$A = a_0 \circ N_{L/K}, \quad X = x_0 \circ N_{L/K}.$$

*For  $r = 1, 2$ , define the relative Eisenstein sums*

$$\mathcal{E}_r(c) = \sum_{\text{Tr}_{L/K}(z)=1} (c \circ N_{L/K})(z) B^r(z). \quad (\text{ES}_1)$$

*For every  $c$  with  $c^2 \neq 1$ , one has*

$$G_L((c \circ N)B^r) = G_K(c^2)\mathcal{E}_r(c), \quad \mathcal{E}_r(c) = \mathcal{E}_{-r}(c), \quad (\text{ES}_2)$$

*and the four-Jacobi antiunit has the exact quadratic-relative form*

$$\boxed{\mathcal{H}_5(c \circ N) = \left( \frac{\mathcal{E}_1(c)}{\mathcal{E}_2(c)} \right)^2}, \quad \boxed{\mathcal{D}_{k,\ell} = \left( \frac{\mathcal{E}_1(a_0)\mathcal{E}_2(a_0x_0)}{\mathcal{E}_2(a_0)\mathcal{E}_1(a_0x_0)} \right)^2}. \quad (\text{ES}_3)$$

*Normalize  $X(z) = \zeta_\ell^{\nu(z)}$ . At the selected prime above  $\ell$ , set*

$$e_r = \overline{\mathcal{E}_r(a_0)}, \quad u_r = \sum_{\text{Tr } z=1} \overline{A(z)B^r(z)} \nu(z). \quad (\text{ES}_4)$$

*Every  $e_r$  is nonzero, and*

$$\mathcal{D}_{k,\ell} \equiv 1 - 2\pi \left( \frac{u_1}{e_1} - \frac{u_2}{e_2} \right) \pmod{\pi^2}. \quad (\text{ES}_5)$$

*Consequently the surviving selected coordinate is exactly the determinant*

$$\boxed{\text{ind}_\ell(d_k) = 2G_1^{-1} \frac{\Delta_{k,\ell}}{e_1 e_2}, \quad \Delta_{k,\ell} := u_1 e_2 - u_2 e_1}, \quad (\text{ES}_6)$$



and therefore

$$D'_k \text{ holds at } \ell \iff \Delta_{k,\ell} \neq 0. \quad (\text{ES}_7)$$

The first-order complete- $B$ -orbit Hasse–Davenport relation does not decide this determinant. Indeed, for any conductor-five-fixed  $C$ , Frobenius gives

$$y_1 = G(CB) = G(CB^{-1}), \quad y_2 = G(CB^2) = G(CB^{-2}),$$

while the order-five product formula has the form

$$G(C)y_1^2y_2^2 = -G(C^5) \prod_{j=0}^4 G(B^j). \quad (\text{ES}_8)$$

Thus it constrains only the invariant product  $y_1y_2$ , and after the  $X$ -linearization only  $u_1/e_1 + u_2/e_2$ . Over the odd residue characteristic, this one symmetric linear relation leaves the anti-invariant difference  $u_1/e_1 - u_2/e_2$  free, and in particular is compatible with its vanishing. Therefore the first-order relation  $(\text{ES}_8)$  cannot prove  $(\text{ES}_7)$ .

*Proof.* One has  $3h, \ell \mid S-1$ , whereas  $S \equiv -1 \pmod{5}$ . Thus  $A$  and  $X$  descend through the norm and  $B$  is trivial on  $K^\times$ . Decompose the Gauss sum over the fibres of  $\text{Tr}_{L/K}$ . On a nonzero fibre, write  $z = tw$  with  $t \in K^\times$  and  $\text{Tr}(w) = 1$ . Since  $N_{L/K}(t) = t^2$ , its contribution separates into  $G_K(c^2)\mathcal{E}_r(c)$ . The trace-zero fibre is  $K$ , and its contribution vanishes because  $c^2$  is nontrivial. This proves the first identity in  $(\text{ES}_2)$ . The map  $z \mapsto z^S$  fixes trace and norm and sends  $B$  to  $B^{-1}$ , proving the second.

Substitute  $(\text{ES}_2)$  into  $(\text{FJ}_4)$ . The base-field Gauss factors cancel and the two Frobenius-paired factors agree, giving the first formula in  $(\text{ES}_3)$ ;  $(\text{FJ}_5)$  gives the second. Since

$$X(z) = (1 + \pi)^{\nu(z)} \equiv 1 + \pi\nu(z) \pmod{\pi^2},$$

one has

$$\mathcal{E}_r(a_0x_0) \equiv \mathcal{E}_r(a_0) + \pi u_r \pmod{\pi^2}.$$

Gauss sums over fields of characteristic two are units at every prime above the odd number  $\ell$ ; hence  $(\text{ES}_2)$  makes each  $e_r$  a unit. Expanding the square in  $(\text{ES}_3)$  proves  $(\text{ES}_5)$ , and comparison with  $(\text{FJ}_1)$  proves  $(\text{ES}_6)$ – $(\text{ES}_7)$ .

For a unit-circle form of the same determinant, every trace-one element of  $L$  is uniquely

$$z = t^{-1}v, \quad v \in \ker N_{L/K} \setminus \{1\}, \quad t = v + v^{-1}.$$

Thus  $e_r$  is a quintic Soto–Andrade sum weighted by  $a_0^{-2}(t)$ , while  $u_r$  inserts the extra weight  $-2\text{ind}_{x_0}(t)$ . Their determinant is genuinely the anti-quintic Wronskian, rather than an unweighted full-field Fourier coefficient.

Finally, apply the order-five Hasse–Davenport formula to the complete  $B$ -orbit of  $C$ . In characteristic two the usual  $C(5^{-5})$  factor is one, which gives  $(\text{ES}_8)$ . Its invariance under  $B \mapsto B^2$  shows directly that its first-order term is the symmetric sum. Since that sum imposes no nonvanishing condition on the alternating difference, the final assertion follows.  $\square$

**Proposition 8.27** (semiprimitive parity obstruction on the mixed line). *Let  $h = 3^k$ , let the prime  $\ell \mid \Psi_k$ , and retain the primitive mixed character of conductor  $M\ell = 15h\ell$  from Proposition 8.24. There is no exponent  $t$  for which*

$$2^t \equiv -1 \pmod{5\ell}. \quad (\text{SMP}_1)$$

In particular, two is not semiprimitive modulo  $M\ell$ . Hence the semiprimitive Gauss–Jacobi evaluations available for the pure  $\ell$ -character and the proper-conductor components do not evaluate the surviving primitive mixed coefficient  $C_1 = G_1L(1)$ . More precisely, the mixed conductor has decomposition index

$$\frac{\varphi(M\ell)}{\text{ord}_{M\ell}(2)} = 2(\ell - 1), \quad (\text{SMP}_2)$$

whereas the pure  $M$ -component has index two and the pure  $\ell$ -component is semiprimitive.

*Proof.* The cyclotomic-divisor criterion gives  $\text{ord}_\ell(2) = 2h$ , and  $2^h \equiv -1 \pmod{\ell}$ . The unique element of order two in  $\langle 2 \rangle$  is therefore reached exactly at exponents

$$t \equiv h \pmod{2h}.$$

Since  $h$  is odd, every such  $t$  is odd. On the other hand,  $\text{ord}_5(2) = 4$ , so  $2^t \equiv -1 \pmod{5}$  forces  $t \equiv 2 \pmod{4}$ , which is even. Thus  $(\text{SMP}_1)$  is impossible. Because  $5\ell \mid M\ell$ , semiprimitivity modulo  $M\ell$  is impossible as well. The final statement follows: the mixed character is precisely the component whose first-order value is  $G_1L(1)$ , while the semiprimitive proper components are the already forced-flat coordinates of Proposition 8.23.

Finally  $\gcd(M, \ell) = 1$ , so

$$\text{ord}_{M\ell}(2) = \text{lcm}(\text{ord}_M(2), \text{ord}_\ell(2)) = \text{lcm}(4h, 2h) = 4h,$$

while  $\varphi(M\ell) = 8h(\ell - 1)$ . This proves  $(\text{SMP}_2)$ . Thus the two explicitly evaluable constituent Gauss sums leave a mixed Gauss sum of growing decomposition index in their Jacobi quotient.  $\square$

**Corollary 8.28** (only a weighted exceptional resolvent sees the first line). *Retain Proposition 8.24, put  $U = (\mathbb{Z}/M\mathbb{Z})^\times$ , and choose integer lifts  $w_j \equiv \lambda(j) \pmod{\ell}$ . Then*

$$\prod_{j \in U} (-J_j) \equiv 1 \pmod{\pi^2}, \quad \prod_{j \in U} (-J_j)^{w_j} \equiv 1 - 8h\pi C_1 \pmod{\pi^2}. \quad (\text{HD}_1)$$

*The same unweighted congruence holds after adjoining all nonunit indices  $1 \leq j < M$ . Consequently every ordinary equal-weight Hasse–Davenport product is flat whether or not  $D'_k$  holds, while the weighted product is nonflat exactly when  $D'_k$  holds.*

*Proof.* Changing variables in the Gauss sum of  $(\text{DJ}_2)$  gives  $G_j = \lambda(j)^{-1}G_1$ , hence  $C_j = \lambda(j)^{-1}C_1$  for  $j \in U$ . Since  $-J_j \equiv 1 - \pi C_j \pmod{\pi^2}$ , the unweighted linear term is

$$-\pi C_1 \sum_{j \in U} \lambda(j)^{-1} = 0,$$

whereas the weighted term is  $-\pi|U|C_1 = -8h\pi C_1$ . Here  $\ell \nmid 8h$ , so Proposition 8.24 proves the detector assertion. For nonunit  $j$ , Proposition 8.23 already gives  $J_j \equiv -1 \pmod{\pi^2}$ , proving the final extension.  $\square$

**Proposition 8.29** (the ordinary cyclotomic derivative is blind). *Retain Proposition 8.24, write  $M = 15h$ , and let  $\tilde{\rho}$  be the complex primitive  $M$ -th root whose reduction at the selected prime above two is  $z$ . Put  $Q = 2^{4h}$ , choose a generator  $g \in \mathbb{F}_Q^\times$ , and normalize*

$$z = g^{(Q-1)/M}, \quad \xi = g^{(Q-1)/\ell}, \quad \text{ind}_\ell(g^v) = v \pmod{\ell}.$$

Then the selected Euler residue is exactly

$$(1 - z)^{(Q-1)/\ell} = \xi^{L(1)}. \quad (\text{ED}_1)$$

By contrast, the reduction of the ordinary cyclotomic derivative is unconditionally an  $\ell$ -th power:

$$\overline{\Phi'_M(\tilde{\rho})} = z^{-1} \frac{z^h - 1}{(z^{3h} - 1)(z^{5h} - 1)} \in (\mathbb{F}_Q^\times)^\ell. \quad (\text{ED}_2)$$

Thus neither  $\Phi'_M(\tilde{\rho})$  nor any formula depending only on its ordinary reduction can evaluate the selected exponent  $L(1)$ .

There is an exact weighted replacement, but it is the original target. For  $a \in U = (\mathbb{Z}/M\mathbb{Z})^\times$ , choose integer lifts  $r_a \equiv \lambda(a)^{-1} \pmod{\ell}$  and put, modulo global  $\ell$ -th powers,

$$\varepsilon_{\lambda^{-1}} = \prod_{a \in U} (1 - \tilde{\rho}^a)^{r_a}.$$

Its selected reduction satisfies

$$\text{ind}_\ell(\overline{\varepsilon_{\lambda^{-1}}}) = 8hL(1). \quad (\text{ED}_3)$$

Consequently  $D'_k$  holds at  $\ell$  exactly when this weighted circular unit has nontrivial  $\ell$ -power residue at the selected prime above two.

Finally, if  $\tilde{\lambda}$  is the characteristic-zero lift of the primitive even nontrivial character  $\lambda$ , then

$$B_{1, \tilde{\lambda}^{-1}} = -L(0, \tilde{\lambda}^{-1}) = 0. \quad (\text{ED}_4)$$

This vanishing is independent of the exceptional target: for example  $D'_2$  holds analytically by Proposition 8.5, so its selected  $L(1)$  is nonzero although  $(\text{ED}_4)$  vanishes. Hence the first generalized Bernoulli value cannot be a nonzero scalar multiple of the selected Kummer index.

*Proof.* Equation  $(\text{ED}_1)$  is the Euler criterion after writing  $1 - z = g^v$ : both sides are  $g^{v(Q-1)/\ell}$ , and  $v \bmod \ell = L(1)$ .

Since  $M = 5 \cdot 3^{k+1} = 15h$ , one has the polynomial identity

$$\Phi_M(X) = \frac{(X^M - 1)(X^h - 1)}{(X^{3h} - 1)(X^{5h} - 1)}.$$

Differentiating at the primitive root  $\tilde{\rho}$  gives

$$\Phi'_M(\tilde{\rho}) = M\tilde{\rho}^{-1} \frac{\tilde{\rho}^h - 1}{(\tilde{\rho}^{3h} - 1)(\tilde{\rho}^{5h} - 1)}.$$

After reduction in characteristic two the odd scalar  $M$  becomes one, proving the displayed equality in  $(\text{ED}_2)$ . The element  $z$  has order  $M$ , prime to  $\ell$ . Moreover  $z^h, z^{3h}, z^{5h}$  have orders 15, 5, 3, respectively, so the three factors involving them lie in  $\mathbb{F}_{16}^\times$  or  $\mathbb{F}_4^\times$ . All four factors therefore have order prime to  $\ell$ , and each is an  $\ell$ -th power in  $\mathbb{F}_Q^\times$ .

The choice of the lifts  $r_a$  changes  $\varepsilon_{\lambda^{-1}}$  only by an  $\ell$ -th power. Using  $(\text{DJ}_2)$  gives

$$\text{ind}_\ell(\overline{\varepsilon_{\lambda^{-1}}}) = \sum_{a \in U} \lambda(a)^{-1} L(a) = \sum_{a \in U} L(1) = 8hL(1),$$

which proves  $(\text{ED}_3)$ ; here  $\ell \nmid 8h$ . Finally, for any nontrivial even character  $\chi$  modulo  $M$ , pairing  $a$  with  $M - a$  in

$$B_{1, \chi} = M^{-1} \sum_{a=1}^M a\chi(a)$$

and using  $\sum_a \chi(a) = 0$  gives  $B_{1,\chi} = 0$ . Apply this to  $\chi = \tilde{\lambda}^{-1}$ . This proves (ED<sub>4</sub>) and the final separation.  $\square$

**Proposition 8.30** (the wild regulator is explicit but homogeneous). *Retain Proposition 8.24. Let  $K_0/\mathbb{Q}_\ell$  be the unramified extension of degree  $f = \text{ord}_M(\ell)$  containing  $\mu_M$ , with residue field  $k_0$ , and for an unramified unit  $y$  define*

$$q_\ell(y) = \frac{y^{\ell^f-1} - 1}{\ell} \pmod{\ell}.$$

For  $\tau \in \mu_M \setminus \{1\}$ , one has the finite truncated-log formula

$$q_\ell(1 - \tau) = -\frac{1}{1 - \tau} \sum_{a=1}^{\ell-1} \frac{\tau^{a\ell^{f-1}}}{a} \quad (\text{WL}_1)$$

in the residue field. For the characteristic-zero antiunit  $d_k = E_1/E_2$  of (DJ<sub>5</sub>), put

$$\begin{aligned} A_d &= q_\ell(1 - \alpha\beta) + q_\ell(1 - \alpha\beta^{-1}) \\ &\quad - q_\ell(1 - \alpha\beta^2) - q_\ell(1 - \alpha\beta^{-2}). \end{aligned} \quad (\text{WL}_2)$$

If  $[\bar{d}_k]$  is its Teichmüller part, then

$$d_k = [\bar{d}_k](1 - \ell A_d + O(\ell^2)). \quad (\text{WL}_3)$$

Now pass to  $K = K_0(\zeta_\ell)$ , with  $\pi = \zeta_\ell - 1$ . For a mixed local selector

$$w = [\bar{w}](1 + \pi B_w + O(\pi^2)),$$

the critical explicit norm-residue formula has the form

$$(d_k, w)_{K,\ell} = \zeta_\ell^{\text{Tr}_{k_0/\mathbb{F}_\ell}(c_\ell A_d B_w)}, \quad (\text{WL}_4)$$

where  $c_\ell$  is a nonzero scalar depending only on the uniformizer normalization. Base-field selectors have  $B_w = 0$ . For the canonical weighted Jacobi resolvent

$$\mathcal{R}_\lambda = \prod_{j \in (\mathbb{Z}/M\mathbb{Z})^\times} (-J_j)^{w_j}, \quad w_j \equiv \lambda(j) \pmod{\ell},$$

one has

$$B_w = -8hC_1 = -8hG_1L(1), \quad \text{ind}_\ell(d_k) = 4L(1). \quad (\text{WL}_5)$$

Thus the wild pairing with the only canonical selector that isolates the exceptional character is homogeneous in the still-unknown  $L(1)$ . When exceptional failure holds, both the selected Kummer symbol and the critical wild term are one. Explicit wild reciprocity therefore computes the local regulator  $A_d$ , but does not force  $\text{ind}_\ell(d_k) \neq 0$ .

*Proof.* Because  $\tau^{\ell^f} = \tau$ , expand  $(1 - \tau)^{\ell^f} - (1 - \tau)$  modulo  $\ell^2$ . The only interior binomial coefficients of valuation one have indices  $a\ell^{f-1}$ ,  $1 \leq a < \ell$ , and

$$\frac{1}{\ell} \binom{\ell^f}{a\ell^{f-1}} \equiv \frac{(-1)^{a-1}}{a} \pmod{\ell}.$$

Since  $\ell^{f-1}$  is odd, multiplication by  $(-\tau)^{a\ell^{f-1}}$ , followed by division by  $1 - \tau$ , proves (WL<sub>1</sub>). The Fermat quotient is additive on products and subtractive on quotients modulo  $\ell$ , which gives (WL<sub>2</sub>). If  $d_k = [\bar{d}_k](1 + \ell c + O(\ell^2))$ , then  $\ell^f - 1 \equiv -1 \pmod{\ell}$ , so  $q_\ell(d_k) = -c$ ; this proves (WL<sub>3</sub>).

In  $K$ , the first entry of the pairing lies in  $U_{\ell-1}$  and the second in  $U_1$ . The critical case of Iwasawa's explicit norm-residue formula [19] is the nondegenerate residue pairing (WL<sub>4</sub>); the unit converting  $\ell$  to  $\pi^{\ell-1}$  is absorbed into  $c_\ell$ . Finally, first-order expansion of the displayed weighted product and (DJ<sub>2</sub>) give

$$\mathcal{R}_\lambda \equiv 1 - \pi \sum_{j \in U} \lambda(j) C_j = 1 - 8h\pi C_1 \pmod{\pi^2}.$$

Equation (DJ<sub>5</sub>) supplies the second identity in (WL<sub>5</sub>), completing the proof.  $\square$

**Proposition 8.31** (binary cyclic-parity form of exceptional failure). *Retain Proposition 8.23, set*

$$E = \frac{Q-1}{\ell}, \quad S_M(X) = 1 + X + \cdots + X^{M-1},$$

*and let  $R_{k,\ell}(X)$  be the degree- $< M$  reduction of  $(1+X)^E$  modulo  $X^M - 1$ . Then failure of  $D'_k$  at  $\ell$  is equivalent to*

$$R_{k,\ell}(X) = 1 + S_M(X) \quad \text{in } \mathbb{F}_2[X]/(X^M - 1). \quad (\text{DP}_1)$$

*Equivalently, among the binary submasks  $u$  of  $E$ , the number with  $u \equiv 0 \pmod{M}$  is even, while the number in every nonzero residue class modulo  $M$  is odd.*

*Moreover, raising (DP<sub>1</sub>) to the  $\ell$ -th power gives only the unconditional identity*

$$(1+X)^{Q-1} = 1 + S_M(X) \quad \text{in } \mathbb{F}_2[X]/(X^M - 1). \quad (\text{DP}_2)$$

*Thus ordinary exponent, determinant, or product consequences of (DP<sub>1</sub>) cannot rule out exceptional failure; the remaining assertion is exactly that this selected binary-residue parity pattern never occurs.*

*Proof.* The proof of Proposition 8.23 established that failure is equivalent to

$$1 - z^a \in (\mathbb{F}_Q^\times)^\ell \quad (1 \leq a < M). \quad (\text{DP}_3)$$

Here  $\ell \mid Q-1$ , and in the cyclic group  $\mathbb{F}_Q^\times$  an element  $v$  is an  $\ell$ -th power exactly when  $v^E = 1$ . Consequently (DP<sub>3</sub>) says that  $R_{k,\ell}(\xi) = 1$  at every nonidentity  $M$ -th root  $\xi$ . At  $\xi = 1$ , it has value 0. Since  $M$  is odd,  $X^M - 1$  is squarefree, and  $1 + S_M$  has exactly these values. This proves (DP<sub>1</sub>), including its converse by evaluation at  $z$ .

Lucas's theorem gives

$$(1+X)^E = \sum_{u \subseteq_2 E} X^u,$$

so cyclic reduction proves the submask-parity formulation. Finally  $S_M^2 = MS_M = S_M$ , hence  $1 + S_M$  is idempotent. The right side of (DP<sub>1</sub>) is therefore unchanged on taking its  $\ell$ -th power. Identity (DP<sub>2</sub>) is unconditional: at  $X = 1$  both sides vanish, while at every other  $M$ -th root  $\xi \in \mathbb{F}_Q$ , Fermat gives  $(1 + \xi)^{Q-1} = 1$ . Squarefree interpolation completes the proof.  $\square$

**Proposition 8.32** (single-deletion words exhaust the code and design bounds). *Let  $m$  be odd and put*

$$A_m = \mathbb{F}_2[X]/(X^m - 1), \quad J_m = 1 + X + \cdots + X^{m-1}, \quad v_r = J_m + X^r.$$

Then

$$v_r = X^r(1 + J_m), \quad \text{supp}(v_r) = \mathbb{Z}/m\mathbb{Z} \setminus \{r\}, \quad (\text{CD}_1)$$

and, at the  $m$ -th roots in a splitting field,

$$v_r(1) = 0, \quad v_r(\xi) = \xi^r \quad (\xi \neq 1). \quad (\text{CD}_2)$$

Consequently

$$\gcd(v_r, X^m - 1) = X + 1, \quad (\text{CD}_3)$$

so the cyclic code generated by  $v_r$  is exactly the binary single-parity-check code  $[m, m - 1, 2]$ . Its coefficient support and its finite-field Fourier support both have size  $m - 1$ .

The same support is the trivial cyclic difference set

$$(m, m - 1, m - 2), \quad (\text{CD}_4)$$

whose nontrivial complex character sums have squared absolute value one. Adding the exceptional monomial gives

$$v_r + X^r = J_m, \quad (\text{CD}_5)$$

the generator of the repetition code, whose Fourier support has size one.

Under cubic failure, Proposition 6.46 gives the word  $v_r$ ; under exceptional failure, Proposition 8.31 gives  $v_0$ . Hence cyclic distance bounds, support-uncertainty inequalities, ordinary difference-set multiplier conditions, and character-magnitude conditions all admit the two failure patterns exactly. After (CD<sub>5</sub>), the associated vanishing sum is merely the full regular polygon  $J_m(\zeta) = 0$ , which decomposes through the proper conductor primes. These methods discard the same primitive CRT coordinate isolated in (CP<sub>7</sub>) and (DP<sub>1</sub>).

*Proof.* Translation fixes  $J_m$ , so  $X^r(1 + J_m) = X^r + J_m = v_r$ , proving (CD<sub>1</sub>). Since  $m$  is odd,  $J_m(1) = 1$  in characteristic two, while  $J_m(\xi) = 0$  for every nonidentity  $m$ -th root. This proves (CD<sub>2</sub>). The polynomial  $X^m - 1$  is squarefree, so one is the only common root and (CD<sub>3</sub>) follows. The ideal generated by  $X + 1$  is the even-weight code, giving the stated parameters.

Every nonzero group element has exactly  $m - 2$  representations as a difference of two elements outside a fixed singleton. This proves (CD<sub>4</sub>); equivalently, a nontrivial complex character has sum  $-\chi(r)$  over the support. Equation (CD<sub>5</sub>) is immediate. The two applications are exactly (CP<sub>3</sub>) and (DP<sub>1</sub>).  $\square$

**Proposition 8.33** (higher two-adic lifts carry no odd Kummer obstruction). *Let  $\mathcal{O}$  be the ring of integers of a finite unramified extension of  $\mathbb{Q}_2$ , with residue field  $K$ , and let  $\ell$  be odd. Then*

$$1 + 2\mathcal{O} \longrightarrow 1 + 2\mathcal{O}, \quad x \longmapsto x^\ell \quad (2K_1)$$

*is an automorphism. Under the Teichmüller splitting*

$$\mathcal{O}^\times = [K^\times] \times (1 + 2\mathcal{O}), \quad (2K_2)$$

*a unit is therefore an  $\ell$ -th power exactly when its residue in  $K^\times$  is an  $\ell$ -th power.*

The statement applies coordinatewise to the unramified lift of  $\mathbb{F}_2[C_m]$  for odd  $m$ , because that group algebra is squarefree and hence a product of finite fields. Consequently modulo-four, modulo-eight, and higher carry refinements of the cubic or exceptional parity identities cannot supply an independent obstruction: the principal-unit lift is uniquely  $\ell$ -divisible once the selected residue Kummer coordinate is fixed. The same conclusion applies to the Fermat and ordinary arms.

*Proof.* For  $u \in 1 + 2\mathcal{O}$ , the polynomial  $Y^\ell - u$  has the root one modulo two and derivative  $\ell Y^{\ell-1}$ , which is a unit. Hensel's lemma gives a unique root in  $1 + 2\mathcal{O}$ . Thus  $(2K_1)$  is bijective. The standard Teichmüller decomposition gives  $(2K_2)$  and the power criterion. The coordinatewise assertion follows from the squarefreeness of  $X^m - 1$  in characteristic two when  $m$  is odd.  $\square$

**Proposition 8.34** (exceptional half-block norm and its automatic symmetries). *Retain Proposition 8.31, and put*

$$A = 2^{2h}, \quad u = \frac{A-1}{\ell}, \quad R = \mathbb{F}_2[X]/(X^M - 1), \quad e = 1 + S_M.$$

Let  $I = eR$ , with identity  $e$ , and set

$$U = 1 + X \in I^\times, \quad \tau(X) = X^A, \quad F = U^u.$$

Then

$$F^\ell = \frac{\tau(U)}{U}, \quad F\tau(F) = U^{u(A+1)}, \quad (\text{DH}_1)$$

and exceptional failure is exactly

$$F\tau(F) = e. \quad (\text{DH}_2)$$

On the selected primitive  $M$ -th-root component of Corollary 8.22, put

$$w = (1 + \rho)(1 + \rho^A) \in \mathbb{F}_A^\times, \quad v = w^u \in \mu_\ell.$$

Then failure is precisely  $v = 1$ .

There is no further Hilbert-90 or quadratic-norm obstruction to  $(\text{DH}_2)$ . Indeed  $\tau(U)/U$  belongs to the norm-one group of  $\mathbb{F}_{A^2}/\mathbb{F}_A$ , whose order is  $A+1$ ; since  $\gcd(\ell, A+1) = 1$ , it has a unique  $\ell$ -th root in that group. Failure says only that the distinguished root  $F = U^u$  is this norm-one root.

The coefficient symmetries are automatic as well. The  $\tau$ -fixed residues modulo  $M$  are exactly the multiples of five. For such a residue  $j$ ,

$$[X^j]F\tau(F) = \sum_{\substack{a \pmod{M} \\ (1+A)a=j}} [X^a]F. \quad (\text{DH}_3)$$

After identifying the fixed subgroup with  $C_{3h}$ , the right side is, up to an invertible multiplier of its indices, the cyclic reduction of  $(1+Y)^u$ ; unconditionally,

$$(1+Y)^u = e_{3h} \quad \text{in } \mathbb{F}_2[C_{3h}]. \quad (\text{DH}_4)$$

Thus the fixed-residue part of the exceptional parity pattern is forced before the selected Kummer question is asked.

*Proof.* Since  $U^A = \tau(U)$ , the first identity in  $(\text{DH}_1)$  is

$$F^\ell = U^{A-1} = \frac{\tau(U)}{U}.$$

The second follows by multiplying  $U^u$  and  $\tau(U)^u = U^{Au}$ . Since

$$u(A+1) = \frac{A^2-1}{\ell} = \frac{Q-1}{\ell},$$

Proposition 8.31 proves (DH<sub>2</sub>). On a primitive component,  $\tau$  is the quadratic Frobenius of  $\mathbb{F}_{A^2}/\mathbb{F}_A$ , and evaluation gives  $F\tau(F) = w^u = v$ . The norm-one assertion and uniqueness then follow from  $\ell \mid A - 1$  and  $\gcd(A - 1, A + 1) = 1$ .

One has

$$\gcd(A - 1, M) = 3h, \quad \gcd(A + 1, M) = 5.$$

Hence the fixed residues of multiplication by  $A$  are precisely the multiples of five. In the convolution coefficient of such a residue, swapping the two selected submasks preserves the residue. Every off-diagonal pair cancels in characteristic two, leaving only the diagonal pairs; this is (DH<sub>3</sub>). Multiplication by  $(A + 1)/5$  is invertible modulo  $3h$ , so the displayed pushforward is the reduction of  $(1 + Y)^u$ , up to a permutation of coefficients.

Put  $q = 2^h$ . If  $\xi \neq 1$  is a  $3h$ -th root of unity, then  $\xi^q = \xi^{-1}$ , and therefore

$$(1 + \xi)^{q-1} = \xi^{-1}.$$

Also  $u = (q - 1)(q + 1)/\ell$ , while  $3h \mid (q + 1)/\ell$ . It follows that  $(1 + \xi)^u = 1$ . At  $\xi = 1$  it vanishes, so squarefree interpolation proves (DH<sub>4</sub>).  $\square$

**Proposition 8.35** (one-coordinate trace form of the exceptional phase). *Retain Proposition 8.34, put  $s = h/3$ , and let*

$$B \equiv 2^s \pmod{\ell}.$$

*Then  $B$  has order six modulo  $\ell$  and  $B^2 - B + 1 = 0$ . For  $v \in \mu_\ell \subset \mathbb{F}_{2^{6s}}$ , define*

$$\mathcal{T}(v) = \text{Tr}_{\mathbb{F}_{2^{6s}}/\mathbb{F}_{2^s}}(v) = \sum_{j=0}^5 v^{B^j}.$$

*One has the exact equivalence*

$$\mathcal{T}(v) = 0 \iff v = 1. \tag{DT_1}$$

*Consequently the exceptional assertion at  $\ell$  is exactly*

$$\mathcal{T}\left(\left((1 + \rho)(1 + \rho^A)\right)^{(A-1)/\ell}\right) \neq 0. \tag{DT_2}$$

*This is a single relative-trace coordinate of the selected residue symbol. It is equivalent to, rather than a generic norm consequence of, the missing nontriviality.*

*Proof.* The order identity  $\text{ord}_\ell(2) = 2h = 6s$  proves the assertions about  $B$ . Put  $x = v$  and  $y = v^B$ . The six powers of  $B$ , together with  $B^2 = B - 1$ , give the elementary factorization

$$xy \mathcal{T}(v) = (x + y)(x + 1)(y + 1). \tag{DT_3}$$

If  $v \neq 1$ , then  $x \neq 1$ ,  $y \neq 1$ , and  $x \neq y$ , because  $\ell$  is prime and  $B \not\equiv 1 \pmod{\ell}$ . Thus the right side is nonzero. Conversely  $\mathcal{T}(1) = 6 = 0$  in characteristic two, proving (DT<sub>1</sub>). Proposition 8.34 identifies failure with  $v = 1$ , so its negation is (DT<sub>2</sub>).  $\square$



## 8.4 Cyclotomic antiunit and regulator bounds

**Proposition 8.36** (quadratic-twist antiunit and the two exceptional zeros). *Retain Corollary 8.22, put  $m = 3h$ , and write the unique coprime-order decomposition*

$$\tilde{\rho} = \alpha\beta, \quad \text{ord}(\alpha) = m, \quad \text{ord}(\beta) = 5.$$

Set

$$s = \beta + \beta^{-1}, \quad s' = \beta^2 + \beta^{-2},$$

and define the cyclotomic antiunit

$$d_k := \frac{(1 - \alpha\beta)(1 - \alpha\beta^{-1})}{(1 - \alpha\beta^2)(1 - \alpha\beta^{-2})} = \frac{\alpha^2 - s\alpha + 1}{\alpha^2 - s'\alpha + 1} = \frac{\alpha + \alpha^{-1} - s}{\alpha + \alpha^{-1} - s'}. \quad (\text{QT}_1)$$

Then  $d_k$  belongs to

$$F_k^{\text{tw}} = \mathbb{Q}(\alpha + \alpha^{-1}, \sqrt{5}),$$

is fixed by complex conjugation, and is inverted by the nontrivial automorphism of  $F_k^{\text{tw}}/\mathbb{Q}(\alpha + \alpha^{-1})$ . In particular its relative norm is one.

The invariant conductor-five norm loses exactly the information carried by  $d_k$ :

$$N_{\mathbb{Q}(\mu_{5m})/\mathbb{Q}(\mu_m)}(1 - \tilde{\rho}) = \Phi_5(\alpha) = \frac{1 - \alpha^5}{1 - \alpha}. \quad (\text{QT}_2)$$

More precisely, reduce at the prime selected in Corollary 8.22, write

$$a = \bar{\alpha}, \quad b = \bar{\beta}, \quad c = b + b^{-1} \in \mathbb{F}_4 \setminus \mathbb{F}_2,$$

and put, in  $K = \mathbb{F}_{2^{2h}}$ ,

$$U_k = a^2 + ca + 1, \quad U'_k = a^2 + c^2a + 1, \quad R_k = \frac{U_k}{U'_k} = \bar{d}_k. \quad (\text{QT}_3)$$

For every prime  $\ell \mid \Psi_k$ , the element  $\Phi_5(a)$  is an  $\ell$ -th power in  $K$ , and

$$D'_k \text{ holds at } \ell \iff U_k \notin (K^\times)^\ell \iff R_k \notin (K^\times)^\ell. \quad (\text{QT}_4)$$

Thus the two conductor-five places over the cubic field carry inverse Kummer classes; their invariant product is identically blind.

This antiunit is an exact quadratic lift of a conjugate of the cubic Gaussian period. If

$$q = 2^h, \quad \gamma = a + a^{-1}, \quad A = \gamma^2 + \gamma + 1,$$

then

$$R_k^q = R_k^{-1}, \quad R_k + R_k^{-1} = A^{-1}, \quad \frac{R_k}{(R_k + 1)^2} = A. \quad (\text{QT}_5)$$

After choosing compatible roots  $a_k^3 = a_{k-1}$  and keeping  $c$  fixed, the degree-three norm swaps the two factors:

$$N(U_k) = U'_{k-1}, \quad N(U'_k) = U_{k-1}, \quad N(R_k) = R_{k-1}^{-1}. \quad (\text{QT}_6)$$

Consequently

$$\mathcal{E}_k := R_k^{(-1)^k} \text{ satisfies } N(\mathcal{E}_k) = \mathcal{E}_{k-1}. \quad (\text{QT}_7)$$

Finally the cubic and exceptional arms are the untwisted and quadratic-modulo-five twists of the same simple exceptional zero. On the current Kummer line the 3-power cyclotomic generator acts as  $+2$  for  $\ell \mid \Phi_h(2)$ , and as  $-2$  for  $\ell \mid \Phi_h(-2) = \Phi_{2h}(2)$ . With  $r = h/3$ , the relative norm polynomial

$$g_r(X) = 1 + X^r + X^{2r}$$

has a simple zero at  $X = 2$  in the cubic case and at  $X = -2$  in the exceptional case. Hence the norm coherence in  $(\text{QT}_7)$  transports old factors but annihilates the current one. It does not prove the still-open nonvanishing in  $(\text{QT}_4)$ ; both arms require the distinguished first-order unit regulator in their respective  $\pm 2$ -eigenspaces.

*Proof.* The coprime orders  $m$  and  $5$  give the unique decomposition  $\tilde{\rho} = \alpha\beta$ . The four automorphisms of  $\mathbb{Q}(\mu_{5m})/\mathbb{Q}(\mu_m)$  fix  $\alpha$  and send  $\beta$  through the four primitive fifth roots. Therefore

$$\prod_{j=1}^4 (1 - \alpha\beta^j) = \Phi_5(\alpha) = \frac{1 - \alpha^5}{1 - \alpha},$$

which proves  $(\text{QT}_2)$ . The two values  $s, s'$  are the roots of  $X^2 + X - 1$ , so their interchange is the nontrivial automorphism of  $\mathbb{Q}(\sqrt{5})$ . It interchanges the numerator and denominator of  $d_k$ . Complex conjugation replaces  $\alpha$  by  $\alpha^{-1}$  and multiplies both quadratic factors by  $\alpha^{-2}$ , so their quotient is fixed. This proves all the characteristic-zero assertions preceding  $(\text{QT}_2)$ .

The residue of  $\alpha$  has order  $m$  and lies in  $K$ , while the residue of  $\beta$  lies in  $\mathbb{F}_{16}$ . Since

$$2^{2h} \equiv 1 \pmod{m}, \quad 2^{2h} \equiv -1 \pmod{5},$$

the local norm from  $L = \mathbb{F}_{2^{4h}}$  to  $K$  gives

$$N_{L/K}(1 + ab) = (1 + ab)(1 + ab^{-1}) = U_k.$$

Also  $2^{2h} + 1 \equiv 5 \pmod{15}$ , so  $N_{L/K}(u) = u^5 = \omega$ . Hence the corrected norm  $N$  of Proposition 8.16 is  $\omega U_k$ . Because  $\ell \neq 3$ , it has the same  $\ell$ -Kummer class as  $U_k$ .

It remains to show that the invariant product is blind. If  $x = 1 + a$ , then  $a^q = a^{-1}$  and

$$x^{q-1} = a^{-1}.$$

Here  $\ell \nmid q - 1$ , whereas  $a$  has order  $m$ , prime to  $\ell$ . Projection to the  $\ell$ -Sylow subgroup of  $K^\times$  therefore shows that  $1 + a$  is an  $\ell$ -th power. The same argument applies to  $1 + a^5$ . Thus

$$U_k U'_k = \Phi_5(a) = \frac{1 + a^5}{1 + a} \in (K^\times)^\ell.$$

In additive notation on the one-dimensional Kummer quotient this says  $[U'_k] = -[U_k]$ , and hence  $[R_k] = 2[U_k]$ . Since  $\ell$  is odd, Proposition 8.16 now proves  $(\text{QT}_4)$ .

As  $h$  is odd,  $q$ -Frobenius sends  $c$  to  $c^2$ . Directly,

$$U_k^q = a^{-2} U'_k, \quad (U'_k)^q = a^{-2} U_k,$$

which gives  $R_k^q = R_k^{-1}$ . Moreover

$$U_k + U'_k = a, \quad U_k U'_k = \Phi_5(a).$$

It follows that

$$R_k + R_k^{-1} = \frac{a^2}{\Phi_5(a)} = \frac{1}{(a + a^{-1})^2 + (a + a^{-1}) + 1}.$$

The last identity in  $(\mathbf{QT}_5)$  is the characteristic-two identity  $R/(R+1)^2 = 1/(R+R^{-1})$ .

The nonvanishing in  $(\mathbf{QT}_4)$  is unchanged when  $a$  is replaced by any primitive  $m$ -th root or  $c$  is interchanged with  $c^2$ : the former roots form one Frobenius orbit, and odd Frobenius powers merely perform the latter interchange. We may therefore choose  $a_k^3 = a_{k-1}$ . Factoring

$$U_k = (1 + a_k b)(1 + a_k b^{-1})$$

and, after the harmless quadratic scalar extension adjoining  $b$ , taking the product over the three conjugates obtained by multiplying  $a_k$  by a cube root of unity gives

$$N(U_k) = (1 + a_{k-1} b^3)(1 + a_{k-1} b^{-3}) = U'_{k-1}.$$

The same calculation with  $b^2$  gives  $N(U'_k) = U_{k-1}$ , proving  $(\mathbf{QT}_6)$ – $(\mathbf{QT}_7)$ .

For the last assertion, let  $\sigma$  be the 3-power cyclotomic generator. In the untwisted real field its reduction is Frobenius, so it acts as 2 on the current  $\ell$ -Kummer line. In the quadratic twist, rational Frobenius is the product of  $\sigma$  and the  $\sqrt{5}$ -involution. The latter reduces to  $q$ -Frobenius and acts as  $q = -1$  because  $\ell \mid q+1$ ; hence  $\sigma$  acts as  $-2$ . If  $\ell \mid \Phi_h(2)$ , then  $2^r$  is a primitive cube root modulo  $\ell$ . If  $\ell \mid \Phi_h(-2)$  and  $\ell \neq 3$ , then  $(-2)^r$  is a primitive cube root instead. Thus  $g_r$  vanishes at the asserted scalar. Its derivative there is

$$g'_r(X) = rX^{r-1}(1 + 2X^r) \not\equiv 0 \pmod{\ell} :$$

one has  $\ell \nmid r$ , and a primitive cube root equal to  $-1/2$  would force  $\ell \mid 9$ . The zero is therefore simple. The final open/non-open distinction follows: the norm operator vanishes on the current line, while its first derivative only identifies the one-dimensional regulator coordinate and does not evaluate  $d_k$  in that coordinate.  $\square$

**Proposition 8.37** (the four binary Euler blocks are the antiunit phase). *Retain Propositions 8.31 and 8.36. Put*

$$q = 2^h, \quad Q = q^4, \quad d = \frac{q+1}{\ell}, \quad E = \frac{Q-1}{\ell},$$

and define

$$A_0 = q - d, \quad B_0 = d - 1.$$

Then  $A_0 + B_0 = q - 1$ , so  $A_0$  and  $B_0$  are complementary  $h$ -bit words, and

$$\boxed{E = A_0 + B_0 q + A_0 q^2 + B_0 q^3, \quad \text{wt}_2(E) = 2h.} \quad (\text{DB}_1)$$

Thus the complete binary expansion of the exceptional Euler exponent has four alternating blocks.

Use the residue notation from  $(\mathbf{QT}_3)$ : in  $L = \mathbb{F}_{q^4}$ , write

$$\rho = ab, \quad \text{ord}(a) = 3h, \quad \text{ord}(b) = 5,$$

and let  $R_k = \overline{d_k} \in K = \mathbb{F}_{q^2}$ . Then the four-block factorization has the exact oriented value

$$\boxed{\begin{aligned} (1 + \rho)^E &= (1 + \rho)^{A_0} (1 + \rho^q)^{B_0} (1 + \rho^{q^2})^{A_0} (1 + \rho^{q^3})^{B_0} \\ &= \left( \frac{(1 + \rho^q)(1 + \rho^{q^3})}{(1 + \rho)(1 + \rho^{q^2})} \right)^d = R_k^{-d}. \end{aligned}} \quad (\text{DB}_2)$$

Consequently

$$\boxed{D'_k \text{ holds at } \ell \iff (1 + \rho)^E \neq 1 \iff R_k^d \neq 1.} \quad (\text{DB}_3)$$

Hence the four binary submask blocks in Proposition 8.31 and the conductor-five antiunit in Proposition 8.36 are inverse presentations of the same selected phase, not independent constraints. The identity does not prove that this phase is nontrivial.

*Proof.* The order identity  $\text{ord}_\ell(2) = 2h$  gives  $\ell \mid q + 1$ . Since  $q + 1 = \ell d$ , direct algebra gives

$$\frac{q^4 - 1}{\ell} = d(q - 1)(1 + q^2) = A_0 + B_0 q + A_0 q^2 + B_0 q^3,$$

which is (DB<sub>1</sub>). The equality  $q - d = (q - 1) - (d - 1)$  says exactly that the two  $h$ -bit blocks are bitwise complements, proving the weight assertion.

In characteristic two, Frobenius and (DB<sub>1</sub>) give the first line of (DB<sub>2</sub>). Put

$$D_0 = (1 + \rho)(1 + \rho^{q^2}), \quad N_0 = (1 + \rho^q)(1 + \rho^{q^3}).$$

These are nonzero because  $\rho$  has order  $15h$ , and  $D_0^q = N_0$ . Therefore

$$D_0^{q-d} N_0^{d-1} = (N_0/D_0)^d,$$

which proves the middle equality.

Now  $a^q = a^{-1}$ ,  $a^{q^2} = a$ , and  $b^q \in \{b^2, b^{-2}\}$ , while  $b^{q^2} = b^{-1}$ . Hence

$$D_0 = U_k, \quad N_0 = a^{-2} U'_k, \quad \frac{N_0}{D_0} = a^{-2} R_k^{-1}.$$

Also  $3h \mid q + 1$ , and  $\ell \nmid 3h$ , so  $3h \mid d$ . Thus  $a^{-2d} = 1$ , proving the last equality in (DB<sub>2</sub>).

Finally  $R_k^q = R_k^{-1}$ , so  $R_k$  lies in the cyclic norm-one group of order  $q + 1 = \ell d$ . Its Kummer class is trivial exactly when  $R_k^d = 1$ . Equations (DP<sub>3</sub>) and (QT<sub>4</sub>) now give (DB<sub>3</sub>).

Lean kernel-checks the four-block exponent identity and its cross-multiplied commutative-group form. The finite-field Frobenius, binary-complement, and antiunit specializations remain paper-level.  $\square$

**Proposition 8.38** (alternating translated root chain and the full-norm boundary). *Put  $E_j = \mathbb{F}_{2^{2 \cdot 3j}}$ , choose compatible primitive  $3^{j+1}$ -st roots  $a_j \in E_j$  with  $a_j^3 = a_{j-1}$ , and put  $\gamma_j = a_j + a_j^{-1}$ , so that  $\gamma_0 = 1$  and  $\gamma_j^3 + \gamma_j = \gamma_{j-1}$ . Fix the orientation  $c_k = c \in \mathbb{F}_4 \setminus \mathbb{F}_2$  used in (QT<sub>3</sub>), and define recursively*

$$c_{j-1} = c_j^2, \quad u_j = \gamma_j + c_j \quad (0 \leq j \leq k).$$

Then, for every  $j \geq 1$ ,

$$\boxed{u_j^3 + c_j u_j^2 + c_j u_j = u_{j-1}, \quad N_{E_j/E_{j-1}}(u_j) = u_{j-1}.} \quad (\text{AT}_1)$$

More precisely,

$$T^3 + c_j T^2 + c_j T + u_{j-1} \quad (\text{AT}_2)$$

is the irreducible minimal polynomial of  $u_j$  over  $E_{j-1}$ . The same translate is an Artin–Schreier root of the actual auxiliary coefficient:

$$u_j^2 + u_j = g_j = \gamma_j^2 + \gamma_j + 1. \quad (\text{AT}_3)$$

Let  $\ell \mid \Psi_k$  be current, so  $\text{ord}_\ell(2) = 2 \cdot 3^k$ . Then exceptional failure at  $\ell$  is exactly

$$u_k \in (E_k^\times)^\ell. \quad (\text{AT}_4)$$

Equivalently, for the normalized root  $w_k$  of  $(\text{CD}_7)$ ,

$$w_k \in (E_k^\times)^\ell \iff u_k \in (E_k^\times)^\ell. \quad (\text{AT}_5)$$

This is an equivalence of the two vanishing tests; it does not identify  $w_k$  with  $u_k$ , nor their Kummer classes with a specified scalar. Their Artin–Schreier coefficients are respectively  $Z_k = g_k^{2^{2 \cdot 3^k - 1} - 2}$  and  $g_k$ .

Under failure choose  $v_k \in E_k^\times$  with  $v_k^\ell = u_k$ . For every  $j < k$ , let  $v_j \in E_j^\times$  be the unique element with  $v_j^\ell = u_j$ . Then

$$\boxed{N_{E_j/E_{j-1}}(v_j) = v_{j-1} \quad (1 \leq j \leq k), \quad v_0 = u_0 = c_0^2.} \quad (\text{AT}_6)$$

Consequently a hypothetical failure is compatible with the complete selected cubic recursion, every downward multiplicative norm, and the unique lower Kummer roots, all the way to  $\mathbb{F}_4$ . Thus no descent which factors only through norms or Kummer corestriction can contradict failure at the bottom. This does not exclude a nonlinear additive–multiplicative relation involving an  $\ell$ -th root and the full twisted cubics  $(\text{AT}_2)$ .

*Proof.* Since  $c_j^2 + c_j = 1$ , substituting  $\gamma_j = u_j + c_j$  into  $\gamma_j^3 + \gamma_j = \gamma_{j-1}$  gives the first identity in  $(\text{AT}_1)$ . The finite-field degree calculation  $[E_{j-1}(\gamma_j) : E_{j-1}] = 3$ —equivalently, the quadratic constant extension is disjoint from the selected cubic extension—shows that  $T^3 + T + \gamma_{j-1}$  remains irreducible over  $E_{j-1}$ . Translation by  $c_j$  then proves  $(\text{AT}_2)$ , and its constant coefficient gives the norm identity. Directly,

$$(\gamma_j + c_j)^2 + (\gamma_j + c_j) = \gamma_j^2 + \gamma_j + (c_j^2 + c_j) = g_j,$$

which is  $(\text{AT}_3)$ .

At the top level,  $(\text{QT}_3)$  has

$$U_k = a_k^2 + c_k a_k + 1 = a_k(a_k + a_k^{-1} + c_k) = a_k u_k.$$

Because  $\ell \nmid 3^{k+1} = \text{ord}(a_k)$ , the element  $a_k$  is an  $\ell$ -th power in  $E_k^\times$ . Hence  $(\text{QT}_4)$  gives  $(\text{AT}_4)$ . Proposition 8.7 gives failure iff  $w_k$  is an  $\ell$ -th power, proving  $(\text{AT}_5)$  without identifying the two roots.

For  $j < k$ , the order identity for  $\ell$  gives  $\ell \nmid 2^{2 \cdot 3^j} - 1$ ; hence  $\ell$ -powering is bijective on  $E_j^\times$ . Multiplicativity of norm now gives

$$N(v_j)^\ell = N(v_j^\ell) = N(u_j) = u_{j-1} = v_{j-1}^\ell,$$

and injectivity downstairs proves  $(\text{AT}_6)$ . Finally  $u_0 = 1 + c_0 = c_0^2$ , while  $\text{ord}_\ell(2) = 2 \cdot 3^k \mid \ell - 1$  implies  $\ell \equiv 1 \pmod{3}$ ; thus  $(c_0^2)^\ell = c_0^2$ , giving the stated terminal root.

Lean kernel-checks the characteristic-two translate, Artin–Schreier, and symmetric product identities, multiplication by a known power, and the norm-of-a-power-root uniqueness step. The finite-field tower, order, and Kummer interpretation remain paper-level.  $\square$

**Proposition 8.39** (alternating twisted-cubic Kummer fibre). *Retain Proposition 8.38, and let  $\ell \mid \Psi_k$  be current. For  $0 \leq i < k$ , let  $v_i \in E_i^\times$  be the unique element satisfying*

$$v_i^\ell = u_i.$$

Thus  $v_0 = u_0 = c_0^2$ , and

$$N_{E_i/E_{i-1}}(v_i) = v_{i-1} \quad (1 \leq i < k). \quad (\text{TF}_1)$$

Use the generalized cubic power map of Proposition 6.54,

$$\Psi_{\ell,E}(C, D) = (T_\ell(C, D, E), T_\ell(D, EC, E^2)).$$

For  $1 \leq j \leq k$ , put  $K_j = E_{j-1}$  and define the fixed-constant coefficient fibre

$$\mathcal{X}_j = \Psi_{\ell, v_{j-1}}^{-1}(c_j, c_j) \subset \mathbb{A}_{K_j}^2. \quad (\text{TF}_2)$$

Then  $\mathcal{X}_j$  is finite reduced étale of geometric degree  $\ell^2$ . Every  $K_j$ -rational point  $(C, D)$  gives an irreducible cubic

$$X^3 + CX^2 + DX + v_{j-1}. \quad (\text{TF}_3)$$

If  $J$  denotes the Jacobian determinant of  $\Psi_{\ell, v_{j-1}}$ , then at every geometric point of  $\mathcal{X}_j$ ,

$$(CD + v_{j-1})J = c_j^2 + u_{j-1} = \gamma_{j-1} \neq 0. \quad (\text{TF}_4)$$

Its rational-point counts are

$$\#\mathcal{X}_j(K_j) = 1 \quad (1 \leq j < k), \quad (\text{TF}_5)$$

where the unique point is the coefficient pair of the minimal polynomial of  $v_j$ , while

$$\#\mathcal{X}_k(E_{k-1}) = \begin{cases} 0, & u_k \notin (E_k^\times)^\ell, \\ \ell, & u_k \in (E_k^\times)^\ell. \end{cases} \quad (\text{TF}_6)$$

More explicitly, put

$$Q = |E_{k-1}| = 2^{2 \cdot 3^{k-1}}, \quad P_k(X) = X^3 + c_k X^2 + c_k X + u_{k-1}.$$

Under failure choose  $v_k^\ell = u_k$  and a primitive  $\rho \in \mu_\ell \subset E_k$ . For  $r \in \mathbb{Z}/\ell\mathbb{Z}$  set

$$\begin{aligned} C_r &= \rho^r v_k + \rho^{rQ} v_k^Q + \rho^{rQ^2} v_k^{Q^2}, \\ D_r &= \rho^{r(1+Q)} v_k^{1+Q} + \rho^{r(Q+Q^2)} v_k^{Q+Q^2} + \rho^{r(Q^2+1)} v_k^{Q^2+1}. \end{aligned} \quad (\text{TF}_7)$$

Then  $C_r, D_r \in E_{k-1}$ , the resulting cubics are pairwise distinct and irreducible, and

$$P_k(X^\ell) = \prod_{r \bmod \ell} (X^3 + C_r X^2 + D_r X + v_{k-1}). \quad (\text{TF}_8)$$

On success  $P_k(X^\ell)$  is irreducible of degree  $3\ell$ . Every factor in (TF<sub>8</sub>) has constant coefficient  $v_{k-1}$ , whose unique norm-coherent descendants are precisely  $v_{k-2}, \dots, v_0 = c_0^2$ . Thus every bad top lift retains the same complete selected lower Kummer ancestry.

*Proof.* For  $i < k$ , the order identity  $\text{ord}_\ell(2) = 2 \cdot 3^k$  gives  $\ell \nmid |E_i^\times|$ , so  $\ell$ -powering is bijective on  $E_i^\times$ . This proves existence and uniqueness of  $v_i$ ; the norm identity in Proposition 8.38 then proves (TF<sub>1</sub>).

Let  $y_1, y_2, y_3$  be the three roots of

$$X^3 + c_j X^2 + c_j X + u_{j-1}.$$

Choose independently an  $\ell$ -th root  $x_1$  of  $y_1$  and an  $\ell$ -th root  $x_2$  of  $y_2$ , and set

$$x_3 = \frac{v_{j-1}}{x_1 x_2}.$$

Since  $y_1 y_2 y_3 = u_{j-1} = v_{j-1}^\ell$ , one has  $x_3^\ell = y_3$ . Hence there are exactly  $\ell^2$  geometric source triples with product  $v_{j-1}$ . The distinct target roots recover the labels after powering, so two such triples give the same coefficient point only when they agree. This proves the geometric degree.

The universal identity

$$(CD + E)J = AB + E^\ell$$

from Proposition 6.54, specialized to  $A = B = c_j$  and  $E = v_{j-1}$ , gives

$$(CD + v_{j-1})J = c_j^2 + v_{j-1}^\ell = c_j^2 + u_{j-1}.$$

But

$$u_{j-1} = \gamma_{j-1} + c_{j-1} = \gamma_{j-1} + c_j^2,$$

so the right-hand side is  $\gamma_{j-1}$ . This is nonzero:  $\gamma_{j-1} = a_{j-1} + a_{j-1}^{-1} = 0$  would imply  $a_{j-1}^2 = 1$ , contradicting its primitive  $3^j$ -order. Thus both  $CD + v_{j-1}$  and  $J$  are nonzero at every point, proving reducedness and étaleness.

A reducible cubic (TF<sub>3</sub>) over the finite field  $K_j$  would have a  $K_j$ -root. Its  $\ell$ -th power would be a  $K_j$ -root of the irreducible selected target cubic, which is impossible. Hence every rational point is irreducible.

For  $j < k$ ,  $\ell$ -powering is bijective on  $E_j^\times$ , so the three target roots have one Frobenius orbit of unique preimages. Their common coefficient pair is therefore the sole  $K_j$ -rational point. Its constant coefficient is  $v_{j-1}$ , because its norm has  $\ell$ -th power  $u_{j-1}$  and powering is injective on  $K_j^\times$ . This proves (TF<sub>5</sub>).

At  $j = k$ , a rational point supplies an  $\ell$ -th root in  $E_k$  of a conjugate of  $u_k$ , hence, after Frobenius conjugation, an  $\ell$ -th root of  $u_k$ . Thus there are no rational points on success. Under failure the  $\ell$  elements  $\rho^r v_k$  are exactly the roots above the chosen root  $u_k$ . Two of them could give the same coefficient pair only if  $\rho^r v_k$  were a  $Q$ -Frobenius conjugate of  $\rho^s v_k$ . Raising to the  $\ell$ -th power would give  $u_k = u_k^{Q^i}$ , forcing  $i = 0$  by the selected cubic's irreducibility, and then  $r = s$ . This proves (TF<sub>6</sub>).

Finally  $\text{ord}_\ell(Q) = 3$ , so  $1 + Q + Q^2 \equiv 0 \pmod{\ell}$ . Frobenius cycles the three roots

$$\rho^r v_k, \quad \rho^{rQ} v_k^Q, \quad \rho^{rQ^2} v_k^{Q^2},$$

and their product is

$$\rho^{r(1+Q+Q^2)} N_{E_k/E_{k-1}}(v_k) = v_{k-1}.$$

Their elementary coefficients are exactly (TF<sub>7</sub>), proving (TF<sub>8</sub>). On success,  $X^\ell - u_k$  is irreducible over  $E_k$  by Kummer theory, and Capelli's theorem makes  $P_k(X^\ell)$  irreducible over  $E_{k-1}$ .  $\square$

**Proposition 8.40** (Fourier diagonalization of the twisted Kummer fibre). *Retain Proposition 8.39, put*

$$K = E_{k-1}, \quad Q = |K|, \quad e = \frac{Q^3 - 1}{\ell},$$

*and choose, in an algebraic closure,  $v^\ell = u_k$  and a primitive  $\rho \in \mu_\ell \subset E_k$ . Define  $C_r, D_r$  by (TF<sub>7</sub>), whether or not they are  $K$ -rational, and put*

$$\widehat{C}_m = \sum_{r \bmod \ell} \rho^{-mr} C_r, \quad \widehat{D}_m = \sum_{r \bmod \ell} \rho^{-mr} D_r.$$

*All indices below are modulo  $\ell$ . The transforms have exactly six nonzero coordinates:*

$$\widehat{C}_m = \begin{cases} v, & m = 1, \\ v^Q, & m = Q, \\ v^{Q^2}, & m = Q^2, \\ 0, & \text{otherwise,} \end{cases} \quad (\text{WF}_1)$$

*and*

$$\widehat{D}_m = \begin{cases} v^{1+Q}, & m = 1 + Q = -Q^2, \\ v^{Q+Q^2}, & m = Q + Q^2 = -1, \\ v^{Q^2+1}, & m = Q^2 + 1 = -Q, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{WF}_2)$$

*Thus the first weighted resolvent is literally  $\widehat{C}_1 = v$ ; the three  $D$ -coordinates contain no further datum.*

*More precisely, put*

$$n = v^{1+Q+Q^2}, \quad \Delta = \frac{n}{v_{k-1}} \in \mu_\ell, \quad \Omega = u_k^e \in \mu_\ell.$$

*Then*

$$n^\ell = u_{k-1}, \quad \boxed{\Delta^{Q-1} = \Omega}. \quad (\text{WF}_3)$$

*The element  $n$ , and hence  $\Delta$ , is unchanged under  $v \mapsto \rho^a v$ . Since  $\gcd(Q-1, \ell) = 1$ ,*

$$\begin{aligned} D'_k \text{ fails at } \ell &\iff \Omega = 1 \iff \Delta = 1 \iff n = v_{k-1} \\ &\iff C_r, D_r \in K \quad \text{for every } r. \end{aligned} \quad (\text{WF}_4)$$

*The last equality retains the complete lower ancestry exactly:  $v_{k-1}$  has the unique norm-coherent descendants  $v_{k-2}, \dots, v_0 = c_0^2$ . No descent of the individual  $C_r, D_r$  is asserted.*

*There is also a sharp rational-eigenweight boundary. Let  $\mu_\ell$  act on  $E_k(V)$  by  $V \mapsto \rho V$ . For every  $m \in \mathbb{Z}/\ell\mathbb{Z}$ ,*

$$\{F : F(\rho V) = \rho^m F(V)\} = V^m E_k(V^\ell). \quad (\text{WF}_5)$$

*If  $F(V) = V^m R(V^\ell)$ , with  $F(v)$  defined and nonzero, then*

$$F(v)^{Q^3} = \Omega^m F(v). \quad (\text{WF}_6)$$

*Consequently, for  $m \neq 0$ ,*

$$F(v) \in E_k \iff \Omega = 1.$$

*Thus every nonzero pure Kummer eigenweight carries an invertible power of the original selected obstruction; it cannot supply a second condition. Mixed-weight cancellations, vanishing specializations, and independently normalized characteristic-zero constructions lie outside this assertion.*



*Proof.* The character orthogonality relation is

$$\sum_{r \bmod \ell} \rho^{r(a-m)} = \begin{cases} 1, & a = m, \\ 0, & a \neq m, \end{cases}$$

because  $\ell = 1$  in characteristic two. Applying it termwise to  $(\text{TF}_7)$  proves  $(\text{WF}_1)$ – $(\text{WF}_2)$ . The congruence  $1 + Q + Q^2 = 0 \pmod{\ell}$  shows that  $n$  is independent of the chosen root  $v$ . The selected norm identity gives

$$n^\ell = u_k^{1+Q+Q^2} = N_{E_k/K}(u_k) = u_{k-1}.$$

Therefore  $n/v_{k-1} \in \mu_\ell$ , and

$$\Delta^{Q-1} = v^{(1+Q+Q^2)(Q-1)} = v^{Q^3-1} = u_k^{(Q^3-1)/\ell} = \Omega,$$

which proves  $(\text{WF}_3)$ . Since  $Q - 1$  is invertible modulo  $\ell$ , the first four equivalences in  $(\text{WF}_4)$  follow. If all  $C_r$  lie in  $K$ , Fourier inversion gives  $v = \widehat{C}_1 \in E_k$ ; the converse follows because  $Q$ -Frobenius then cycles each displayed triple. This proves the last equivalence.

Finally, dividing a weight- $m$  rational function by  $V^m$  gives a weight-zero function, proving  $(\text{WF}_5)$ . Since  $v^{Q^3} = \Omega v$ , specialization gives  $(\text{WF}_6)$ .

Lean kernel-checks the pseudonorm root, root-choice invariance, normalized defect–Euler identity, and pure-eigenweight monodromy. The finite-field Fourier transform, fixed-field description, and selected specialization remain paper-level.  $\square$

**Proposition 8.41** (mixed-weight saturation of the twisted Kummer fibre). *Retain Proposition 8.40, put*

$$F = E_k, \quad u = u_k, \quad \mathcal{A} = F[T]/(T^\ell - u),$$

and let  $t$  be the class of  $T$ . Root scaling  $\tau(t) = \rho t$  gives the exact grading

$$\mathcal{A} = \bigoplus_{m \bmod \ell} \mathcal{A}_m, \quad \mathcal{A}_m = Ft^m, \quad \mathcal{A}^{(\tau)} = F. \quad (\text{MW}_1)$$

Under the Fourier identification of  $(\text{WF}_1)$ – $(\text{WF}_2)$ , the six nonzero modes are

$$t, t^Q, t^{Q^2}, t^{1+Q}, t^{Q+Q^2}, t^{Q^2+1}. \quad (\text{MW}_2)$$

They generate  $\mathcal{A}$ , and the quadratic modes are just products of the linear ones:

$$\widehat{D}_{1+Q} = \widehat{C}_1 \widehat{C}_Q, \quad \widehat{D}_{Q+Q^2} = \widehat{C}_Q \widehat{C}_{Q^2}, \quad \widehat{D}_{Q^2+1} = \widehat{C}_{Q^2} \widehat{C}_1. \quad (\text{MW}_3)$$

The three complementary mixed products all equal the pseudonorm

$$\boxed{\widehat{C}_1 \widehat{D}_{Q+Q^2} = \widehat{C}_Q \widehat{D}_{Q^2+1} = \widehat{C}_{Q^2} \widehat{D}_{1+Q} = n = t^{1+Q+Q^2}.} \quad (\text{MW}_4)$$

Writing  $s = (1+Q+Q^2)/\ell$ , one has  $n = u^s$ ; its normalized defect is the  $\Delta$  of  $(\text{WF}_3)$ , with  $\Delta^{Q-1} = \Omega$ .

More generally, if a monomial has total exponent  $W = \ell a + m$ ,  $0 \leq m < \ell$ , then

$$t^W = u^a t^m. \quad (\text{MW}_5)$$

Every polynomial expression in the coefficient family therefore has a unique weight decomposition  $\sum_m a_m t^m$ , and every expression invariant under the choice  $t \mapsto \rho t$  has weight zero. The statement

is unchanged after adjoining the exact Artin–Schreier coefficient  $g_k = u^2 + u$  and all lower  $v_i, c_i, g_i$ , since they are fixed coefficients in  $F$ . It also covers rational expressions regular on the whole Kummer fibre, because their denominators are units of  $\mathcal{A}$ .

If  $D'_k$  holds,  $X^\ell - u$  is irreducible and evaluation into the Kummer field is injective, so an  $F$ -valued mixed specialization forces every nonzero-weight coefficient to vanish separately. If  $D'_k$  fails,  $\mathcal{A} \simeq \prod_{r \bmod \ell} F$ , and evaluation at one root is a projection. A cross-weight cancellation then lies in the kernel of a chosen projection and depends on selecting one  $\ell$ -th root, a choice absent from the Conway ancestry. Hence mixed Fourier resolvents, adjacent twisted cubics, and  $g_k$  supply no second root-choice-invariant condition. A surviving proof must directly evaluate a weight-zero arithmetic function of  $u$ , in particular  $\Omega = u^{(Q^3-1)/\ell}$ , or provide genuinely external  $\ell$ -root orientation.

*Proof.* Monicity of  $T^\ell - u$  gives the basis  $1, t, \dots, t^{\ell-1}$ . The scaling action has the distinct eigenvalues  $1, \rho, \dots, \rho^{\ell-1}$ , which proves (MW<sub>1</sub>). Substitution from (WF<sub>1</sub>)–(WF<sub>2</sub>) proves (MW<sub>2</sub>)–(MW<sub>4</sub>). Euclidean division of  $W$  and  $t^\ell = u$  give (MW<sub>5</sub>), and uniqueness follows from the power basis. Fourier inversion identifies expressions in  $(C_r, D_r)_r$  with expressions in the six modes. Fixed ancestry scalars do not alter weights. The final field/split alternatives are the prime Kummer criterion.

The assertion covers polynomial expressions and rational expressions regular on every branch. A rational expression with poles on other branches is not covered: its domain has already selected a factor. Nor does the proposition evaluate a special coefficient  $a_m(u, g_k)$ ; proving one nonzero would be direct selected arithmetic. Lean kernel-checks the three complementary products, the collapse  $t^{\ell a} = u^a$ , and the abstract power-basis no-cancellation step. The quotient-algebra grading and invariant-ring specialization remain paper-level.  $\square$

**Proposition 8.42** (immediate cubic trace of the selected phase). *Retain Proposition 8.40, and write*

$$t = 2^{3^{k-1}}, \quad Q = t^2, \quad E_k = \mathbb{F}_{Q^3}, \quad E_{k-1} = \mathbb{F}_Q, \quad e = \frac{Q^3 - 1}{\ell}.$$

*For the selected weight-zero phase*

$$\Omega = u_k^e \in \mu_\ell$$

*put*

$$\mathcal{S}(\Omega) = \text{Tr}_{E_k/E_{k-1}}(\Omega) = \Omega + \Omega^Q + \Omega^{Q^2}.$$

*Then*

$$\boxed{D'_k \text{ fails at } \ell \iff \Omega = 1 \iff \mathcal{S}(\Omega) = 1.} \quad (\text{ICT}_1)$$

*Equivalently,*

$$D'_k \text{ holds at } \ell \iff \mathcal{S}(\Omega) \neq 1. \quad (\text{ICT}_2)$$

*This trace retains the selected twisted-cubic ancestry explicitly. Writing  $u = u_k$ ,  $c = c_k$ , and  $u_- = u_{k-1}$ , define*

$$P_n = \text{Tr}_{E_k/E_{k-1}}(u^n).$$

*The selected cubic*

$$u^3 + cu^2 + cu = u_-$$

*gives*

$$\begin{aligned} P_0 &= 1, & P_1 &= c, & P_2 &= c^2, \\ P_n &= cP_{n-1} + cP_{n-2} + u_-P_{n-3} \quad (n \geq 3), \end{aligned} \quad (\text{ICT}_3)$$

*and*

$$\mathcal{S}(\Omega) = P_e. \quad (\text{ICT}_4)$$

*Thus the remaining assertion is the exact selected value  $P_e \neq 1$ .*

*Proof.* Since  $\ell \mid \Psi_k$ , one has

$$t^2 - t + 1 \equiv 0 \pmod{\ell}.$$

Hence  $t^3 \equiv -1 \pmod{\ell}$ , while

$$1 + Q + Q^2 = 1 + t^2 + t^4 = (t^2 - t + 1)(t^2 + t + 1) \equiv 0 \pmod{\ell}.$$

Therefore

$$\Omega^{1+Q+Q^2} = 1. \quad (\text{ICT}_5)$$

Suppose that  $\mathcal{S}(\Omega) = 1$ . Applying the  $t$ -power Frobenius and using

$$t \equiv -Q^2, \quad t^3 \equiv -1, \quad t^5 \equiv -Q \pmod{\ell}$$

gives

$$1 = \mathcal{S}(\Omega)^t = \Omega^{-Q^2} + \Omega^{-1} + \Omega^{-Q}. \quad (\text{ICT}_6)$$

Put

$$x = \Omega, \quad y = \Omega^Q, \quad z = \Omega^{Q^2}.$$

Equations (ICT<sub>5</sub>)–(ICT<sub>6</sub>) say

$$xyz = 1, \quad x^{-1} + y^{-1} + z^{-1} = 1.$$

Multiplying the latter equality by  $xyz$  yields

$$xy + xz + yz = 1.$$

Together with  $x + y + z = 1$ , all three elementary symmetric functions of  $x, y, z$  are one. Their cubic is therefore

$$(X + x)(X + y)(X + z) = X^3 + X^2 + X + 1 = (X + 1)^3.$$

Since  $E_k$  is a field,  $x = \Omega = 1$ . Conversely,  $\mathcal{S}(1) = 1 + 1 + 1 = 1$  in characteristic two. Proposition 8.40 identifies  $\Omega = 1$  with failure, proving (ICT<sub>1</sub>)–(ICT<sub>2</sub>).

Finally, summing the selected cubic equation over the three  $Q$ -conjugates gives the recurrence in (ICT<sub>3</sub>), with its initial values obtained from the first two power sums. Since  $\Omega = u^e$ , equation (ICT<sub>4</sub>) follows.  $\square$

**Proposition 8.43** (the immediate cubic phase trace is lossless). *Retain Proposition 8.42. For  $z \in \mu_\ell$ , put*

$$\mathcal{S}(z) = z + z^Q + z^{Q^2} \in \mathbb{F}_Q.$$

*The congruences*

$$t \equiv -Q^2, \quad Qt \equiv -1, \quad Q^2t \equiv -Q \pmod{\ell}$$

*give*

$$\boxed{\mathcal{S}(z)^t = \mathcal{S}(z^{-1}) = z^{-1} + z^{-Q} + z^{-Q^2}.} \quad (\text{ICO}_1)$$

*Consequently the trace separates complete  $Q$ -Frobenius orbits:*

$$\boxed{\mathcal{S}(z) = \mathcal{S}(w) \iff w \in \{z, z^Q, z^{Q^2}\}.} \quad (\text{ICO}_2)$$

Indeed the orbit polynomial is

$$\prod_{i=0}^2 (X + z^{Q^i}) = X^3 + \mathcal{S}(z)X^2 + \mathcal{S}(z)^t X + 1. \quad (\text{ICO}_3)$$

The quadratic norm of this lossless coordinate has the exact factorization

$$\mathcal{S}(z)^{t+1} + 1 = (z + z^Q)(z + z^{Q^2})(z^Q + z^{Q^2}). \quad (\text{ICO}_4)$$

Because  $\text{ord}_\ell(Q) = 3$ , it follows that

$$N_{\mathbb{F}_Q/\mathbb{F}_t}(\mathcal{S}(z)) = 1 \iff z = 1. \quad (\text{ICO}_5)$$

For  $z \neq 1$ , the cubic in  $(\text{ICO}_3)$  is irreducible over  $\mathbb{F}_Q$  and has nonzero discriminant

$$\text{disc}(X^3 + \mathcal{S}(z)X^2 + \mathcal{S}(z)^t X + 1) = (\mathcal{S}(z)^{t+1} + 1)^2. \quad (\text{ICO}_6)$$

For  $z = 1$ , it is  $(X + 1)^3$ .

Specializing  $z = \Omega = u_k^e$  and  $P_e = \mathcal{S}(\Omega)$  gives

$$D'_k \text{ fails at } \ell \iff P_e^{t+1} = 1. \quad (\text{ICO}_7)$$

Moreover the complete shifted power-sum recurrence supplies no further condition:

$$P_e = 1 \iff P_{n+e} = P_n \text{ for every } n \geq 0. \quad (\text{ICO}_8)$$

Thus  $\mathcal{S}$  is a complete invariant of the  $Q$ -Frobenius orbit. The named norm, discriminant, irreducibility, and shifted-periodicity conditions are equivalent tests only for the binary event  $\Omega = 1$ . Any phase-only orbit invariant factors through  $\mathcal{S}$ , but need not itself be injective. None of these tests evaluates  $P_e$ , and hence none proves the selected nonvanishing. Nonlinear expressions involving  $u_k$  before Euler projection or genuinely external ancestry input remain outside this phase-saturation statement.

*Proof.* The congruences preceding  $(\text{ICO}_1)$  follow from  $t^2 - t + 1 \equiv 0 \pmod{\ell}$ , and give that identity after raising the three summands to the  $t$ -th power. The three conjugates have product one. If  $\mathcal{S}(z) = \mathcal{S}(w)$ , then  $(\text{ICO}_1)$  also gives equality of their inverse traces, so the two norm-one triples have all three elementary symmetric functions equal. Their orbit cubics coincide, proving  $(\text{ICO}_2)$ ; the converse is immediate.

Expansion in characteristic two gives  $(\text{ICO}_4)$ . The right side vanishes exactly when two members of the  $Q$ -orbit coincide. Since  $Q$  has order three modulo  $\ell$ , this happens exactly at  $z = 1$ , proving  $(\text{ICO}_5)$ . The same root-difference product is the square root of the cubic discriminant, which proves  $(\text{ICO}_6)$ ; for  $z \neq 1$ , the three-element Frobenius orbit also proves irreducibility.

Proposition 8.42 gives  $P_e = 1 \iff \Omega = 1$ , and  $(\text{ICO}_5)$  gives  $(\text{ICO}_7)$ . If  $P_e = 1$ , then  $\Omega = u_k^e = 1$ , so every conjugate of  $u_k$  has  $e$ -th power one and  $P_{n+e} = P_n$  for every  $n$ . Conversely the case  $n = 0$  gives  $P_e = P_0 = 1$ . This proves  $(\text{ICO}_8)$ .

Lean kernel-checks the characteristic-two half-discriminant/root-difference factorization and collision criterion, distinguished-root membership from equal norm-one trace and inverse trace, and the forward shifted power-sum periodicity once all powered roots are one. The polynomial-discriminant identity, finite-field Frobenius congruences, orbit-separation specialization, irreducibility, converse periodicity implication, and identification with the selected phase remain paper-level.  $\square$

**Remark 8.44** (the cubic trace is a compression, not an evaluation). The six-term trace of Proposition 8.35 is the quadratic trace of  $\mathcal{S}(\Omega)$ :

$$\mathcal{T}(\Omega) = \text{Tr}_{\mathbb{F}_Q/\mathbb{F}_t}(\mathcal{S}(\Omega)) = \mathcal{S}(\Omega) + \mathcal{S}(\Omega)^t.$$

Thus Proposition 8.42 replaces the six-term detector by the immediate three-term trace in the preceding twisted-cubic field, but it supplies no independent nonvanishing. Evaluating  $P_e$  uniformly is another presentation of the already isolated selected terminal Dickson problem; norm coherence determines the recurrence, not its terminal value.

**Proposition 8.45** (conductor-five reflection descent). *Retain Proposition 8.36, put*

$$x = \alpha + \alpha^{-1}, \quad F = \mathbf{Q}(x), \quad K = F(\sqrt{5}) = F_k^{\text{tw}},$$

and let the prime  $\ell \mid \Psi_k$ . Set

$$B = F(\zeta_\ell), \quad M = K(\zeta_\ell).$$

Because  $\text{ord}_\ell(2) = 2h$ , one has  $\ell \nmid 15h$ . The extension  $K/\mathbf{Q}$  has conductor dividing  $15h$ , so coprime-conductor disjointness gives  $K \cap \mathbf{Q}(\zeta_\ell) = \mathbf{Q}$ . After base change from  $\mathbf{Q}$  to  $F$ , this gives

$$K \cap B = F, \quad [M : B] = 2, \quad [M : K] = \ell - 1.$$

Equivalently, one may see the first equality from ramification:  $K/F$  is ramified above five, whereas  $B/F$  is unramified there. Suppose first that  $d_k \notin (M^\times)^\ell$ , choose  $\theta^\ell = d_k$ , and put  $\mathcal{E}_D = M(\theta)$ . The nontrivial conductor-five involution  $j \in \text{Gal}(K/F)$  extends by

$$j(\zeta_\ell) = \zeta_\ell, \quad j(\theta) = \theta^{-1}.$$

For  $L_D = \mathcal{E}_D^{(j)}$  and  $t = \theta + \theta^{-1}$ , one has

$$\text{Gal}(\mathcal{E}_D/B) \simeq \text{the dihedral group of order } 2\ell, \quad [L_D : B] = \ell,$$

and the reflection field has the explicit lower-coordinate equation

$$\boxed{L_D = B(t), \quad D_\ell^{\text{cl}}(t) = 2 + \frac{5}{x^2 + x - 1}.} \quad (\text{DHR}_1)$$

Moreover

$$\boxed{D'_k \text{ fails at } \ell \iff 2 \in \text{N}_{L_D/B}(L_D^\times).} \quad (\text{DHR}_2)$$

The degree- $\ell$  extension  $L_D/B$  satisfies the Hasse norm principle, and its norm-one torus has weak approximation. At every prime  $\mathfrak{r} \mid 2$  of  $B$ , the local algebra is  $B_\ell^\ell$  on failure and the unramified degree- $\ell$  field on success. Every local norm condition for the element two away from the primes over two is automatic.

If  $d_k \in (M^\times)^\ell$ , then  $D'_k$  already fails and the dihedral construction degenerates. Thus  $\text{DHR}_2$  is an exact reflection-norm normal form for the exceptional arm, not a new proof of nonvanishing.

*Proof.* In the notation of  $\text{QT}_1$ ,  $d_k = (x - s)/(x - s')$ , with  $s + s' = -1$  and  $ss' = -1$ . Direct algebra gives

$$(x - s)(x - s') = x^2 + x - 1$$

and

$$(x - s)^2 + (x - s')^2 = 2(x^2 + x - 1) + 5.$$

The two denominators are nonzero: otherwise  $\alpha$  would have order five, contrary to  $\text{ord}(\alpha) = 3h$ . Therefore

$$d_k + d_k^{-1} = 2 + \frac{5}{x^2 + x - 1}. \quad (\text{DHR}_3)$$

The involution  $j$  interchanges  $s, s'$ , hence sends  $d_k$  to  $d_k^{-1}$ . If  $\sigma(\theta) = \zeta_\ell \theta$ , then  $j\sigma j^{-1} = \sigma^{-1}$ , proving the dihedral assertion. The classical power-sum identity gives  $\text{DHR}_1$ . Its  $\ell$  conjugates are distinct: a collision would imply  $\zeta_\ell^{a+b}\theta^2 = 1$ , hence  $d_k^2 = 1$ ; since  $\ell$  is odd, that would make  $d_k$  an  $\ell$ -th power.

The reflection symmetrization used in Corollary 7.73 works verbatim here, over both global fields and their completions. Thus, for every  $a \in B^\times$ ,

$$a \in N_{\mathcal{E}_D/M}(\mathcal{E}_D^\times) \iff a \in N_{L_D/B}(L_D^\times). \quad (\text{DHR}_4)$$

It remains to identify the selected norm. Each  $1 - \alpha\beta^r$  occurring in  $\text{QT}_1$  has exact conductor  $15h$ ; since this conductor has two distinct prime divisors,  $\Phi_{15h}(1) = 1$ , so these factors and their quotient  $d_k$  are global units. Let  $\mathfrak{p} \mid 2$  be the prime selected in Corollary 8.22, let  $\mathfrak{P} = \mathfrak{p} \cap K$ , and choose a compatible prime  $\mathfrak{q}_1 \mid \mathfrak{P}$  of  $M$ . Then  $k(\mathfrak{P}) = k(\mathfrak{q}_1) = \mathbb{F}_{2^{2h}}$ , this field contains  $\mu_\ell$ , and tame reciprocity gives

$$(d_k, 2)_{\mathfrak{q}_1, \ell} = R_k^{(2^{2h}-1)/\ell}. \quad (\text{DHR}_5)$$

By  $\text{QT}_4$ , this symbol is one exactly when  $D'_k$  fails. The prime  $\mathfrak{P}$  splits completely in  $M/K$ . Label its primes  $\mathfrak{q}_a = \tau_a \mathfrak{q}_1$ . Galois covariance gives

$$(d_k, 2)_{\mathfrak{q}_a, \ell} = (d_k, 2)_{\mathfrak{q}_1, \ell}^a,$$

while  $j$  pairs  $\mathfrak{q}_a$  with  $\mathfrak{q}_{-a}$ . Thus all two-adic symbols are one exactly in that same case. At every finite place away from  $2\ell$ , both arguments are units and the tame symbol is one. Because  $\text{ord}_\ell(2) = 2h$ , the prime  $\ell$  is coprime to the conductor  $15h$ ; hence  $K$  is unramified there, while  $M/K$  is cyclotomically totally ramified. Each prime above  $\ell$  is fixed by the full cyclotomic group. Covariance fixes the two inputs and raises the value in  $\mu_\ell$  by every  $a \in \mathbb{F}_\ell^\times$ , so the local symbol is one. The infinite places of  $M$  are complex.

The cyclic Hasse norm theorem for  $\mathcal{E}_D/M$ , followed by  $\text{DHR}_4$ , proves  $\text{DHR}_2$ . Completionwise, a prime  $v \mid 2$  of  $F$  has residue degree  $h$ . For every  $\mathfrak{r} \mid v$  of  $B$ , both  $K_{\mathfrak{P}}/F_v$  and  $B_{\mathfrak{r}}/F_v$  are the unique unramified quadratic extension. Hence  $M/B$  splits at  $\mathfrak{r}$ . Trivial selected Frobenius gives the split local  $L_D$ -algebra, while a nontrivial translation gives the unramified degree- $\ell$  field, whose norm valuations are divisible by  $\ell$  and therefore exclude  $v(2) = 1$ . This proves the local assertions. Finally, Hoshi–Yamasaki’s odd-dihedral norm-torus theorem supplies the Hasse norm principle and weak approximation independently of the explicit descent [18, Theorem 3.2 and Example 4.6].

Lean checks the two denominator-free identities, their quotient form  $\text{DHR}_3$ , and the generic reflection symmetrization. The number-field, local-symbol, Hasse, and weak-approximation interpretations remain paper-level.  $\square$

**Proposition 8.46** (selected exceptional sextic and the Dickson-factor boundary). *Retain Propositions 8.34, 8.35, and 8.36. Put*

$$s = 3^{k-1}, \quad A_0 = 2^s, \quad q = A_0^3 = 2^h, \quad \Lambda = A_0^2 - A_0 + 1 = 3\Psi_k,$$

and write

$$\gamma = \gamma_k, \quad g = \gamma_{k-1}, \quad C = g^2 + g + 1.$$

Choose the orientation  $c \in \mathbb{F}_4 \setminus \mathbb{F}_2$  from  $(\mathbf{QT}_3)$ , and define

$$R = \frac{\gamma + c}{\gamma + c^2}, \quad r = R^{A_0+1}, \quad y = r + r^{-1}.$$

Then  $r^\Lambda = 1$ . If  $\ell \mid \Psi_k$  is prime and  $e = \Lambda/\ell$ , the half-block phase is exactly

$$v = (r^e)^{-1}. \quad (\text{DS}_1)$$

Put  $z = \gamma^{A_0^2}$ . The selected trace coordinate collapses to

$$y = \frac{(g+1)z + g}{C}, \quad (\text{DS}_2)$$

and has the irreducible cubic over  $\mathbb{F}_{A_0}$

$$G(Y) = C^2Y^3 + gCY^2 + Y + g^2. \quad (\text{DS}_3)$$

The element  $r$  has degree six over  $\mathbb{F}_{A_0}$ , with scaled irreducible reciprocal polynomial

$$S(T) = T^3G(T + T^{-1}) = C^2T^6 + gCT^5 + (C^2 + 1)T^4 + g^2T^3 + (C^2 + 1)T^2 + gCT + C^2. \quad (\text{DS}_4)$$

Writing  $D_n(Y) = \mathcal{D}_n(Y, 1)$ , exceptional failure at  $\ell$  is equivalent to each of

$$D_e(y) = 0, \quad G(Y) \mid D_e(Y), \quad S(T) \mid T^e - 1. \quad (\text{DS}_5)$$

Moreover the trace in Proposition 8.35 satisfies

$$\mathcal{T}(v) = \text{Tr}_{\mathbb{F}_{A_0^3}/\mathbb{F}_{A_0}}(D_e(y)). \quad (\text{DS}_6)$$

There is an equivalent reverse-Dickson form. Put

$$\mathcal{A} = \gamma^2 + \gamma + 1.$$

Then  $\mathcal{A}$  and  $\mathcal{A}^{-1}$  have the irreducible cubics

$$\begin{aligned} \mathcal{A}^3 + \mathcal{A}^2 + (g+1)\mathcal{A} + C &= 0, \\ CZ^3 + (g+1)Z^2 + Z + 1 &= 0. \end{aligned} \quad (\text{DS}_7)$$

and

$$y = D_{A_0+1}(\mathcal{A}^{-1}).$$

Thus, with  $n = (q+1)/\ell = (A_0+1)e$ , failure is also equivalent to

$$CZ^3 + (g+1)Z^2 + Z + 1 \mid D_n(Z). \quad (\text{DS}_8)$$

Finally the ambient Dickson polynomial has the exact factorization

$$D_e(Y) = Y(Y+1)^2 \prod_{i=1}^{(e-3)/6} Q_i(Y)^2, \quad (\text{DS}_9)$$

where every  $Q_i$  is an irreducible cubic over  $\mathbb{F}_{A_0}$ . When  $e = 3$ , this independently recovers the prime- $\Psi_k$  case already proved in Proposition 8.5. For composite  $\Psi_k$ , the recursion selects the explicit cubic  $G$  among  $(e-3)/6$  structurally admissible cubics; excluding that selected factor remains exactly the exceptional theorem.

*Proof.* Use the notation in the proof of Proposition 8.36. Thus, for the order- $3h$  element  $\alpha$ ,

$$U = \alpha^2 + c\alpha + 1 = \alpha(\gamma + c), \quad U' = \alpha^2 + c^2\alpha + 1 = \alpha(\gamma + c^2),$$

and  $R = U/U'$ . The half-block element  $w$  is  $U$ . Put

$$u = \frac{A_0^6 - 1}{\ell}, \quad \xi = r^e = R^{(q+1)/\ell}.$$

Since  $UU'$  is an  $\ell$ -th power,  $(UU')^u = 1$ . If  $v' = (U')^u$ , then  $vv' = 1$ , and hence

$$R^u = \frac{v}{v'} = v^2.$$

On the other hand  $R^u = \xi^{q-1} = \xi^{-2}$ , because  $\xi \in \mu_\ell$  and  $q \equiv -1 \pmod{\ell}$ . The group  $\mu_\ell$  has odd order, so  $(v\xi)^2 = 1$  proves (DS<sub>1</sub>). Also  $R^{q+1} = 1$ , which gives  $r^\Lambda = 1$ .

Let  $x = \gamma$ ,  $x_1 = \gamma^{A_0}$ , and  $z = \gamma^{A_0^2}$ . The Gaussian-period recursion gives

$$x + x_1 + z = 0, \quad z^3 + z + g = 0.$$

Since  $c^{A_0} = c^2$ ,

$$r = RR^{A_0} = \frac{(x+c)(x_1+c^2)}{(x+c^2)(x_1+c)} = \frac{N}{D}.$$

Here  $N + D = z$ . Also

$$ND = (x^2 + x + 1)(x_1^2 + x_1 + 1).$$

For  $f(T) = T^3 + T + g$ ,

$$\prod_{t \in \{x, x_1, z\}} (t^2 + t + 1) = f(c)f(c^2) = (g+c)(g+c^2) = C.$$

Therefore

$$y = \frac{(N+D)^2}{ND} = \frac{z^2(z^2+z+1)}{C} = \frac{(g+1)z+g}{C},$$

proving (DS<sub>2</sub>). Both  $C$  and  $g+1$  are nonzero, so  $y$  generates the same cubic extension of  $\mathbb{F}_{A_0}$  as  $z$ . Substitution in  $z^3 + z + g = 0$  gives (DS<sub>3</sub>).

The quadratic  $T^2 + yT + 1$  of  $r$  over  $\mathbb{F}_{A_0}(y)$  is irreducible. Otherwise  $r \in \mathbb{F}_{A_0^3}$ ; but  $r^q = r^{-1}$ , so  $r = 1$ , which would force  $y = 0$ , contrary to (DS<sub>3</sub>). This proves the degree-six assertion and (DS<sub>4</sub>).

Dickson's identity gives

$$D_e(y) = r^e + r^{-e}.$$

As  $\text{ord}(r)$  is odd, this vanishes exactly when  $r^e = 1$ , which by (DS<sub>1</sub>) is failure. Minimal-polynomial divisibility proves (DS<sub>5</sub>). Pairing the six  $A_0$ -Frobenius conjugates of  $v = \xi^{-1}$  under the inversion  $A_0^3$ -Frobenius gives (DS<sub>6</sub>).

The three conjugates  $x_i$  of  $\gamma$  satisfy

$$\sum_i (x_i^2 + x_i + 1) = 1, \quad \sum_{i < j} (x_i^2 + x_i + 1)(x_j^2 + x_j + 1) = g + 1,$$

and the product of the three transformed conjugates  $x_i^2 + x_i + 1$  is  $C$ . This proves the first cubic in (DS<sub>7</sub>). Since  $\gamma$  has degree three over  $\mathbb{F}_{A_0}$ , while it is at most quadratic over  $\mathbb{F}_{A_0}(\mathcal{A})$ , the odd



tower law forces  $\mathbb{F}_{A_0}(\mathcal{A}) = \mathbb{F}_{A_0}(\gamma)$ ; hence the cubic is irreducible. Inversion gives the second cubic. Directly

$$\frac{R}{(R+1)^2} = (\gamma + c)(\gamma + c^2) = \mathcal{A},$$

so the semiconjugacy in Proposition 7.48 gives

$$\Phi(R) = \mathcal{A}, \quad \Phi(r) = F_{A_0+1}(\mathcal{A}) = \frac{1}{D_{A_0+1}(\mathcal{A}^{-1})},$$

and  $\Phi(r) = 1/y$ . Therefore

$$y = D_{A_0+1}(\mathcal{A}^{-1}).$$

The Dickson composition law, with  $n = (A_0 + 1)e$ , now proves (DS<sub>8</sub>).

It remains to factor  $D_e$ . One has  $v_3(\Lambda) = 1$ , while  $\ell \neq 3$ ; hence  $e \equiv 3 \pmod{6}$ . Over an algebraic closure,

$$D_e(Y) = Y \prod_{\substack{\{t, t^{-1}\} \\ t \in \mu_e \setminus \{1\}}} (Y - (t + t^{-1}))^2. \quad (\text{DS}_{10})$$

The pair of nontrivial cube roots gives  $(Y + 1)^2$ . For every remaining  $t$ , its order has a prime divisor other than three. Since  $A_0^3 \equiv -1 \pmod{\Lambda}$ , while  $\gcd(A_0 - 1, \Lambda) = 1$  and  $\gcd(A_0 + 1, \Lambda) = 3$ , its inverse-pair Frobenius orbit has length three. Grouping the remaining roots in (DS<sub>10</sub>) proves (DS<sub>9</sub>).  $\square$

**Proposition 8.47** (the exceptional phase normalizations are identical). *Retain Propositions 8.24, 8.34, 8.38, 8.46, and 8.43. Put*

$$h = 3^k, \quad t = 2^{h/3}, \quad Q = t^2, \quad A_D = Q^3 = 2^{2h}, \quad e_c = \frac{A_D - 1}{\ell}.$$

*In the mixed-Jacobi normalization choose*

$$z = ab = \mathfrak{g}^{(A_D^2 - 1)/(15h)}, \quad \xi_\ell = \mathfrak{g}^{(A_D^2 - 1)/\ell}, \quad V = 1 + z,$$

*where  $\mathfrak{g}$  generates  $\mathbb{F}_{A_D}^\times$ , and choose the conductor-five orientation so that  $c_k = b + b^{-1}$ . The residue indices*

$$L(1), \quad G_1^{-1}C_1, \quad 4^{-1} \text{ind}_\ell(d_k),$$

*and hence the corresponding exponents of  $\xi_\ell$ , are read in  $\mathbb{F}_\ell$ ;  $e_c$ ,  $d_s$ , and  $t + 1$  are ordinary integers. Then the selected weighted-fibre phase has the exact forms*

$$\boxed{\begin{aligned} \Omega = u_k^{e_c} &= \xi_\ell^{L(1)} = \xi_\ell^{G_1^{-1}C_1} \\ &= \xi_\ell^{4^{-1} \text{ind}_\ell(d_k)} = v = (r^{d_s})^{-1}, \end{aligned}} \quad (\text{PI}_1)$$

*where  $d_s = \Lambda/\ell$  is the exponent in Proposition 8.46. In particular the inverse orientation of the selected reciprocal sextic is fixed, rather than only determined up to inversion.*

*For*

$$\mathcal{S}(x) = x + x^Q + x^{Q^2},$$

*the selected cubic recurrence has terminal value*

$$\boxed{P_{e_c} = \mathcal{S}(\Omega) = \mathcal{S}\left(\xi_\ell^{G_1^{-1}C_1}\right) = \mathcal{S}\left((r^{d_s})^{-1}\right).} \quad (\text{PI}_2)$$

Consequently

$$\boxed{D'_k \text{ holds at } \ell \iff L(1) \neq 0 \iff C_1 \neq 0 \iff \text{ind}_\ell(d_k) \neq 0} \\ \iff r^{d_s} \neq 1 \iff \Omega \neq 1 \iff P_{e_c}^{t+1} \neq 1. \quad (\text{Pl}_3)$$

Thus the mixed Jacobi coefficient, conductor-five regulator, half-block phase, selected sextic, and cubic terminal recurrence are not independent constraints. After fixing  $\mathfrak{g}$  and  $\xi_\ell$ , they are all functions of the one selected scalar  $L(1)$ , while the primitive root  $\xi_\ell$  makes  $\xi_\ell^{L(1)}$  recover that scalar. The surviving theorem is still the direct nonvanishing  $L(1) \neq 0$ .

*Proof.* The  $A_D$ -power Frobenius fixes the order- $3h$  factor  $a$  and inverts the order-five factor  $b$ . Hence

$$U := N_{\mathbb{F}_{A_D^2}/\mathbb{F}_{A_D}}(V) = (1 + ab)(1 + ab^{-1}) = a(a + a^{-1} + c_k) = a u_k. \quad (\text{Pl}_4)$$

Both  $3h$  and  $\ell$  divide  $A_D - 1$ , and  $\gcd(3h, \ell) = 1$ , so

$$a^{e_c} = 1.$$

Since

$$(A_D + 1)e_c = \frac{A_D^2 - 1}{\ell},$$

the norm and the definition of the power-residue index give

$$\Omega = u_k^{e_c} = U^{e_c} = V^{(A_D^2 - 1)/\ell} = \xi_\ell^{L(1)}. \quad (\text{Pl}_5)$$

Equations (DJ<sub>2</sub>)–(DJ<sub>3</sub>) give  $L(1) = G_1^{-1}C_1$ , while (DJ<sub>5</sub>) gives  $L(1) = 4^{-1}\text{ind}_\ell(d_k)$ . On the selected primitive conductor- $15h$  component, the  $U$  in (Pl<sub>4</sub>) is exactly the half-block element  $w$ , so (Pl<sub>5</sub>) is  $\Omega = v$ . Finally (DS<sub>1</sub>) gives  $v = (r^{d_s})^{-1}$ . This proves (Pl<sub>1</sub>).

Equation (ICT<sub>4</sub>) now gives (Pl<sub>2</sub>). The first three equivalences in (Pl<sub>3</sub>) are (DJ<sub>4</sub>)–(DJ<sub>6</sub>); the sextic equivalence is (DS<sub>5</sub>), and the final two are (ICT<sub>1</sub>) and (ICO<sub>7</sub>).

Lean kernel-checks the abstract norm–Euler exponent transport, removal of the Euler-trivial ancestry factor, the antiunit quotient phase, and formal inversion of an assumed inverse-phase identity. The finite-field norm identification, residue indices, and specialization to the selected Jacobi, cubic, and sextic coordinates—including the selected sextic orientation—remain paper-level.  $\square$

**Proposition 8.48** (the selected Dickson phase has trivial downward norm). *Retain Proposition 8.46. Thus*

$$A_0 = 2^{3^{k-1}}, \quad \Lambda = A_0^2 - A_0 + 1 = 3\Psi_k, \quad r \in \mathbb{F}_{A_0^6}^\times, \quad r^\Lambda = 1.$$

Then

$$N_{\mathbb{F}_{A_0^6}/\mathbb{F}_{A_0^2}}(r) = 1. \quad (\text{NC}_1)$$

More strongly, if  $\ell^a \parallel \Psi_k$ , the norm annihilates the full  $\ell$ -primary subgroup of  $\mathbb{F}_{A_0^6}^\times$ , while  $\mathbb{F}_{A_0^2}^\times$  has no  $\ell$ -torsion. Hence the induced corestriction

$$\mathbb{F}_{A_0^6}^\times / (\mathbb{F}_{A_0^6}^\times)^{\ell^a} \longrightarrow \mathbb{F}_{A_0^2}^\times / (\mathbb{F}_{A_0^2}^\times)^{\ell^a} \quad (\text{NC}_2)$$

is zero on the entire current primary line. In particular, the exact antiunit recurrence cannot transport any part of the assertion

$$r^{\Lambda/\ell} \neq 1$$

from level  $k - 1$ .

*Proof.* The norm exponent factors as

$$\frac{A_0^6 - 1}{A_0^2 - 1} = A_0^4 + A_0^2 + 1 = (A_0^2 - A_0 + 1)(A_0^2 + A_0 + 1) = \Lambda(A_0^2 + A_0 + 1).$$

Equation  $r^\Lambda = 1$  proves (NC<sub>1</sub>). If  $\ell^a \parallel \Psi_k$ , then  $\ell^a \mid \Lambda$ , whereas  $\text{ord}_\ell(2) = 6 \cdot 3^{k-1}$  makes  $A_0$  have exact order six modulo  $\ell$ , and hence

$$\ell \nmid (A_0^2 - 1)(A_0^2 + A_0 + 1).$$

Thus the displayed norm exponent kills the full current  $\ell$ -primary subgroup, and raising to the  $\ell^a$ -th power is an automorphism of the lower unit group. Its Kummer quotient is therefore trivial, proving (NC<sub>2</sub>).  $\square$

**Proposition 8.49** (the trace-constant shape of every Dickson cubic). *Let  $Q$  be a power of two with  $Q \equiv 2 \pmod{3}$ , and put  $\Lambda = Q^2 - Q + 1$ . If  $t \in \mu_\Lambda \setminus \mu_3$  and  $y = t + t^{-1}$ , then  $[\mathbb{F}_Q(y) : \mathbb{F}_Q] = 3$ , and the monic minimal polynomial of  $y$  has the form*

$$Y^3 + aY^2 + bY + a^2, \tag{CS_1}$$

where  $a = \text{Tr}_{\mathbb{F}_{Q^3}/\mathbb{F}_Q}(y)$ .

Consequently every irreducible cubic  $Q_i$  in (DS<sub>9</sub>) has this trace-constant shape. The selected cubic  $G/C^2$  has

$$a = \frac{g}{C}, \quad b = \frac{1}{C^2}, \quad \frac{g^2}{C^2} = a^2. \tag{CS_2}$$

Thus its trace and constant term satisfy the universal Dickson-cubic identity; coefficient shape cannot exclude the selected factor.

*Proof.* The congruence  $Q^3 \equiv -1 \pmod{\Lambda}$  shows that  $y \in \mathbb{F}_{Q^3}$ . If  $y \in \mathbb{F}_Q$ , equality of the inverse pairs  $\{t^Q, t^{-Q}\} = \{t, t^{-1}\}$  would give  $t^{Q-1} = 1$  or  $t^{Q+1} = 1$ . But

$$\gcd(\Lambda, Q - 1) = 1, \quad \gcd(\Lambda, Q + 1) = 3,$$

contrary to  $t \notin \mu_3$ . Hence the degree is three.

Put  $y_i = t^{Q^i} + t^{-Q^i}$  for  $0 \leq i < 3$ , and  $a = y_0 + y_1 + y_2$ . Since  $Q^2 \equiv Q - 1 \pmod{\Lambda}$ , expanding the eight terms of the product and pairing inverse exponents gives

$$\begin{aligned} y_0 y_1 y_2 &= (t^2 + t^{-2}) + (t^{2Q} + t^{-2Q}) + (t^{2Q^2} + t^{-2Q^2}) + 1 + 1 \\ &= a^2. \end{aligned} \tag{CS_3}$$

The trace and product are respectively the quadratic and constant coefficients of the monic cubic in characteristic two, proving (CS<sub>1</sub>). Dividing (DS<sub>3</sub>) by  $C^2$  gives (CS<sub>2</sub>).  $\square$

**Proposition 8.50** (depressing the selected cubic recovers the Gaussian recursion). *Over a field of characteristic two, put*

$$f(Y) = Y^3 + aY^2 + bY + a^2, \quad \delta^2 = a^2 + b.$$

Then

$$f(\delta U + a) = \delta^3 \left( U^3 + U + \frac{ab + a^2}{\delta^3} \right). \tag{DC_1}$$

For the recursively selected cubic  $f = G/C^2$ , so that

$$a = \frac{g}{C}, \quad b = \frac{1}{C^2}, \quad C = g^2 + g + 1,$$

one may take  $\delta = (g + 1)/C$ , and the constant in (DC<sub>1</sub>) is exactly  $g$ . Thus

$$\boxed{(G/C^2) \left( \frac{g+1}{C} U + \frac{g}{C} \right) = \left( \frac{g+1}{C} \right)^3 (U^3 + U + g).} \quad (\text{DC}_2)$$

In particular, depressing and scaling the selected Dickson cubic is a lossless recovery of the preceding Gaussian-tower equation; it supplies no invariant independent of the recursion.

*Proof.* Direct expansion in characteristic two cancels the quadratic and constant terms coming from  $(\delta U + a)^3 + a(\delta U + a)^2$ , and the linear coefficient is  $\delta(a^2 + b) = \delta^3$ . This proves (DC<sub>1</sub>). In the selected case,

$$a^2 + b = \frac{(g+1)^2}{C^2}, \quad ab + a^2 = \frac{g(1+gC)}{C^3}.$$

The identity  $1 + gC = (g + 1)^3$  now gives  $(ab + a^2)/\delta^3 = g$ , proving (DC<sub>2</sub>).  $\square$

**Proposition 8.51** (the first Dickson Hasse jets are universal). *Let  $F$  have characteristic two, let  $e$  be odd, and let  $D_e$  be characterized by*

$$D_e(T + T^{-1}) = T^e + T^{-e}.$$

*Suppose that  $t^e = 1$ ,  $t \neq 1$ , and put  $y = t + t^{-1}$ . If  $D_e^{[j]}$  denotes the  $j$ -th Hasse derivative, then*

$$\begin{aligned} D_e^{[1]}(y) &= 0, & D_e^{[2]}(y) &= y^{-2}, & D_e^{[3]}(y) &= y^{-3}, \\ D_e^{[4]}(y) &= (y^2 + 1)y^{-6}, & D_e^{[5]}(y) &= (y^2 + 1)y^{-7}. \end{aligned} \quad (\text{HJ}_1)$$

*The first place at which the exponent can enter the local expansion is order six. More precisely,*

$$D_e^{[6]}(y) = \epsilon_e y^{-6}, \quad \epsilon_e = \begin{cases} 1, & e \equiv 1, 7 \pmod{8}, \\ 0, & e \equiv 3, 5 \pmod{8}. \end{cases} \quad (\text{HJ}_2)$$

*Consequently the derivative, multiplicity, and Hasse-jet data through degree five impose no condition beyond the root coordinate itself: they are fixed rational functions of  $y$ , independent of  $e$ . Even the sixth jet adds only  $e \bmod 8$ , common to every root of the same Dickson polynomial. Thus higher multiplicity does not furnish a new exponent-sensitive ancestry invariant for the recursively selected cubic in Proposition 8.46.*

*Proof.* The element  $y$  is nonzero: otherwise  $t^2 = 1$ , and the odd order of  $t$  would force  $t = 1$ . Work in the completed local ring and write

$$T = t(1 + u), \quad x = (T + T^{-1}) - y, \quad a = t^{-1}.$$

Then

$$x = yu + a(u^2 + u^3 + u^4 + u^5) + O(u^6). \quad (\text{HJ}_3)$$

Since  $t^e = 1$ ,

$$D_e(y+x) = (1+u)^e + (1+u)^{-e}.$$

For odd  $e$ , the coefficient

$$\binom{e}{j} + \binom{e+j-1}{j} \pmod{2}$$

is zero for  $j = 0, 1$ , is one for  $2 \leq j \leq 5$ , and at  $j = 6$  is  $\epsilon_e$ . Hence

$$D_e(y+x) = u^2 + u^3 + u^4 + u^5 + \epsilon_e u^6 + O(u^7). \quad (\text{HJ}_4)$$

Successively inverting (HJ<sub>3</sub>) gives

$$u = px + qx^2 + rx^3 + sx^4 + O(x^5),$$

where

$$p = y^{-1}, \quad q = ay^{-3}, \quad r = ay^{-4}, \quad s = a(1+y^2)y^{-7}. \quad (\text{HJ}_5)$$

Substitution in (HJ<sub>4</sub>) gives the coefficients of  $x^2, x^3, x^4, x^5$  as

$$p^2, \quad p^3, \quad q^2 + p^2q + p^4, \quad p^2r + pq^2 + p^5.$$

Using  $a^2 + ay = 1$ , these are exactly the four values in (HJ<sub>1</sub>). The coefficient of  $x^6$  is

$$r^2 + p^2s + q^3 + p^4q + \epsilon_e p^6 = \epsilon_e y^{-6};$$

the non- $\epsilon_e$  terms cancel again by  $a^2 + ay = 1$ . By the defining Taylor identity for Hasse derivatives, these local coefficients are  $D_e^{[j]}(y)$ , proving both formulas and the stated no-go.  $\square$

**Proposition 8.52** (the Gaussian depressed class is not a selected invariant). *Let  $Q = 2^s$  with  $s$  odd, let  $g \in \mathbb{F}_Q$ , and suppose that*

$$h(U) = U^3 + U + g$$

*is irreducible over  $\mathbb{F}_Q$ . For every  $\xi \in \mathbb{F}_Q^\times$ , put*

$$H_\xi = \xi^3 + \xi + g, \quad \delta_\xi = \frac{\xi^2}{H_\xi}, \quad a_\xi = \xi \delta_\xi, \quad b_\xi = a_\xi^2 + \delta_\xi^2,$$

*and*

$$f_\xi(Y) = Y^3 + a_\xi Y^2 + b_\xi Y + a_\xi^2.$$

*Then  $H_\xi \neq 0$ , the cubics  $f_\xi$  are irreducible and pairwise distinct, and*

$$f_\xi(\delta_\xi U + a_\xi) = \delta_\xi^3(U^3 + U + g). \quad (\text{GM}_1)$$

*Put  $\Lambda = Q^2 - Q + 1$ . If*

$$\text{Tr}_{\mathbb{F}_Q/\mathbb{F}_2}(\xi^{-1}) = 0, \quad (\text{GM}_2)$$

*then  $f_\xi \mid D_\Lambda$ . There are exactly  $Q/2 - 1$  such parameters  $\xi$ . Consequently the single depressed recursion  $U^3 + U + g$  occurs in  $Q/2 - 1$  distinct irreducible cubic factors of the ambient Dickson polynomial  $D_\Lambda$ .*

For the exceptional arm, take  $Q = A_0$ ,  $g = \gamma_{k-1}$ . The recursively selected cubic of Proposition 8.50 is the member with

$$\xi = \frac{a}{\delta} = \frac{g}{g+1}.$$

Thus, for  $k \geq 2$ , at least  $Q/2 - 2$  nonselected Dickson cubics have exactly the same affine depressed equation. Excluding the selected factor from  $D_e$  must use its multiplicative placement in the subgroup  $\mu_e$ , not its full affine cubic class.

*Proof.* Irreducibility of  $h$  gives  $H_\xi \neq 0$ . Put  $a = a_\xi$ ,  $b = b_\xi$ , and  $\delta = \delta_\xi$ . Since  $a/\delta = \xi$  and  $\delta H_\xi = \xi^2$ , direct substitution in Proposition 8.50 gives

$$\frac{ab + a^2}{\delta^3} = \xi^3 + \xi + \frac{\xi^2}{\delta} = g,$$

which proves (GM<sub>1</sub>). Hence every  $f_\xi$  is irreducible. Moreover  $\delta$  is the unique square root of  $a^2 + b$ , so the parameter  $\xi = a/\delta$  is recovered from the coefficients of  $f_\xi$ ; the cubics are pairwise distinct.

Let  $y$  be a root of  $f_\xi$  in  $\mathbb{F}_{Q^3}$ . The quadratic

$$T^2 + yT + 1$$

is irreducible over  $\mathbb{F}_{Q^3}$  exactly when

$$\mathrm{Tr}_{\mathbb{F}_{Q^3}/\mathbb{F}_2}(y^{-2}) = 1.$$

If  $y_0, y_1, y_2$  are the three conjugates of  $y$ , then

$$\sum_i y_i^{-1} = \frac{b}{a^2},$$

so invariance of absolute trace under squaring gives

$$\begin{aligned} \mathrm{Tr}_{\mathbb{F}_{Q^3}/\mathbb{F}_2}(y^{-2}) &= \mathrm{Tr}_{\mathbb{F}_Q/\mathbb{F}_2}\left(\frac{b}{a^2}\right) \\ &= \mathrm{Tr}_{\mathbb{F}_Q/\mathbb{F}_2}(1 + \xi^{-2}) = 1 + \mathrm{Tr}_{\mathbb{F}_Q/\mathbb{F}_2}(\xi^{-1}). \end{aligned} \tag{GM_3}$$

Here  $\mathrm{Tr}(1) = 1$  because  $s$  is odd. Thus (GM<sub>2</sub>) makes the quadratic irreducible. Choose a root  $t \in \mathbb{F}_{Q^6}$ , so  $y = t + t^{-1}$  and  $t^{Q^3} = t^{-1}$ .

It remains to recover the smaller torus  $\mu_\Lambda$ , rather than only  $\mu_{Q^3+1}$ . Put

$$x = t, \quad u = t^Q, \quad v = t^{Q^2}.$$

The trace-constant identity for  $f_\xi$  has the exact factorization

$$\begin{aligned} \prod_{w \in \{x, u, v\}} (w + w^{-1}) + \left( \sum_{w \in \{x, u, v\}} (w + w^{-1}) \right)^2 \\ = \frac{(u + xv)(v + xu)(x + uv)(1 + xuv)}{x^2 u^2 v^2} = 0. \end{aligned} \tag{GM_4}$$

The four factors respectively assert that the order of  $t$  divides

$$Q^2 - Q + 1, \quad Q^2 - Q - 1, \quad Q^2 + Q - 1, \quad Q^2 + Q + 1.$$

On the other hand it divides  $Q^3 + 1$ . The last three displayed integers are coprime to  $Q^3 + 1$ : for the first two of them this follows by Euclidean reduction to the coprime pairs  $(Q+1, Q^2 - Q - 1)$  and  $(Q, Q^2 + Q - 1)$ , while  $Q^2 + Q + 1 \mid Q^3 - 1$  and  $\gcd(Q^3 - 1, Q^3 + 1) = 1$  because both relevant orders are odd. Since  $t \neq 1$ , only the first factor can vanish. Hence  $t^\Lambda = 1$ , and Dickson's identity gives

$$D_\Lambda(y) = t^\Lambda + t^{-\Lambda} = 0.$$

Thus  $f_\xi \mid D_\Lambda$ . Finally inversion permutes  $\mathbb{F}_Q^\times$ , and the absolute-trace-zero hyperplane has  $Q/2$  elements, one of which is zero. Therefore exactly  $Q/2 - 1$  nonzero parameters satisfy  $(\text{GM}_2)$ . The specialization to the selected cubic follows from  $a = g/C$  and  $\delta = (g+1)/C$ ; the reciprocal sextic is irreducible by Proposition 8.46, so  $(\text{GM}_3)$  also places the selected parameter among the  $Q/2 - 1$  counted above.  $\square$

**Proposition 8.53** (the selected cubic's absolute-trace fingerprint). *Put  $K_j = \mathbb{F}_{2^{3j}}$ , set  $\gamma_0 = 1$ , and choose the compatible Gaussian periods so that  $\gamma_j^3 + \gamma_j = \gamma_{j-1}$ . Define*

$$\phi(X) = \frac{X}{X^2 + X + 1}, \quad \psi(X) = \frac{1}{X^2 + X + 1}.$$

Then, for every  $j \geq 1$ ,

$$\begin{aligned} \text{Tr}_{K_j/K_{j-1}}(\phi(\gamma_j)) &= \phi(\gamma_{j-1}), \\ \text{Tr}_{K_j/K_{j-1}}(\psi(\gamma_j)) &= \phi(\gamma_{j-1}) + \psi(\gamma_{j-1}). \end{aligned} \tag{CF_1}$$

Consequently

$$\text{Tr}_{K_j/\mathbb{F}_2}(\phi(\gamma_j)) = 1, \quad \text{Tr}_{K_j/\mathbb{F}_2}(\psi(\gamma_j)) = j + 1 \pmod{2}. \tag{CF_2}$$

In the notation of Proposition 8.46, this gives the selected coefficient fingerprint

$$\text{Tr}_{\mathbb{F}_{A_0}/\mathbb{F}_2}\left(\frac{g}{C}\right) = 1, \quad \text{Tr}_{\mathbb{F}_{A_0}/\mathbb{F}_2}\left(\frac{1}{C^2}\right) = k \pmod{2}. \tag{CF_3}$$

These identities distinguish two absolute coordinates of  $G$ , but the factorization  $(\text{DS}_9)$  supplies no incompatible coordinate identity; they do not by themselves exclude  $G \mid D_e$ .

*Proof.* Let  $x_0, x_1, x_2$  be the roots of  $f(T) = T^3 + T + b$ , where  $b = \gamma_{j-1}$ , and work after the harmless scalar extension adjoining  $c \in \mathbb{F}_4 \setminus \mathbb{F}_2$ . The denominators below are nonzero:  $\gamma_j$  has odd degree  $3^j$ , so it cannot be a nontrivial element of  $\mathbb{F}_4$ . The partial fractions

$$\phi(x) = \frac{c}{x+c} + \frac{c^2}{x+c^2}, \quad \psi(x) = \frac{1}{x+c} + \frac{1}{x+c^2}$$

and the logarithmic-derivative identity

$$\sum_{i=0}^2 \frac{1}{x_i + a} = \frac{f'(a)}{f(a)}$$

apply. Since

$$f(c) = b + c^2, \quad f(c^2) = b + c, \quad \text{quad} f'(c) = c, \quad f'(c^2) = c^2,$$

direct simplification gives (CF<sub>1</sub>).

The first identity iterates to  $\phi(\gamma_0) = \phi(1) = 1$ . If  $U_j = \text{Tr}_{K_j/\mathbb{F}_2}(\psi(\gamma_j))$ , the second gives  $U_j = 1 + U_{j-1}$ , with  $U_0 = \psi(1) = 1$ ; this proves (CF<sub>2</sub>). Finally take  $j = k - 1$ , use  $C = g^2 + g + 1$ , and use invariance of the absolute trace under squaring to obtain (CF<sub>3</sub>).  $\square$

**Proposition 8.54** (the joint fingerprint admits a nonselected factor). *At  $k = 4$ , the two absolute traces in Proposition 8.53 are simultaneously realized by an irreducible cubic factor of the bad Dickson polynomial which is not the selected cubic. More precisely, put*

$$A = 2^{27}, \quad \Psi_4 = 163 \cdot 135433 \cdot 272010961, \quad \Lambda = A^2 - A + 1 = 3\Psi_4,$$

and let  $P_4$  be the projected element of Proposition 8.5. In the exact trinomial model  $\mathbb{F}_2[X]/(X^{162} + X^{81} + 1)$ , one has  $\text{ord}(P_4) = \Psi_4$ . Set

$$t = P_4^{163 \cdot 7}, \quad e = \frac{\Lambda}{163} = 3 \cdot 135433 \cdot 272010961, \quad y_t = t + t^{-1}.$$

Then

$$\text{ord}(t) = 135433 \cdot 272010961,$$

so  $t \in \mu_e \setminus \mu_3$ . If

$$a_t = \sum_{i=0}^2 y_t^{A^i}, \quad b_t = \sum_{0 \leq i < j \leq 2} y_t^{A^i} y_t^{A^j},$$

then

$$Q_t(Y) = Y^3 + a_t Y^2 + b_t Y + a_t^2$$

is an irreducible cubic factor of  $D_e(Y)$ , distinct from  $G$ , and the exact finite-field certificate gives

$$\text{Tr}_{\mathbb{F}_A/\mathbb{F}_2}(a_t) = 1, \quad \text{Tr}_{\mathbb{F}_A/\mathbb{F}_2}(b_t) = 0. \quad (\text{CF}_4)$$

This is exactly the selected pair  $(1, k \bmod 2) = (1, 0)$ . Thus even the two trace coordinates jointly cannot exclude the selected factor.

*Proof.* The order and trace statements in this proposition are an exact finite certificate, not an analytical instance of  $D'_k$ . They use the same bit-polynomial arithmetic as `experiments/exception_column_m4.py`: represent  $\zeta$  by  $X$ , reduce with  $X^{162} = X^{81} + 1$ , form  $P_4$  from (D<sub>2</sub>), and use repeated squaring for Frobenius and absolute trace. Exact powering first gives

$$P_4^{\Psi_4} = 1, \quad P_4^{\Psi_4/r} \neq 1 \quad (r = 163, 135433, 272010961).$$

Since  $\text{gcd}(163 \cdot 7, \Psi_4) = 163$ , this proves the displayed order of  $t$ , hence  $t^e = 1$ . Proposition 8.49 then gives the irreducible cubic  $Q_t$ , and the root factorization (DS<sub>10</sub>) gives  $Q_t \mid D_e$ .

The same exact reduction evaluates the two Frobenius sums defining  $a_t, b_t$ ; summing their 27 absolute Frobenius conjugates gives respectively 1 and 0, which is (CF<sub>4</sub>). Finally the certified selected element at  $k = 4$  retains its 163-primary order, whereas  $t$  does not. Their inverse-pair Frobenius orbits therefore cannot agree, so  $Q_t \neq G$ .  $\square$

**Remark 8.55** (the plus and minus current tori have disjoint orders). At the same cubic step  $Q = 2^{3^{k-1}}$ , the current rational-point groups in arms (C) and (D) have respective orders

$$Q^2 + Q + 1, \quad Q^2 - Q + 1.$$



Their gcd divides both  $2Q$  and two odd integers coprime to  $Q$ , so it is one. Hence every homomorphism between these two finite rational-point groups is trivial. The common antiunit formalism does not transfer a current primary order assertion from one arm to the other; the two selected Kummer classes require separate evaluations.

**Proposition 8.56** (the actual  $C$  and  $D$  Kummer fields are disjoint). *Let  $k \geq 2$ ,  $h = 3^k$ , and let  $\xi$  be a primitive  $3^{k+1}$ -st root of unity. Put*

$$F = \mathbb{Q}(\xi + \xi^{-1}), \quad K = F(\sqrt{5}),$$

*and retain the actual selected radicals*

$$c_k = (1 + \xi)(1 + \xi^{-1}) \in F^\times, \quad d_k = \frac{x - s}{x - s'} \in K^\times$$

*from Propositions 6.14 and 8.45. Choose current primes*

$$\ell_+ \mid \Phi_h(2), \quad \ell_- \mid \Psi_k.$$

*Then*

$$\text{ord}_{\ell_+}(2) = h, \quad \text{ord}_{\ell_-}(2) = 2h, \quad \ell_+ \neq \ell_-. \quad (\text{KID}_1)$$

*More precisely, for  $Q = 2^{h/3}$ ,*

$$\gcd\left(Q^2 + Q + 1, \frac{Q^2 - Q + 1}{3}\right) = 1. \quad (\text{KID}_2)$$

*Assume both reflection constructions are nondegenerate, and put*

$$\Omega = K(\mu_{\ell_+}, \mu_{\ell_-}).$$

*For roots  $u^{\ell_+} = c_k$  and  $v^{\ell_-} = d_k$ , one has*

$$[\Omega(u) : \Omega] = \ell_+, \quad [\Omega(v) : \Omega] = \ell_-, \quad (\text{KID}_3)$$

*and hence*

$$\boxed{\text{Gal}(\Omega(u, v)/\Omega) \simeq C_{\ell_+} \times C_{\ell_-}.} \quad (\text{KID}_4)$$

*At a Conway-compatible prime of  $\Omega$  above two, the two Frobenius coordinates detect exactly the selected  $C$ - and  $D$ -failure bits: the  $C$ -coordinate is an invertible power of its original Euler value after the quadratic residue-field base change, and the  $D$ -coordinate is the value in  $(\text{DHR}_5)$ . Thus the local product structure admits independently the four split/inert patterns. Chebotarev realizes every pair of translation coordinates at infinitely many other unramified primes of  $\Omega$ .*

*Consequently no implication  $C_k \Rightarrow D'_k$  can follow solely from an intersection or common quotient of the actual Kummer fields, an abelian reflection-field descent, or cancellation in one shared reciprocity product. Any such implication must use the special additive Artin–Schreier coupling of Proposition 8.7, not only the two global Kummer extensions.*

*Proof.* The order assertions are the cyclotomic-divisor criteria in the two arms. Also

$$\gcd(Q^2 + Q + 1, Q^2 - Q + 1) \mid 2Q.$$

Both displayed factors are odd and coprime to  $Q$ , so their gcd is one; division of the second by three preserves this conclusion.

The field  $\Omega$  lies in an abelian cyclotomic compositum over  $\mathbb{Q}$ , hence  $\Omega/F$  and  $\Omega/K$  are abelian. Suppose that  $c_k$  became an  $\ell_+$ -th power in  $\Omega$ . Nondegeneracy makes  $X^{\ell_+} - c_k$  irreducible over  $F$ , so  $F(u)/F$  has prime degree  $\ell_+$ . It is not Galois:  $F$  is totally real and contains no nontrivial  $\ell_+$ -th roots of unity, while the other roots are  $\zeta_{\ell_+}^j u$ . But  $F(u) \subset \Omega$  would be an intermediate field of an abelian extension and therefore Galois, a contradiction. This proves the first equality in (KID<sub>3</sub>). The same argument over the totally real field  $K$ , using  $d_k$ , proves the second. The two cyclic Kummer extensions over  $\Omega$  have distinct prime degrees, so they are linearly disjoint, proving (KID<sub>4</sub>).

The residue field of the selected prime of  $K$  above two is  $\mathbb{F}_{2^{2h}}$ , which contains both root-of-unity groups by (KID<sub>1</sub>). Therefore the added cyclotomic base splits completely there. Base change squares the  $C$ -Frobenius coordinate, an automorphism of its odd prime-order group, and leaves the exact  $D$ -coordinate described in (DHR<sub>5</sub>). This proves the selected-place assertion. The product Galois group gives the remaining structural and Chebotarev claims.

Finally, if  $r_+ \in \mu_{\ell_+}$  and  $r_- \in \mu_{\ell_-}$ , then

$$r_+ r_- = 1 \iff r_+ = r_- = 1$$

because their orders are coprime. This abstract product-separation lemma is kernel-checked in `Ogdoad.Excess`; the number-field Kummer, local-Frobenius, and Chebotarev interpretations remain paper-level.  $\square$

**Proposition 8.57** (partition regulator sieve for the exceptional arm). *Retain Proposition 8.5. Let*

$$H_k = \left\lfloor \frac{h-1}{8} \right\rfloor$$

*and let  $B_-(h)$  be the number of partitions into distinct positive parts congruent to 1 (mod 3) whose total is at most  $H_k$ . Then*

$$\text{ord}(P_k) \geq B_-(h) = \exp\left(\left(\frac{\pi}{6\sqrt{2}} + o(1)\right) \sqrt{h}\right). \quad (\text{PR}_1)$$

*Consequently, if  $\ell \mid \Psi_k$  and*

$$\ell > \frac{\Lambda_k}{B_-(h)}, \quad (\text{PR}_2)$$

*then  $P_k$ , and hence  $M_k$ , contains the full  $\ell$ -primary part of  $\Lambda_k$  in its order. Thus  $D'_k$  holds at every current prime satisfying (PR<sub>2</sub>). This replaces the linear denominator in (D<sub>6</sub>) by an unconditional quantity exponential in  $\sqrt{h}$ ; current factors at or below the new threshold remain open.*

*Proof.* The element  $\zeta$  has degree  $h$  over  $\mathbb{F}_4$ , because  $\text{ord}_{3h}(4) = h$ . The subgroup generated by 4 in  $(\mathbb{Z}/3h\mathbb{Z})^\times$  consists exactly of the classes congruent to 1 (mod 3). Hence, for every positive  $j \equiv 1 \pmod{3}$  with  $j < 3h$ , there is  $z_j$  such that  $4^{z_j} \equiv j \pmod{3h}$ .

Put

$$A_j(X) = X^{4j} + X^j + \omega, \quad B_j(X) = \omega^2 X^{4j} + X^{3j} + 1.$$

Equation (D<sub>2</sub>), followed by  $4^{z_j}$ -Frobenius, gives

$$P_k^{4^{z_j}} = \frac{A_j(\zeta)}{B_j(\zeta)}. \quad (\text{PR}_3)$$

For a partition  $I$  counted by  $B_-(h)$ , set  $E_I = \sum_{j \in I} 4^{z_j}$ . We claim that the powers  $P_k^{E_I}$  are pairwise distinct.

Suppose two of them, indexed by  $I$  and  $J$ , were equal. Cancel the common parts and cross-multiply in (PR<sub>3</sub>). The resulting polynomial in  $\mathbb{F}_4[X]$  has degree at most

$$4 \left( \sum_{j \in I} j + \sum_{j \in J} j \right) \leq 8H_k < h. \quad (\text{PR}_4)$$

It is nonzero. Indeed  $A_j(0)/B_j(0) = \omega$ , so different cardinalities modulo three are already separated by the constant term. After dividing each factor by  $\omega$ , its expansion at zero is

$$\omega^{-1} \frac{A_j(X)}{B_j(X)} = 1 + \omega^2 X^j + O(X^{2j}). \quad (\text{PR}_5)$$

If the constant terms agree, the coefficient belonging to the least part in  $I \triangle J$  therefore separates the two products. A nonzero polynomial of degree below  $h$  cannot vanish at  $\zeta$ , proving the claim. The  $B_-(h)$  distinct powers give the first inequality in (PR<sub>1</sub>).

For the asymptotic, the generating function for distinct parts congruent to one modulo three is

$$\prod_{r \geq 0} (1 + x^{3r+1}).$$

Its logarithm at  $x = e^{-t}$  is  $\pi^2/(36t) + O(\log(1/t))$ . The usual Meinardus/saddle-point estimate [28] therefore gives  $\log B_-(h) \sim (\pi/3)\sqrt{H_k}$ , which is the second assertion of (PR<sub>1</sub>).

Finally  $P_k$  lies in the cyclic subgroup of order  $\Lambda_k$ . If its order missed the full  $\ell$ -primary part, then  $P_k$  would lie in the subgroup of  $\ell$ -th powers, of order  $\Lambda_k/\ell$ . All the powers just constructed would lie there, so  $B_-(h) \leq \Lambda_k/\ell$ , contrary to (PR<sub>2</sub>). Equation (D<sub>1</sub>) transfers the conclusion to  $M_k$ .  $\square$

## 8.5 Exact bad-locus exhaustions

**Proposition 8.58** (semiprimitive exhaustion and the saturated planes). *Retain Proposition 8.18, put*

$$h = 3^k, \quad q = 2^h, \quad K = \mathbb{F}_{q^2}, \quad L = \mathbb{F}_{q^4},$$

and, for a prime  $\ell \mid \Psi_k$ , set

$$H_K = (K^\times)^\ell, \quad H_L = (L^\times)^\ell.$$

Every nontrivial order- $\ell$  character  $\chi$  on  $K^\times$  has, for the canonical additive character,

$$G_K(\chi^j) = q \quad (1 \leq j < \ell). \quad (\text{SP}_1)$$

Every order- $\ell$  character on  $L^\times$  is the norm lift of one on  $K^\times$ , and quadratic Hasse–Davenport gives

$$G_L(\tilde{\chi}^j) = -q^2 \quad (1 \leq j < \ell). \quad (\text{SP}_2)$$

Consequently, whenever  $a, b, a + b \not\equiv 0 \pmod{\ell}$ ,

$$J_K(\chi^a, \chi^b) = q, \quad J_L(\tilde{\chi}^a, \tilde{\chi}^b) = -q^2, \quad (\text{SP}_3)$$

while  $J(\chi^a, \chi^{-a}) = -1$  in either field.

Failure of  $D'_k$  at  $\ell$  has the following two equivalent additive forms:

$$W := \mathbb{F}_{16} + \mathbb{F}_{16}\zeta, \quad W \setminus \{0\} \subset H_L, \quad (\text{SP}_4)$$

and

$$V := \mathbb{F}_4 + \mathbb{F}_4\zeta + \mathbb{F}_4\zeta^2, \quad V \setminus \{0\} \subset H_K. \quad (\text{SP}_5)$$

Thus a bad translate saturates an eight-dimensional  $\mathbb{F}_2$ -space in  $L$ , and its norm-down form saturates a six-dimensional  $\mathbb{F}_2$ -space in  $K$ , not merely the original affine line.

The exact semiprimitive Fourier equations nevertheless permit both planes. If  $W^\perp$  is the additive annihilator of  $W$ , put

$$A_L = |(W^\perp \setminus \{0\}) \cap H_L|, \quad R = q^2, \quad w = 256.$$

Then orthogonality and  $(\text{SP}_2)$  force

$$A_L = \frac{(R-1)(R+w) - \ell R(w-1)}{\ell w}, \quad |W^\perp \setminus (H_L \cup \{0\})| = \frac{(R-1)(R+w)(\ell-1)}{\ell w}. \quad (\text{SP}_6)$$

Likewise, for  $w_0 = 64$ ,

$$A_K := |(V^\perp \setminus \{0\}) \cap H_K| = \frac{(q+1)(q-w_0) + \ell q(w_0-1)}{\ell w_0}, \quad (\text{SP}_7)$$

and

$$|V^\perp \setminus (H_K \cup \{0\})| = \frac{(\ell-1)(q+1)(q-w_0)}{\ell w_0}. \quad (\text{SP}_8)$$

All four quantities are nonnegative integers with the required total sizes. Hence the complete semiprimitive Gauss and Jacobi table, even together with the dimensions of the saturated spaces, gives no contradiction. A proof of  $D'_k$  must use the distinguished cyclotomic placement of  $W$  or  $V$ , not only the semiprimitive character table or additive-subspace incidence.

*Proof.* Since  $\text{ord}_\ell(2) = 2h$ , one has  $q \equiv -1 \pmod{\ell}$ . Thus every order- $\ell$  character of  $K^\times$  is trivial on  $\mathbb{F}_q^\times$ . Decompose  $K^\times$  into its  $q+1$  projective  $\mathbb{F}_q$ -lines. On the trace-zero line  $\mathbb{F}_q$ , the additive sum is  $q-1$ ; on every other line it is  $-1$ . The induced nontrivial projective character sums to zero, giving  $G_K(\chi) = q$ . The same argument applies to every nontrivial power. The norm induces an isomorphism on the order- $\ell$  character groups, so Hasse–Davenport proves  $(\text{SP}_2)$ . The standard quotient formula for Jacobi sums proves  $(\text{SP}_3)$ , with the inverse case evaluated separately.

If  $D'_k$  fails, Proposition 8.18 puts every  $\zeta + c$ ,  $c \in \mathbb{F}_{16}$ , in  $H_L$ . Since  $\mathbb{F}_{16}^\times \subset H_L$ , every nonzero  $a\zeta + b$  is an  $\ell$ -th power, proving  $(\text{SP}_4)$ ; the converse follows by taking  $a = 1, b = u$ .

For  $(\text{SP}_5)$ , every polynomial of degree at most two over  $\mathbb{F}_4$  is a product of linear factors or is an irreducible quadratic. In the first case its value at  $\zeta$  is an  $\ell$ -th power by  $(\text{SP}_4)$ . In the second, choose a root  $c \in \mathbb{F}_{16} \setminus \mathbb{F}_4$ ; its value is

$$(\zeta + c)(\zeta + c^4) = N_{L/K}(\zeta + c),$$

again an  $\ell$ -th power. Conversely the corrected norm  $N = \zeta^2 + \zeta + \omega$  belongs to  $V$ ; membership in  $H_K$  is exactly failure of  $D'_k$ .

Finally the two additive Fourier values of  $H_L$  are

$$\frac{-1 - R(\ell - 1)}{\ell} \quad \text{on } H_L, \quad \frac{R - 1}{\ell} \quad \text{off } H_L,$$

by (SP<sub>2</sub>). Fourier inversion applied to  $|W \cap H_L| = w - 1$  gives (SP<sub>6</sub>). The corresponding values for  $H_K$  are

$$\frac{-1 + q(\ell - 1)}{\ell} \quad \text{on } H_K, \quad \frac{-1 - q}{\ell} \quad \text{off } H_K,$$

and  $|V \cap H_K| = w_0 - 1$  gives (SP<sub>7</sub>)–(SP<sub>8</sub>). Integrality follows from  $\ell \mid q + 1$ ,  $w \mid q^2$ , and  $w_0 \mid q$ ; nonnegativity is immediate from  $\ell \leq q + 1$  and  $q \geq 2^9$ .  $\square$

**Proposition 8.59** (cyclotomic Dirichlet coefficient and subspace-polynomial exhaustion). *Retain  $k \geq 2$ , put  $h = 3^k$ , and write*

$$K = \mathbb{F}_{4^h} = \mathbb{F}_4(\zeta), \quad \zeta^h = \omega.$$

*Let  $\ell \mid \Psi_k$ , let  $\chi : K^\times \rightarrow \mu_\ell$  have exact order  $\ell$ , realized complexly, and let  $\tilde{\chi} = \chi \circ \text{Norm}_{\mathbb{F}_{16^h}/K}$ . Put*

$$s = h/3, \quad A = 2^s \pmod{\ell},$$

*choose  $u \in \mathbb{F}_{16}$  of order fifteen with  $u^5 = \omega$ , and set  $t = \tilde{\chi}(\zeta + u)$ . For  $0 \leq r < h$ , define*

$$\mathcal{A}_r(\chi) = \sum_{\substack{f \in \mathbb{F}_4[X] \text{ monic} \\ \deg f = r}} \chi(f(\zeta)).$$

*Then*

$$\mathcal{A}_0 = 1, \quad \mathcal{A}_1 = 4, \quad \mathcal{A}_2 = 10 + \sum_{j=0}^5 t^{A^j}. \quad (\text{DP}_1)$$

*Failure of  $D'_k$  at  $\ell$  is equivalent to the single extremal coefficient*

$$\mathcal{A}_2 = 16. \quad (\text{DP}_2)$$

*If  $\mathcal{A}'_1$  is the degree-one coefficient after the quadratic constant-field extension to  $\mathbb{F}_{16}$ , then*

$$\mathcal{A}'_1 = 2\mathcal{A}_2 - \mathcal{A}_1^2 = 2\mathcal{A}_2 - 16. \quad (\text{DP}_3)$$

*Thus the sixteen-term  $\mathbb{F}_{16}$ -line extremum and the six-term irreducible-quadratic extremum are the same constraint under constant-field base change, not two independent conditions.*

*More precisely, let*

$$L_\chi(T) = \sum_{r=0}^{h-1} \mathcal{A}_r T^r = (1 - T)L_\chi^*(T), \quad L_\chi^*(T) = \sum_{r=0}^d b_r T^r, \quad d = h - 2,$$

*so  $b_r = \sum_{j=0}^r \mathcal{A}_j$ . The functional equation has root number  $+1$  and gives*

$$b_{d-r} = 2^{d-2r} \overline{b_r}. \quad (\text{DP}_4)$$

*If  $D'_k$  fails, then*

$$(b_0, b_1, b_2) = (1, 5, 21), \quad b_{d-2} = 21 \cdot 2^{h-6}. \quad (\text{DP}_5)$$

*For*

$$V = \langle 1, \zeta, \zeta^2 \rangle_{\mathbb{F}_4},$$

the distinguished trace annihilator is

$$V^\perp = \zeta \{f(\zeta) : f \in \mathbb{F}_4[X], \deg f \leq h-4\}. \quad (\text{DP}_6)$$

Consequently, in the bad case,

$$\sum_{x \in V^\perp} \chi(x) = 3b_{d-2} = 63 \cdot 2^{h-6}, \quad (\text{DP}_7)$$

exactly the value forced independently by additive Poisson summation and  $G_K(\chi) = 2^h$ . Thus the first two Dirichlet coefficients, the exact cyclotomic trace-dual placement, the semiprimitive Gauss sign, the functional equation, and the Riemann hypothesis are mutually compatible with failure.

Finally put

$$D = \zeta^4 + \zeta, \quad E = D^5(D^3 + 1).$$

The  $\mathbb{F}_4$ -subspace polynomial of  $V$  is

$$\begin{aligned} P_V(Z) &= Z^{64} + ((1 + D^3)^4 + E^3)Z^{16} \\ &\quad + (D^{12} + E^3(1 + D^3))Z^4 + E^3D^3Z. \end{aligned} \quad (\text{DP}_8)$$

Both  $D$  and  $E$  are  $\ell$ -th powers unconditionally. Hence the derivative coefficient  $E^3D^3$  supplies no missing nonresidue; a subspace-polynomial proof would have to use an additive relation among the  $Z^{16}$ - and  $Z^4$ -coefficients rather than merely products of the saturated elements.

*Proof.* The four monic linears  $X + c$ ,  $c \in \mathbb{F}_4$ , evaluate to  $\ell$ -th powers, so  $\mathcal{A}_1 = 4$ . Of the sixteen monic quadratics over  $\mathbb{F}_4$ , ten are reducible and therefore also have character value one. The six irreducibles are

$$f_c(X) = (X + c)(X + c^4),$$

indexed by the six Frobenius pairs in  $\mathbb{F}_{16} \setminus \mathbb{F}_4$ . Since

$$\tilde{\chi}(\zeta + c) = \chi(f_c(\zeta)),$$

the orbit table of Proposition 8.18 gives their six values exactly  $t^{A^j}$ ,  $0 \leq j < 6$ . This proves  $(\text{DP}_1)$ . The six summands have modulus one, so  $\mathcal{A}_2 = 16$  if and only if all are one; the  $j = 0$  term then gives  $t = 1$ , which is exactly failure of  $D'_k$ .

The reciprocal roots of  $L_\chi$  are squared under quadratic constant-field extension. Newton's identity therefore gives

$$\mathcal{A}'_1 = 2\mathcal{A}_2 - \mathcal{A}_1^2,$$

proving  $(\text{DP}_3)$ .

The character is primitive modulo  $X^h + \omega$  and even because it is trivial on  $\mathbb{F}_4^\times$ . Standard function-field Dirichlet theory gives  $L_\chi = (1 - T)L_\chi^*$ , with  $\deg L_\chi^* = d = h - 2$ , and the coefficient relation in  $(\text{DP}_4)$  up to its root number. We determine that sign directly. Let  $U$  be the  $\mathbb{F}_4$ -space of polynomials in  $\zeta$  of degree at most  $h - 2$ . Newton sums for  $X^h + \omega$  give

$$\text{Tr}(\zeta^r) = 0 \quad (1 \leq r < h),$$

so  $U^\perp = \mathbb{F}_4\zeta$ . Additive Poisson summation and  $\chi(\mathbb{F}_4^\times\zeta) = 1$  give

$$3b_d = \sum_{x \in U} \chi(x) = \frac{3}{4}G_K(\chi).$$

Proposition 8.58 gives  $G_K(\chi) = 2^h$ , so

$$b_d = 2^{h-2} = 2^d.$$

The root number is therefore +1. Under failure,  $b_2 = 1 + 4 + 16 = 21$ , and  $(\text{DP}_4)$  at  $r = 2$  gives  $(\text{DP}_5)$ .

For  $x = \sum_{j=0}^{h-1} x_j \zeta^j$ , the same Newton-sum calculation gives

$$\text{Tr}(x) = x_0, \quad \text{Tr}(x\zeta) = \omega x_{h-1}, \quad \text{Tr}(x\zeta^2) = \omega x_{h-2}.$$

This proves  $(\text{DP}_6)$ . Every nonzero polynomial of degree at most  $h - 4$  is a unique  $\mathbb{F}_4^\times$ -multiple of a monic one, while  $\chi(\zeta) = 1$ . Hence

$$\sum_{x \in V^\perp} \chi(x) = 3b_{h-4} = 3b_{d-2}.$$

Direct Poisson summation under  $V \setminus \{0\} \subset \ker \chi$  gives  $(63/64)G_K(\chi)$ , the same value  $63 \cdot 2^{h-6}$ . This proves  $(\text{DP}_7)$ .

It remains to derive the subspace polynomial. First put  $U_1 = \langle 1, \zeta \rangle_{\mathbb{F}_4}$ . Since  $\prod_{c \in \mathbb{F}_4} (Z + c) = Z^4 + Z$ ,

$$\begin{aligned} P_{U_1}(Z) &= (Z^4 + Z)^4 + D^3(Z^4 + Z) \\ &= Z^{16} + (1 + D^3)Z^4 + D^3Z. \end{aligned}$$

Evaluating at  $\zeta^2$  gives

$$E = P_{U_1}(\zeta^2) = D^8 + D^5 = D^5(D^3 + 1).$$

Because  $P_{U_1}$  is  $\mathbb{F}_4$ -linearized,

$$P_V(Z) = P_{U_1}(Z)^4 + E^3 P_{U_1}(Z),$$

which expands to  $(\text{DP}_8)$ .

Finally,

$$D = \prod_{c \in \mathbb{F}_4} (\zeta + c),$$

so  $\chi(D) = 1$ . Also  $E$  is the product of all sixteen monic-quadratic values. The ten reducible values have character one, and the other six multiply to

$$t^{\sum_{j=0}^5 A^j} = 1$$

because  $A$  has order six modulo  $\ell$ . Thus  $\chi(E) = 1$  without assuming failure, which finishes the proof. □

**Proposition 8.60** (relative Dickson form and its degree ceiling). *Retain*

$$s = 3^{k-1}, \quad a = 2^s, \quad E = \mathbb{F}_a, \quad K = \mathbb{F}_{a^6}, \quad \Lambda_k = a^2 - a + 1,$$

and set  $T_k = \{x \in K^\times : x^{\Lambda_k} = 1\}$ . The element  $P_k \in T_k$  has degree six over  $E$ . Put

$$Y_k = P_k + P_k^{-1} \in \mathbb{F}_{a^3}, \quad G_k(T) = \prod_{j=0}^2 (T + Y_k^{a^j}) \in E[T].$$

Then  $G_k$  is irreducible of degree three and  $X^3 G_k(X + X^{-1})$  is the irreducible reciprocal minimal polynomial of  $P_k$  over  $E$ . If  $\mathcal{D}_\ell$  is characterized by

$$\mathcal{D}_\ell(v + v^{-1}) = v^\ell + v^{-\ell},$$

then, for every prime  $\ell \mid \Psi_k$ ,

$$P_k \notin T_k^\ell \iff G_k(\mathcal{D}_\ell(X)) \text{ is irreducible over } E. \quad (\text{RD})$$

In the bad case the composition splits over  $\mathbb{F}_{a^3}$ , and all its irreducible factors over  $E$  have degree dividing three. Moreover

$$T_k \cap \mathbb{F}_{a^2}^\times = \mu_3, \quad T_k \cap \mathbb{F}_{a^3}^\times = \{1\}.$$

Thus the elements of  $T_k$  of relative degree below six are exactly  $\mu_3$ , while the bad subgroup  $T_k^\ell$  contains exactly

$$\frac{\Lambda_k}{\ell} - 3$$

elements of exact degree six. Except in the already settled case  $\Lambda_k/\ell = 3$ , full relative degree and reciprocal-torus structure cannot prove  $(D'_k)$ .

*Proof.* Proposition 8.5 gives  $[\mathbb{F}_2(P_k) : \mathbb{F}_2] = 6s$ , so  $E(P_k) = K$ . Since  $a^3 + 1 = (a + 1)\Lambda_k$ , one has  $P_k^{a^3} = P_k^{-1}$ , whence  $Y_k \in \mathbb{F}_{a^3}$ . The element  $P_k$  is quadratic over  $E(Y_k)$ ; its relative degree six forces  $[E(Y_k) : E] = 3$ , proving the minimal-polynomial assertions.

Suppose  $P_k \notin T_k^\ell$ . The whole current  $\ell$ -primary part of  $K^\times$  lies in  $T_k$ , so this is equivalent to  $P_k \notin (K^\times)^\ell$ . Kummer theory and Capelli make

$$X^{3\ell} G_k(X^\ell + X^{-\ell})$$

irreducible of degree  $6\ell$ . If  $\alpha^\ell = P_k$  and  $t = \alpha + \alpha^{-1}$ , then  $\mathcal{D}_\ell(t) = Y_k$ . Since  $\alpha$  is at most quadratic over  $E(t)$ , the degree of  $t$  is at least  $3\ell$ ; it is a root of the degree- $3\ell$  polynomial  $G_k(\mathcal{D}_\ell(X))$ , proving the forward implication in (RD).

Conversely, if  $P_k = \alpha^\ell$  with  $\alpha \in T_k$ , then  $\alpha^{a^3} = \alpha^{-1}$ , so  $t \in \mathbb{F}_{a^3}$ . The same holds after multiplying  $\alpha$  by any  $\ell$ -th root of unity and after applying the three  $E$ -conjugates. Hence every root of the Dickson composition lies in  $\mathbb{F}_{a^3}$ , proving the bad-case factorization statement and the reverse implication.

Finally the two displayed intersections have orders

$$\gcd(\Lambda_k, a^2 - 1) = 3, \quad \gcd(\Lambda_k, a^3 - 1) = 1.$$

Every proper intermediate field of  $K/E$  lies in  $\mathbb{F}_{a^2}$  or  $\mathbb{F}_{a^3}$ , so only  $\mu_3$  has relative degree below six. The cyclic group  $T_k^\ell$  has order  $\Lambda_k/\ell$  and contains  $\mu_3$ , because  $\ell \neq 3$ . Removing those three elements gives the asserted exact count.  $\square$

**Proposition 8.61** (trace-degree ceiling for the fibotomic criterion). *Retain Proposition 8.6, and let*

$$\mathcal{I}_\ell := \pi(U^\ell \setminus \{1\}) \subseteq \mathbb{F}_q \quad (\ell \text{ a prime divisor of } \Psi_k).$$

*Every element of  $\mathcal{I}_\ell$  has absolute trace one, and*

$$|\mathcal{I}_\ell| = \frac{(q+1)/\ell - 1}{2} \geq \frac{3a+2}{2} > a. \quad (\text{T})$$



At least  $a/2 + 1$  elements of  $\mathcal{I}_\ell$  have exact degree  $h$  over  $\mathbb{F}_2$ . Consequently, absolute trace one together with full degree  $h$  cannot prove that the distinguished  $Z$  lies outside  $\pi(U^\ell)$ .

*Proof.* The two-to-one trace-one parametrization in Proposition 8.6 restricts to  $U^\ell$ . This cyclic subgroup has order  $(q+1)/\ell$ , which is odd, so its only self-inverse element is 1. This proves the equality in (T). Since  $\ell \leq \Psi_k = \Lambda_k/3$  and  $q+1 = (a+1)\Lambda_k$ , it also gives

$$|\mathcal{I}_\ell| \geq \frac{3(a+1) - 1}{2} = \frac{3a+2}{2}.$$

Every proper subfield of  $\mathbb{F}_q$  lies in  $\mathbb{F}_{2^{h/3}} = \mathbb{F}_a$ , because  $h = 3^k$ . Hence at most  $a$  elements of  $\mathcal{I}_\ell$  have degree smaller than  $h$ , leaving at least  $(3a+2)/2 - a = a/2 + 1$  of exact degree  $h$ .

For completeness, the distinguished  $Z$  itself has exact degree  $h$ . Indeed,  $(D_3)$  and  $P_k \in \mathbb{F}_2(M_k)$  force  $[\mathbb{F}_2(M_k) : \mathbb{F}_2] = 2h$ . The element  $M_k$  is quadratic over  $\mathbb{F}_2(Z)$ , satisfying

$$X^2 + Z^{-1}X + 1 = 0,$$

while  $Z \in \mathbb{F}_q$ . Therefore  $[\mathbb{F}_2(Z) : \mathbb{F}_2] = h$ . □

## 8.6 The limits of norm induction

**Proposition 8.62** (all proper-subfield norms are blind). *Retain  $k \geq 2$  and the notation of Proposition 8.3. Let  $\ell$  be a prime divisor of  $\Psi_k$ , and let  $d$  be any proper divisor of  $2h$ . Then*

$$N_{\mathbb{F}_{2^{2h}}/\mathbb{F}_{2^d}}$$

*annihilates the entire  $\ell$ -Sylow subgroup of  $\mathbb{F}_{2^{2h}}^\times$ . Equivalently, multiplying an element by any current  $\ell$ -primary component leaves every multiplicative norm to a proper subfield unchanged.*

*Proof.* The norm exponent is

$$E_d = \frac{2^{2h} - 1}{2^d - 1}.$$

Since  $\text{ord}_\ell(2) = 2h$  and  $d < 2h$ , one has  $\ell \nmid 2^d - 1$ . Therefore

$$v_\ell(E_d) = v_\ell(2^{2h} - 1),$$

so exponentiation by  $E_d$  kills the full  $\ell$ -Sylow subgroup. □

For  $k = 1$ ,  $\Psi_1 = 1$  and  $D'_1$  is vacuous. Exact finite evidence for the subsequent levels, together with its factorization and primality dependencies, is recorded in Appendix A.4.

**Proposition 8.63** (twisted norm tower). *With  $\eta = \zeta^3 = \kappa_{3^{k-1}}$ ,*

$$N_{\mathbb{F}_{2^{2 \cdot 3^k}}/\mathbb{F}_{2^{2 \cdot 3^{k-1}}}}(N_k) = \eta^2 + \omega^2\eta + 1,$$

*which is neither  $N_{k-1} = \eta^2 + \eta + \omega$ , nor one of its Frobenius conjugates (their exponent pattern is  $(2^{j+1}, 2^j)$ ), nor a scaling  $N_{k-1}(\lambda\eta)$  with  $\lambda \in \mu_3$ .*

*Proof.* The generator  $\tau$  of the degree-3 step sends  $\zeta \mapsto \zeta^c$  with  $c = 2^{2 \cdot 3^{k-1}}$  a primitive cube root of unity modulo  $3^{k+1}$ ; the cube roots are  $1 + a3^k$ , so  $\tau(\zeta) = \omega^a\zeta$  ( $a \neq 0$ ) while  $\tau(\omega) = \omega$ . Hence the norm is  $\prod_{r^3=\eta} (r^2 + r + \omega)$  over the three cube roots  $r$  of  $\eta$ . Writing  $\alpha, \beta$  for the two roots of

$T^2 + T + \omega$  (so  $\alpha + \beta = 1$ ,  $\alpha\beta = \omega$ ), each factor is  $(r + \alpha)(r + \beta)$ , and the product collapses by the resolvent

$$\prod_{r^3=\eta} (r^2 + r + \omega) = (\alpha^3 + \eta)(\beta^3 + \eta) = \eta^2 + (\alpha^3 + \beta^3)\eta + (\alpha\beta)^3 = \eta^2 + \omega^2\eta + 1,$$

since  $\alpha^3 + \beta^3 = (\alpha + \beta)^3 + \alpha\beta(\alpha + \beta) = 1 + \omega = \omega^2$  and  $(\alpha\beta)^3 = \omega^3 = 1$ . Machine-checked at  $k = 2, 3, 4$ .  $\square$

**Proposition 8.64** (trace form and norm blindness). *Retain the notation of Proposition 8.3, and put*

$$\gamma_k := \zeta + \zeta^{-1}, \quad A_k := \gamma_k^2 + \gamma_k + 1, \quad Q := 2^{2 \cdot 3^{k-1}} = 2^{2h/3}.$$

Set  $\gamma_0 = A_0 = 1$ . Then, with  $\overline{N} = N^{2^h}$ ,

$$N + \overline{N} = A_k, \quad N\overline{N} = A_k^Q, \quad M_k + M_k^{-1} = A_k^{2-Q}, \quad Z = A_k^{Q-2}. \quad (4)$$

Moreover

$$N_{\mathbb{F}_{2^{3k}}/\mathbb{F}_{2^{3^{k-1}}}}(A_k) = A_{k-1}, \quad N_{\mathbb{F}_{2^{3k}}/\mathbb{F}_{2^{3^{k-1}}}}(Z) = A_{k-1}^{-1}. \quad (5)$$

For every prime  $\ell \mid \Psi_k$ , however, this relative norm is blind to the required  $\ell$ -power residue class.

*Proof.* Since  $2^h \equiv -1 \pmod{3h}$  and  $\zeta^h + \zeta^{2h} = 1$ , direct expansion gives

$$N + \overline{N} = \zeta^2 + \zeta^{-2} + \zeta + \zeta^{-1} + 1 = A_k.$$

The congruence  $Q \equiv h + 1 \pmod{3h}$  follows inductively by cubing  $1 + 3^r$  modulo  $3^{r+2}$ . Hence  $\gamma_k^Q = \zeta^{h+1} + \zeta^{-(h+1)}$ , and another direct expansion yields  $N\overline{N} = A_k^Q$ . Therefore

$$M_k + M_k^{-1} = \frac{(N + \overline{N})^2}{N\overline{N}} = A_k^{2-Q},$$

and  $Z = \pi(M_k) = 1/(M_k + M_k^{-1})$ , which proves (4).

Over  $B = \mathbb{F}_{2^{h/3}}$ , the element  $\gamma_k$  has minimal polynomial  $X^3 + X + \gamma_{k-1}$ . Base-change this cubic extension by  $B_2 = B\mathbb{F}_4 = \mathbb{F}_{2^{2h/3}}$ ; its compositum is  $\mathbb{F}_{2^{2h}}$ , and the degree remains three. With  $\omega = \zeta^h \in \mathbb{F}_4$  one has  $A_k = (\gamma_k + \omega)(\gamma_k + \omega^2)$ . Evaluating that minimal polynomial at  $\omega$  and  $\omega^2$  gives

$$N_{\mathbb{F}_{2^{2h}}/B_2}(\gamma_k + \omega) = \gamma_{k-1} + \omega^2, \quad N_{\mathbb{F}_{2^{2h}}/B_2}(\gamma_k + \omega^2) = \gamma_{k-1} + \omega.$$

Their product is  $A_{k-1}$ . Since norm commutes with this scalar extension, this is precisely  $N_{\mathbb{F}_{2^h}/B}(A_k)$ , proving the first identity in (5). Furthermore  $Z = A_k^{Q-2}$ , so

$$N_{\mathbb{F}_{2^h}/B}(Z) = A_{k-1}^{Q-2} = A_{k-1}^{-1},$$

because  $Q - 2 \equiv -1 \pmod{2^{h/3} - 1}$ . This proves the second identity.

If  $\ell \mid \Psi_k$ , then  $\ell \nmid 2^h - 1$  and  $\ell \nmid 2^{h/3} - 1$ . Thus both  $A_k$  and  $A_{k-1}$  lie in multiplicative groups on which raising to the  $\ell$ -th power is an automorphism, so (5) is tautological on  $\ell$ -power classes. Equivalently,  $\text{ord}_\ell(Q) = 3$  and  $1 + Q + Q^2 \equiv 0 \pmod{\ell}$ : the degree-three norm annihilates the relevant character. The surviving current-level condition is the Capelli test  $R_k(X^\ell)$  irreducible from Proposition 8.3.  $\square$

So, unlike the  $\gamma$  tower of Proposition 6.7, old-prime full-order parts do *not propagate by that argument*, and  $D'_k$  is genuinely per-level for the present method: the certification at each  $k$  stands on its own. Structurally, at the degree-3 step with  $Q = 2^{2 \cdot 3^{k-1}}$ , the relative norm kernel has order

$$Q^2 + Q + 1 = \Phi_{3^k}(2)\Phi_{2 \cdot 3^k}(2).$$

Its gcd with  $Q - 1$  is 3, so there is no direct-product splitting at the 3-part as in Proposition 6.8; nevertheless, for every  $\ell \neq 3$  the new  $\ell$ -Sylow subgroups of both columns lie entirely in the norm kernel. This is the prime-to-3 analogue of Proposition 6.8 and explains why a lower-level norm identity cannot determine  $D'_k$ .

**Remark 8.65** (the 3-part). Proposition 8.5 and  $v_3(a + 1) = k$  imply

$$v_3(\text{ord } M_k) = k + 1 \quad (k \text{ odd}), \quad v_3(\text{ord } M_k) \leq k \quad (k \text{ even}).$$

Thus the odd-level full 3-part is now explained analytically. *Computational evidence.* At even levels  $2 \leq k \leq 6$ , the sharper observed value is  $v_3(\text{ord } M_k) = k - 1$ ; the remaining one-step deficiency is not determined by the projection theorem.

## 8.7 The exceptional-spine conjecture ( $D$ )

The exceptional arm is precisely Conjecture 8.4: for each  $k$ , the distinguished antiunit  $M_k$  must contain the full current-level factor  $\Psi_k$  in its order. The projection and partition-regulator theorems settle explicit large primary factors, while the translated-binomial, semiprimitive, Dirichlet, Dickson, and norm calculations prove that the remaining smaller factors survive all present generic criteria. What remains is the single arithmetic evaluation  $\text{ind}_\ell(d_k) \neq 0$ . The explicit wild regulator in Proposition 8.30 and the code/design and two-adic lifting no-go results in Propositions 8.32 and 8.33 show that neither a derived local symbol nor a higher lift supplies an independent value. This is a distinguished cyclotomic Kummer-class problem, not a further degree or norm identity.

**Proposition 8.66** (common selected-prime reciprocity boundary). *Let  $\ell$  be an odd prime, let  $E$  be a number field containing  $\mu_\ell$ , let  $u \in \mathcal{O}_E^\times$ , and let  $\mathfrak{P} \nmid \ell$ . Fix a finite set  $S$  of primes, disjoint from  $\{\mathfrak{P}\}$  and the primes over  $\ell$ . Suppose that  $d \in \mathbb{Z}_{>0}$ ,  $n_\tau \in \mathbb{Z}$ , and  $\beta \in E^\times$  principalize the selected prime up to  $S$ , as an equality of fractional ideals:*

$$(\beta) = \mathfrak{P}^d \prod_{\tau \in S} \tau^{n_\tau}. \quad (\text{PR}_1)$$

Use the convention

$$(u, b)_{v, \ell} := \frac{\text{rec}_v(b)(\sqrt[\ell]{u})}{\sqrt[\ell]{u}},$$

where local reciprocity sends a uniformizer to arithmetic Frobenius. Then

$$\left(\frac{u}{\mathfrak{P}}\right)_\ell^d = \left[ \prod_{\tau \in S} \left(\frac{u}{\tau}\right)_\ell^{n_\tau} \prod_{\mathfrak{s} \mid \ell} (u, \beta)_{\mathfrak{s}, \ell} \right]^{-1}. \quad (\text{PR}_2)$$

Here a negative exponent  $n_\tau$  means the corresponding inverse residue symbol. If  $\ell \nmid d$ , this determines the selected symbol from the displayed local product after raising the right side to  $d^{-1} \bmod \ell$ . If  $\ell \mid d$ , its left side is identically one and the formula gives no selected information.

Suppose in addition that  $E/K$  is Galois, that  $u \in \mathcal{O}_K^\times$ , and that a prime  $\mathfrak{q} \mid 2$  of  $K$  splits into the  $\ell - 1$  primes  $\mathfrak{P}_a = \tau_a \mathfrak{P}_1$ ,  $a \in \mathbb{F}_\ell^\times$ , where  $\tau_a(\zeta_\ell) = \zeta_\ell^a$ . If

$$A = \left( \frac{u}{\mathfrak{P}_1} \right)_\ell,$$

then

$$\left( \frac{u}{\mathfrak{P}_a} \right)_\ell = A^a, \quad \prod_{a \in \mathbb{F}_\ell^\times} \left( \frac{u}{\mathfrak{P}_a} \right)_\ell = A^{\ell(\ell-1)/2} = 1. \quad (\text{PR}_3)$$

Consequently an element of the base field, whose valuations are equal at all the  $\mathfrak{P}_a$ , can recover only the tautological orbit product. Isolating one Conway-selected lift requires a genuinely mixed principalizer in the  $K$ - and  $\ell$ -cyclotomic directions.

*Proof.* At a finite place away from  $\ell$ , adjoining an  $\ell$ -th root of the unit  $u$  gives either the split extension or an unramified Kummer extension. Local units therefore act trivially. Equation  $(\text{PR}_1)$  gives

$$(u, \beta)_{\mathfrak{P}, \ell} = \left( \frac{u}{\mathfrak{P}} \right)_\ell^d, \quad (u, \beta)_{\mathfrak{r}, \ell} = \left( \frac{u}{\mathfrak{r}} \right)_\ell^{n_{\mathfrak{r}}}.$$

All other finite places away from  $\ell$  contribute one, and the archimedean symbols are trivial because  $\ell$  is odd. The global Hilbert product formula is exactly  $(\text{PR}_2)$ .

For the second assertion, Galois covariance of the Hilbert symbol acts on its value through the cyclotomic character, giving the first formula in  $(\text{PR}_3)$ . The sum of the nonzero residue classes is  $\ell(\ell - 1)/2$ , a multiple of  $\ell$ , while  $A^\ell = 1$ . This proves the orbit cancellation. Equal valuations of a base-field element at conjugate lifts cannot replace that product by one selected factor.  $\square$

**Corollary 8.67** (the cyclotomic-unit arms share one wild ray boundary). *The cyclotomic-unit specializations of the cubic arm, the ordinary arm, the exact two-component zero reduction, and the exceptional arm all meet the same principalizer boundary:*

- in the cubic arm one may take  $u = (1 + \xi)(1 + \xi^{-1})$ , whose selected residue symbol is the root of unity determined by the coefficient  $L(1)$  of Proposition 6.41;
- the ordinary arm and exact two-component zero reduction use respectively their norm-one quotient and  $1 + \zeta = (1 - \zeta^2)/(1 - \zeta)$ , both global cyclotomic units;
- in the exceptional arm, writing  $\rho = \alpha\beta$ , with  $\text{ord}(\alpha) = 3h$  and  $\text{ord}(\beta) = 5$ , the antiunit

$$d_k = \frac{(1 - \alpha\beta)(1 - \alpha\beta^{-1})}{(1 - \alpha\beta^2)(1 - \alpha\beta^{-2})}$$

has the same selected  $\ell$ -class as  $1 - \rho$ , up to the invertible factor four. At a prime over five one has  $d_k \equiv 1$ . At a prime over three,

$$\bar{d}_k = \frac{\beta}{(1 + \beta)^2} \in \mathbb{F}_{3^4}^\times,$$

whose order divides eighty; hence it and the principal-unit correction are  $\ell$ -th powers for every current  $\ell$ .

In each case choose  $d$  to be the order of the selected prime in the ordinary ideal class group, so that  $(\text{PR}_1)$  has  $S = \emptyset$ . Formula  $(\text{PR}_2)$  then identifies the mixed  $\ell$ -adic local product with the  $d$ -th power of the selected symbol. If  $\ell \nmid d$ , this transports the selected ray coordinate but does not evaluate it; if  $\ell \mid d$ , the formula is tautological. Equivalently, the surviving quantities  $\Theta_{q,s}$ , the cubic  $L(1)$ , and the exceptional antiunit index are distinct instances of the same selected-prime ray-coordinate problem, not consequences of conductor-local reciprocity.

The abstract proposition also applies to a chosen characteristic-zero lift of the singleton-even Conway minimal polynomial, whose constant coefficient is one and whose root is therefore a global unit. That lift is noncanonical and the resulting mixed local coordinate is still unevaluated. No claim here replaces the separate synchronized-phase problem for zero arms with three or more components.

*Proof.* Finiteness of the ordinary ideal class group supplies the stated choice of  $d$  and a generator of  $\mathfrak{P}^d$ . Only the exceptional local calculation is not already part of the arm-specific propositions. At five,  $\beta$  reduces to one, so the two numerator and two denominator factors agree. At three,  $\alpha$  reduces to one and direct cancellation gives the displayed element. Its residue order divides  $3^4 - 1 = 80$ ; the current primes are different from two, three, and five. Raising to the  $\ell$ -th power is therefore an automorphism on this residue group, and also on the pro-three and pro-five principal units. The remaining assertions are the respective selected-symbol identifications and Proposition 8.66.  $\square$

**Proposition 8.68** (Jacobi failure is the second ramified torsion layer). *Let  $M$  and  $\ell$  be coprime odd integers, with  $\ell$  prime, and suppose  $M\ell \mid Q - 1$ . Over  $\mathbb{F}_Q$ , let*

$$X_{M,\ell} : \quad y^\ell = 1 - x^M, \quad J = \text{Jac}(X_{M,\ell}), \quad \pi = \zeta_\ell - 1.$$

For nontrivial characters of orders dividing  $M$  and  $\ell$ , put

$$J_{j,t} = \sum_{a \in \mathbb{F}_Q} \chi_M(a)^j \chi_\ell(1 - a)^t \quad (1 \leq j < M, \ 1 \leq t < \ell).$$

Then

$$g(X_{M,\ell}) = \frac{(M-1)(\ell-1)}{2}, \quad P_X(T) = \prod_{j=1}^{M-1} \prod_{t=1}^{\ell-1} (1 + J_{j,t}T). \quad (\text{JT}_1)$$

In particular, the Frobenius eigenvalues are  $\alpha_{j,t} = -J_{j,t}$ .

The simultaneous first-order flatness condition

$$J_{j,1} \equiv -1 \pmod{\pi^2} \quad (1 \leq j < M) \quad (\text{JT}_2)$$

is equivalent to

$$J[\pi^2] \subseteq J(\mathbb{F}_Q). \quad (\text{JT}_3)$$

Consequently it implies the exact divisibility

$$\ell^{2(M-1)} \mid \#J(\mathbb{F}_Q). \quad (\text{JT}_4)$$

The congruence modulo  $\pi$ , and hence rationality of  $J[\pi]$ , is unconditional. Failure is exactly rationality through the second ramified layer and may lift further; it does not assert rationality of the full  $\ell$ -torsion.

For the cubic arm take

$$M = 3^{k+1}, \quad Q = 2^{2 \cdot 3^k};$$

for the exceptional arm, with  $k \geq 2$ , take

$$M = 15 \cdot 3^k, \quad Q = 2^{4 \cdot 3^k}.$$

Their respective mixed-Jacobi failure criteria are exactly (JT<sub>2</sub>). In both cases

$$\ell < \sqrt{Q} + 1. \quad (\text{JT}_5)$$

Thus even the much stronger hypothetical inclusion  $J[\ell] \subseteq J(\mathbb{F}_Q)$ , which would force the divisor  $\ell^{2g}$ , is compatible with the Weil bound

$$\#J(\mathbb{F}_Q) \leq (\sqrt{Q} + 1)^{2g}. \quad (\text{JT}_6)$$

Neither the point-count bound nor the polarization excludes the actual second-layer failure pattern.

*Proof.* The polynomial  $1 - x^M$  is squarefree and  $(M, \ell) = 1$ , so the superelliptic genus formula gives the first identity in (JT<sub>1</sub>). To count points, expand the number of  $\ell$ -th roots of  $1 - x^M$  by  $\ell$ -characters, and expand the number of  $M$ -th roots of a field element by  $M$ -characters. The two nontrivial character sums are exactly the  $J_{j,t}$ ; the unique point at infinity supplies the remaining one. Hence

$$\#X_{M,\ell}(\mathbb{F}_Q) = Q + 1 + \sum_{j,t} J_{j,t}.$$

The same calculation over every finite extension, together with Hasse–Davenport, gives the second identity in (JT<sub>1</sub>).

Let  $\mathcal{O} = \mathbb{Z}_\ell[\zeta_\ell]$  and

$$R = \mathcal{O}[T] / ((T^M - 1)/(T - 1)).$$

Because  $\ell \nmid M$ , this is finite étale of rank  $M - 1$  over  $\mathcal{O}$ . The automorphisms  $(x, y) \mapsto (\zeta_M x, y)$  and  $(x, y) \mapsto (x, \zeta_\ell y)$  act on  $J$ . After the unramified extension adjoining  $\zeta_M$ , the quotient by either deck group is genus zero, so neither the  $M$ - nor the  $\ell$ -invariant part of the Tate module occurs. Because  $\ell \nmid M$ , the  $M$ -action is  $\ell$ -adically semisimple;  $T_\ell J$  is  $\mathcal{O}$ -torsion-free, and formula (JT<sub>1</sub>) shows that every nontrivial  $M$ -character component is free of  $\mathcal{O}$ -rank one. Descent from this unramified splitting then makes  $T_\ell J$  rank-one projective over  $R$ . Frobenius acts on its character lines by the  $\alpha_{j,1}$ , up to harmless inversion of  $j$  according to the eigenspace convention.

Every automorphism  $\zeta_\ell \mapsto \zeta_\ell^t$  sends  $\pi$  to a unit multiple of  $\pi$  and sends  $J_{j,1}$  to  $J_{j,t}$ . Therefore (JT<sub>2</sub>) says exactly that Frobenius is the identity modulo  $\pi^2$  on  $T_\ell J$ . This is equivalent to (JT<sub>3</sub>). Since

$$|R/\pi^2 R| = \ell^{2(M-1)},$$

equation (JT<sub>4</sub>) follows. Reduction modulo  $\pi$  makes every  $\ell$ -character value one away from zero, giving  $J_{j,t} \equiv -1 \pmod{\pi}$  without any failure hypothesis.

For the cubic arm, put  $n = 3^k$  and  $x = 2^{n/3}$ . A current prime satisfies

$$\ell \leq x^2 + x + 1 < x^3 + 1 = \sqrt{Q} + 1.$$

For the exceptional arm, with  $h = 3^k$ ,

$$\ell \leq \Psi_k < 2^{2h} = \sqrt{Q}.$$

This proves (JT<sub>5</sub>), and raising it to the power  $2g$  proves compatibility with (JT<sub>6</sub>). Finally  $Q \equiv 1 \pmod{\ell}$ , so the Weil pairing permits rational  $\ell$ -torsion. More sharply, for the present  $\ell \geq 7$ , the established  $\mathcal{O}$ -freeness gives  $J[\pi^2] = \pi^{\ell-3} J[\ell]$ . The Rosati involution sends  $\zeta_\ell$  to  $\zeta_\ell^{-1}$ , so  $\pi^\dagger$  is a unit times  $\pi$ . For  $x = \pi^{\ell-3} x_0$  and  $y = \pi^{\ell-3} y_0$ , the product  $(\pi^{\ell-3})^\dagger \pi^{\ell-3}$  has  $\pi$ -valuation  $2\ell - 6 \geq \ell - 1 = v_\pi(\ell)$ ; the Weil pairing of  $x$  and  $y$  is therefore one. Thus  $J[\pi^2]$  is totally isotropic, and its rank respects the polarization bound. This completes the claimed geometric no-go.  $\square$

## 9 Consequences and the Conjectural Frontier

### 9.1 Boundedness

Lenstra conjectured that  $m_p$  is globally bounded [22]. His unconditional rules give positive excess on singleton odd component sets and the lower bound  $m_p \geq 4$  when  $f(p) = 2 \cdot 3^k$ , but do not give an absolute upper bound. The four-arm equivalence of Theorem 1.2 shows exactly what would settle the question: Conjecture 1.1 implies

$$m_p \in \{0, 1, 4\} \quad \text{for every odd prime } p.$$

There is also an exact formulation of the weaker boundedness question which does not ask for any maximal-order arm. It identifies precisely what a direct fixed-window proof would still have to evaluate.

**Proposition 9.1** (finite Conway-line criterion for absolute boundedness). *For  $a \geq 0$ , let*

$$C_a = \mathbb{F}_{2^{2^a}} = \{0, 1, \dots, 2^{2^a} - 1\}$$

*be the corresponding finite Conway subfield. For an odd prime  $p$ , put  $x_p = \kappa_{f(p)}$  and*

$$L_{p,a} = \mathbb{F}_2(x_p)C_a.$$

*Then*

$$m_p < 2^{2^a} \iff \text{some } c \in C_a \text{ is such that } x_p + c \notin L_{p,a}^p, \quad (\text{AB}_1)$$

*where  $L^p = \{y^p : y \in L\}$ . Consequently*

$$\sup_p m_p < \infty \iff \text{for some fixed } a, \quad \forall p \exists c \in C_a : x_p + c \notin L_{p,a}^p. \quad (\text{AB}_2)$$

*Thus absolute boundedness is exactly uniform nonsaturation of one fixed finite-Conway-subfield affine line.*

*Proof.* Fix  $c \in C_a$ , put  $\beta = x_p + c$ , and let  $F_c = \mathbb{F}_2(\beta)$ . Since  $x_p = \beta + c$ , one has

$$L_{p,a} = F_c C_a.$$

Hence  $[L_{p,a} : F_c]$  divides  $[C_a : \mathbb{F}_2] = 2^a$ , and is therefore prime to the odd prime  $p$ . Extension from  $F_c$  to  $L_{p,a}$  is injective on  $p$ -Kummer quotients: restriction followed by norm is multiplication by  $[L_{p,a} : F_c]$ , which is invertible modulo  $p$ . Thus  $\beta$  is a  $p$ -th power in its defining finite field exactly when it is a  $p$ -th power in the common field  $L_{p,a}$ .

The finite numbers below  $2^{2^a}$  are exactly  $C_a$ , so the definition of the excess gives  $(\text{AB}_1)$ . A fixed  $a$  therefore gives the uniform bound  $2^{2^a} - 1$ . Conversely, from a uniform bound  $B$ , choose  $a$  with  $2^{2^a} > B$ ; every witnessing translate then lies in  $C_a$ . This proves  $(\text{AB}_2)$ .  $\square$

The theorem also explains why a generic finite-field argument is insufficient. Proposition 5.27 constructs arbitrarily large exact-degree elements whose translates in any fixed finite window are all  $p$ -th powers. Any proof must therefore use the distinguished Conway-tower identities.

The available uniform results are substantial but incomplete. The zero arm is proved for primitive support and completely at the first two power-of-two two-spine levels  $h = 12, 24$ ; the ordinary odd-spine arm has projective reductions and an absolute-Frobenius signed-ball sieve; the cubic and

exceptional arms have explicit partition-regulator bounds. On the exceptional column, Proposition 8.21 also gives the direct effective estimate

$$m_\ell < 16(3^k)^4 \quad (f(\ell) = 2 \cdot 3^k, \ k \geq 2).$$

Its proof shows why this does not become absolute: a fixed Conway line is one fixed constant-extension power sum of a selected Dirichlet  $L$ -polynomial, and the Riemann–Weil bound is eventually compatible with complete saturation of that line. Each leaves smaller current primary factors untreated. Consequently neither global boundedness nor the nonexistence of an excess  $m_p \geq 5$  is presently proved.

## 9.2 The four remaining order problems

The proof frontier is most cleanly stated in the same four coordinates as Theorem 1.2.

**A. Zero arm.** There are two exact maximal-order problems. At a singleton-even level, prove

$$\delta_n = F_n$$

in Proposition 7.1. The fibotomic recursion, elliptic correspondence, and equivariant five-point generalized Jacobian reduce this to the nonvanishing of one explicit distinguished Kummer value; they also prove that the available trace, norm, and boundary-character data are circular. Proposition 7.31 pushes the minimal-counterexample hypothesis through unique  $\ell$ -th roots: every lower trace, norm, and maximal-order identity survives automatically, while top failure is exactly one Dickson power-sum equation. Corollary 7.32 removes even the auxiliary root choice: for  $d = (q + 1)/\ell$  the lower coefficient is  $(a_{n-1}^d)^{q/2}$ . The resulting equation is still the same selected nonvanishing target, rather than a lower-level contradiction. Proposition 7.41 proves that its Dickson polynomial has no intermediate factorization: failure is complete splitting and success is irreducibility, with the same Kummer scalar appearing in the exact resultant. Proposition 7.42 then transfers failure to the selected divisibility  $\mathcal{P}_{\ell,n} \mid H_m$ : the absolute norm of a reverse-Dickson polynomial, containing the actual minimal polynomial of  $a_{n-1}$ , must divide the trace-one polynomial. Its factorization is one irreducible of degree  $m\ell$  on success and  $\ell$  irreducibles of degree  $m$  on failure. Proposition 7.43 identifies the selected ancestry more rigidly: absolute trace in the Conway bit basis is exactly the top-bit functional, and  $a_{n-1} = q/2$  is the least trace-one number. After translation by that literal top bit, failure is the selected divisibility  $P_{\ell,a}(Z + a) \mid \sum_{i < m} Z^{2^i}$  inside the trace-zero linearized polynomial. This centers the obstruction but does not exclude it; transported nonselected Kummer cosets yield the same type of divisor. Proposition 7.44 makes the limitation sharp: after an adapted binary labeling, nonselected full-degree Kummer cosets simultaneously realize the top-bit trace rule, the centered divisor, all Moore-span bounds, and the adjoint trace identities. Such a labeling need not respect Conway multiplication, so the construction is not a counterexample; it proves that additive linearization cannot replace the missing multiplicative-ancestry argument. Corollary 7.61 and Proposition 7.62 retain the literal least upper Conway interval. Its complete character histogram, factor degrees, discriminants, pairwise resultants, and root-difference data are universal after one common translation. The marked endpoint is the only remaining datum: its trivial-label test is  $B_1(0) = S_d(a_{n-1})$ . Every individual coefficient or Hasse-jet test is a shifted Fibonacci value, and its selected norm is the corresponding shifted resultant. Thus the entire labelled block supplies no selector-blind second invariant; any gain must add genuinely ancestry-sensitive input, possibly through a relation among shifted values.



Proposition 7.5 keeps two consecutive selected levels at once: its sparse quartic composition is irreducible on success and splits into exactly  $\ell$  irreducible quartics on failure. Thus even two-level degree data preserve, rather than exclude, the bad Kummer pattern. Proposition 7.6 gives every bad quartic's exact coefficient conic and an inverse formula recovering its selected Dickson trace. It is therefore a lossless coordinate chart on the same failure fibre, not a new constraint capable of excluding that fibre. Proposition 7.3 also rules out the most direct transfer from Wiedemann's tower: although both targets are norm-one Fermat-order assertions, their recursive drivers have incompatible multiplicative orders and cannot be related by Frobenius. Proposition 7.25 is stronger than a predecessor-only warning. Its degree-sixteen predecessor has order  $(2^{16} - 1)/3$ , and the exact resultant successor has order  $(2^{32} - 1)/3 = F_1 F_2 F_3 F_4$ ; nevertheless the following norm-one quotient has order  $F_5/641$ . Thus even maximal propagation of every old Fermat factor through the successor does not determine its new Fermat component. The proof uses the internal direct product of the base-field and norm-one factors, avoiding the spurious factor-three ambiguity left by an order-lcm argument. Proposition 7.18 now restores the complete actual ancestry: dividing any candidate section by  $X^2 + YX + Y^3$  pushes selected divisibility to two exact coefficient divisibilities at the preceding minimal polynomial, and iteration terminates at  $A_{-1} = Y + 1$ . Thus the preceding countermodel's alternative degree-sixteen factor does not certify divisibility on the actual path. However, the resulting leaf nonvanishing is equivalent to the original selected Conway–Fermat obstruction; the descent itself supplies no coprimality. Proposition 7.19 makes the remaining Euclidean boundary exact. Recursive coordinate splitting is an  $\mathbb{F}_2$ -linear isomorphism from the reduced residue to all  $2^n$  leaf bits, and the complete Euclidean/subresultant tail after the first division is information-equivalent to that residue. Across unrestricted polynomials the raw first quotient and the complete leaf vector are independent. Thus a surviving quotient argument must use a special Fibonacci coefficient identity relating the actual quotient  $Q_{S_d}$  to its residue, rather than formal division or ancestry alone. Proposition 7.20 treats the first special derivative/next-value identity: its reduced defects are exactly the odd and even coefficient blocks of the selected remainder and together reconstruct it. Corollary 7.34 then shows that under failure the first  $A$ -adic quotient digit is the existing choice-free Kummer root divided by  $aA'(a)$ . A surviving coefficient proof must use higher  $A$ -adic digits, a genuinely global relation in the unreduced quotient, or directly force one parity block nonzero. Proposition 7.35 now makes that remaining local boundary exact. The first  $4M$  low coefficients and first  $M$  leading coefficients of the hypothetical quotient agree with the unconditional full-Fermat cofactor, while in the completed local ring  $S_d$  is merely the parameter  $A$  multiplied by a unit. All principal-ideal powers and unmarked completed-local invariants are therefore unchanged by a formal-coordinate automorphism. Fixed-coordinate higher digits, middle/global quotient coefficients, and genuinely ancestry-sensitive relations remain outside this collapse. Corollary 7.36 sharpens both endpoint windows to their exact factor-dependent boundaries. If  $R = 2^{v_2(\ell-1)}$ , the quotients agree through the first  $R$  low and  $R/4$  leading coefficients, then differ by one at both next coefficients. Thus even the first exposed endpoint defects encode only the two-adic valuation of the arithmetic factor; later, middle, cross-endpoint, and fixed- $A$ -coordinate relations remain the genuine quotient frontier. Proposition 7.37 resolves the first two fixed-coordinate Hasse blocks as well. Every canonical digit through  $q_{2R-2}$  is forced by  $A$ , the factor valuation, and the existing first digit: the apparent order- $R$  escape folds back through the unconditional identity  $X^R H = S_d + (1 + T_s)S'_d$ . The first term not reduced by this identity occurs at Hasse order  $2R$  and is a provably nonzero partial-trace prefactor times an explicit half-index Fibonacci value. That value is automatically nonzero in the even half-index branch. In the odd branch its vanishing is equivalent to the unknown selected order dividing  $\Gamma_{n,\ell} = \gcd(d, 3R + 1)$ ; each prime in this gcd is also

a primitive divisor of an explicit smaller base-three generalized Fermat number and is one modulo  $12M$ . This yields the sharper  $(12M + 1)^2$  factor window and an unconditional exclusion when  $M - s \leq n + 3$ . Proposition 7.39 further proves that a vanished odd  $2R$ -jet is followed by an explicit nonzero Hasse jet unless the selected cofactor is exactly  $d = 3R + 1$ ; only that endpoint can remain Hasse-flat. The gcd is one for every published factor in the pinned  $F_{12}, \dots, F_{18}$  screen, so this first free jet is nonzero there even under hypothetical failure. This is still only higher-jet information: the later nonzero jet does not contradict divisibility, and odd-branch gcds at untested levels, later digits, and global relations remain open. Proposition 7.40 pushes the surviving odd collision through the complete selected ancestry to a lower exact resultant. Writing  $v = v_2(s)$ , hypothetical failure forces one explicit iterated quadratic resultant to vanish modulo  $A_{v-1}$ ; a vanished odd order- $2R$  jet forces the factor-independent divisibility  $A_{v-1} \mid D_{n-v}$ , with  $D_j \neq 0$  and  $\deg D_j < 3^j$ . This excludes the collision whenever  $3^{n-v} \leq 2^v$ . It is a genuine lower selected-ancestry obstruction, not a maximal-order theorem: a nonzero Hasse jet remains compatible with  $A_{n-1} \mid S_d$ , and the descended condition is automatic when  $v = 0$ . Proposition 7.83 makes the generalized-Jacobian boundary sharp at the actual  $n = 9$  ordinal degree: a connected four-function Kummer cover and its quarter-turn twist contain points satisfying the complete equivariant Kummer orbit with  $\mathfrak{c}(P)$  an  $\ell$ -th power. What those points lack is precisely the full selected Conway ancestry. Proposition 7.45 saturates that failure fibre under translation by the half-field. It gives the sharper conditional sieve  $\ell \leq \sqrt{q}/2$ ; the exact obstruction is a collision of the selected roots under  $z \mapsto z^{\sqrt{q}} + z$ . Proposition 7.47 factors the collision equation completely. After centering a pair, it says that the ratio of one selected transported value and one fibotomic value lies either in the half-field multiplicative group or in its relative norm-one torus; the two alternatives are disjoint. Equivalently,  $a^{-1}$  must be the value of  $\mathcal{D}_\ell(\tau(s + s^{-1}), 1)$  for a fibotomic  $s$  and an axis scalar  $\tau$ ; full Conway ancestry forces the corresponding  $J$ -value to be  $a_{n-2}$ , but does not evaluate it. The proposition also constructs nonselected failure cosets with the first kind of collision whenever  $\ell \leq \sqrt{q}$ , including the interval  $\sqrt{q}/2 < \ell \leq \sqrt{q}$ . Thus proving the actual Conway ratio avoids both axes remains additional selected content, rather than a consequence of the one-step trace identities or the square-root count. Remark 7.50 proves that the resulting degree and coefficient-gap constraints remain compatible with every composite Fermat case. Proposition 7.48 identifies  $H_m$  with the inversion quotient of the norm-one torus and  $P_{\ell,a}$  with one transported  $\mu_\ell$ -coset. Remark 7.49 gives the exact fibotomic strata of that coset and proves their capacity constraints automatic. Thus the selected divisor is sharper but still unevaluated: additive and support data are realized by many nonselected cosets. For the complete trace-one factorization, Proposition 7.23 proves that  $S_d$  and its cofactor  $J_{N,d}$  give complementary selected resultants whose sum is one. The differentiated factorization supplies an inverse on the nonvanishing side, while the complete fibotomic factorization has exactly one zero selected resultant. Its position is the same unknown order  $\delta_n$ ; factor/cofactor splitting and derivative resultants therefore add the opposite terminal bit, not a second constraint. This rules out that proof class while leaving open an ancestry-sensitive identity involving a polynomial outside the factorization. Proposition 7.24 strengthens this terminal boundary. The differentiated factors are the two canonical CRT support idempotents, independent of the chosen final Bézout or subresultant certificate. Every proper additive trace through the literal quadratic ancestry kills both selected bits; every norm and multiplication determinant returns exactly the original bit; and multiplication trace and discriminant vanish in both branches. Thus terminal Euclidean support-projector linear algebra carries no hidden orientation. Intermediate quotient-polynomial data, before the support collapse, remain outside this no-go. Proposition 7.33 independently closes the shifted-Fibonacci variant. Under failure every  $d$ -shifted block is a scalar copy of the initial

block, and its scalar  $S_{d+1}(a)$  is exactly the existing choice-free lower  $\ell$ -th root. Thus coupled consecutive-value, Cassini, continued-fraction, and post-reduction Euclidean identities add no selected condition. Coefficient-sensitive intermediate quotients before passage to the failure factor remain outside this collapse. For multicomponent zero sets, Proposition 4.6 proves that the Conway class is nontrivial in the primitive-support quotient under its stated two-spine hypotheses, but generation of a composite quotient remains open. Proposition 4.7 turns the exact Hilbert–90 representative into a cyclic-variation order lower bound and proves every primary factor above its explicit complementary threshold; the smaller factors remain unevaluated.

- B. Ordinary odd-spine arm.** Evaluate the selected power-residue class  $\Theta_{q,s}$  of Proposition 5.1 and prove its full projective order. This includes the 11-chain and the 23, 29, 47, 179,  $\dots$  components. The relative-unit formulation and Proposition 5.26 prove every primary factor above its sharper explicit signed-ball threshold, so the unresolved content is confined to the smaller factors. The formerly source-pinned  $m_{359} = 1$  row is now independently certified by the exact Hilbert-root calculation below. Proposition 5.32 generalizes its last reduction to every safe-prime component: nontriviality is one absolute trace bit, but the selected Conway norm still has to be evaluated. Remark 5.19 shows that the unweighted orbit product in global reciprocity still cancels over the  $p - 1$  lifts for every selector from the base cyclotomic field. Proposition 5.20 proves that pure order- $p$  selectors also have zero weight. The mixed Jacobi family escapes both cancellations, but Corollary 5.22 identifies the target with one inverse-Fourier coordinate of its first local phase. Proving that coordinate nonzero remains the theorem. Proposition 5.23 computes the previously implicit wild side at the actual full conductor. Its truncated-log regulator  $A_\lambda$  vanishes exactly when the relative cyclotomic unit is locally a  $p$ -th power; otherwise the local character has conductor exactly  $\lambda^2$ . Global reciprocity then pairs the explicit Stickelberger valuation vector with  $A_\lambda C_{j,\lambda}$ . This is a genuine conductor-sensitive refinement, but it still contains the same selected first Jacobi phase. If all  $A_\lambda$  vanish, the extension is everywhere unramified and is either trivial or, if nontrivial, defines a  $p$ -class-group character. Proposition 5.9 gives an equivalent characteristic-two boundary: failure means that every nonconstant binary section of  $(1+x)^{\mathcal{L}_{r,a}/p}$  vanishes at  $x_{a-1}$ , or, after descent, that the actual Conway minimal polynomial  $P_{\alpha_r}$  divides every selected section polynomial  $R_{e,c}$ . Corollary 5.15 proves that norm and complement symmetry are automatic under this pattern. The remaining theorem is an explicit selected-polynomial nondivisibility, not a generic binomial-step statement. Proposition 5.29 and Remark 5.30 prove this distinction sharp: the simultaneous divisibilities occur in arbitrarily many full-degree irreducible factors while preserving the complete  $r$ -power ancestry, smooth degree, multiplicative orders, and full  $p$ -primary test. They also occur among scalings inside any fixed bottom Kummer coset. Only the actual Conway scaling  $c = 1$  remains unevaluated. Proposition 5.42 proves that this mark cannot be recovered from the surrounding affine orbit. Rescaling a Kummer generator permutes the complete translate set and its full character histogram, while moving the distinguished value from  $\chi(1+x)$  to  $\chi(1+ux)$ . Hence every symmetric affine-orbit invariant is blind; the now-certified  $p = 359$  trace bit is precisely the missing mark  $u = 1$  fixed by Conway ancestry. Proposition 5.31 removes even fixed translation and ordinal-minimal labeling as generic escape routes: one exact degree-twenty model has  $c = 1$ , full Kummer degree, simultaneous binary-section failure, trivial norm-one reciprocity symbols, and a lower element labeled as the least fifth-power nonresidue. It is not a Conway counterexample because that lower element has order five rather than  $\text{ord}(\alpha_5) = 15$ , changing the cyclotomic conductor from 75 to 25. Thus the surviving theorem must use the full multiplicative ancestry and the Conway-selected prime inside that conductor. Proposition 5.10 makes that boundary exact. The full  $r$ -valuation in the Conway Kummer carry gives  $P_{x_s}(X) = P_{\alpha_r}(X^{r^s})$ , so

wrong-conductor models are cyclotomically disjoint from the selected factor. Proposition 5.34 specializes the whole actual dependency chain at the first independent safe-prime row. It gives one degree-3504820, 3447-term polynomial for the marked  $\kappa_{179}$ , and identifies  $m_{359} = 1$  with its exact Euler gcd, Capelli composition, and safe-prime trace. Preserving that polynomial and the mark leaves no countermodel freedom: marked power status is invariant under every finite-field isomorphism. Sparsity therefore packages the selected evaluation but does not decide it. Proposition 5.35 makes a short independent verification target exact: one degree-19580 factor over  $\mathbb{F}_{2^{179}}$ , together with a 171-bit exponentiation at its value one, would give a roughly 438-KB certificate. Proposition 5.36 now instantiates that target in an equivalent root-coordinate form. Its checked-in 438103-byte element satisfies  $y^{179} = A$ ; an exact 19579-step resultant computes  $\theta = N(1 + y)$ , whose 359-phase is nontrivial. This proves the actual selected value  $m_{359} = 1$  by a locally replayable finite calculation, with the algebraic reduction proved in the text. The pinned Peeters commit can be followed through its mixed-radix reductions to the literal marked element  $1 + \kappa_{179}$ ; its  $\{359, 1\}$  row and 177379-second uncached timing are therefore evidence of an executed exact fixed-mark calculation, not merely a copied table entry. The historical repository additionally retains two complete progress traces, with elapsed times 177379 and 177717 seconds, that both finish with a 1743227-term nonidentity result. The checkout's source-audit script verifies the historical blob hashes and both endpoint pairs. Since no final coefficient vector was retained upstream, these runs corroborate rather than constitute the local certificate. Proposition 5.37 gives the exact next-row boundary. The certified 359-class constructs a root  $y^{359} = 1 + \kappa_{179}$ ; its inner degree-179 norm reduces  $m_{719} = 1$  to one Euler test in  $\mathbb{F}_{2^{7029220}}$ . Equivalently, one may take any degree-179 factor of  $(T^{359} + 1)^{179} + (\kappa_{89} + 1)$  over that field and evaluate it at one. The two lower norm shadows are explicit, but twisting inside their fixed norm fibre realizes every 719-phase, so they cannot decide the row. Proposition 5.38 compresses a replayable certificate from a roughly 158-MB factor over  $\mathbb{F}_{2^{7029220}}$  to two 438103-byte payloads over  $\mathbb{F}_{2^{179}}$  and one final Euler test in  $\mathbb{F}_{2^{64261}}$ . The first payload  $a$  is checked exactly by Proposition 5.39, while Proposition 5.40 supplies the monic minimal-polynomial payload, recomputes the crossed norm, and obtains a nontrivial 719-phase. Thus  $m_{719} = 1$  is now proved by a locally replayable exact finite computation together with the paper-level crossed tower reduction. Proposition 5.41 advances the exact runtime boundary once more without a large payload. Here  $\text{ord}_{727}(2) = 121 = 11^2$ , so the actual selected polynomial  $P_{\alpha_{11}}(X^{121})$  has degree only 2420 and five terms. A self-contained binary-polynomial calculation proves it irreducible, checks that translate zero is still a 727-th power, and gives a nontrivial 727-phase at translate one. Hence  $m_{727} = 1$ ; this finite row does not alter the still-open universal ordinary-spine assertion. It is also the deliberate endpoint of row-by-row certification: the implementation keeps  $p = 733$  as an explicit refusal boundary, but the mathematical target from here is a uniform selected-Conway nondivisibility proof, not the next finite row. Proposition 5.11 then pushes any hypothetical failure through each odd-Kummer singleton 0/1-excess dependency satisfying its hypotheses by taking all  $Q$  polynomial sections. This retains the actual selected ancestry without loss, but ends in the same unevaluated selected-factor divisibilities. Multicomponent and excess-4 nodes still require their synchronized orbit resultants, while a two-primary node uses its separate quadratic ancestry.

C. *Cubic arm.* Prove  $C_k$  for every  $k$ : equivalently, for each  $\ell^a \mid \Phi_{3^k}(2)$ , prove

$$v_\ell(\text{ord } \gamma_k) = a.$$

The norm tower propagates all old factors, and Proposition 6.35 settles primary factors above its explicit threshold. The derivative regulator, secant saturation, and reciprocity calculations

all recover the same one-dimensional Kummer class without evaluating it. Proposition 6.17 proves that the apparent  $3^k$ -determinant of the conjugate circular-unit lattice acts as one after reduction and admits equivariant abstract models of every current-factor index, even while preserving maximal lower-field norm. Meanwhile Proposition 6.10 proves that the coefficient recursion does not preserve primitivity even when it preserves irreducibility. Proposition 6.46 makes the remaining assertion purely combinatorial in the cyclic group algebra: the binary subset sums of  $(2^{3^k} - 1)/\ell$  cannot have parity one at every residue except a single prescribed residue. This universal inequality, equivalently direct Kummer nonvanishing at the smaller current factors, remains open. Proposition 6.21 subsumes every Euler derivative or weighted Frobenius-word escape: evaluating the word at two either kills the selected power-residue class or returns an invertible power of exactly that class. It also proves that a scalar identity for  $L(1)$  with a fixed integer Euler quotient is normalization-ill-posed; changing the compatible primitive generator while fixing the selected root rescales  $L(1)$  arbitrarily. Only its vanishing is intrinsic. Proposition 6.41 gives the exact characteristic-zero counterpart: every first-order mixed Jacobi coefficient outside the primitive top component vanishes, and every primitive coefficient is a known nonzero multiple of  $L(1)$ . Thus a single selected Fermat quotient decides the arm:

$$J_1 \not\equiv -1 \pmod{(\zeta_\ell - 1)^2}.$$

Remark 6.42 proves that higher Jacobi congruences cannot bootstrap this first coefficient: their terms have distinct local valuations, so every deeper condition begins only after the still-unknown  $L(1)$  has vanished. Stickelberger ideal data see  $J_1$ , but not this additive first-order congruence for  $J_1 + 1$ . Corollary 6.43 packages it into one weighted product  $\mathcal{R}_\lambda \not\equiv 1 \pmod{(\zeta_\ell - 1)^2}$ , while proving that the corresponding unweighted product is identically flat. Proposition 8.68 gives the exact geometric meaning of this congruence: cubic failure rationalizes  $J[\pi^2]$  on the associated mixed Kummer-curve Jacobian, whereas  $J[\pi]$  is rational unconditionally. The theorem forces only the second ramified layer (higher layers may also lift), and even full rational  $\ell$ -torsion would fit inside the Weil and polarization bounds. Corollary 6.47 proves that the exceptional residue is one of the three lifts  $3^{k+1} - 1, 3^k - 1, 2 \cdot 3^k - 1$ , and isolates all success/failure information in the top  $\Phi_{3^{k+1}}$ -component. Thus every proper cyclic projection is provably blind; its weight and mod-four consequences give exact screens but do not evaluate the selected top coordinate. Corollary 6.48 shows that both the full Frobenius product and the apparent  $\ell$ -th-power exponent test are unconditional: failure is an extra proper-subproduct relation, so those identities alone cannot exclude it. Proposition 6.49 compresses that relation to one reciprocal unit and proves that mask complementation forces the prescribed exceptional residue rather than contradicting it. Proposition 6.50 gives the remaining exact factor-selection problem: failure is the occurrence of the distinguished minimal polynomial of  $\eta_k$ , together with its reciprocal translate, in a binary Wendt gcd. Every Wendt factor already has the required degree and a translation partner, so a closure must exclude this selected pair rather than all such factors. Proposition 6.26 gives a complementary degree-three descent: failure is exactly the existence over the preceding field of one irreducible reciprocal coefficient pair satisfying two explicit Dickson power-sum equations, and the degree- $3\ell$  Capelli composition then factors into  $\ell$  selected irreducible cubics. This is lossless compression, not a contradiction: the two equations are the same missing Kummer fibre. Proposition 6.27 proves that irreducibility is already forced by those two equations. Their geometric fibre is reduced étale of size  $\ell^2$ , with exactly zero or  $\ell$  rational points according as the selected Kummer class succeeds or fails. The universal Jacobian identity, discriminant ratio, reciprocal root, selected  $\mathbb{F}_4$ -norm product, and Berlekamp trace all collapse to that same class. Hence ramification and cubic-discriminant

methods supply no second obstruction capable of emptying the rational fibre. Proposition 6.51 factors the gcd through Singer coset intersections and gives their exact first two moments. In the range  $\ell - 1 < \sqrt{q} + 1/\sqrt{q}$ , every coset contains degree- $n$  Wendt factors, proving that generic factor exclusion is impossible there. Proposition 6.52 descends the selected pair to the degree- $3^{k-1}$  minimal polynomial of  $\eta_{k-1}^2 + \eta_{k-1} + 1$ ; deciding whether that trace fiber lies in the chosen multiplicative coset is the same Kummer evaluation. Corollary 6.53 identifies this quotient selector with  $\gamma_{k-1}^{-2}$ , gives its irreducible cubic recursion, and proves the exact order seam

$$\text{ord}(z_k) = \text{ord}(z_{k-1}) \text{ord}(\eta_k).$$

Thus the descent is a lossless reciprocal form of  $C_k$ , not an induction from  $C_{k-1}$ . Proposition 6.54 carries this descent through the entire selected lower tower. The unique lower  $\ell$ -th roots are norm coherent; after fixing that norm, every geometric coefficient fibre has  $\ell^2$  reduced points, each lower rational fibre is a singleton, and the top rational fibre has zero or  $\ell$  points. Proposition 6.55 gives the complementary literal-trace version at every edge. Its target is the actual reciprocal coefficient pair  $(\eta_{j-1}, \eta_{j-1} + 1)$ ; the Jacobian numerator is  $z_{j-1} \neq 0$ , every lower rational fibre is again a singleton, and the top fibre has size zero or  $\ell$  according to the same Euler bit. Thus the full trace recursion, symmetric source coefficients, discriminants, and their ordinary norms are jointly failure-compatible. A surviving full-ancestry argument must be nonsymmetric across edges or introduce a mixed additive-multiplicative correlation not fixed by these fibres. Proposition 6.63 closes the homogeneous part of that nonsymmetric derivative escape. Every partial canonical square-root Jacobian lies on the single lower-field line generated by  $\gamma_k + 1$ , and every partial Jacobian lies on its square. Hence any nonzero homogeneous expression with arbitrary separate lower-ancestry coefficients has either trivial current weight or an invertible power of the original Euler phase. Proposition 6.64 closes the zero and inhomogeneous part of the same derivative route. The full formal mixed ancestral algebra has the explicit presentation  $K[T_1, \dots, U_k]/I \simeq K[z] = L_k$ , with a unique quadratic normal form; in particular nonzero formal mixed polynomials can vanish, but every such cancellation is generated by the selected translated cubic and the linear/square relations. Since  $z + 1 = \gamma_k$ , the algebra contains the original Euler endpoint rather than a second obstruction. What remains is direct selected nonvanishing of one quadratic remainder or a correlation using genuinely different top-point data. Proposition 6.56 closes the algebraic chosen-root escape. Compatible roots of  $\eta_k, \varepsilon_k, \beta_k, \gamma_k$ , together with every lower selected radical, generate one Kummer extension and span one Kummer-class line rather than independent Kummer classes. Its three top weights are units modulo  $\ell$ , and all root-choice invariants return to the coefficient field. What remains is a nonzero-weight correlation uniform on that one torsor, an external character-sum estimate, or a direct weight-zero evaluation. Proposition 6.57 closes the regular pure-weight version of that escape. Every evaluated nonzero weight has full-field monodromy an invertible power of the original Euler phase, so its field-of-definition test is precisely the same failure bit. The root-orbit product of a mixed polynomial is the weight-zero resultant  $\text{Res}(X^\ell - \tau_k, P)$ : it detects whether some branch vanishes but forgets which one. Thus a mixed regular construction can only move the problem to a selected weight-zero resultant that still requires direct evaluation. Proposition 6.58 identifies the top fibre with one  $T[\ell]$ -torsor and its complete unresolved coordinate with

$$\Omega_{k,\ell} = \frac{z_k^{(q^2+q+1)/\ell}}{r_{k-1}}.$$

Its exact affine Frobenius cycle structure shows that Jacobians, ordinary norms, scalar char-

acters, and linear determinant data erase or merely re-express this one translation class; they do not create a second obstruction. Proposition 6.65 makes the last coordinate one extremal value  $\Theta_\chi(\eta_{k-1}) = 3$ . Other Singer trace fibres genuinely attain 3, so uniform character bounds cannot close the selected fibre. Proposition 6.66 shows that this conclusion is unchanged after retaining every literal lower trace in the Conway chain. The complete selected flag is exactly the same three-point immediate trace fibre, and arbitrary additive multi-Fourier weights formed from those iterated trace coordinates collapse to one immediate-edge character. Combined with the Wendt criterion, failure remains the single equality  $\Theta_\chi(\eta_{k-1}) = 3$ , equivalently  $P_{\eta,k} \mid G_{N/\ell}$ . This closes recursive trace-projector refinements, but not top-point weights depending on non-trace data or genuinely mixed nonsymmetric correlations with separate lower ancestors. Proposition 6.67 makes one genuinely mixed additive–multiplicative correlation exact. The selected trace hyperplane is the projective line spanned by  $1, \gamma_k$ , and the ratio of its character sum to the unshifted hyperplane sum is precisely  $\rho(\gamma_k)^{-1}$ . Thus equality of the two line sums is exactly the original failure condition; all lower-ancestry affine translates span the same line. This recovers the selected phase without evaluating it. Proposition 6.28 also proves that both selected period coordinates have absolute trace zero and are not normal. Thus primitive-normal and normal-Gaussian-period theorems do not apply to the actual cubic sequence; normality is not a missing route to the selected Kummer value. Proposition 6.68 further proves that every point of the preceding Singer set has exactly one current three-point trace fibre. Thus preceding Singer membership and lower-level primitivity do not single out the selected fibre. The full cubic recursion remains rigid up to absolute Frobenius, so this is a sharp obstruction to one-step descent rather than a recursive countermodel. Propositions 6.25 and 6.30 nevertheless supply two selected normal coordinates joined by an explicit determinant-one half-circulant. Their full Frobenius orbits lie in the current Kummer kernel simultaneously, so normal-basis, Moore, and subspace-polynomial data again carry only the original selected phase. Proposition 6.31 makes this compatibility exact at the actual degree (81) and current prime 2593: the Singer, normalized inverse, and half-circulant identities, absolute normality of  $\varepsilon$ , and complete normality of  $\beta$  all survive failure. The model breaks exactly at the next lower selected trace equation. Thus a proof must use the full recursive Conway ancestry, not the nonlinear incidence of these two normal bases alone.

**D. Exceptional arm.** Prove  $D'_k$  for every  $k$ , equivalently

$$\Psi_k \mid \text{ord}(M_k)$$

in Conjecture 8.4. Proposition 8.3 gives Capelli and resultant formulations, while Proposition 8.57 treats the large factors. The antiunit, translated-binomial, semiprimitive, Dirichlet, Dickson, and norm-blindness results show precisely why the remaining smaller factors are not consequences of the present generic identities. Proposition 8.7 transports the complete fibotomic coefficient through the selected cubic ancestry:

$$Z = \pi(M_k) = (\gamma_k^2 + \gamma_k + 1)^{2^{2h/3}-2}.$$

The intermediate  $g_k = \gamma_k^2 + \gamma_k + 1$  form a norm-coherent tower, and the corrected exceptional norm becomes one selected Artin–Schreier root pair  $w, w + 1$ . This still does not transfer  $C_k$  to  $D'_k$ : their current primes are disjoint, every base-field coefficient is automatically an  $\ell$ -th power on the  $D$  line, and the unevaluated datum is the unordered opposite Kummer-class pair  $\{[w], -[w]\}$ . Proposition 8.12 proves that the coarse  $C$ -tower invariants alone do not select this coefficient. The alternative  $Z_k^\circ = \gamma_k^{-1}$  has trace one, full degree, norm-coherent actual

$C$ -period ancestry, and is primitive under  $C_k$ , yet its Artin–Schreier root misses the entire current  $\ell$ -primary factor. This does not disprove a special-form implication  $C_k \Rightarrow D'_k$ ; such an implication remains open and must use the actual formula for  $Z$ . Proposition 8.11 strengthens that boundary. At the actual  $k = 4$  degrees and current prime 163, both a lower coordinate and its cubic child are primitive, the exact equation  $\gamma^3 + \gamma = a$  holds, and consequently all three selected symmetric coordinates of  $g = \gamma^2 + \gamma + 1$  have the exact one-step selected form; nevertheless the resulting Artin–Schreier root is a 163-rd power. The sole mismatch is that  $a$  is not the Conway-selected preceding coordinate. Hence even one exact recursive step plus primitivity cannot transfer  $C$  to  $D$ ; the complete lower ancestry is indispensable. Proposition 8.45 puts that actual formula into the same non-Galois reflection-norm normal form as the singleton-even arm:

$$D'_k \text{ fails at } \ell \iff 2 \in N_{L_D/F(\zeta_\ell)}(L_D^\times), \quad D_\ell^{\text{cl}}(t) = 2 + \frac{5}{x^2 + x - 1}.$$

This reflection field satisfies the Hasse norm principle and weak approximation. Its sole local norm obstruction is the original selected prime over two, so the descent imports the exact  $C$ -coordinate but does not couple the disjoint  $+2$  and  $-2$  current eigenspaces. Proposition 8.13 computes the actual root's exact next lower norm; current  $\ell$ -powering is bijective in that lower field, so even this stronger selected norm remains blind. Proposition 8.38 makes this full-chain boundary exact. After alternating the  $\mathbb{F}_4$ -orientation, the actual exceptional Kummer test is represented by  $u_j = \gamma_j + c_j$ , which obeys a norm-coherent twisted cubic at every selected level. Under hypothetical failure its  $\ell$ -th roots descend uniquely and norm-coherently through the complete Conway chain to the consistent terminal root  $c_0^2 \in \mathbb{F}_4$ . Hence even full recursive ancestry cannot close the arm through norms or corestriction alone; any surviving proof must use a nonlinear interaction between the top Kummer root and the twisted cubic equations. Proposition 8.39 exhausts the resulting unweighted nonlinear coefficient route. The complete alternating ancestry defines a reduced étale degree- $\ell^2$  fibre with nonzero Jacobian numerator  $\gamma_{j-1}$  at every level, singleton lower rational fibres, and a top  $0/\ell$  rational-point dichotomy. Under failure the Capelli composition splits into  $\ell$  distinct selected irreducible cubics, all carrying the same lower root chain. Hence discriminant, resultant, and every root-choice-invariant rational construction remain compatible with failure. Proposition 8.40 closes the formal Kummer-eigenweight escape as well: Fourier transformation recovers exactly the chosen root and its conjugates, while every pure nonzero weight has monodromy an invertible power of the original Euler symbol. Proposition 8.41 closes the formal mixed-weight escape too. The six Fourier modes generate only the standard Kummer algebra and its root-scaling grading separates all weights. Complementary mixed products recover the pseudonorm, while every weight-zero monomial reduces to a power of  $u_k$ . Since the Artin–Schreier coefficient and the complete lower root chain are fixed scalars, they do not couple distinct weights. Cancellation at one bad rational root is projection to a chosen split factor and imports exactly the missing  $\ell$ -root orientation. The remaining theorem is direct nonvanishing of a selected weight-zero phase, not another Fourier or ancestry relation. Proposition 8.42 compresses that phase once more:  $\Omega = 1$  exactly when its immediate cubic trace is one. Along the selected twisted cubic this trace is the terminal value of an explicit third-order recurrence with coefficients  $c_k, c_k, u_{k-1}$ . This replaces the earlier six-term detector by three terms but does not evaluate the terminal recurrence value; it is the same selected Dickson obstruction. Proposition 8.43 proves that this compression is lossless on the entire current phase orbit. Its quadratic norm, orbit-cubic discriminant, irreducibility, and shifted power-sum periodicity all distinguish  $\Omega = 1$  from  $\Omega \neq 1$ , but each is an equivalent restatement of the binary failure condition on  $P_e$ , not a



determination of that value. The complete trace  $\mathcal{S}$  records the phase orbit, while the named binary tests do not refine the detector without evaluating it. Proposition 8.47 then closes the normalization gap between this detector and the earlier arithmetic constructions. With the conductor-five orientation fixed, the weighted-fibre phase is exactly  $\xi_\ell^{L(1)}$ , equivalently the  $G_1^{-1}C_1$  Jacobi phase, the  $4^{-1}\text{ind}_\ell(d_k)$  antiunit phase, the half-block phase, and the inverse selected-sextic phase. Its cubic trace is  $P_e$ . Consequently these are exactly coordinated presentations of one scalar, not independent constraints: the phase-level presentations recover  $L(1)$ , while  $P_e$  records its  $Q$ -Frobenius orbit and the reciprocal polynomial records the corresponding selected factor test. The remaining assertion is still  $L(1) \neq 0$ . Proposition 8.23 adds an exact first-order boundary: failure is simultaneous  $\pi^2$ -flatness of all mixed Jacobi sums of conductor  $15h$ . Proposition 8.24 then proves that this vector has rank one: the single coefficient  $C_1$  is a nonzero Gauss factor times  $\text{ind}_\ell(d_k)$ . Thus the remaining first-order theorem is exactly nonvanishing of the selected antiunit regulator, not a family of independent Jacobi congruences. Proposition 8.25 compresses it further to the canonical four-term cross-resolvent  $J_1J_4/(J_2J_3)$ . It is the unique inversion-invariant, zero-augmentation detector on the conductor-five phases, equivalently the selected  $\ell$ -character tangent of one real quintic Jacobi antiunit. The order-five norm sends that antiunit to one and therefore discards the tangent rather than evaluating it. Proposition 8.26 descends this tangent through the quadratic subfield and identifies it exactly with the alternating determinant  $u_1e_2 - u_2e_1$  of two relative Eisenstein sums. The complete order-five Hasse–Davenport product has only the complementary symmetric first-order coordinate  $u_1/e_1 + u_2/e_2$ ; that relation is compatible with zero determinant and cannot prove the required nonvanishing. Proposition 8.68 also identifies simultaneous flatness with rationality of the second  $\pi$ -torsion layer of the mixed Kummer-curve Jacobian. Its exact size divides the Weil envelope by a wide margin, so point counting and polarization do not exclude the failure fibre. Proposition 8.27 proves that the surviving primitive mixed character cannot enter the semiprimitive Gauss–Jacobi regime: its current  $\ell$ -component requires an odd minus-one exponent while its conductor-five component requires an even one. Hence the explicit semiprimitive tables stop exactly before  $C_1$ ; the mixed decomposition index is in fact  $2(\ell - 1)$ , not two. Corollary 8.28 gives the complementary product obstruction: every equal-weight Hasse–Davenport product is automatically flat, and only the  $\lambda$ -weighted resolvent retains the selected first line. Proposition 8.29 closes the ordinary cyclotomic-derivative escape: its reduction is unconditionally an  $\ell$ -th power, while the  $\lambda^{-1}$ -weighted circular unit has index exactly  $8hL(1)$ . The first generalized Bernoulli value is identically zero for the relevant even character, including levels where  $L(1) \neq 0$ . Thus the weighted construction is an exact detector but not an independent evaluation. Proposition 8.31 gives the parallel characteristic-two boundary: failure is a complement-of-zero submask parity pattern modulo  $15h$ , while its  $\ell$ -th-power consequence is an unconditional idempotent identity. Proposition 8.37 identifies the complete binary exponent before cyclic reduction. Its four alternating  $h$ -bit blocks evaluate at the selected conductor root to  $R_k^{-d}$ , exactly the inverse antiunit phase. Thus the binary submask and conductor-five regulator routes retain the same selected scalar and cannot constrain one another; their common nontriviality is still open. Proposition 8.34 splits the exponent into two Frobenius-related blocks and proves that the resulting quadratic norm equation, its Hilbert–90 condition, and its involution-fixed coefficients are all compatible with failure. The exact surviving coordinate is Proposition 8.35: the selected residue symbol must have nonzero relative trace. Proposition 8.46 then identifies the phase with one explicit reciprocal sextic over  $\mathbb{F}_{2^{3k-1}}$ . Equivalently, failure is the selected divisibility

$$C^2Y^3 + gCY^2 + Y + g^2 \mid D_{(2^{2 \cdot 3^{k-1}} - 2^{3^{k-1}} + 1)/\ell}(Y).$$

The Dickson polynomial factors completely as  $Y(Y+1)^2$  times squares of irreducible cubics. Thus degree, reciprocity, and factor shape admit many false positives. Proposition 8.48 proves that even the exact selected phase has trivial downward norm: corestriction annihilates its full current  $\ell$ -primary line, while the lower field has no  $\ell$ -torsion. The recursive antiunit norm therefore cannot transport a single bit of the required current order. Proposition 8.49 proves more: every admissible cubic has constant coefficient equal to the square of its trace coefficient, exactly as  $G/C^2$  does. Proposition 8.50 then shows that even the full affine depressed form is recursion-neutral: the selected cubic is carried exactly back to  $U^3 + U + g$ . Thus depressing the cubic does not produce an independent discriminant or coefficient obstruction. Proposition 8.51 closes the suggested higher-derivative escape: the Hasse jets through degree five are universal rational functions of the root, and the sixth adds only the common residue  $e \bmod 8$ . More decisively, Proposition 8.52 constructs symbolically  $Q/2 - 1$  distinct ambient Dickson cubics whose full affine depressed equation is the same  $U^3 + U + g$ . The selected cubic is only one member; its unresolved content is placement in  $\mu_e$ , not its affine cubic class. Proposition 8.53 evaluates the two absolute traces

$$\mathrm{Tr}(g/C) = 1, \quad \mathrm{Tr}(C^{-2}) = k \pmod{2},$$

and Proposition 8.54 proves conclusively that their joint value is not an exclusion invariant: at  $k = 4$  a nonselected irreducible factor of the bad Dickson polynomial has the same pair  $(1, 0)$ . Hence no argument using only these two absolute traces can close the exceptional arm. Proposition 8.21 treats the weaker boundedness question directly. Its constant-extension moment proves  $m_\ell < 16(3^k)^4$  throughout this column, but the bound grows with the level. At any fixed Conway-subfield dimension, Riemann–Weil is compatible with full line saturation; obtaining an absolute window still requires uniform nonextremality of that selected moment. The complete-line product is unavailable as a substitute because Proposition 8.20 proves it is unconditionally an  $\ell$ -th power. Proposition 8.19 additionally reduces the D arm, below Euler projection and in the original corrected-norm field, to the selected adjacent-power assertion

$$D = \zeta^4 + \zeta \in (K^\times)^\ell, \quad D + 1 \notin (K^\times)^\ell.$$

Equivalently,  $X^{3\ell} + X^{2\ell} + X^\ell + \Phi_5(\zeta^3)$  is irreducible over  $\mathbb{F}_{4^{3k-1}}$ . Its norm is automatically an  $\ell$ -th power, so this exact full-mark compression still does not prove the required nonvanishing. Remark 8.55 also shows that the cubic and exceptional current groups have coprime orders, so no rational-point homomorphism can transfer arm (C) to arm (D). The remaining theorem is precisely nonvanishing of the recursively selected antiunit index, equivalently exclusion of the recursively selected cubic.

Appendix A records finite computations and external factorization dependencies. Those data are valuable for falsification and audit, but none is used to promote a uniform conjecture to a theorem.

### 9.3 Conclusion

The finite excess problem is therefore no longer diffuse. The global 0/1/4 prediction is equivalent to four selected-value statements, each with an exact algebraic detector and an explicit range of primary factors already controlled by order bounds. The remaining difficulty is not a missing generic estimate: it is the arithmetic of the distinguished Conway-compatible selector in each family. Proving those four nonvanishing assertions would establish the observed rule and hence a uniform bound. Conversely, a failure in any arm would identify the first mechanism by which Lenstra excess escapes  $\{0, 1, 4\}$ .

## A Computational Evidence and Reproducibility

This appendix separates exact finite computation from theorem and from external data. A row is called *locally certified* only when the checkout contains enough exact arithmetic and primality information to verify it without trusting a probable-prime test or an external database. A *source-assisted* row has exact local field arithmetic but relies on an identified external primality or factorization record. A *consistent* row verifies only the currently known factors and does not control an unfactored cofactor. None of these finite labels asserts a uniform theorem in the level.

### A.1 Source-pinned tower data

The implementation vendors the complete OEIS A380496 b-file for the 126 odd primes  $3 \leq p \leq 709$  and diffs it in full against the source-pinned copy. These integer rows determine the implemented Kummer carries below the stated boundary; selected resulting ordinals receive independent value tests. In particular  $m_{359} = 1$  is a source-pinned table value and now also has the independent exact Hilbert-root certificate of Proposition 5.36. The first two rows beyond that b-file are independently certified. The value  $m_{719} = 1$  is backed by the crossed certificate of Propositions 5.39 and 5.40; the self-contained degree-2420 binary-field certificate of Proposition 5.41 gives  $m_{727} = 1$ . The implementation includes both rows separately and therefore crosses  $\alpha_{727}$ . The next unsupported carry is  $\alpha_{733}$ , where it refuses to continue. This is an intentional computational cutoff rather than an invitation to extend the table prime by prime; the remaining ordinary arm is the global assertion in the main-results table.

The contributor-linked extended A380496 auxiliary table [30, 32] also records  $m_{1093} = m_{3511} = 0$ . These are the two known base-two Wieferich rows:  $v_{1093}(2^{364} - 1) = v_{3511}(2^{1755} - 1) = 2$ . The external calculator applies Lenstra’s exact exponent  $(2^E - 1)/p$ , so each zero asserts the full  $p^2$ -primary order, not merely divisibility by  $p$ . The two rows lie outside the approved 126-term b-file, however, and the external repository stores no compact nonidentity remainder certificate for them. They are therefore source-assisted external computations, not locally certified rows or analytical instances of Proposition 4.1.

### A.2 Independent small-row oracle

`experiments/ordinal_excess_probe.py` is a local term-algebra oracle independent of the Rust production tower. It implements the impartial term algebra used by the calculators, multiplicative-order tests for small component fields, and fixed-base root tests for larger selected rows. Its audited sample is:

| $p$ | $m$ | $\mathcal{Q}$ | $p$ -th root? |
|-----|-----|---------------|---------------|
| 7   | 0   | {3}           | yes           |
| 7   | 1   | {3}           | no            |
| 19  | 1   | {9}           | yes           |
| 19  | 4   | {9}           | no            |
| 73  | 1   | {9}           | no            |
| 89  | 1   | {11}          | no            |
| 47  | 1   | {23}          | no            |
| 179 | 1   | {89}          | no            |

These are exact finite checks, not evidence for an all-level maximal-order theorem.

### A.3 Cubic finite evidence

Proposition 6.73 proves  $C_1, C_2, C_3$  without an order census. The maintained script `experiments/cyclotomic_3k_family.py` then performs exact factor reconstruction and power tests.

For  $k = 4, 5$ , every required factor is locally proved prime and the script certifies that  $\gamma_k$  contains the full primary part of  $2^{3^k} - 1$ ; hence  $C_4$  and  $C_5$  are locally certified. At  $k = 6$ , the same exact field computations cover all listed factors, but the 42- and 90-digit factors are only probable primes under the local Miller–Rabin path. FactorDB records them as proven primes, but no certificates are vendored here. Thus  $C_6$  is *source-assisted*, not locally certified.

At  $k = 7, 8$ , every currently known prime factor of  $\Phi_{3^k}(2)$  passes the required order test, while unfactored composite cofactors remain. These levels are only consistent with  $C_k$ . The distinctions are summarized as follows:

| levels            | status            | basis                                           |
|-------------------|-------------------|-------------------------------------------------|
| $1 \leq k \leq 3$ | theorem           | prime-step argument                             |
| $k = 4, 5$        | locally certified | exact factorization, primality, and order       |
| $k = 6$           | source-assisted   | exact order; external primality for P42 and P90 |
| $k = 7, 8$        | consistent        | known factors only; unfactored cofactors        |

### A.4 Exceptional finite evidence

The maintained script `experiments/exception_column_m4.py` checks the corrected norm, the projected antiunit, and the current-factor power tests. The fully factored levels  $k = 2, \dots, 5$  are locally certified and give  $m_p = 4$  for every prime with

$$f(p) \in \{18, 54, 162, 486\}.$$

At  $k = 6$ , the field computations are exact but the 78-digit factor is only a local probable prime; FactorDB records it as proven. The complete  $k = 6$  conclusion is therefore source-assisted unless a primality certificate is supplied. At  $k = 7, 8$ , all currently known prime factors pass, but the unfactored cofactors and their multiplicities remain unknown; these checks are merely consistent with  $D'_k$ .

The small anchor rows  $m_{19} = m_{163} = m_{1459} = 4$  are reproduced rather than assumed. An independent Artin–Schreier compositum model also verifies  $\sigma(4) = 5$ , Proposition 8.1, and the direct power test on the smallest factor of each checked level.

### A.5 Reproducibility

The four maintained entry points are `experiments/ordinal_excess_probe.py`, `experiments/cyclotomic_3k_family.py`, and `experiments/exception_column_m4.py`, together with the focused certificate `experiments/cubic_two_normal_countermodel.py`. The additional programs under `experiments/excess/` are exploratory and are not evidence for the theorem statements in this paper. Those statements depend only on the displayed proofs and cited results; the maintained programs serve finite certification, falsification, and audit.

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